

37 High-Resolution Timer (HRTIM)

37.1 Introduction

The high-resolution timer can generate up to 10 digital signals with highly accurate timings. It is primarily intended to drive power conversion systems such as switch mode power supplies or lighting systems, but can be of general purpose usage, whenever a very fine timing resolution is expected.

Its modular architecture allows to generate either independent or coupled waveforms. The wave-shape is defined by self-contained timings (using counters and compare units) and a broad range of external events, such as analog or digital feedbacks and synchronization signals. This allows to produce a large variety of control signal (PWM, phase-shifted, constant Ton,...) and address most of conversion topologies.

For control and monitoring purposes, the timer has also timing measure capabilities and links to built-in ADC and DAC converters. Last, it features light-load management mode and is able to handle various fault schemes for safe shut-down purposes.

37.2 Main features

- Multiple timing units
 - Full-resolution available on all outputs, possibility to adjust duty-cycle, frequency and pulse width in triggered one-pulse mode
 - 6 16-bit timing units (each one with an independent counter and 4 compare units)
 - 10 outputs that can be controlled by any timing unit, up to 32 set/reset sources per channel
 - Modular architecture to address either multiple independent converters with 1 or 2 switches or few large multi-switch topologies
- Up to 10 external events, available for any timing unit
 - Programmable polarity and edge sensitivity
 - 5 events with a fast asynchronous mode
 - 5 events with a programmable digital filter
 - Spurious events filtering with blanking and windowing modes
- Multiple links to built-in analog peripherals
 - 4 triggers to ADC converters
 - 3 triggers to DAC converters
 - 3 comparators for analog signal conditioning
- Versatile protection scheme
 - 5 fault inputs can be combined and associated to any timing unit
 - Programmable polarity, edge sensitivity, and programmable digital filter
 - dedicated delayed protections for resonant converters
- Multiple HRTIM instances can be synchronized with external synchronization inputs/outputs
- Versatile output stage
 - Full-resolution Deadtime insertion
 - Programmable output polarity
 - Chopper mode
- Burst mode controller to handle light-load operation synchronously on multiple converters
- 7 interrupt vectors, each one with up to 14 sources
- 6 DMA requests with up to 14 sources, with a burst mode for multiple registers update

37.3 Functional description

37.3.1 General description

The HRTIM can be partitioned into several sub entities:

- The master timer
- The timing units (Timer A to Timer E)
- The output stage
- The burst mode controller
- An external event and fault signal conditioning logic that is shared by all timers
- The system interface

The master timer is based on a 16-bit up counter. It can set/reset any of the 10 outputs via 4 compare units and it provides synchronization signals to the 5 timer units. Its main purpose is to have the timer units controlled by a unique source. An interleaved buck converter is a typical application example where the master timer manages the phase-shifts between the multiple units.

The timer units are working either independently or coupled with the other timers including the master timer. Each timer contains the controls for two outputs. The outputs set/reset events are triggered either by the timing units compare registers or by events coming from the master timer, from the other timers or from external events.

The output stage has several duties

- Addition of deadtime when the 2 outputs are configured in complementary PWM mode
- Addition of a carrier frequency on top of the modulating signal
- Management of fault events, by asynchronously asserting the outputs to a predefined safe level

The burst mode controller can take over the control of one or multiple timers in case of light-load operation. The burst length and period can be programmed, as well as the idle state of the outputs.

The external event and fault signal conditioning logic includes:

- The input selection MUXes (for instance for selecting a digital input or an on-chip source for a given external event channel)
- Polarity and edge-sensitivity programming
- Digital filtering (for 5 channels out of 10)

The system interface allows the HRTIM to interact with the rest of the MCU:

- Interrupt requests to the CPU
- DMA controller for automatic accesses to/from the memories, including an HRTIM specific burst mode
- Triggers for the ADC and DAC converters

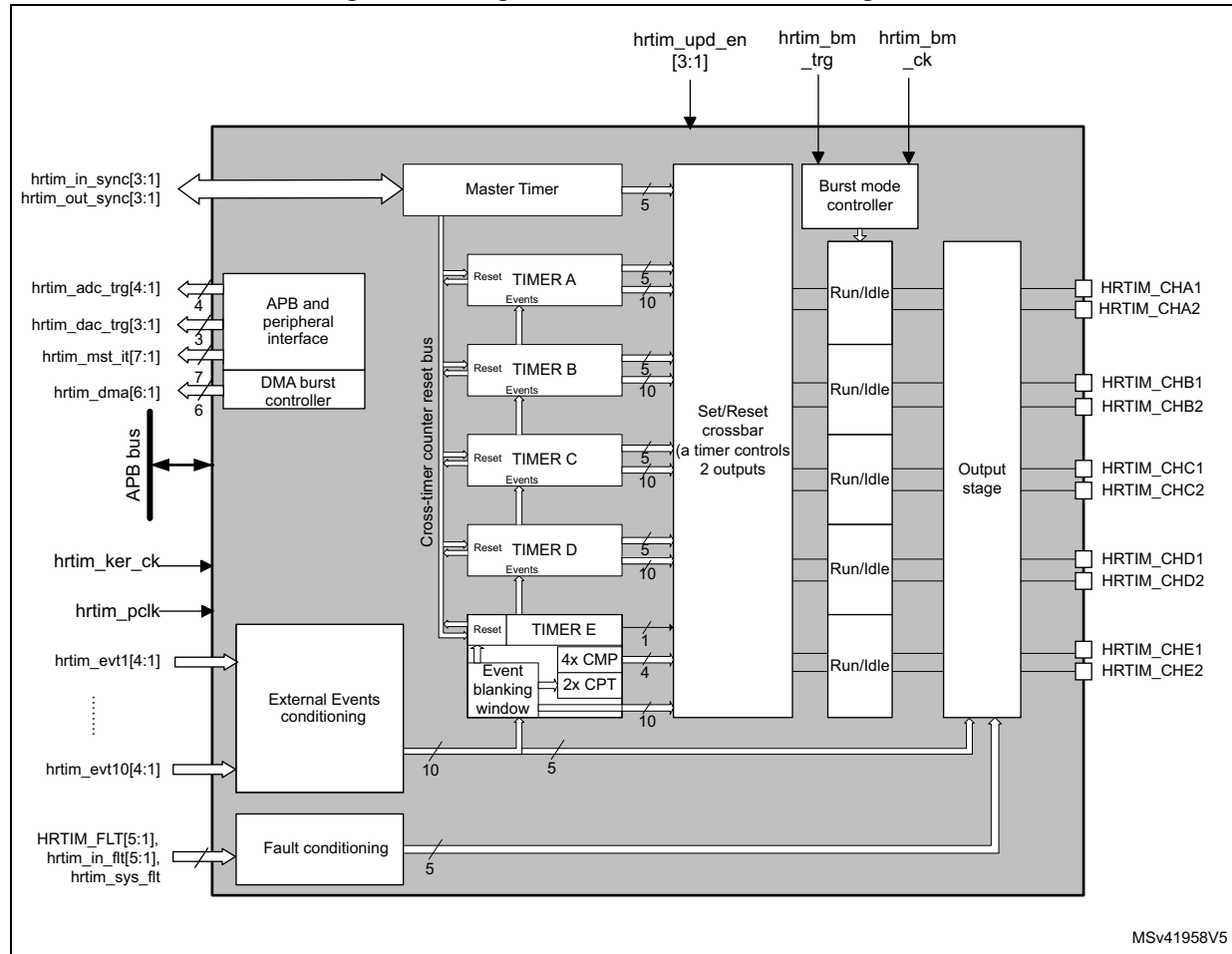
The HRTIM registers are split into 7 groups:

- Master timer registers
- Timer A to Timer E registers
- Common registers for features shared by all timer units

Note: As a writing convention, references to the 5 timing units in the text and in registers are generalized using the “x” letter, where x can be any value from A to E.

The block diagram of the timer is shown in [Figure 281](#).

Figure 281. High-resolution timer block diagram



37.3.2 HRTIM pins and internal signals

The table here below summarizes the HRTIM inputs and outputs, both on-chip and off-chip.

Table 299. HRTIM Input/output summary

Signal name	Signal type	Description
HRTIM_CHA1, HRTIM_CHA2, HRTIM_CHB1, HRTIM_CHB2, HRTIM_CHC1, HRTIM_CHC2, HRTIM_CHD1, HRTIM_CHD2, HRTIM_CHE1, HRTIM_CHE2	Outputs	Main HRTIM timer outputs. They can be coupled by pairs (HRTIM_CHx1 & HRTIM_CHx2) with deadtime insertion or work independently.
HRTIM_FLT[5:1], hrtim_inflt[5:1]	Digital input	Fault inputs: immediately disable the HRTIM outputs when asserted (5 on-chip inputs and 5 off-chip HRTIM_FLTx inputs).
hrtim_sysflt	Digital input	System fault gathering MCU internal fault events (Clock security system, SRAM parity error, Cortex®-M7 lockup (HardFault), PVD output).
hrtim_in_sync[3:1]	Digital Input	Synchronization inputs to synchronize the whole HRTIM with other internal or external timer resources: hrtim_in_sync1: reserved hrtim_in_sync2: the source is a regular TIMx timer (via on-chip interconnect) hrtim_in_sync3: the source is an external HRTIM (via the HRTIM_SCIN input pins)
hrtim_out_sync[2:1]	Digital output	The purpose of this output is to cascade or synchronize several HRTIM instances, either on-chip or off-chip: hrtim_out_sync1: reserved hrtim_out_sync2: the destination is an off-chip HRTIM or peripheral (via HRTIM_SCOUT output pins)
hrtim_evt1[4:1] hrtim_evt2[4:1] hrtim_evt3[4:1] hrtim_evt4[4:1] hrtim_evt5[4:1] hrtim_evt6[4:1] hrtim_evt7[4:1] hrtim_evt8[4:1] hrtim_evt9[4:1] hrtim_evt10[4:1]	Digital input	External events. Each of the 10 events can be selected among 4 sources, either on-chip (from other built-in peripherals: comparator, ADC analog watchdog, TIMx timers, trigger outputs) or off-chip (HRTIM_EEVx input pins)
hrtim_upd_en[3:1]	Digital input	HRTIM register update enable inputs (on-chip interconnect) trigger the transfer from shadow to active registers

Table 299. HRTIM Input/output summary (continued)

Signal name	Signal type	Description
hrtim_bm_trg	Digital input	Burst mode trigger event (on-chip interconnect)
hrtim_bm_ck[4:1]	Digital input	Burst mode clock (on-chip interconnect)
hrtim_adc_trg[4:1]	Digital output	ADC start of conversion triggers
hrtim_dac_trg[3:1]	Digital output	DAC conversion update triggers
hrtim_mst_it[7:1]	Digital output	Interrupt requests
hrtim_dma[6:1]	Digital output	DMA requests
hrtim_pclk	Digital input	APB clock
hrtim_ker_ck	Digital input	HRTIM kernel clock (hereafter mentioned as f_{HRTIM}).

37.3.3 Clocks

The HRTIM must be supplied by the t_{HRTIM} system clock to offer a full resolution. All clocks present in the HRTIM are derived from this reference clock.

Definition of terms

f_{HRTIM} : main HRTIM clock (hrtim_ker_ck). All subsequent clocks are derived and synchronous with this source.

f_{DTG} : deadtime generator clock. For convenience, only the t_{DTG} period ($t_{\text{DTG}} = 1/f_{\text{DTG}}$) is used in this document.

f_{CHPFREQ} : chopper stage clock source.

$f_{1\text{STPW}}$: clock source defining the length of the initial pulse in chopper mode. For convenience, only the $t_{1\text{STPW}}$ period ($t_{1\text{STPW}} = 1/f_{1\text{STPW}}$) is used in this document.

f_{BRST} : burst mode controller counter clock.

f_{SAMPLING} : clock needed to sample the fault or the external events inputs.

f_{FLTS} : clock derived from f_{HRTIM} which is used as a source for f_{SAMPLING} to filter fault events.

f_{EEVS} : clock derived from f_{HRTIM} which is used as a source for f_{SAMPLING} to filter external events.

f_{pclk} (hrtim_pclk): APB bus clock, needed for register read/write accesses

Timer clock and prescaler

Each timer in the HRTIM has its own individual clock prescaler, which allows you to adjust the timer resolution. (See [Table 300](#)).

Table 300. Timer resolution and min. PWM frequency for $f_{\text{HRTIM}} = 400 \text{ MHz}$

CKPSC[2:0] ⁽¹⁾	Prescaling ratio	f_{COUNTER}	Resolution	Min PWM frequency
101	1	400 MHz	2.5 ns	6.1 kHz
110	2	400/2 MHz = 200 MHz	5 ns	3.05 kHz
111	4	400/4 MHz = 100MHz	10 ns	1.5 kHz

1. CKPSC[2:0] values from 000 to 100 are reserved.

The Full-resolution is available for edge positioning, PWM period adjustment and externally triggered pulse duration.

Initialization

At start-up, it is mandatory to initialize first the prescaler bitfields before writing the compare and period registers. Once the timer is enabled (MCEN or TxCEN bit set in the HRTIM_MCR register), the prescaler cannot be modified.

When multiple timers are enabled, the prescalers are synchronized with the prescaler of the timer that was started first.

Warning: It is possible to have different prescaling ratios in the master and TIMA..E timers only if the counter and output behavior does not depend on other timers' information and signals. It is mandatory to configure identical prescaling ratios in these timers when one of the following events is propagated from one timing unit (or master timer) to another: output set/reset event, counter reset event, update event, external event filter or capture triggers. Prescaler factors not equal will yield to unpredictable results.

Deadtime generator clock

The deadtime prescaler is supplied by $f_{\text{HRTIM}} / 8 / 2^{(\text{DTPRSC}[2:0])}$, programmed with DTPRSC[2:0] bits in the HRTIM_DTxR register.

t_{DTG} ranges from 2.5 ns to 20 ns for $f_{\text{HRTIM}} = 400 \text{ MHz}$.

Chopper stage clock

The chopper stage clock source f_{CHPFRQ} is derived from f_{HRTIM} with a division factor ranging from 16 to 256, so that $1.56 \text{ MHz} \leq f_{\text{CHPFRQ}} \leq 25 \text{ MHz}$ for $f_{\text{HRTIM}} = 400 \text{ MHz}$.

$t_{1\text{STPW}}$ is the length of the initial pulse in chopper mode, programmed with the STRPW[3:0] bits in the HRTIM_CHPxR register, as follows:

$$t_{1\text{STPW}} = (\text{STRPW}[3:0] + 1) \times 16 \times t_{\text{HRTIM}}$$

It uses $f_{\text{HRTIM}} / 16$ as clock source (25 MHz for $f_{\text{HRTIM}} = 400 \text{ MHz}$).

Burst Mode Prescaler

The burst mode controller counter clock f_{BRST} can be supplied by several sources, among which one is derived from f_{HRTIM} .

In this case, f_{BRST} ranges from f_{HRTIM} to $f_{\text{HRTIM}} / 32768$ (12.2 kHz for $f_{\text{HRTIM}} = 400 \text{ MHz}$).

Fault input sampling clock

The fault input noise rejection filter has a time constant defined with f_{SAMPLING} which can be either f_{HRTIM} or f_{FLTS} .

f_{FLTS} is derived from f_{HRTIM} and ranges from 400 MHz to 50 MHz for $f_{\text{HRTIM}} = 400 \text{ MHz}$.

External Event input sampling clock

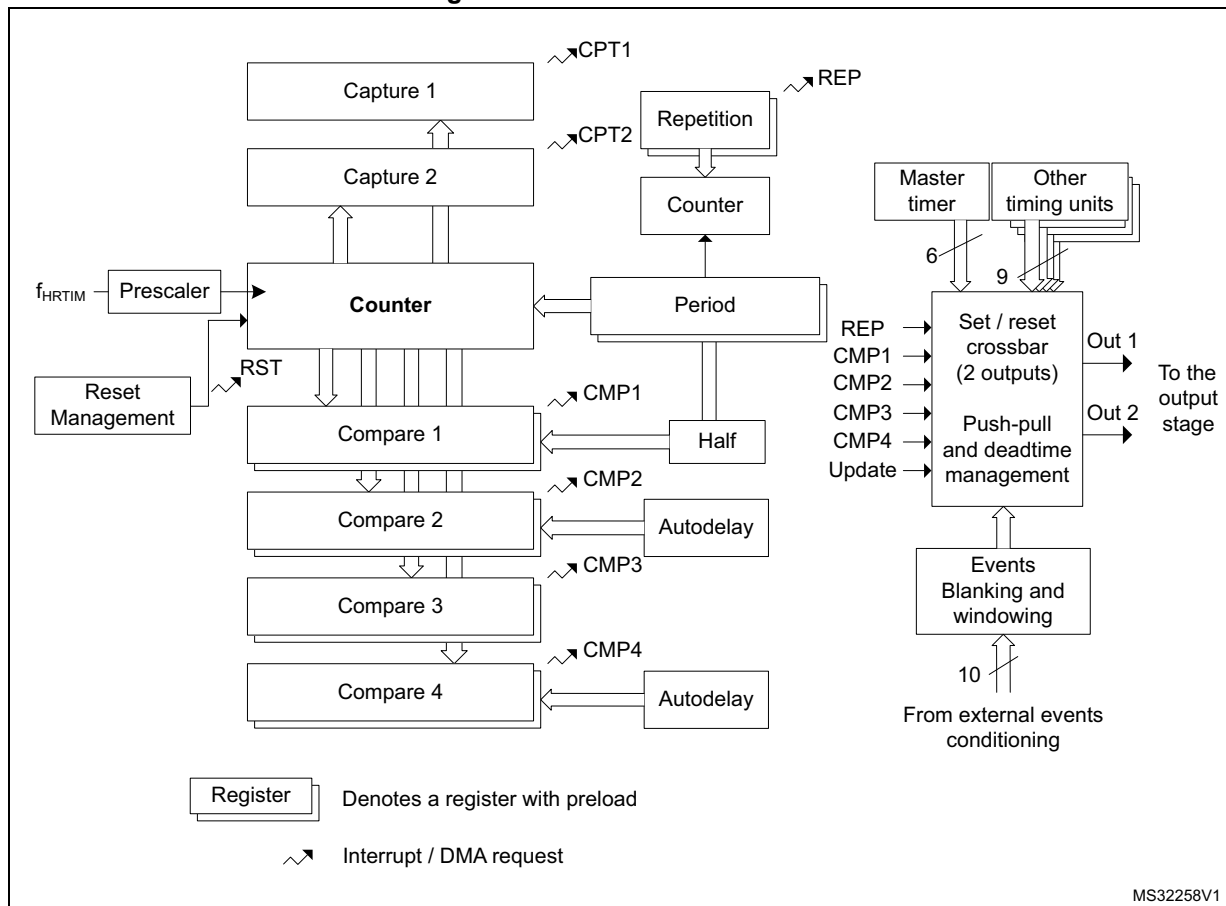
The fault input noise rejection filter has a time constant defined with f_{SAMPLING} which can be either f_{HRTIM} or f_{EEVS} .

f_{EEVS} is derived from f_{HRTIM} and ranges from 400 MHz to 50 MHz for $f_{\text{HRTIM}} = 400 \text{ MHz}$.

37.3.4 Timer A..E timing units

The HRTIM embeds 5 identical timing units made of a 16-bit up-counter with an auto-reload mechanism to define the counting period, 4 compare and 2 capture units, as per [Figure 282](#). Each unit includes all control features for 2 outputs, so that it can operate as a standalone timer.

Figure 282. Timer A..E overview



The period and compare values must be within a lower and an upper limit related to the high-resolution implementation and listed in [Table 301](#):

- The minimum value must be greater than or equal to 3 periods of the f_{HRTIM} clock
- The maximum value must be less than or equal to $0xFFFF - 1$ periods of the f_{HRTIM} clock

Table 301. Period and Compare registers min and max values

CKPSC[2:0] value ⁽¹⁾	Min	Max
≥ 5	0x0003	0xFFFFD

1. CKPSC[2:0] values < 5 are reserved.

Note: A compare value greater than the period register value will not generate a compare match event.

Counter operating mode

Timer A..E can operate in continuous (free-running) mode or in single-shot manner where counting is started by a reset event, using the CONT bit in the HRTIM_TIMxCR control register. An additional RETRIG bit allows you to select whether the single-shot operation is retriggerable or non-retriggerable. Details of operation are summarized on [Table 302](#) and on [Figure 283](#) and [Figure 284](#).

Table 302. Timer operating modes

CONT	RETRIG	Operating mode	Start / Stop conditions Clocking and event generation
0	0	Single-shot Non-retriggerable	Setting the TxEN bit enables the timer but does not start the counter. A first reset event starts the counting and any subsequent reset is ignored until the counter reaches the PER value. The PER event is then generated and the counter is stopped. A reset event re-starts the counting operation from 0x0000.
0	1	Single-shot Retriggerable	Setting the TxEN bit enables the timer but does not start the counter. A reset event starts the counting if the counter is stopped, otherwise it clears the counter. When the counter reaches the PER value, the PER event is generated and the counter is stopped. A reset event re-starts the counting operation from 0x0000.
1	X	Continuous mode	Setting the TxEN bit enables the timer and starts the counter simultaneously. When the counter reaches the PER value, it rolls-over to 0x0000 and resumes counting. The counter can be reset at any time.

The TxEN bit can be cleared at any time to disable the timer and stop the counting.

Figure 283. Continuous timer operation

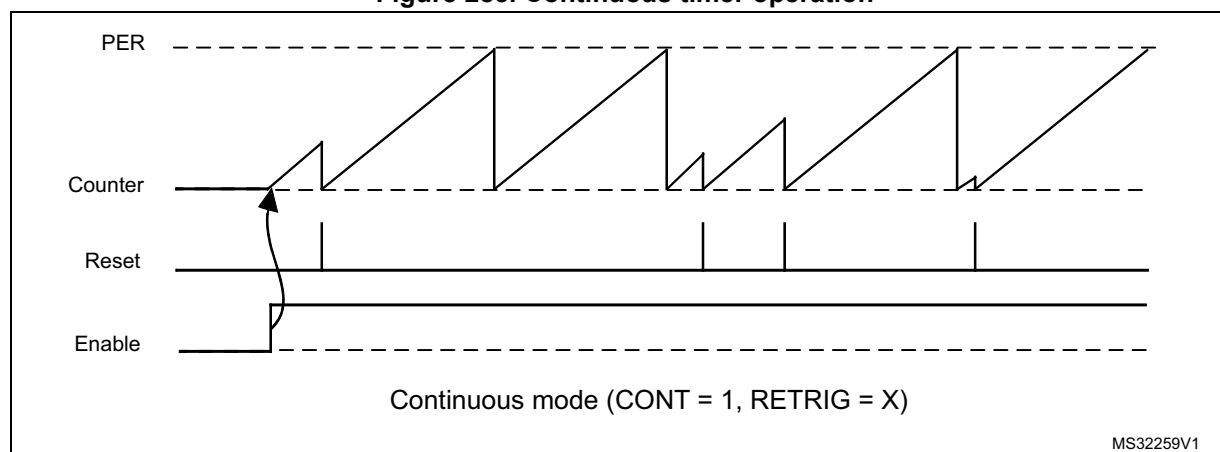
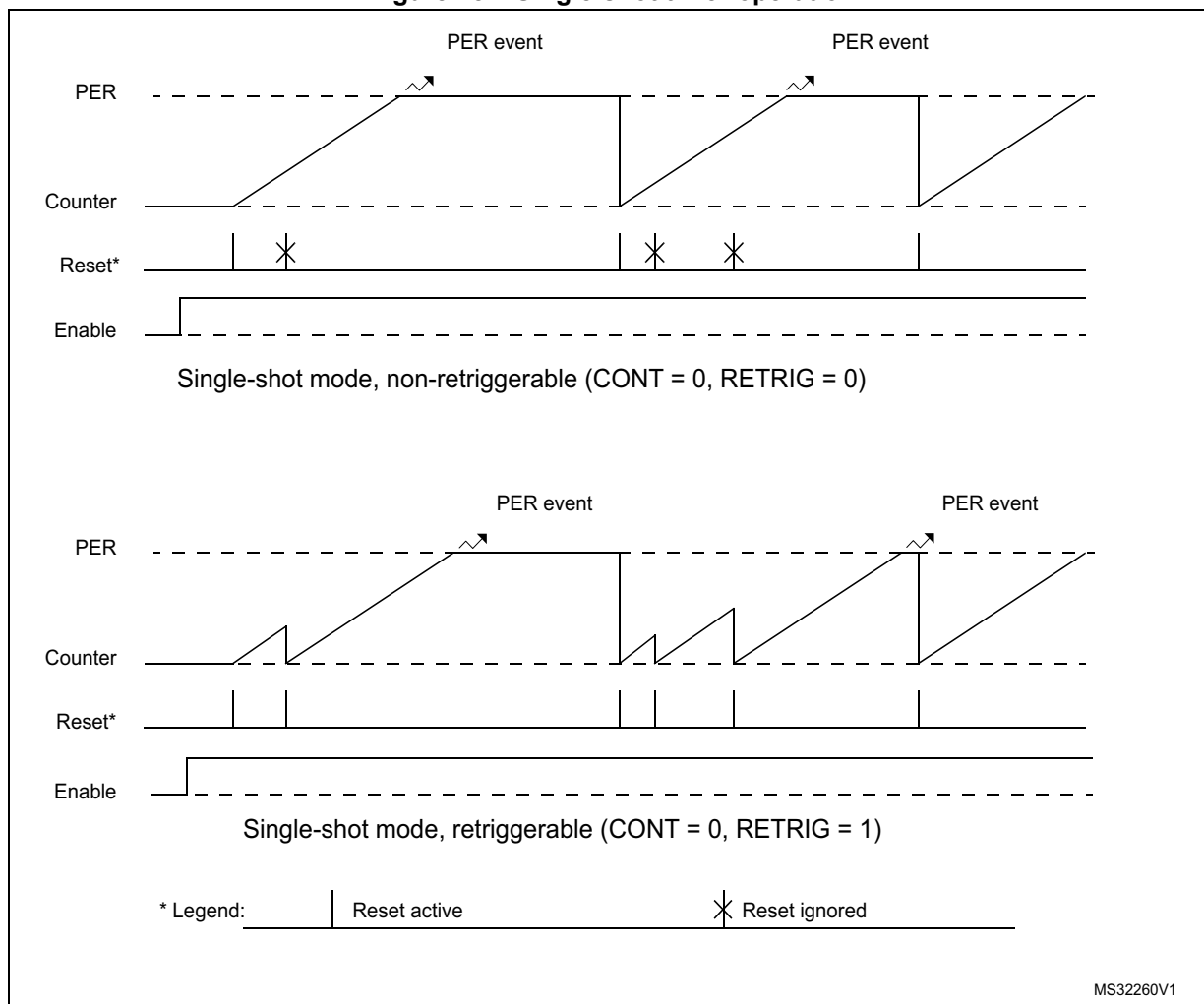


Figure 284. Single-shot timer operation



Roll-over event

A counter roll-over event is generated when the counter goes back to 0 after having reached the period value set in the HRTIM_PERxR register in continuous mode.

This event is used for multiple purposes in the HRTIM:

- To set/reset the outputs
- To trigger the register content update (transfer from preload to active)
- To trigger an IRQ or a DMA request
- To serve as a burst mode clock source or a burst start trigger
- as an ADC trigger
- To decrement the repetition counter

If the initial counter value is above the period value when the timer is started, or if a new period is set while the counter is already above this value, the counter is not reset: it will overflow at the maximum period value and the repetition counter will not decrement.

Timer reset

The reset of the timing unit counter can be triggered by up to 30 events that can be selected simultaneously in the HRTIM_RSTxR register, among the following sources:

- The timing unit: Compare 2, Compare 4 and Update (3 events)
- The master timer: Reset and Compare 1..4 (5 events)
- The external events EXTEVNT1..10 (10 events)
- All other timing units (e.g. Timer B..E for timer A): Compare 1, 2 and 4 (12 events)

Several events can be selected simultaneously to handle multiple reset sources. In this case, the multiple reset requests are ORed. When 2 counter reset events are generated within the same f_{HRTIM} clock cycle, the last counter reset is taken into account.

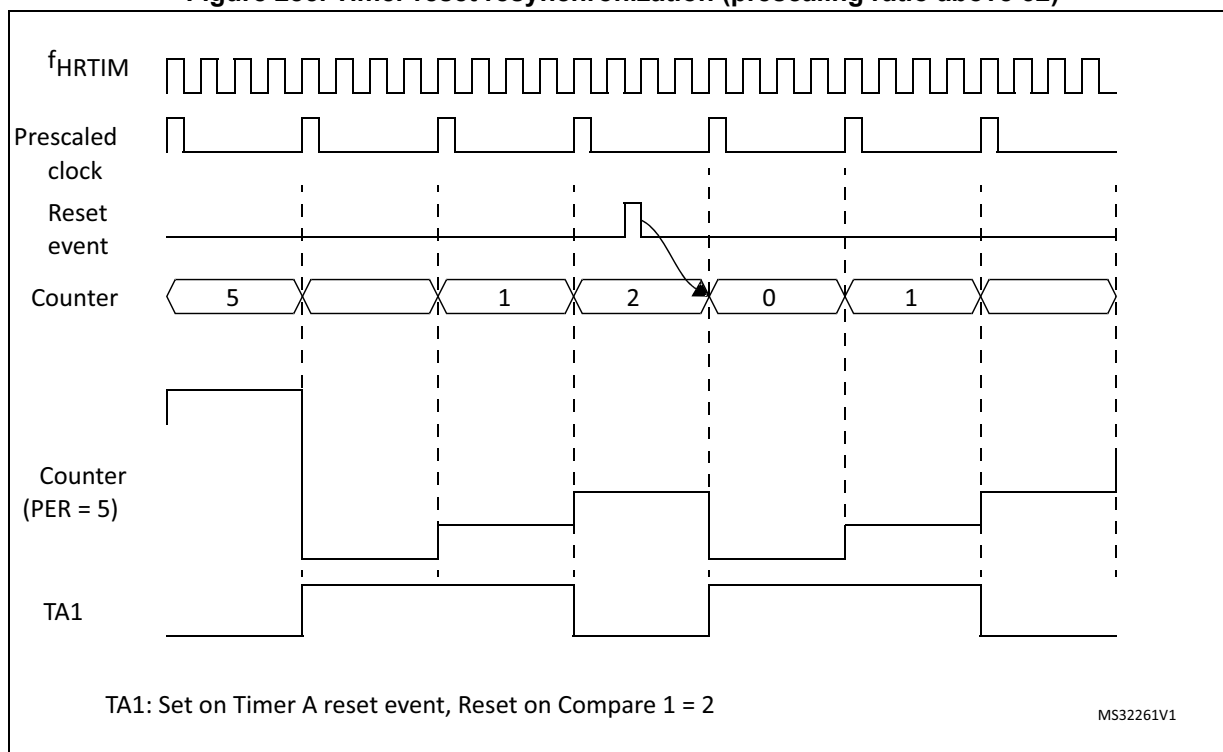
Additionally, it is possible to do a software reset of the counter using the TxRST bits in the HRTIM_CR2 register. These control bits are grouped into a single register to allow the simultaneous reset of several counters.

The reset requests are taken into account only once the related counters are enabled (TxCEN bit set).

When the f_{HRTIM} clock prescaling ratio is above 1, the counter reset event is delayed to the next active edge of the prescaled clock. This allows to maintain a jitterless waveform generation when an output transition is synchronized to the reset event (typically a constant Ton time converter).

[Figure 285](#) shows how the reset is handled for a clock prescaling ratio of 4 (f_{HRTIM} divided by 4).

Figure 285. Timer reset resynchronization (prescaling ratio above 32)



Repetition counter

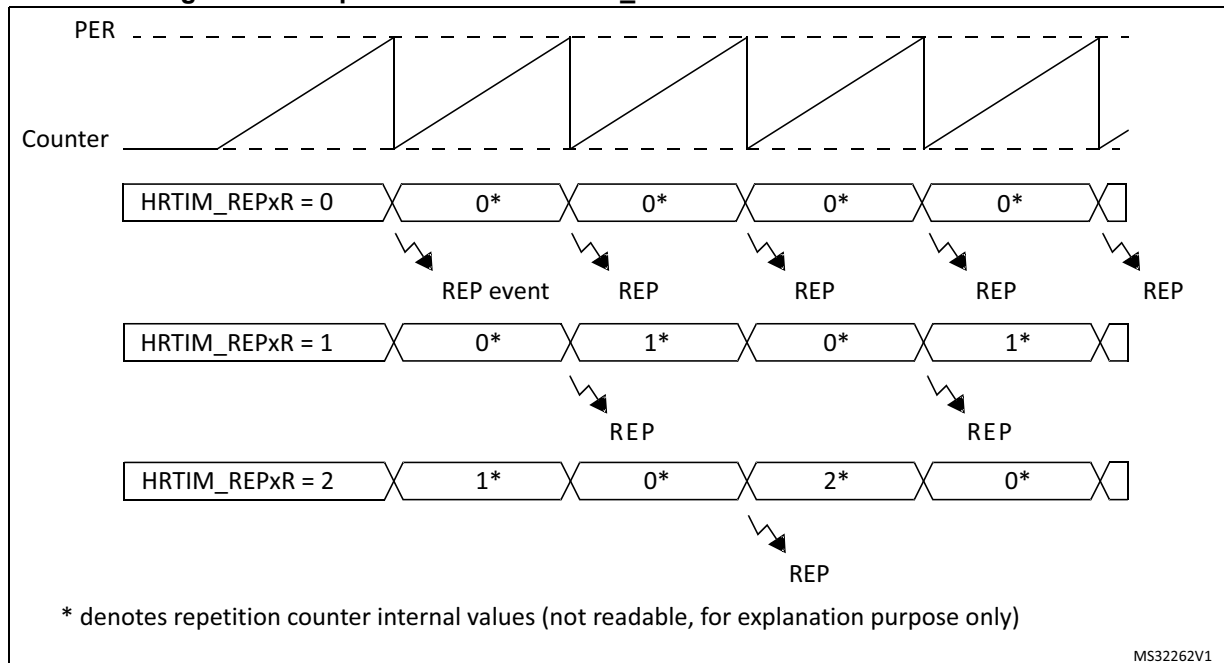
A common software practice is to have an interrupt generated when the period value is reached, so that the maximum amount of time is left for processing before the next period begins. The main purpose of the repetition counter is to adjust the period interrupt rate and off-load the CPU by decoupling the switching frequency and the interrupt frequency.

The timing units have a repetition counter. This counter cannot be read, but solely programmed with an auto-reload value in the HRTIM_REPxR register.

The repetition counter is initialized with the content of the HRTIM_REPxR register when the timer is enabled (TXCEN bit set). Once the timer has been enabled, any time the counter is cleared, either due to a reset event or due to a counter roll-over, the repetition counter is decreased. When it reaches zero, a REP interrupt or a DMA request is issued if enabled (REPIE and REPDE bits in the HRTIM_DIER register).

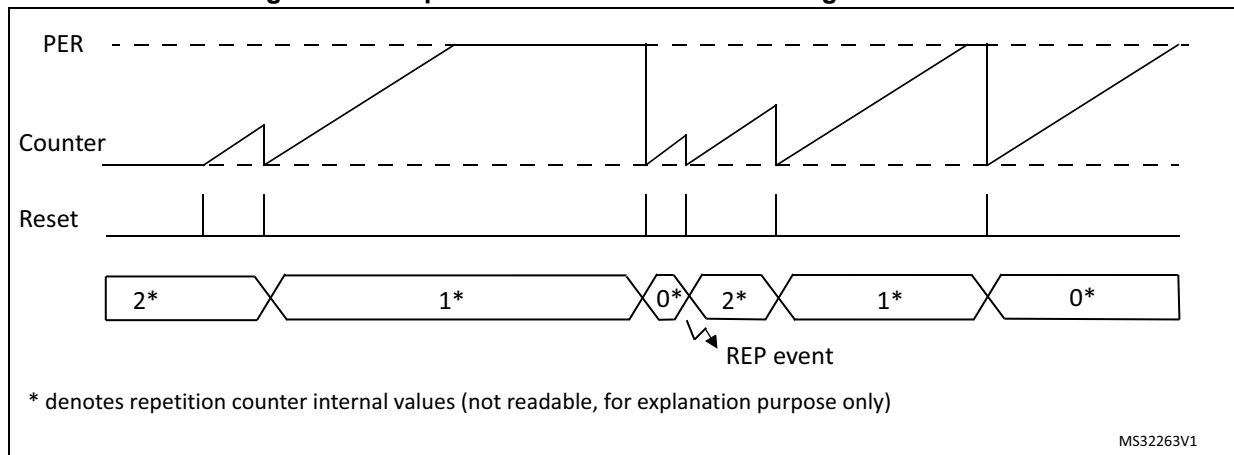
If the HRTIM_REPxR register is set to 0, an interrupt is generated for each and every period. For any value above 0, a REP interrupt is generated after (HRTIM_REPxR + 1) periods. [Figure 286](#) presents the repetition counter operation for various values, in continuous mode.

Figure 286. Repetition rate vs HRTIM_REPxR content in continuous mode



The repetition counter can also be used when the counter is reset before reaching the period value (variable frequency operation) either in continuous or in single-shot mode ([Figure 287](#) here-below). The reset causes the repetition counter to be decremented, at the exception of the very first start following counter enable (TxCEN bit set).

Figure 287. Repetition counter behavior in single-shot mode



A reset or start event from the `hrtim_in_sync[3:1]` source causes the repetition to be decremented as any other reset. However, in SYNCIN-started single-shot mode (SYNCSTRTx bit set in the HRTIM_TIMxCR register), the repetition counter will be decremented only on the 1st reset event following the period. Any subsequent reset will not alter the repetition counter until the counter is re-started by a new request on `hrtim_in_sync[3:1]` inputs.

Set / reset crossbar

A “set” event correspond to a transition to the output active state, while a “reset” event corresponds to a transition to the output inactive state.

The polarity of the waveform is defined in the output stage to accommodate positive or negative logic external components: an active level corresponds to a logic level 1 for a positive polarity ($POLx = 0$), and to a logic level 0 for a negative polarity ($POLx = 1$).

Each of the timing units handles the set/reset crossbar for two outputs. These 2 outputs can be set, reset or toggled by up to 32 events that can be selected among the following sources:

- The timing unit: Period, Compare 1..4, register update (6 events)
- The master timer: Period, Compare 1..4, HRTIM synchronization (6 events)
- All other timing units (e.g. Timer B..E for timer A): TIMEVNT1..9 (9 events described in [Table 303](#))
- The external events EXTEVNT1..10 (10 events)
- A software forcing (1 event)

The event sources are ORed and multiple events can be simultaneously selected.

Each output is controlled by two 32-bit registers, one coding for the set (HRTIM_SETxyR) and another one for the reset (HRTIM_RSTxyR), where x stands for the timing unit: A..E and y stands for the output 1or 2 (e.g. HRTIM_SETA1R, HRTIM_RSTC2R,...).

If the same event is selected for both set and reset, it will toggle the output. It is not possible to toggle the output state more than one time per t_{HRTIM} period: in case of two consecutive toggling events within the same cycle, only the first one is considered.

The set and reset requests are taken into account only once the counter is enabled (TxCEN bit set), except if the software is forcing a request to allow the prepositioning of the outputs at timer start-up.

[Table 303](#) summarizes the events from other timing units that can be used to set and reset the outputs. The number corresponds to the timer events (such as TIMEVNTx) listed in the register, and empty locations are indicating non-available events.

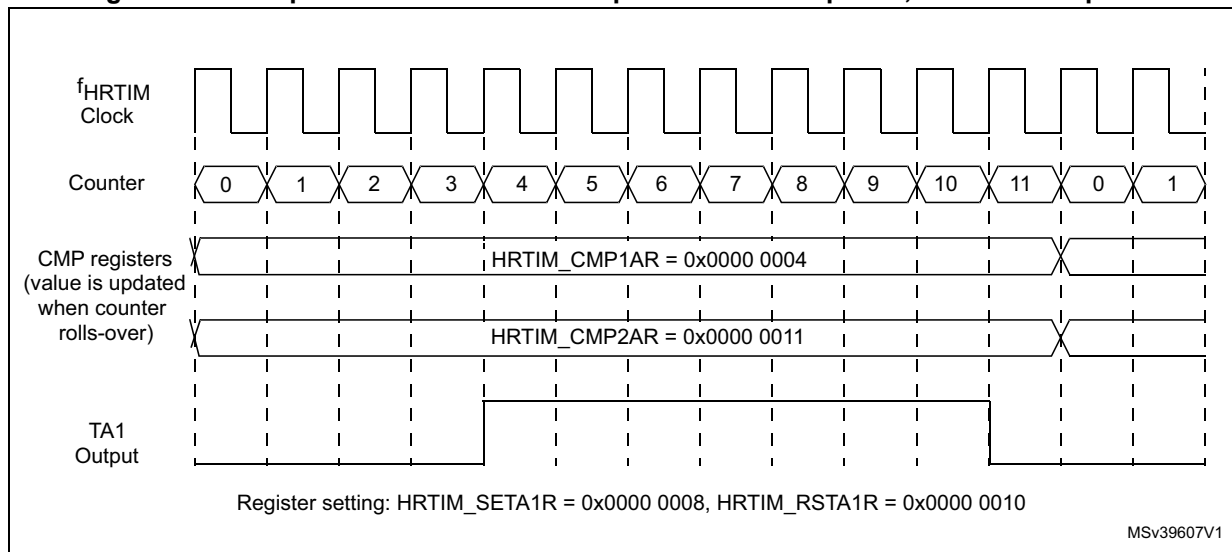
For instance, Timer A outputs can be set or reset by the following events: Timer B Compare1, 2 and 4, Timer C Compare 2 and 3,... and Timer E Compare 3 will be listed as TIMEVNT8 in HRTIM_SETA1R.

Table 303. Events mapping across Timer A to E

Source		Timer A				Timer B				Timer C				Timer D				Timer E			
		CMP1	CMP2	CMP3	CMP4	CMP1	CMP2	CMP3	CMP4	CMP1	CMP2	CMP3	CMP4	CMP1	CMP2	CMP3	CMP4	CMP1	CMP2	CMP3	CMP4
Destination	Timer A	-	-	-	-	1	2	-	3	-	4	5	-	6	7	-	-	-	-	8	9
	Timer B	1	2	-	3	-	-	-	-	-	-	4	5	-	-	6	7	8	9	-	-
	Timer C	-	1	2	-	-	3	4	-	-	-	-	-	-	5	-	6	-	7	8	9
	Timer D	1	-	-	2	-	3	-	4	5	-	6	7	-	-	-	-	8	-	-	9
	Timer E	-	-	1	2	-	-	3	4	5	6	-	-	7	8	-	9	-	-	-	-

[Figure 288](#) represents how a PWM signal is generated using two compare events.

Figure 288. Compare events action on outputs: set on compare 1, reset on compare 2



Set/Reset on Update events

Half mode

This mode aims at generating square signal with fixed 50% duty cycle and variable frequency (typically for converters using resonant topologies). It allows to have the duty cycle automatically forced to half of the period value when a new period is programmed.

This mode is enabled by writing HALF bit to 1 in the HRTIM_TIMxCR register. When the HRTIM_PERxR register is written, it causes an automatic update of the Compare 1 value with HRTIM_PERxR/2 value.

The output on which a square wave is generated must be programmed to have one transition on CMP1 event, and one transition on the period event, as follows:

- HRTIM_SETxyR = 0x0000 0008, HRTIM_RSTxyR = 0x0000 0004, or
- HRTIM_SETxyR = 0x0000 0004, HRTIM_RSTxyR = 0x0000 0008

The HALF mode overrides the content of the HRTIM_CMP1xR register. The access to the HRTIM_PERxR register only causes Compare 1 internal register to be updated. The user-accessible HRTIM_CMP1xR register is not updated with the HRTIM_PERxR / 2 value.

When the preload is enabled (PREEN = 1, MUDIS, TxUDIS), Compare 1 active register is refreshed on the Update event. If the preload is disabled (PREEN= 0), Compare 1 active register is updated as soon as HRTIM_PERxR is written.

The period must be greater than or equal to 6 periods of the f_{HRTIM} clock when the HALF mode is enabled.

Capture

The timing unit has the capability to capture the counter value, triggered by internal and external events. The purpose is to:

- measure events arrival timings or occurrence intervals
- update Compare 2 and Compare 4 values in auto-delayed mode (see [Auto-delayed mode](#)).

The capture is done with f_{HRTIM} resolution.

The timer has 2 capture registers: HRTIM_CPT1xR and HRTIM_CPT2xR. The capture triggers are programmed in the HRTIM_CPT1xCR and HRTIM_CPT2xCR registers.

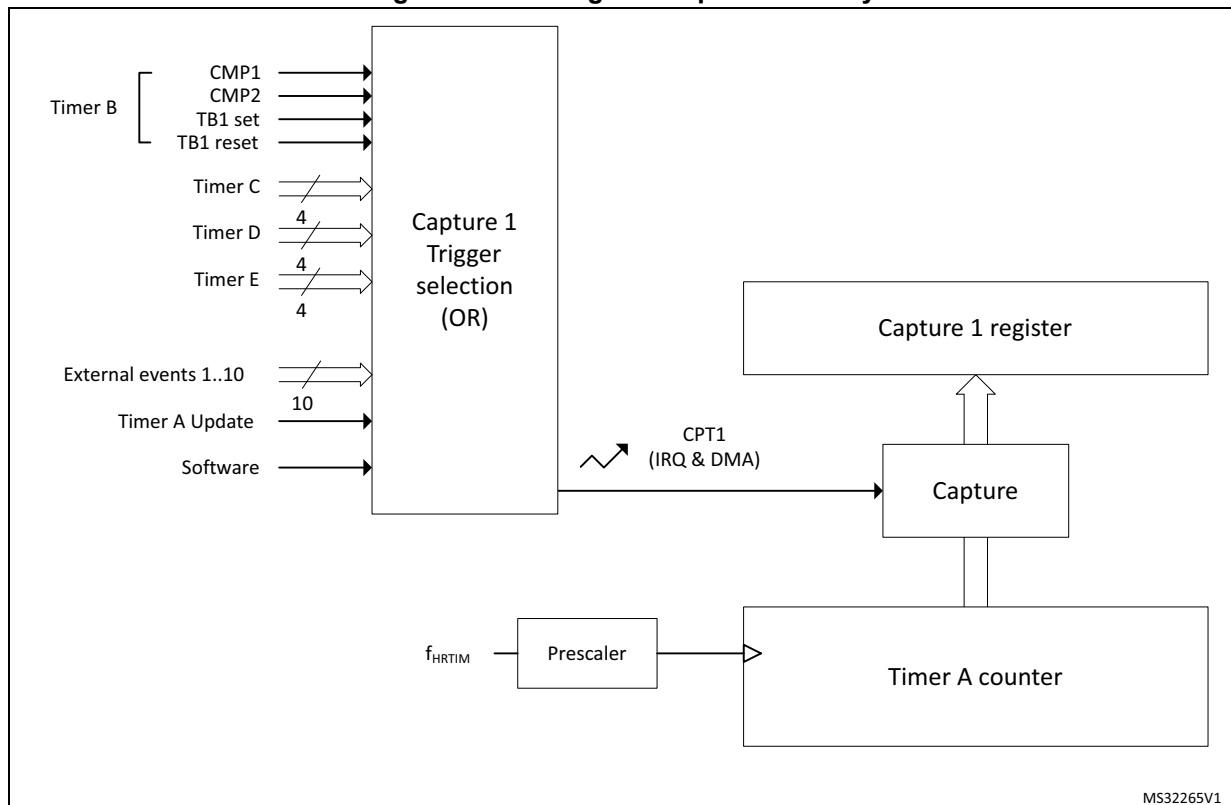
The capture of the timing unit counter can be triggered by up to 28 events that can be selected simultaneously in the HRTIM_CPT1xCR and HRTIM_CPT2xCR registers, among the following sources:

- The external events, EXTEVNT1..10 (10 events)
- All other timing units (e.g. Timer B..E for timer A): Compare 1, 2 and output 1 set/reset events (16 events)
- The timing unit: Update (1 event)
- A software capture (1 event)

Several events can be selected simultaneously to handle multiple capture triggers. In this case, the concurrent trigger requests are ORed. The capture can generate an interrupt or a DMA request when CPTxIE and CPTxDE bits are set in the HRTIM_TIMxDIER register.

Over-capture is not prevented by the circuitry: a new capture is triggered even if the previous value was not read, or if the capture flag was not cleared.

Figure 289. Timing unit capture circuitry



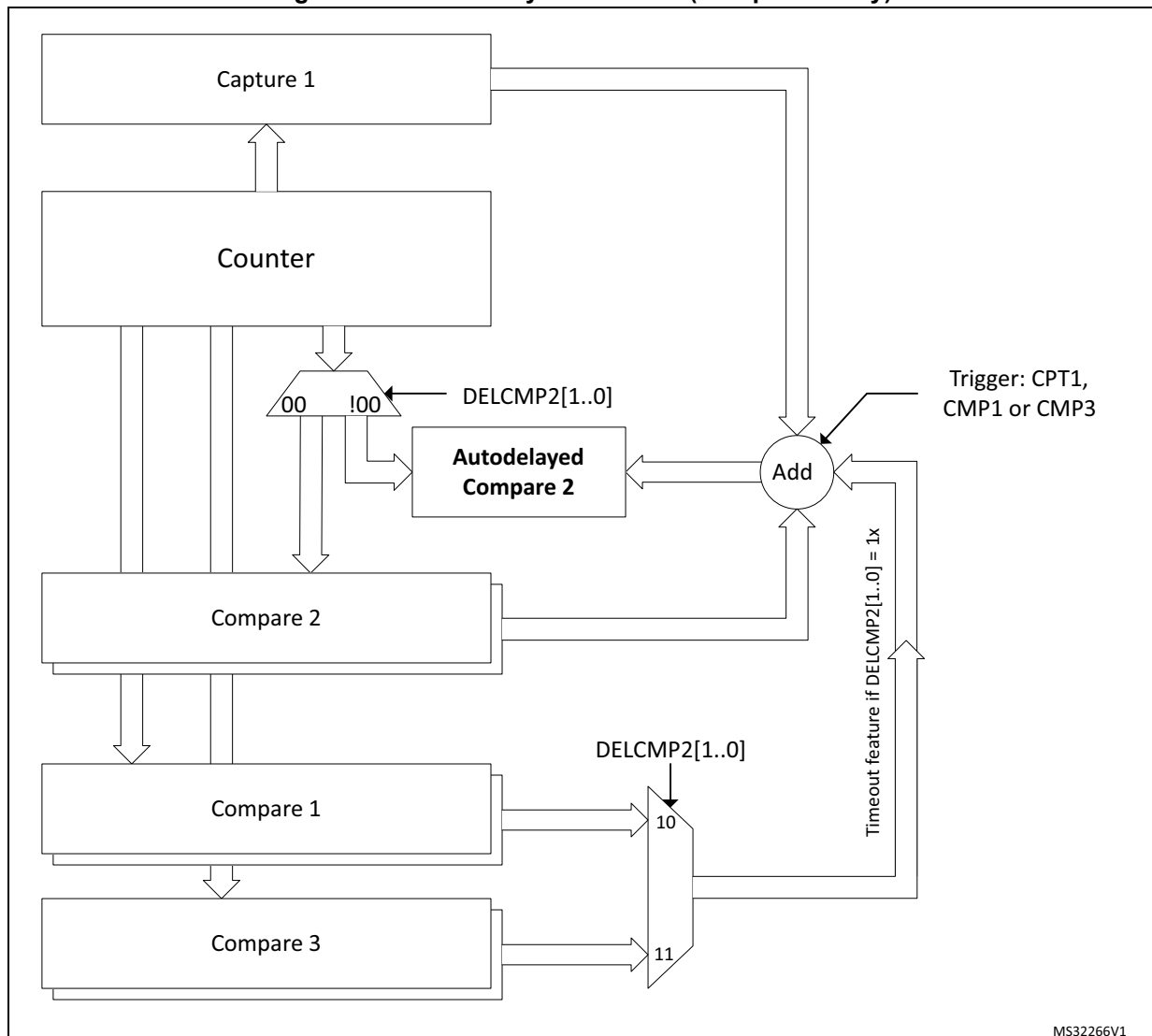
Auto-delayed mode

This mode allows to have compare events generated relatively to capture events, so that for instance an output change can happen with a programmed timing following a capture. In this case, the compare match occurs independently from the timer counter value. It enables the generation of waveforms with timings synchronized to external events without the need of software computation and interrupt servicing.

As long as no capture is triggered, the content of the `HRTIM_CMPxR` register is ignored (no compare event is generated when the counter value matches the Compare value). Once the capture is triggered, the compare value programmed in `HRTIM_CMPxR` is summed with the captured counter value in `HRTIM_CPTxyR`, and it updates the internal auto-delayed compare register, as seen on [Figure 290](#). The auto-delayed compare register is internal to the timing unit and cannot be read. The `HRTIM_CMPxR` preload register is not modified after the calculation.

This feature is available only for Compare 2 and Compare 4 registers. Compare 2 is associated with capture 1, while Compare 4 is associated with capture 2. `HRTIM_CMP2xR` and `HRTIM_CMP4xR` Compares cannot be programmed with a value below $3 f_{\text{HRTIM}}$ clock periods, as in the regular mode.

Figure 290. Auto-delayed overview (Compare 2 only)



The auto-delayed Compare is only valid from the capture up to the period event: once the counter has reached the period value, the system is re-armed with Compare disabled until a capture occurs.

DELCMP2[1:0] and DELCMP4[1:0] bits in HRTIM_TIMxCR register allow to configure the auto-delayed mode as follows:

- 00
Regular compare mode: HRTIM_CMP2xR and HRTIM_CMP4xR register contents are directly compared with the counter value.
- 01
Auto-delayed mode: Compare 2 and Compare 4 values are recomputed and used for comparison with the counter after a capture 1/2 event.

- 1X
Auto-delayed mode with timeout: Compare 2 and Compare 4 values are recomputed and used for comparison with the counter after a capture 1/2 event or after a Compare 1 match (DELCMPx[1:0]= 10) or a Compare 3 match (DELCMPx[1:0]= 11) to have a timeout function if capture 1/2 event is missing.

When the capture occurs, the comparison is done with the (HRTIM_CMP2/4xR + HRTIM_CPT1/2xR) value. If no capture is triggered within the period, the behavior depends on the DELCMPx[1:0] value:

- DELCMPx[1:0] = 01: the compare event is not generated
- DELCMPx[1:0] = 10 or 11: the comparison is done with the sum of the 2 compares (for instance HRTIM_CMP2xR + HRTIM_CMP1xR). The captures are not taken into account if they are triggered after CMPx + CMP1 (resp. CMPx + CMP3).

The captures are enabled again at the beginning of the next PWM period.

If the result of the auto-delayed summation is above 0xFFFF (overflow), the value is ignored and no compare event will be generated until a new period is started.

Note: *DELCMPx[1:0] bitfield must be reset when reprogrammed from one value to the other to re-initialize properly the auto-delayed mechanism, for instance:*

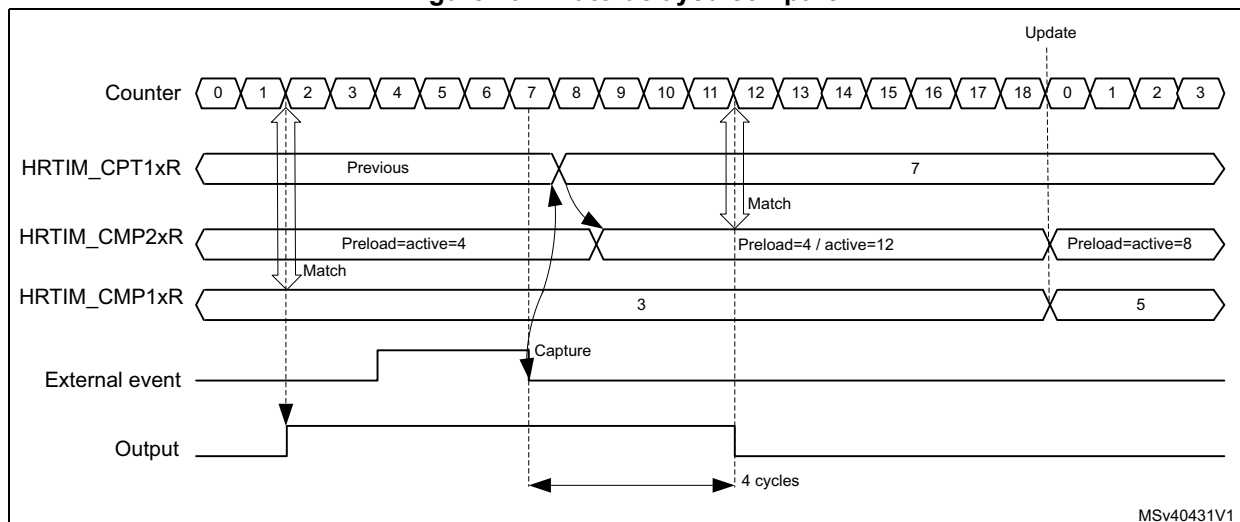
- DELCMPx[1:0] = 10
- DELCMPx[1:0] = 00
- DELCMPx[1:0] = 11

As an example, [Figure 291](#) shows how the following signal can be generated:

- Output set when the counter is equal to Compare 1 value
- Output reset 4 cycles after a falling edge on a given external event

Note: *To simplify the figure, the external event signal is shown without any resynchronization delay: practically, there is a delay of 1 to 2 f_{HRTIM} clock periods between the falling edge and the capture event due to an internal resynchronization stage which is necessary to process external input signals.*

Figure 291. Auto-delayed compare



A regular compare channel (e.g. Compare 1) is used for the output set: as soon as the counter matches the content of the compare register, the output goes to its active state.

A delayed compare is used for the output reset: the compare event can be generated only if a capture event has occurred. No event is generated when the counter matches the delayed compare value (counter = 4). Once the capture event has been triggered by the external event, the content of the capture register is summed to the delayed compare value to have the new compare value. In the example, the auto-delayed value 4 is summed to the capture equal to 7 to give a value of 12 in the auto-delayed compare register. From this time on, the compare event can be generated and will happen when the counter is equal to 12, causing the output to be reset.

Overcapture management in auto-delayed mode

Overcapture is prevented when the auto-delayed mode is enabled (DELCMPx[1:0] = 01, 10, 11).

When multiple capture requests occur within the same counting period, only the first capture is taken into account to compute the auto-delayed compare value. A new capture is possible only:

- Once the auto-delayed compare has matched the counter value (compare event)
- Once the counter has rolled over (period)
- Once the timer has been reset

Changing auto-delayed compare values

When the auto-delayed compare value is preloaded (PREEN bit set), the new compare value is taken into account on the next coming update event (for instance on the period event), regardless of when the compare register was written and if the capture occurred (see [Figure 291](#), where the delay is changed when the counter rolls over).

When the preload is disabled (PREEN bit reset), the new compare value is taken into account immediately, even if it is modified after the capture event has occurred, as per the example below:

1. At t1, DELCMP2 = 1.
2. At t2, CMP2_act = 0x40 => comparison disabled
3. At t3, a capture event occurs capturing the value CPTR1 = 0x20. => comparison enabled, compare value = 0x60
4. At t4, CMP2_act = 0x100 (before the counter reached value CPTR1 + 0x40) => comparison still enabled, new compare value = 0x120
5. At t5, the counter reaches the period value => comparison disabled, cmp2_act = 0x100

Similarly, if the CMP1(CMP3) value changes while DELCMPx = 10 or 11, and preload is disabled:

1. At t1, DELCMP2 = 2.
2. At t2, CMP2_act = 0x40 => comparison disabled
3. At t3, CMP3 event occurs - CMP3_act = 0x50 before capture 1 event occurs => comparison enabled, compare value = 0x90
4. At t4, CMP3_act = 0x100 (before the counter reached value 0x90) => comparison still enabled, Compare 2 event will occur at = 0x140

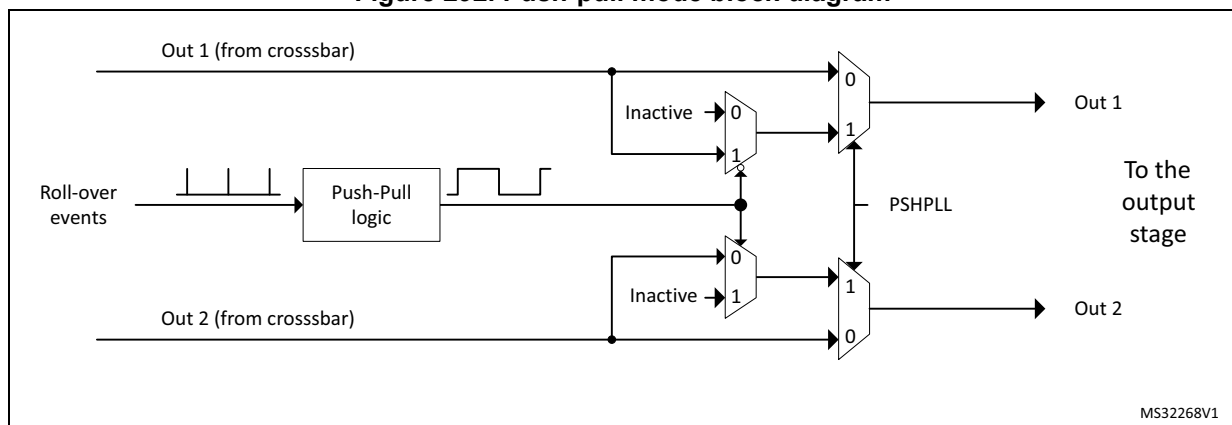
Push-pull mode

This mode primarily aims at driving converters using push-pull topologies. It also needs to be enabled when the delayed idle protection is required, typically for resonant converters (refer to [Section 37.3.9: Delayed Protection](#)).

The push-pull mode is enabled by setting PSHPLL bit in the HRTIM_TIMxCR register.

It applies the signals generated by the crossbar to output 1 and output 2 alternatively, on the period basis, maintaining the other output to its inactive state. The redirection rate (push-pull frequency) is defined by the timer's period event, as shown on [Figure 292](#). The push-pull period is twice the timer counting period.

Figure 292. Push-pull mode block diagram



The push-pull mode is only available when the timer operates in continuous mode: the counter must not be reset once it has been enabled (TxCEN bit set). It is necessary to disable the timer to stop a push-pull operation and to reset the counter before re-enabling it.

The signal shape is defined using HRTIM_SETxyR and HRTIM_RSTxyR for both outputs. It is necessary to have HRTIM_SETx1R = HRTIM_SETx2R and HRTIM_RSTx1R = HRTIM_RSTx2R to have both outputs with identical waveforms and to achieve a balanced operation. Still, it is possible to have different programming on both outputs for other uses.

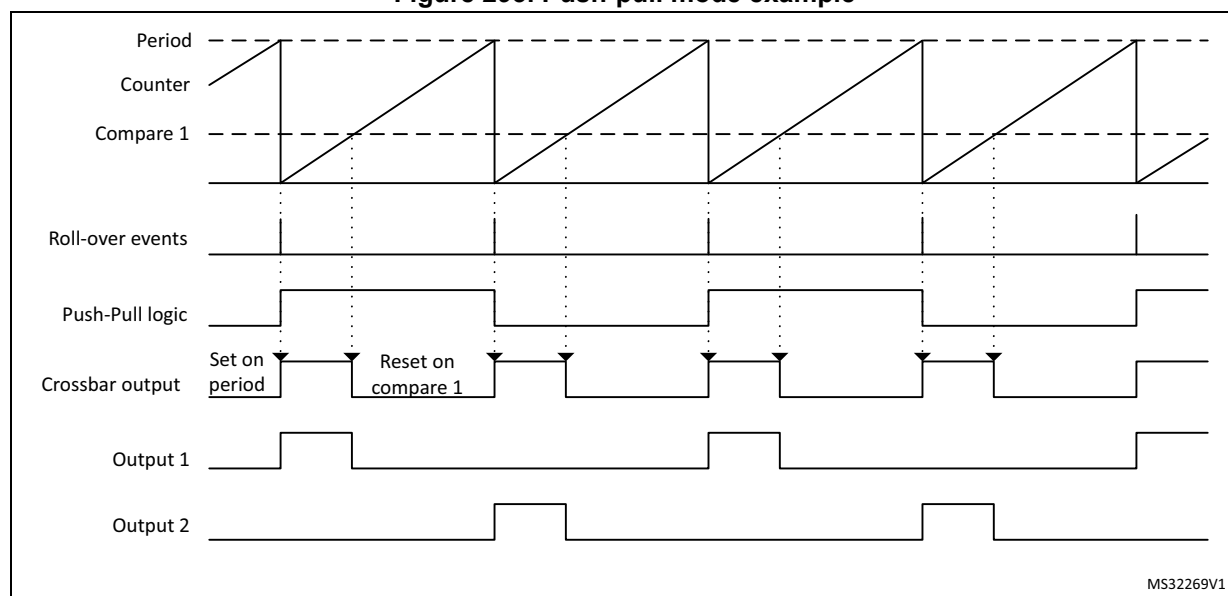
Note: *The push-pull operation cannot be used when a deadtime is enabled (mutually exclusive functions).*

The CPPSAT status bit in HRTIM_TIMxISR indicates on which output the signal is currently active. CPPSTAT is reset when the push-pull mode is disabled.

In the example given on [Figure 293](#), the timer internal waveform is defined as follows:

- Output set on period event
- Output reset on Compare 1 match event

Figure 293. Push-pull mode example



Deadtime

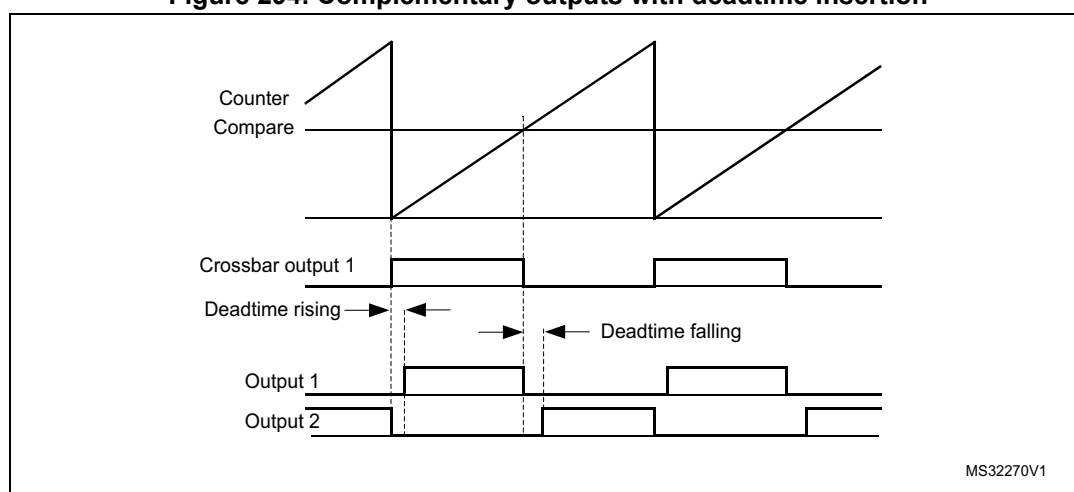
A deadtime insertion unit allows to generate a couple of complementary signals from a single reference waveform, with programmable delays between active state transitions. This is commonly used for topologies using half-bridges or full bridges. It simplifies the software: only 1 waveform is programmed and controlled to drive two outputs.

The Dead time insertion is enabled by setting DTEN bit in HRTIM_OUTxR register. The complementary signals are built based on the reference waveform defined for output 1, using HRTIM_SETx1R and HRTIM_RSTx1R registers: HRTIM_SETx2R and HRTIM_RSTx2R registers are not significant when DTEN bit is set.

Note: *The deadtime cannot be used simultaneously with the push-pull mode.*

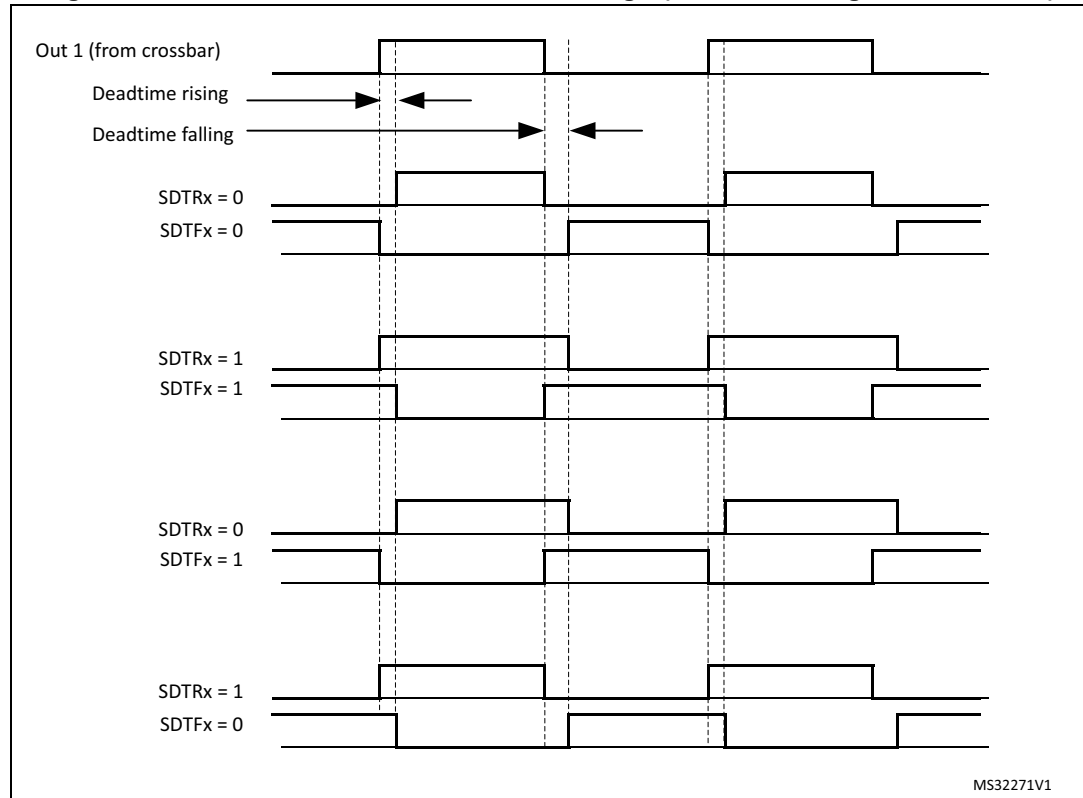
Two deadtimes can be defined in relationship with the rising edge and the falling edge of the reference waveform, as in [Figure 294](#).

Figure 294. Complementary outputs with deadtime insertion



Negative deadtime values can be defined when some control overlap is required. This is done using the deadtime sign bits (SDTFx and SDTRx bits in HRTIM_DTxR register). [Figure 295](#) shows complementary signal waveforms depending on respective signs.

Figure 295. Deadtime insertion vs deadtime sign (1 indicates negative deadtime)



The deadtime values are defined with DTFx[8:0] and DTRx[8:0] bitfields and based on a specific clock prescaled according to DTPRSC[2:0] bits, as follows:

$$t_{DTx} = \pm DTFx[8:0] \times t_{DTG}$$

$$\text{where } x \text{ is either R or F and } t_{DTG} = (2^{DTPRSC[2:0]}) \times t_{HRTIM}$$

[Table 304](#) gives the resolution and maximum absolute values depending on the prescaler value.

Table 304. Deadtime resolution and max absolute values

DTPRSC[2:0] ⁽¹⁾	t_{DTG}	$t_{DTx} \text{ max}$	$f_{HRTIM} = 400 \text{ MHz}$	
			$t_{DTG} \text{ (ns)}$	$ t_{DTx} \text{ max } (\mu\text{s})$
011	t_{HRTIM}	$511 \times t_{DTG}$	2.5	1.28
100	$2 \times t_{HRTIM}$		5	2.56
101	$4 \times t_{HRTIM}$		10	5.11
110	$8 \times t_{HRTIM}$		20	10.22
111	$16 \times t_{HRTIM}$		40	20.44

1. DTPRSC[2:0] values 000, 001, 010 are reserved.

Figure 296 to Figure 299 present how the deadtime generator behaves for reference waveforms with pulsewidth below the deadtime values, for all deadtime configurations.

Figure 296. Complementary outputs for low pulse width (SDTRx = SDTFx = 0)

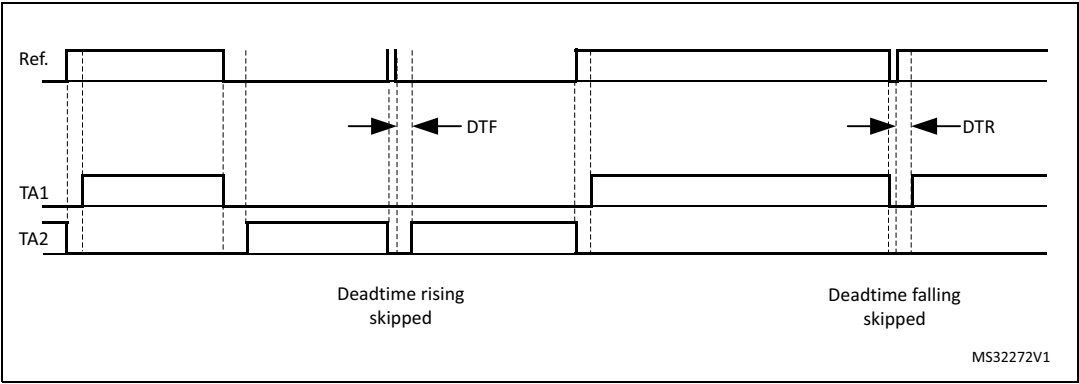


Figure 297. Complementary outputs for low pulse width (SDTRx = SDTFx = 1)

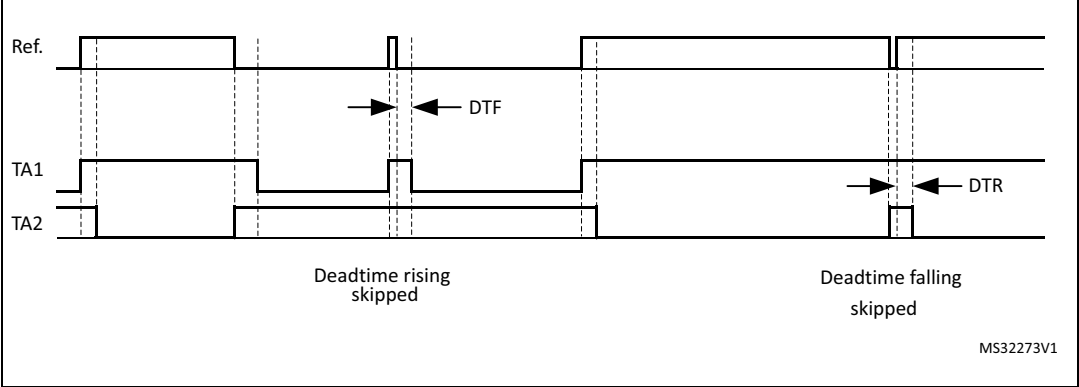


Figure 298. Complementary outputs for low pulse width (SDTRx = 0, SDTFx = 1)

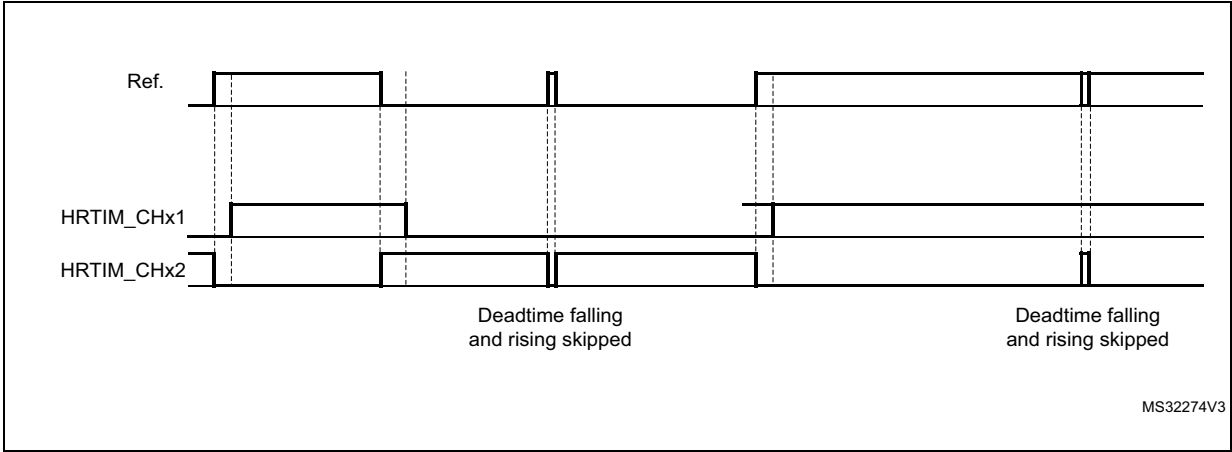
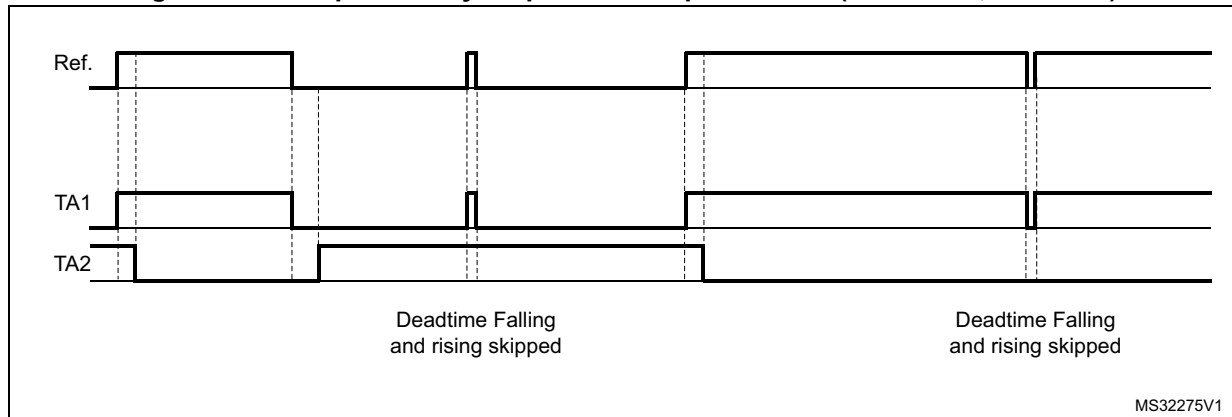


Figure 299. Complementary outputs for low pulse width (SDTRx = 1, SDTFx=0)

For safety purposes, it is possible to prevent any spurious write into the deadtime registers by locking the sign and/or the value of the deadtime using DTFLKx, DTRLKx, DTFSLKx and DTRSLKx. Once these bits are set, the related bits and bitfields are becoming read only until the next system reset.

Caution: DTEN bit must not be changed in the following cases:

- When the timer is enabled (TxEN bit set)
- When the timer outputs are set/reset by another timer (while TxEN is reset)

Otherwise, an unpredictable behavior would result.
It is therefore necessary to disable the timer (TxCEN bit reset) and have the corresponding outputs disabled.

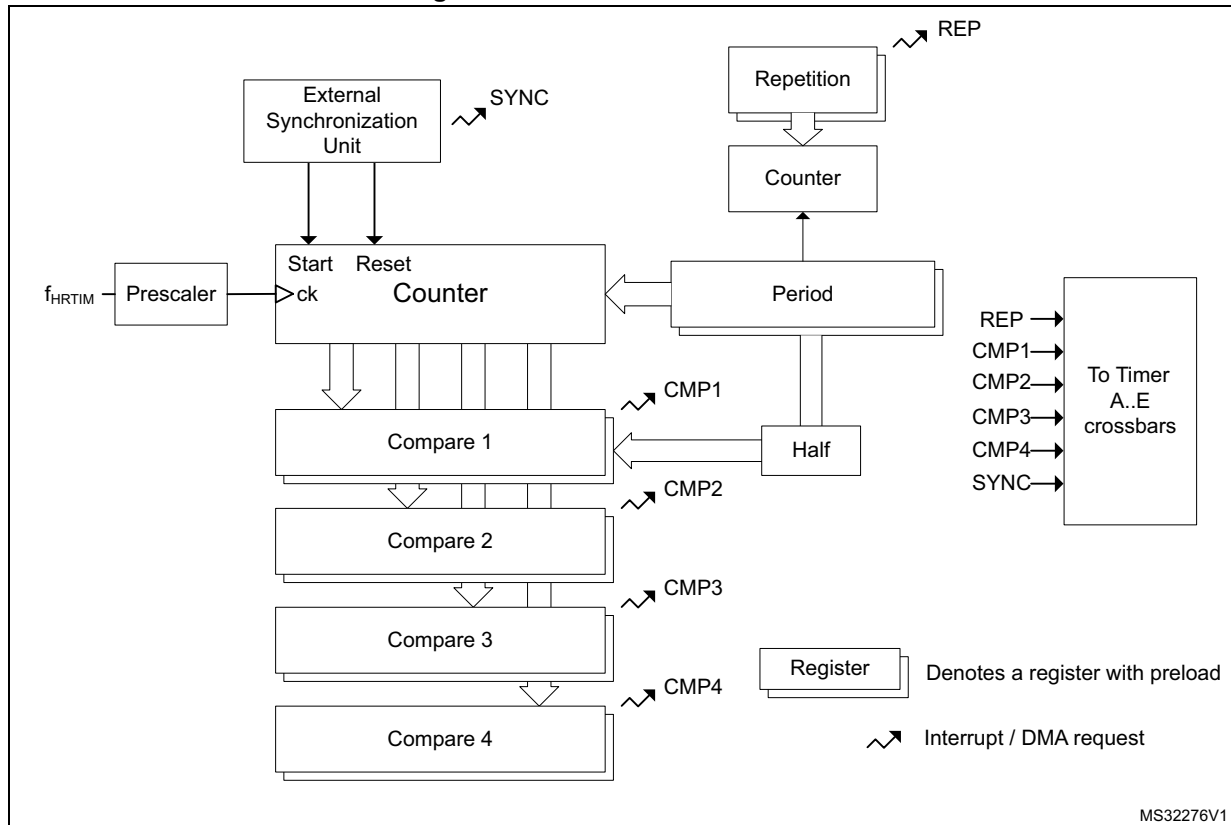
For the particular case where DTEN must be set while the burst mode is enabled with a deadtime upon entry (BME = 1, DIDL = 1, IDLEM = 1), it is necessary to force the two outputs in their IDLES state by software commands (SST, RST bits) before setting DTEN bit. This is to avoid any side effect resulting from a burst mode entry that would happen immediately before a deadtime enable.

37.3.5 Master timer

The main purpose of the master timer is to provide common signals to the 5 timing units, either for synchronization purpose or to set/reset outputs. It does not have direct control over any outputs, but still can be used indirectly by the set/reset crossbars.

Figure 300 provides an overview of the master timer.

Figure 300. Master timer overview



The master timer is based on the very same architecture as the timing units, with the following differences:

- It does not have outputs associated with, nor output related control
- It does not have its own crossbar unit, nor push-pull or deadtime mode
- It can only be reset by the external synchronization circuitry
- It does not have a capture unit, nor the auto-delayed mode
- It does not include external event blanking and windowing circuitry
- It has a limited set of interrupt / DMA requests: Compare 1..4, repetition, register update and external synchronization event.

The master timer control register includes all the timer enable bits, for the master and Timer A..E timing units. This allows to have all timer synchronously started with a single write access.

It also handles the external synchronization for the whole HRTIM timer (see [Section 37.3.17: Synchronizing the HRTIM with other timers or HRTIM instances](#)), with both MCU internal and external (inputs/outputs) resources.

Master timer control registers are mapped with the same offset as the timing units' registers.

37.3.6 Set/reset events priorities and narrow pulses management

This section describes how the output waveform is generated when several set and/or reset requests are occurring within 3 consecutive t_{HRTIM} periods.

An arbitration is performed during each t_{HRTIM} period, in 2 steps:

1. For each active event, the desired output transition is determined (set, reset or toggle).
2. A predefined arbitration is performed among the active events (from highest to lowest priority $\text{CMP4} \rightarrow \text{CMP3} \rightarrow \text{CMP2} \rightarrow \text{CMP1} \rightarrow \text{PER}$, see [Concurrent set request / Concurrent reset requests](#)).

When set and reset requests from two different sources are simultaneous, the reset action has the highest priority.

Concurrent set request / Concurrent reset requests

When multiple sources are selected for a set event, an arbitration is performed when the set requests occur within the same f_{HRTIM} clock period.

In case of multiple requests from adjacent timers (TIMEVNT1..9), the request which occurs first is taken into account. The arbitration is done in 2 steps, depending on the source (from the highest to the lowest priority): $\text{CMP4} \rightarrow \text{CMP3} \rightarrow \text{CMP2} \rightarrow \text{CMP1}$.

If multiple requests from the master timer occur within the same f_{HRTIM} clock period, a predefined arbitration is applied and a single request will be taken into account (from the highest to the lowest priority):

$\text{MSTCMP4} \rightarrow \text{MSTCMP3} \rightarrow \text{MSTCMP2} \rightarrow \text{MSTCMP1} \rightarrow \text{MSTCMPEP}$

When multiple requests internal to the timer occur within the same f_{HRTIM} clock period, a predefined arbitration is applied and the requests are taken with the following priority, whatever the effective timing (from highest to lowest):

$\text{CMP4} \rightarrow \text{CMP3} \rightarrow \text{CMP2} \rightarrow \text{CMP1} \rightarrow \text{PER}$

Note: Practically, this is of a primary importance only when using auto-delayed Compare 2 and Compare 4 simultaneously (i.e. when the effective set/reset cannot be determined a priori because it is related to an external event). In this case, the highest priority signal must be affected to the CMP4 event.

Last, the highest priority is given to non timing-related: EXTEVNT1..10 , RESYNC (coming from SYNC event if SYNCRSTx or SYNCSRTx is set or from a software reset), update and software set (SST).

As a summary, in case of simultaneous events, the effective set (reset) event will be arbitrated between:

- Any TIMEVNT1..9 event
- A single source from the master (as per the fixed arbitration given above)
- A single source from the timer
- The “non timing-related events”.

The same arbitration principle applies for concurrent reset requests. In this case, the reset request has the highest priority.

A set or reset event occurring within the prescaler clock cycle is delayed to the next active edge of the prescaled clock (as for a counter reset), even if the arbitration is still performed every t_{HRTIM} cycle.

If a reset event is followed by a set event within the same prescaler clock cycle, the latest event will be considered.

37.3.7 External events global conditioning

The HRTIM timer can handle events not generated within the timer, referred to as “external event”. These external events come from multiple sources, either on-chip or off-chip:

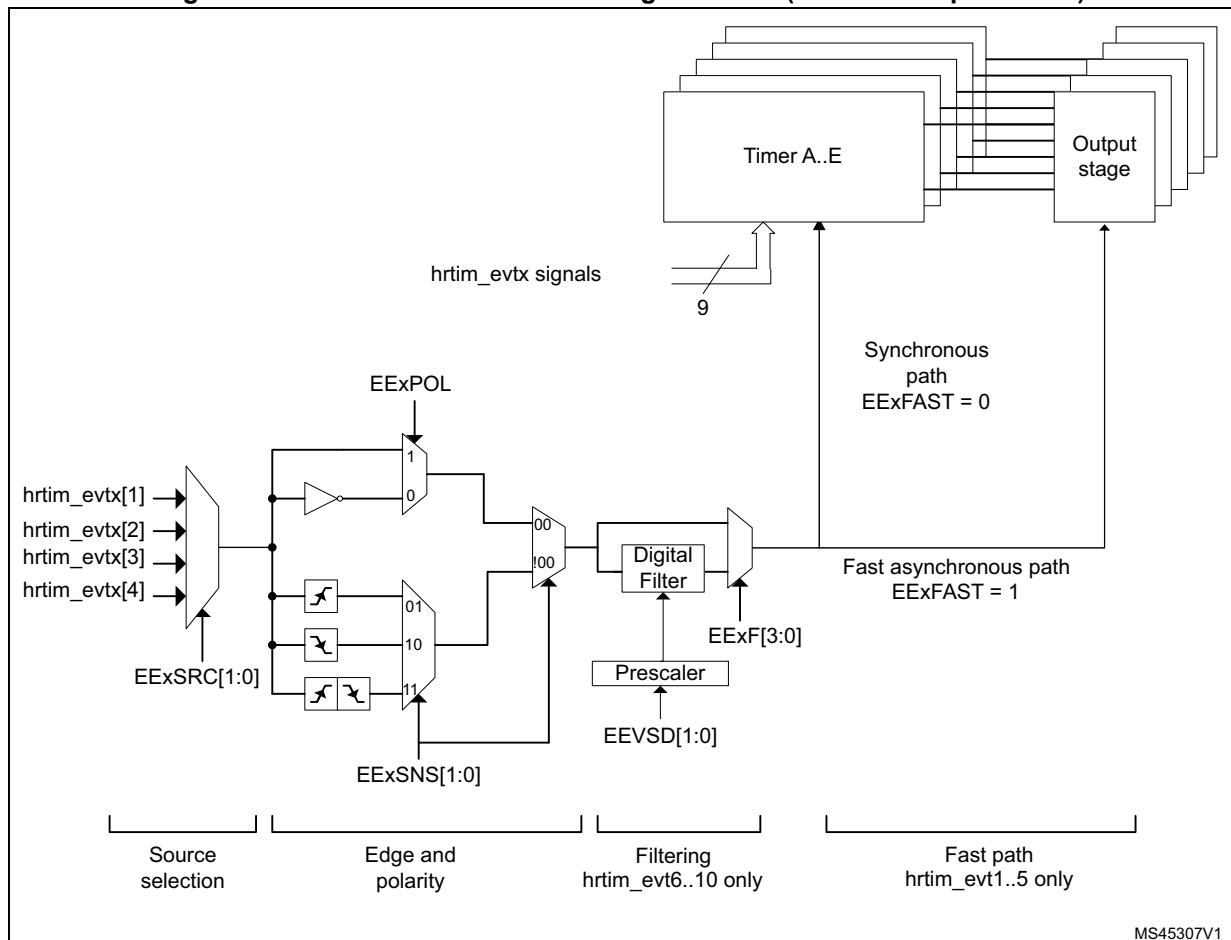
- built-in comparators,
- digital input pins (typically connected to off-chip comparators and zero-crossing detectors),
- on-chip events for other peripheral (ADC’s analog watchdogs and general purpose timer trigger outputs).

The external events conditioning circuitry allows to select the signal source for a given channel (with a 4:1 multiplexer) and to convert it into an information that can be processed by the crossbar unit (for instance, to have an output reset triggered by a falling edge detection on an external event channel).

Up to 10 external event channels can be conditioned and are available simultaneously for any of the 5 timers. This conditioning is common to all timers, since this is usually dictated by external components (such as a zero-crossing detector) and environmental conditions (typically the filter set-up will be related to the applications noise level and signature).

[Figure 301](#) presents an overview of the conditioning logic for a single channel.

Figure 301. External event conditioning overview (1 channel represented)



The 10 external events are initialized using the HRTIM_EECR1 and HRTIM_EECR2 registers:

- to select up to 4 sources with the EExSRC[1:0] bits,
- to select the sensitivity with EExSNS[1:0] bits, to be either level-sensitive or edge-sensitive (rising, falling or both),
- to select the polarity, in case of a level sensitivity, with EExPOL bit,
- to have a low latency mode, with EExFAST bits (see [Latency to external events](#)), for external events 1 to 5.

Note: The external events used as triggers for reset, capture, burst mode, ADC triggers and delayed protection are edge-sensitive even if EESNS bit is reset (level-sensitive selection): if POL = 0 the trigger is active on external event rising edge, while if POL = 1 the trigger is active on external event falling edge.

The external events are discarded as long as the counters are disabled (TxCEN bit reset) to prevent any output state change and counter reset, except if they are used as ADC triggers.

Additionally, it is possible to enable digital noise filters, for external events 6 to 10, using EExF[3:0] bits in the HRTIM_EECR3 register.

A digital filter is made of a counter in which a number N of valid samples is needed to validate a transition on the output. If the input value changes before the counter has

reached the value N, the counter is reset and the transition is discarded (considered as a spurious event). If the counter reaches N, the transition is considered as valid and transmitted as a correct external event. Consequently, the digital filter adds a latency to the external events being filtered, depending on the sampling clock and on the filter length (number of valid samples expected).

The sampling clock is either the f_{HRTIM} clock or a specific prescaled clock f_{EEVS} derived from f_{HRTIM} , defined with EEVSD[1:0] bits in HRTIM_EECR3 register.

[Table 305](#) summarizes the available sources and features associated with each of the 10 external events channels.

Table 305. External events mapping and associated features

External event channel	Fast mode	Digital filter	Balanced fault timer A,B,C	Balanced fault timer D,E	Src1	Src 2	Src3	Src4
1	Yes	-	-	-	PC10	COMP1	TIM1_TRGO	ADC1_AWD1
2	Yes	-	-	-	PC12	COMP2	TIM2_TRGO	ADC1_AWD2
3	Yes	-	-	-	PD5	-	TIM3_TRGO	ADC1_AWD3
4	Yes	-	-	-	PG11	OPAMP1 ⁽¹⁾	TIM7_TRGO	ADC2_AWD1
5	Yes	-	-	-	PG12	-	LPTIM1 OUT	ADC2_AWD2
6	-	Yes	Yes	-	PB4	COMP1	TIM6_TRGO	ADC2_AWD3
7	-	Yes	Yes	-	PB5	COMP2	TIM7_TRGO	-
8	-	Yes	-	Yes	PB6	-	TIM6_TRGO	TTCAN_TMP
9	-	Yes	-	Yes	PB7	OPAMP1 ⁽¹⁾	TIM15_TRGO	TTCAN_RTP
10	-	Yes	-	-	PG13	-	LPTIM2 OUT	TTCAN_SOC

1. OPAMP1_VOUT can be used as High-resolution timer internal event source. In this case, OPAMP1_VOUT (PC4) pin must be configured in input mode. The data from the GPIO pin is redirect to the HRTIM external events through the pin Schmitt trigger. If OPAMP1 is disabled, PC4 pin, configured in input mode, can be used as HRTIM external events.

Latency to external events

The external event conditioning gives the possibility to adjust the external event processing time (and associated latency) depending on performance expectations:

- A regular operating mode, in which the external event is resampled with the clock before acting on the output crossbar. This adds some latency but gives access to all crossbar functionalities. It enables the generation of an externally triggered high-resolution pulse.
- A fast operating mode, in which the latency between the external event and the action on the output is minimized. This mode is convenient for ultra-fast over-current protections, for instance.

EExFAST bits in the HRTIM_EECR1 register allow to define the operating for channels 1 to 5. This influences the latency and the jitter present on the output pulses, as summarized in the table below.

Table 306. Output set/reset latency and jitter vs external event operating mode

EExFAST	Response time latency	Response time jitter	Jitter on output pulse (counter reset by ext. event)
0	5 to 6 cycles of f_{HRTIM} clock	1 cycles of f_{HRTIM} clock	No jitter, pulse width maintained with high-resolution
1	Minimal latency (depends whether the comparator or digital input is used)	Minimal jitter	1 cycle of f_{HRTIM} clock jitter pulse width resolution down to t_{HRTIM}

The EExFAST mode is only available with level-sensitive programming (EExSNS[1:0] = 00); the edge-sensitivity cannot be programmed.

It is possible to apply event filtering to external events (both blanking and windowing with EExFLTR[3:0] != 0000, see [Section 37.3.8](#)). In this case, EExLTCHx bit must be reset: the postponed mode is not supported, neither the windowing timeout feature.

Note: *The external event configuration (source and polarity) must not be modified once the related EExFAST bit is set.*

A fast external event cannot be used to toggle an output: it must be enabled either in HRTIM_SETxyR or HRTIM_RSTxyR registers, not in both.

When a set and a reset event - from 2 independent fast external events - occur simultaneously, the reset has the highest priority in the crossbar and the output becomes inactive.

When EExFAST bit is set, the output cannot be changed during the 11 f_{HRTIM} clock periods following the external event.

[Figure 302](#) and [Figure 303](#) give practical examples of the reaction time to external events, for output set/reset and counter reset.

Figure 302. Latency to external events falling edge (counter reset and output set)

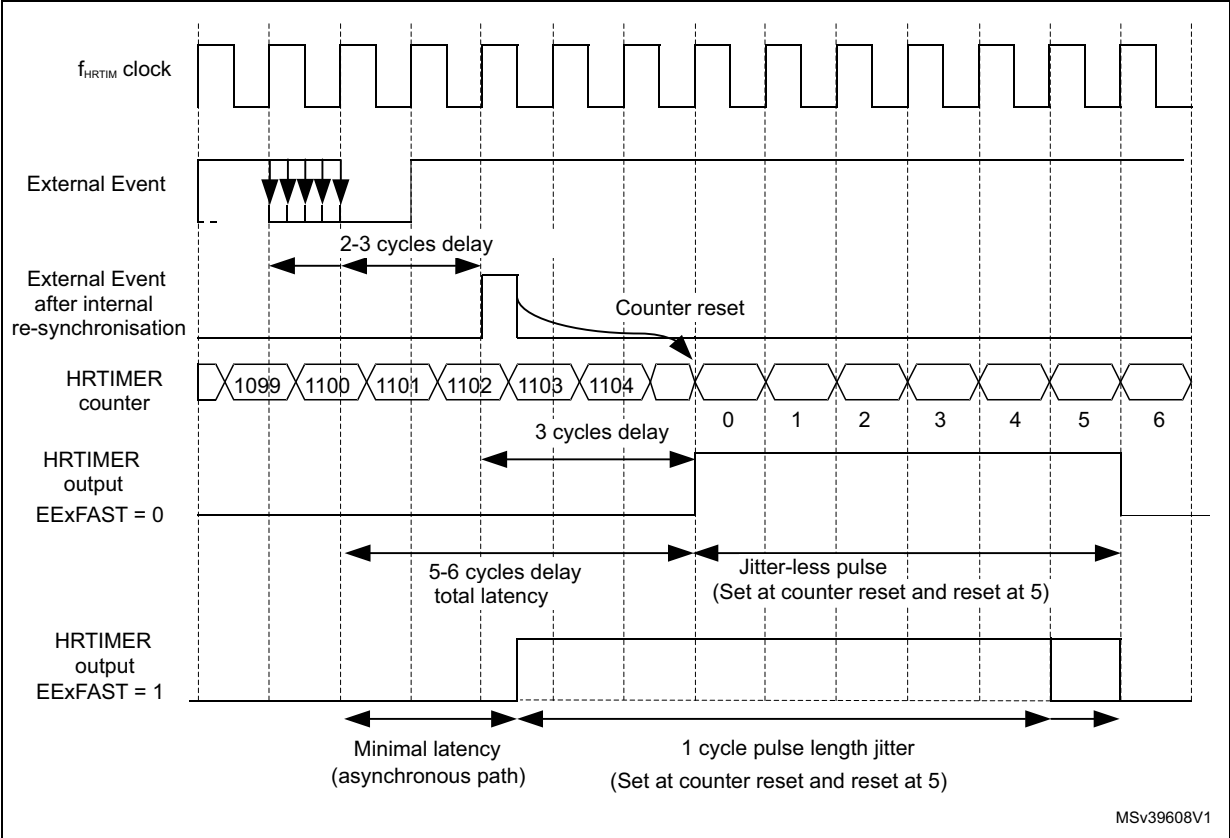
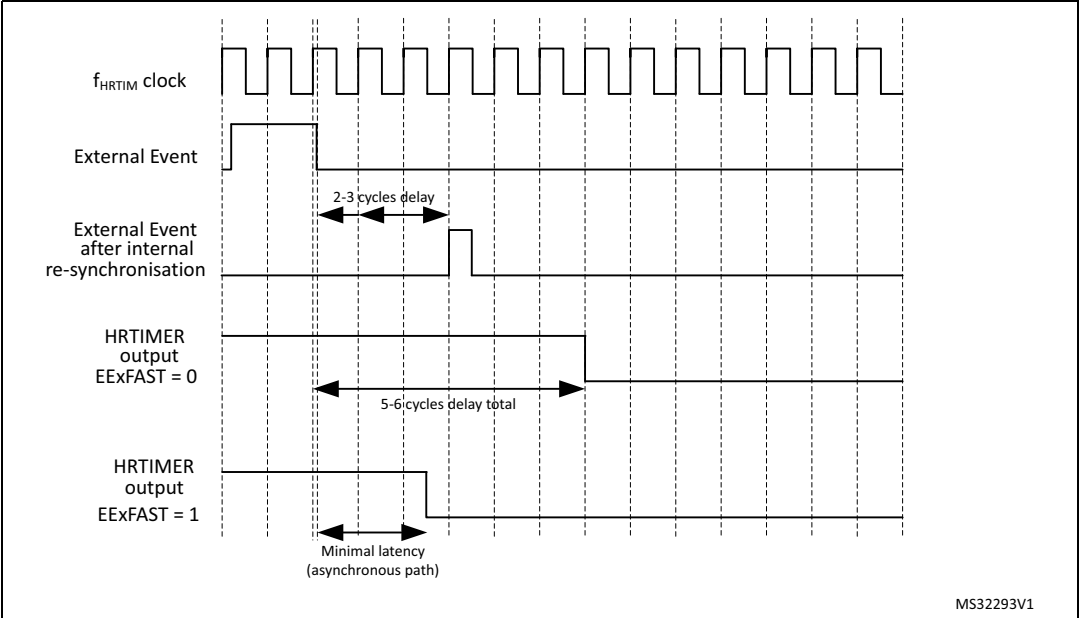


Figure 303. Latency to external events (output reset on external event)



37.3.8 External event filtering in timing units

Once conditioned, the 10 external events are available for all timing units.

They can be used directly and are active as soon as the timing unit counter is enabled (TxCEN bit set).

They can also be filtered to have an action limited in time, usually related to the counting period. Two operations can be performed:

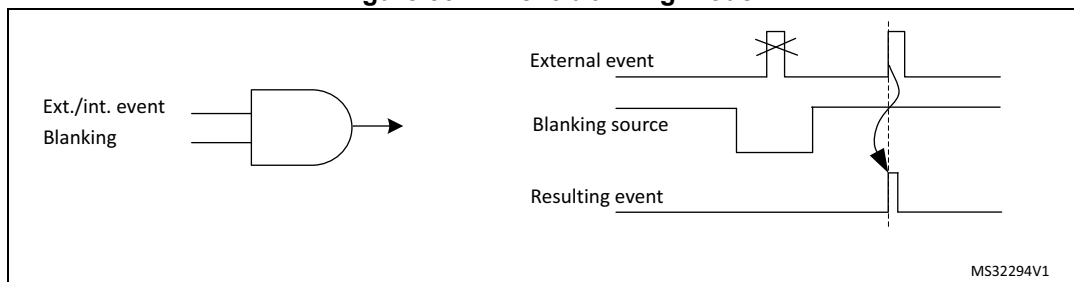
- blanking, to mask external events during a defined time period,
- windowing, to enable external events only during a defined time period.

These modes are enabled using HRTIM_EEFLTR[3:0] bits in the HRTIM_EEFxR1 and HRTIM_EEFxR2 registers. Each of the 5 TimerA..E timing units has its own programmable filter settings for the 10 external events.

Blanking mode

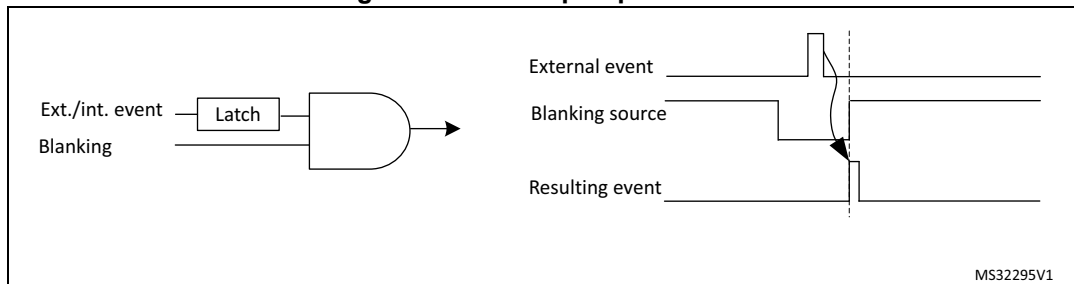
In event blanking mode (see [Figure 304](#)), the external event is ignored if it happens during a given blanking period. This is convenient, for instance, to avoid a current limit to trip on switching noise at the beginning of a PWM period. This mode is active for EEFLTR[3:0] bitfield values ranging from 0001 to 1100.

Figure 304. Event blanking mode



In event postpone mode, the external event is not taken into account immediately but is memorized (latched) and generated as soon as the blanking period is completed, as shown on [Figure 305](#). This mode is enabled by setting EExLTCH bit in HRTIM_EEFxR1 and HRTIM_EEFxR2 registers.

Figure 305. Event postpone mode



The blanking signal comes from several sources:

- the timer itself: the blanking lasts from the counter reset to the compare match (EExFLTR[3:0] = 0001 to 0100 for Compare 1 to Compare 4)
- from other timing units (EExFLTR[3:0] = 0101 to 1100): the blanking lasts from the selected timing unit counter reset to one of its compare match, or can be fully programmed as a waveform on Tx2 output. In this case, events are masked as long as the Tx2 signal is inactive (it is not necessary to have the output enabled, the signal is taken prior to the output stage).

The EEXFLTR[3:0] configurations from 0101 to 1100 are referred to as TIMFLTR1 to TIMFLTR8 in the bit description, and differ from one timing unit to the other. [Table 307](#) gives the 8 available options per timer: CMPx refers to blanking from counter reset to compare match, Tx2 refers to the timing unit TIMx output 2 waveform defined with HRTIM_SETx2 and HRTIM_RSTx2 registers. For instance, Timer B (TIMFLTR6) is Timer C output 2 waveform.

Table 307. Filtering signals mapping per time

	Source	Timer A				Timer B				Timer C				Timer D				Timer E			
		CMP 1	CMP 2	CMP 4	TA2	CMP 1	CMP 2	CMP 4	TB2	CMP 1	CMP 2	CMP 4	TC2	CMP 1	CMP 2	CMP 4	TD2	CMP 1	CMP 2	CMP 4	TE2
Destination	Timer A	-	-	-	-	1	-	2	3	4	-	5	6	7	-	-	-	-	8	-	-
	Timer B	1	-	2	3	-	-	-	-	4	5	-	6	-	7	-	-	8	-	-	-
	Timer C	-	1	-	-	2	-	3	4	-	-	-	-	5	-	6	7	-	-	8	-
	Timer D	1	-	-	-	-	2	-	-	3	4	-	5	-	-	-	-	6	-	7	8
	Timer E	-	1	-	-	2	-	-	-	3	-	4	5	6	-	7	8	-	-	-	-

[Figure 306](#) and [Figure 307](#) give an example of external event blanking for all edge and level sensitivities, in regular and postponed modes.

Figure 306. External trigger blanking with edge-sensitive trigger

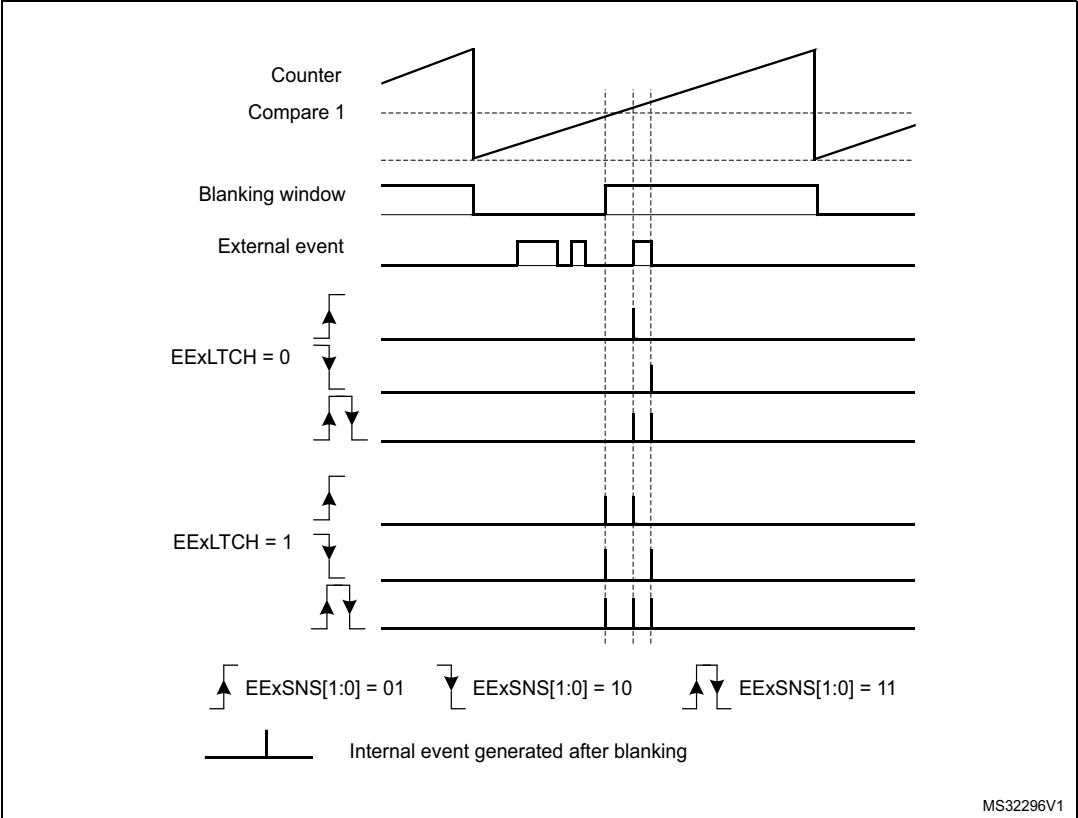
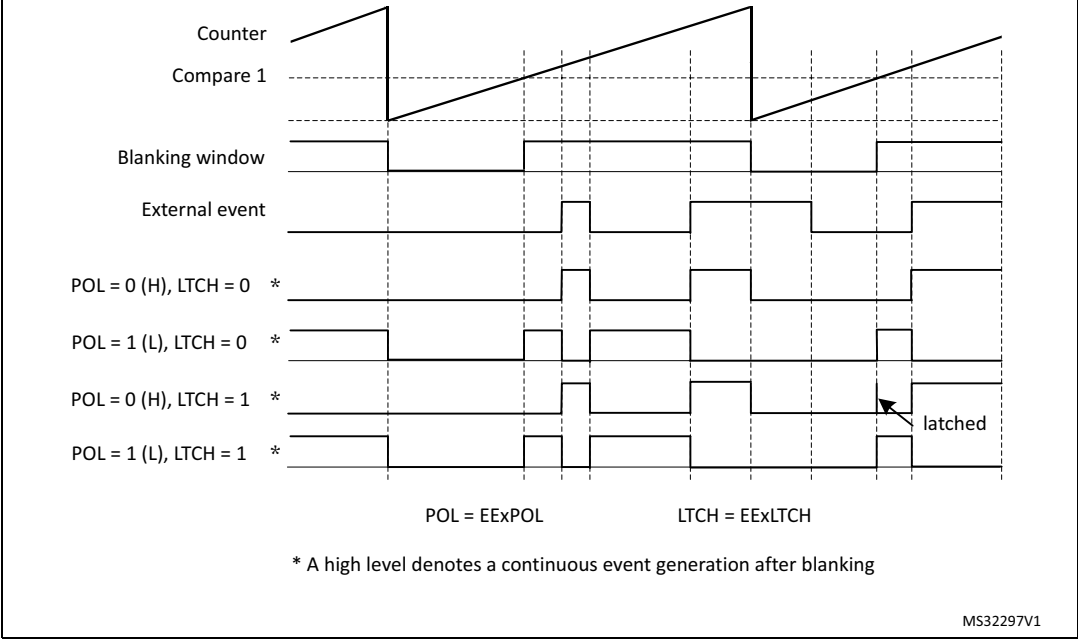


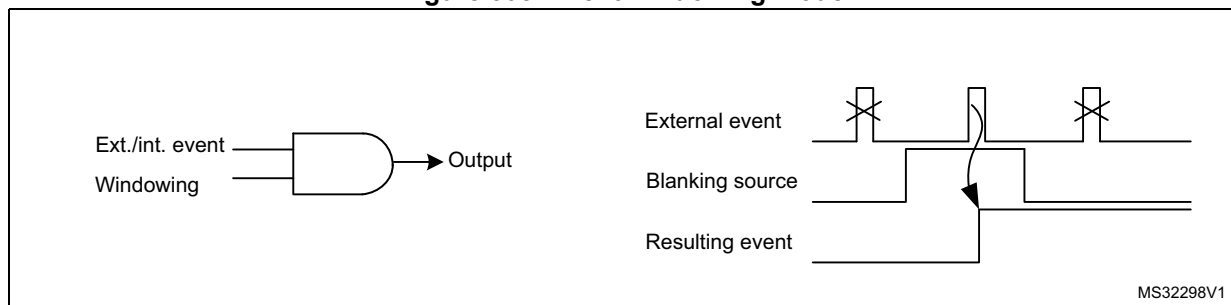
Figure 307. External trigger blanking, level sensitive triggering



Windowing mode

In event windowing mode, the event is taken into account only if it occurs within a given time window, otherwise it is ignored. This mode is active for EExFLTR[3:0] ranging from 1101 to 1111.

Figure 308. Event windowing mode



EExLTCH bit in EEExR1 and EEExR2 registers allows to latch the signal, if set to 1: in this case, an event is accepted if it occurs during the window but is delayed at the end of it.

- If EExLTCH bit is reset and the signal occurs during the window, it is passed through directly.
- If EExLTCH bit is reset and no signal occurs, a timeout event is generated at the end of the window.

A use case of the windowing mode is to filter synchronization signals. The timeout generation allows to force a default synchronization event, when the expected synchronization event is lacking (for instance during a converter start-up).

There are 3 sources for each external event windowing, coded as follows:

- 1101 and 1110: the windowing lasts from the counter reset to the compare match (respectively Compare 2 and Compare 3)
- 1111: the windowing is related to another timing unit and lasts from its counter reset to its Compare 2 match. The source is described as TIMWIN in the bit description and is given in [Table 308](#). As an example, the external events in timer B can be filtered by a window starting from timer A counter reset to timer A Compare 2.

Table 308. Windowing signals mapping per timer (EEFLTR[3:0] = 1111)

Destination	Timer A	Timer B	Timer C	Timer D	Timer E
TIMWIN (source)	Timer B CMP2	Timer A CMP2	Timer D CMP2	Timer C CMP2	Timer D CMP2

Note: The timeout event generation is not supported if the external event is programmed in fast mode.

[Figure 309](#) and [Figure 310](#) present how the events are generated for the various edge and level sensitivities, as well as depending on EExLTCH bit setting. Timeout events are specifically mentioned for clarity reasons.

Figure 309. External trigger windowing with edge-sensitive trigger

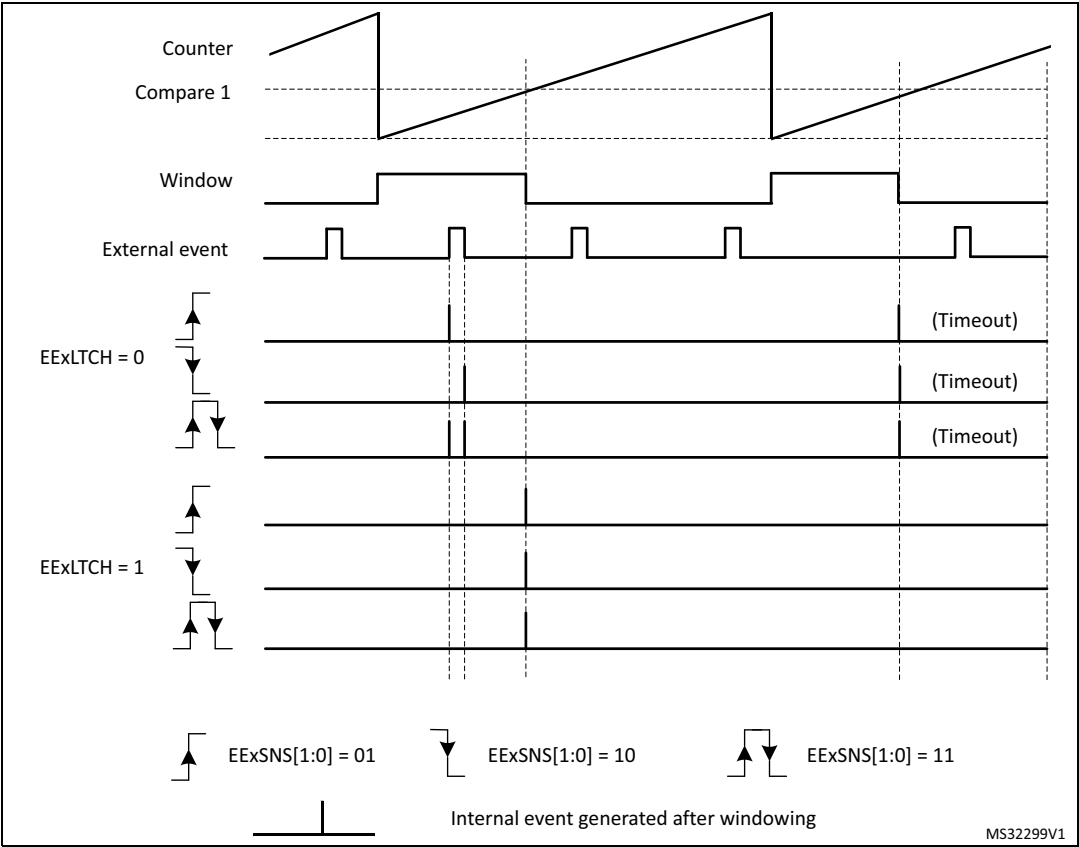
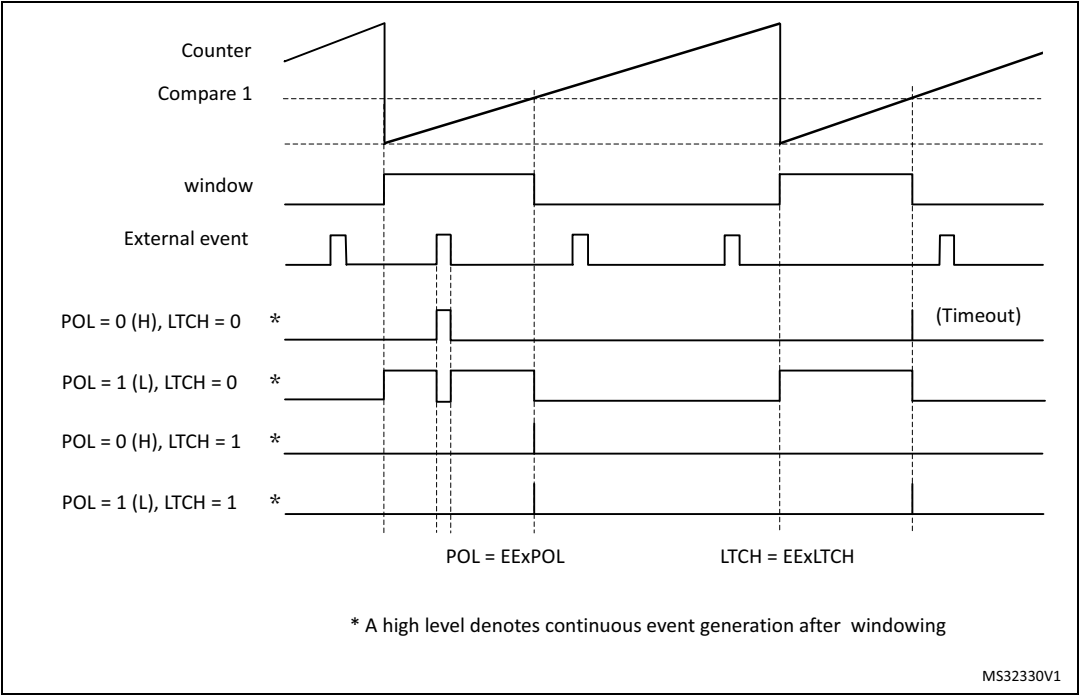


Figure 310. External trigger windowing, level sensitive triggering



37.3.9 Delayed Protection

The HRTIM features specific protection schemes, typically for resonant converters when it is necessary to shut down the PWM outputs in a delayed manner, either once the active pulse is completed or once a push-pull period is completed. These features are enabled with DLYPRTEN bit in the HRTIM_OUTxR register, and are using specific external event channels.

Delayed Idle

In this mode, the active pulse is completed before the protection is activated. The selected external event causes the output to enter in idle mode at the end of the active pulse (defined by an output reset event in HRTIM_RSTx1R or HRTIM_RSTx2R).

Once the protection is triggered, the idle mode is permanently maintained but the counter continues to run, until the output is re-enabled. Tx1OEN and Tx2OEN bits are not affected by the delayed idle entry. To exit from delayed idle and resume operation, it is necessary to overwrite Tx1OEN and Tx2OEN bits to 1. The output state will change on the first transition to an active state following the output enable command.

Note: The delayed idle mode cannot be exited immediately after having been entered, before the active pulse is completed: it is mandatory to make sure that the outputs are in idle state before resuming the run mode. This can be done by waiting up to the next period, for instance, or by polling the O1CPY and/or O2CPY status bits in the TIMxISR register.

The delayed idle mode can be applied to a single output (DLYPRT[2:0] = x00 or x01) or to both outputs (DLYPRT[2:0] = x10).

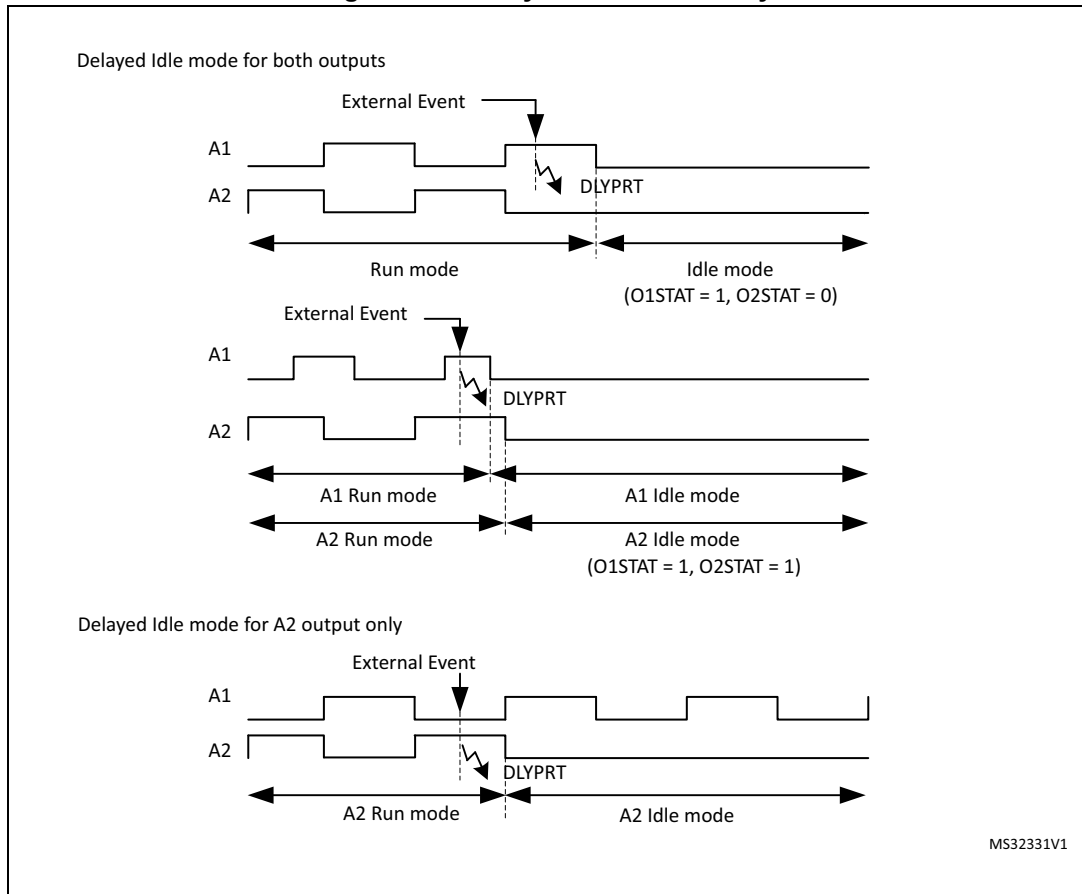
An interrupt or a DMA request can be generated in response to a Delayed Idle mode entry. The DLYPRT flag in HRTIM_TIMxISR is set as soon as the external event arrives, independently from the end of the active pulse on output.

When the Delayed Idle mode is triggered, the output states can be determined using O1STAT and O2STAT in HRTIM_TIMxISR. Both status bits are updated even if the delayed idle is applied to a single output. When the push-pull mode is enabled, the IPPSTAT flag in HRTIM_TIMxISR indicates during which period the delayed protection request occurred.

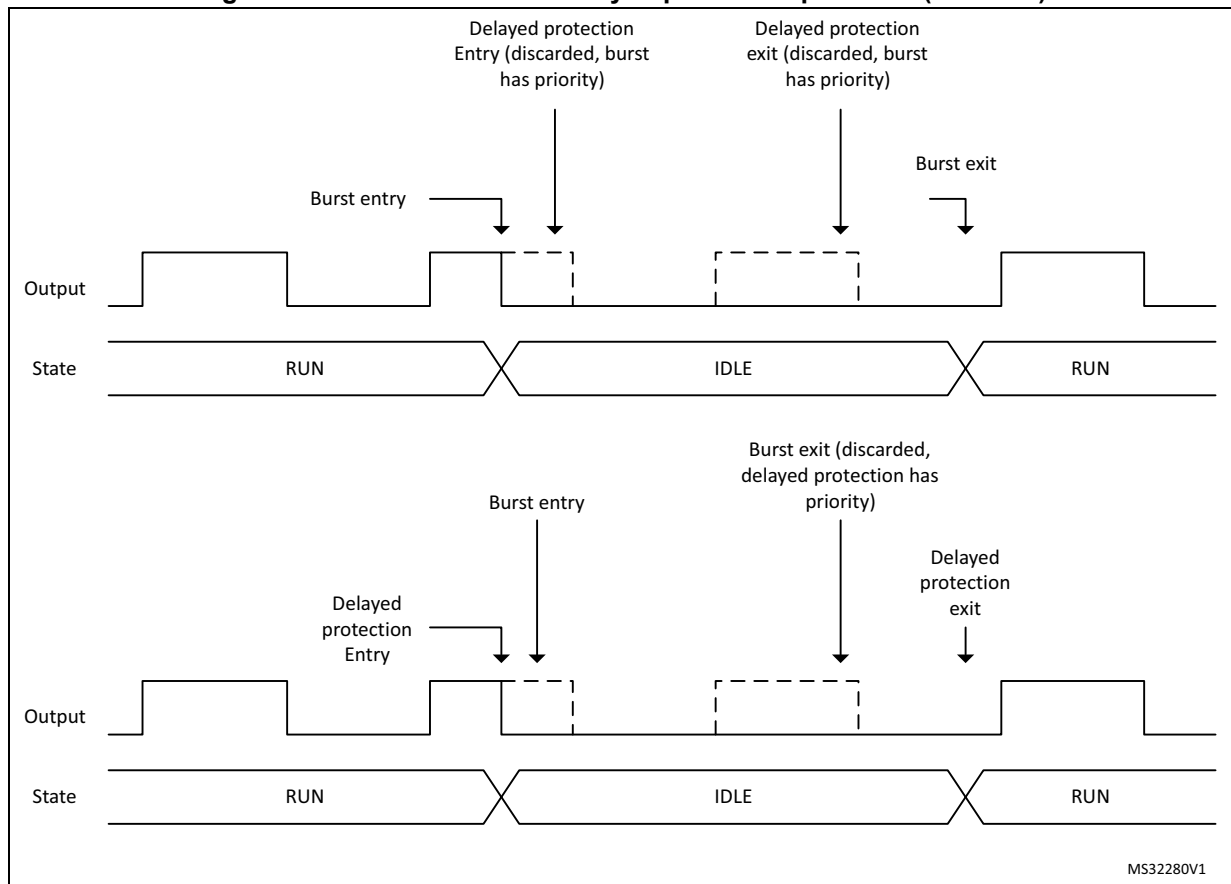
This mode is available whatever the timer operating mode (regular, push-pull, deadtime). It is available with 2 external events only:

- hrtim_evt6 and hrtim_evt7 for Timer A, B and C
- hrtim_evt8 and hrtim_evt9 for Timer D and E

The delayed protection mode can be triggered only when the counter is enabled (TxCEN bit set). It remains active even if the TxEN bit is reset, until the TxyOEN bits are set.

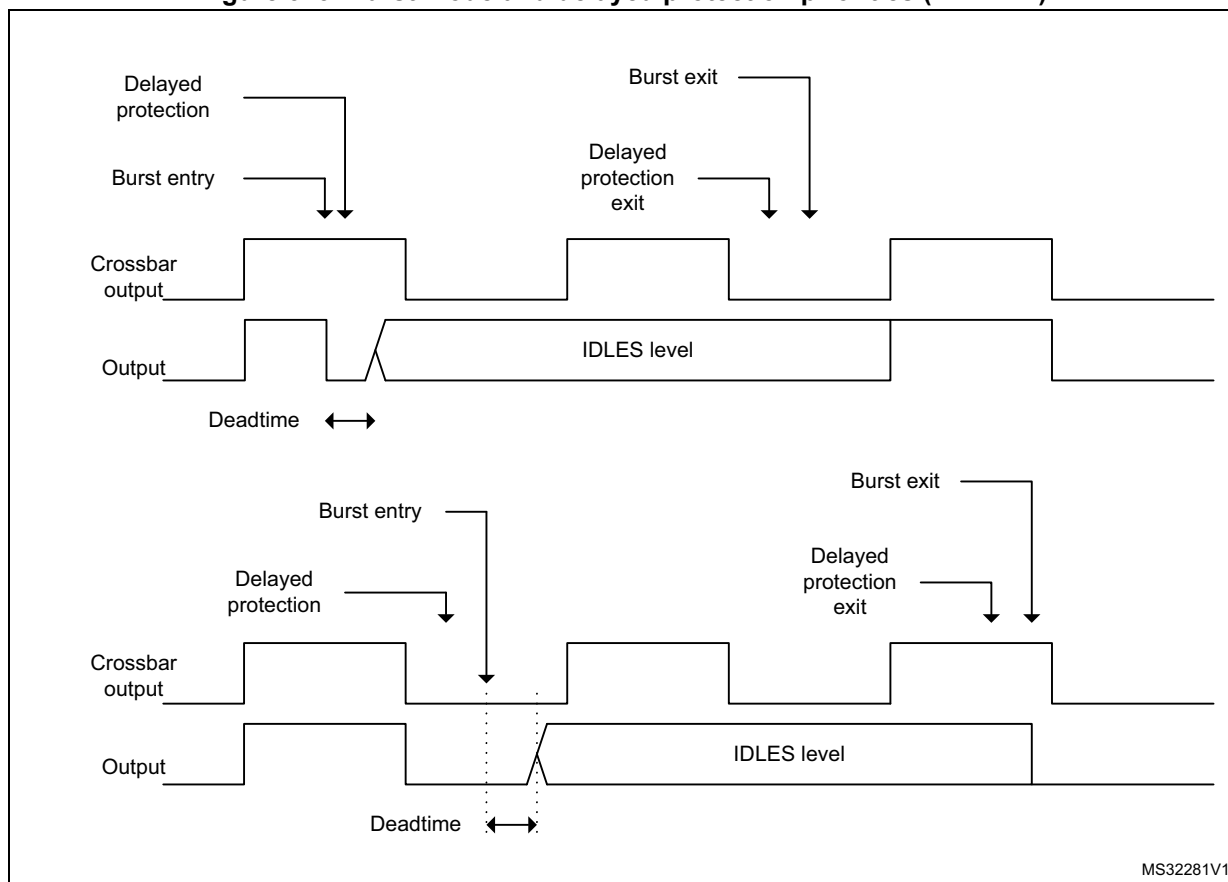
Figure 311. Delayed Idle mode entry

The delayed idle mode has a higher priority than the burst mode: any burst mode exit request is discarded once the delayed idle protection has been triggered. On the contrary, If the delayed protection is exited while the burst mode is active, the burst mode will be resumed normally and the output will be maintained in the idle state until the burst mode exits. [Figure 312](#) gives an overview of these different scenarios.

Figure 312. Burst mode and delayed protection priorities (DIDL = 0)

The same priorities are applied when the delayed burst mode entry is enabled (DIDL bit set), as shown on [Figure 313](#) below.

Figure 313. Burst mode and delayed protection priorities (DIDL = 1)



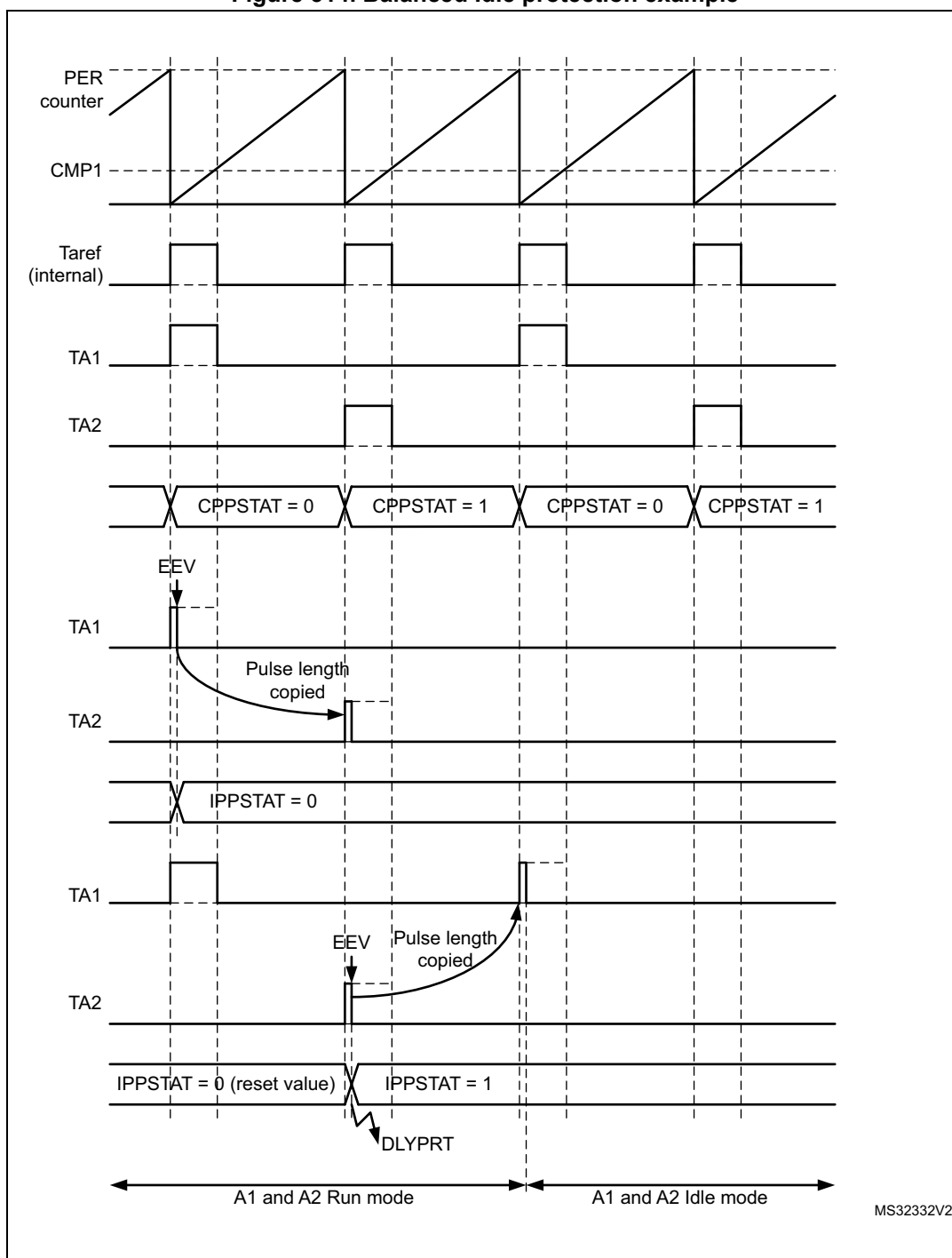
Balanced Idle

Only available in push-pull mode, it allows to have balanced pulsewidth on the two outputs when one of the active pulse is shortened due to a protection. The pulsewidth, which was terminated earlier than programmed, is copied on the alternate output and the two outputs are then put in idle state, until the normal operation is resumed by software. This mode is enabled by writing x11 in DLYPRT[2:0] bitfield in HRTIM_OUTxR.

This mode is available with 2 external events only:

- hrtim_evt6 and hrtim_evt7 for Timer A, B and C
- hrtim_evt8 and hrtim_evt9 for Timer D and E

Figure 314. Balanced Idle protection example



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When the balanced Idle mode is enabled, the selected external event triggers a capture of the counter value into the Compare 4 active register (this value is not user-accessible). The push-pull is maintained for one additional period so that the shortened pulse can be repeated: a new output reset event is generated while the regular output set event is maintained.

The Idle mode is then entered and the output takes the level defined by IDLESx bits in the HRTIM_OUTxR register. The balanced idle mode entry is indicated by the DLYPRT flag, while the IPPSTAT flag indicates during which period the external event occurred, to determine the sequence of shorten pulses (HRTIM_CHA1 then HRTIM_CHA2 or vice versa).

The timer operation is not interrupted (the counter continues to run).

To enable the balanced idle mode, it is necessary to have the following initialization:

- timer operating in continuous mode (CONT = 1)
- Push-pull mode enabled
- HRTIM_CMP4xR must be set to 0 and the content transferred into the active register (for instance by forcing a software update)
- DELCMP4[1:0] bit field must be set to 00 (auto-delayed mode disabled)
- DLYPRT[2:0] = x11 (delayed protection enable)

Note: *The HRTIM_CMP4xR register must not be written during a balanced idle operation. The CMP4 event is reserved and cannot be used for another purpose.*

In balanced idle mode, it is recommended to avoid multiple external events or software-based reset events causing an output reset. If such an event arrives before a balanced idle request within the same period, it will cause the output pulses to be unbalanced (1st pulse length defined by the external event or software reset, while the 2nd pulse is defined by the balanced idle mode entry).

The minimum pulsewidth that can be handled in balanced idle mode is $4 f_{\text{HRTIM}}$ clock periods.

If the capture occurs before the counter has reached this minimum value, the current pulse is extended up to $4 f_{\text{HRTIM}}$ clock periods before being copied into the secondary output. In any case, the pulsewidths are always balanced.

Tx1OEN and Tx2OEN bits are not affected by the balanced idle entry. To exit from balanced idle and resume the operation, it is necessary to overwrite Tx1OEN and Tx2OEN bits to 1 simultaneously. The output state will change on the first active transition following the output enable.

It is possible to resume operation similarly to the delayed idle entry. For instance, if the external event arrives while output 1 is active (delayed idle effective after output 2 pulse), the re-start sequence can be initiated for output 1 first. To do so, it is necessary to poll CPPSTAT bit in the HRTIM_TIMxISR register. Using the above example (IPPSTAT flag equal to 0), the operation will be resumed when CPPSTAT bit is 0.

In order to have a specific re-start sequence, it is possible to poll the CPPSTAT to know which output will be active first. This allows, for instance, to re-start with the same sequence as the idle entry sequence: if EEV arrives during output 1 active, the re-start sequence will be initiated when the output 1 is active (CPPSTAT = 0).

Note: *The balanced idle mode must not be disabled while a pulse balancing sequence is on-going. It is necessary to wait until the CMP4 flag is set, thus indicating that the sequence is completed, to reset the DLYPRTEN bit.*

The balanced idle protection mode can be triggered only when the counter is enabled (TxCEN bit set). It remains active even if the TxCEN bit is reset, until TxyOEN bits are set.

Balanced idle can be used together with the burst mode under the following conditions:

- TxBM bit must be reset (counter clock maintained during the burst, see [Section 37.3.13](#)),
- No balanced idle protection must be triggered while the outputs are in a burst idle state.

The balanced idle mode has a higher priority than the burst mode: any burst mode exit request is discarded once the balanced idle protection has been triggered. On the contrary, if the delayed protection is exited while the burst mode is active, the burst mode will be resumed normally.

Note: Although the output state is frozen in idle mode, a number of events are still generated on the auxiliary outputs (see [Section 37.3.16](#)) during the idle period following the delayed protection:

- Output set/reset interrupt or DMA requests
- External event filtering based on output signal
- Capture events triggered by set/reset

37.3.10 Register preload and update management

Most of HRTIM registers are buffered and can be preloaded if needed. Typically, this allows to prevent the waveforms from being altered by a register update not synchronized with the active events (set/reset).

When the preload mode is enabled, accessed registers are shadow registers. Their content is transferred into the active register after an update request, either software or synchronized with an event.

By default, PREEN bits in HRTIM_MCR and HRTIM_TIMxCR registers are reset and the registers are not preloaded: any write directly updates the active registers. If PREEN bit is reset while the timer is running and preload was enabled, the content of the preload registers is directly transferred into the active registers.

Each timing unit and the master timer have their own PREEN bit. If PREEN is set, the preload registers are enabled and transferred to the active register only upon an update event.

There are two options to initialize the timer when the preload feature is needed:

- Enable PREEN bit at the very end of the timer initialization to have the preload registers transferred into the active registers before the timer is enabled (by setting MCEN and TxCEN bits).
- enable PREEN bit at any time during the initialization and force a software update immediately before starting.

[Table 309](#) lists the registers which can be preloaded, together with a summary of available update events.

Table 309. HRTIM preloadable control registers and associated update sources

Timer	Preloadable registers	Preload enable	Update sources
Master Timer	HRTIM_DIER HRTIM_MPER HRTIM_MREP HRTIM_MCMP1R HRTIM_MCMP2R HRTIM_MCMP3R HRTIM_MCMP4R	PREEN bit in HRTIM_MCR	Software Repetition event Burst DMA event Repetition event following a burst DMA event
Timer x x = A..E	HRTIM_TIMxDIER HRTIM_TIMxPER HRTIM_TIMxREP HRTIM_TIMxCMP1R HRTIM_TIMxCMP1CR HRTIM_TIMxCMP2R HRTIM_TIMxCMP3R HRTIM_TIMxCMP4R HRTIM_DTxDIER HRTIM_SETx1R HRTIM_RSTx1R HRTIM_SETx2R HRTIM_RSTx2R HRTIM_RSTxR	PREEN bit in HRTIM_TIMxCR	Software TIMx Repetition event TIMx Reset Event Burst DMA event Update event from other timers (TIMy, Master) Update event following a burst DMA event Update enable input 1..3 Update event following an update enable input 1..3
HRTIM Common	HRTIM_ADC1R HRTIM_ADC2R HRTIM_ADC3R HRTIM_ADC4R	TIMx or Master timer Update, depending on ADxUSRC[2:0] bits in HRTIM_CR1, if PREEN = 1 in the selected timer	

The master timer has 4 update options:

1. Software: writing 1 into MSWU bit in HRTIM_CR2 forces an immediate update of the registers. In this case, any pending hardware update request is cancelled.
2. Update done when the master counter rolls over and the master repetition counter is equal to 0. This is enabled when MREPU bit is set in HRTIM_MCR.
3. Update done once Burst DMA is completed (see [Section 37.3.21](#) for details). This is enabled when BRSTDMA[1:0] = 01 in HRTIM_MCR. It is possible to have both MREPU=1 and BRSTDMA=01.
Note: The update can take place immediately after the end of the burst sequence if SWU bit is set (i.e. forced update mode). If SWU bit is reset, the update will be done on the next update event following the end of the burst sequence.
4. Update done when the master counter rolls over following a Burst DMA completion. This is enabled when BRSTDMA[1:0] = 10 in HRTIM_MCR.

An interrupt or a DMA request can be generated by the master update event.

Each timer (TIMA..E) can also have the update done as follows:

- By software: writing 1 into TxSWU bit in HRTIM_CR2 forces an immediate update of the registers. In this case, any pending hardware update request is canceled.
- Update done when the counter rolls over and the repetition counter is equal to 0. This is enabled when TxREPU bit is set in HRTIM_TIMxCR.
- Update done when the counter is reset or rolls over in continuous mode. This is enabled when TxRSTU bit is set in HRTIM_TIMxCR. This is used for a timer operating in single-shot mode, for instance.
- Update done once a Burst DMA is completed. This is enabled when UPDGAT[3:0] = 0001 in HRTIM_TIMxCR.
- Update done on the update event following a Burst DMA completion (the event can be enabled with TxREPU, MSTU or TxU). This is enabled when UPDGAT[3:0] = 0010 in HRTIM_TIMxCR.
- Update done when receiving a request on the update enable input 1..3. This is enabled when UPDGAT[3:0] = 0011, 0100, 0101 in HRTIM_TIMxCR.
- Update done on the update event following a request on the update enable input 1..3 (the event can be enabled with TxREPU, MSTU or TxU). This is enabled when UPDGAT[3:0] = 0110, 0111, 1000 in HRTIM_TIMxCR.
- Update done synchronously with any other timer or master update (for instance TIMA can be updated simultaneously with TIMB). This is used for converters requiring several timers, and is enabled by setting bits MSTU and TxU in HRTIM_TIMxCR register.

The update enable inputs 1..3 allow to have an update event synchronized with on-chip events coming from the general-purpose timers. These inputs are rising-edge sensitive.

[Table 310](#) lists the connections between update enable inputs and the on-chip sources.

Table 310. Update enable inputs and sources

Update enable input	Update source
Update enable input 1	TIM16_OC
Update enable input 2	TIM17_OC
Update enable input 3	TIM6_TRGO

This allows to synchronize low frequency update requests with high-frequency signals (for instance an update on the counter roll-over of a 100 kHz PWM that has to be done at a 100 Hz rate).

Note: *The update events are synchronized to the prescaler clock when CKPSC[2:0] > 5.*

An interrupt or a DMA request can be generated by the Timx update event.

MUDIS and TxUDIS bits in the HRTIM_CR1 register allow to temporarily disable the transfer from preload to active registers, whatever the selected update event. This allows to modify several registers in multiple timers. The regular update event takes place once these bits are reset.

MUDIS and TxUDIS bits are all grouped in the same register. This allows the update of multiple timers (not necessarily synchronized) to be disabled and resumed simultaneously.

The following example is a practical use case. A first power converter is controlled with the master, TIMB and TIMC. TIMB and TIMC must be updated simultaneously with the master timer repetition event. A second converter works in parallel with TIMA, TIMD and TIME, and TIMD, TIME must be updated with TIMA repetition event.

First converter

In HRTIM_MCR, MREPU bit is set: the update will occur at the end of the master timer counter repetition period. In HRTIM_TIMBCR and HRTIM_TIMCCR, MSTU bits are set to have TIMB and TIMC timers updated simultaneously with the master timer.

When the power converter set-point has to be adjusted by software, MUDIS, TBUDIS and TCUDIS bits of the HRTIM_CR register must be set prior to write accessing registers to update the values (for instance the compare values). From this time on, any hardware update request is ignored and the preload registers can be accessed without any risk to have them transferred into the active registers. Once the software processing is over, MUDIS, TBUDIS and TCUDIS bits must be reset. The transfer from preload to active registers will be done as soon as the master repetition event occurs.

Second converter

In HRTIM_TIMACR, TAREPU bit is set: the update will occur at the end of the Timer A counter repetition period. In HRTIM_TIMDCR and HRTIM_TIMECR, TAU bits are set to have TIMD and TIME timers updated simultaneously with Timer A.

When the power converter set-point has to be adjusted by software, TAUDIS, TDUDIS and TEUDIS bits of the HRTIM_CR register must be set prior to write accessing the registers to update the values (for instance the compare values). From this time on, any hardware update request is ignored and the preload registers can be accessed without any risk to have them transferred into the active registers. Once the software processing is over, TAUDIS, TDUDIS and TEUDIS bits can be reset: the transfer from preload to active registers will be done as soon as the Timer A repetition event occurs.

37.3.11 Events propagation within or across multiple timers

The HRTIM offers many possibilities for cascading events or sharing them across multiple timing units, including the master timer, to get full benefits from its modular architecture. These are key features for converters requiring multiple synchronized outputs.

This section summarizes the various options and specifies whether and how an event is propagated within the HRTIM.

TIMx update triggered by the Master timer update

The sources listed in [Table 311](#) are generating a master timer update. The table indicates if the source event can be used to trigger a simultaneous update in any of TIMx timing units.

Operating condition: MSTU bit is set in HRTIM_TIMxCR register.

Table 311. Master timer update event propagation

Source	Condition	Propagation	Comment
Burst DMA end	BRSTDMA[1:0] = 01	No	Must be done in TIMxCR (UPDGAT[3:0] = 0001)
Roll-over event following a Burst DMA end	BRSTDMA[1:0] = 10	Yes	-
Repetition event caused by a counter roll-over	MREPU = 1	Yes	-
Repetition event caused by a counter reset (from HRTIM_SCIN or software)		No	-
Software update	MSWU = 1	No	All software update bits (TxSWU) are grouped in the HRTIM_CR2 register and can be used for a simultaneous update

TIMx update triggered by the TIMy update

The sources listed in [Table 312](#) are generating a TIMy update. The table indicates if the given event can be used to trigger a simultaneous update in another or multiple TIMx timers.

Operating condition: TyU bit set in HRTIM_TIMxCR register (source = TIMy and destination = TIMx).

Table 312. TIMx update event propagation

Source	Condition	Propagation	Comment
Burst DMA end	UPDGAT[3:0] = 0001	No	Must be done directly in HRTIM_TIMxCR (UPDGAT[3:0] = 0001)
Update caused by the update enable input	UPDGAT[3:0] = 0011, 0100, 0101	No	Must be done directly in HRTIM_TIMxCR (UPDGAT[3:0] = 0011, 0100, 0101)
Master update	MSTU = 1 in HRTIM_TIMyCR	No	Must be done with MSTU = 1 in HRTIM_TIMxCR
Another TIMx update (TIMz > TIMy > TIMx)	TzU=1 in HRTIM_TIMyCR TyU=1 in TIMxCR	No	Must be done with TzU=1 in HRTIM_TIMxCR TzU=1 in HRTIM_TIMyCR
Repetition event caused by a counter roll-over	TyREPU = 1	Yes	-
Repetition event caused by a counter reset	TyREPU = 1	-	Refer to counter reset cases below
Counter roll-over	TyRSTU = 1	Yes	-
Counter software reset	TyRST=1 in HRTIM_CR2	No	Can be done simultaneously with update in HRTIM_CR2 register
Counter reset caused by a TIMz compare	TIMzCMPn in HRTIM_RSTyR	No	Must be done using TIMzCMPn in HRTIM_RSTxR
Counter reset caused by external events	EXTEVNTn in HRTIM_RSTyR	Yes	-

Table 312. TIMx update event propagation (continued)

Source	Condition	Propagation	Comment
Counter reset caused by a master compare or a master period	MSTCMPn or MSTPER in HRTIM_RSTyR	No	-
Counter reset caused by a TIMy compare	CMPn in HRTIM_RSTyR	Yes	-
Counter reset caused by an update	UPDT in HRTIM_RSTyR	No	Propagation would result in a lock-up situation (update causing reset causing update)
Counter reset caused by HRTIM_SCIN	SYNCRSTy in HRTIM_TIMyCR	No	-
Software update	TySWU = 1	No	All software update bits (TxSWU) are grouped in the HRTIM_CR2 register and can be used for a simultaneous update

TIMx Counter reset causing a TIMx update

[Table 313](#) lists the counter reset sources and indicates whether they can be used to generate an update.

Operating condition: TxRSTU bit in HRTIM_TIMxCR register.

Table 313. Reset events able to generate an update

Source	Condition	Propagation	Comment
Counter roll-over		Yes	
Update event	UPDT in HRTIM_RSTxR	No	Propagation would result in a lock-up situation (update causing a reset causing an update)
External Event	EXTEVNTn in HRTIM_RSTxR	Yes	-
TIMy compare	TIMyCMPn in HRTIM_RSTxR	Yes	-
Master compare	MSTCMPn in HRTIM_RSTxR	Yes	-
Master period	MSTPER in HRTIM_RSTxR	Yes	-
Compare 2 and 4	CMPn in HRTIM_RSTxR	Yes	-
Software	TxRST=1 in HRTIM_CR2	Yes	-
HRTIM_SCIN	SYNCRSTx in HRTIM_TIMxCR	Yes	-

TIMx update causing a TIMx counter reset

Table 314 lists the update event sources and indicates whether they can be used to generate a counter reset.

Operating condition: UPDT bit set in HRTIM_RSTxR.

Table 314. Update event propagation for a timer reset

Source	Condition	Propagation	Comment
Burst DMA end	UPDGAT[3:0] = 0001	Yes	-
Update caused by the update enable input	UPDGAT[3:0] = 0011, 0100, 0101	Yes	-
Master update caused by a roll-over after a Burst DMA	MSTU = 1 in HRTIM_TIMxCR BRSTDMA[1:0] = 10 in HRTIM_MCR	Yes	-
Master update caused by a repetition event following a roll-over	MSTU = 1 in HRTIM_TIMxCR	Yes	-
Master update caused by a repetition event following a counter reset (software or due to HRTIM_SCIN)	MREPU = 1 in HRTIM_MCR	No	-
Software triggered master timer update	MSTU = 1 in HRTIM_TIMxCR MSWU = 1 in HRTIM_CR2	No	All software update bits (TxSWU) are grouped in the HRTIM_CR2 register and can be used for a simultaneous update
TIMy update caused by a TIMy counter roll-over	TyU = 1 in HRTIM_TIMxCR TyRSTU = 1 in HRTIM_TIMyCR	Yes	-
TIMy update caused by a TIMy repetition event	TyU = 1 in HRTIM_TIMxCR TyREPU = 1 in HRTIM_TIMyCR	Yes	-
TIMy update caused by an external event or a TIMy compare (through a TIMy reset)	TyU = 1 in HRTIM_TIMxCR TyRSTU = 1 in HRTIM_TIMyCR EXTEVNTn or CMP4/2 in HRTIM_RSTyCR	Yes	-
TIMy update caused by sources other than those listed above	TyU = 1 in HRTIM_TIMxCR	No	-

Table 314. Update event propagation for a timer reset (continued)

Source	Condition	Propagation	Comment
Repetition event following a roll-over	TxREPU = 1 in HRTIM_TIMxCR	Yes	-
Repetition event following a counter reset		No	-
Timer reset	TxRSTU = 1 in HRTIM_TIMxCR	No	Propagation would result in a lock-up situation (reset causing an update causing a reset)
Software	TxSWU in HRTIM_CR2	No	-

37.3.12 Output management

Each timing unit controls a pair of outputs. The outputs have three operating states:

- **RUN:** this is the main operating mode, where the output can take the active or inactive level as programmed in the crossbar unit.
- **IDLE:** this state is the default operating state after an HRTIM reset, when the outputs are disabled by software or during a burst mode operation (where outputs are temporary disabled during a normal operating mode; refer to [Section 37.3.13](#) for more details). It is either permanently active or inactive.
- **FAULT:** this is the safety state, entered in case of a shut-down request on FAULTx inputs. It can be permanently active, inactive or Hi-Z.

The output status is indicated by TxyOEN bit in HRTIM_OENR register and TxyODS bit in HRTIM_ODSR register, as in [Table 315](#).

Table 315. Output state programming, x= A..E, y = 1 or 2

TxyOEN (control/status) (set by software, cleared by hardware)	TxyODS (status)	Output operating state
1	x	RUN
0	0	IDLE
0	1	FAULT

TxyOEN bit is both a control and a status bit: it must be set by software to have the output in RUN mode. It is cleared by hardware when the output goes back in IDLE or FAULT mode. When TxyOEN bit is cleared, TxyODS bit indicates whether the output is in the IDLE or FAULT state. A third bit in the HRTIM_ODISR register allows to disable the output by software.

Figure 315. Output management overview

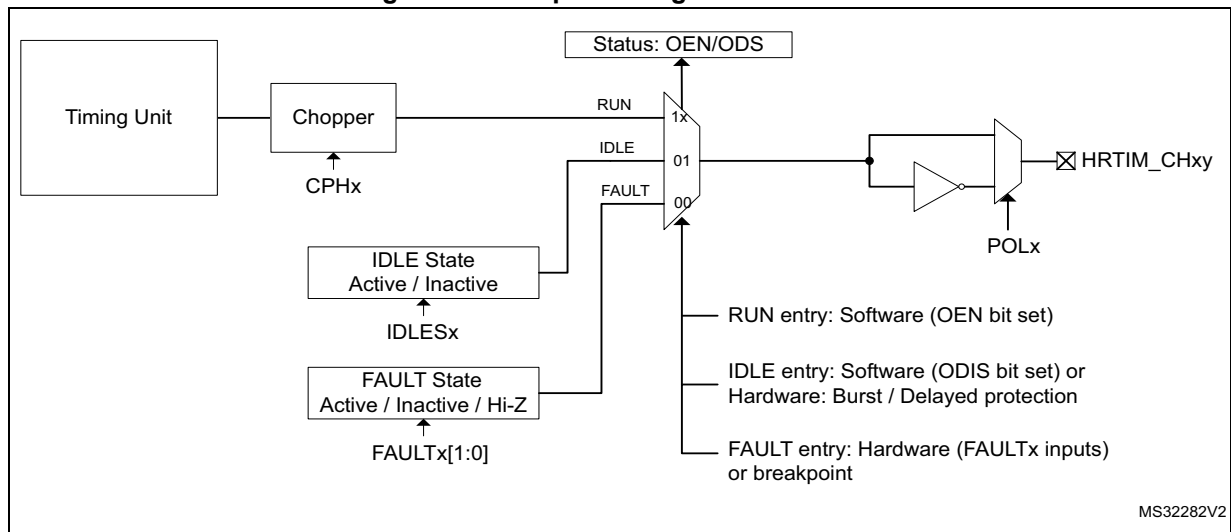
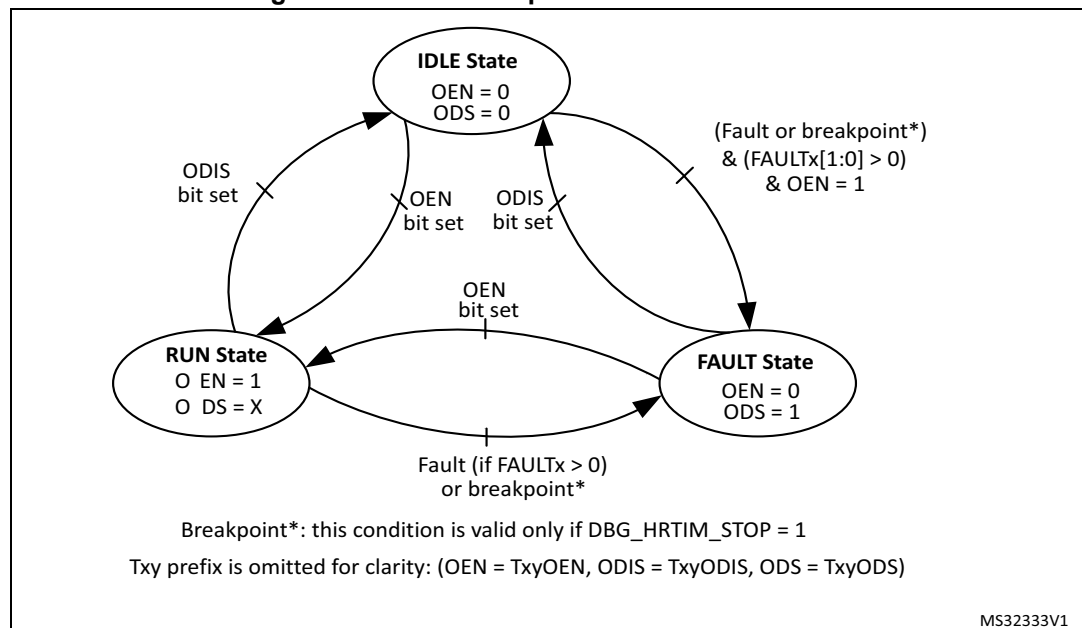


Figure 316 summarizes the bit values for the three states and how the transitions are triggered. Faults can be triggered by any external or internal fault source, as listed in Section 37.3.15, while the Idle state can be entered when the burst mode or delayed protections are active.

Figure 316. HRTIM output states and transitions



The FAULT and IDLE levels are defined as active or inactive. Active (or inactive) refers to the level on the timer output that causes a power switch to be closed (or opened for an inactive state).

The IDLE state has the highest priority: the transition FAULT → IDLE is possible even if the FAULT condition is still valid, triggered by ODIS bit set.

The FAULT state has priority over the RUN state: if TxyOEN bit is set simultaneously with a Fault event, the FAULT state will be entered. The condition is given on the transition IDLE →

FAULT, as in [Figure 316](#): fault protection needs to be enabled (FAULTx[1:0] bits = 01, 10, 11) and the Txy OEN bit set with a fault active (or during a breakpoint if DBG_HRTIM_STOP = 1).

The output polarity is programmed using POLx bits in HRTIM_OUTxR. When POLx = 0, the polarity is positive (output active high), while it is active low in case of a negative polarity (POLx = 1). Practically, the polarity is defined depending on the power switch to be driven (PMOS vs. NMOS) or on a gate driver polarity.

The output level in the FAULT state is configured using FAULTx[1:0] bits in HRTIM_OUTxR, for each output, as follows:

- 00: output never enters the fault state and stays in RUN or IDLE state
- 01: output at active level when in FAULT
- 10: output at inactive level when in FAULT
- 11: output is tri-stated when in FAULT. The safe state must be forced externally with pull-up or pull-down resistors, for instance.

Note: *FAULTx[1:0] bits must not be changed as long as the outputs are in FAULT state.*

The level of the output in IDLE state is configured using IDLESx bit in HRTIM_OUTxR, as follows:

- 0: output at inactive level when in IDLE
- 1: output at active level when in IDLE

When TxyOEN bit is set to enter the RUN state, the output is immediately connected to the crossbar output. If the timer clock is stopped, the level will either be inactive (after an HRTIM reset) or correspond to the RUN level (when the timer was stopped and the output disabled).

During the HRTIM initialization, the output level can be prepositioned prior to have it in RUN mode, using the software forced output set and reset in the HRTIM_SETx1R and HRTIM_RSTx1R registers.

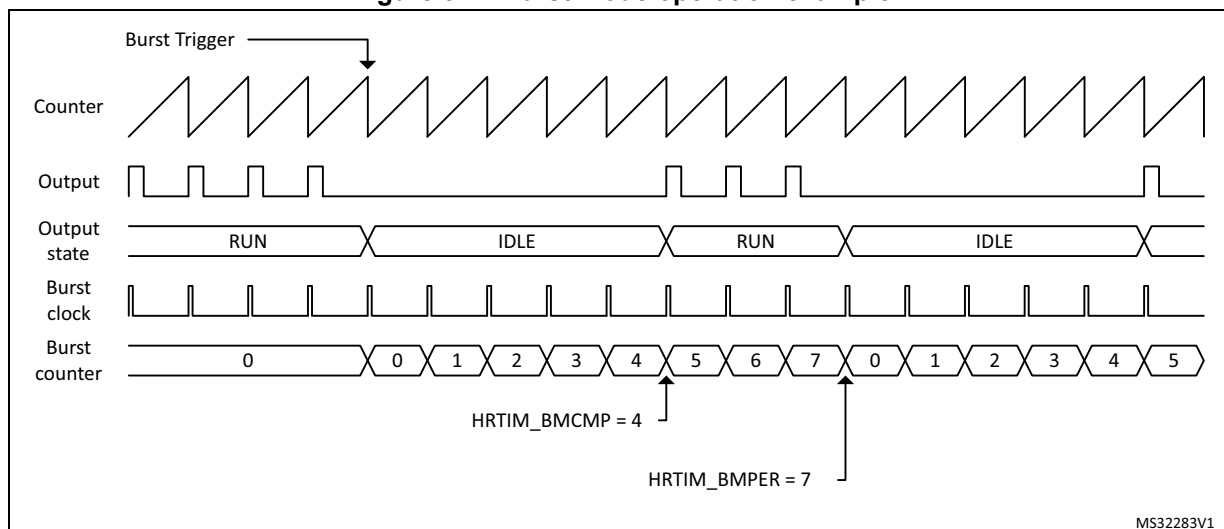
37.3.13 Burst mode controller

The burst mode controller allows to have the outputs alternatively in IDLE and RUN state, by hardware, so as to skip some switching periods with a programmable periodicity and duty cycle.

Burst mode operation is of common use in power converters when operating under light loads. It can significantly increase the efficiency of the converter by reducing the number of transitions on the outputs and the associated switching losses.

When operating in burst mode, one or a few pulses are outputs followed by an idle period equal to several counting periods, typically, where no output pulses are produced, as shown in the example on [Figure 317](#).

Figure 317. Burst mode operation example



The burst mode controller consists of:

- A counter that can be clocked by various sources, either within or outside the HRTIM (typically the end of a PWM period).
- A compare register to define the number of idle periods: HRTIM_BMCMP.
- A period register to define the burst repetition rate (corresponding to the sum of the idle and run periods): HRTIM_BMPER.

The burst mode controller is able to take over the control of any of the 10 PWM outputs. The state of each output during a burst mode operation is programmed using IDLESx and IDLEMx bits in the HRTIM_OUTxR register, as in [Table 316](#).

Table 316. Timer output programming for burst mode

IDLEMx	IDLESx	Output state during burst mode
0	X	No action: the output is not affected by the burst mode operation.
1	0	Output inactive during the burst
1	1	Output active during the burst

Note: IDLEMx bit must not be changed while the burst mode is active.

The burst mode controller only acts on the output stage. A number of events are still generated during the idle period:

- Output set/reset interrupt or DMA requests
- External event filtering based on Tx2 output signal
- Capture events triggered by output set/reset

During the burst mode, neither start nor reset events are generated on the HRTIM_SCOUT output, even if TxBM bit is set.

Operating mode

It is necessary to have the counter enabled (TxCEN bit set) before using the burst mode on a given timing unit. The burst mode is enabled with BME bit in the HRTIM_BMCR register.

It can operate in continuous or single-shot mode, using BMOM bit in the HRTIM_BMCR register. The continuous mode is enabled when BMOM = 1. The Burst operation is maintained until BMSTAT bit in HRTIM_BMCR is reset to terminate it.

In single-shot mode (BMOM = 0), the idle sequence is executed once, following the burst mode trigger, and the normal timer operation is resumed immediately after.

The duration of the idle and run periods is defined with a burst mode counter and 2 registers. The HRTIM_BMCMPR register defines the number of counts during which the selected timer(s) are in an idle state (idle period). HRTIM_BMPER defines the overall burst mode period (sum of the idle and run periods). Once the initial burst mode trigger has occurred, the idle period length is HRTIM_BMCMPR+1, the overall burst period is HRTIM_BMPER+1.

Note: *The burst mode period must not be less than or equal to the deadtime duration defined with DTRx[8:0] and DTFx[8:0] bitfields.*

The counters of the timing units and the master timer can be stopped and reset during the burst mode operation. HRTIM_BMCR holds 6 control bits for this purpose: MTBM (master) and TABM..TEBM for Timer A..E.

When MTBM or TxBM bit is reset, the counter clock is maintained. This allows to keep a phase relationship with other timers in multiphase systems, for instance.

When MTBM or TxBM bit is set, the corresponding counter is stopped and maintained in reset state during the burst idle period. This allows to have the timer restarting a full period when exiting from idle. If SYNCSRC[1:0] = 00 or 10 (synchronization output on the master start or timer A start), a pulse is sent on the HRTIM_SCOUT output when exiting the burst mode.

Note: *TxBM bit must not be set when the balanced idle mode is active (DLYPRT[1:0] = 0x11).*

Burst mode clock

The burst mode controller counter can be clocked by several sources, selected with BMCLK[3:0] bits in the HRTIM_BMCR register:

- BMCLK[3:0] = 0000 to 0101: Master timer and TIMA..E reset/roll-over events. This allows to have burst mode idle and run periods aligned with the timing unit counting period (both in free-running and counter reset mode).
- BMCLK[3:0] = 0110 to 1001: The clocking is provided by the general purpose timers, as in [Table 317](#). In this case, the burst mode idle and run periods are not necessarily aligned with timing unit counting period (a pulse on the output may be interrupted, resulting a waveform with modified duty cycle for instance).
- BMCLK[3:0] = 1010: The f_{HRTIM} clock prescaled by a factor defined with BMPRSC[3:0] bits in HRTIM_BMCR register. In this case, the burst mode idle and run periods are not necessarily aligned with the timing unit counting period (a pulse on the output may be interrupted, resulting in a waveform with a modified duty cycle, for instance).

Table 317. Burst mode clock sources from general purpose timer

BMCLK[3:0]	Clock source
0110	hrtim_bm_ck1: TIM16 OC
0111	hrtim_bm_ck2: TIM17 OC
1000	hrtim_bm_ck3: TIM7 TRGO
1001	hrtim_bm_ck4: Reserved

The pulsewidth on TIMxx OC output must be at least $N f_{\text{HRTIM}}$ clock cycles long to be detected by the HRTIM burst mode controller.

Burst mode triggers

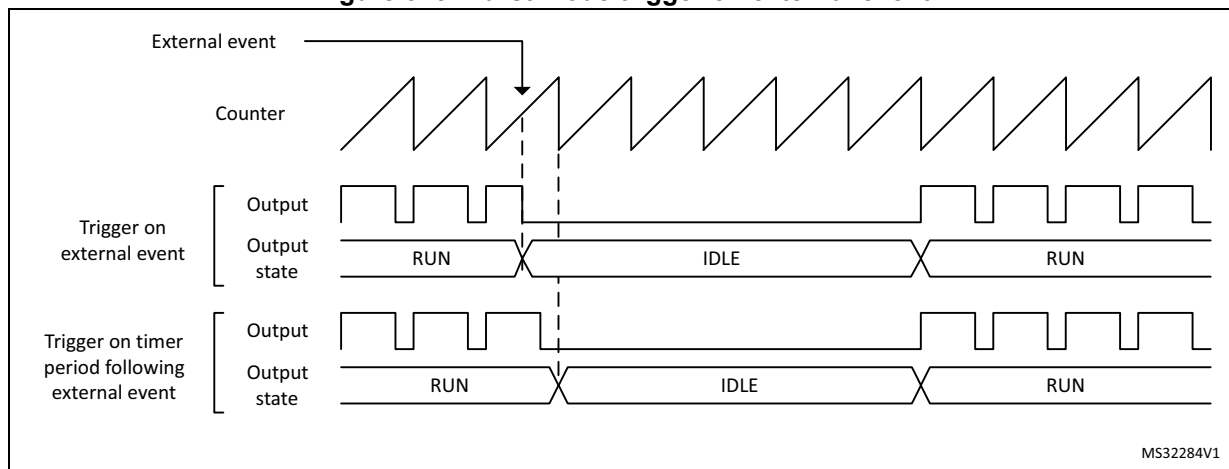
To trigger the burst operation, 32 sources are available and are selected using the HRTIM_BMTRGR register:

- Software trigger (set by software and reset by hardware)
- 6 Master timer events: repetition, reset/roll-over, Compare 1 to 4
- 5 x 4 events from timers A..E: repetition, reset/roll-over, Compare 1 and 2
- hrtim_evt7 (including TIMA event filtering) and hrtim_evt8 (including TIMD event filtering)
- Timer A period following hrtim_evt7 (including TIMA event filtering)
- Timer D period following hrtim_evt8 (including TIMD event filtering)
- On-chip events coming from other general purpose timer (hrtim_bm_trg output:TIM7_TRGO output)

These sources can be combined to have multiple concurrent triggers.

Burst mode is not re-triggerable. In continuous mode, new triggers are ignored until the burst mode is terminated, while in single-shot mode, the triggers are ignored until the current burst completion including run periods (HRTIM_BMPER+1 cycles). This is also valid for software trigger (the software bit is reset by hardware even if it is discarded).

Figure 318 shows how the burst mode is started in response to an external event, either immediately or on the timer period following the event.

Figure 318. Burst mode trigger on external event

For TAEV7 and TDEEV8 combined triggers (trigger on a Timer period following an external event), the external event detection is always active, regardless of the burst mode programming and the on-going burst operation:

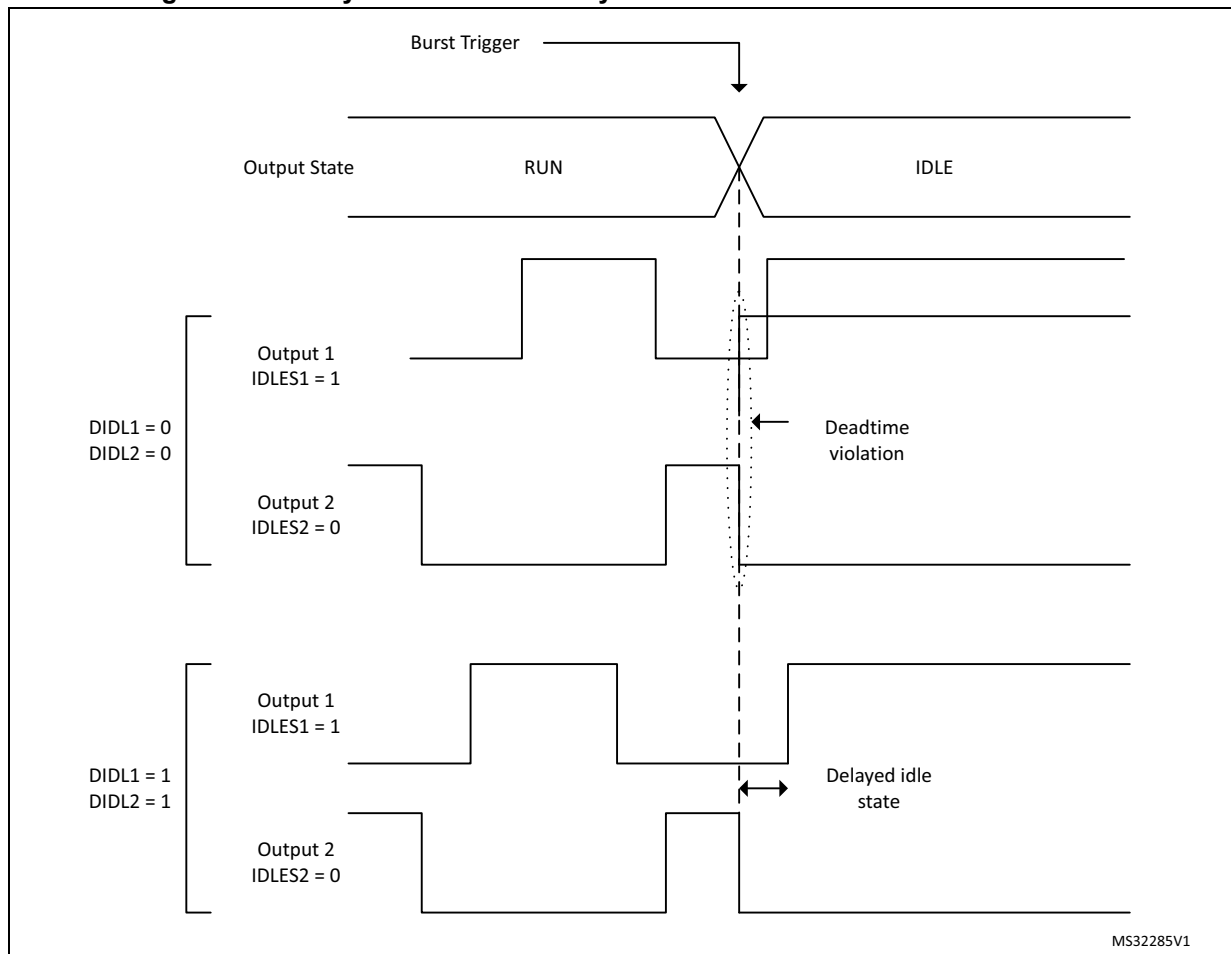
- When the burst mode is enabled (BME=1) or the trigger is enabled (TAEV7 or TDEEV8 bit set in the BMTRG register) in between the external event and the timer period event, the burst is triggered.
- The single-shot burst mode is re-triggered even if the external event occurs before the burst end (as long as the corresponding period happens after the burst).

Note: TAEV7 and TDEEV8 triggers are valid only after a period event. If the counter is reset before the period event, the pending hrtim_evt7/8 event is discarded.

Burst mode delayed entry

By default, the outputs are taking their idle level (as per IDLES1 and IDLES2 setting) immediately after the burst mode trigger.

It is also possible to delay the burst mode entry and force the output to an inactive state during a programmable period before the output takes its idle state. This is useful when driving two complementary outputs, one of them having an active idle state, to avoid a deadtime violation as shown on [Figure 319](#). This prevents any risk of shoot through current in half-bridges, but causes a delayed response to the burst mode entry.

Figure 319. Delayed burst mode entry with deadtime enabled and IDLESx = 1

The delayed burst entry mode is enabled with DIDLx bit in the HRTIM_OUTxR register (one enable bit per output). It forces a deadtime insertion before the output takes its idle state. Each TIMx output has its own deadtime value:

- DTRx[8:0] on output 1 when DIDL1 = 1
- DTFx[8:0] on output 2 when DIDL2 = 1

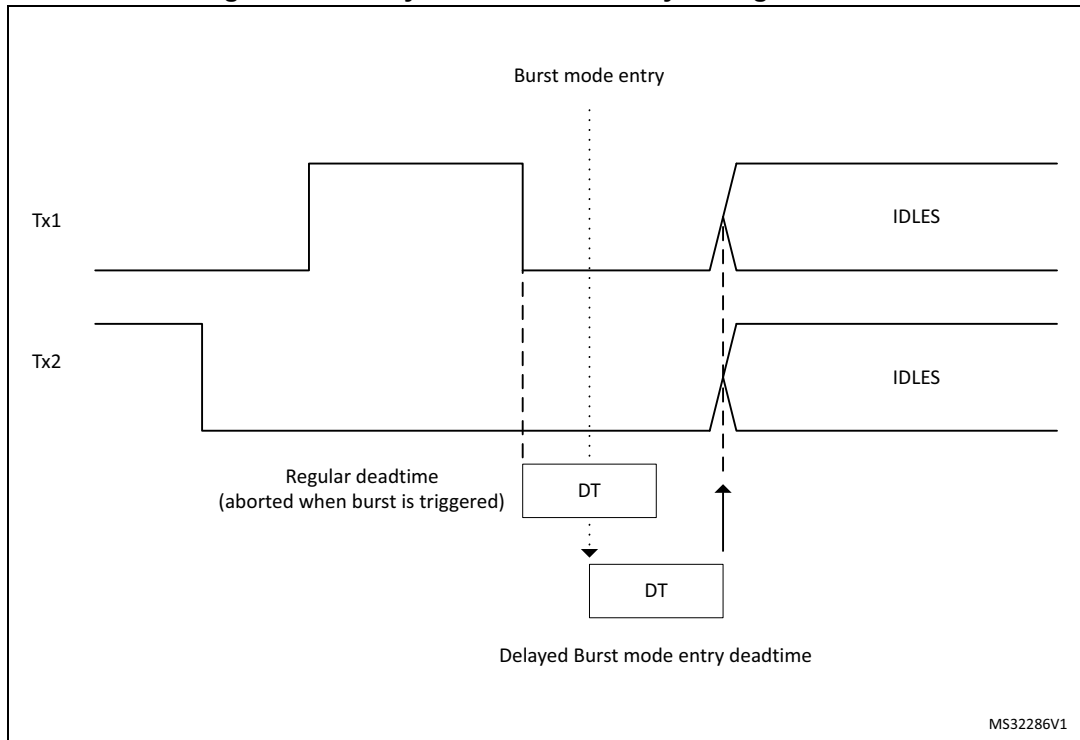
DIDLx bits can be set only if one of the outputs has an active idle level during the burst mode (IDLES = 1) and only when positive deadtimes are used (SDTR/SDTF set to 0).

Note: *The delayed burst entry mode uses deadtime generator resources. Consequently, when any of the 2 DIDLx bits is set and the corresponding timing unit uses the deadtime insertion (DTEN bit set in HRTIM_OUTxR), it is not possible to use the timerx output 2 as a filter for external events (Tx2 filtering signal is not available).*

When durations defined by DTRx[8:0] and DTFx[8:0] are lower than 3 f_{HRTIM} clock cycle periods, the limitations related to the narrow pulse management listed in [Section 37.3.6](#) must be applied.

When the burst mode entry arrives during the regular deadtime, it is aborted and a new deadtime is re-started corresponding to the inactive period, as on [Figure 320](#).

Figure 320. Delayed Burst mode entry during deadtime



Burst mode exit

The burst mode exit is either forced by software (in continuous mode) or once the idle period is elapsed (in single-shot mode). In both cases, the counter is re-started immediately (if it was hold in a reset state with MTBM or TxBM bit = 1), but the effective output state transition from the idle to active mode only happens after the programmed set/reset event.

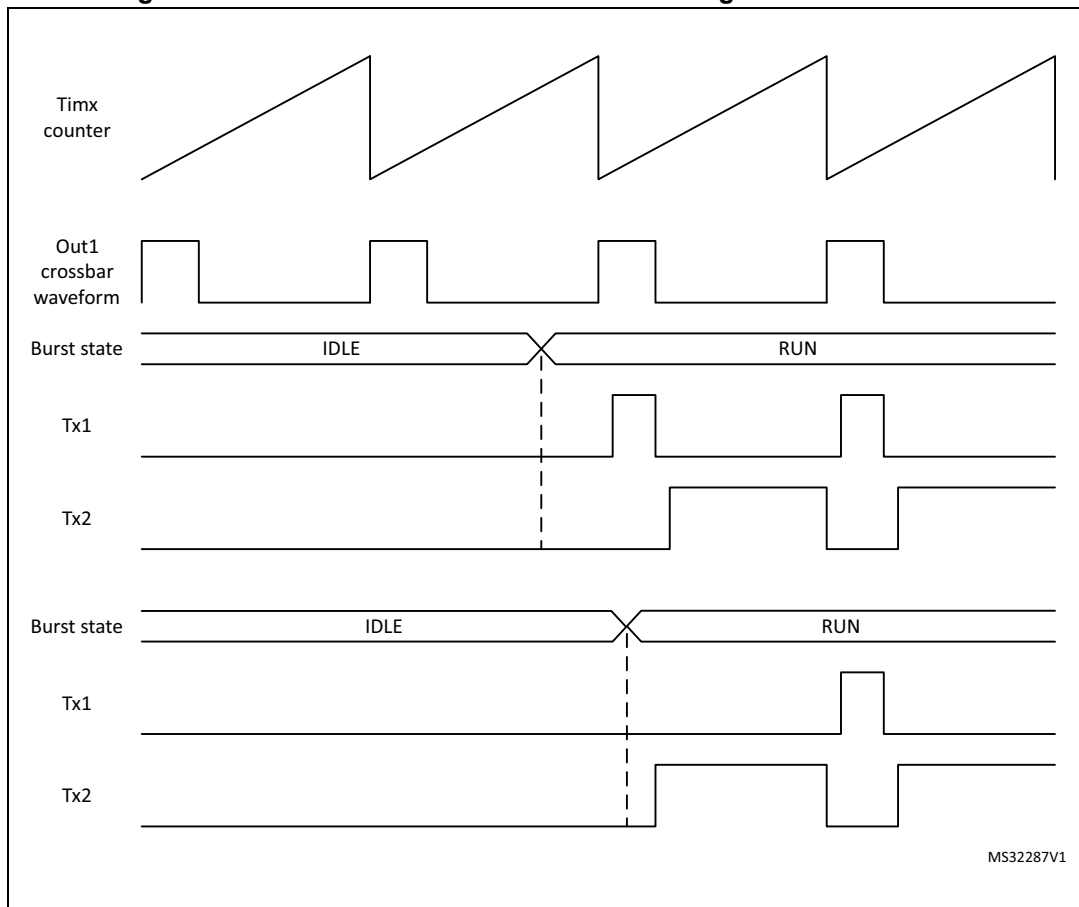
A burst period interrupt is generated in single-shot and continuous modes when BMPERIE enable bit is set in the HRTIM_IER register. This interrupt can be used to synchronize the burst mode exit with a burst period in continuous burst mode.

[Figure 321](#) shows how a normal operation is resumed when the deadtime is enabled.

Although the burst mode exit is immediate, this is only effective on the first set event on any of the complementary outputs.

Two different cases are presented:

1. The burst mode ends while the signal is inactive on the crossbar output waveform. The active state is resumed on Tx1 and Tx2 on the set event for the Tx1 output, and the Tx2 output does not take the complementary level on burst exit.
2. The burst mode ends while the crossbar output waveform is active: the activity is resumed on the set event of Tx2 output, and Tx1 does not take the active level immediately on burst exit.

Figure 321. Burst mode exit when the deadtime generator is enabled

The behavior described above is slightly different when the push-pull mode is enabled. The push-pull mode forces an output reset at the beginning of the period if the output is inactive, or symmetrically forces an active level if the output was high during the preceding period.

Consequently, an output with an active idle state can be reset at the time the burst mode is exited even if no transition is explicitly programmed. For symmetrical reasons, an output can be set at the time the burst mode is exited even if no transition is explicitly programmed, in case it was active when it entered in idle state.

Burst mode registers preloading and update

BMPREN bit (Burst mode Preload Enable) allows to have the burst mode compare and period registers preloaded (HRTIM_BMCMP and HRTIM_BMPER).

When BMPREN is set, the transfer from preload to active register happens:

- when the burst mode is enabled (BME = 1),
- at the end of the burst mode period.

A write into the HRTIM_BMPER period register disables the update temporarily, until the HRTIM_BMCMP compare register is written, to ensure the consistency of the two registers when they are modified.

If the compare register only needs to be changed, a single write is necessary. If the period only needs to be changed, it is also necessary to re-write the compare to have the new values taken into account.

When BMPREN bits is reset, the write access into BMCMPR and BMPER directly updates the active register. In this case, it is necessary to consider when the update is done during the overall burst period, for the 2 cases below:

a) Compare register update

If the new compare value is above the current burst mode counter value, the new compare is taken into account in the current period.

If the new compare value is below the current burst mode counter value, the new compare is taken into account in the next burst period in continuous mode, and ignored in single-shot mode (no compare match will occur and the idle state will last until the end of the idle period).

b) Period register update

If the new period value is above the current burst mode counter value, the change is taken into account in the current period.

Note: *If the new period value is below the current burst mode counter value, the new period will not be taken into account, the burst mode counter will overflow (at 0xFFFF) and the change will be effective in the next period. In single-shot mode, the counter will roll over at 0xFFFF and the burst mode will re-start for another period up to the new programmed value.*

Burst mode emulation using a compound register

The burst mode controller only controls one or a set of timers for a single converter. When the burst mode is necessary for multiple independent timers, it is possible to emulate a simple burst mode controller using the DMA and the HRTIM_CMP1CxR compound register, which holds aliases of both the repetition and the Compare 1 registers.

This is applicable to a converter which only requires a simple PWM (typically a buck converter), where the duty cycle only needs to be updated. In this case, the CMP1 register is used to reset the output (and define the duty cycle), while it is set on the period event.

In this case, a single 32-bit write access in CMP1CxR is sufficient to define the duty cycle (with the CMP1 value) and the number of periods during which this duty cycle is maintained (with the repetition value). To implement a burst mode, it is then only necessary to transfer by DMA (upon repetition event) two 32-bit data in continuous mode, organized as follows:

CMPC1xR = {REP_Run; CMP1 = Duty_Cycle}, {REP_Idle; CMP1 = 0}

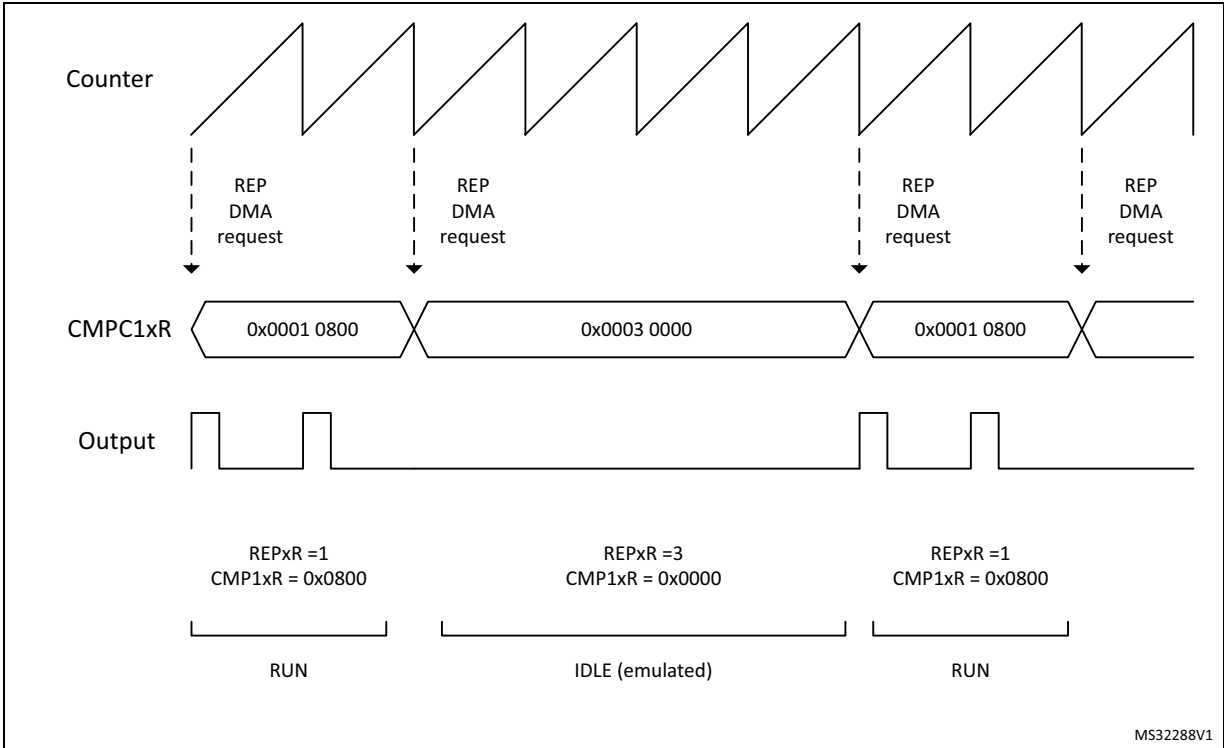
For instance, the values:

{0x0003 0000}: CMP1 = 0 for 3 periods

{0x0001 0800}: CMP1 = 0x0800 for 1 period

will provide a burst mode with 2 periods active every 6 PWM periods, as shown on [Figure 322](#).

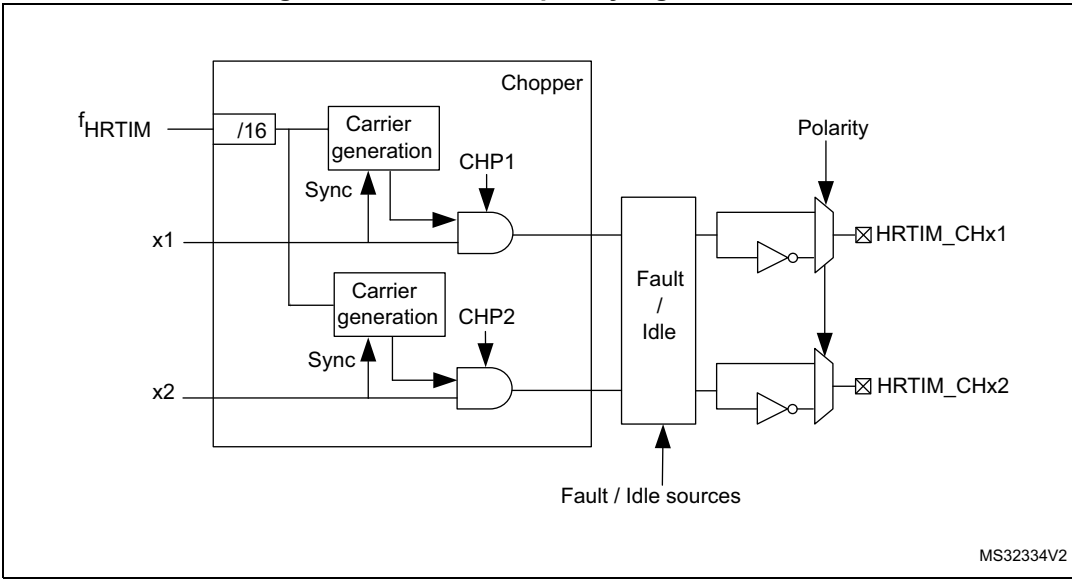
Figure 322. Burst mode emulation example



37.3.14 Chopper

A high-frequency carrier can be added on top of the timing unit output signals to drive isolation transformers. This is done in the output stage before the polarity insertion, as shown on [Figure 323](#), using CHP1 and CHP2 bits in the HRTIM_OUTxR register, to enable chopper on outputs 1 and 2, respectively.

Figure 323. Carrier frequency signal insertion



The chopper parameters can be adjusted using the HRIM_CHPxR register, with the possibility to define a specific pulsewidth at the beginning of the pulse, to be followed by a carrier frequency with programmable frequency and duty cycle, as in [Figure 324](#).

CARFRQ[3:0] bits define the frequency, ranging from 156 MHz to 25 MHz (for $f_{\text{HRTIM}} = 400 \text{ MHz}$) following the formula $F_{\text{CHPFRQ}} = f_{\text{HRTIM}} / (16 \times (\text{CARFRQ}[3:0] + 1))$.

The duty cycle can be adjusted by 1/8 step with CARDTY[2:0], from 0/8 up to 7/8 duty cycle. When CARDTY[2:0] = 000 (duty cycle = 0/8), the output waveform only contains the starting pulse following the rising edge of the reference waveform, without any added carrier.

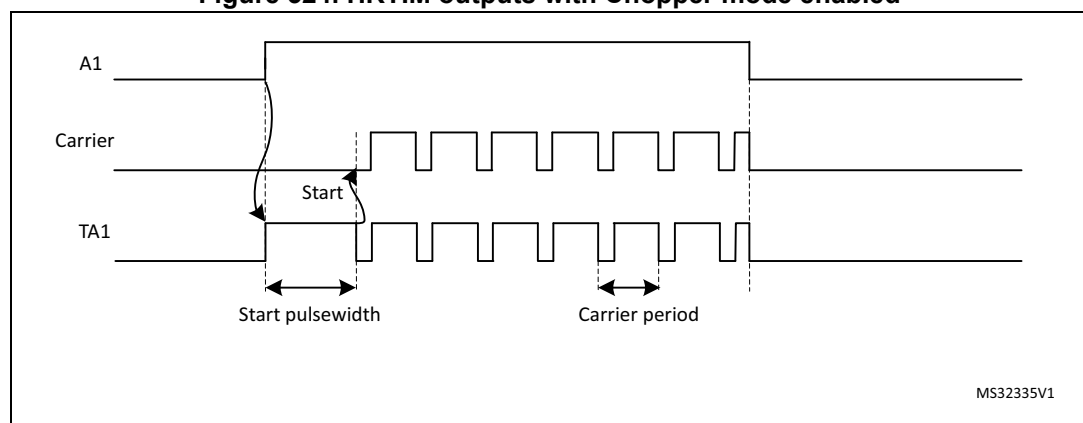
The pulsewidth of the initial pulse is defined using the STRPW[3:0] bitfield as follows: $t_{\text{1STPW}} = (\text{STRPW}[3:0] + 1) \times 16 \times t_{\text{HRTIM}}$ and ranges from 40 ns to 0.63 μs (for $f_{\text{HRTIM}} = 400 \text{ MHz}$).

The carrier frequency parameters are defined based on the f_{HRTIM} frequency, and are not dependent from the CKPSC[2:0] setting.

In chopper mode, the carrier frequency and the initial pulsewidth are combined with the reference waveform using an AND function. A synchronization is performed at the end of the initial pulse to have a repetitive signal shape.

The chopping signal is stopped at the end of the output waveform active state, without waiting for the current carrier period to be completed. It can thus contain shorter pulses than programmed.

Figure 324. HRTIM outputs with Chopper mode enabled



Note: *CHP1 and CHP2 bits must be set prior to the output enable done with TxyOEN bits in the HRTIM_OENR register.*

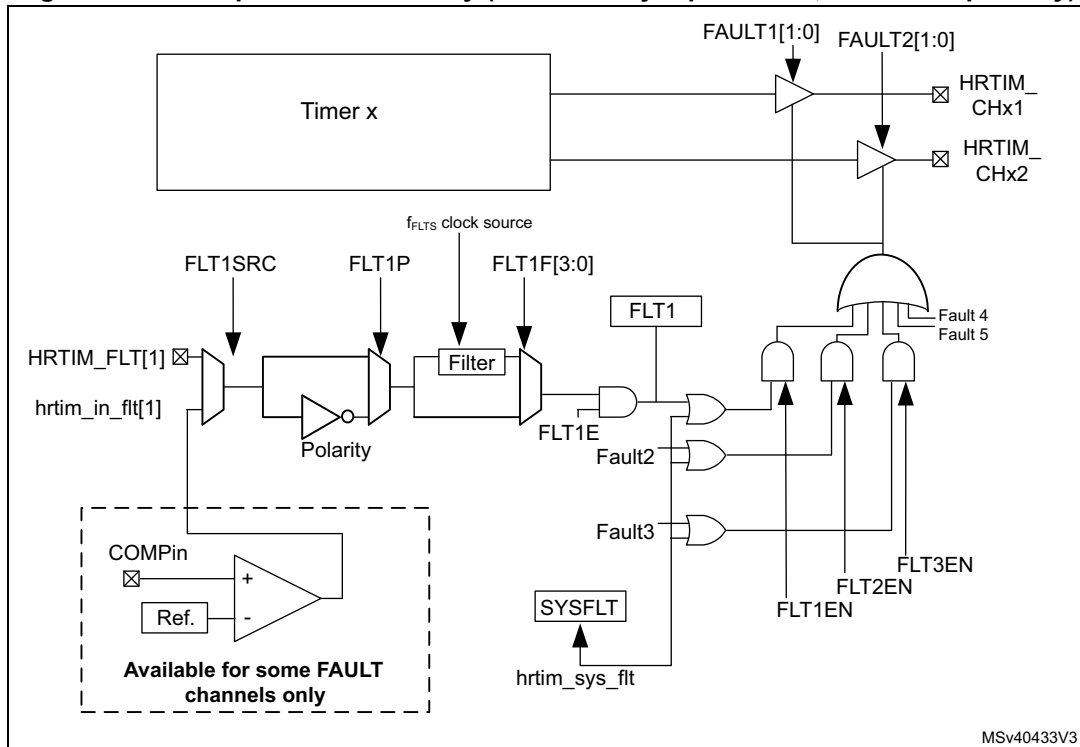
CARFRQ[2:0], CARDTY[2:0] and STRPW[3:0] bitfields cannot be modified while the chopper mode is active (at least one of the two CHPx bits is set).

37.3.15 Fault protection

The HRTIMER has a versatile fault protection circuitry to disable the outputs in case of an abnormal operation. Once a fault has been triggered, the outputs take a predefined safe state. This state is maintained until the output is re-enabled by software. In case of a permanent fault request, the output will remain in its fault state, even if the software attempts to re-enable them, until the fault source disappears.

The HRTIM has 5 FAULT input channels; all of them are available and can be combined for each of the 5 timing units, as shown on [Figure 325](#).

Figure 325. Fault protection circuitry (FAULT1 fully represented, FAULT2..5 partially)



Each fault channel is fully configurable using HRTIM_FLTINR1 and HRTIM_FLTINR2 registers before being routed to the timing units. FLTxsRC bit selects the source of the Fault signal, that can be either a digital input or an internal event (built-in comparator output).

Table 318 summarizes the available sources for each of the 10 faults channels:

Table 318. Fault inputs

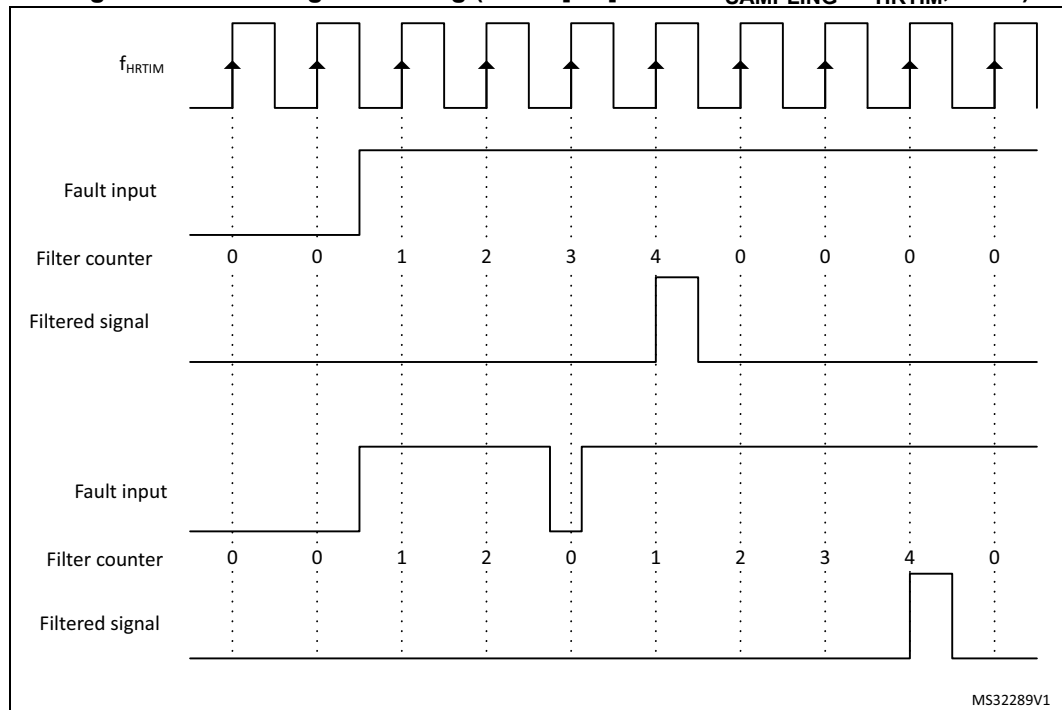
Fault channel	External Input (FLTxsRC = 0)	On-chip source (FLTxsRC = 1)
FAULT 1	PA15	COMP1
FAULT 2	PC11	COMP2
FAULT 3	PD4	NC
FAULT 4	PB3	NC
FAULT 5	PG10	NC

The polarity of the signal can be selected to define the active level, using the FLTxF polarity bit in HRTIM_FLTINRx registers. If FLTxF = 0, the signal is active at low level; if FLTxF = 1, it is active when high.

The fault information can be filtered after the polarity setting. If FLTxF[3:0] bitfield is set to 0000, the signal is not filtered and will act asynchronously, independently from the f_{HRTIM} clock. For all other FLTxF[3:0] bitfield values, the signal is digitally filtered. The digital filter is made of a counter in which a number N of valid samples is needed to validate a transition on the output. If the input value changes before the counter has reached the value N, the counter is reset and the transition is discarded (considered as a spurious event). If the counter reaches N, the transition is considered as valid and transmitted as a correct external

event. Consequently, the digital filter adds a latency to the external events being filtered, depending on the sampling clock and on the filter length (number of valid samples expected). [Figure 326](#) shows how a spurious fault signal is filtered.

Figure 326. Fault signal filtering (FLTxF[3:0]= 0010: $f_{\text{SAMPLING}} = f_{\text{HRTIM}}$, N = 4)



The filtering period ranges from 2 cycles of the f_{HRTIM} clock up to 8 cycles of the f_{FLTS} clock divided by 32. f_{FLTS} is defined using FLTSD[1:0] bits in the HRTIM_FLTINR2 register.

[Table 319](#) summarizes the sampling rate and the filter length. A jitter of 1 sampling clock period must be subtracted from the filter length to take into account the uncertainty due to the sampling and have the effective filtering.

Table 319. Sampling rate and filter length vs FLTxF[3:0] and clock setting

FLTxF[3:0]	f_{FLTS} vs FLTSD[1:0]				Filter length for $f_{\text{HRTIM}} = 400 \text{ MHz}$	
	00	01	10	11	Min	Max
0001, 0010, 0011	f_{HRTIM}	f_{HRTIM}	f_{HRTIM}	f_{HRTIM}	f_{HRTIM} , N = 2 5 ns	f_{HRTIM} , N = 8 20 ns
0100, 0101	$f_{\text{HRTIM}} / 2$	$f_{\text{HRTIM}} / 4$	$f_{\text{HRTIM}} / 8$	$f_{\text{HRTIM}} / 16$	$f_{\text{HRTIM}} / 2$, N = 6 30 ns	$f_{\text{HRTIM}} / 16$, N = 8 320 ns
0110, 0111	$f_{\text{HRTIM}} / 4$	$f_{\text{HRTIM}} / 8$	$f_{\text{HRTIM}} / 16$	$f_{\text{HRTIM}} / 32$	$f_{\text{HRTIM}} / 4$, N = 6 60 ns	$f_{\text{HRTIM}} / 32$, N = 8 640 ns
1000, 1001	$f_{\text{HRTIM}} / 8$	$f_{\text{HRTIM}} / 16$	$f_{\text{HRTIM}} / 32$	$f_{\text{HRTIM}} / 64$	$f_{\text{HRTIM}} / 8$, N = 6 120 ns	$f_{\text{HRTIM}} / 64$, N = 8 1.28 μs

Table 319. Sampling rate and filter length vs FLTxF[3:0] and clock setting (continued)

FLTxF[3:0]	f_{FLTS} vs FLTSD[1:0]				Filter length for $f_{\text{HRTIM}} = 400 \text{ MHz}$	
	00	01	10	11	Min	Max
1010, 1011, 1100	$f_{\text{HRTIM}} / 16$	$f_{\text{HRTIM}} / 32$	$f_{\text{HRTIM}} / 64$	$f_{\text{HRTIM}} / 128$	$f_{\text{HRTIM}} / 16$, N = 5 200 ns	$f_{\text{HRTIM}} / 128$, N = 8 2.56 μs
1101, 1110, 1111	$f_{\text{HRTIM}} / 32$	$f_{\text{HRTIM}} / 64$	$f_{\text{HRTIM}} / 128$	$f_{\text{HRTIM}} / 256$	$f_{\text{HRTIM}} / 32$, N = 5 400 ns	$f_{\text{HRTIM}} / 256$, N = 8 5.12 μs

System fault input (hrtim_sysflt)

This fault is provided by the MCU Class B circuitry (see the System configuration controller (SYSCFG) section for details) and corresponds to a system fault coming from:

- the Clock Security System
- the SRAM parity checker
- the Cortex[®]-M7-lockup signal
- the PVD detector

This input overrides the FAULT inputs and disables all outputs having FAULTy[1:0] = 01, 10, 11.

For each FAULT channel, a write-once FLTxDLCK bit in the HRTIM_FLTxDL register allows to lock FLTxE, FLTxD, FLTxDL, FLTxDL[3:0] bits (it renders them read-only), for functional safety purpose. If enabled, the fault conditioning set-up is frozen until the next HRTIM or system reset.

Once the fault signal is conditioned as explained above, it is routed to the timing units. For any of them, the 5 fault channels are enabled using bits FLT1EN to FLT5EN in the HRTIM_FLTxDL register, and they can be selected simultaneously (the sysfault is automatically enabled as long as the output is protected by the fault mechanism). This allows to have, for instance:

- One fault channel simultaneously disabling several timing units
- Multiple fault channels being ORed to disable a single timing unit

A write-once FLTDLCK bit in the HRTIM_FLTxDL register allows to lock FLTxDL bits (it renders them read-only) until the next reset, for functional safety purpose. If enabled, the timing unit fault-related set-up is frozen until the next HRTIM or system reset.

For each of the timers, the output state during a fault is defined with FAULT1[1:0] and FAULT2[1:0] bits in the HRTIM_OUTxDL register (see [Section 37.3.12](#)).

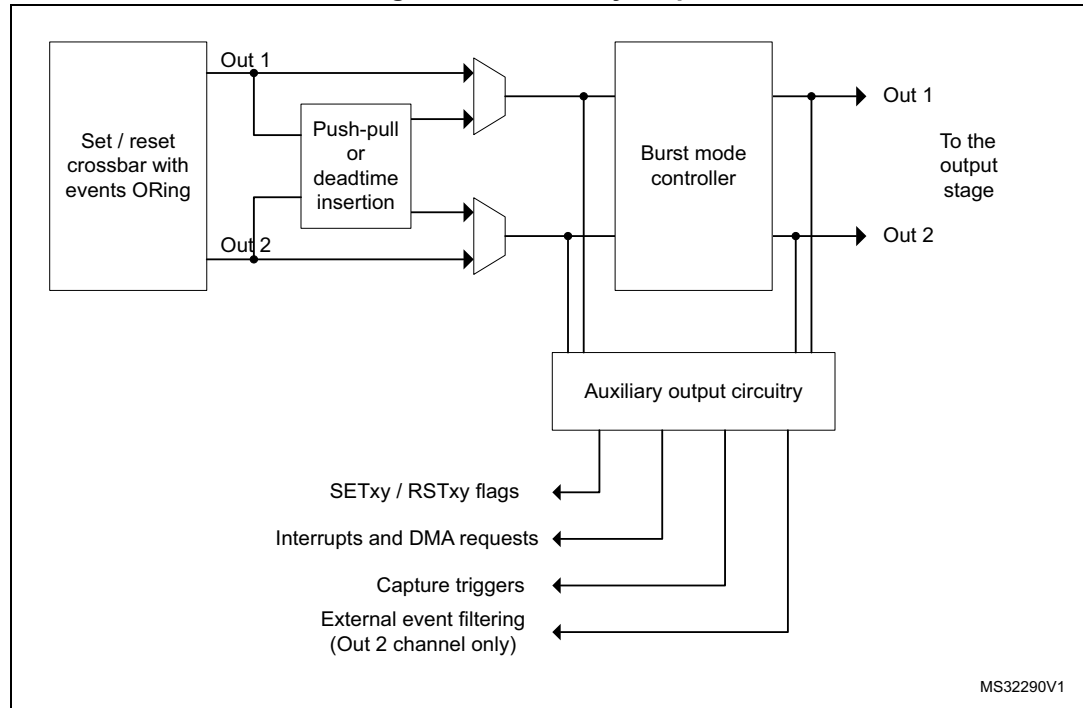
37.3.16 Auxiliary outputs

Timer A to E have auxiliary outputs in parallel with the regular outputs going to the output stage. They provide the following internal status, events and signals:

- SETxy and RSTxy status flags, together with the corresponding interrupts and DMA requests
- Capture triggers upon output set/reset
- External event filters following a Tx2 output copy (see details in [Section 37.3.8](#))

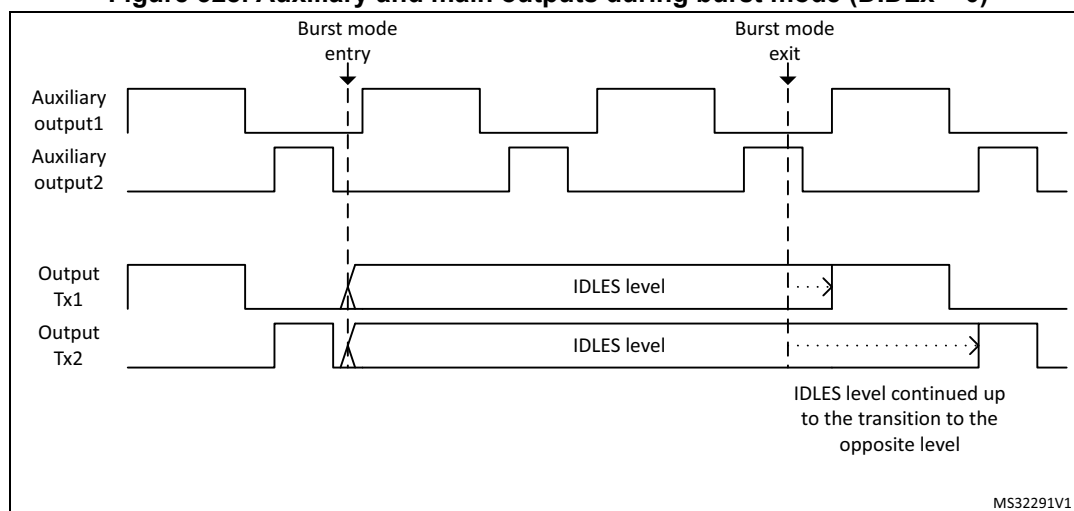
The auxiliary outputs are taken either before or after the burst mode controller, depending on the HRTIM operating mode. An overview is given on [Figure 327](#).

Figure 327. Auxiliary outputs

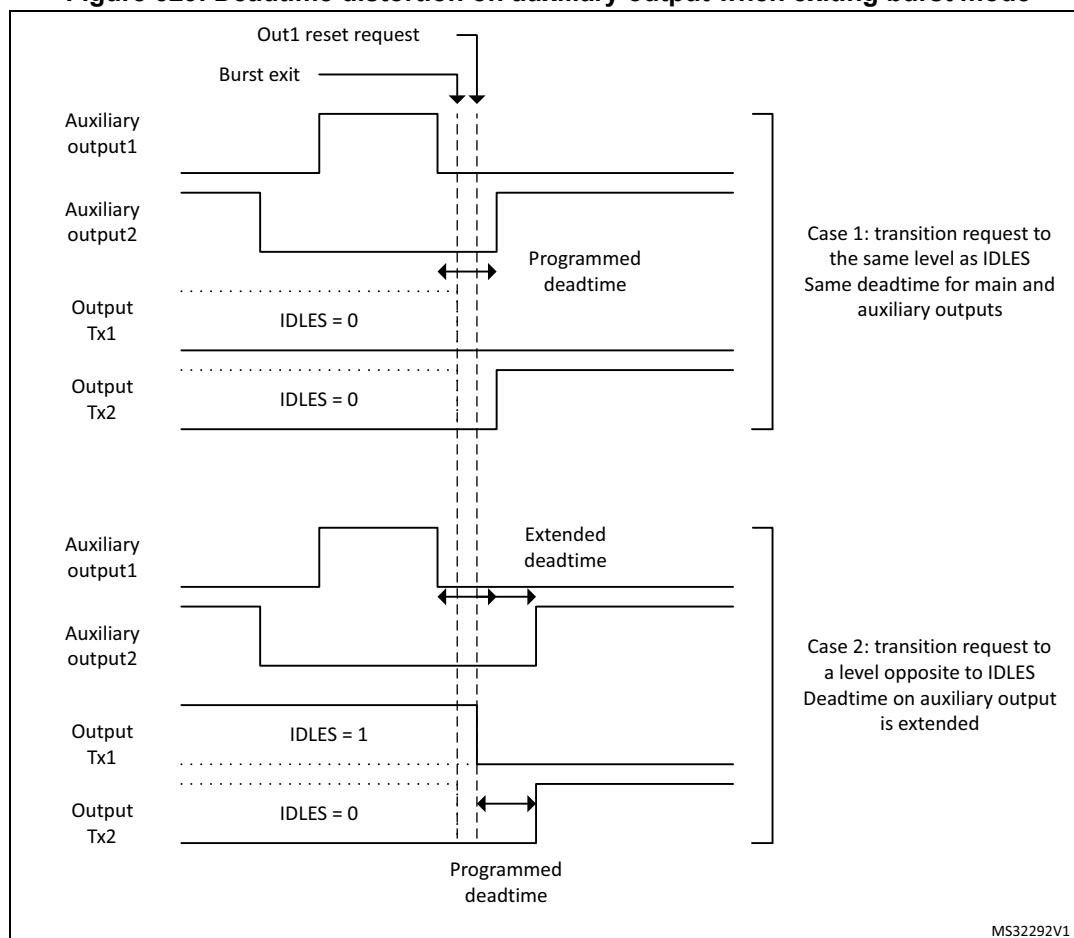


By default, the auxiliary outputs are copies of outputs Tx1 and Tx2. The exceptions are:

- The delayed idle and the balanced idle protections, when the deadtime is disabled (DTEN = 0). When the protection is triggered, the auxiliary outputs are maintained and follow the signal coming out of the crossbar. On the contrary, if the deadtime is enabled (DTEN = 1), both main and auxiliary outputs are forced to an inactive level.
- The burst mode (TCEN=1, IDLEMx=1); there are 2 cases:
 - a) If DTEN=0 or DIDLx=0, the auxiliary outputs are not affected by the burst mode entry and continue to follow the reference signal coming out of the crossbar (see [Figure 328](#)).
 - b) If the deadtime is enabled (DTEN=1) together with the delayed burst mode entry (DIDLx=1), the auxiliary outputs have the same behavior as the main outputs. They are forced to the IDLES level after a deadtime duration, then they keep this level during all the burst period. When the burst mode is terminated, the IDLES level is maintained until a transition occurs to the opposite level, similarly to the main output.

Figure 328. Auxiliary and main outputs during burst mode (DIDLx = 0)

The signal on the auxiliary output can be slightly distorted when exiting from the burst mode or when re-enabling the outputs after a delayed protection, if this happens during a deadtime. In this case, the deadtime applied to the auxiliary outputs is extended so that the deadtime on the main outputs is respected. [Figure 329](#) gives some examples.

Figure 329. Deadtime distortion on auxiliary output when exiting burst mode

37.3.17 Synchronizing the HRTIM with other timers or HRTIM instances

The HRTIM provides options for synchronizing multiple HRTIM instances, as a master unit (generating a synchronization signal) or as a slave (waiting for a trigger to be synchronized). This feature can also be used to synchronize the HRTIM with other timers, either external or on-chip. The synchronization circuitry is controlled inside the master timer.

Synchronization output

This section explains how the HRTIM must be configured to synchronize external resources and act as a master unit.

Four events can be selected as the source to be sent to the synchronization output. This is done using SYNC_SRC[1:0] bits in the HRTIM_MCR register, as follows:

- 00: Master timer Start
This event is generated when MCEN bit is set or when the timer is re-started after having reached the period value in single-shot mode. It is also generated on a reset which occurs during the counting (when CONT or RETRIG bits are set).
- 01: Master timer Compare 1 event
- 10: Timer A start
This event is generated when TACEN bit is set or when the counter is reset and re-starts counting in response to this reset. The following counter reset events are not propagated to the synchronization output: counter roll-over in continuous mode, and discarded reset request in single-shot non-retriggerable mode. The reset is only taken into account when it occurs during the counting (CONT or RETRIG bits are set).
- 11: Timer A Compare 1 event

SYNC_OUT[1:0] bits in the HRTIM_MCR register specify how the synchronization event is generated.

The synchronization pulses are generated on the HRTIM_SCOUT output pin, with SYNC_OUT[1:0] = 1x. SYNC_OUT[0] bit specifies the polarity of the synchronization signal. If SYNC_OUT[0] = 0, the HRTIM_SCOUT pin has a low idle level and issues a positive pulse of $16 f_{\text{HRTIM}}$ clock cycles length for the synchronization). If SYNC_OUT[0] = 1, the idle level is high and a negative pulse is generated.

Note: *The synchronization pulse is followed by an idle level of $16 f_{\text{HRTIM}}$ clock cycles during which any new synchronization request is discarded. Consequently, the maximum synchronization frequency is $f_{\text{HRTIM}}/32$.*

The idle level on the HRTIM_SCOUT pin is applied as soon as the SYNC_OUT[1:0] bits are enabled (i.e. the bitfield value is different from 00).

The synchronization output initialization procedure must be done prior to the configuration of the MCU outputs and counter enable, in the following order:

1. SYNC_OUT[1:0] and SYNC_SRC[1:0] bitfield configuration in HRTIM_MCR
2. HRTIM_SCOUT pin configuration (see the General-purpose I/Os section)
3. Master or Timer A counter enable (MCEN or TACEN bit set)

When the synchronization input mode is enabled and starts the counter (using SYNC_STRTM/SYNC_STRTx bits) simultaneously with the synchronization output mode (SYNC_SRC[1:0] = 00 or 10), the output pulse is generated only when the counter is starting or is reset while running. Any reset request clearing the counter without causing it to start will not affect the synchronization output.

Synchronization input

The HRTIM can be synchronized by external sources, as per the programming of the SYNCIN[1:0] bits in the HRTIM_MCR register:

- 00: synchronization input is disabled
- 01: reserved configuration
- 10: the on-chip TIM1 general purpose timer (TIM1 TRGO output)
- 11: a positive pulse on the HRTIM_SCIN input pin

This bitfield cannot be changed once the destination timer (master timer or timing unit) is enabled (MCEN and/or TxCEN bit set).

The HRTIM_SCIN input is rising-edge sensitive. The timer behavior is defined with the following bits present in HRTIM_MCR and HRTIM_TIMxCR registers (see [Table 320](#) for details):

- Synchronous start: the incoming signal starts the timer's counter (SYNCSTRM and/or SYNCSTRTx bits set). TxCEN (MCEN) bits must be set to have the timer enabled and the counter ready to start. In continuous mode, the counter will not start until the synchronization signal is received.
- Synchronous reset: the incoming signal resets the counter (SYNCRSTM and/or SYNCRSTx bits set). This event decrements the repetition counter as any other reset event.

The synchronization events are taken into account only once the related counters are enabled (MCEN or TxCEN bit set). A synchronization request triggers a SYNC interrupt.

Note: *A synchronized start event resets the counter if the current counter value is above the active period value.*

The effect of the synchronization event depends on the timer operating mode, as summarized in [Table 320](#).

Table 320. Effect of sync event vs timer operating modes

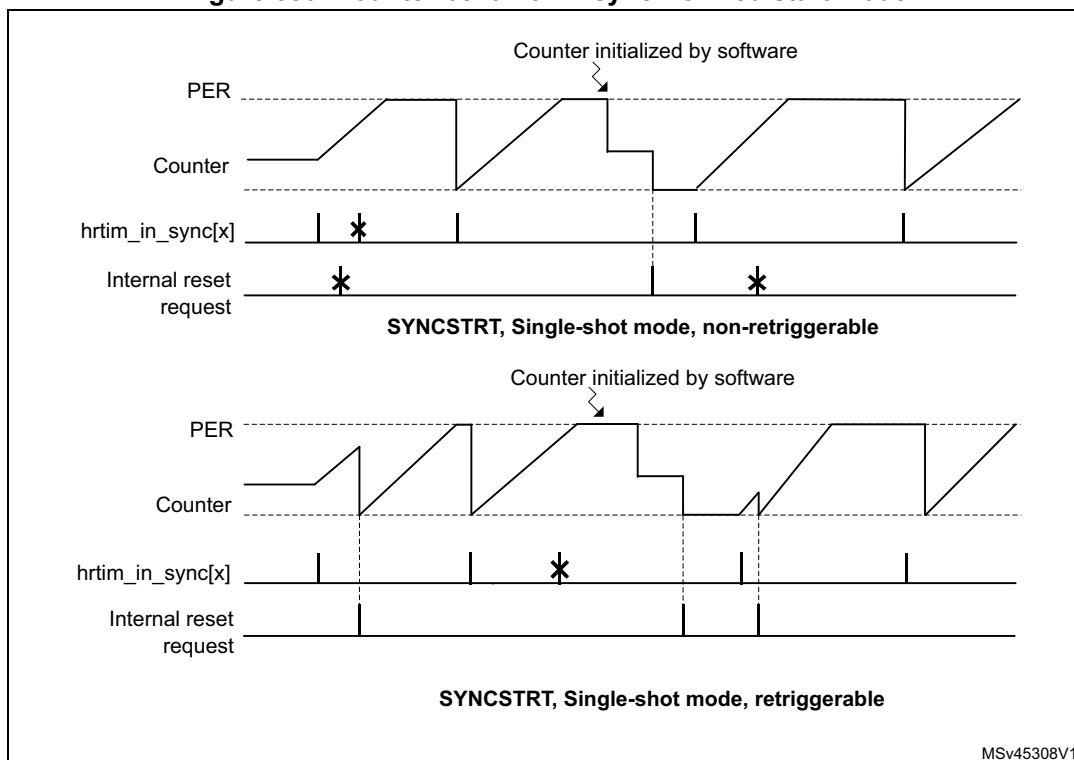
Operating mode	SYNC RSTx	SYNC STRTx	Behavior following a SYNC reset or start event
Single-shot non-retriggerable	0	1	Start events are taken into account when the counter is stopped and: – once the MCEN or TxCEN bits are set – once the period has been reached. A start occurring when the counter is stopped at the period value resets the counter. A reset request clears the counter but does not start it (the counter can solely be re-started with the synchronization). Any reset occurring during the counting is ignored (as during regular non-retriggerable mode).
	1	X	Reset events are starting the timer counting. They are taken into account only if the counter is stopped and: – once the MCEN or TxCEN bits are set – once the period has been reached. When multiple reset requests are selected (from HRTIM_SCIN and from internal events), only the first arriving request is taken into account.

Table 320. Effect of sync event vs timer operating modes (continued)

Operating mode	SYNC RSTx	SYNC STRTx	Behavior following a SYNC reset or start event
Single-shot retriggerable	0	1	The counter start is effective only if the counter is not started or period is elapsed. Any synchronization event occurring after counter start has no effect. A start occurring when the counter is stopped at the period value resets the counter. A reset request clears the counter but does not start it (the counter can solely be started by the synchronization). A reset occurring during counting is taken into account (as during regular retriggerable mode).
	1	X	The reset from HRTIM_SCIN is taken into account as any HRTIM counter reset from internal events and is starting or re-starting the timer counting. When multiple reset requests are selected, the first arriving request is taken into account.
Continuous mode	0	1	The timer is enabled (MCEN or TxCEN bit set) and is waiting for the synchronization event to start the counter. Any synchronization event occurring after the counter start has no effect (the counter can solely be started by the synchronization). A reset request clears the counter but does not start it.
	1	X	The reset from HRTIM_SCIN is taken into account as any HRTIM counter reset from internal events and is starting or re-starting the timer counting. When multiple reset requests are selected, the first arriving request is taken into account.

Figure 330 presents how the synchronized start is done in single-shot mode.

Figure 330. Counter behavior in synchronized start mode



37.3.18 ADC triggers

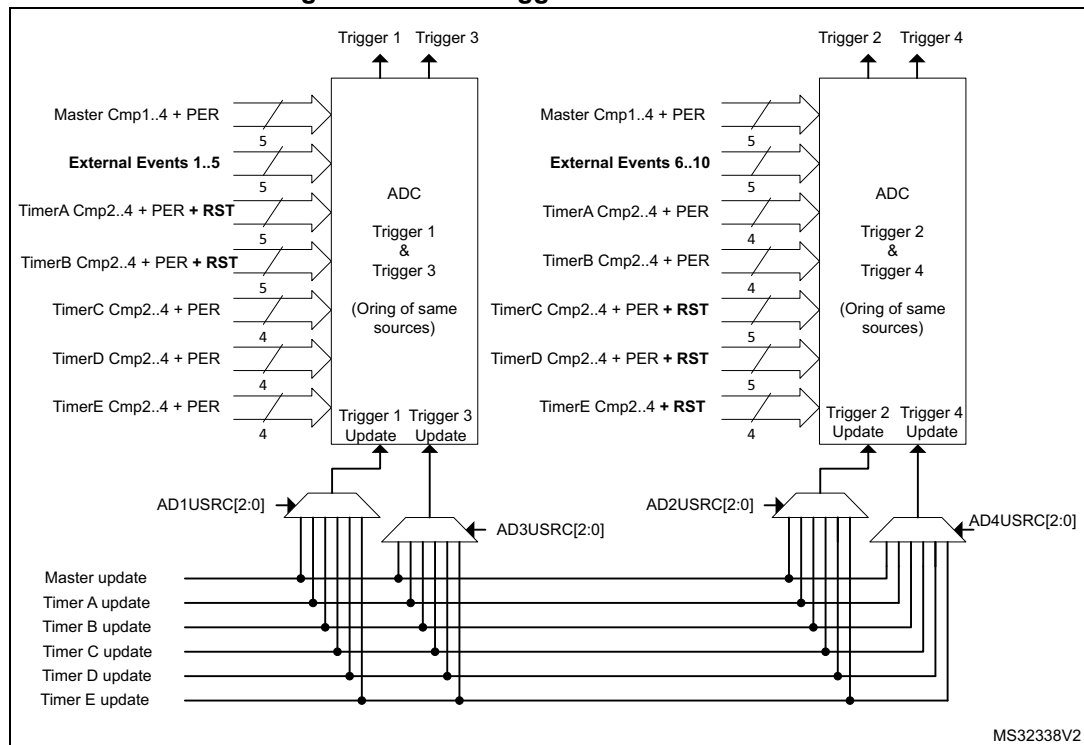
The ADCs can be triggered by the master and the 5 timing units.

4 independent triggers are available to start both the regular and the injected sequencers of the 2 ADCs. Up to 32 events can be combined (ORed) for each trigger output, in registers HRTIM_ADC1R to HRTIM_ADC4R, as shown on [Figure 331](#). Triggers 1/3 and 2/4 are using the same source set.

The external events can be used as a trigger. They are taken right after the conditioning defined in HRTIM_EECRx registers, and are not depending on EEFR1 and EEFR2 register settings.

Multiple triggering is possible within a single switching period by selecting several sources simultaneously. A typical use case is for a non-overlapping multiphase converter, where all phases can be sampled in a row using a single ADC trigger output.

Figure 331. ADC trigger selection overview



HRTIM_ADC1R to HRTIM_ADC4R registers are preloaded and can be updated synchronously with the timer they are related to. The update source is defined with ADxUSRC[2:0] bits in the HRTIM_CR1 register.

For instance, if ADC trigger 1 outputs Timer A CMP2 events (HRTIM_ADC1R = 0x0000 0400), HRTIM_ADC1R will be typically updated simultaneously with Timer A (AD1USRC[2:0] = 001).

When the preload is disabled (PREEN bit reset) in the source timer, the HRTIM_ADCxR registers are not preloaded either: a write access will result in an immediate update of the trigger source.

37.3.19 DAC triggers

The HRTIMER allows to have the embedded DACs updated synchronously with the timer updates.

The update events from the master timer and the timer units can generate DAC update triggers on any of the 3 hrtim_dac_trgx outputs.

Note: Each timer has its own DAC-related control register.

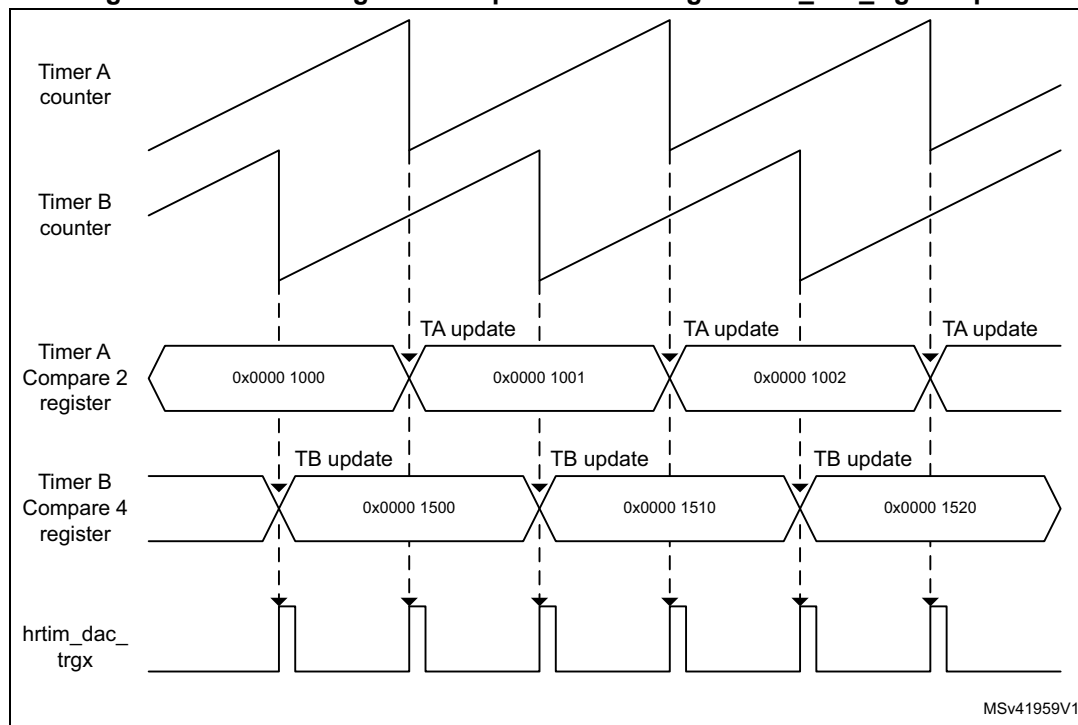
DACSYNC[1:0] bits of the HRTIM_MCR and HRTIM_TIMxCR registers are programmed as follows:

- 00: No update generated
- 01: Update generated on hrtim_dac_trg1
- 10: Update generated on hrtim_dac_trg2
- 11: Update generated on hrtim_dac_trg3

An output pulse of 1 f_{HRTIM} clock periods is generated on the hrtim_dac_trgx output.

When DACSYNC[1:0] bits are enabled in multiple timers, the `hrtim_dac_trgx` output will consist of an OR of all timers' update events. For instance, if `DACSYNC = 1` in timer A and in timer B, the update event in timer A will be ORed with the update event in timer B to generate a DAC update trigger on the corresponding `hrtim_dac_trgx` output, as shown on [Figure 332](#).

Figure 332. Combining several updates on a single `hrtim_dac_trgx` output



`hrtim_dac_trgx` pins are connected to the DACs as follows:

- `hrtim_dac_trg1`: DAC1_CH1 trigger input 9 (TSEL1[2:0] = 1001 in DAC_CR of DAC1 peripheral)
- `hrtim_dac_trg2`: DAC1_CH2 trigger input 10 (TSEL1[2:0] = 1010 in DAC_CR of DAC1 peripheral)
- `hrtim_dac_trg3`: not connected

37.3.20 HRTIM Interrupts

7 interrupts can be generated by the master timer:

- Master timer registers update
- Synchronization event received
- Master timer repetition event
- Master Compare 1 to 4 event

14 interrupts can be generated by each timing unit:

- Delayed protection triggered
- Counter reset or roll-over event
- Output 1 and output 2 reset (transition active to inactive)
- Output 1 and output 2 set (transition inactive to active)
- Capture 1 and 2 events
- Timing unit registers update
- Repetition event
- Compare 1 to 4 event

8 global interrupts are generated for the whole HRTIM:

- System fault and Fault 1 to 5 (regardless of the timing unit attribution)
- Burst mode period completed

The interrupt requests are grouped in 7 vectors as follows:

- `hrtim_mst_it`: Master timer interrupts (Master Update, Sync Input, Repetition, MCMP1..4) and global interrupt except faults (Burst mode period)
- `hrtim_tima_it`: TIMA interrupts
- `hrtim_timb_it`: TIMB interrupts
- `hrtim_timc_it`: TIMC interrupts
- `hrtim_timd_it`: TIMD interrupts
- `hrtim_time_it`: TIME interrupts
- `hrtim_fault_it`: Dedicated vector all fault interrupts to allow high-priority interrupt handling

[Table 321](#) is a summary of the interrupt requests, their mapping and associated control, and status bits.

Table 321. HRTIM interrupt summary

Interrupt vector	Interrupt event	Event flag	Enable control bit	Flag clearing bit
hrtim_mst_it	Burst mode period completed	BMPER	BMPERIE	BMPERC
	Master timer registers update	MUPD	MUPDIE	MUPDC
	Synchronization event received	SYNC	SYNCIE	SYNCC
	Master timer repetition event	MREP	MREPIE	MREPC
	Master Compare 1 to 4 event	MCMP1	MCMP1IE	MCP1C
		MCMP2	MCMP2IE	MCP2C
		MCMP3	MCMP3IE	MCP3C
		MCMP4	MCMP4IE	MCP4C
hrtim_tima_it hrtim_timb_it hrtim_timc_it hrtim_timd_it hrtim_time_it	Delayed protection triggered	DLYPRT	DLYPRTIE	DLYPRTC
	Counter reset or roll-over event	RST	RSTIE	RSTC
	Output 1 and output 2 reset (transition active to inactive)	RSTx1	RSTx1IE	RSTx1C
		RSTx2	RSTx2IE	RSTx2C
	Output 1 and output 2 set (transition inactive to active)	SETx1	SETx1IE	SETx1C
		SETx2	SETx2IE	SETx2C
	Capture 1 and 2 events	CPT1	CPT1IE	CPT1C
		CPT2	CPT2IE	CPT2C
	Timing unit registers update	UPD	UPDIE	UPDC
	Repetition event	REP	REPIE	REPC
	Compare 1 to 4 event	CMP1	CMP1IE	CMP1C
		CMP2	CMP2IE	CMP2C
		CMP3	CMP3IE	CMP3C
		CMP4	CMP4IE	CMP4C
hrtim_fault_it	System fault	SYSFLT	SYSFLTIE	SYSFLTC
	Fault 1 to 5	FLT1	FLT1IE	FLT1C
		FLT2	FLT2IE	FLT2C
		FLT3	FLT3IE	FLT3C
		FLT4	FLT4IE	FLT4C
		FLT5	FLT5IE	FLT5C

37.3.21 DMA

Most of the events able to generate an interrupt can also generate a DMA request, even both simultaneously. Each timer (master, TIMA...E) has its own DMA enable register.

The individual DMA requests are ORed into 6 channels as follows:

- 1 channel for the master timer
- 1 channel per timing unit

Note: Before disabling a DMA channel (DMA enable bit reset in TIMxDIER), it is necessary to disable first the DMA controller.

Table 322 is a summary of the events with their associated DMA enable bits.

Table 322. HRTIM DMA request summary

DMA Channel	Event	DMA capable	DMA enable bit
hrtim_dma1 (Master timer)	Burst mode period completed	No	N/A
	Master timer registers update	Yes	MUPDDE
	Synchronization event received	Yes	SYNCDE
	Master timer repetition event	Yes	MREPDE
	Master Compare 1 to 4 event	Yes	MCMP1DE
		Yes	MCMP2DE
		Yes	MCMP3DE
		Yes	MCMP4DE
hrtim_dma2 (Timer A) hrtim_dma3 (Timer B) hrtim_dma4 (Timer C) hrtim_dma5 (Timer D) hrtim_dma6 (Timer E)	Delayed protection triggered	Yes	DLYPRTDE
	Counter reset or roll-over event	Yes	RSTDE
	Output 1 and output 2 reset (transition active to inactive)	Yes	RSTx1DE
		Yes	RSTx2DE
	Output 1 and output 2 set (transition inactive to active)	Yes	SETx1DE
		Yes	SETx2DE
	Capture 1 and 2 events	Yes	CPT1DE
		Yes	CPT2DE
	Timing unit registers update	Yes	UPDDE
	Repetition event	Yes	REPDE
	Compare 1 to 4 event	Yes	CMP1DE
		Yes	CMP2DE
		Yes	CMP3DE
		Yes	CMP4DE
N/A	System fault	No	N/A
	Fault 1 to 5	No	N/A
	Burst mode period completed	No	N/A

Burst DMA transfers

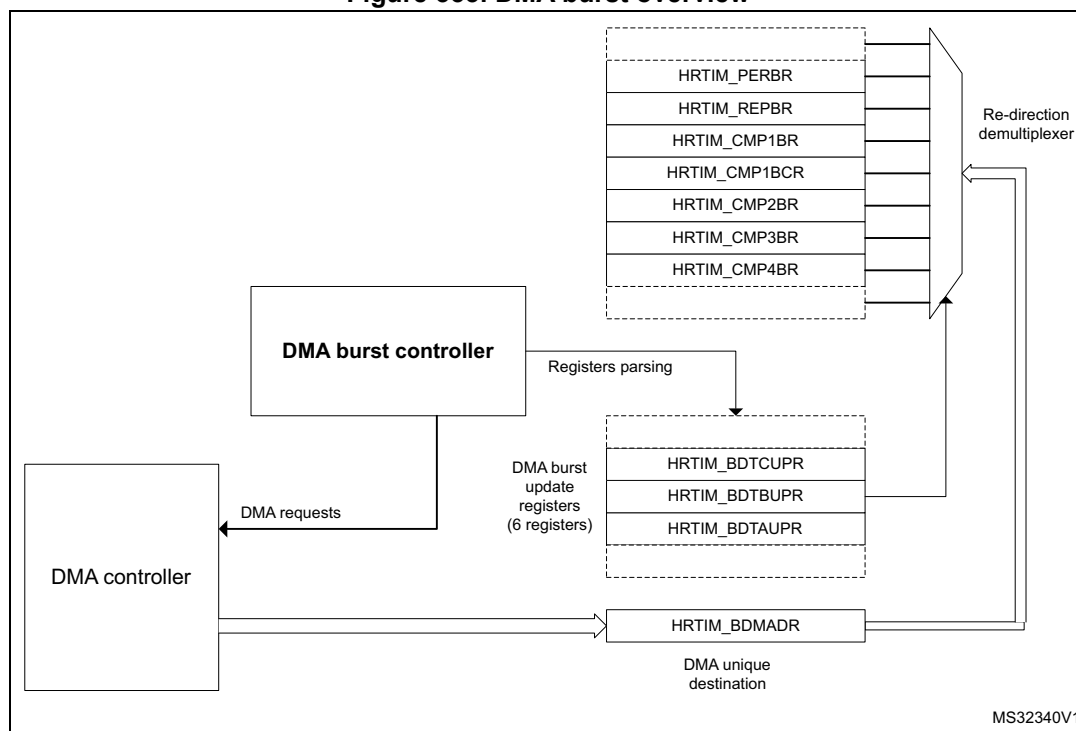
In addition to the standard DMA requests, the HRTIM features a DMA burst controller to have multiple registers updated with a single DMA request. This allows to:

- update multiple data registers with one DMA channel only,
- reprogram dynamically one or several timing units, for converters using multiple timer outputs.

The burst DMA feature is only available for one DMA channel, but any of the 6 channels can be selected for burst DMA transfers.

The principle is to program which registers are to be written by DMA. The master timer and TIMA..E have the burst DMA update register, where most of their control and data registers are associated with a selection bit: HRTIM_BDMUPR, HRTIM_BDTAUPR to HRTIM_BDTEUPR (this is applicable only for registers with write accesses). A redirection mechanism allows to forward the DMA write accesses to the HRTIM registers automatically, as shown on [Figure 333](#).

Figure 333. DMA burst overview



When the DMA trigger occurs, the HRTIM generates multiple 32-bit DMA requests and parses the update register. If the control bit is set, the write access is redirected to the associated register. If the bit is reset, the register update is skipped and the register parsing is resumed until a new bit set is detected, to trigger a new request. Once the 6 update registers (HRTIM_BDMUPR, 5x HRTIM_BDTxUPR) are parsed, the burst is completed and the system is ready for another DMA trigger (see the flowchart on [Figure 334](#)).

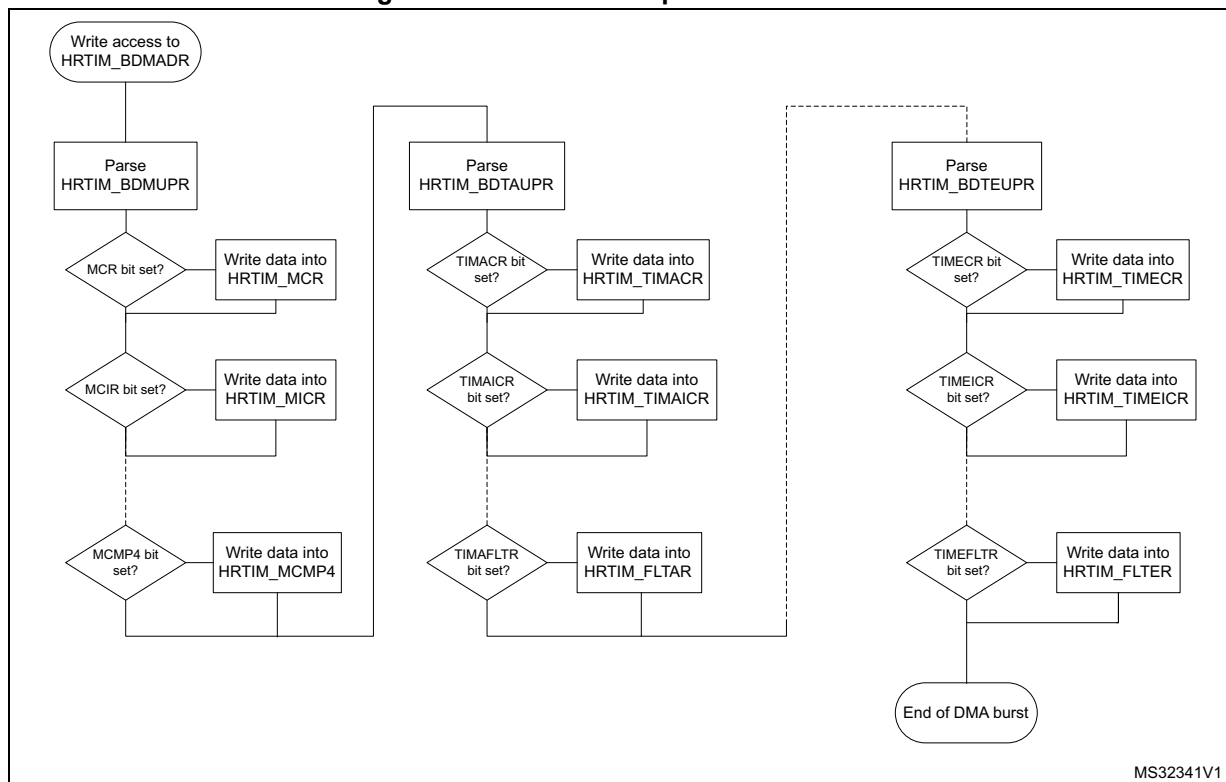
Note: Any trigger occurring while the burst is on-going is discarded, except if it occurs during the very last data transfer.

The burst DMA mode is permanently enabled (there is no enable bit). A burst DMA operation is started by the first write access into the HRTIM_BDMADR register.

It is only necessary to have the DMA controller pointing to the HRTIM_BDMADR register as the destination, in the memory, to the peripheral configuration with the peripheral increment mode disabled (the HRTIM handles internally the data re-routing to the final destination register).

To re-initialize the burst DMA mode if it was interrupted during a transaction, it is necessary to write at least to one of the 6 update registers.

Figure 334. Burst DMA operation flowchart



Several options are available once the DMA burst is completed, depending on the register update strategy.

If the PREEN bit is reset (preload disabled), the value written by the DMA is immediately transferred into the active register and the registers are updated sequentially, following the DMA transaction pace.

When the preload is enabled (PREEN bit set), there are 3 use cases:

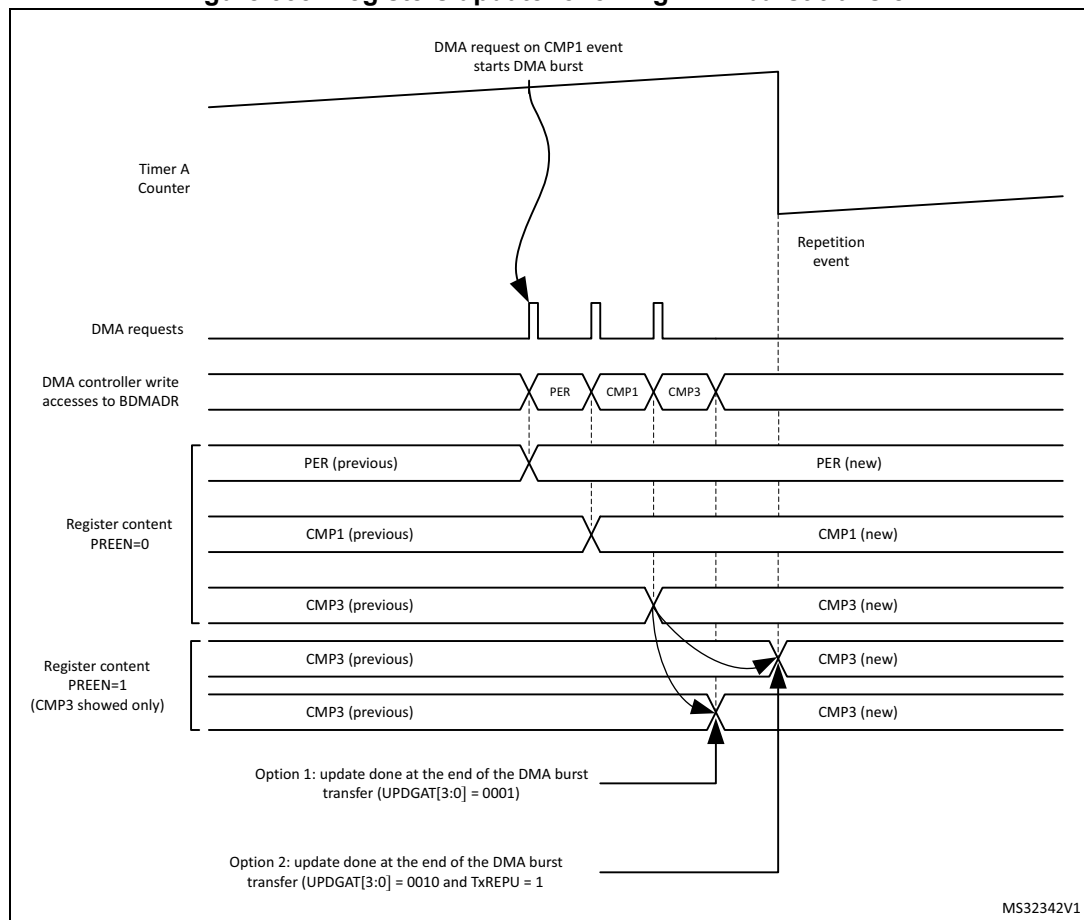
1. The update is done independently from DMA burst transfers (UPDGAT[3:0] = 0000 in HRTIM_TIMxCR and BRSTDMA[1:0] = 00 in HRTIM_MCR). In this case, and if it is necessary to have all transferred data taken into account simultaneously, the user must check that the DMA burst is completed before the update event takes place. On the contrary, if the update event happens while the DMA transfer is on-going, only part of the registers will be loaded and the complete register update will require 2 consecutive update events.
2. The update is done when the DMA burst transfer is completed (UPDGAT[3:0] = 0000 in HRTIM_TIMxCR and BRSTDMA[1:0] = 01 in HRTIM_MCR). This mode guarantees that all new register values are transferred simultaneously. This is done independently

from the counter value and can be combined with regular update events, if necessary (for instance, an update on a counter reset when TxRSTU is set).

- The update is done on the update event following the DMA burst transfer completion (UPDGAT[3:0] = 0010 in HRTIM_TIMxCR and BRSTDMA[1:0] = 10 in HRTIM_MCR). This mode guarantees both a coherent update of all transferred data and the synchronization with regular update events, with the timer counter. In this case, if a regular update request occurs while the transfer is on-going, it will be discarded and the effective update will happen on the next coming update request.

The chronogram on [Figure 335](#) presents the active register content for 3 cases: PREEN=0, UPDGAT[3:0] = 0001 and UPDGAT[3:0] = 0001 (when PREEN = 1).

Figure 335. Registers update following DMA burst transfer



37.3.22 HRTIM initialization

This section describes the recommended HRTIM initialization procedure, including other related MCU peripherals.

The HRTIM clock source must be enabled in the Reset and Clock control unit (RCC).

The HRTIM control registers can be initialized as per the power converter topology and the timing units use case. All inputs have to be configured (source, polarity, edge-sensitivity).

The HRTIM outputs must be set up eventually, with the following sequence:

- the polarity must be defined using POLx bits in HRTIM_OUTxR
- the FAULT and IDLE states must be configured using FAULTx[1:0] and IDLESx bits in HRTIM_OUTxR

The HRTIM outputs are ready to be connected to the MCU I/Os. In the GPIO controller, the selected HRTIM I/Os have to be configured as per the alternate function mapping table in the product datasheet.

From this point on, the HRTIM controls the outputs, which are in the IDLE state.

The outputs are configured in RUN mode by setting TxyOEN bits in the HRTIM_OENR register. The 2 outputs are in the inactive state until the first valid set/reset event in RUN mode. Any output set/reset event (except software requests using SST, SRT) are ignored as long as Tx Cen bit is reset, as well as burst mode requests (IDLEM bit value is ignored). Similarly, any counter reset request coming from the burst mode controller is ignored (if Tx BM bit is set).

Note: When the deadtime insertion is enabled (DTEN bit set), it is necessary to force the output state by software, using SST and RST bits, to have the outputs in a complementary state as soon as the RUN mode is entered.

The HRTIM operation can eventually be started by setting Tx Cen or MCEN bits in HRTIM_MCR.

If the HRTIM peripheral is reset with the Reset and Clock Controller, the HRTIM outputs are put in IDLE mode with a low level. It is recommended to first disconnect the HRTIMER from the outputs (using the GPIO controller) before performing a peripheral reset.

37.3.23 Debug

When a microcontroller enters the debug mode (Cortex®-M7 core halted), the TIMx counter either continues to work normally or stops, depending on DBG_HRTIM_STOP configuration bit in DBG module:

- DBG_HRTIM_STOP = 0: no behavior change, the HRTIM continues to operate.
- DBG_HRTIM_STOP = 1: all HRTIM timers, including the master, are stopped. The outputs in RUN mode enter the FAULT state if FAULTx[1:0] = 01, 10, 11, or keep their current state if FAULTx[1:0] = 00. The outputs in idle state are maintained in this state. This is permanently maintained even if the MCU exits the halt mode. This allows to maintain a safe state during the execution stepping. The outputs can be enabled again by settings TxyOEN bit (requires the use of the debugger).

Timer behavior during MCU halt when DBG_HRTIM_STOP = 1

The set/reset crossbar, the dead-time and push-pull unit, the idle/balanced fault detection and all the logic driving the normal output in RUN mode are not affected by debug. The output will keep on toggling internally, so as to retrieve regular signals of the outputs when TxyOEN will be set again (during or after the MCU halt). Associated triggers and filters are also following internal waveforms when the outputs are disabled.

FAULT inputs and events (any source) are enabled during the MCU halt.

Fault status bits can be set and TxyOEN bits reset during the MCU halt if a fault occurs at that time (TxyOEN and TxyODS are not affected by DBG_HRTIM_STOP bit state).

Synchronization, counter reset, start and reset-start events are discarded in debug mode, as well as capture events. This is to keep all related registers stable as long as the MCU is halted.

The counter stops counting when a breakpoint is reached. However, the counter enable signal is not reset; consequently no start event will be emitted when exiting from debug. All counter reset and capture triggers are disabled, as well as external events (ignored as long as the MCU is halted). The outputs SET and RST flags are frozen, except in case of forced software set/reset. A level-sensitive event is masked during the debug but will be active again as soon as the debug will be exited. For edge-sensitive events, if the signal is maintained active during the MCU halt, a new edge is not generated when exiting from debug.

The update events are discarded. This prevents any update trigger on `hrtim_upd_en[3:1]` inputs. DMA triggers are disabled. The burst mode circuit is frozen: the triggers are ignored and the burst mode counter stopped.

37.4 Application use cases

37.4.1 Buck converter

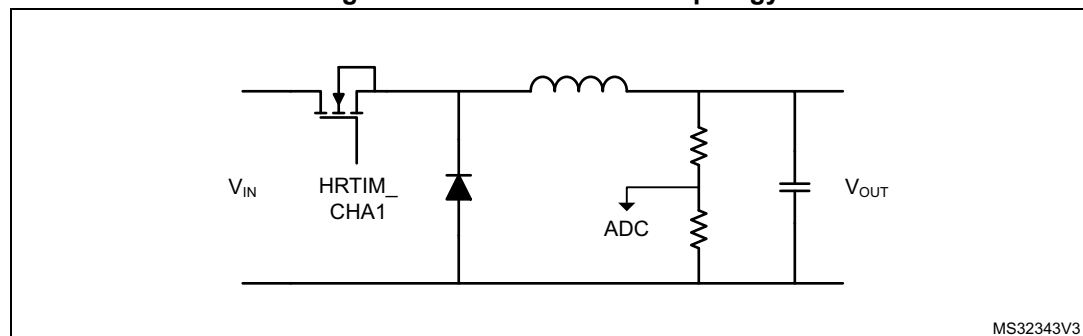
Buck converters are of common use as step-down converters. The HRTIM can control up to 10 buck converters with 6 independent switching frequencies.

The converter usually operates at a fixed frequency and the V_{in}/V_{out} ratio depends on the duty cycle D applied to the power switch:.

$$V_{out} = D \times V_{in}$$

The topology is given on [Figure 336](#) with the connection to the ADC for voltage reading.

Figure 336. Buck converter topology

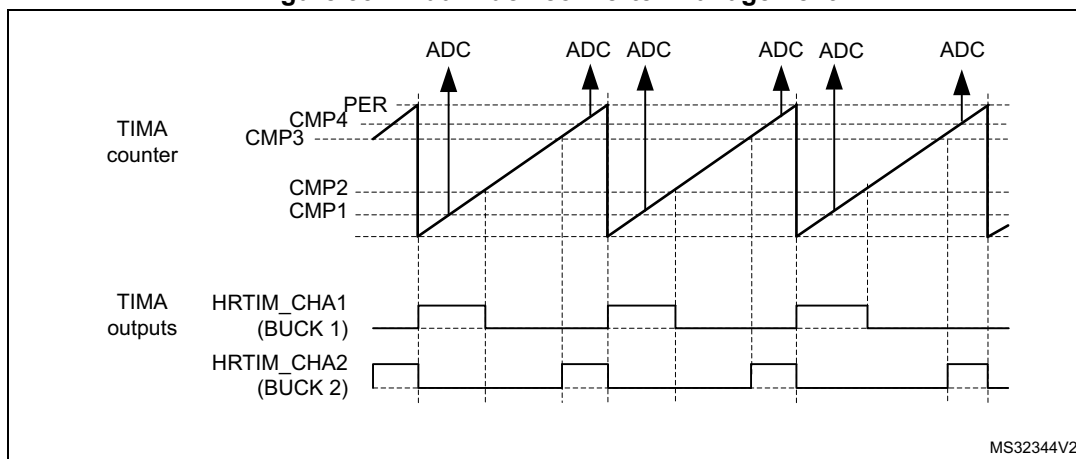


[Figure 337](#) presents the management of two converters with identical frequency PWM signals. The outputs are defined as follows:

- HRTIM_CHA1 set on period, reset on CMP1
- HRTIM_CHA2 set on CMP3, reset on PER

The ADC is triggered twice per period, precisely in the middle of the ON time, using CMP2 and CMP4 events.

Figure 337. Dual Buck converter management

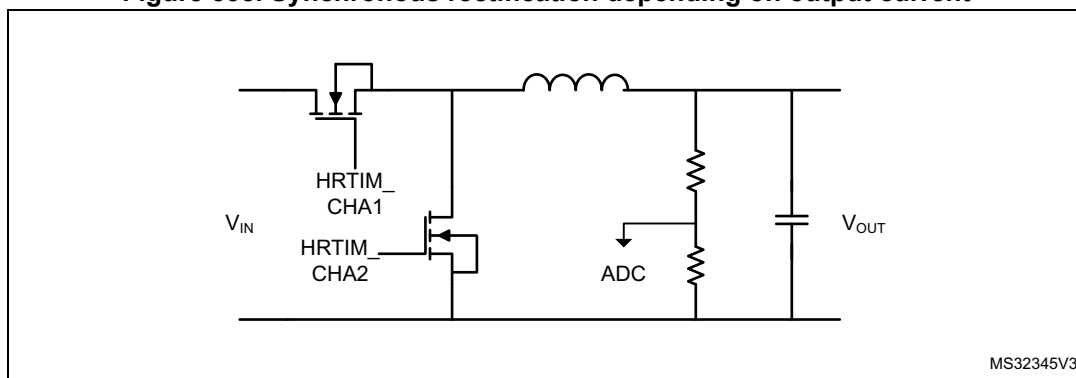


Timers A..E provide either 10 buck converters coupled by pairs (both with identical switching frequencies) or 6 completely independent converters (each of them having a different switching frequency), using the master timer as the 6th time base.

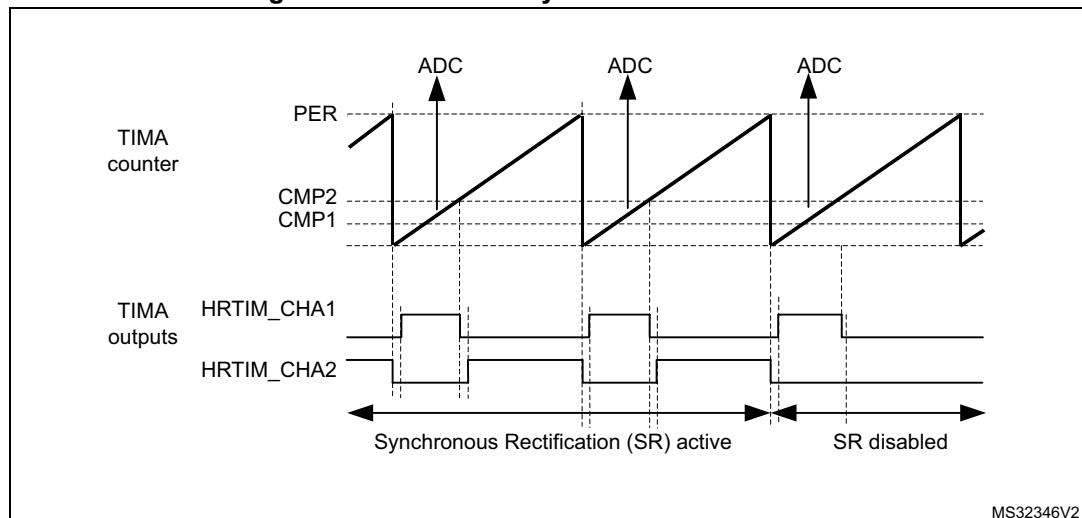
37.4.2 Buck converter with synchronous rectification

Synchronous rectification allows to minimize losses in buck converters, by means of a FET replacing the freewheeling diode. Synchronous rectification can be turned on or off on the fly depending on the output current level, as shown on [Figure 338](#).

Figure 338. Synchronous rectification depending on output current



The main difference vs. a single-switch buck converter is the addition of a deadtime for an almost complementary waveform generation on HRTIM_CHA2, based on the reference waveform on HRTIM_CHA1 (see [Figure 339](#)).

Figure 339. Buck with synchronous rectification

MS32346V2

37.4.3 Multiphase converters

Multiphase techniques can be applied to multiple power conversion topologies (buck, flyback). Their main benefits are:

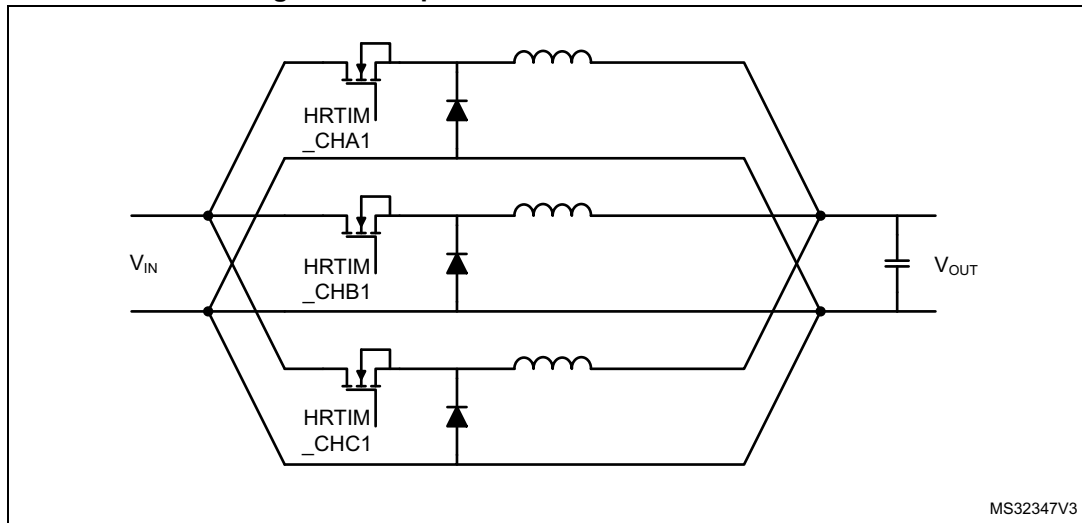
- Reduction of the current ripple on the input and output capacitors
- Reduced EMI
- Higher efficiency at light load by dynamically changing the number of phases (phase shedding)

The HRTIM is able to manage multiple converters. The number of converters that can be controlled depends on the topologies and resources used (including the ADC triggers):

- 5 buck converters with synchronous rectification (SR), using the master timer and the 5 timers
- 4 buck converters (without SR), using the master timer and 2 timers
- ...

[Figure 341](#) presents the topology of a 3-phase interleaved buck converter.

Figure 340. 3-phase interleaved buck converter



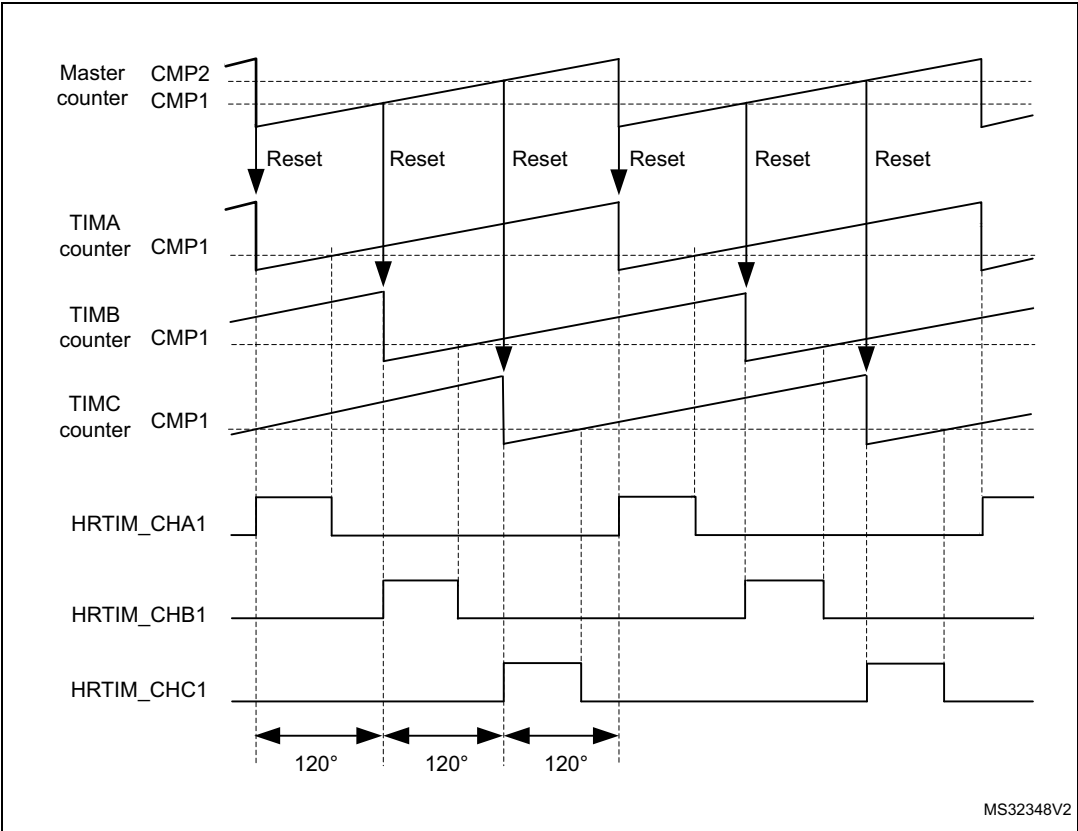
The master timer is responsible for the phase management: it defines the phase relationship between the converters by resetting the timers periodically. The phase-shift is 360° divided by the number of phases, 120° in the given example.

The duty cycle is then programmed into each of the timers. The outputs are defined as follows:

- HRTIM_CHA1 set on master timer period, reset on TACMP1
- HRTIM_CHB1 set on master timer MCMP1, reset on TBCMP1
- HRTIM_CHC1 set on master timer MCMP2, reset on TCCMP1

The ADC trigger can be generated on TxCMP2 compare event. Since all ADC trigger sources are phase-shifted because of the converter topology, it is possible to have all of them combined into a single ADC trigger to save ADC resources (for instance 1 ADC regular channel for the full multi-phase converter).

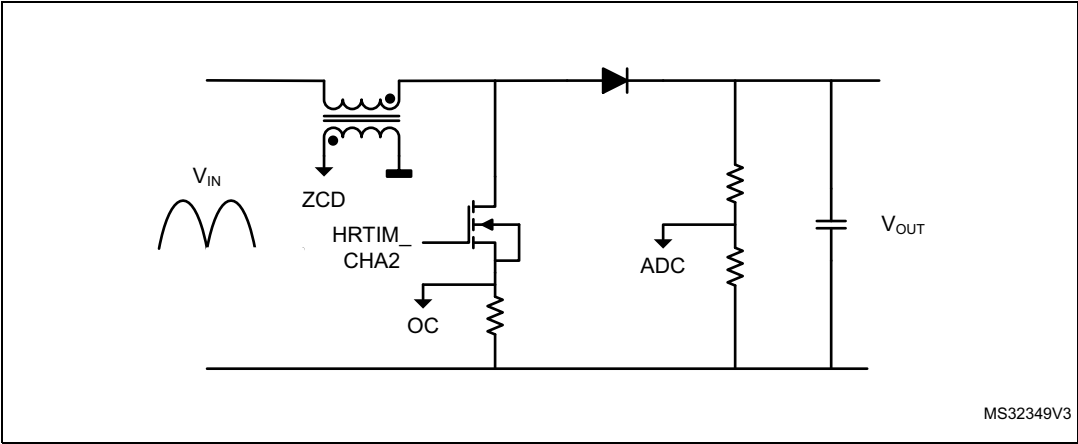
Figure 341. 3-phase interleaved buck converter control



37.4.4 Transition mode Power Factor Correction

The basic operating principle is to build up current into an inductor during a fixed T_{on} time. This current will then decay during the T_{off} time, and the period will be re-started when it becomes null. This is detected using a Zero Crossing Detection circuitry (ZCD), as shown on [Figure 342](#). With a constant T_{on} time, the peak current value in the inductor is proportional to the rectified AC input voltage, which provides the power factor correction.

Figure 342. Transition mode PFC



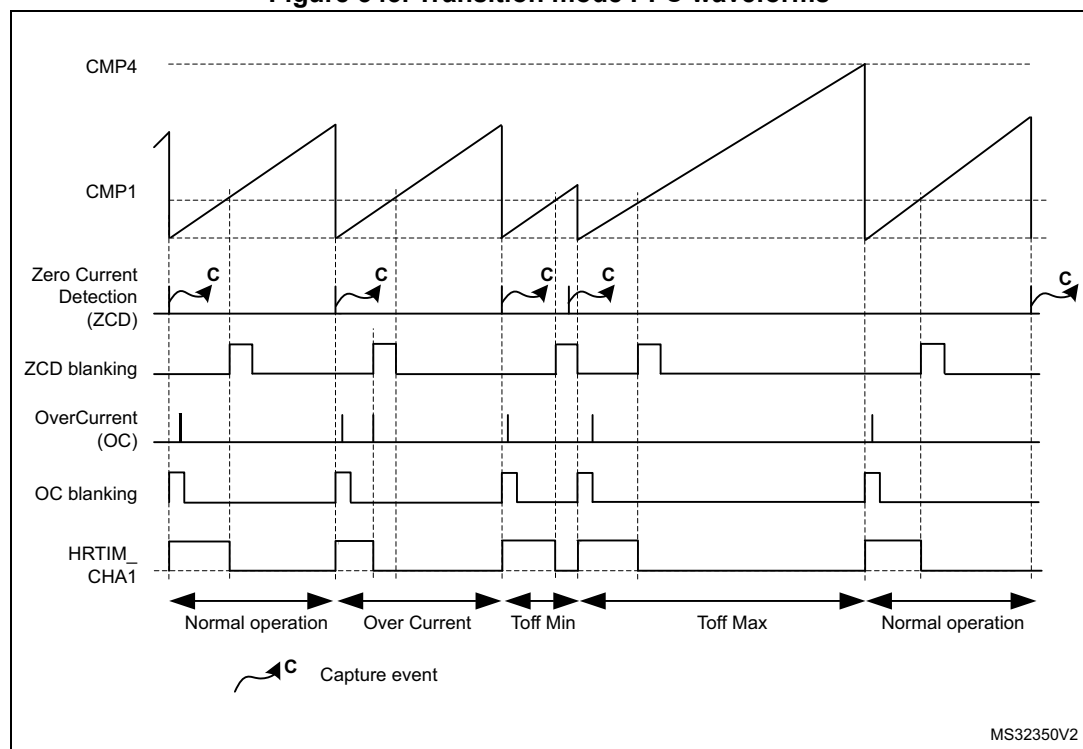
This converter is operating with a constant Ton time and a variable frequency due the Toff time variation (depending on the input voltage). It must also include some features to operate when no zero-crossing is detected, or to limit the Ton time in case of over-current (OC). The OC feedback is usually conditioned with the built-in comparator and routed onto an external event input.

Figure 343 presents the waveform during the various operating modes, with the following parameters defined:

- Ton Min: masks spurious overcurrent (freewheeling diode recovery current), represented as OC blanking
- Ton Max: practically, the converter set-point. It is defined by CMP1
- Toff Min: limits the frequency when the current limit is close to zero (demagnetization is very fast). It is defined with CMP2.
- Toff Max: prevents the system to be stuck if no ZCD occurs. It is defined with CMP4 in auto-delayed mode.

Both Toff values are auto-delayed since the value must be relative to the output falling edge.

Figure 343. Transition mode PFC waveforms



37.5 HRTIM registers

37.5.1 HRTIM Master Timer Control Register (HRTIM_MCR)

Address offset: 0x0000h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BRSTDMA[1:0]		MREPU	Res.	PREEN	DACSINC[1:0]		Res.	Res.	Res.	TECEN	TDCEN	TCCEN	TBCEN	TACEN	MCEN
rw	rw	rw		rw	rw	rw				rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SYNCSRC[1:0]		SYNCOU[1:0]		SYNCS TRTM	SYNCR STM	SYNCSIN[1:0]		Res.	Res.	HALF	RETRI G	CONT	CKPSC[2:0]		
rw	rw	rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	rw	rw

Bits 31:30 BRSTDMA[1:0]: Burst DMA Update

These bits define how the update occurs relatively to a burst DMA transaction.

00: Update done independently from the DMA burst transfer completion

01: Update done when the DMA burst transfer is completed

10: Update done on master timer roll-over following a DMA burst transfer completion. This mode only works in continuous mode.

11: reserved

Bit 29 MREPU: Master Timer Repetition update

This bit defines whether an update occurs when the master timer repetition period is completed (either due to roll-over or reset events). MREPU can be set only if BRSTDMA[1:0] = 00 or 01.

0: Update on repetition disabled

1: Update on repetition enabled

Bit 28 Reserved, must be kept at reset value.

Bit 27 PREEN: Preload enable

This bit enables the registers preload mechanism and defines whether the write accesses to the memory mapped registers are done into HRTIM active or preload registers.

0: Preload disabled: the write access is directly done into the active register

1: Preload enabled: the write access is done into the preload register

Bits 26:25 DACSYNC[1:0] DAC Synchronization

A DAC synchronization event can be enabled and generated when the master timer update occurs. These bits are defining on which output the DAC synchronization is sent (refer to [Section 37.3.19: DAC triggers](#) for connections details).

00: No DAC trigger generated

01: Trigger generated on hrtim_dac_trg1

10: Trigger generated on hrtim_dac_trg2

11: Trigger generated on hrtim_dac_trg3

Bits 24:22 Reserved, must be kept at reset value.

Bit 21 TECEN: Timer E counter enable

This bit starts the Timer E counter.

0: Timer E counter disabled

1: Timer E counter enabled

Note: This bit must not be changed within a minimum of 8 cycles of f_{HRTIM} clock.

Bit 20 **TDCEN**: Timer D counter enable

This bit starts the Timer D counter.

0: Timer D counter disabled

1: Timer D counter enabled

Note: This bit must not be changed within a minimum of 8 cycles of f_{HRTIM} clock.

Bit 19 **TCCEN**: Timer C counter enable

This bit starts the Timer C counter.

0: Timer C counter disabled

1: Timer C counter enabled

Note: This bit must not be changed within a minimum of 8 cycles of f_{HRTIM} clock.

Bit 18 **TBCEN**: Timer B counter enable

This bit starts the Timer B counter.

0: Timer B counter disabled

1: Timer B counter enabled

Note: This bit must not be changed within a minimum of 8 cycles of f_{HRTIM} clock.

Bit 17 **TACEN**: Timer A counter enable

This bit starts the Timer A counter.

0: Timer A counter disabled

1: Timer A counter enabled

Note: This bit must not be changed within a minimum of 8 cycles of f_{HRTIM} clock.

Bit 16 **MCEN**: Master timer counter enable

This bit starts the Master timer counter.

0: Master counter disabled

1: Master counter enabled

Note: This bit must not be changed within a minimum of 8 cycles of f_{HRTIM} clock.

Bits 15:14 **SYNCSRC[1:0]**: Synchronization source

These bits define the source and event to be sent on the synchronization outputs SYNCOUT[2:1]

00: Master timer Start

01: Master timer Compare 1 event

10: Timer A start/reset

11: Timer A Compare 1 event

Bits 13:12 **SYNCOUT[1:0]**: Synchronization output

These bits define the routing and conditioning of the synchronization output event.

00: disabled

01: Reserved.

10: Positive pulse on HRTIM_SCOUT output (16x f_{HRTIM} clock cycles)

11: Negative pulse on HRTIM_SCOUT output (16x f_{HRTIM} clock cycles)

Note: This bitfield must not be modified once the counter is enabled (TxCEN bit set)

Bit 11 **SYNCSTRM**: Synchronization Starts Master

This bit enables the Master timer start when receiving a synchronization input event:

0: No effect on the Master timer

1: A synchronization input event starts the Master timer

Bit 10 **SYNCRSTM**: Synchronization Resets Master

This bit enables the Master timer reset when receiving a synchronization input event:

0: No effect on the Master timer

1: A synchronization input event resets the Master timer

Bits 9:8 **SYNCIN[1:0]** *Synchronization input*

These bits are defining the synchronization input source.

00: disabled. HRTIM is not synchronized and runs in standalone mode.

01: Reserved.

10: Internal event: the HRTIM is synchronized with the on-chip timer (see [Synchronization input](#)).

11: External event (input pin). A positive pulse on HRTIM_SCIN input triggers the HRTIM.

Note: This parameter cannot be changed once the impacted timers are enabled.

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 **HALF**: *Half mode*

This bit enables the half duty-cycle mode: the HRTIM_MCMP1xR active register is automatically updated with HRTIM_MPER/2 value when HRTIM_MPER register is written.

0: Half mode disabled

1: Half mode enabled

Bit 4 **RETRIG**: *Re-triggerable mode*

This bit defines the behavior of the master timer counter in single-shot mode.

0: The timer is not re-triggerable: a counter reset can be done only if the counter is stopped (period elapsed)

1: The timer is re-triggerable: a counter reset is done whatever the counter state (running or stopped)

Bit 3 **CONT**: *Continuous mode*

0: The timer operates in single-shot mode and stops when it reaches the MPER value

1: The timer operates in continuous (free-running) mode and rolls over to zero when it reaches the MPER value

Bits 2:0 **CKPSC[2:0]**: *Clock prescaler*

These bits define the master timer clock prescaler ratio.

The counter clock equivalent frequency (f_{COUNTER}) is equal to $f_{\text{HRCK}} / 2^{(\text{CKPSC}[2:0]-5)}$.

The prescaling ratio cannot be modified once the timer is enabled.

000: Reserved

001: Reserved

010: Reserved

011: Reserved

100: Reserved

101: $f_{\text{COUNTER}} = f_{\text{HRTIM}}$

110: $f_{\text{COUNTER}} = f_{\text{HRTIM}} / 2$

111: $f_{\text{COUNTER}} = f_{\text{HRTIM}} / 4$

37.5.2 HRTIM Master Timer Interrupt Status Register (HRTIM_MISR)

Address offset: 0x0004h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MUPD	SYNC	MREP	MCMP4	MCMP3	MCMP2	MCMP1
									r	r	r	r	r	r	r

Bits 31:7 Reserved, must be kept at reset value.

Bit 6 MUPD: Master Update Interrupt Flag

This bit is set by hardware when the Master timer registers are updated.

0: No Master Update interrupt occurred

1: Master Update interrupt occurred

Bit 5 SYNC: Sync Input Interrupt Flag

This bit is set by hardware when a synchronization input event is received.

0: No Sync input interrupt occurred

1: Sync input interrupt occurred

Bit 4 MREP: Master Repetition Interrupt Flag

This bit is set by hardware when the Master timer repetition period has elapsed.

0: No Master Repetition interrupt occurred

1: Master Repetition interrupt occurred

Bit 3 MCMP4: Master Compare 4 Interrupt Flag

Refer to MCMP1 description

Bit 2 MCMP3: Master Compare 3 Interrupt Flag

Refer to MCMP1 description

Bit 1 MCMP2: Master Compare 2 Interrupt Flag

Refer to MCMP1 description

Bit 0 MCMP1: Master Compare 1 Interrupt Flag

This bit is set by hardware when the Master timer counter matches the value programmed in the master Compare 1 register.

0: No Master Compare 1 interrupt occurred

1: Master Compare 1 interrupt occurred

37.5.3 HRTIM Master Timer Interrupt Clear Register (HRTIM_MICR)

Address offset: 0x0008h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MUPD C	SYNCC	MREP C	MCMP 4C	MCMP 3C	MCMP 2C	MCMP 1C
									w	w	w	w	w	w	w

Bits 31:7 Reserved, must be kept at reset value.

Bit 6 **MUPDC**: Master update Interrupt flag clear

Writing 1 to this bit clears the MUPDC flag in HRTIM_MISR register

Bit 5 **SYNCC**: Sync Input Interrupt flag clear

Writing 1 to this bit clears the SYNC flag in HRTIM_MISR register

Bit 4 **MREPC**: Repetition Interrupt flag clear

Writing 1 to this bit clears the MREP flag in HRTIM_MISR register

Bit 3 **MCMP4C**: Master Compare 4 Interrupt flag clear

Writing 1 to this bit clears the MCMP4 flag in HRTIM_MISR register

Bit 2 **MCMP3C**: Master Compare 3 Interrupt flag clear

Writing 1 to this bit clears the MCMP3 flag in HRTIM_MISR register

Bit 1 **MCMP2C**: Master Compare 2 Interrupt flag clear

Writing 1 to this bit clears the MCMP2 flag in HRTIM_MISR register

Bit 0 **MCMP1C**: Master Compare 1 Interrupt flag clear

Writing 1 to this bit clears the MCMP1 flag in HRTIM_MISR register

37.5.4 HRTIM Master Timer DMA / Interrupt Enable Register (HRTIM_MDIER)

Address offset: 0x000Ch

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MUPD DE	SYNCD E	MREP DE	MCMP 4DE	MCMP 3DE	MCMP 2DE	MCMP 1DE
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MUPDI E	SYNCI E	MREPI E	MCMP 4IE	MCMP 3IE	MCMP 2IE	MCMP 1IE
									rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bit 22 **MUPDDE**: Master Update DMA request Enable

This bit is set and cleared by software to enable/disable the Master update DMA requests.

0: Master update DMA request disabled

1: Master update DMA request enabled

Bit 21 **SYNCDE**: Sync Input DMA request Enable

This bit is set and cleared by software to enable/disable the Sync input DMA requests.

0: Sync input DMA request disabled

1: Sync input DMA request enabled

Bit 20 **MREPDE**: Master Repetition DMA request Enable

This bit is set and cleared by software to enable/disable the Master timer repetition DMA requests.

0: Repetition DMA request disabled

1: Repetition DMA request enabled

Bit 19 **MCMP4DE**: Master Compare 4 DMA request Enable

Refer to MCMP1DE description

Bit 18 **MCMP3DE**: Master Compare 3 DMA request Enable

Refer to MCMP1DE description

Bit 17 **MCMP2DE**: Master Compare 2 DMA request Enable

Refer to MCMP1DE description

Bit 16 **MCMP1DE**: Master Compare 1 DMA request Enable

This bit is set and cleared by software to enable/disable the Master timer Compare 1 DMA requests.

0: Compare 1 DMA request disabled

1: Compare 1 DMA request enabled

Bits 15:6 Reserved, must be kept at reset value.

Bit 6 **MUPDIE**: Master Update Interrupt Enable

This bit is set and cleared by software to enable/disable the Master timer registers update interrupts

0: Master update interrupts disabled

1: Master update interrupts enabled

Bit 5 **SYNCIE**: Sync Input Interrupt Enable

This bit is set and cleared by software to enable/disable the Sync input interrupts

0: Sync input interrupts disabled

1: Sync input interrupts enabled

Bit 4 **MREPIE**: Master Repetition Interrupt Enable

This bit is set and cleared by software to enable/disable the Master timer repetition interrupts

0: Master repetition interrupt disabled

1: Master repetition interrupt enabled

Bit 3 **MCMP4IE**: Master Compare 4 Interrupt Enable

Refer to MCMP1IE description

Bit 2 **MCMP3IE**: Master Compare 3 Interrupt Enable

Refer to MCMP1IE description

Bit 1 **MCMP2IE**: Master Compare 2 Interrupt Enable

Refer to MCMP1IE description

Bit 0 **MCMP1IE**: Master Compare 1 Interrupt Enable

This bit is set and cleared by software to enable/disable the Master timer Compare 1 interrupt

0: Compare 1 interrupt disabled

1: Compare 1 interrupt enabled

37.5.5 HRTIM Master Timer Counter Register (HRTIM_MCNT)

Address offset: 0x0010h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MCNT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **MCNT[15:0]**: Counter value

Holds the master timer counter value. This register can only be written when the master timer is stopped (MCEN = 0 in HRTIM_MCR).

Note: The timer behavior is not guaranteed if the counter value is set above the HRTIM_MPER register value.

37.5.6 HRTIM Master Timer Period Register (HRTIM_MPER)

Address offset: 0x0014h

Reset value: 0x0000 FFDF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MPER[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **MPER[15:0]**: Master Timer Period value

This register defines the counter overflow value.

The period value must be above or equal to 3 periods of the f_{HRTIM} clock.

The maximum value is 0x0000 FFDF.

37.5.7 HRTIM Master Timer Repetition Register (HRTIM_MREP)

Address offset: 0x0018h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MREP[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **MREP[7:0]**: *Master Timer Repetition period value*

This register holds the repetition period value for the master counter. It is either the preload register or the active register if preload is disabled.

37.5.8 HRTIM Master Timer Compare 1 Register (HRTIM_MCMP1R)

Address offset: 0x001Ch

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MCMP1[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **MCMP1[15:0]**: *Master Timer Compare 1 value*

This register holds the master timer Compare 1 value. It is either the preload register or the active register if preload is disabled.

The compare value must be above or equal to 3 periods of the f_{HRTIM} clock.

37.5.9 HRTIM Master Timer Compare 2 Register (HRTIM_MCMP2R)

Address offset: 0x0024h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MCMP2[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **MCMP2[15:0]: Master Timer Compare 2 value**

This register holds the master timer Compare 2 value. It is either the preload register or the active register if preload is disabled.

The compare value must be above or equal to 3 periods of the f_{HRTIM} clock.

37.5.10 HRTIM Master Timer Compare 3 Register (HRTIM_MCMP3R)

Address offset: 0x0028h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MCMP3[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **MCMP3[15:0]: Master Timer Compare 3 value**

This register holds the master timer Compare 3 value. It is either the preload register or the active register if preload is disabled.

The compare value must be above or equal to 3 periods of the f_{HRTIM} clock.

37.5.11 HRTIM Master Timer Compare 4 Register (HRTIM_MCMP4R)

Address offset: 0x002Ch

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MCMP4[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **MCMP4[15:0]: Master Timer Compare 4 value**

This register holds the master timer Compare 4 value. It is either the preload register or the active register if preload is disabled.

The compare value must be above or equal to 3 periods of the f_{HRTIM} clock.



37.5.12 HRTIM Timerx Control Register (HRTIM_TIMxCR)

Address offset: 0x0000h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UPDGAT[3:0]				PREEN	DACSINC[1:0]		MSTU	TEU	TDU	TCU	TBU	Res.	TxRST U	TxREP U	Res.
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DELCMP4[1:0]		DELCMP2[1:0]		SYNCS TRTx	SYNCR STx	Res.	Res.	Res.	PSHPL L	HALF	RETRI G	CONT	CKPSCx[2:0]		
rw	rw	rw	rw	rw	rw				rw	rw	rw	rw	rw	rw	rw

Bits 31:28 UPDGAT[3:0]: Update Gating

These bits define how the update occurs relatively to the burst DMA transaction and the external update request on update enable inputs 1 to 3 (see [Table 310: Update enable inputs and sources](#))
The update events, as mentioned below, can be: MSTU, TEU, TDU, TCU, TBU, TAU, TxRSTU, TxREPU.

0000: the update occurs independently from the DMA burst transfer

0001: the update occurs when the DMA burst transfer is completed

0010: the update occurs on the update event following the DMA burst transfer completion

0011: the update occurs on a rising edge of HRTIM update enable input 1 (hrtim_upd_en1)

0100: the update occurs on a rising edge of HRTIM update enable input 2 (hrtim_upd_en2)

0101: the update occurs on a rising edge of HRTIM update enable input 3 (hrtim_upd_en3)

0110: the update occurs on the update event following a rising edge of HRTIM update enable input 1 (hrtim_upd_en1)

0111: the update occurs on the update event following a rising edge of HRTIM update enable input 2 (hrtim_upd_en2)

1000: the update occurs on the update event following a rising edge of HRTIM update enable input 3 (hrtim_upd_en3)

Other codes: reserved

Note: This bitfield must be reset before programming a new value.

For UPDGAT[3:0] values equal to 0001, 0011, 0100, 0101, it is possible to have multiple concurrent update source (for instance RSTU and DMA burst).

Bit 27 PREEN: Preload enable

This bit enables the registers preload mechanism and defines whether a write access into a preloadable register is done into the active or the preload register.

0: Preload disabled: the write access is directly done into the active register

1: Preload enabled: the write access is done into the preload register

Bits 26:25 DACSYNC[1:0] DAC Synchronization

A DAC synchronization event is generated when the timer update occurs. These bits are defining on which output the DAC synchronization is sent (refer to [Section 37.3.19: DAC triggers](#) for connections details).

00: No DAC trigger generated

01: Trigger generated on hrtim_dac_trg1

10: Trigger generated on hrtim_dac_trg2

11: Trigger generated on hrtim_dac_trg3

- Bit 24 **MSTU**: *Master Timer update*
Register update is triggered by the master timer update.
0: Update by master timer disabled
1: Update by master timer enabled
- Bit 23 In HRTIM_TIMACR, HRTIM_TIMBCR, HRTIM_TIMCCR, HRTIM_TIMDCR:
TEU: *Timer E update*
Register update is triggered by the timer E update
0: Update by timer E disabled
1: Update by timer E enabled
- In HRTIM_TIMECR:
Reserved, must be kept at reset value
- Bit 22 In HRTIM_TIMACR, HRTIM_TIMBCR, HRTIM_TIMCCR, HRTIM_TIMECR:
TDU: *Timer D update*
Register update is triggered by the timer D update
0: Update by timer D disabled
1: Update by timer D enabled
- In HRTIM_TIMDCR:
Reserved, must be kept at reset value
- Bit 21 In HRTIM_TIMACR, HRTIM_TIMBCR, HRTIM_TIMDCR, HRTIM_TIMECR:
TCU: *Timer C update*
Register update is triggered by the timer C update
0: Update by timer C disabled
1: Update by timer C enabled
- In HRTIM_TIMCCR:
Reserved, must be kept at reset value
- Bit 20 In HRTIM_TIMACR, HRTIM_TIMCCR, HRTIM_TIMDCR, HRTIM_TIMECR:
TBU: *Timer B update*
Register update is triggered by the timer B update
0: Update by timer B disabled
1: Update by timer B enabled
- In HRTIM_TIMBCR:
Reserved, must be kept at reset value
- Bit 19 In HRTIM_TIMBCR, HRTIM_TIMCCR, HRTIM_TIMDCR, HRTIM_TIMECR:
TAU: *Timer A update*
Register update is triggered by the timer A update
0: Update by timer A disabled
1: Update by timer A enabled
- In HRTIM_TIMACR:
Reserved, must be kept at reset value

Bit 18 **TxRSTU**: *Timerx reset update*

Register update is triggered by Timerx counter reset or roll-over to 0 after reaching the period value in continuous mode.

- 0: Update by timer x reset / roll-over disabled
- 1: Update by timer x reset / roll-over enabled

Bit 17 **TxREPU**: *Timer x Repetition update*

Register update is triggered when the counter rolls over and HRTIM_REPx = 0

- 0: Update on repetition disabled
- 1: Update on repetition enabled

Bit 16 Reserved, must be kept at reset value.

Bits 15:14 **DELCMP4[1:0]**: *CMP4 auto-delayed mode*

This bitfield defines whether the compare register is behaving in standard mode (compare match issued as soon as counter equal compare), or in auto-delayed mode (see [Auto-delayed mode](#)).

- 00: CMP4 register is always active (standard compare mode)
- 01: CMP4 value is recomputed and is active following a capture 2 event
- 10: CMP4 value is recomputed and is active following a capture 2 event, or is recomputed and active after Compare 1 match (timeout function if capture 2 event is missing)
- 11: CMP4 value is recomputed and is active following a capture event, or is recomputed and active after Compare 3 match (timeout function if capture event is missing)

Note: This bitfield must not be modified once the counter is enabled (TxCEN bit set)

Bits 13:12 **DELCMP2[1:0]**: *CMP2 auto-delayed mode*

This bitfield defines whether the compare register is behaving in standard mode (compare match issued as soon as counter equal compare), or in auto-delayed mode (see [Auto-delayed mode](#)).

- 00: CMP2 register is always active (standard compare mode)
- 01: CMP2 value is recomputed and is active following a capture 1 event
- 10: CMP2 value is recomputed and is active following a capture 1 event, or is recomputed and active after Compare 1 match (timeout function if capture event is missing)
- 11: CMP2 value is recomputed and is active following a capture 1 event, or is recomputed and active after Compare 3 match (timeout function if capture event is missing)

Note: This bitfield must not be modified once the counter is enabled (TxCEN bit set)

Bit 11 **SYNCSTRTx**: *Synchronization Starts Timer x*

This bit defines the Timer x behavior following the synchronization event:

- 0: No effect on Timer x
- 1: A synchronization input event starts the Timer x

Bit 10 **SYNCRSTx**: *Synchronization Resets Timer x*

This bit defines the Timer x behavior following the synchronization event:

- 0: No effect on Timer x
- 1: A synchronization input event resets the Timer x

Bits 9:7 Reserved, must be kept at reset value.

Bit 6 **PSHPLL**: *Push-Pull mode enable*

This bit enables the push-pull mode.

- 0: Push-Pull mode disabled
- 1: Push-Pull mode enabled

Note: This bitfield must not be modified once the counter is enabled (TxCEN bit set)

Bit 5 **HALF**: *Half mode enable*

This bit enables the half duty-cycle mode: the HRTIM_CMP1xR active register is automatically updated with HRTIM_PERxR/2 value when HRTIM_PERxR register is written.

- 0: Half mode disabled
- 1: Half mode enabled

Bit 4 **RETRIG**: *Re-triggerable mode*

This bit defines the counter behavior in single shot mode.

- 0: The timer is not re-triggerable: a counter reset is done if the counter is stopped (period elapsed in single-shot mode or counter stopped in continuous mode)
- 1: The timer is re-triggerable: a counter reset is done whatever the counter state.

Bit 3 **CONT**: *Continuous mode*

This bit defines the timer operating mode.

- 0: The timer operates in single-shot mode and stops when it reaches TIMxPER value
- 1: The timer operates in continuous mode and rolls over to zero when it reaches TIMxPER value

Bits 2:0 **CKPSCx[2:0]**: *HRTIM Timer x Clock prescaler*

These bits define the master timer clock prescaler ratio.

The counter clock equivalent frequency (f_{COUNTER}) is equal to $f_{\text{HRCK}} / 2^{(\text{CKPSC}[2:0]-5)}$.

The prescaling ratio cannot be modified once the timer is enabled.

- 000: Reserved
- 001: Reserved
- 010: Reserved
- 011: Reserved
- 100: Reserved
- 101: $f_{\text{COUNTER}} = f_{\text{HRTIM}}$
- 110: $f_{\text{COUNTER}} = f_{\text{HRTIM}} / 2$
- 111: $f_{\text{COUNTER}} = f_{\text{HRTIM}} / 4$

37.5.13 HRTIM Timerx Interrupt Status Register (HRTIM_TIMxISR)

Address offset: 0x0004h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	O2CPY	O1CPY	O2STAT	O1STAT	IPPSTAT	CPPSTAT
										r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	DLYPRT	RST	RSTx2	SETx2	RSTx1	SETx1	CPT2	CPT1	UPD	Res.	REP	CMP4	CMP3	CMP2	CMP1
	r	r	r	r	r	r	r	r	r		r	r	r	r	r

Bits 31:22 Reserved, must be kept at reset value.

Bit 21 **O2CPY**: Output 2 Copy

This status bit is a raw copy of the output 2 state, before the output stage (chopper, polarity). It allows to check the current output state before re-enabling the output after a delayed protection.

0: Output 2 is inactive

1: Output 2 is active

Bit 20 **O1CPY**: Output 1 Copy

This status bit is a raw copy of the output 1 state, before the output stage (chopper, polarity). It allows to check the current output state before re-enabling the output after a delayed protection.

0: Output 1 is inactive

1: Output 1 is active

Bit 19 **O2STAT**: Output 2 Status

This status bit indicates the output 2 state when the delayed idle protection was triggered. This bit is updated upon any new delayed protection entry. This bit is not updated in balanced idle.

0: Output 2 was inactive

1: Output 2 was active

Bit 18 **O1STAT**: Output 1 Status

This status bit indicates the output 1 state when the delayed idle protection was triggered. This bit is updated upon any new delayed protection entry. This bit is not updated in balanced idle.

0: Output 1 was inactive

1: Output 1 was active

Bit 17 **IPPSTAT**: Idle Push Pull Status

This status bit indicates on which output the signal was applied, in push-pull mode balanced fault mode or delayed idle mode, when the protection was triggered (whatever the output state, active or inactive).

0: Protection occurred when the output 1 was active and output 2 forced inactive

1: Protection occurred when the output 2 was active and output 1 forced inactive

Bit 16 **CPPSTAT**: Current Push Pull Status

This status bit indicates on which output the signal is currently applied, in push-pull mode. It is only significant in this configuration.

0: Signal applied on output 1 and output 2 forced inactive

1: Signal applied on output 2 and output 1 forced inactive

Bit 15 Reserved

Bit 14 **DLYPRT**: Delayed Protection Flag

This bit indicates delayed idle or the balanced idle mode entry.

- Bit 13 **RST**: Reset and/or roll-over Interrupt Flag
This bit is set by hardware when the timer x counter is reset or rolls over in continuous mode.
0: No TIMx counter reset/roll-over interrupt occurred
1: TIMX counter reset/roll-over interrupt occurred
- Bit 12 **RSTx2**: Output 2 Reset Interrupt Flag
Refer to RSTx1 description
- Bit 11 **SETx2**: Output 2 Set Interrupt Flag
Refer to SETx1 description
- Bit 10 **RSTx1**: Output 1 Reset Interrupt Flag
This bit is set by hardware when the Tx1 output is reset (goes from active to inactive mode).
0: No Tx1 output reset interrupt occurred
1: Tx1 output reset interrupt occurred
- Bit 9 **SETx1**: Output 1 Set Interrupt Flag
This bit is set by hardware when the Tx1 output is set (goes from inactive to active mode).
0: No Tx1 output set interrupt occurred
1: Tx1 output set interrupt occurred
- Bit 8 **CPT2**: Capture2 Interrupt Flag
Refer to CPT1 description
- Bit 7 **CPT1**: Capture1 Interrupt Flag
This bit is set by hardware when the timer x capture 1 event occurs.
0: No timer x Capture 1 reset interrupt occurred
1: Timer x output 1 reset interrupt occurred
- Bit 6 **UPD**: Update Interrupt Flag
This bit is set by hardware when the timer x update event occurs.
0: No timer x update interrupt occurred
1: Timer x update interrupt occurred
- Bit 5 Reserved, must be kept at reset value.
- Bit 4 **REP**: Repetition Interrupt Flag
This bit is set by hardware when the timer x repetition period has elapsed.
0: No timer x repetition interrupt occurred
1: Timer x repetition interrupt occurred
- Bit 3 **CMP4**: Compare 4 Interrupt Flag
Refer to CMP1 description
- Bit 2 **CMP3**: Compare 3 Interrupt Flag
Refer to CMP1 description
- Bit 1 **CMP2**: Compare 2 Interrupt Flag
Refer to CMP1 description
- Bit 0 **CMP1**: Compare 1 Interrupt Flag
This bit is set by hardware when the timer x counter matches the value programmed in the Compare 1 register.
0: No Compare 1 interrupt occurred
1: Compare 1 interrupt occurred

37.5.14 HRTIM Timerx Interrupt Clear Register (HRTIM_TIMxICR)

Address offset: 0x0008h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	DLYPRTC	RSTC	RSTx2C	SETx2C	RSTx1C	SETx1C	CPT2C	CPT1C	UPDC	Res.	REPC	CMP4C	CMP3C	CMP2C	CMP1C
	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:15 Reserved, must be kept at reset value.

Bit 14 **DLYPRTC**: Delayed Protection Flag Clear

Writing 1 to this bit clears the DLYPRT flag in HRTIM_TIMxISR register

Bit 13 **RSTC**: Reset Interrupt flag Clear

Writing 1 to this bit clears the RST flag in HRTIM_TIMxISR register

Bit 12 **RSTx2C**: Output 2 Reset flag Clear

Writing 1 to this bit clears the RSTx2 flag in HRTIM_TIMxISR register

Bit 11 **SETx2C**: Output 2 Set flag Clear

Writing 1 to this bit clears the SETx2 flag in HRTIM_TIMxISR register

Bit 10 **RSTx1C**: Output 1 Reset flag Clear

Writing 1 to this bit clears the RSTx1 flag in HRTIM_TIMxISR register

Bit 9 **SETx1C**: Output 1 Set flag Clear

Writing 1 to this bit clears the SETx1 flag in HRTIM_TIMxISR register

Bit 8 **CPT2C**: Capture2 Interrupt flag Clear

Writing 1 to this bit clears the CPT2 flag in HRTIM_TIMxISR register

Bit 7 **CPT1C**: Capture1 Interrupt flag Clear

Writing 1 to this bit clears the CPT1 flag in HRTIM_TIMxISR register

Bit 6 **UPDC**: Update Interrupt flag Clear

Writing 1 to this bit clears the UPD flag in HRTIM_TIMxISR register

Bit 5 Reserved, must be kept at reset value.

Bit 4 **REPC**: Repetition Interrupt flag Clear

Writing 1 to this bit clears the REP flag in HRTIM_TIMxISR register

Bit 3 **CMP4C**: Compare 4 Interrupt flag Clear

Writing 1 to this bit clears the CMP4 flag in HRTIM_TIMxISR register

Bit 2 **CMP3C**: Compare 3 Interrupt flag Clear

Writing 1 to this bit clears the CMP3 flag in HRTIM_TIMxISR register

Bit 1 **CMP2C**: Compare 2 Interrupt flag Clear

Writing 1 to this bit clears the CMP2 flag in HRTIM_TIMxISR register

Bit 0 **CMP1C**: Compare 1 Interrupt flag Clear

Writing 1 to this bit clears the CMP1 flag in HRTIM_TIMxISR register

37.5.15 HRTIM Timerx DMA / Interrupt Enable Register (HRTIM_TIMxDIER)

Address offset: 0x000Ch (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	DLYPR TDE	RSTDE	RSTx2 DE	SETx2 DE	RSTx1 DE	SETx1 DE	CPT2D E	CPT1D E	UPDDE	Res.	REPDE	CMP4D E	CMP3D E	CMP2D E	CMP1D E
	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	DLYPR TIE	RSTIE	RSTx2I E	SETx2I E	RSTx1I E	SETx1I E	CPT2IE	CPT1IE	UPDIE	Res.	REPIE	CMP4I E	CMP3I E	CMP2I E	CMP1I E
	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw

Bit 31 Reserved

Bit 30 **DLYPRTDE**: Delayed Protection DMA request Enable

This bit is set and cleared by software to enable/disable DMA requests on delayed protection.

0: Delayed protection DMA request disabled

1: Delayed protection DMA request enabled

Bit 29 **RSTDE**: Reset/roll-over DMA request Enable

This bit is set and cleared by software to enable/disable DMA requests on timer x counter reset or roll-over in continuous mode.

0: Timer x counter reset/roll-over DMA request disabled

1: Timer x counter reset/roll-over DMA request enabled

Bit 28 **RSTx2DE**: Output 2 Reset DMA request Enable

Refer to RSTx1DE description

Bit 27 **SETx2DE**: Output 2 Set DMA request Enable

Refer to SETx1DE description

Bit 26 **RSTx1DE**: Output 1 Reset DMA request Enable

This bit is set and cleared by software to enable/disable Tx1 output reset DMA requests.

0: Tx1 output reset DMA request disabled

1: Tx1 output reset DMA request enabled

Bit 25 **SETx1DE**: Output 1 Set DMA request Enable

This bit is set and cleared by software to enable/disable Tx1 output set DMA requests.

0: Tx1 output set DMA request disabled

1: Tx1 output set DMA request enabled

Bit 24 **CPT2DE**: Capture 2 DMA request Enable

Refer to CPT1DE description

Bit 23 **CPT1DE**: Capture 1 DMA request Enable

This bit is set and cleared by software to enable/disable Capture 1 DMA requests.

0: Capture 1 DMA request disabled

1: Capture 1 DMA request enabled

Bit 22 **UPDDE**: Update DMA request Enable

This bit is set and cleared by software to enable/disable DMA requests on update event.

0: Update DMA request disabled

1: Update DMA request enabled

- Bit 21 Reserved, must be kept at reset value.
- Bit 20 **REPDE**: Repetition DMA request Enable
This bit is set and cleared by software to enable/disable DMA requests on repetition event.
0: Repetition DMA request disabled
1: Repetition DMA request enabled
- Bit 19 **CMP4DE**: Compare 4 DMA request Enable
Refer to CMP1DE description
- Bit 18 **CMP3DE**: Compare 3 DMA request Enable
Refer to CMP1DE description
- Bit 17 **CMP2DE**: Compare 2 DMA request Enable
Refer to CMP1DE description
- Bit 16 **CMP1DE**: Compare 1 DMA request Enable
This bit is set and cleared by software to enable/disable the Compare 1 DMA requests.
0: Compare 1 DMA request disabled
1: Compare 1 DMA request enabled
- Bit 15 Reserved
- Bit 14 **DLYPTIE**: Delayed Protection Interrupt Enable
This bit is set and cleared by software to enable/disable interrupts on delayed protection.
0: Delayed protection interrupts disabled
1: Delayed protection interrupts enabled
- Bit 13 **RSTIE**: Reset/roll-over Interrupt Enable
This bit is set and cleared by software to enable/disable interrupts on timer x counter reset or roll-over in continuous mode.
0: Timer x counter reset/roll-over interrupt disabled
1: Timer x counter reset/roll-over interrupt enabled
- Bit 12 **RSTx2IE**: Output 2 Reset Interrupt Enable
Refer to RSTx1IE description
- Bit 11 **SETx2IE**: Output 2 Set Interrupt Enable
Refer to SETx1IE description
- Bit 10 **RSTx1IE**: Output 1 Reset Interrupt Enable
This bit is set and cleared by software to enable/disable Tx1 output reset interrupts.
0: Tx1 output reset interrupts disabled
1: Tx1 output reset interrupts enabled
- Bit 9 **SETx1IE**: Output 1 Set Interrupt Enable
This bit is set and cleared by software to enable/disable Tx1 output set interrupts.
0: Tx1 output set interrupts disabled
1: Tx1 output set interrupts enabled
- Bit 8 **CPT2IE**: Capture Interrupt Enable
Refer to CPT1IE description
- Bit 7 **CPT1IE**: Capture Interrupt Enable
This bit is set and cleared by software to enable/disable Capture 1 interrupts.
0: Capture 1 interrupts disabled
1: Capture 1 interrupts enabled

Bit 6 **UPDIE**: Update Interrupt Enable

This bit is set and cleared by software to enable/disable update event interrupts.

0: Update interrupts disabled

1: Update interrupts enabled

Bit 5 Reserved, must be kept at reset value.

Bit 4 **REPIE**: Repetition Interrupt Enable

This bit is set and cleared by software to enable/disable repetition event interrupts.

0: Repetition interrupts disabled

1: Repetition interrupts enabled

Bit 3 **CMP4IE**: Compare 4 Interrupt Enable

Refer to CMP1IE description

Bit 2 **CMP3IE**: Compare 3 Interrupt Enable

Refer to CMP1IE description

Bit 1 **CMP2IE**: Compare 2 Interrupt Enable

Refer to CMP1IE description

Bit 0 **CMP1IE**: Compare 1 Interrupt Enable

This bit is set and cleared by software to enable/disable the Compare 1 interrupts.

0: Compare 1 interrupt disabled

1: Compare 1 interrupt enabled

37.5.16 HRTIM Timerx Counter Register (HRTIM_CNTxR)

Address offset: 0x0010h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNTx[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **CNTx[15:0]**: *Timerx Counter value*

This register holds the Timerx counter value. It can only be written when the timer is stopped (TxCEN = 0 in HRTIM_TIMxCR).

Note: The timer behavior is not guaranteed if the counter value is above the HRTIM_PERxR register value.

37.5.17 HRTIM Timerx Period Register (HRTIM_PERxR)

Address offset: 0x14h (this offset address is relative to timer x base address)

Reset value: 0x0000 FFDF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PERx[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **PERx[15:0]**: *Timerx Period value*

This register holds timer x period value.

This register holds either the content of the preload register or the content of the active register if preload is disabled.

The period value must be above or equal to 3 periods of the f_{HRTIM} clock.

The maximum value is 0x0000 FFDF.

37.5.18 HRTIM Timerx Repetition Register (HRTIM_REPxR)

Address offset: 0x18h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REPx[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits31:8 Reserved, must be kept at reset value.

Bits 7:0 **REPx[7:0]**: *Timerx Repetition period value*

This register holds the repetition period value.

This register holds either the content of the preload register or the content of the active register if preload is disabled.

37.5.19 HRTIM Timerx Compare 1 Register (HRTIM_CMP1xR)

Address offset: 0x1Ch (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMP1x[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **CMP1x[15:0]**: *Timerx Compare 1 value*

This register holds the compare 1 value.

This register holds either the content of the preload register or the content of the active register if preload is disabled.

The compare value must be above or equal to 3 periods of the f_{HRTIM} clock.

37.5.20 HRTIM Timerx Compare 1 Compound Register (HRTIM_CMP1CxR)

Address offset: 0x20h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REP _x [7:0]							
								r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMP1 _x [15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:24 Reserved, must be kept at reset value.

Bits 23:16 **REP_x[7:0]**: *Timerx Repetition value (aliased from HRTIM_REP_x register)*

This bitfield is an alias from the REP_x[7:0] bitfield in the HRTIM_x_REP_xR register.

Bits 15:0 **CMP1_x[15:0]**: *Timerx Compare 1 value*

This bitfield is an alias from the CMP1_x[15:0] bitfield in the HRTIM_x_CMP1_xR register.

37.5.21 HRTIM Timerx Compare 2 Register (HRTIM_CMP2xR)

Address offset: 0x24h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMP2 _x [15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **CMP2_x[15:0]**: *Timerx Compare 2 value*

This register holds the Compare 2 value.

This register holds either the content of the preload register or the content of the active register if preload is disabled.

The compare value must be above or equal to 3 periods of the f_{HRTIM} clock.

This register can behave as an auto-delayed compare register, if enabled with DELCMP2[1:0] bits in HRTIM_TIMxCR.

37.5.22 HRTIM Timerx Compare 3 Register (HRTIM_CMP3xR)

Address offset: 0x28h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMP3x[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **CMP3x[15:0]**: *Timerx Compare 3 value*

This register holds the Compare 3 value.

This register holds either the content of the preload register or the content of the active register if preload is disabled.

The compare value must be above or equal to 3 periods of the f_{HRTIM} clock.

37.5.23 HRTIM Timerx Compare 4 Register (HRTIM_CMP4xR)

Address offset: 0x2Ch (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMP4x[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **CMP4x[15:0]**: *Timerx Compare 4 value*

This register holds the Compare 4 value.

This register holds either the content of the preload register or the content of the active register if preload is disabled.

The compare value must be above or equal to 3 periods of the f_{HRTIM} clock.

This register can behave as an auto-delayed compare register, if enabled with DELCMP4[1:0] bits in HRTIM_TIMxCR.

37.5.24 HRTIM Timerx Capture 1 Register (HRTIM_CPT1xR)

Address offset: 0x30h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CPT1x[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **CPT1x[15:0]**: *Timerx Capture 1 value*

This register holds the counter value when the capture 1 event occurred.

37.5.25 HRTIM Timerx Capture 2 Register (HRTIM_CPT2xR)

Address offset: 0x34h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CPT2x[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **CPT2x[15:0]**: *Timerx Capture 2 value*

This register holds the counter value when the capture 2 event occurred.

37.5.26 HRTIM Timerx Deadtime Register (HRTIM_DTxR)

Address offset: 0x38h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DTFLK _x	DTFSL _{Kx}	Res.	Res.	Res.	Res.	SDTFx	DTFx[8:0]								
rwo	rwo					rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DTRLK _x	DTRSL _{Kx}	Res.	DTPRSC[1:0]			SDTRx	DTRx[8:0]								
rwo	rwo		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 DTFLK_x: Deadtime Falling Lock

This write-once bit prevents the deadtime (sign and value) to be modified, if enabled.

0: Deadtime falling value and sign is writable

1: Deadtime falling value and sign is read-only

Note: This bit is not preloaded

Bit 30 DTFSLK_x: Deadtime Falling Sign Lock

This write-once bit prevents the sign of falling deadtime to be modified, if enabled.

0: Deadtime falling sign is writable

1: Deadtime falling sign is read-only

Note: This bit is not preloaded

Bits 29:26 Reserved, must be kept at reset value.

Bit 25 SDTFx: Sign Deadtime Falling value

This register determines whether the deadtime is positive (signals not overlapping) or negative (signals overlapping).

0: Positive deadtime on falling edge

1: Negative deadtime on falling edge

Bits 24:16 DTFx[8:0]: Deadtime Falling value

This register holds the value of the deadtime following a falling edge of reference PWM signal.

$$t_{DTF} = DTFx[8:0] \times t_{DTG}$$

Bit 15 DTRLK_x: Deadtime Rising Lock

This write-once bit prevents the deadtime (sign and value) to be modified, if enabled

0: Deadtime rising value and sign is writable

1: Deadtime rising value and sign is read-only

Note: This bit is not preloaded

Bit 14 DTRSLK_x: Deadtime Rising Sign Lock

This write-once bit prevents the sign of deadtime to be modified, if enabled

0: Deadtime rising sign is writable

1: Deadtime rising sign is read-only

Note: This bit is not preloaded

Bit 13 Reserved, must be kept at reset value.

Bits 12:10 **DTPRSC[2:0]: Deadtime Prescaler**

This register holds the value of the deadtime clock prescaler.

$$t_{DTG} = (2^{(DTPRSC[2:0]-3)}) \times t_{HRTIM}$$

000: Reserved

001: Reserved

010: Reserved

011: $t_{DTG} = t_{HRTIM}$

100: $t_{DTG} = t_{HRTIM} \times 2$

101: $t_{DTG} = t_{HRTIM} \times 4$

110: $t_{DTG} = t_{HRTIM} \times 8$

111: $t_{DTG} = t_{HRTIM} \times 16$

This bitfield is read-only as soon as any of the lock bit is enabled (DTFLKs, DTFSLKx, DTRLKx, DTRSLKx).

Bit 9 **SDTRx: Sign Deadtime Rising value**

This register determines whether the deadtime is positive or negative (overlapping signals)

0: Positive deadtime on rising edge

1: Negative deadtime on rising edge

Bits 8:0 **DTRx[8:0]: Deadtime Rising value**

This register holds the value of the deadtime following a rising edge of reference PWM signal.

$$t_{DTR} = DTRx[8:0] \times t_{DTG}$$

37.5.27 HRTIM Timerx Output1 Set Register (HRTIM_SETx1R)

Address offset: 0x3Ch (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UPDATE	EXT EVNT10	EXT EVNT9	EXT EVNT8	EXT EVNT7	EXT EVNT6	EXT EVNT5	EXT EVNT4	EXT EVNT3	EXT EVNT2	EXT EVNT1	TIM EVNT9	TIM EVNT8	TIM EVNT7	TIM EVNT6	TIM EVNT5
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIM EVNT4	TIM EVNT3	TIM EVNT2	TIM EVNT1	MST CMP4	MST CMP3	MST CMP2	MST CMP1	MST PER	CMP4	CMP3	CMP2	CMP1	PER	RESYNC	SST
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **UPDATE**: Registers update (transfer preload to active)

Register update event forces the output to its active state.

Bit 30 **EXTEVNT10**: External Event 10

Refer to EXTEVNT1 description

Bit 29 **EXTEVNT9**: External Event 9

Refer to EXTEVNT1 description

Bit 28 **EXTEVNT8**: External Event 8

Refer to EXTEVNT1 description

Bit 27 **EXTEVNT7**: External Event 7

Refer to EXTEVNT1 description

Bit 26 **EXTEVNT6**: External Event 6

Refer to EXTEVNT1 description

Bit 25 **EXTEVNT5**: External Event 5

Refer to EXTEVNT1 description

Bit 24 **EXTEVNT4**: External Event 4

Refer to EXTEVNT1 description

Bit 23 **EXTEVNT3**: External Event 3

Refer to EXTEVNT1 description

Bit 22 **EXTEVNT2**: External Event 2

Refer to EXTEVNT1 description

Bit 21 **EXTEVNT1**: External Event 1

External event 1 forces the output to its active state.

Bit 20 **TIMEVNT9**: Timer Event 9

Refer to TIMEVNT1 description

Bit 19 **TIMEVNT8**: Timer Event 8

Refer to TIMEVNT1 description

Bit 18 **TIMEVNT7**: Timer Event 7

Refer to TIMEVNT1 description

Bit 17 **TIMEVNT6**: Timer Event 6

Refer to TIMEVNT1 description

- Bit 16 **TIMEVNT5**: *Timer Event 5*
Refer to TIMEVNT1 description
- Bit 15 **TIMEVNT4**: *Timer Event 4*
Refer to TIMEVNT1 description
- Bit 14 **TIMEVNT3**: *Timer Event 3*
Refer to TIMEVNT1 description
- Bit 13 **TIMEVNT2**: *Timer Event 2*
Refer to TIMEVNT1 description
- Bit 12 **TIMEVNT1**: *Timer Event 1*
Timers event 1 forces the output to its active state (refer to [Table 303](#) for Timer Events assignments)
- Bit 11 **MSTCMP4**: *Master Compare 4*
Master Timer Compare 4 event forces the output to its active state.
- Bit 10 **MSTCMP3**: *Master Compare 3*
Master Timer Compare 3 event forces the output to its active state.
- Bit 9 **MSTCMP2**: *Master Compare 2*
Master Timer Compare 2 event forces the output to its active state.
- Bit 8 **MSTCMP1**: *Master Compare 1*
Master Timer compare 1 event forces the output to its active state.
- Bit 7 **MSTPER**: *Master Period*
The master timer counter roll-over in continuous mode, or to the master timer reset in single-shot mode forces the output to its active state.
- Bit 6 **CMP4**: *Timer x Compare 4*
Timer A compare 4 event forces the output to its active state.
- Bit 5 **CMP3**: *Timer x Compare 3*
Timer A compare 3 event forces the output to its active state.
- Bit 4 **CMP2**: *Timer x Compare 2*
Timer A compare 2 event forces the output to its active state.
- Bit 3 **CMP1**: *Timer x Compare 1*
Timer A compare 1 event forces the output to its active state.
- Bit 2 **PER**: *Timer x Period*
Timer A Period event forces the output to its active state.
- Bit 1 **RESYNC**: *Timer A resynchronization*
Timer A reset event coming solely from software or SYNC input forces the output to its active state.
Note: Other timer reset are not affecting the output when RESYNC=1
- Bit 0 **SST**: *Software Set trigger*
This bit forces the output to its active state. This bit can only be set by software and is reset by hardware.
Note: This bit is not preloaded

37.5.28 HRTIM Timerx Output1 Reset Register (HRTIM_RSTx1R)

Address offset: 0x40h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UPDAT E	EXT EVNT1 0	EXT EVNT9	EXT EVNT8	EXT EVNT7	EXT EVNT6	EXT EVNT5	EXT EVNT4	EXT EVNT3	EXT EVNT2	EXT EVNT1	TIM EVNT9	TIM EVNT8	TIM EVNT7	TIM EVNT6	TIM EVNT5
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIM EVNT4	TIM EVNT3	TIM EVNT2	TIM EVNT1	MST CMP4	MST CMP3	MST CMP2	MST CMP1	MST PER	CMP4	CMP3	CMP2	CMP1	PER	RESYN C	SRT
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 Refer to HRTIM_SETx1R bits description.

These bits are defining the source which can force the Tx1 output to its inactive state.

37.5.29 HRTIM Timerx Output2 Set Register (HRTIM_SETx2R)

Address offset: 0x44h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UPDAT E	EXT EVNT1 0	EXT EVNT9	EXT EVNT8	EXT EVNT7	EXT EVNT6	EXT EVNT5	EXT EVNT4	EXT EVNT3	EXT EVNT2	EXT EVNT1	TIM EVNT9	TIM EVNT8	TIM EVNT7	TIM EVNT6	TIM EVNT5
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIM EVNT4	TIM EVNT3	TIM EVNT2	TIM EVNT1	MST CMP4	MST CMP3	MST CMP2	MST CMP1	MST PER	CMP4	CMP3	CMP2	CMP1	PER	RESYN C	SST
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 Refer to HRTIM_SETx1R bits description.

These bits are defining the source which can force the Tx2 output to its active state.

37.5.30 HRTIM Timerx Output2 Reset Register (HRTIM_RSTx2R)

Address offset: 0x48h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UPDAT E	EXT EVNT1 0	EXT EVNT9	EXT EVNT8	EXT EVNT7	EXT EVNT6	EXT EVNT5	EXT EVNT4	EXT EVNT3	EXT EVNT2	EXT EVNT1	TIM EVNT9	TIM EVNT8	TIM EVNT7	TIM EVNT6	TIM EVNT5
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIM EVNT4	TIM EVNT3	TIM EVNT2	TIM EVNT1	MST CMP4	MST CMP3	MST CMP2	MST CMP1	MST PER	CMP4	CMP3	CMP2	CMP1	PER	RESYN C	SRT
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 Refer to HRTIM_SETx1R bits description.

These bits are defining the source which can force the Tx2 output to its inactive state.

37.5.31 HRTIM Timerx External Event Filtering Register 1 (HRTIM_EEFxR1)

Address offset: 0x4Ch (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	EE5FLTR[3:0]				EE5LTCH	Res.	EE4FLTR[3:0]				EE4LTCH	Res.	EE3FLTR[3]
			rw	rw	rw	rw	rw		rw	rw	rw	rw	rw		rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EE3FLTR[2:0]			EE3LTCH	Res.	EE2FLTR[3:0]				EE2LTCH	Res.	EE1FLTR[3:0]				EE1LTCH
rw	rw	rw	rw		rw	rw	rw	rw	rw		rw	rw	rw	rw	rw

Bits 31:29 Reserved, must be kept at reset value.

Bits 28:25 **EE5FLTR[3:0]**: *External Event 5 filter*
Refer to EE1FLTR[3:0] description

Bit 24 **EE5LTCH**: *External Event 5 latch*
Refer to EE1LTCH description

Bit 23 Reserved, must be kept at reset value.

Bits 22:19 **EE4FLTR[3:0]**: *External Event 4 filter*
Refer to EE1FLTR[3:0] description

Bit 18 **EE4LTCH**: *External Event 4 latch*
Refer to EE1LTCH description

Bit 17 Reserved, must be kept at reset value.

Bits 16:13 **EE3FLTR[3:0]**: *External Event 3 filter*
Refer to EE1FLTR[3:0] description

Bit 12 **EE3LTCH**: *External Event 3 latch*
Refer to EE1LTCH description

Bit 11 Reserved, must be kept at reset value.

Bits 10:7 **EE2FLTR[3:0]**: *External Event 2 filter*
Refer to EE1FLTR[3:0] description

Bit 6 **EE2LTCH**: *External Event 2 latch*
Refer to EE1LTCH description

Bit 5 Reserved, must be kept at reset value.

Bits 4:1 **EE1FLTR[3:0]**: *External Event 1 filter*

- 0000: No filtering
- 0001: Blanking from counter reset/roll-over to Compare 1
- 0010: Blanking from counter reset/roll-over to Compare 2
- 0011: Blanking from counter reset/roll-over to Compare 3
- 0100: Blanking from counter reset/roll-over to Compare 4
- 0101: Blanking from another timing unit: TIMFLTR1 source (see [Table 307](#) for details)
- 0110: Blanking from another timing unit: TIMFLTR2 source (see [Table 307](#) for details)
- 0111: Blanking from another timing unit: TIMFLTR3 source (see [Table 307](#) for details)
- 1000: Blanking from another timing unit: TIMFLTR4 source (see [Table 307](#) for details)
- 1001: Blanking from another timing unit: TIMFLTR5 source (see [Table 307](#) for details)
- 1010: Blanking from another timing unit: TIMFLTR6 source (see [Table 307](#) for details)
- 1011: Blanking from another timing unit: TIMFLTR7 source (see [Table 307](#) for details)
- 1100: Blanking from another timing unit: TIMFLTR8 source (see [Table 307](#) for details)
- 1101: Windowing from counter reset/roll-over to Compare 2
- 1110: Windowing from counter reset/roll-over to Compare 3
- 1111: Windowing from another timing unit: TIMWIN source (see [Table 308](#) for details)

Note: Whenever a compare register is used for filtering, the value must be strictly above 0.

This bitfield must not be modified once the counter is enabled (TxCEN bit set)

Bit 0 **EE1LTCH**: *External Event 1 latch*

- 0: Event 1 is ignored if it happens during a blank, or passed through during a window.
- 1: Event 1 is latched and delayed till the end of the blanking or windowing period.

Note: A timeout event is generated in window mode (EE1FLTR[3:0]=1101, 1110, 1111) if EE1LTCH = 0, except if the External event is programmed in fast mode (EEFAST = 1).

This bitfield must not be modified once the counter is enabled (TxCEN bit set)

37.5.32 HRTIM Timerx External Event Filtering Register 2 (HRTIM_EEFxR2)

Address offset: 0x50h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	EE10FLTR[3:0]				EE10LTCH	Res.	EE9FLTR[3:0]				EE9LTCH	Res.	EE8FLTR[3]
			rw	rw	rw	rw	rw		rw	rw	rw	rw	rw		rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EE8FLTR[2:0]			EE8LTCH	Res.	EE7FLTR[3:0]				EE7LTCH	Res.	EE6FLTR[3:0]				EE6LTCH
rw	rw	rw	rw		rw	rw	rw	rw	rw		rw	rw	rw	rw	rw

Bits 31:29 Reserved, must be kept at reset value.

Bits 28:25 **EE10FLTR[3:0]**: *External Event 10 filter*
Refer to EE1FLTR[3:0] description

Bit 24 **EE10LTCH**: *External Event 10 latch*
Refer to EE1LTCH description

Bit 23 Reserved, must be kept at reset value.

Bits 22:19 **EE9FLTR[3:0]**: *External Event 9 filter*
Refer to EE1FLTR[3:0] description

Bit 18 **EE9LTCH**: *External Event 9 latch*
Refer to EE1LTCH description

Bit 17 Reserved, must be kept at reset value.

Bits 16:13 **EE8FLTR[3:0]**: *External Event 8 filter*
Refer to EE1FLTR[3:0] description

Bit 12 **EE8LTCH**: *External Event 8 latch*
Refer to EE1LTCH description

Bit 11 Reserved, must be kept at reset value.

Bits 10:7 **EE7FLTR[3:0]**: *External Event 7 filter*
Refer to EE1FLTR[3:0] description

Bit 6 **EE7LTCH**: *External Event 7 latch*
Refer to EE1LTCH description

Bit 5 Reserved, must be kept at reset value.

Bits 4:1 **EE6FLTR[3:0]**: *External Event 6 filter*
Refer to EE1FLTR[3:0] description

Bit 0 **EE6LTCH**: *External Event 6 latch*
Refer to EE1LTCH description

37.5.33 HRTIM Timerx Reset Register (HRTIM_RSTxR)**HRTIM TimerA Reset Register (HRTIM_RSTAR)**

Address offset: 0xD4h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	TIME CMP4	TIME CMP2	TIME CMP1	TIMD CMP4	TIMD CMP2	TIMD CMP1	TIMC CMP4	TIMC CMP2	TIMC CMP1	TIMB CMP4	TIMB CMP2	TIMB CMP1	EXTEV NT10	EXTEV NT9	EXTEV NT8
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTEV NT7	EXTEV NT6	EXTEV NT5	EXTEV NT4	EXTEV NT3	EXTEV NT2	EXTEV NT1	MSTC MP4	MSTC MP3	MSTC MP2	MSTC MP1	MSTPE R	CMP4	CMP2	UPDT	Res.
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	

Bit 31 Reserved, must be kept at reset value.

Bit 30 **TECPM4**: *Timer E Compare 4*

The timer A counter is reset upon timer E Compare 4 event.

Bit 29 **TECMP2**: *Timer E Compare 2*

The timer A counter is reset upon timer E Compare 2 event.

Bit 28 **TECMP1**: *Timer E Compare 1*

The timer A counter is reset upon timer E Compare 1 event.

Bit 27 **TDCMP4**: *Timer D Compare 4*

The timer A counter is reset upon timer D Compare 4 event.

Bit 26 **TDCMP2**: *Timer D Compare 2*

The timer A counter is reset upon timer D Compare 2 event.

Bit 25 **TDCMP1**: *Timer D Compare 1*

The timer A counter is reset upon timer D Compare 1 event.

Bit 24 **TCCMP4**: *Timer C Compare 4*

The timer A counter is reset upon timer C Compare 4 event.

Bit 23 **TCCMP2**: *Timer C Compare 2*

The timer A counter is reset upon timer C Compare 2 event.

Bit 22 **TCCMP1**: *Timer C Compare 1*

The timer A counter is reset upon timer C Compare 1 event.

Bit 21 **TBCMP4**: *Timer B Compare 4*

The timer A counter is reset upon timer B Compare 4 event.

Bit 20 **TBCMP2**: *Timer B Compare 2*

The timer A counter is reset upon timer B Compare 2 event.

Bit 19 **TBCMP1**: *Timer B Compare 1*

The timer A counter is reset upon timer B Compare 1 event.

Bit 18 **EXTEVNT10**: *External Event*

The timer A counter is reset upon external event 10.

Bit 17 **EXTEVNT9**: *External Event 9*

The timer A counter is reset upon external event 9.

- Bit 16 **EXTEVNT8**: *External Event 8*
The timer A counter is reset upon external event 8.
- Bit 15 **EXTEVNT7**: *External Event 7*
The timer A counter is reset upon external event 7.
- Bit 14 **EXTEVNT6**: *External Event 6*
The timer A counter is reset upon external event 6.
- Bit 13 **EXTEVNT5**: *External Event 5*
The timer A counter is reset upon external event 5.
- Bit 12 **EXTEVNT4**: *External Event 4*
The timer A counter is reset upon external event 4.
- Bit 11 **EXTEVNT3**: *External Event 3*
The timer A counter is reset upon external event 3.
- Bit 10 **EXTEVNT2**: *External Event 2*
The timer A counter is reset upon external event 2.
- Bit 9 **EXTEVNT1**: *External Event 1*
The timer A counter is reset upon external event 1.
- Bit 8 **MSTCMP4**: *Master compare 4*
The timer A counter is reset upon master timer Compare 4 event.
- Bit 7 **MSTCMP3**: *Master compare 3*
The timer A counter is reset upon master timer Compare 3 event.
- Bit 6 **MSTCMP2**: *Master compare 2*
The timer A counter is reset upon master timer Compare 2 event.
- Bit 5 **MSTCMP1**: *Master compare 1*
The timer A counter is reset upon master timer Compare 1 event.
- Bit 4 **MSTPER**: *Master timer Period*
The timer A counter is reset upon master timer period event.
- Bit 3 **CMP4**: *Timer A compare 4 reset*
The timer A counter is reset upon Timer A Compare 4 event.
- Bit 2 **CMP2**: *Timer A compare 2 reset*
The timer A counter is reset upon Timer A Compare 2 event.
- Bit 1 **UPDT**: *Timer A Update reset*
The timer A counter is reset upon update event.
- Bit 0 Reserved, must be kept at reset value.

HRTIM TimerB Reset Register (HRTIM_RSTBR)

Address offset: 0x154h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	TIME CMP4	TIME CMP2	TIME CMP1	TIMD CMP4	TIMD CMP2	TIMD CMP1	TIMC CMP4	TIMC CMP2	TIMC CMP1	TIMA CMP4	TIMA CMP2	TIMA CMP1	EXTEV NT10	EXTEV NT9	EXTEV NT8
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTEV NT7	EXTEV NT6	EXTEV NT5	EXTEV NT4	EXTEV NT3	EXTEV NT2	EXTEV NT1	MSTC MP4	MSTC MP3	MSTC MP2	MSTC MP1	MSTPE R	CMP4	CMP2	UPDT	Res.
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	

Bits 30:1 Refer to HRTIM_RSTAR bits description.

Bits 30:19 differ (reset signals come from TIMA, TIMC, TIMD and TIME)

HRTIM TimerC Reset Register (HRTIM_RSTCR)

Address offset: 0x1D4h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	TIME CMP4	TIME CMP2	TIME CMP1	TIMD CMP4	TIMD CMP2	TIMD CMP1	TIMB CMP4	TIMB CMP2	TIMB CMP1	TIMA CMP4	TIMA CMP2	TIMA CMP1	EXTEV NT10	EXTEV NT9	EXTEV NT8
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTEV NT7	EXTEV NT6	EXTEV NT5	EXTEV NT4	EXTEV NT3	EXTEV NT2	EXTEV NT1	MSTC MP4	MSTC MP3	MSTC MP2	MSTC MP1	MSTPE R	CMP4	CMP2	UPDT	Res.
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	

Bits 30:1 Refer to HRTIM_RSTAR bits description.

Bits 30:19 differ (reset signals come from TIMA, TIMB, TIMD and TIME)

HRTIM TimerD Reset Register (HRTIM_RSTDR)

Address offset: 0x254h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	TIME CMP4	TIME CMP2	TIME CMP1	TIMC CMP4	TIMC CMP2	TIMC CMP1	TIMB CMP4	TIMB CMP2	TIMB CMP1	TIMA CMP4	TIMA CMP2	TIMA CMP1	EXTEV NT10	EXTEV NT9	EXTEV NT8
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTEV NT7	EXTEV NT6	EXTEV NT5	EXTEV NT4	EXTEV NT3	EXTEV NT2	EXTEV NT1	MSTC MP4	MSTC MP3	MSTC MP2	MSTC MP1	MSTPE R	CMP4	CMP2	UPDT	Res.
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	

Bits 30:1 Refer to HRTIM_RSTAR bits description.

Bits 30:19 differ (reset signals come from TIMA, TIMB, TIMC and TIME)

HRTIM Timerx Reset Register (HRTIM_RSTER)

Address offset: 0x2D4h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	TIMD CMP4	TIMD CMP2	TIMD CMP1	TIMC CMP4	TIMC CMP2	TIMC CMP1	TIMB CMP4	TIMB CMP2	TIMB CMP1	TIMA CMP4	TIMA CMP2	TIMA CMP1	EXTEV NT10	EXTEV NT9	EXTEV NT8
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTEV NT7	EXTEV NT6	EXTEV NT5	EXTEV NT4	EXTEV NT3	EXTEV NT2	EXTEV NT1	MSTC MP4	MSTC MP3	MSTC MP2	MSTC MP1	MSTPE R	CMP4	CMP2	UPDT	Res.
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	

Bits 30:1 Refer to HRTIM_RSTAR bits description.

Bits 30:19 differ (reset signals come from TIMA, TIMB, TIMC and TIMD)

37.5.34 HRTIM Timerx Chopper Register (HRTIM_CHPxR)

Address offset: 0x58h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	STRTPW[3:0]				CARDTY[2:0)			CARFRQ[3:0]			
					rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:11 Reserved, must be kept at reset value.

Bits 10:7 **STRPW[3:0]**: *Timerx start pulsewidth*

This register defines the initial pulsewidth following a rising edge on output signal.

This bitfield cannot be modified when one of the CHPx bits is set.

$$t_{1STPW} = (STRPW[3:0] + 1) \times 16 \times t_{HRTIM}$$

0000: 40 ns (1/25 MHz)

...

1111: 640 ns (16/25 MHz)

Bits 6:4 **CARDTY[2:0]**: *Timerx chopper duty cycle value*

This register defines the duty cycle of the carrier signal. This bitfield cannot be modified when one of the CHPx bits is set.

000: 0/8 (i.e. only 1st pulse is present)

...

111: 7/8

Bits 3:0 **CARFRQ[3:0]**: *Timerx carrier frequency value*

This register defines the carrier frequency $F_{CHPFRQ} = f_{HRTIM} / (16 \times (CARFRQ[3:0] + 1))$.

This bitfield cannot be modified when one of the CHPx bits is set.

0000: 25 MHz ($f_{HRTIM} / 16$)

...

1111: 1.56 MHz ($f_{HRTIM} / 256$)

37.5.35 HRTIM Timerx Capture 1 Control Register (HRTIM_CPT1xCR)

Address offset: 0x5Ch (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved (for TIME only)				Reserved (for TIMD only)				Reserved (for TIMC only)				Reserved (for TIMB only)			
TECMP 2	TECMP 1	TE1RS T	TE1SE T	TDCM P2	TDCM P1	TD1RS T	TD1SE T	TCCM P2	TCCM P1	TC1RS T	TC1SE T	TBCMP 2	TBCMP 1	TB1RS T	TB1SE T
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved (for TIMA only)				EXEV1 0CPT	EXEV9 CPT	EXEV8 CPT	EXEV7 CPT	EXEV6 CPT	EXEV5 CPT	EXEV4 CPT	EXEV3 CPT	EXEV2 CPT	EXEV1 CPT	UPDCP T	SWCP T
TACMP 2	TACMP 1	TA1RS T	TA1SE T	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 Refer to HRTIM_CPT2xCR bit description

37.5.36 HRTIM Timerx Capture 2 Control Register (HRTIM_CPT2xCR)

Address offset: 0x60h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved (for TIME only)				Reserved (for TIMD only)				Reserved (for TIMC only)				Reserved (for TIMB only)			
TECMP 2	TECMP 1	TE1RS T	TE1SE T	TDCM P2	TDCM P1	TD1RS T	TD1SE T	TCCM P2	TCCM P1	TC1RS T	TC1SE T	TBCMP 2	TBCMP 1	TB1RS T	TB1SE T
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved (for TIMA only)				EXEV1 0CPT	EXEV9 CPT	EXEV8 CPT	EXEV7 CPT	EXEV6 CPT	EXEV5 CPT	EXEV4 CPT	EXEV3 CPT	EXEV2 CPT	EXEV1 CPT	UPDCP T	SWCP T
TACMP 2	TACMP 1	TA1RS T	TA1SE T	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

- Bit 31 **TECMP2**: Timer E Compare 2
Refer to TACMP1 description
Note: This bit is reserved for Timer E
- Bit 30 **TECMP1**: Timer E Compare 1
Refer to TACMP1 description
Note: This bit is reserved for Timer E
- Bit 29 **TE1RST**: Timer E output 1 Reset
Refer to TA1RST description
Note: This bit is reserved for Timer E
- Bit 28 **TE1SET**: Timer E output 1 Set
Refer to TA1SET description
Note: This bit is reserved for Timer E
- Bit 27 **TDCMP2**: Timer D Compare 2
Refer to TACMP1 description
Note: This bit is reserved for Timer D
- Bit 26 **TDCMP1**: Timer D Compare 1
Refer to TACMP1 description
Note: This bit is reserved for Timer D
- Bit 25 **TD1RST**: Timer D output 1 Reset
Refer to TA1RST description
Note: This bit is reserved for Timer D
- Bit 24 **TD1SET**: Timer D output 1 Set
Refer to TA1SET description
Note: This bit is reserved for Timer D
- Bit 23 **TCCMP2**: Timer C Compare 2
Refer to TACMP1 description
Note: This bit is reserved for Timer C

- Bit 22 **TCCMP1**: Timer C Compare 1
Refer to TACMP1 description
Note: This bit is reserved for Timer C
- Bit 21 **TC1RST**: Timer C output 1 Reset
Refer to TA1RST description
Note: This bit is reserved for Timer C
- Bit 20 **TC1SET**: Timer C output 1 Set
Refer to TA1SET description
Note: This bit is reserved for Timer C
- Bit 19 **TBCMP2**: Timer B Compare 2
Refer to TACMP1 description
Note: This bit is reserved for Timer B
- Bit 18 **TBCMP1**: Timer B Compare 1
Refer to TACMP1 description
Note: This bit is reserved for Timer B
- Bit 17 **TB1RST**: Timer B output 1 Reset
Refer to TA1RST description
Note: This bit is reserved for Timer B
- Bit 16 **TB1SET**: Timer B output 1 Set
Refer to TA1SET description
Note: This bit is reserved for Timer B
- Bit 15 **TACMP2**: Timer A Compare 2
Timer A Compare 2 triggers Capture 2.
Note: This bit is reserved for Timer A
- Bit 14 **TACMP1**: Timer A Compare 1
Timer A Compare 1 triggers Capture 2.
Note: This bit is reserved for Timer A
- Bit 13 **TA1RST**: Timer B output 1 Reset
Capture 2 is triggered by HRTIM_CHA1 output active to inactive transition.
Note: This bit is reserved for Timer A
- Bit 12 **TA1SET**: Timer B output 1 Set
Capture 2 is triggered by HRTIM_CHA1 output inactive to active transition.
Note: This bit is reserved for Timer A
- Bit 11 **EXEV10CPT**: External Event 10 Capture
Refer to EXEV1CPT description
- Bit 10 **EXEV9CPT**: External Event 9 Capture
Refer to EXEV1CPT description
- Bit 9 **EXEV8CPT**: External Event 8 Capture
Refer to EXEV1CPT description
- Bit 8 **EXEV7CPT**: External Event 7 Capture
Refer to EXEV1CPT description
- Bit 7 **EXEV6CPT**: External Event 6 Capture
Refer to EXEV1CPT description

- Bit 6 **EXEV5CPT**: *External Event 5 Capture*
Refer to EXEV1CPT description
- Bit 5 **EXEV4CPT**: *External Event 4 Capture*
Refer to EXEV1CPT description
- Bit 4 **EXEV3CPT**: *External Event 3 Capture*
Refer to EXEV1CPT description
- Bit 3 **EXEV2CPT**: *External Event 2 Capture*
Refer to EXEV1CPT description
- Bit 2 **EXEV1CPT**: *External Event 1 Capture*
The External event 1 triggers the Capture 2.
- Bit 1 **UPDCPT**: *Update Capture*
The update event triggers the Capture 2.
- Bit 0 **SWCPT**: *Software Capture*
This bit forces the Capture 2 by software. This bit is set only, reset by hardware

37.5.37 HRTIM Timerx Output Register (HRTIM_OUTxR)

Address offset: 0x64h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DIDL2	CHP2	FAULT2[1:0]		IDLES2	IDLEM2	POL2	Res.
								rw	rw	rw	rw	rw	rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	DLYPRT[2:0]			DLYPR TEN	DTEN	DIDL1	CHP1	FAULT1[1:0]		IDLES1	IDLEM1	POL1	Res.
			rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 DIDL2: Output 2 Deadtime upon burst mode Idle entry

This bit can delay the idle mode entry by forcing a deadtime insertion before switching the outputs to their idle state. This setting only applies when entering in idle state during a burst mode operation.

0: The programmed Idle state is applied immediately to the Output 2

1: Deadtime (inactive level) is inserted on output 2 before entering the idle mode. The deadtime value is set by DTFx[8:0].

Note: This parameter cannot be changed once the timer x is enabled.

DIDL=1 can be set only if one of the outputs is active during the burst mode (IDLES=1), and with positive deadtimes (SDTR/SDTF set to 0).

Bit 22 CHP2: Output 2 Chopper enable

This bit enables the chopper on output 2

0: Output signal is not altered

1: Output signal is chopped by a carrier signal

Note: This parameter cannot be changed once the timer x is enabled.

Bits 21:20 FAULT2[1:0]: Output 2 Fault state

These bits select the output 2 state after a fault event

00: No action: the output is not affected by the fault input and stays in run mode.

01: Active

10: Inactive

11: High-Z

Note: This parameter cannot be changed once the timer x is enabled (TxCEN bit set), if FLTENx bit is set or if the output is in FAULT state.

Bit 19 IDLES2: Output 2 Idle State

This bit selects the output 2 idle state

0: Inactive

1: Active

Note: This parameter must be set prior to have the HRTIM controlling the outputs.

Bit 18 IDLEM2: Output 2 Idle mode

This bit selects the output 2 idle mode

0: No action: the output is not affected by the burst mode operation

1: The output is in idle state when requested by the burst mode controller.

Note: This bit is preloaded and can be changed during run-time, but must not be changed while the burst mode is active.

Bit 17 POL2: Output 2 polarity

This bit selects the output 2 polarity
 0: positive polarity (output active high)
 1: negative polarity (output active low)

Note: This parameter cannot be changed once the timer x is enabled.

Bits 16:12 Reserved, must be kept at reset value.

Bits 12:10 DLYPRT[2:0]: Delayed Protection

These bits define the source and outputs on which the delayed protection schemes are applied.

In HRTIM_OUTAR, HRTIM_OUTBR, HRTIM_OUTCR:

000: Output 1 delayed Idle on external Event 6
 001: Output 2 delayed Idle on external Event 6
 010: Output 1 and output 2 delayed Idle on external Event 6
 011: Balanced Idle on external Event 6
 100: Output 1 delayed Idle on external Event 7
 101: Output 2 delayed Idle on external Event 7
 110: Output 1 and output 2 delayed Idle on external Event 7
 111: Balanced Idle on external Event 7

In HRTIM_OUTDR, HRTIM_OUTER:

000: Output 1 delayed Idle on external Event 8
 001: Output 2 delayed Idle on external Event 8
 010: Output 1 and output 2 delayed Idle on external Event 8
 011: Balanced Idle on external Event 8
 100: Output 1 delayed Idle on external Event 9
 101: Output 2 delayed Idle on external Event 9
 110: Output 1 and output 2 delayed Idle on external Event 9
 111: Balanced Idle on external Event 9

Note: This bitfield must not be modified once the delayed protection is enabled (DLYPRTEN bit set)

Bit 9 DLYPRTEN: Delayed Protection Enable

This bit enables the delayed protection scheme
 0: No action
 1: Delayed protection is enabled, as per DLYPRT[2:0] bits

Note: This parameter cannot be changed once the timer x is enabled (TxEN bit set).

Bit 8 DTEN: Deadtime enable

This bit enables the deadtime insertion on output 1 and output 2
 0: Output 1 and output 2 signals are independent.
 1: Deadtime is inserted between output 1 and output 2 (reference signal is output 1 signal generator)

Note: This parameter cannot be changed once the timer is operating (TxEN bit set) or if its outputs are enabled and set/reset by another timer.

Bit 7 DIDL1: Output 1 Deadtime upon burst mode Idle entry

This bit can delay the idle mode entry by forcing a deadtime insertion before switching the outputs to their idle state. This setting only applies when entering the idle state during a burst mode operation.

0: The programmed Idle state is applied immediately to the Output 1
 1: Deadtime (inactive level) is inserted on output 1 before entering the idle mode. The deadtime value is set by DTRx[8:0].

Note: This parameter cannot be changed once the timer x is enabled.

DIDL=1 can be set only if one of the outputs is active during the burst mode (IDLES=1), and with positive deadtimes (SDTR/SDTF set to 0).

Bit 6 **CHP1**: Output 1 Chopper enable

This bit enables the chopper on output 1

0: Output signal is not altered

1: Output signal is chopped by a carrier signal

Note: This parameter cannot be changed once the timer x is enabled.

Bits 5:4 **FAULT1[1:0]**: Output 1 Fault state

These bits select the output 1 state after a fault event

00: No action: the output is not affected by the fault input and stays in run mode.

01: Active

10: Inactive

11: High-Z

Note: This parameter cannot be changed once the timer x is enabled (TxCEN bit set), if FLTENx bit is set or if the output is in FAULT state.

Bit 3 **IDLES1**: Output 1 Idle State

This bit selects the output 1 idle state

0: Inactive

1: Active

Note: This parameter must be set prior to HRTIM controlling the outputs.

Bit 2 **IDLEM1**: Output 1 Idle mode

This bit selects the output 1 idle mode

0: No action: the output is not affected by the burst mode operation

1: The output is in idle state when requested by the burst mode controller.

Note: This bit is preloaded and can be changed during runtime, but must not be changed while burst mode is active.

Bit 1 **POL1**: Output 1 polarity

This bit selects the output 1 polarity

0: positive polarity (output active high)

1: negative polarity (output active low)

Note: This parameter cannot be changed once the timer x is enabled.

Bit 0 Reserved

37.5.38 HRTIM Timerx Fault Register (HRTIM_FLTxR)

Address offset: 0x68h (this offset address is relative to timer x base address)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FTLCK K	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rwo															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FLT5E N	FLT4E N	FLT3E N	FLT2E N	FLT1E N
											rw	rw	rw	rw	rw

Bit 31 **FTLCK**: *Fault sources Lock*

0: FLT1EN..FLT5EN bits are read/write

1: FLT1EN..FLT5EN bits are read only

The FTLCK bit is write-once. Once it has been set, it cannot be modified till the next system reset.

Bits 30:5 Reserved, must be kept at reset value.

Bit 4 **FLT5EN**: *Fault 5 enable*

0: Fault 5 input ignored

1: Fault 5 input is active and can disable HRTIM outputs.

Bit 3 **FLT4EN**: *Fault 4 enable*

0: Fault 4 input ignored

1: Fault 4 input is active and can disable HRTIM outputs.

Bit 2 **FLT3EN**: *Fault 3 enable*

0: Fault 3 input ignored

1: Fault 3 input is active and can disable HRTIM outputs.

Bit 1 **FLT2EN**: *Fault 2 enable*

0: Fault 2 input ignored

1: Fault 2 input is active and can disable HRTIM outputs.

Bit 0 **FLT1EN**: *Fault 1 enable*

0: Fault 1 input ignored

1: Fault 1 input is active and can disable HRTIM outputs.

37.5.39 HRTIM Control Register 1 (HRTIM_CR1)

Address offset: 0x380h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	AD4USRC[2:0]			AD3USRC[2:0]			AD2USRC[2:0]			AD1USRC[2:0]		
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TEUDIS	TDUDIS	TCUDIS	TBUDIS	TAUDIS	MUDIS
										rw	rw	rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:25 **AD4USRC[2:0]**: *ADC Trigger 4 Update Source*
Refer to AD1USRC[2:0] description

Bits 24:22 **AD3USRC[2:0]**: *ADC Trigger 3 Update Source*
Refer to AD1USRC[2:0] description

Bits 21:19 **AD2USRC[2:0]**: *ADC Trigger 2 Update Source*
Refer to AD1USRC[2:0] description

Bits 18:16 **AD1USRC[2:0]**: *ADC Trigger 1 Update Source*

These bits define the source which will trigger the update of the HRTIM_ADC1R register (transfer from preload to active register). It only defines the source timer. The precise condition is defined within the timer itself, in HRTIM_MCR or HRTIM_TIMxCR.

000: Master Timer

001: Timer A

010: Timer B

011: Timer C

100: Timer D

101: Timer E

110, 111: Reserved

Bits 15:6 Reserved, must be kept at reset value.

Bit 5 **TEUDIS**: *Timer E Update Disable*
Refer to TAUDIS description

Bit 4 **TDUDIS**: *Timer D Update Disable*
Refer to TAUDIS description

Bit 3 **TCUDIS**: *Timer C Update Disable*
Refer to TAUDIS description

Bit 2 **TBUDIS**: *Timer B Update Disable*

Refer to TAUDIS description

Bit 1 **TAUDIS**: *Timer A Update Disable*

This bit is set and cleared by software to enable/disable an update event generation temporarily on Timer A.

0: update enabled. The update occurs upon generation of the selected source.

1: update disabled. The updates are temporarily disabled to allow the software to write multiple registers that have to be simultaneously taken into account.

Bit 0 **MUDIS**: *Master Update Disable*

This bit is set and cleared by software to enable/disable an update event generation temporarily.

0: update enabled.

1: update disabled. The updates are temporarily disabled to allow the software to write multiple registers that have to be simultaneously taken into account.

37.5.40 HRTIM Control Register 2 (HRTIM_CR2)

Address offset: 0x384h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	TERST	TDRST	TCRST	TBRST	TARST	MRST	Res.	Res.	TESW U	TDSW U	TCSW U	TBSW U	TASWU	MSWU
		rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	rw	rw

Bits 31:14 Reserved, must be kept at reset value.

Bit 13 **TERST**: *Timer E counter software reset*
Refer to TARST description

Bit 12 **TDRST**: *Timer D counter software reset*
Refer to TARST description

Bit 11 **TCRST**: *Timer C counter software reset*
Refer to TARST description

Bit 10 **TBRST**: *Timer B counter software reset*
Refer to TARST description

Bit 9 **TARST**: *Timer A counter software reset*
Setting this bit resets the TimerA counter.
The bit is automatically reset by hardware.

Bit 8 **MRST**: *Master Counter software reset*
Setting this bit resets the Master timer counter.
The bit is automatically reset by hardware.

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 **TESWU**: *Timer E Software Update*
Refer to TASWU description

Bit 4 **TDSWU**: *Timer D Software Update*
Refer to TASWU description

Bit 3 **TCSWU**: *Timer C Software Update*
Refer to TASWU description

Bit 2 **TBSWU**: *Timer B Software Update*
Refer to TASWU description

Bit 1 **TASWU**: *Timer A Software update*
This bit is set by software and automatically reset by hardware. It forces an immediate transfer from the preload to the active register and any pending update request is cancelled.

Bit 0 **MSWU**: *Master Timer Software update*
This bit is set by software and automatically reset by hardware. It forces an immediate transfer from the preload to the active register in the master timer and any pending update request is cancelled.

37.5.41 HRTIM Interrupt Status Register (HRTIM_ISR)

Address offset: 0x388h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BMPER	Res.
														r	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SYSFLT	FLT5	FLT4	FLT3	FLT2	FLT1
											r	r	r	r	r

Bits 31:18 Reserved, must be kept at reset value.

Bit 17 **BMPER**: Burst mode Period Interrupt Flag

This bit is set by hardware when a single-shot burst mode operation is completed or at the end of a burst mode period in continuous mode. It is cleared by software writing it at 1.

0: No Burst mode period interrupt occurred

1: Burst mode period interrupt occurred

Bits 16:6 Reserved, must be kept at reset value.

Bit 5 **SYSFLT**: System Fault Interrupt Flag

Refer to FLT1 description

Bit 4 **FLT5**: Fault 5 Interrupt Flag

Refer to FLT1 description

Bit 3 **FLT4**: Fault 4 Interrupt Flag

Refer to FLT1 description

Bit 2 **FLT3**: Fault 3 Interrupt Flag

Refer to FLT1 description

Bit 1 **FLT2**: Fault 2 Interrupt Flag

Refer to FLT1 description

Bit 0 **FLT1**: Fault 1 Interrupt Flag

This bit is set by hardware when Fault 1 event occurs. It is cleared by software writing it at 1.

0: No Fault 1 interrupt occurred

1: Fault 1 interrupt occurred

37.5.42 HRTIM Interrupt Clear Register (HRTIM_ICR)

Address offset: 0x38Ch

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BMPERC	Res.
														w	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SYSFLTC	FLT5C	FLT4C	FLT3C	FLT2C	FLT1C
											w	w	w	w	w

Bits 31:18 Reserved, must be kept at reset value.

Bit 17 **BMPERC**: Burst mode period flag Clear

Writing 1 to this bit clears the BMPER flag in HRTIM_ISR register.

Bits 16:6 Reserved, must be kept at reset value.

Bit 5 **SYSFLTC**: System Fault Interrupt Flag Clear

Writing 1 to this bit clears the SYSFLT flag in HRTIM_ISR register.

Bit 4 **FLT5C**: Fault 5 Interrupt Flag Clear

Writing 1 to this bit clears the FLT5 flag in HRTIM_ISR register.

Bit 3 **FLT4C**: Fault 4 Interrupt Flag Clear

Writing 1 to this bit clears the FLT4 flag in HRTIM_ISR register.

Bit 2 **FLT3C**: Fault 3 Interrupt Flag Clear

Writing 1 to this bit clears the FLT3 flag in HRTIM_ISR register.

Bit 1 **FLT2C**: Fault 2 Interrupt Flag Clear

Writing 1 to this bit clears the FLT2 flag in HRTIM_ISR register.

Bit 0 **FLT1C**: Fault 1 Interrupt Flag Clear

Writing 1 to this bit clears the FLT1 flag in HRTIM_ISR register.

37.5.43 HRTIM Interrupt Enable Register (HRTIM_IER)

Address offset: 0x390h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BMPERIE	Res.
														rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SYSFLTIE	FLT5IE	FLT4IE	FLT3IE	FLT2IE	FLT1IE
											rw	rw	rw	rw	rw

Bits 31:18 Reserved, must be kept at reset value.

Bit 17 **BMPERIE**: Burst mode period Interrupt Enable

This bit is set and cleared by software to enable/disable the Burst mode period interrupt.

0: Burst mode period interrupt disabled

1: Burst mode period interrupt enabled

Bits 16:6 Reserved, must be kept at reset value.

Bit 5 **SYSFLTIE**: System Fault Interrupt Enable

Refer to FLT1IE description

Bit 4 **FLT5IE**: Fault 5 Interrupt Enable

Refer to FLT1IE description

Bit 3 **FLT4IE**: Fault 4 Interrupt Enable

Refer to FLT1IE description

Bit 2 **FLT3IE**: Fault 3 Interrupt Enable

Refer to FLT1IE description

Bit 1 **FLT2IE**: Fault 2 Interrupt Enable

Refer to FLT1IE description

Bit 0 **FLT1IE**: Fault 1 Interrupt Enable

This bit is set and cleared by software to enable/disable the Fault 1 interrupt.

0: Fault 1 interrupt disabled

1: Fault 1 interrupt enabled

37.5.44 HRTIM Output Enable Register (HRTIM_OENR)

Address offset: 0x394h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	TE2O EN	TE1O EN	TD2O EN	TD1O EN	TC2O EN	TC1O EN	TB2O EN	TB1O EN	TA2O EN	TA1O EN
						rs	rs	rs	rs	rs	rs	rs	rs	rs	rs

Bits 31:10 Reserved, must be kept at reset value.

Bit 9 **TE2OEN**: Timer E Output 2 Enable

Refer to TA1OEN description

Bit 8 **TE1OEN**: Timer E Output 1 Enable

Refer to TA1OEN description

Bit 7 **TD2OEN**: Timer D Output 2 Enable

Refer to TA1OEN description

Bit 6 **TD1OEN**: Timer D Output 1 Enable

Refer to TA1OEN description

Bit 5 **TC2OEN**: Timer C Output 2 Enable

Refer to TA1OEN description

Bit 4 **TC1OEN**: Timer C Output 1 Enable

Refer to TA1OEN description

Bit 3 **TB2OEN**: Timer B Output 2 Enable

Refer to TA1OEN description

Bit 2 **TB1OEN**: Timer B Output 1 Enable

Refer to TA1OEN description

Bit 1 **TA2OEN**: Timer A Output 2 Enable

Refer to TA1OEN description

Bit 0 **TA1OEN**: Timer A Output 1 (HRTIM_CHA1) Enable

Setting this bit enables the Timer A output 1. Writing "0" has no effect.

Reading the bit returns the output enable/disable status.

This bit is cleared asynchronously by hardware as soon as the timer-related fault input(s) is (are) active.

0: output HRTIM_CHA1 disabled. The output is either in Fault or Idle state.

1: output HRTIM_CHA1 enabled

Note: The disable status corresponds to both idle and fault states. The output disable status is given by TA1ODS bit in the HRTIM_ODSR register.

37.5.45 HRTIM Output Disable Register (HRTIM_ODISR)

Address offset: 0x398h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	TE2OD IS	TE1OD IS	TD2OD IS	TD1OD IS	TC2OD IS	TC1OD IS	TB2OD IS	TB1OD IS	TA2OD IS	TA1OD IS
						w	w	w	w	w	w	w	w	w	w

Bits 31:10 Reserved, must be kept at reset value.

Bit 9 **TE2ODIS**: Timer E Output 2 disable

Refer to TA1ODIS description

Bit 8 **TE1ODIS**: Timer E Output 1 disable

Refer to TA1ODIS description

Bit 7 **TD2ODIS**: Timer D Output 2 disable

Refer to TA1ODIS description

Bit 6 **TD1ODIS**: Timer D Output 1 disable

Refer to TA1ODIS description

Bit 5 **TC2ODIS**: Timer C Output 2 disable

Refer to TA1ODIS description

Bit 4 **TC1ODIS**: Timer C Output 1 disable

Refer to TA1ODIS description

Bit 3 **TB2ODIS**: Timer B Output 2 disable

Refer to TA1ODIS description

Bit 2 **TB1ODIS**: Timer B Output 1 disable

Refer to TA1ODIS description

Bit 1 **TA2ODIS**: Timer A Output 2 disable

Refer to TA1ODIS description

Bit 0 **TA1ODIS**: Timer A Output 1 (HRTIM_CHA1) disable

Setting this bit disables the Timer A output 1. The output enters the idle state, either from the run state or from the fault state.

Writing "0" has no effect.

37.5.46 HRTIM Output Disable Status Register (HRTIM_ODSR)

Address offset: 0x39Ch

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	TE2ODS	TE1ODS	TD2ODS	TD1ODS	TC2ODS	TC1ODS	TB2ODS	TB1ODS	TA2ODS	TA1ODS
						r	r	r	r	r	r	r	r	r	r

Bits 31:10 Reserved, must be kept at reset value.

Bit 9 **TE2ODS**: Timer E Output 2 disable status

Refer to TA1ODS description

Bit 8 **TE1ODS**: Timer E Output 1 disable status

Refer to TA1ODS description

Bit 7 **TD2ODS**: Timer D Output 2 disable status

Refer to TA1ODS description

Bit 6 **TD1ODS**: Timer D Output 1 disable status

Refer to TA1ODS description

Bit 5 **TC2ODS**: Timer C Output 2 disable status

Refer to TA1ODS description

Bit 4 **TC1ODS**: Timer C Output 1 disable status

Refer to TA1ODS description

Bit 3 **TB2ODS**: Timer B Output 2 disable status

Refer to TA1ODS description

Bit 2 **TB1ODS**: Timer B Output 1 disable status

Refer to TA1ODS description

Bit 1 **TA2ODS**: Timer A Output 2 disable status

Refer to TA1ODS description

Bit 0 **TA1ODS**: Timer A Output 1 disable status

Reading the bit returns the output disable status. It is not significant when the output is active (Tx1OEN or Tx2OEN = 1).

0: output HRTIM_CHA1 disabled, in Idle state.

1: output HRTIM_CHA1 disabled, in Fault state.

37.5.47 HRTIM Burst Mode Control Register (HRTIM_BMCR)

Address offset: 0x3A0h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BMSTAT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TEBM	TDBM	TCBM	TBBM	TABM	MTBM
rc_w0										rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	BMPREN	BMPRSC[3:0]				BMCLK[3:0]				BMOM	BME
					rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 BMSTAT: Burst Mode Status

This bit gives the current operating state.

0: Normal operation

1: Burst operation on-going. Writing this bit to 0 causes a burst mode early termination.

Bits 30:22 Reserved, must be kept at reset value.

Bit 21 TEBM: Timer E Burst Mode

Refer to TABM description

Bit 20 TDBM: Timer D Burst Mode

Refer to TABM description

Bit 19 TCBM: Timer C Burst Mode

Refer to TABM description

Bit 18 TBBM: Timer B Burst Mode

Refer to TABM description

Bit 17 TABM: Timer A Burst Mode

This bit defines how the timer behaves during a burst mode operation. This bitfield cannot be changed while the burst mode is enabled.

0: Timer A counter clock is maintained and the timer operates normally

1: Timer A counter clock is stopped and the counter is reset

Note: This bit must not be set when the balanced idle mode is active (DLYPRT[2:0] = 0x11)

Bit 16 MTBM: Master Timer Burst Mode

This bit defines how the timer behaves during a burst mode operation. This bitfield cannot be changed while the burst mode is enabled.

0: Master Timer counter clock is maintained and the timer operates normally

1: Master Timer counter clock is stopped and the counter is reset

Bits 15:11 Reserved, must be kept at reset value.

Bit 10 BMPREN: Burst Mode Preload Enable

This bit enables the registers preload mechanism and defines whether a write access into a preloadable register (HRTIM_BMCMR, HRTIM_BMPER) is done into the active or the preload register.

0: Preload disabled: the write access is directly done into active registers

1: Preload enabled: the write access is done into preload registers

Bits 9:6 **BMPRSC[3:0]: Burst Mode Prescaler**

Defines the prescaling ratio of the f_{HRTIM} clock for the burst mode controller. This bitfield cannot be changed while the burst mode is enabled.

0000: Clock not divided

0001: Division by 2

0010: Division by 4

0011: Division by 8

0100: Division by 16

0101: Division by 32

0110: Division by 64

0111: Division by 128

1000: Division by 256

1001: Division by 512

1010: Division by 1024

1011: Division by 2048

1100: Division by 4096

1101: Division by 8192

1110: Division by 16384

1111: Division by 32768

Bits 5:2 **BMCLK[3:0]: Burst Mode Clock source**

This bitfield defines the clock source for the burst mode counter. It cannot be changed while the burst mode is enabled (refer to [Table 317](#) for on-chip events 1..4 connections details).

0000: Master timer counter reset/roll-over

0001: Timer A counter reset/roll-over

0010: Timer B counter reset/roll-over

0011: Timer C counter reset/roll-over

0100: Timer D counter reset/roll-over

0101: Timer E counter reset/roll-over

0110: On-chip Event 1 (hrtim_bm_ck1), acting as a burst mode counter clock

0111: On-chip Event 2 (hrtim_bm_ck2) acting as a burst mode counter clock

1000: On-chip Event 3 (hrtim_bm_ck3) acting as a burst mode counter clock

1001: On-chip Event 4 (hrtim_bm_ck4) acting as a burst mode counter clock

1010: Prescaled f_{HRTIM} clock (as per BMPRSC[3:0] setting)

Other codes reserved

Bit 1 **BMOM: Burst Mode operating mode**

This bit defines if the burst mode is entered once or if it is continuously operating.

0: Single-shot mode

1: Continuous operation

Bit 0 **BME: Burst Mode enable**

This bit starts the burst mode controller which becomes ready to receive the start trigger.

Writing this bit to 0 causes a burst mode early termination.

0: Burst mode disabled

1: Burst mode enabled

37.5.48 HRTIM Burst Mode Trigger Register (HRTIM_BMTRGR)

Address offset: 0x3A4h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OCHPEV	EEV8	EEV7	TDEEV8	TAEV7	TECMP2	TECMP1	TEREP	TERST	TDCMP2	TDCMP1	TDREP	TDRST	TCCMP2	TCCMP1	TCREP
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TCRST	TBCMP2	TBCMP1	TBREP	TBRST	TACMP2	TACMP1	TAREP	TARST	MSTCMP4	MSTCMP3	MSTCMP2	MSTCMP1	MSTREP	MSTRST	SW
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **OCHPEV**: On-chip Event

A rising edge on an on-chip Event (see [Section : Burst mode triggers](#)) triggers a burst mode entry.

Bit 30 **EEV8**: External Event 8 (TIMD filters applied)

The external event 8 conditioned by TIMD filters is starting the burst mode operation.

Bit 29 **EEV7**: External Event 7 (TIMA filters applied)

The external event 7 conditioned by TIMA filters is starting the burst mode operation.

Bit 28 **TDEEV8**: Timer D period following External Event 8

The timer D period following an external event 8 (conditioned by TIMD filters) is starting the burst mode operation.

Bit 27 **TAEV7**: Timer A period following External Event 7

The timer A period following an external event 7 (conditioned by TIMA filters) is starting the burst mode operation.

Bit 26 **TECMP2**: Timer E Compare 2 event

Refer to TACMP1 description

Bit 25 **TECMP1**: Timer E Compare 1 event

Refer to TACMP1 description

Bit 24 **TEREP**: Timer E repetition

Refer to TAREP description

Bit 23 **TERST**: Timer E counter reset or roll-over

Refer to TARST description

Bit 22 **TDCMP2**: Timer D Compare 2 event

Refer to TACMP1 description

Bit 21 **TDCMP1**: Timer D Compare 1 event

Refer to TACMP1 description

Bit 20 **TDREP**: Timer D repetition

Refer to TAREP description

Bit 19 **TDRST**: Timer D reset or roll-over

Refer to TARST description

Bit 18 **TCCMP2**: Timer C Compare 2 event

Refer to TACMP1 description

- Bit 17 **TCCMP1**: *Timer C Compare 1 event*
Refer to TACMP1 description
- Bit 16 **TCREP**: *Timer C repetition*
Refer to TAREP description
- Bit 15 **TCRST**: *Timer C reset or roll-over*
Refer to TARST description
- Bit 14 **TBCMP2**: *Timer B Compare 2 event*
Refer to TACMP1 description
- Bit 13 **TBCMP1**: *Timer B Compare 1 event*
Refer to TACMP1 description
- Bit 12 **TBREP**: *Timer B repetition*
Refer to TAREP description
- Bit 11 **TBRST**: *Timer B reset or roll-over*
Refer to TARST description
- Bit 10 **TACMP2**: *Timer A Compare 2 event*
Refer to TACMP1 description
- Bit 9 **TACMP1**: *Timer A Compare 1 event*
The timer A compare 1 event is starting the burst mode operation.
- Bit 8 **TAREP**: *Timer A repetition*
The Timer A repetition event is starting the burst mode operation.
- Bit 7 **TARST**: *Timer A reset or roll-over*
The Timer A reset or roll-over event is starting the burst mode operation.
- Bit 6 **MSTCMP4**: *Master Compare 4*
Refer to MSTCMP1 description
- Bit 5 **MSTCMP3**: *Master Compare 3*
Refer to MSTCMP1 description
- Bit 4 **MSTCMP2**: *Master Compare 2*
Refer to MSTCMP1 description
- Bit 3 **MSTCMP1**: *Master Compare 1*
The master timer Compare 1 event is starting the burst mode operation.
- Bit 2 **MSTREP**: *Master repetition*
The master timer repetition event is starting the burst mode operation.
- Bit 1 **MSTRST**: *Master reset or roll-over*
The master timer reset and roll-over event is starting the burst mode operation.
- Bit 0 **SW**: *Software start*
This bit is set by software and automatically reset by hardware.
When set, It starts the burst mode operation immediately.
This bit is not active if the burst mode is not enabled (BME bit is reset).

37.5.49 HRTIM Burst Mode Compare Register (HRTIM_BMCMPR)

Address offset: 0x3A8h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BMCMP[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **BMCMP[15:0]**: *Burst mode compare value*

Defines the number of periods during which the selected timers are in idle state.

This register holds either the content of the preload register or the content of the active register if the preload is disabled.

Note: BMCMP[15:0] cannot be set to 0x0000 when using the f_{HRTIM} clock without a prescaler as the burst mode clock source (BMCLK[3:0] = 1010 and BMPRESC[3:0] = 0000).

37.5.50 HRTIM Burst Mode Period Register (HRTIM_BMPER)

Address offset: 0x3ACh

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BMPER[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **BMPER[15:0]**: *Burst mode Period*

Defines the burst mode repetition period.

This register holds either the content of the preload register or the content of the active register if preload is disabled.

Note: The BMPER[15:0] must not be null when the burst mode is enabled.

37.5.51 HRTIM Timer External Event Control Register 1 (HRTIM_EECR1)

Address offset: 0x3B0h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	EE5FAST	EE5SNS[1:0]		EE5POL	EE5SRC[1:0]		EE4FAST	EE4SNS[1:0]		EE4POL	EE4SRC[1:0]		EE3FAST	EE3SNS[1]
		r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EE3SNS[0]	EE3POL	EE3SRC[1:0]		EE2FAST	EE2SNS[1:0]		EE2POL	EE2SRC[1:0]		EE1FAST	EE1SNS[1:0]		EE1POL	EE1SRC[1:0]	
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **EE5FAST**: *External Event 5 Fast mode*
Refer to EE1FAST description

Bits 28:27 **EE5SNS[1:0]**: *External Event 5 Sensitivity*
Refer to EE1SNS[1:0] description

Bit 26 **EE5POL**: *External Event 5 Polarity*
Refer to EE1POL description

Bits 25:24 **EE5SRC[1:0]**: *External Event 5 Source*
Refer to EE1SRC[1:0] description

Bit 23 **EE4FAST**: *External Event 4 Fast mode*
Refer to EE1FAST description

Bits 22:21 **EE4SNS[1:0]**: *External Event 4 Sensitivity*
Refer to EE1SNS[1:0] description

Bit 20 **EE4POL**: *External Event 4 Polarity*
Refer to EE1POL description

Bits 19:18 **EE4SRC[1:0]**: *External Event 4 Source*
Refer to EE1SRC[1:0] description

Bit 17 **EE3FAST**: *External Event 3 Fast mode*
Refer to EE1FAST description

Bits 16:15 **EE3SNS[1:0]**: *External Event 3 Sensitivity*
Refer to EE1SNS[1:0] description

Bit 14 **EE3POL**: *External Event 3 Polarity*
Refer to EE1POL description

Bits 13:12 **EE3SRC[1:0]**: *External Event 3 Source*
Refer to EE1SRC[1:0] description

Bit 11 **EE2FAST**: *External Event 2 Fast mode*
Refer to EE1FAST description

Bits 10:9 **EE2SNS[1:0]**: *External Event 2 Sensitivity*
Refer to EE1SNS[1:0] description

Bit 8 **EE2POL**: *External Event 2 Polarity*
Refer to EE1POL description

Bits 7:6 **EE2SRC[1:0]**: *External Event 2 Source*

Refer to EE1SRC[1:0] description

Bit 5 **EE1FAST**: *External Event 1 Fast mode*

0: External Event 1 is re-synchronized by the HRTIM logic before acting on outputs, which adds a f_{HRTIM} clock-related latency

1: External Event 1 is acting asynchronously on outputs (low latency mode)

Note: This bit must not be modified once the counter in which the event is used is enabled (TxCEN bit set)

Bits 4:3 **EE1SNS[1:0]**: *External Event 1 Sensitivity*

00: On active level defined by EE1POL bit

01: Rising edge, whatever EE1POL bit value

10: Falling edge, whatever EE1POL bit value

11: Both edges, whatever EE1POL bit value

Bit 2 **EE1POL**: *External Event 1 Polarity*

This bit is only significant if EE1SNS[1:0] = 00.

0: External event is active high

1: External event is active low

Note: This parameter cannot be changed once the timer x is enabled. It must be configured prior to setting EE1FAST bit.

Bits 1:0 **EE1SRC[1:0]**: *External Event 1 Source*

00: hrtim_evt11

01: hrtim_evt12

10: hrtim_evt13

11: hrtim_evt14

Note: This parameter cannot be changed once the timer x is enabled. It must be configured prior to setting EE1FAST bit.

37.5.52 HRTIM Timer External Event Control Register 2 (HRTIM_EECR2)

Address offset: 0x3B4h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	EE10SNS[1:0]		EE10POL	EE10SRC[1:0]		Res.	EE9SNS[1:0]		EE9POL	EE9SRC[1:0]		Res.	EE8SNS[1]
			rw	rw	rw	rw	rw		rw	rw	rw	rw	rw		rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EE8SNS[0]	EE8POL	EE8SRC[1:0]		Res.	EE7SNS[1:0]		EE7POL	EE7SRC[1:0]		Res.	EE6SNS[1:0]		EE6POL	EE6SRC[1:0]	
rw	rw	rw	rw		rw	rw	rw	rw	rw		rw	rw	rw	rw	rw

Bits 31:29 Reserved, must be kept at reset value.

Bits 28:27 **EE10SNS[1:0]**: *External Event 10 Sensitivity*

Refer to EE1SNS[1:0] description

Bit 26 **EE10POL**: *External Event 10 Polarity*

Refer to EE1POL description

Bits 25:24 **EE10SRC[1:0]**: *External Event 10 Source*

Refer to EE1SRC[1:0] description

Bit 23 Reserved, must be kept at reset value.

Bits 22:21 **EE9SNS[1:0]**: *External Event 9 Sensitivity*

Refer to EE1SNS[1:0] description

Bit 20 **EE9POL**: *External Event 9 Polarity*

Refer to EE1POL description

Bits 19:18 **EE9SRC[1:0]**: *External Event 9 Source*

Refer to EE1SRC[1:0] description

Bit 17 Reserved, must be kept at reset value.

Bits 16:15 **EE8SNS[1:0]**: *External Event 8 Sensitivity*

Refer to EE1SNS[1:0] description

Bit 14 **EE8POL**: *External Event 8 Polarity*

Refer to EE1POL description

Bits 13:12 **EE8SRC[1:0]**: *External Event 8 Source*

Refer to EE1SRC[1:0] description

Bit 11 Reserved, must be kept at reset value.

Bits 10:9 **EE7SNS[1:0]**: *External Event 7 Sensitivity*

Refer to EE1SNS[1:0] description

Bit 8 **EE7POL**: *External Event 7 Polarity*

Refer to EE1POL description

Bits 7:6 **EE7SRC[1:0]**: *External Event 7 Source*

Refer to EE1SRC[1:0] description

Bit 5 Reserved, must be kept at reset value.

Bits 4:3 **EE6SNS[1:0]**: *External Event 6 Sensitivity*

Refer to EE1SNS[1:0] description

Bit 2 **EE6POL**: *External Event 6 Polarity*

Refer to EE1POL description

Bits 1:0 **EE6SRC[1:0]**: *External Event 6 Source*

Refer to EE1SRC[1:0] description

37.5.53 HRTIM Timer External Event Control Register 3 (HRTIM_EECR3)

Address offset: 0x3B8h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EEVSD[1:0]		Res.	Res.	EE10F[3:0]				Res.	Res.	EE9F[3:0]				Res.	Res.
rw	rw			rw	rw	rw	rw			rw	rw	rw	rw		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EE8F[3:0]				Res.	Res.	EE7F[3:0]				Res.	Res.	EE6F[3:0]			
rw	rw	rw	rw			rw	rw	rw	rw			rw	rw	rw	rw

Bits 31:30 **EEVSD[1:0]**: External Event Sampling clock division

This bitfield indicates the division ratio between the timer clock frequency (f_{HRTIM}) and the External Event signal sampling clock (f_{EEVS}) used by the digital filters.

00: $f_{EEVS}=f_{HRTIM}$

01: $f_{EEVS}=f_{HRTIM} / 2$

10: $f_{EEVS}=f_{HRTIM} / 4$

11: $f_{EEVS}=f_{HRTIM} / 8$

Bits 29:28 Reserved, must be kept at reset value.

Bits 27:24 **EE10F[3:0]**: External Event 10 filter

Refer to EE6F[3:0] description

Bits 23:22 Reserved, must be kept at reset value.

Bits 21:18 **EE9F[3:0]**: External Event 9 filter

Refer to EE6F[3:0] description

Bits 17:16 Reserved, must be kept at reset value.

Bits 15:12 **EE8F[3:0]**: External Event 8 filter

Refer to EE6F[3:0] description

Bits 11:10 Reserved, must be kept at reset value.

Bits 9:6 **EE7F[3:0]**: External Event 7 filter
Refer to EE6F[3:0] description

Bits 4:5 Reserved, must be kept at reset value.

Bits 3:0 **EE6F[3:0]**: External Event 6 filter

This bitfield defines the frequency used to sample External Event 6 input and the length of the digital filter applied to `hrtim_evt6`. The digital filter is made of a counter in which N valid samples are needed to validate a transition on the output.

0000: Filter disabled

0001: $f_{\text{SAMPLING}} = f_{\text{HRTIM}}$, N=2

0010: $f_{\text{SAMPLING}} = f_{\text{HRTIM}}$, N=4

0011: $f_{\text{SAMPLING}} = f_{\text{HRTIM}}$, N=8

0100: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/2$, N=6

0101: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/2$, N=8

0110: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/4$, N=6

0111: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/4$, N=8

1000: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/8$, N=6

1001: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/8$, N=8

1010: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/16$, N=5

1011: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/16$, N=6

1100: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/16$, N=8

1101: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/32$, N=5

1110: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/32$, N=6

1111: $f_{\text{SAMPLING}} = f_{\text{EEVS}}/32$, N=8

37.5.54 HRTIM ADC Trigger 1 Register (HRTIM_ADC1R)

Address offset: 0x3BCh

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AD1TE PER	AD1TE C4	AD1TE C3	AD1TE C2	AD1TD PER	AD1TD C4	AD1TD C3	AD1TD C2	AD1TC PER	AD1TC C4	AD1TC C3	AD1TC C2	AD1TB RST	AD1TB PER	AD1TB C4	AD1TB C3
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AD1TB C2	AD1TA RST	AD1TA PER	AD1TA C4	AD1TA C3	AD1TA C2	AD1EE V5	AD1EE V4	AD1EE V3	AD1EE V2	AD1EE V1	AD1MP ER	AD1MC 4	AD1MC 3	AD1MC 2	AD1MC 1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 These bits select the trigger source for the ADC Trigger 1 output (`hrtim_adc_trg1`). Refer to HRTIM_ADC3R bits description for details

37.5.55 HRTIM ADC Trigger 2 Register (HRTIM_ADC2R)

Address offset: 0x3C0h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AD2TE RST	AD2TE C4	AD2TE C3	AD2TE C2	AD2TD RST	AD2TD PER	AD2TD C4	AD2TD C3	AD2TD C2	AD2TC RST	AD2TC PER	AD2TC C4	AD2TC C3	AD2TC C2	AD2TB PER	AD2TB C4
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AD2TB C3	AD2TB C2	AD2TA PER	AD2TA C4	AD2TA C3	AD2TA C2	AD2EE V10	AD2EE V9	AD2EE V8	AD2EE V7	AD2EE V6	AD2MP ER	AD2MC 4	AD2MC 3	AD2MC 2	AD2MC 1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 These bits select the trigger source for the ADC Trigger 2 output (hrtim_adc_trg2). Refer to HRTIM_ADC4R bits description for details

37.5.56 HRTIM ADC Trigger 3 Register (HRTIM_ADC3R)

Address offset: 0x3C4h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADC3TEPER	ADC3TEC4	ADC3TEC3	ADC3TEC2	ADC3TDPER	ADC3TDC4	ADC3TDC3	ADC3TDC2	ADC3TCPER	ADC3TCC4	ADC3TCC3	ADC3TCC2	ADC3TBRST	ADC3TBPER	ADC3TBC4	ADC3TBC3
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADC3TBC2	ADC3TARST	ADC3TAPER	ADC3TAC4	ADC3TAC3	ADC3TAC2	ADC3EEV5	ADC3EEV4	ADC3EEV3	ADC3EEV2	ADC3EEV1	ADC3MPER	ADC3MC4	ADC3MC3	ADC3MC2	ADC3MC1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **ADC3TEPER**: ADC trigger 3 on Timer E Period

Refer to ADC3TAPER description

Bit 30 **ADC3TEC4**: ADC trigger 3 on Timer E Compare 4

Refer to ADC3TAC2 description

Bit 29 **ADC3TEC3**: ADC trigger 3 on Timer E Compare 3

Refer to ADC3TAC2 description

Bit 28 **ADC3TEC2**: ADC trigger 3 on Timer E Compare 2

Refer to ADC3TAC2 description

Bit 27 **ADC3TDPER**: ADC trigger 3 on Timer D Period

Refer to ADC3TAPER description

Bit 26 **ADC3TDC4**: ADC trigger 3 on Timer D Compare 4

Refer to ADC3TAC2 description

Bit 25 **ADC3TDC3**: ADC trigger 3 on Timer D Compare 3

Refer to ADC3TAC2 description

Bit 24 **ADC3TDC2**: ADC trigger 3 on Timer D Compare 2

Refer to ADC3TAC2 description

Bit 23 **ADC3TCPER**: ADC trigger 3 on Timer C Period

Refer to ADC3TAPER description

Bit 22 **ADC3TCC4**: ADC trigger 3 on Timer C Compare 4

Refer to ADC3TAC2 description

Bit 21 **ADC3TCC3**: ADC trigger 3 on Timer C Compare 3

Refer to ADC3TAC2 description

Bit 20 **ADC3TCC2**: ADC trigger 3 on Timer C Compare 2

Refer to ADC3TAC2 description

Bit 19 **ADC3TBRST**: ADC trigger 3 on Timer B Reset and counter roll-over

Refer to ADC3TBRST description

Bit 18 **ADC3TBPER**: ADC trigger 3 on Timer B Period

Refer to ADC3TAPER description

Bit 17 **ADC3TBC4**: ADC trigger 3 on Timer B Compare 4

Refer to ADC3TAC2 description

- Bit 16 **ADC3TBC3**: *ADC trigger 3 on Timer B Compare 3*
Refer to ADC3TAC2 description
- Bit 15 **ADC3TBC2**: *ADC trigger 3 on Timer B Compare 2*
Refer to ADC3TAC2 description
- Bit 14 **ADC3TARST**: *ADC trigger 3 on Timer A Reset and counter roll-over*
This bit enables the generation of an ADC Trigger upon Timer A reset and roll-over event, on ADC Trigger 1 output.
- Bit 13 **ADC3TAPER**: *ADC trigger 3 on Timer A Period*
This bit enables the generation of an ADC Trigger upon Timer A period event, on ADC Trigger 3 output (hrtim_adc_trg3).
- Bit 12 **ADC3TAC4**: *ADC trigger 3 on Timer A Compare 4*
Refer to ADC3TAC2 description
- Bit 11 **ADC3TAC3**: *ADC trigger 3 on Timer A Compare 3*
Refer to ADC3TAC2 description
- Bit 10 **ADC3TAC2**: *ADC trigger 3 on Timer A Compare 2*
This bit enables the generation of an ADC Trigger upon Timer A Compare 2 event, on ADC Trigger 3 output (hrtim_adc_trg3).
- Bit 9 **ADC3EEV5**: *ADC trigger 3 on External Event 5*
Refer to ADC3EEV1 description
- Bit 8 **ADC3EEV4**: *ADC trigger 3 on External Event 4*
Refer to ADC3EEV1 description
- Bit 7 **ADC3EEV3**: *ADC trigger 3 on External Event 3*
Refer to ADC3EEV1 description
- Bit 6 **ADC3EEV2**: *ADC trigger 3 on External Event 2*
Refer to ADC3EEV1 description
- Bit 5 **ADC3EEV1**: *ADC trigger 3 on External Event 1*
This bit enables the generation of an ADC Trigger upon External event 1, on ADC Trigger 3 output (hrtim_adc_trg3).
- Bit 4 **ADC3MPER**: *ADC trigger 3 on Master Period*
This bit enables the generation of an ADC Trigger upon Master timer period event, on ADC Trigger 3 output (hrtim_adc_trg3).
- Bit 3 **ADC3MC4**: *ADC trigger 3 on Master Compare 4*
Refer to ADC3MC1 description
- Bit 2 **ADC3MC3**: *ADC trigger 3 on Master Compare 3*
Refer to ADC3MC1 description
- Bit 1 **ADC3MC2**: *ADC trigger 3 on Master Compare 2*
Refer to ADC3MC1 description
- Bit 0 **ADC3MC1**: *ADC trigger 3 on Master Compare 1*
This bit enables the generation of an ADC Trigger upon Master Compare 1 event, on ADC Trigger 3 output (hrtim_adc_trg3).

37.5.57 HRTIM ADC Trigger 4 Register (HRTIM_ADC4R)

Address offset: 0x3C8h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADC4T ERST	ADC4T EC4	ADC4T EC3	ADC4T EC2	ADC4T DRST	ADC4T DPER	ADC4T DC4	ADC4T DC3	ADC4T DC2	ADC4T CRST	ADC4T CPER	ADC4T CC4	ADC4T CC3	ADC4T CC2	ADC4T BPER	ADC4T BC4
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADC4T BC3	ADC4T BC2	ADC4T APER	ADC4T AC4	ADC4T AC3	ADC4T AC2	ADC4E EV10	ADC4E EV9	ADC4E EV8	ADC4E EV7	ADC4E EV6	ADC4M PER	ADC4M C4	ADC4M C3	ADC4M C2	ADC4M C1
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **ADC4TERST**: ADC trigger 4 on Timer E Reset and counter roll-over ⁽¹⁾

Refer to ADC4TCRST description

Bit 30 **ADC4TEC4**: ADC trigger 4 on Timer E Compare 4

Refer to ADC4TAC2 description

Bit 29 **ADC4TEC3**: ADC trigger 4 on Timer E Compare 3

Refer to ADC4TAC2 description

Bit 28 **ADC4TEC2**: ADC trigger 4 on Timer E Compare 2

Refer to ADC4TAC2 description

Bit 27 **ADC4TDRST**: ADC trigger 4 on Timer D Reset and counter roll-over ⁽¹⁾

Refer to ADC4TCRST description

Bit 26 **ADC4TDPER**: ADC trigger 4 on Timer D Period

Refer to ADC4TAPER description

Bit 25 **ADC4TDC4**: ADC trigger 4 on Timer D Compare 4

Refer to ADC4TAC2 description

Bit 24 **ADC4TDC3**: ADC trigger 4 on Timer D Compare 3

Refer to ADC4TAC2 description

Bit 23 **ADC4TDC2**: ADC trigger 2 on Timer D Compare 2

Refer to ADC4TAC2 description

Bit 22 **ADC4TCRST**: ADC trigger 4 on Timer C Reset and counter roll-over ⁽¹⁾

This bit enables the generation of an ADC Trigger upon Timer C reset and roll-over event, on ADC Trigger 4 output (hrtim_adc_trg4).

Bit 21 **ADC4TCPER**: ADC trigger 4 on Timer C Period

Refer to ADC4TAPER description

Bit 20 **ADC4TCC4**: ADC trigger 4 on Timer C Compare 4

Refer to ADC4TAC2 description

Bit 19 **ADC4TCC3**: ADC trigger 4 on Timer C Compare 3

Refer to ADC4TAC2 description

Bit 18 **ADC4TCC2**: ADC trigger 4 on Timer C Compare 2

Refer to ADC4TAC2 description

- Bit 17 **ADC4TBPER**: *ADC trigger 4 on Timer B Period*
Refer to ADC4TAPER description
- Bit 16 **ADC4TBC4**: *ADC trigger 4 on Timer B Compare 4*
Refer to ADC4TAC2 description
- Bit 15 **ADC4TBC3**: *ADC trigger 4 on Timer B Compare 3*
Refer to ADC4TAC2 description
- Bit 14 **ADC4TBC2**: *ADC trigger 4 on Timer B Compare 2*
Refer to ADC4TAC2 description
- Bit 13 **ADC4TAPER**: *ADC trigger 4 on Timer A Period*
This bit enables the generation of an ADC Trigger upon Timer A event, on ADC Trigger 4 output (hrtim_adc_trg4).
- Bit 12 **ADC4TAC4**: *ADC trigger 4 on Timer A Compare 4*
Refer to ADC4TAC2 description
- Bit 11 **ADC4TAC3**: *ADC trigger 4 on Timer A Compare 3*
Refer to ADC4TAC2 description
- Bit 10 **ADC4TAC2**: *ADC trigger 4 on Timer A Compare 2*
This bit enables the generation of an ADC Trigger upon Timer A Compare 2, on ADC Trigger 4 output (hrtim_adc_trg4).
- Bit 9 **ADC4EEV10**: *ADC trigger 4 on External Event 10* ⁽¹⁾
Refer to ADC4EEV6 description
- Bit 8 **ADC4EEV9**: *ADC trigger 4 on External Event 9* ⁽¹⁾
Refer to ADC4EEV6 description
- Bit 7 **ADC4EEV8**: *ADC trigger 4 on External Event 8* ⁽¹⁾
Refer to ADC4EEV6 description
- Bit 6 **ADC4EEV7**: *ADC trigger 4 on External Event 7* ⁽¹⁾
Refer to ADC4EEV6 description
- Bit 5 **ADC4EEV6**: *ADC trigger 4 on External Event 6* ⁽¹⁾
This bit enables the generation of an ADC Trigger upon external event 6, on ADC Trigger 4 output (hrtim_adc_trg4).
- Bit 4 **ADC4MPER**: *ADC trigger 4 on Master Period*
This bit enables the generation of an ADC Trigger upon Master period event, on ADC Trigger 4 output (hrtim_adc_trg4).
- Bit 3 **ADC4MC4**: *ADC trigger 4 on Master Compare 4*
Refer to ADC4MC1 description
- Bit 2 **ADC4MC3**: *ADC trigger 4 on Master Compare 3*
Refer to ADC4MC1 description
- Bit 1 **ADC4MC2**: *ADC trigger 4 on Master Compare 2*
Refer to ADC4MC1 description
- Bit 0 **ADC4MC1**: *ADC trigger 4 on Master Compare 1*
This bit enables the generation of an ADC Trigger upon Master Compare 1 event, on ADC Trigger 4 output (hrtim_adc_trg4).

1. These triggers are differing from HRTIM_ADC1R/HRTIM_ADC3R to HRTIM_ADC2R/HRTIM_ADC4R.

37.5.58 HRTIM Fault Input Register 1 (HRTIM_FLTINR1)

Address offset: 0x3D0h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FLT4L CK	FLT4F[3:0]				FLT4S RC	FLT4P	FLT4E	FLT3L CK	FLT3F[3:0]				FLT3S RC	FLT3P	FLT3E
rwo	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLT2L CK	FLT2F[3:0]				FLT2S RC	FLT2P	FLT2E	FLT1L CK	FLT1F[3:0]				FLT1S RC	FLT1P	FLT1E
rwo	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bit 31 **FLT4LCK**: Fault 4 Lock

Refer to FLT5LCK description in HRTIM_FLTINR2 register

Bits 30:27 **FLT4F[3:0]**: Fault 4 filter

Refer to FLT5F[3:0] description in HRTIM_FLTINR2 register

Bit 26 **FLT4SRC**: Fault 4 source

Refer to FLT5SRC description in HRTIM_FLTINR2 register

Bit 25 **FLT4P**: Fault 4 polarity

Refer to FLT5P description in HRTIM_FLTINR2 register

Bit 24 **FLT4E**: Fault 4 enable

Refer to FLT5E description in HRTIM_FLTINR2 register

Bit 23 **FLT3LCK**: Fault 3 Lock

Refer to FLT5LCK description in HRTIM_FLTINR2 register

Bits 22:19 **FLT3F[3:0]**: Fault 3 filter

Refer to FLT5F[3:0] description in HRTIM_FLTINR2 register

Bit 18 **FLT3SRC**: Fault 3 source

Refer to FLT5SRC description in HRTIM_FLTINR2 register

Bit 17 **FLT3P**: Fault 3 polarity

Refer to FLT5P description in HRTIM_FLTINR2 register

Bit 16 **FLT3E**: Fault 3 enable

Refer to FLT5E description in HRTIM_FLTINR2 register

Bit 15 **FLT2LCK**: Fault 2 Lock

Refer to FLT5LCK description in HRTIM_FLTINR2 register

Bits 14:11 **FLT2F[3:0]**: Fault 2 filter

Refer to FLT5F[3:0] description in HRTIM_FLTINR2 register

Bit 10 **FLT2SRC**: Fault 2 source

Refer to FLT5SRC description in HRTIM_FLTINR2 register

Bit 9 **FLT2P**: Fault 2 polarity

Refer to FLT2P description in HRTIM_FLTINR2 register

Bit 8 **FLT2E**: Fault 2 enable

Refer to FLT5E description in HRTIM_FLTINR2 register

- Bit 7 **FLT1LCK**: Fault 1 Lock
Refer to FLT5LCK description in HRTIM_FLTINR2 register
- Bits 6:3 **FLT1F[3:0]**: Fault 1 filter
Refer to FLT5F[3:0] description in HRTIM_FLTINR2 register
- Bit 2 **FLT1SRC**: Fault 1 source
Refer to FLT5SRC description in HRTIM_FLTINR2 register
- Bit 1 **FLT1P**: Fault 1 polarity
Refer to FLT5P description in HRTIM_FLTINR2 register
- Bit 0 **FLT1E**: Fault 1 enable
Refer to FLT5E description in HRTIM_FLTINR2 register

37.5.59 HRTIM Fault Input Register 2 (HRTIM_FLTINR2)

Address offset: 0x3D4h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	FLTSD[1:0]		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
						rw	rw								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FLT5LCK	FLT5F[3:0]				FLT5SRC	FLT5P	FLT5E
								rwo	rw	rw	rw	rw	rw	rw	rw

Bits 31:26 Reserved, must be kept at reset value.

Bits 25:24 **FLTSD[1:0]**: Fault Sampling clock division

This bitfield indicates the division ratio between the timer clock frequency (f_{HRTIM}) and the fault signal sampling clock (f_{FLTS}) used by the digital filters.

00: $f_{FLTS}=f_{HRTIM}$

01: $f_{FLTS}=f_{HRTIM} / 2$

10: $f_{FLTS}=f_{HRTIM} / 4$

11: $f_{FLTS}=f_{HRTIM} / 8$

Note: This bitfield must be written prior to any of the FLTxE enable bits.

Bits 23:8 Reserved, must be kept at reset value.

Bit 7 **FLT5LCK**: Fault 5 Lock

The FLT5LCK bit modifies the write attributes of the fault programming bit, so that they can be protected against spurious write accesses.

This bit is write-once. Once it has been set, it cannot be modified till the next system reset.

0: FLT5E, FLT5P, FLT5SRC, FLT5F[3:0] bits are read/write.

1: FLT5E, FLT5P, FLT5SRC, FLT5F[3:0] bits can no longer be written (read-only mode)

Bits 6:3 **FLT5F[3:0]**: Fault 5 filter

This bitfield defines the frequency used to sample FLT5 input and the length of the digital filter applied to FLT5. The digital filter is made of an event counter in which N events are needed to validate a transition on the output:

0000: No filter, FLT5 acts asynchronously

0001: $f_{\text{SAMPLING}} = f_{\text{HRTIM}}$, N = 2

0010: $f_{\text{SAMPLING}} = f_{\text{HRTIM}}$, N = 4

0011: $f_{\text{SAMPLING}} = f_{\text{HRTIM}}$, N = 8

0100: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/2$, N = 6

0101: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/2$, N = 8

0110: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/4$, N = 6

0111: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/4$, N = 8

1000: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/8$, N = 6

1001: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/8$, N = 8

1010: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/16$, N = 5

1011: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/16$, N = 6

1100: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/16$, N = 8

1101: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/32$, N = 5

1110: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/32$, N = 6

1111: $f_{\text{SAMPLING}} = f_{\text{FLTS}}/32$, N = 8

Note: This bitfield can be written only when FLT5E enable bit is reset.

This bitfield cannot be modified when FLT5LOCK has been programmed.

Bit 2 **FLT5SRC**: Fault 5 source

This bit selects the FAULT5 input source (refer to [Table 318](#) for connection details).

0: Fault 1 input is HRTIM_FLT5 input pin

1: Fault 1 input is hrtim_in_ftl5 signal

Note: This bitfield can be written only when FLT5E enable bit is reset

Bit 1 **FLT5P**: Fault 5 polarity

This bit selects the FAULT5 input polarity.

0: Fault 5 input is active low

1: Fault 5 input is active high

Note: This bitfield can be written only when FLT5E enable bit is reset

Bit 0 **FLT5E**: Fault 5 enable

This bit enables the global FAULT5 input circuitry.

0: Fault 5 input disabled

1: Fault 5 input enabled

37.5.60 HRTIM Burst DMA Master timer update Register (HRTIM_BDMUPR)

Address offset: 0x3D8h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	MCMP4	MCMP3	MCMP2	MCMP1	MREP	MPER	MCNT	MDIER	MICR	MCR
						rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:10 Reserved, must be kept at reset value.

Bit 9 **MCMP4**: *MCMP4R register update enable*
Refer to MCR description

Bit 8 **MCMP3**: *MCMP3R register update enable*
Refer to MCR description

Bit 7 **MCMP2**: *MCMP2R register update enable*
Refer to MCR description

Bit 6 **MCMP1**: *MCMP1R register update enable*
Refer to MCR description

Bit 5 **MREP**: *MREP register update enable*
Refer to MCR description

Bit 4 **MPER**: *MPER register update enable*
Refer to MCR description

Bit 3 **MCNT**: *MCNTR register update enable*
Refer to MCR description

Bit 2 **MDIER**: *MDIER register update enable*
Refer to MCR description

Bit 1 **MICR**: *MICR register update enable*
Refer to MCR description

Bit 0 **MCR**: *MCR register update enable*

This bit defines if the master timer MCR register is part of the list of registers to be updated by the Burst DMA.

0: MCR register is not updated by Burst DMA accesses

1: MCR register is updated by Burst DMA accesses

37.5.61 HRTIM Burst DMA Timerx update Register (HRTIM_BDTxUPR)

Address offset: 0x3DCh-0x3ECh

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIMxFLTR	TIMxOUTR	TIMxCHPR	TIMxRSTR	TIMxEEFR2
											rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIMxEEFR1	TIMxRST2R	TIMxSET2R	TIMxRST1R	TIMxSET1R	TIMxDTR	TIMxCMP4	TIMxCMP3	TIMxCMP2	TIMxCMP1	TIMxREP	TIMxPNT	TIMxDIER	TIMxICR	TIMxICR	TIMxICR
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:21 Reserved, must be kept at reset value.

- Bit 20 **TIMxFLTR**: HRTIM_FLTxR register update enable
Refer to TIMxCR description
- Bit 19 **TIMxOUTR**: HRTIM_OUTxR register update enable
Refer to TIMxCR description
- Bit 18 **TIMxCHPR**: HRTIM_CHPxR register update enable
Refer to TIMxCR description
- Bit 17 **TIMxRSTR**: HRTIM_RSTxR register update enable
Refer to TIMxCR description
- Bit 16 **TIMxEEFR2**: HRTIM_EEFxR2 register update enable
Refer to TIMxCR description
- Bit 15 **TIMxEEFR1**: HRTIM_EEFxR1 register update enable
Refer to TIMxCR description
- Bit 14 **TIMxRST2R**: HRTIM_RST2xR register update enable
Refer to TIMxCR description
- Bit 13 **TIMxSET2R**: HRTIM_SET2xR register update enable
Refer to TIMxCR description
- Bit 12 **TIMxRST1R**: HRTIM_RST1xR register update enable
Refer to TIMxCR description
- Bit 11 **TIMxSET1R**: HRTIM_SET1xR register update enable
Refer to TIMxCR description
- Bit 10 **TIMxDTR**: HRTIM_DTxD register update enable
Refer to TIMxCR description
- Bit 9 **TIMxCMP4**: HRTIM_CMP4xR register update enable
Refer to TIMxCR description
- Bit 8 **TIMxCMP3**: HRTIM_CMP3xR register update enable
Refer to TIMxCR description
- Bit 7 **TIMxCMP2**: HRTIM_CMP2xR register update enable
Refer to TIMxCR description

- Bit 6 **TIMxCMP1**: HRTIM_CMP1xR register update enable
Refer to TIMxCR description
- Bit 5 **TIMxREP**: HRTIM_REPxR register update enable
Refer to TIMxCR description
- Bit 4 **TIMxPER**: HRTIM_PERxR register update enable
Refer to TIMxCR description
- Bit 3 **TIMxCNT**: HRTIM_CNTxR register update enable
Refer to TIMxCR description
- Bit 2 **TIMxDIER**: HRTIM_TIMxDIER register update enable
Refer to TIMxCR description
- Bit 1 **TIMxICR**: HRTIM_TIMxICR register update enable
Refer to TIMxCR description
- Bit 0 **TIMxCR**: HRTIM_TIMxCR register update enable
This bit defines if the master timer MCR register is part of the list of registers to be updated by the Burst DMA.
0: HRTIM_TIMxCR register is not updated by Burst DMA accesses
1: HRTIM_TIMxCR register is updated by Burst DMA accesses

37.5.62 HRTIM Burst DMA Data Register (HRTIM_BDMADR)

Address offset: 0x3F0h

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BDMADR[31:16]															
wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BDMADR[15:0]															
wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo	wo

Bits 31:0 **BDMADR[31:0]**: Burst DMA Data register

Write accesses to this register triggers:

- the copy of the data value into the registers enabled in BDTxUPR and BDMUPR register bits
- the increment of the register pointer to the next location to be filled

37.5.63 HRTIM register map

The tables below summarize the HRTIM registers mapping. The address offsets in [Table 324](#) and [Table 325](#) are referred to in the base address offsets given in [Table 323](#).

Table 323. RTIM global register map

Base address offset	Register
0x000 - 0x07F	Master timer
0x080 - 0x0FF	Timer A
0x100 - 0x17F	Timer B
0x180 - 0x1FF	Timer C
0x200 - 0x27F	Timer D
0x280 - 0x2FF	Timer E
0x300 - 0x37F	Reserved
0x380 - 0x3FF	Common registers

Table 324. HRTIM Register map and reset values: Master timer

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x0000	HRTIM_MCR	BRSTDMA[1:0]		MREPU	Res.	PREEN	DACSYNC[1:0]		Res.	Res.	Res.	TECEN	TDCEN	TCCEN	TBCEN	TACEN	MCEN	SYNCSRC[1:0]		SYNCOUT[1:0]		SYNCSSTRM	SYNCSRSTM	SYNCIN[1:0]		Res.	Res.	HALF	RETRIG	CONT	CKPSC[2:0]		
	Reset value	0	0	0		0	0	0				0	0	0	0	0	0	0	0	0	0	0	0	0	0			0	0	0	0	0	0
0x0004	HRTIM_MISR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MUPD	SYNC	MREP	MCMP4	MCMP3	MCMP2	MCMP1
	Reset value																										0	0	0	0	0	0	0
0x0008	HRTIM_MICR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MUPDC	SYNCC	MREPC	MCMP4C	MCMP3C	MCMP2C	MCMP1C
	Reset value																										0	0	0	0	0	0	0
0x000C	HRTIM_MDIER ⁽⁷⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MUPDDE	SYNCDE	MREPDE	MCMP4DE	MCMP3DE	MCMP2DE	MCMP1DE	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MUPDIE	SYNCIE	MREPIE	MCMP4IE	MCMP3IE	MCMP2IE	MCMP1IE
	Reset value										0	0	0	0	0	0	0										0	0	0	0	0	0	0
0x0010	HRTIM_MCNT_R	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCNT[15:0]																					
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 324. HRTIM Register map and reset values: Master timer (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0												
0x0014	HRTIM_MPER ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MPER[15:0]																											
	Reset value																	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1												
0x0018	HRTIM_MREP ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MREP[7:0]																			
	Reset value																									0	0	0	0	0	0	0	0	0											
0x001C	HRTIM_MCMP1R ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCMP1[15:0]																											
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0												
0x0020	Reserved	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.											
	Reset value																																												
0x0024	HRTIM_MCMP2R ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCMP2[15:0]																											
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0												
0x0028	HRTIM_MCMP3R ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCMP3[15:0]																											
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0												
0x002C	HRTIM_MCMP4R ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCMP4[15:0]																											
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0												

1. This register can be preloaded (see [Table 309 on page 1415](#)).

Table 325. HRTIM Register map and reset values: TIMx (x= A..E)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x0000	HRTIM_TIMxCR	UPDGAT [3:0]				PREEN	DACSYNC[1:0]				MSTU	TEU	TDU	TCU	TBU	TxRSTU	TxREPU	Res.	DELCMP4[1:0]		DELCMP2[1:0]		SYNCSTRTx	SYNCRSTx	Res.	Res.	Res.	PSHPLL	HALF	RETRIG	CONT	CKPSCx[2:0]			
	Reset value	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				0	0	0	0	0	0	0	
0x0004	HRTIM_TIMxISR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	O2CPY	O1CPY	O2STAT	O1STAT	IPPSTAT	CPPSTAT	Res.	DLYPRT	RST	RSTx2	SETx2	RSTx1	SETx1	CPT2	CPT1	UPD	Res.	REP	OMP4	OMP3	OMP2	OMP1		
	Reset value											0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0	0	
0x0008	HRTIM_TIMxICR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DLYPRTC	RSTC	RSTx2C	SET2xC	RSTx1C	SET1xC	CPT2C	CPT1C	UPDC	Res.	REPC	CMP4C	CMP3C	CMP2C	CMP1C		
	Reset value																	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0	0	
0x000C	HRTIM_TIMxDIER ⁽¹⁾	Res.	DLYPRTDE	RSTDE	RSTx2DE	SETx2DE	RSTx1DE	SET1xDE	CPT2DE	CPT1DE	UPDDE	Res.	REPDE	CMP4DE	CMP3DE	CMP2DE	CMP1DE	Res.	DLYPRTIE	RSTIE	RSTx2IE	SETx2IE	RSTx1IE	SET1xIE	CPT2IE	CPT1IE	UPDIE	Res.	REPIE	CMP4IE	CMP3IE	CMP2IE	CMP1IE		
	Reset value		0	0	0	0	0	0	0	0	0		0	0	0	0	0		0	0	0	0	0	0	0	0			0	0	0	0	0	0	
0x0010	HRTIM_CNTxR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNTx[15:0]																	
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x0014	HRTIM_PERxR ^(T)	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PERx[15:0]																	
	Reset value																	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1		
0x0018	HRTIM_REPxR ^(T)	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REPx[7:0]									
	Reset value																									0	0	0	0	0	0	0	0	0	
0x001C	HRTIM_CMP1xR ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CMP1x[15:0]																	
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x0020	HRTIM_CMP1CxR ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REPx[7:0]								CMP1x[15:0]																	
	Reset value									0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x0024	HRTIM_CMP2xR ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CMP2x[15:0]																	
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x0028	HRTIM_CMP3xR ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CMP3x[15:0]																	
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x002C	HRTIM_CMP4xR ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CMP4x[15:0]																	
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x0030	HRTIM_CPT1xR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CPT1x[15:0]																	
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		

Table 325. HRTIM Register map and reset values: TIMx (x= A..E) (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
0x0034	HRTIM_CPT2xR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CPT2x[15:0]																			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x0038	HRTIM_DTxR ⁽¹⁾	DTFLKx	DTFSLKx	Res.	Res.	Res.	Res.	SDTFx	DTFx[8:0]								DTRLKx	DTRSLKx	Res.	DTPRSC[2:0]		SDTRx		DTRx[8:0]													
	Reset value	0	0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x003C	HRTIM_SETx1R ⁽¹⁾	UPDATE	EXTEVNT10	EXTEVNT9	EXTEVNT8	EXTEVNT7	EXTEVNT6	EXTEVNT5	EXTEVNT4	EXTEVNT3	EXTEVNT2	EXTEVNT1	TIMEVNT9	TIMEVNT8	TIMEVNT7	TIMEVNT6	TIMEVNT5	TIMEVNT4	TIMEVNT3	TIMEVNT2	TIMEVNT1	MSTCMP4	MSTCMP3	MSTCMP2	MSTCMP1	MSTPER	CMP4	CMP3	CMP2	CMP1	PER	RESYNC	SST				
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x0040	HRTIM_RSTx1R ⁽¹⁾	UPDATE	EXTEVNT10	EXTEVNT9	EXTEVNT8	EXTEVNT7	EXTEVNT6	EXTEVNT5	EXTEVNT4	EXTEVNT3	EXTEVNT2	EXTEVNT1	TIMEVNT9	TIMEVNT8	TIMEVNT7	TIMEVNT6	TIMEVNT5	TIMEVNT4	TIMEVNT3	TIMEVNT2	TIMEVNT1	MSTCMP4	MSTCMP3	MSTCMP2	MSTCMP1	MSTPER	CMP4	CMP3	CMP2	CMP1	PER	RESYNC	SRT				
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x0044	HRTIM_SETx2R ⁽¹⁾	UPDATE	EXTEVNT10	EXTEVNT9	EXTEVNT8	EXTEVNT7	EXTEVNT6	EXTEVNT5	EXTEVNT4	EXTEVNT3	EXTEVNT2	EXTEVNT1	TIMEVNT9	TIMEVNT8	TIMEVNT7	TIMEVNT6	TIMEVNT5	TIMEVNT4	TIMEVNT3	TIMEVNT2	TIMEVNT1	MSTCMP4	MSTCMP3	MSTCMP2	MSTCMP1	MSTPER	CMP4	CMP3	CMP2	CMP1	PER	RESYNC	SST				
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x0048	HRTIM_RSTx2R ⁽¹⁾	UPDATE	EXTEVNT10	EXTEVNT9	EXTEVNT8	EXTEVNT7	EXTEVNT6	EXTEVNT5	EXTEVNT4	EXTEVNT3	EXTEVNT2	EXTEVNT1	TIMEVNT9	TIMEVNT8	TIMEVNT7	TIMEVNT6	TIMEVNT5	TIMEVNT4	TIMEVNT3	TIMEVNT2	TIMEVNT1	MSTCMP4	MSTCMP3	MSTCMP2	MSTCMP1	MSTPER	CMP4	CMP3	CMP2	CMP1	PER	RESYNC	SRT				
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x004C	HRTIM_EEFxR1	Res.	Res.	Res.	EE5FLTR[3:0]			EE5LTC	Res.	EE4FLTR[3:0]			EE4LTC	Res.	EE3FLTR[3:0]			EE3LTC	Res.	EE2FLTR[3:0]			EE2LTC	Res.	EE1FLTR[3:0]			EE1LTC									
	Reset value				0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0		0	0	0	0	0	0			
0x0050	HRTIM_EEFxR2	Res.	Res.	Res.	EE10FLTR[3:0]			EE10LTC	Res.	EE9FLTR[3:0]			EE9LTC	Res.	EE8FLTR[3:0]			EE8LTC	Res.	EE7FLTR[3:0]			EE7LTC	Res.	EE6FLTR[3:0]			EE6LTC									
	Reset value				0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0		0	0	0	0	0	0			
0x0054	HRTIM_RSTAR ⁽¹⁾	Res.	TIMECMP4	TIMECMP2	TIMECMP1	TIMDCMP4	TIMDCMP2	TIMDCMP1	TIMCCMP4	TIMCCMP2	TIMCCMP1	TIMBCMP4	TIMBCMP2	TIMBCMP1	EXTx1R[0]	EXTx1R[9]	EXTx1R[8]	EXTx1R[7]	EXTx1R[6]	EXTx1R[5]	EXTx1R[4]	EXTx1R[3]	EXTx1R[2]	EXTx1R[1]	EXTx1R[0]	EXTx1R[23]	EXTx1R[22]	EXTx1R[21]	EXTx1R[20]	EXTx1R[19]	EXTx1R[18]	EXTx1R[17]	EXTx1R[16]	EXTx1R[15]			
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			

Table 325. HRTIM Register map and reset values: TIMx (x= A..E) (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x0054	HRTIM_RSTBR ⁽⁷⁾	Res.	TIMECMP4	TIMECMP2	TIMECMP1	TIMDCMP4	TIMDCMP2	TIMDCMP1	TIMCCMP4	TIMCCMP2	TIMCCMP1	TIMBCMP4	TIMBCMP2	TIMBCMP1	EXTEVNT10	EXTEVNT9	EXTEVNT8	EXTEVNT7	EXTEVNT6	EXTEVNT5	EXTEVNT4	EXTEVNT3	EXTEVNT2	EXTEVNT1	MSTCMP4	MSTCMP3	MSTCMP2	MSTCMP1	MSTPER	CMP4	CMP2	UPDT	Res.
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0054	HRTIM_RSTCR ⁽⁷⁾	Res.	TIMECMP4	TIMECMP2	TIMECMP1	TIMDCMP4	TIMDCMP2	TIMDCMP1	TIMBCMP4	TIMBCMP2	TIMBCMP1	TIMACMP4	TIMACMP2	TIMACMP1	EXTEVNT10	EXTEVNT9	EXTEVNT8	EXTEVNT7	EXTEVNT6	EXTEVNT5	EXTEVNT4	EXTEVNT3	EXTEVNT2	EXTEVNT1	MSTCMP4	MSTCMP3	MSTCMP2	MSTCMP1	MSTPER	CMP4	CMP2	UPDT	Res.
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0054	HRTIM_RSTDR ⁽⁷⁾	Res.	TIMECMP4	TIMECMP2	TIMECMP1	TIMCCMP4	TIMCCMP2	TIMCCMP1	TIMBCMP4	TIMBCMP2	TIMBCMP1	TIMACMP4	TIMACMP2	TIMACMP1	EXTEVNT10	EXTEVNT9	EXTEVNT8	EXTEVNT7	EXTEVNT6	EXTEVNT5	EXTEVNT4	EXTEVNT3	EXTEVNT2	EXTEVNT1	MSTCMP4	MSTCMP3	MSTCMP2	MSTCMP1	MSTPER	CMP4	CMP2	UPDT	Res.
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0054	HRTIM_RSTER ⁽⁷⁾	Res.	TIMDCMP4	TIMDCMP2	TIMDCMP1	TIMCCMP4	TIMCCMP2	TIMCCMP1	TIMBCMP4	TIMBCMP2	TIMBCMP1	TIMACMP4	TIMACMP2	TIMACMP1	EXTEVNT10	EXTEVNT9	EXTEVNT8	EXTEVNT7	EXTEVNT6	EXTEVNT5	EXTEVNT4	EXTEVNT3	EXTEVNT2	EXTEVNT1	MSTCMP4	MSTCMP3	MSTCMP2	MSTCMP1	MSTPER	CMP4	CMP2	UPDT	Res.
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0058	HRTIM_CHPxR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	STRTPW [3:0]			CARDTY [2:0]		CARFRQ [3:0]					
	Reset value																						0	0	0	0	0	0	0	0	0	0	0
0x005C	HRTIM_CPT1ACR	TECMP2	TECMP1	TE1RST	TE1SET	TDCMP2	TDCMP1	TD1RST	TD1SET	TCCMP2	TCCMP1	TC1RST	TC1SET	TBCMP2	TBCMP1	TB1RST	TB1SET	Res.	Res.	Res.	Res.	EXEV10CPT	EXEV9CPT	EXEV8CPT	EXEV7CPT	EXEV6CPT	EXEV5CPT	EXEV4CPT	EXEV3CPT	EXEV2CPT	EXEV1CPT	UPDCPT	SWCPT
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0
0x005C	HRTIM_CPT1BCR	TECMP2	TECMP1	TE1RST	TE1SET	TDCMP2	TDCMP1	TD1RST	TD1SET	TCCMP2	TCCMP1	TC1RST	TC1SET	Res.	Res.	Res.	Res.	TACMP2	TACMP1	TA1RST	TA1SET	EXEV10CPT	EXEV9CPT	EXEV8CPT	EXEV7CPT	EXEV6CPT	EXEV5CPT	EXEV4CPT	EXEV3CPT	EXEV2CPT	EXEV1CPT	UPDCPT	SWCPT
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x005C	HRTIM_CPT1CCR	TECMP2	TECMP1	TE1RST	TE1SET	TDCMP2	TDCMP1	TD1RST	TD1SET	Res.	Res.	Res.	Res.	TBCMP2	TBCMP1	TB1RST	TB1SET	TACMP2	TACMP1	TA1RST	TA1SET	EXEV10CPT	EXEV9CPT	EXEV8CPT	EXEV7CPT	EXEV6CPT	EXEV5CPT	EXEV4CPT	EXEV3CPT	EXEV2CPT	EXEV1CPT	UPDCPT	SWCPT
	Reset value	0	0	0	0	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x005C	HRTIM_CPT1DCR	TECMP2	TECMP1	TE1RST	TE1SET	Res.	Res.	Res.	Res.	TCCMP2	TCCMP1	TC1RST	TC1SET	TBCMP2	TBCMP1	TB1RST	TB1SET	TACMP2	TACMP1	TA1RST	TA1SET	EXEV10CPT	EXEV9CPT	EXEV8CPT	EXEV7CPT	EXEV6CPT	EXEV5CPT	EXEV4CPT	EXEV3CPT	EXEV2CPT	EXEV1CPT	UPDCPT	SWCPT
	Reset value	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 325. HRTIM Register map and reset values: TIMx (x= A..E) (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x005C	HRTIM_CPT1ECR	Res.	Res.	Res.	Res.																													
	Reset value					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0060	HRTIM_CPT2ACR	TECMP2	TECMP1	TE1RST	TE1SET	TDCMP2	TDCMP1	TD1RST	TD1SET	TCOMP2	TCOMP1	TC1RST	TC1SET	TBCMP2	TBCMP1	TB1RST	TB1SET	Res.	Res.	Res.	Res.	EXEV10CPT	EXEV9CPT	EXEV8CPT	EXEV7CPT	EXEV6CPT	EXEV5CPT	EXEV4CPT	EXEV3CPT	EXEV2CPT	EXEV1CPT	UPDCPT	SWCPT	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0	0
0x0060	HRTIM_CPT2BCR	TECMP2	TECMP1	TE1RST	TE1SET	TDCMP2	TDCMP1	TD1RST	TD1SET	TCOMP2	TCOMP1	TC1RST	TC1SET	Res.	Res.	Res.	Res.	TACMP2	TACMP1	TACMP1	TA1RST	TA1SET	EXEV10CPT	EXEV9CPT	EXEV8CPT	EXEV7CPT	EXEV6CPT	EXEV5CPT	EXEV4CPT	EXEV3CPT	EXEV2CPT	EXEV1CPT	UPDCPT	SWCPT
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0060	HRTIM_CPT2CCR	TECMP2	TECMP1	TE1RST	TE1SET	TDCMP2	TDCMP1	TD1RST	TD1SET	Res.	Res.	Res.	Res.	TBCMP2	TBCMP1	TB1RST	TB1SET	TACMP2	TACMP1	TA1RST	TA1SET	EXEV10CPT	EXEV9CPT	EXEV8CPT	EXEV7CPT	EXEV6CPT	EXEV5CPT	EXEV4CPT	EXEV3CPT	EXEV2CPT	EXEV1CPT	UPDCPT	SWCPT	
	Reset value	0	0	0	0	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0060	HRTIM_CPT2DCR	TECMP2	TECMP1	TE1RST	TE1SET	Res.	Res.	Res.	Res.	TCOMP2	TCOMP1	TC1RST	TC1SET	TBCMP2	TBCMP1	TB1RST	TB1SET	TACMP2	TACMP1	TA1RST	TA1SET	EXEV10CPT	EXEV9CPT	EXEV8CPT	EXEV7CPT	EXEV6CPT	EXEV5CPT	EXEV4CPT	EXEV3CPT	EXEV2CPT	EXEV1CPT	UPDCPT	SWCPT	
	Reset value	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0060	HRTIM_CPT2ECR	Res.	Res.	Res.	Res.	TDCMP2	TDCMP1	TD1RST	TD1SET	TCOMP2	TCOMP1	TC1RST	TC1SET	TBCMP2	TBCMP1	TB1RST	TB1SET	TACMP2	TACMP1	TA1RST	TA1SET	EXEV10CPT	EXEV9CPT	EXEV8CPT	EXEV7CPT	EXEV6CPT	EXEV5CPT	EXEV4CPT	EXEV3CPT	EXEV2CPT	EXEV1CPT	UPDCPT	SWCPT	
	Reset value					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0064	HRTIM_OUTxR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DIDL2	CHP2	FAULT2[1:0]		IDLES2	IDLEM2	POL2	Res.	Res.	Res.	Res.	DLYPRT[2:0]			DLYPRTEN			DTEN	DIDL1	CHP1	FAULT1[1:0]			Res.	
	Reset value									0	0	0	0	0	0	0									0	0	0	0	0	0	0	0	0	0
0x0068	HRTIM_FLTxR	FLTCLK	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FLT5EN	FLT4EN	FLT3EN	FLT2EN	FLT1EN	
	Reset value	0																										0	0	0	0	0	0	0

1. This register can be preloaded (see [Table 309 on page 1415](#)).

Table 326. HRTIM Register map and reset values: Common functions

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0			
0x0000	HRTIM_CR1	Res.	Res.	Res.	Res.		AD4USRC[2:0]			AD3USRC[2:0]			AD2USRC[2:0]			AD1USRC[2:0]		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TEUDIS	TDUDIS	TCUDIS	TBUDIS	TAUDIS	MUDIS			
	Reset value					0	0	0	0	0	0	0	0	0	0	0	0											0	0	0	0	0	0			
0x0004	HRTIM_CR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TERST	TDRST	TCRST	TBRST	TARST	MRST	Res.	Res.	Res.	TESWU	TDSWU	TCSWU	TBSWU	TASWU	MSWU		
	Reset value																			0	0	0	0	0	0				0	0	0	0	0	0		
0x0008	HRTIM_ISR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BMPER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SYSFLT	FLT5	FLT4	FLT3	FLT2	FLT1			
	Reset value															0												0	0	0	0	0	0	0		
0x000C	HRTIM_ICR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BMPERC	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SYSFLTC	FLT5C	FLT4C	FLT3C	FLT2C	FLT1C			
	Reset value															0												0	0	0	0	0	0	0		
0x0010	HRTIM_IER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BMPERIE	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SYSFLTIE	FLT5IE	FLT4IE	FLT3IE	FLT2IE	FLT1IE			
	Reset value															0												0	0	0	0	0	0	0		
0x0014	HRTIM_OENR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TC2OEN	TC1OEN	TB2OEN	TB1OEN	TA2OEN	TA1OEN			
	Reset value																										0	0	0	0	0	0	0	0		
0x0018	HRTIM_DISR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TC2ODIS	TC1ODIS	TB2ODIS	TB1ODIS	TA2ODIS	TA1ODIS			
	Reset value																										0	0	0	0	0	0	0	0		
0x001C	HRTIM_ODSR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TC2ODS	TC1ODS	TB2ODS	TB1ODS	TA2ODS	TA1ODS			
	Reset value																										0	0	0	0	0	0	0	0		
0x0020	HRTIM_BMCR	BMSTAT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TEBM	TDBM	TCBM	TBBM	TABM	MTBM	Res.	Res.	Res.	Res.	Res.	BMPREN	BMPRSC[3:0]			BMCLK[3:0]									
	Reset value	0										0	0	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0			
0x0024	HRTIM_BMTRG	OCHPEV	Res.	Res.	Res.	Res.	TECMP2	TECMP1	TEREP	TERST	TDCMP2	TDCMP1	TDREP	TDRST	TCCMP2	TCCMP1	TCREP	TCRST	TBCMP2	TBCMP1	TBREP	TBRST	TACMP2	TACMP1	TAREP	TARST	MSTCMP4	MSTCMP3	MSTCMP2	MSTCMP1	MSTREP	MSTRST	SW			
	Reset value	0				0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x0028	HRTIM_BMCMPR ⁽¹⁾	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BMCMP[15:0]																		
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		

Table 326. HRTIM Register map and reset values: Common functions (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0			
0x002C	HRTIM_BMPER ⁽¹⁾	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	BMPER[15:0]																		
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x0030	HRTIM_EECR1	Res	Res	EE5FAST	EE5SNS[1:0]		EE5POL	EE5SRC[1:0]		EE4FAST	EE4SNS[1:0]		EE4POL	EE4SRC[1:0]		EE3FAST	EE3SNS[1:0]		EE3POL	EE3SRC[1:0]		EE2FAST	EE2SNS[1:0]		EE2POL	EE2SRC[1:0]		EE1FAST	EE1SNS[1:0]		EE1POL	EE1SRC[1:0]				
	Reset value			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x0034	HRTIM_EECR2	Res	Res	Res	EE10SNS[1:0]		EE10POL	EE10SRC[1:0]		Res	EE9SNS[1:0]		EE9POL	EE9SRC[1:0]		Res	EE8SNS[1:0]		EE8POL	EE8SRC[1:0]		Res	EE7SNS[1:0]		EE7POL	EE7SRC[1:0]		Res	EE6SNS[1:0]		EE6POL	EE6SRC[1:0]				
	Reset value				0	0	0	0	0		0	0	0	0	0		0	0	0	0	0		0	0	0	0		0	0	0	0	0	0			
0x0038	HRTIM_EECR3	Res	Res	Res	EE10SNS[1:0]		EE10POL	EE10SRC[1:0]		Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	EE6SRC[1:0]			
	Reset value				0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x003C	HRTIM_ADC1R ⁽¹⁾	AD1TEPER	AD1TEC4	AD1TEC3	AD1TEC2	AD1TDPER	AD1TDC4	AD1TDC3	AD1TDC2	AD1TCPER	AD1TCC4	AD1TCC3	AD1TCC2	AD1TBRST	AD1TBPER	AD1TBC4	AD1TBC3	AD1TBC2	AD1TARST	AD1TAPER	AD1TAC4	AD1TAC3	AD1TAC2	AD1TEEV5	AD1TEEV4	AD1TEEV3	AD1TEEV2	AD1TEEV1	AD1MPER	AD1MC4	AD1MC3	AD1MC2	AD1MC1			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x0040	HRTIM_ADC2R ⁽¹⁾	AD2TERST	AD2TEC4	AD2TEC3	AD2TEC2	AD2TDRST	AD2TDPER	AD2TDC4	AD2TDC3	AD2TDC2	AD2TCRST	AD2TCC4	AD2TCC3	AD2TCC2	AD2TBC4	AD2TBC3	AD2TBC2	AD2TBC2	AD2TAPER	AD2TAC4	AD2TAC3	AD2TAC2	AD2TEEV10	AD2TEEV9	AD2TEEV8	AD2TEEV7	AD2TEEV6	AD2MMPER	AD2MC4	AD2MC3	AD2MC2	AD2MC1				
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x0044	HRTIM_ADC3R ⁽¹⁾	ADC3TEPER	AD1TEC4	AD1TEC3	AD1TEC2	AD1TDPER	AD1TDC4	AD1TDC3	AD1TDC2	AD1TCPER	AD1TCC4	AD1TCC3	AD1TCC2	AD1TBRST	AD1TBPER	AD1TBC4	AD1TBC3	AD1TBC2	AD1TARST	AD1TAPER	AD1TAC4	AD1TAC3	AD1TAC2	AD1TEEV5	AD1TEEV4	AD1TEEV3	AD1TEEV2	AD1TEEV1	AD1MPER	AD1MC4	AD1MC3	AD1MC2	AD1MC1			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x0048	HRTIM_ADC4R ⁽¹⁾	AD2TERST	AD2TEC4	AD2TEC3	AD2TEC2	AD2TDRST	AD2TDPER	AD2TDC4	AD2TDC3	AD2TDC2	AD2TCRST	AD2TC4	AD2TCC3	AD2TCC2	AD2TBC4	AD2TBC3	AD2TBC2	AD2TBC2	AD2TAPER	AD2TAC4	AD2TAC3	AD2TAC2	AD2TEEV10	AD2TEEV9	AD2TEEV8	AD2TEEV7	AD2TEEV6	AD2MMPER	AD2MC4	AD2MC3	AD2MC2	AD2MC1				
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x004C	Reserved	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res		
0x0050	HRTIM_FLTINxR1	FLT4LCK	FLT4F[3:0]				FLT4SRC	FLT4P	FLT4E	FLT3LCK	FLT3F[3:0]				FLT3SRC	FLT3P	FLT3E	FLT2LCK	FLT2F[3:0]				FLT2SRC	FLT2P	FLT2E	FLT1LCK				FLT1F[3:0]				FLT1SRC	FLT1P	FLT1E
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Table 326. HRTIM Register map and reset values: Common functions (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x0054	HRTIM_FLTINxR2	Res.	Res.	Res.	Res.	Res.	Res.	0	0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FLT5LCK	FLT5F[3:0]				FLT5SRC	FLT5P	FLT5E
	Reset value							0	0																	0	0	0	0	0	0	0	0	0
0x0058	HRTIM_BDMUPDR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MCMP4	MCMP3	MCMP2	MCMP1	MREP	MPER	MCNT	MDIER	MICR	MCR	
	Reset value																							0	0	0	0	0	0	0	0	0	0	
0x005C	HRTIM_BDTAUPR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIMAFLTR	TIMAOUTR	TIMACHPR	TIMARSTR	TIMAEEFR2	TIMAEEFR1	TIMARST2R	TIMASET2R	TIMARST1R	TIMASET1R	TIMADTxR	TIMACMP4	TIMACMP3	TIMACMP2	TIMACMP1	TIMAREP	TIMAPER	TIMACNT	TIMADIER	TIMAICR	TIMACR	
	Reset value												0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x0060	HRTIM_BDTBUPR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIMBFLTR	TIMBOUTR	TIMBCHPR	TIMBRSTR	TIMBEEFR2	TIMBEEFR1	TIMBRST2R	TIMBSET2R	TIMBRST1R	TIMBSET1R	TIMBDTxR	TIMBCMP4	TIMBCMP3	TIMBCMP2	TIMBCMP1	TIMBREP	TIMBPER	TIMBCNT	TIMBDIER	TIMBICR	TIMBCR	
	Reset value												0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x0064	HRTIM_BDTCUPR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIMCFLTR	TIMCOUTR	TIMCCHPR	TIMCRSTR	TIMCEEFR2	TIMCEEFR1	TIMCRST2R	TIMCSET2R	TIMCRST1R	TIMCSET1R	TIMCDTxR	TIMCCMP4	TIMCCMP3	TIMCCMP2	TIMCCMP1	TIMCREP	TIMCPR	TIMCCNT	TIMCDIER	TIMCICR	TIMCCR	
	Reset value												0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x0068	HRTIM_BDTPUPR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIMDFLTR	TIMDOUTr	TIMDCHPR	TIMDRSTR	TIMDEEFR2	TIMDEEFR1	TIMDRST2R	TIMDSET2R	TIMDRST1R	TIMDSET1R	TIMDDTxR	TIMDCMP4	TIMDCMP3	TIMDCMP2	TIMDCMP1	TIMDREP	TIMDPER	TIMDCNT	TIMDDIER	TIMDICR	TIMDCR	
	Reset value												0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x006C	HRTIM_BDTEUPR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIMEFLTR	TIMEOUTR	TIMECHPR	TIMERSTR	TIMEEEFR2	TIMEEEFR1	TIMERST2R	TIMESET2R	TIMERST1R	TIMESET1R	TIMEDTxR	TIMECMP4	TIMECMP3	TIMECMP2	TIMECMP1	TIMEREP	TIMEPER	TIMECNT	TIMEDIER	TIMEICR	TIMECR	
	Reset value												0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x0070	HRTIM_BDMADR	BDMADR[31:0]																																
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

1. This register can be preloaded (see [Table 309 on page 1415](#)).

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.

38 Advanced-control timers (TIM1/TIM8)

38.1 TIM1/TIM8 introduction

The advanced-control timers (TIM1/TIM8) consist of a 16-bit auto-reload counter driven by a programmable prescaler.

It may be used for a variety of purposes, including measuring the pulse lengths of input signals (input capture) or generating output waveforms (output compare, PWM, complementary PWM with dead-time insertion).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

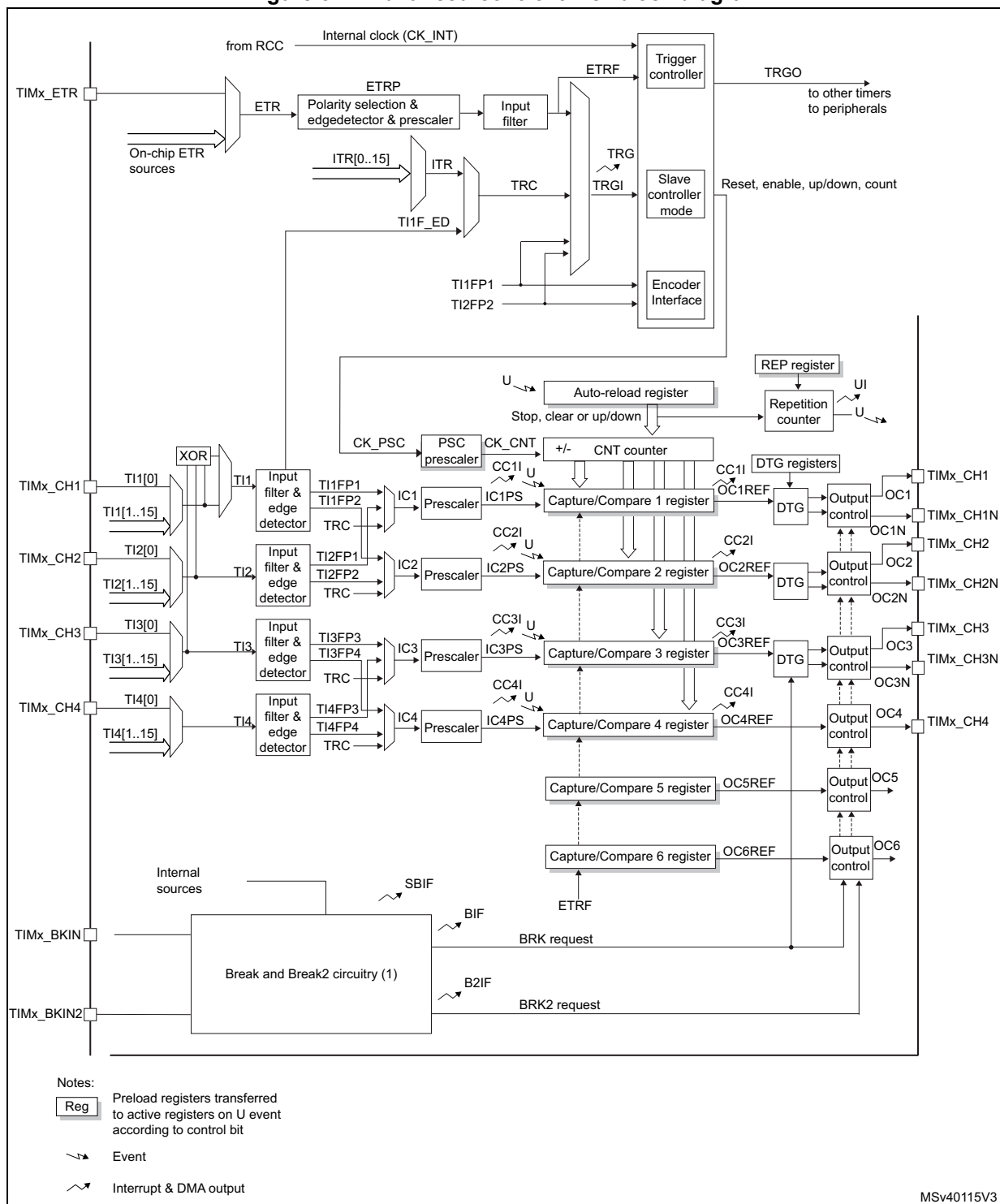
The advanced-control (TIM1/TIM8) and general-purpose (TIMy) timers are completely independent, and do not share any resources. They can be synchronized together as described in [Section 38.3.26: Timer synchronization](#).

38.2 TIM1/TIM8 main features

TIM1/TIM8 timer features include:

- 16-bit up, down, up/down auto-reload counter.
- 16-bit programmable prescaler allowing dividing (also “on the fly”) the counter clock frequency either by any factor between 1 and 65536.
- Up to 6 independent channels for:
 - Input Capture (but channels 5 and 6)
 - Output Compare
 - PWM generation (Edge and Center-aligned Mode)
 - One-pulse mode output
- Complementary outputs with programmable dead-time
- Synchronization circuit to control the timer with external signals and to interconnect several timers together.
- Repetition counter to update the timer registers only after a given number of cycles of the counter.
- 2 break inputs to put the timer’s output signals in a safe user selectable configuration.
- Interrupt/DMA generation on the following events:
 - Update: counter overflow/underflow, counter initialization (by software or internal/external trigger)
 - Trigger event (counter start, stop, initialization or count by internal/external trigger)
 - Input capture
 - Output compare
- Supports incremental (quadrature) encoder and Hall-sensor circuitry for positioning purposes
- Trigger input for external clock or cycle-by-cycle current management

Figure 344. Advanced-control timer block diagram



1. See [Figure 387: Break and Break2 circuitry overview](#) for details

38.3 TIM1/TIM8 functional description

38.3.1 Time-base unit

The main block of the programmable advanced-control timer is a 16-bit counter with its related auto-reload register. The counter can count up, down or both up and down. The counter clock can be divided by a prescaler.

The counter, the auto-reload register and the prescaler register can be written or read by software. This is true even when the counter is running.

The time-base unit includes:

- Counter register (TIMx_CNT)
- Prescaler register (TIMx_PSC)
- Auto-reload register (TIMx_ARR)
- Repetition counter register (TIMx_RCR)

The auto-reload register is preloaded. Writing to or reading from the auto-reload register accesses the preload register. The content of the preload register are transferred into the shadow register permanently or at each update event (UEV), depending on the auto-reload preload enable bit (ARPE) in TIMx_CR1 register. The update event is sent when the counter reaches the overflow (or underflow when downcounting) and if the UDIS bit equals 0 in the TIMx_CR1 register. It can also be generated by software. The generation of the update event is described in detailed for each configuration.

The counter is clocked by the prescaler output CK_CNT, which is enabled only when the counter enable bit (CEN) in TIMx_CR1 register is set (refer also to the slave mode controller description to get more details on counter enabling).

Note that the counter starts counting 1 clock cycle after setting the CEN bit in the TIMx_CR1 register.

Prescaler description

The prescaler can divide the counter clock frequency by any factor between 1 and 65536. It is based on a 16-bit counter controlled through a 16-bit register (in the TIMx_PSC register). It can be changed on the fly as this control register is buffered. The new prescaler ratio is taken into account at the next update event.

Figure 345 and *Figure 346* give some examples of the counter behavior when the prescaler ratio is changed on the fly:

Figure 345. Counter timing diagram with prescaler division change from 1 to 2

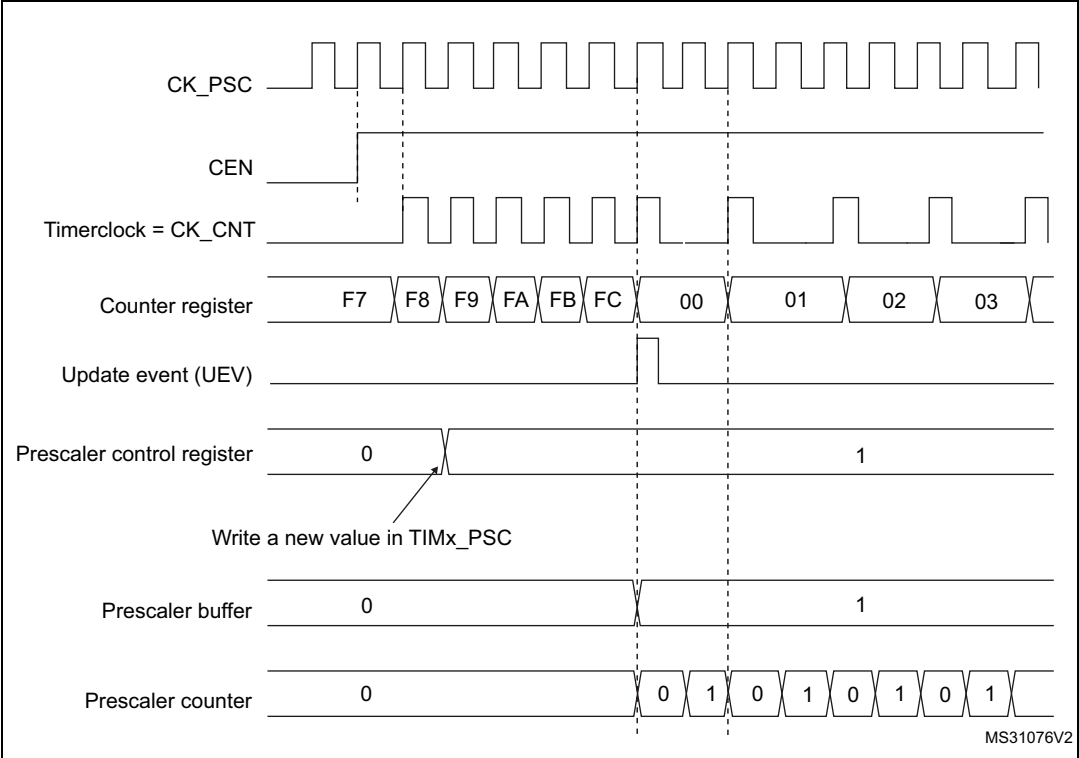
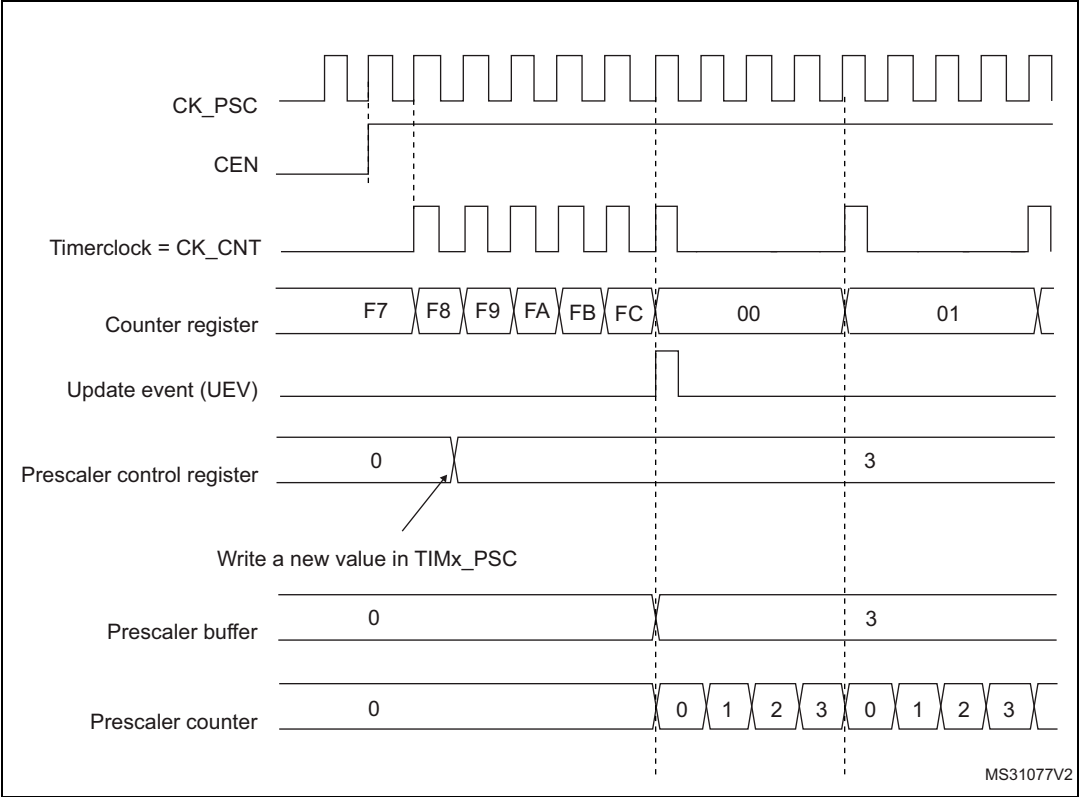


Figure 346. Counter timing diagram with prescaler division change from 1 to 4



38.3.2 Counter modes

Upcounting mode

In upcounting mode, the counter counts from 0 to the auto-reload value (content of the TIMx_ARR register), then restarts from 0 and generates a counter overflow event.

If the repetition counter is used, the update event (UEV) is generated after upcounting is repeated for the number of times programmed in the repetition counter register (TIMx_RCR) + 1. Else the update event is generated at each counter overflow.

Setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller) also generates an update event.

The UEV event can be disabled by software by setting the UDIS bit in the TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until the UDIS bit has been written to 0. However, the counter restarts from 0, as well as the counter of the prescaler (but the prescale rate does not change). In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt or DMA request is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The repetition counter is reloaded with the content of TIMx_RCR register,
- The auto-reload shadow register is updated with the preload value (TIMx_ARR),
- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register).

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR=0x36.

Figure 347. Counter timing diagram, internal clock divided by 1

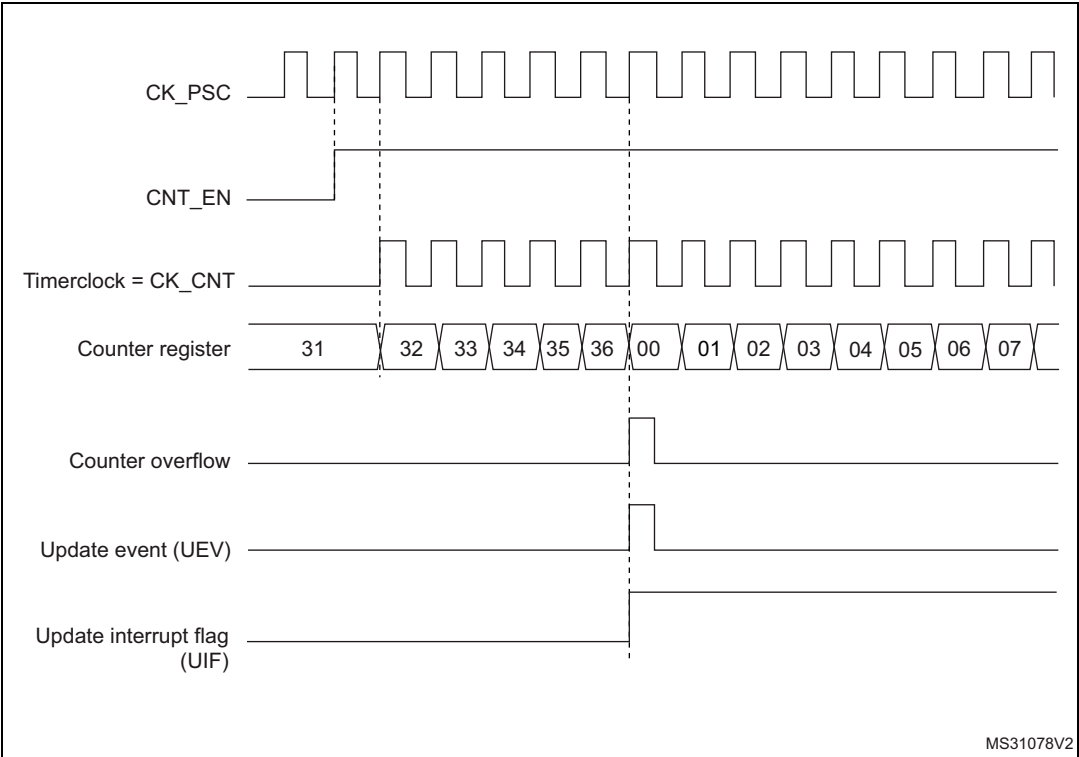


Figure 348. Counter timing diagram, internal clock divided by 2

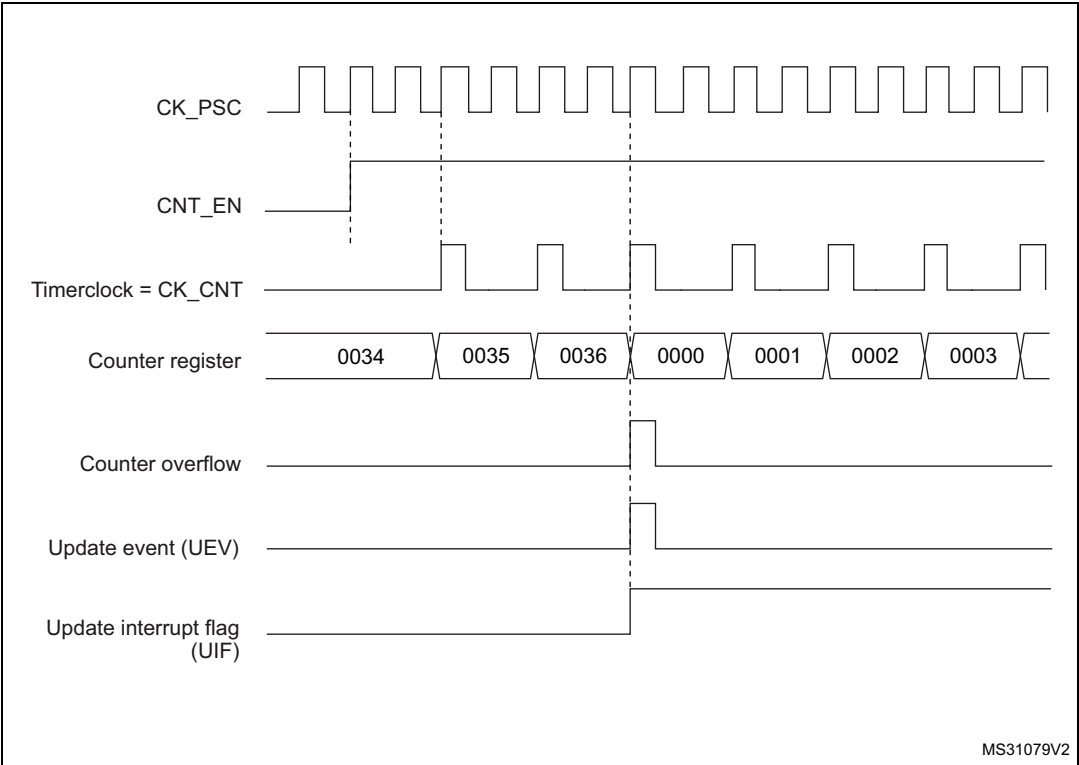


Figure 349. Counter timing diagram, internal clock divided by 4

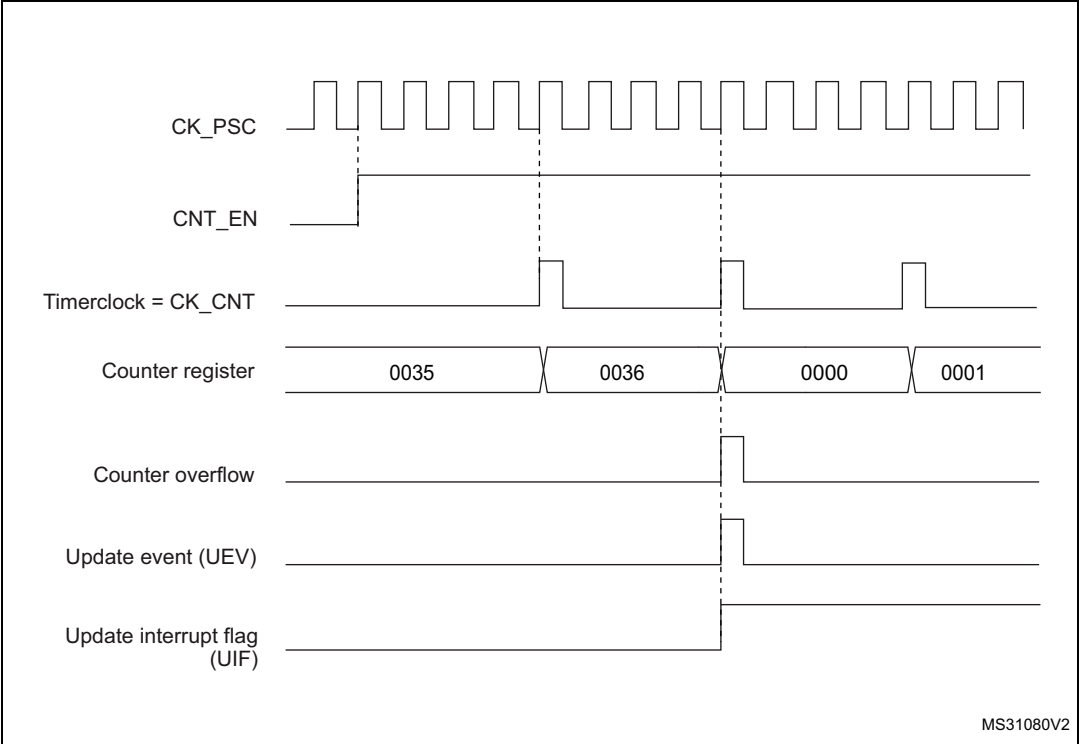


Figure 350. Counter timing diagram, internal clock divided by N

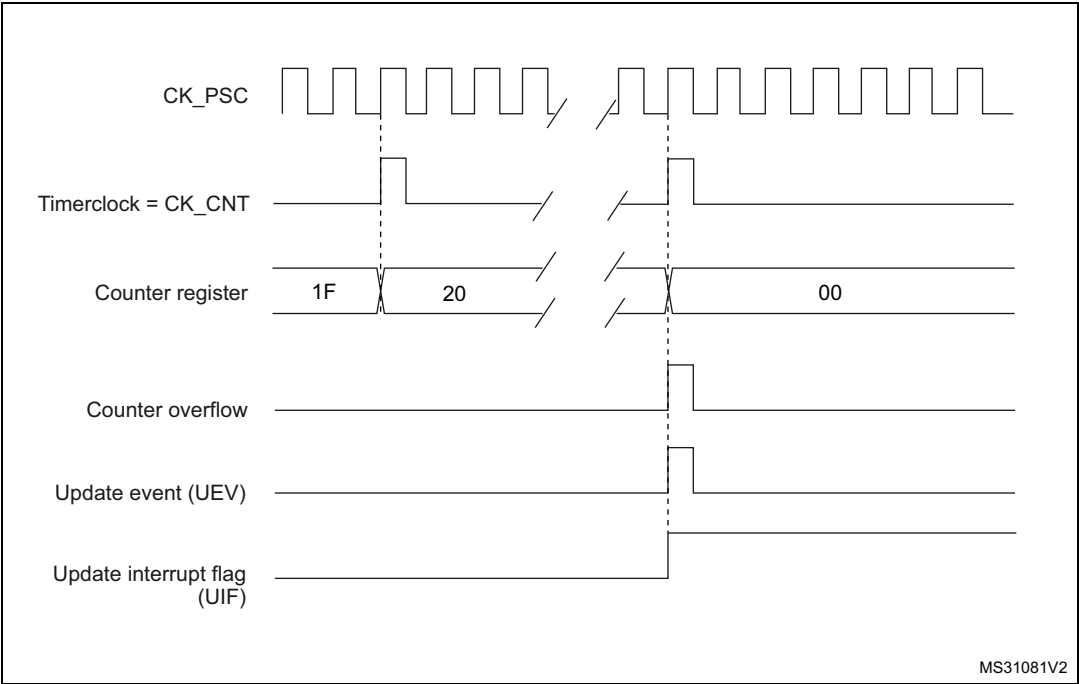


Figure 351. Counter timing diagram, update event when ARPE=0 (TIMx_ARR not preloaded)

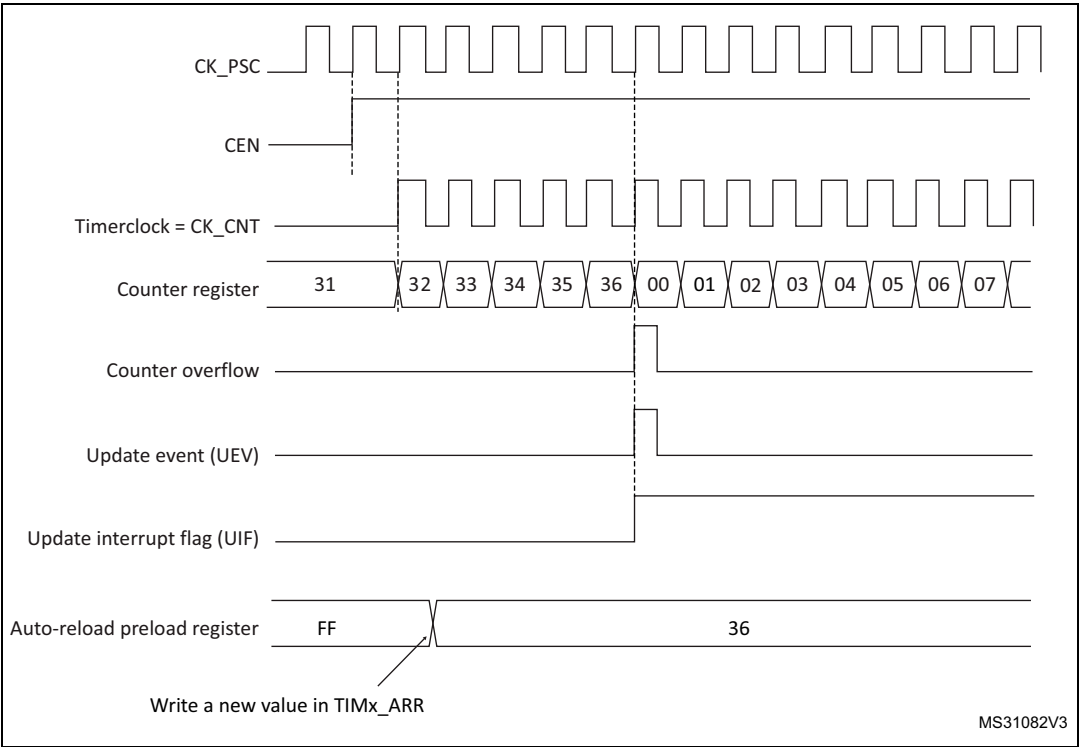
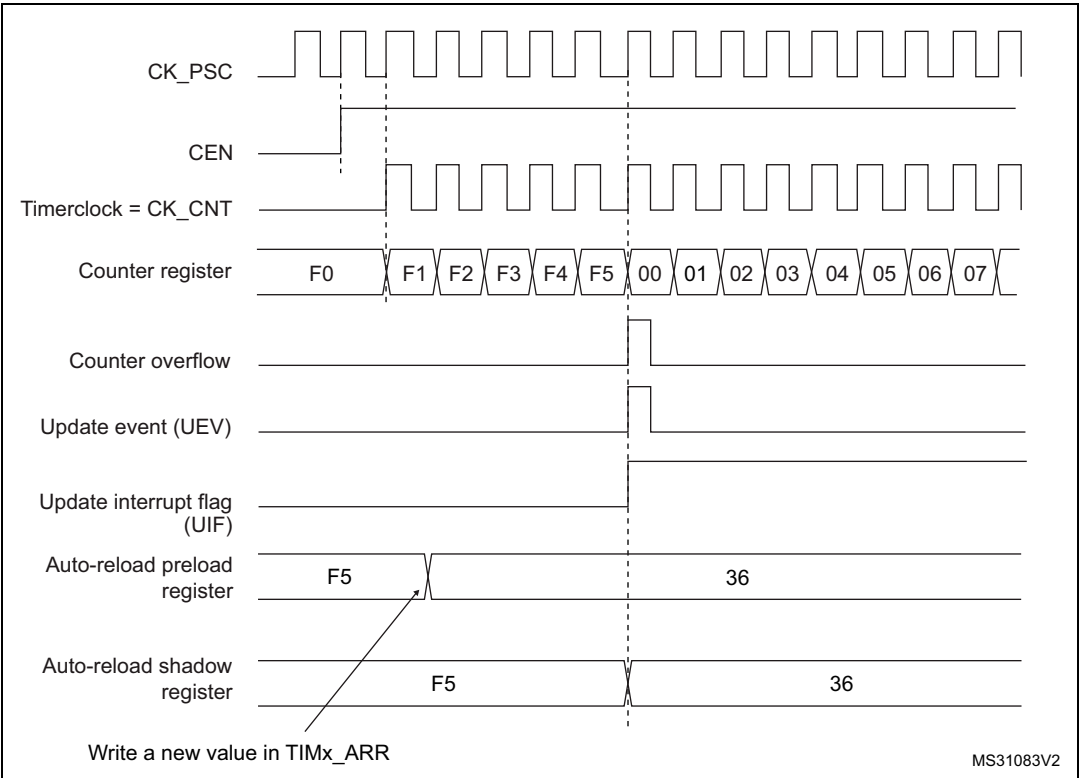


Figure 352. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded)



Downcounting mode

In downcounting mode, the counter counts from the auto-reload value (content of the TIMx_ARR register) down to 0, then restarts from the auto-reload value and generates a counter underflow event.

If the repetition counter is used, the update event (UEV) is generated after downcounting is repeated for the number of times programmed in the repetition counter register (TIMx_RCR) + 1. Else the update event is generated at each counter underflow.

Setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller) also generates an update event.

The UEV update event can be disabled by software by setting the UDIS bit in TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until UDIS bit has been written to 0. However, the counter restarts from the current auto-reload value, whereas the counter of the prescaler restarts from 0 (but the prescale rate doesn't change).

In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt or DMA request is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The repetition counter is reloaded with the content of TIMx_RCR register.
- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register).
- The auto-reload active register is updated with the preload value (content of the TIMx_ARR register). Note that the auto-reload is updated before the counter is reloaded, so that the next period is the expected one.

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR=0x36.

Figure 353. Counter timing diagram, internal clock divided by 1

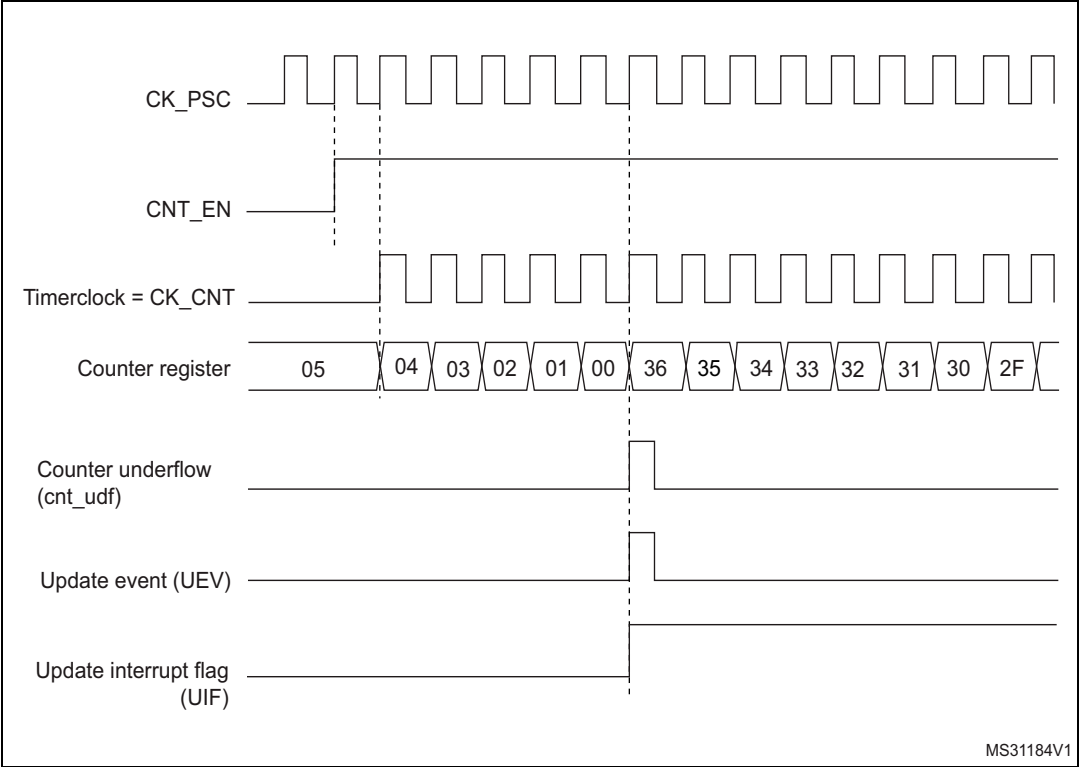


Figure 354. Counter timing diagram, internal clock divided by 2

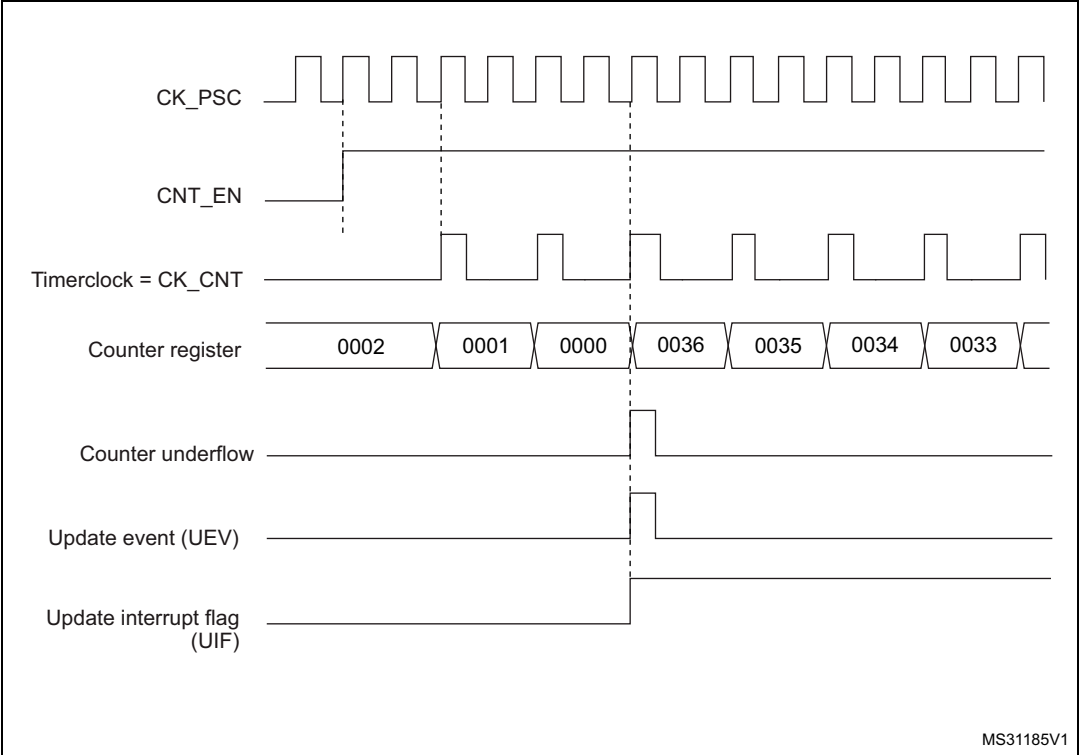


Figure 355. Counter timing diagram, internal clock divided by 4

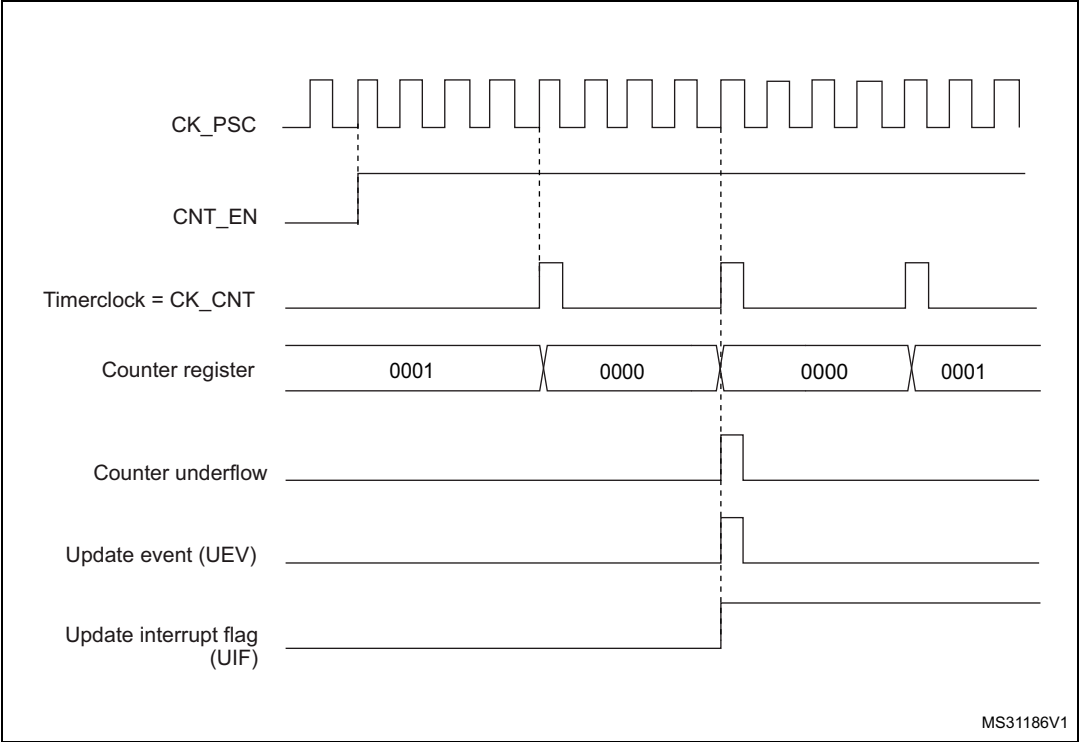


Figure 356. Counter timing diagram, internal clock divided by N

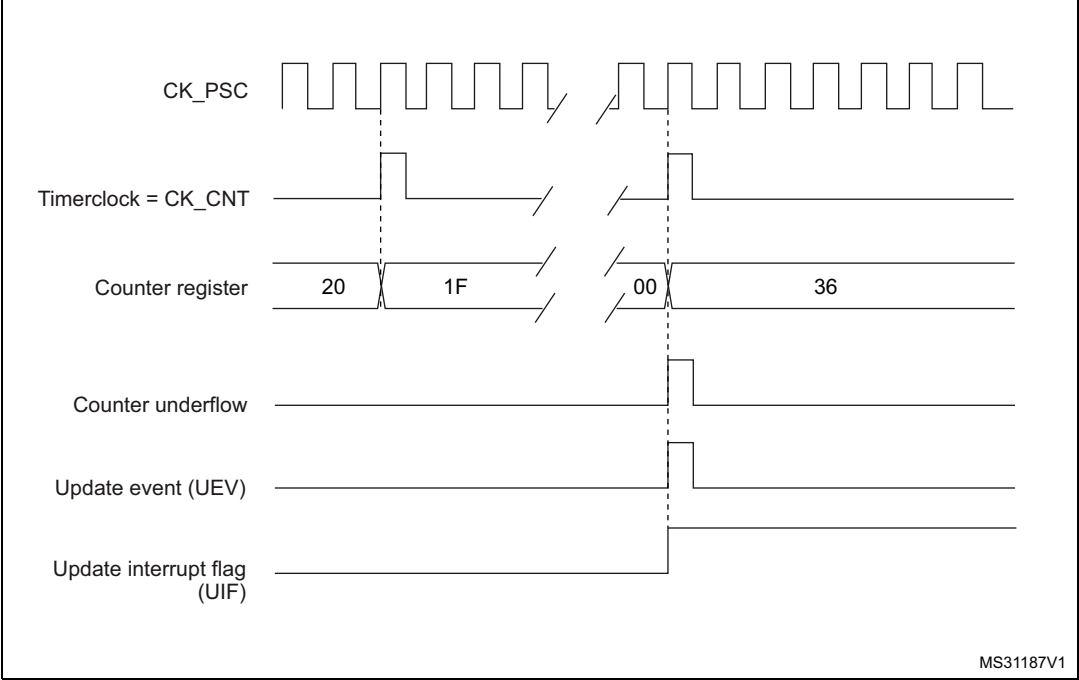
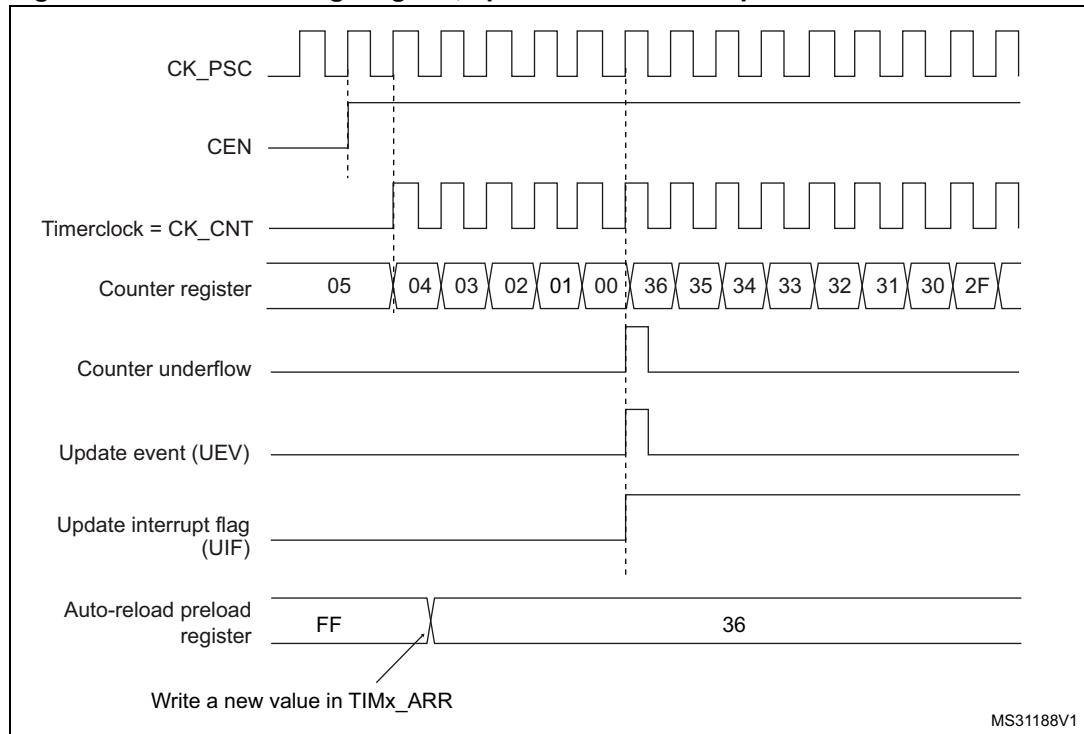


Figure 357. Counter timing diagram, update event when repetition counter is not used

Center-aligned mode (up/down counting)

In center-aligned mode, the counter counts from 0 to the auto-reload value (content of the TIMx_ARR register) – 1, generates a counter overflow event, then counts from the auto-reload value down to 1 and generates a counter underflow event. Then it restarts counting from 0.

Center-aligned mode is active when the CMS bits in TIMx_CR1 register are not equal to '00'. The Output compare interrupt flag of channels configured in output is set when: the counter counts down (Center aligned mode 1, CMS = "01"), the counter counts up (Center aligned mode 2, CMS = "10") the counter counts up and down (Center aligned mode 3, CMS = "11").

In this mode, the DIR direction bit in the TIMx_CR1 register cannot be written. It is updated by hardware and gives the current direction of the counter.

The update event can be generated at each counter overflow and at each counter underflow or by setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller) also generates an update event. In this case, the counter restarts counting from 0, as well as the counter of the prescaler.

The UEV update event can be disabled by software by setting the UDIS bit in the TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until UDIS bit has been written to 0. However, the counter continues counting up and down, based on the current auto-reload value.

In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an UEV update event but without setting the UIF flag (thus no interrupt or

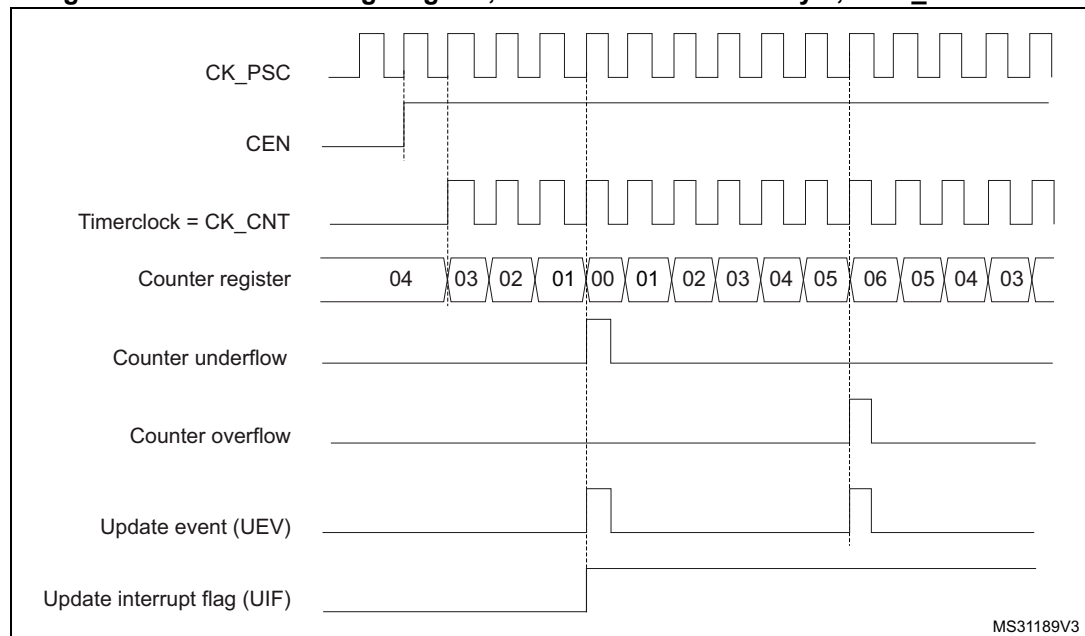
DMA request is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The repetition counter is reloaded with the content of TIMx_RCR register
- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register)
- The auto-reload active register is updated with the preload value (content of the TIMx_ARR register). Note that if the update source is a counter overflow, the auto-reload is updated before the counter is reloaded, so that the next period is the expected one (the counter is loaded with the new value).

The following figures show some examples of the counter behavior for different clock frequencies.

Figure 358. Counter timing diagram, internal clock divided by 1, TIMx_ARR = 0x6



1. Here, center-aligned mode 1 is used (for more details refer to [Section 38.4: TIM1/TIM8 registers](#)).

Figure 359. Counter timing diagram, internal clock divided by 2

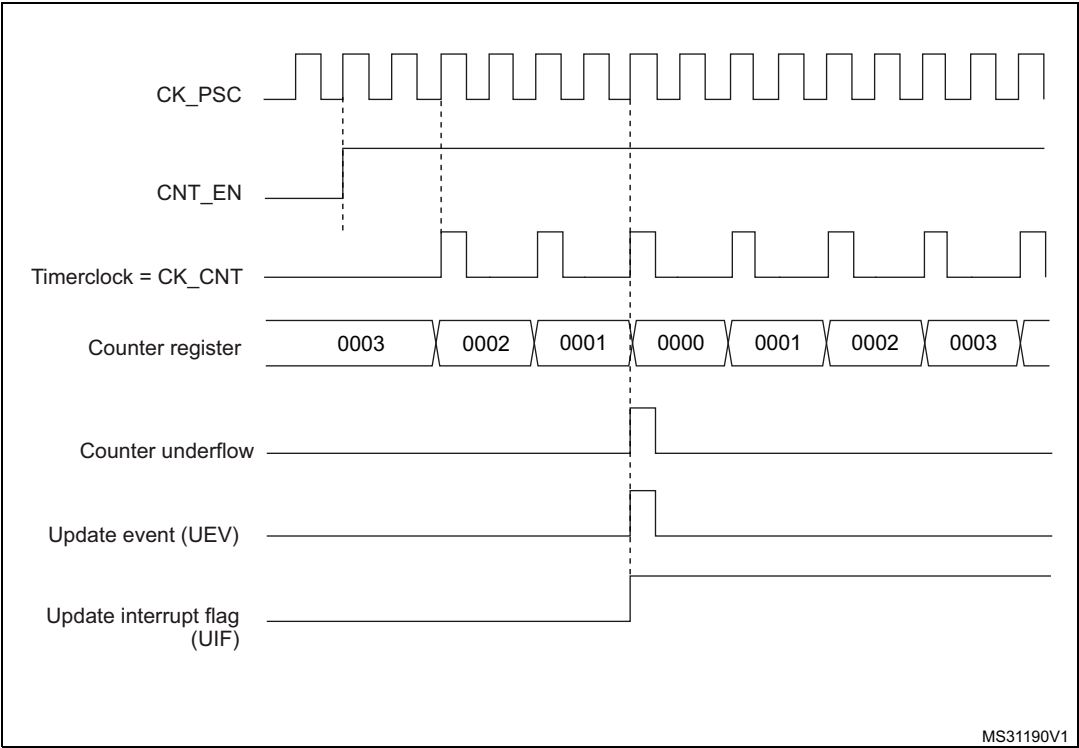


Figure 360. Counter timing diagram, internal clock divided by 4, TIMx_ARR=0x36

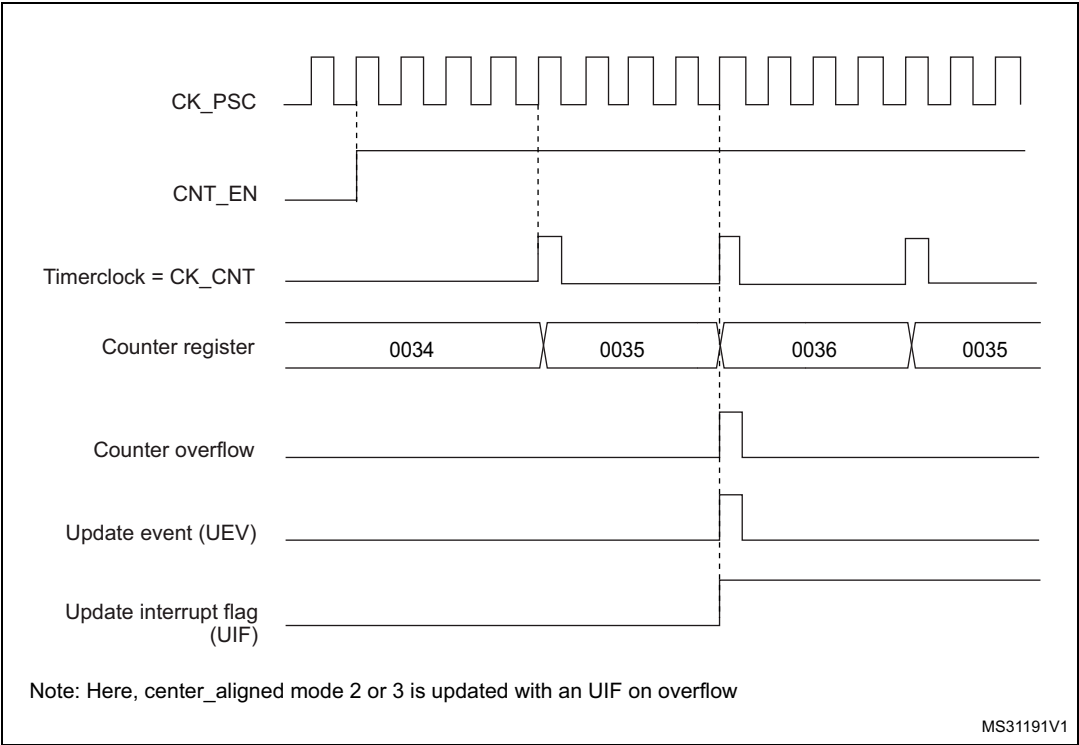


Figure 361. Counter timing diagram, internal clock divided by N

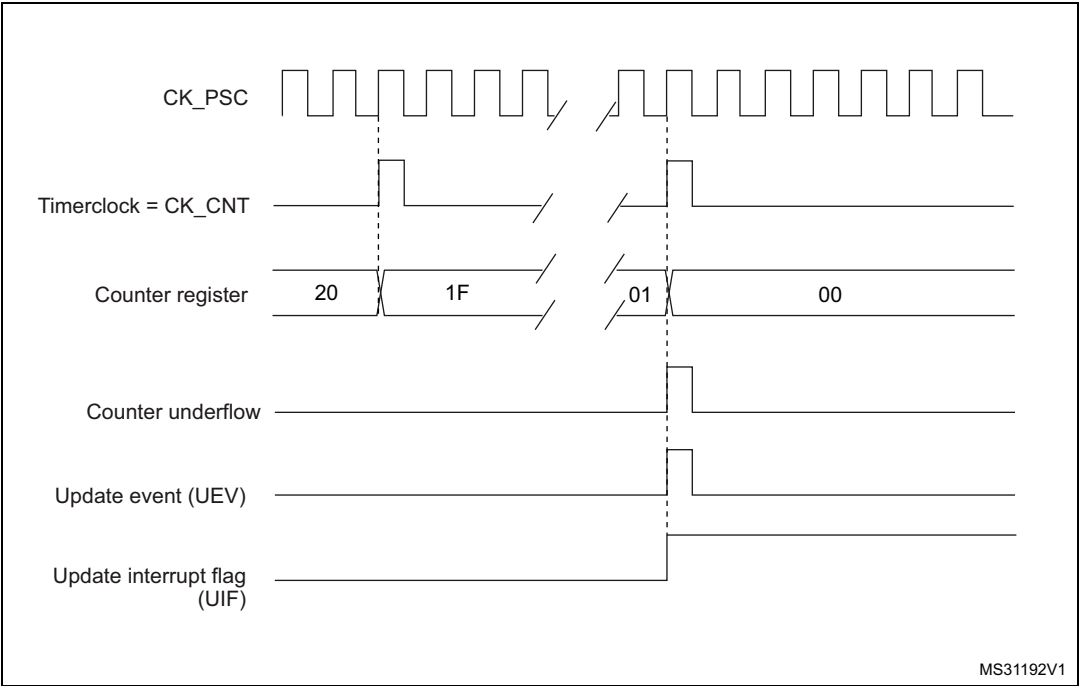


Figure 362. Counter timing diagram, update event with ARPE=1 (counter underflow)

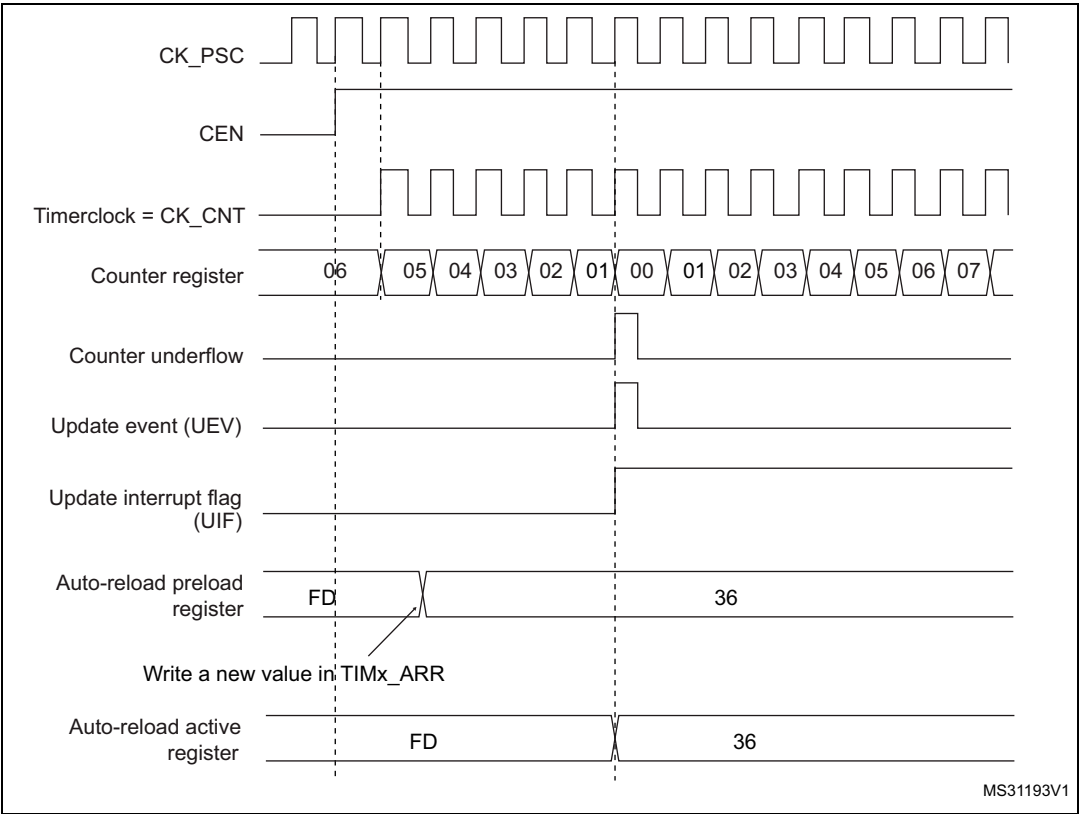
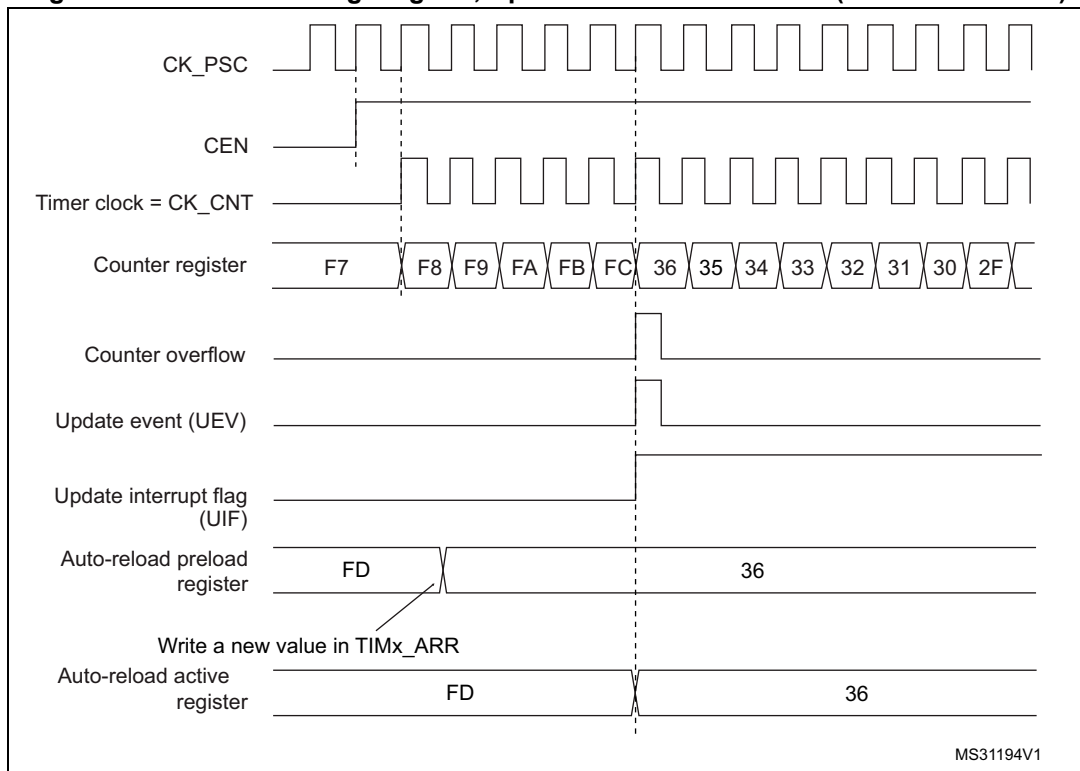


Figure 363. Counter timing diagram, Update event with ARPE=1 (counter overflow)

38.3.3 Repetition counter

[Section 38.3.1: Time-base unit](#) describes how the update event (UEV) is generated with respect to the counter overflows/underflows. It is actually generated only when the repetition counter has reached zero. This can be useful when generating PWM signals.

This means that data are transferred from the preload registers to the shadow registers (TIMx_ARR auto-reload register, TIMx_PSC prescaler register, but also TIMx_CCRx capture/compare registers in compare mode) every N+1 counter overflows or underflows, where N is the value in the TIMx_RCR repetition counter register.

The repetition counter is decremented:

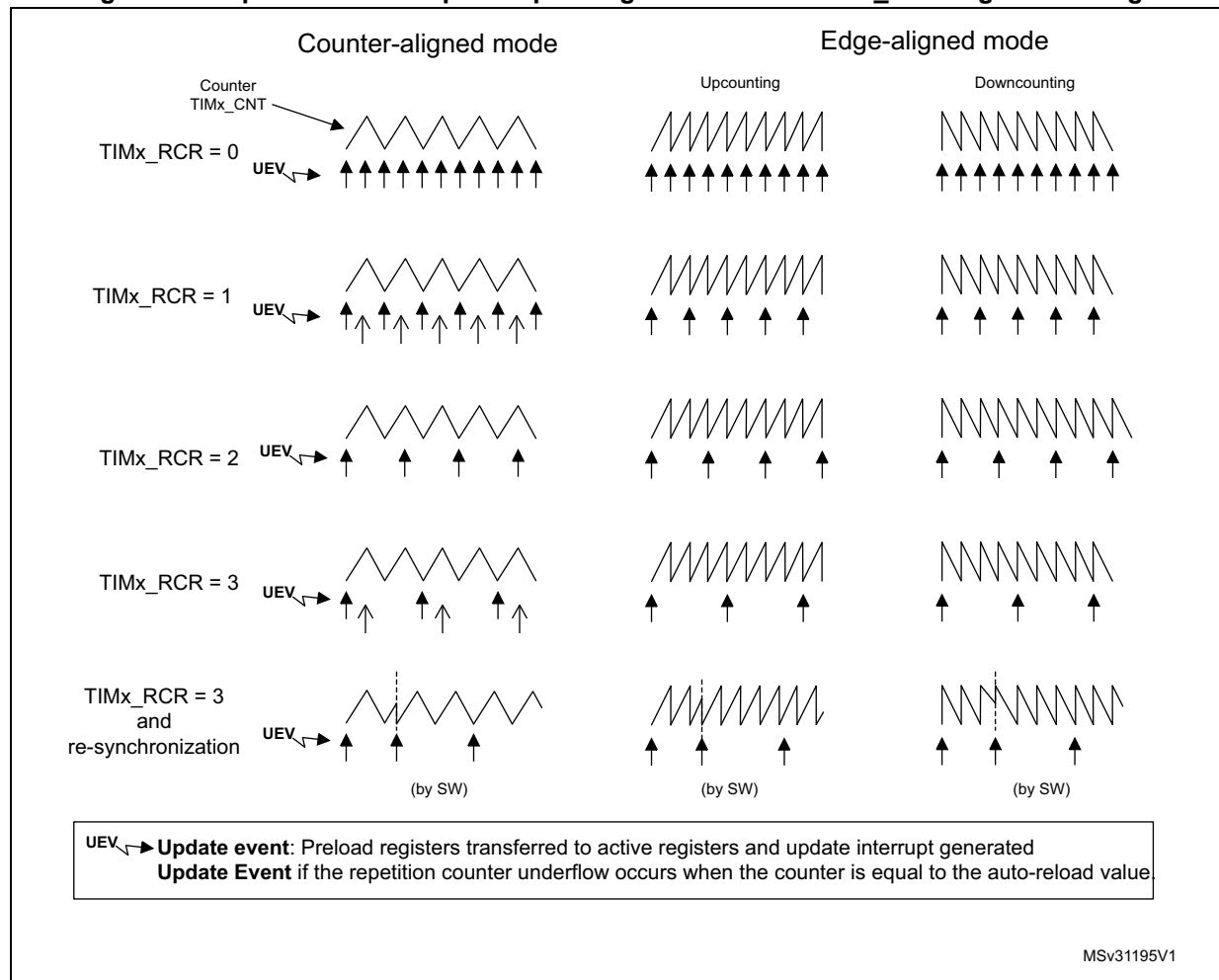
- At each counter overflow in upcounting mode,
 - At each counter underflow in downcounting mode,
 - At each counter overflow and at each counter underflow in center-aligned mode.
- Although this limits the maximum number of repetition to 32768 PWM cycles, it makes it possible to update the duty cycle twice per PWM period. When refreshing compare registers only once per PWM period in center-aligned mode, maximum resolution is $2 \times T_{ck}$, due to the symmetry of the pattern.

The repetition counter is an auto-reload type; the repetition rate is maintained as defined by the TIMx_RCR register value (refer to [Figure 364](#)). When the update event is generated by software (by setting the UG bit in TIMx_EGR register) or by hardware through the slave mode controller, it occurs immediately whatever the value of the repetition counter is and the repetition counter is reloaded with the content of the TIMx_RCR register.

In Center aligned mode, for odd values of RCR, the update event occurs either on the overflow or on the underflow depending on when the RCR register was written and when the counter was launched: if the RCR was written before launching the counter, the UEV occurs on the underflow. If the RCR was written after launching the counter, the UEV occurs on the overflow.

For example, for RCR = 3, the UEV is generated each 4th overflow or underflow event depending on when the RCR was written.

Figure 364. Update rate examples depending on mode and TIMx_RCR register settings



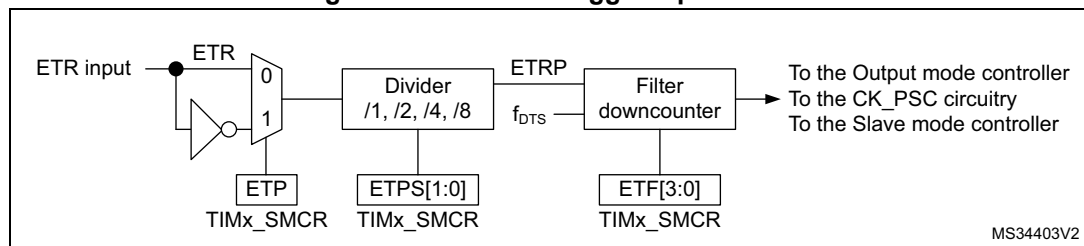
38.3.4 External trigger input

The timer features an external trigger input ETR. It can be used as:

- external clock (external clock mode 2, see [Section 38.3.5](#))
- trigger for the slave mode (see [Section 38.3.26](#))
- PWM reset input for cycle-by-cycle current regulation (see [Section 38.3.7](#))

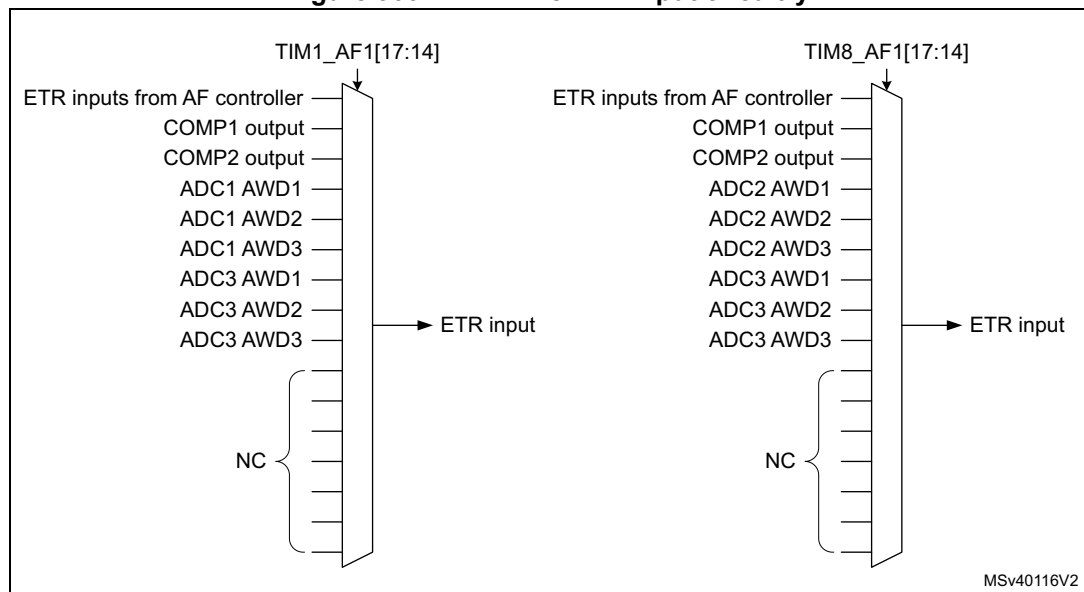
[Figure 365](#) below describes the ETR input conditioning. The input polarity is defined with the ETP bit in TIMxSMCR register. The trigger can be prescaled with the divider programmed by the ETPS[1:0] bitfield and digitally filtered with the ETF[3:0] bitfield.

Figure 365. External trigger input block



The ETR input comes from multiple sources: input pins (default configuration), comparator outputs and analog watchdogs. The selection is done with the ETRSEL[3:0] bitfield.

Figure 366. TIM1/TIM8 ETR input circuitry



38.3.5 Clock selection

The counter clock can be provided by the following clock sources:

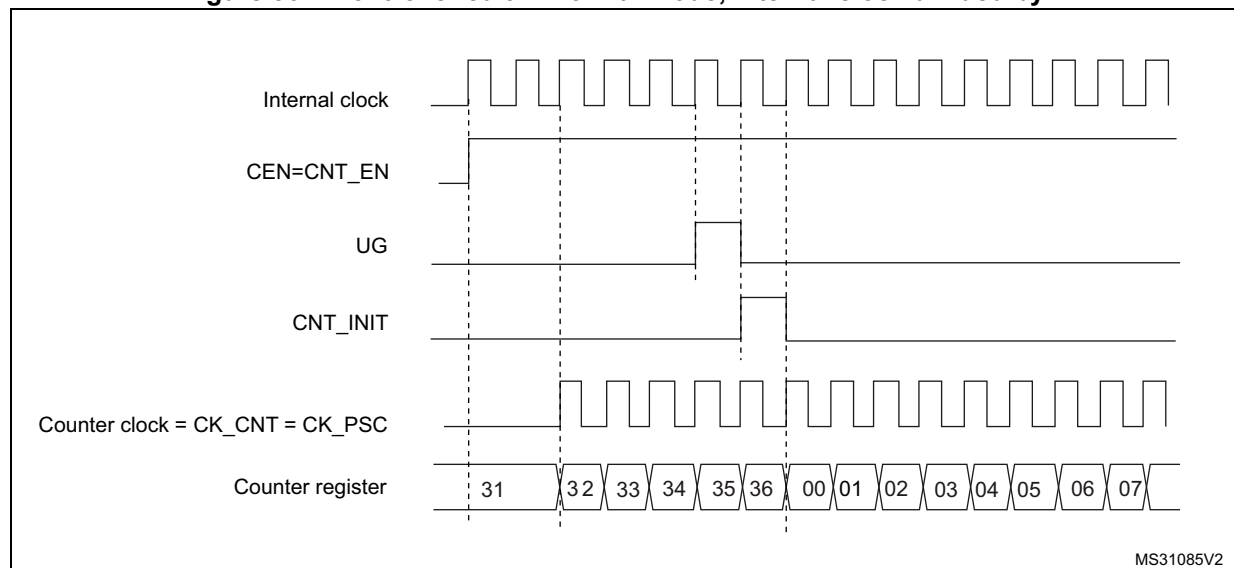
- Internal clock (CK_INT)
- External clock mode1: external input pin
- External clock mode2: external trigger input ETR
- Encoder mode

Internal clock source (CK_INT)

If the slave mode controller is disabled (SMS=000), then the CEN, DIR (in the TIMx_CR1 register) and UG bits (in the TIMx_EGR register) are actual control bits and can be changed only by software (except UG which remains cleared automatically). As soon as the CEN bit is written to 1, the prescaler is clocked by the internal clock CK_INT.

Figure 367 shows the behavior of the control circuit and the upcounter in normal mode, without prescaler.

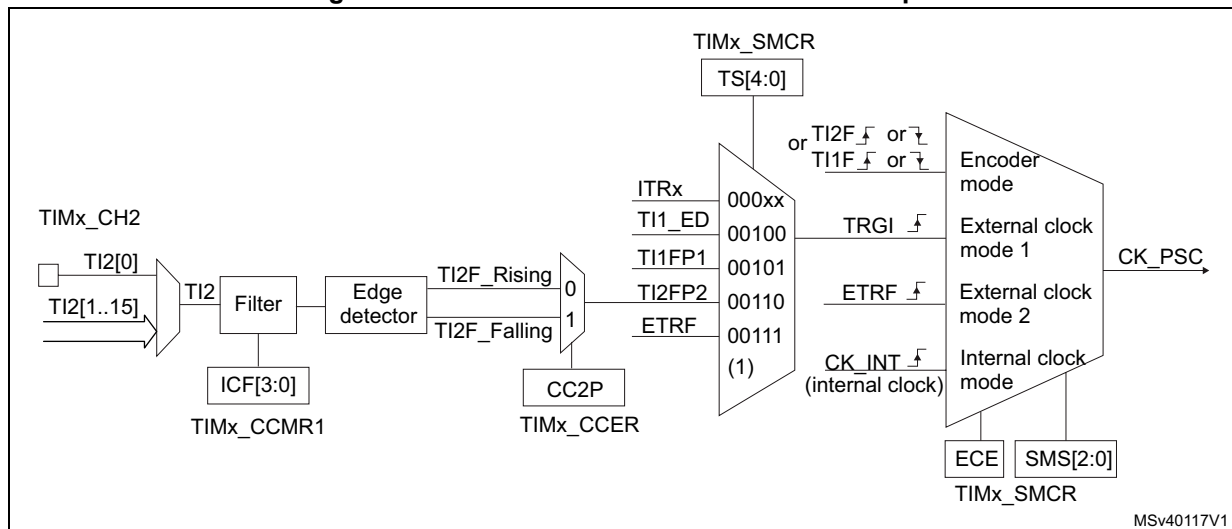
Figure 367. Control circuit in normal mode, internal clock divided by 1



External clock source mode 1

This mode is selected when SMS=111 in the TIMx_SMCR register. The counter can count at each rising or falling edge on a selected input.

Figure 368. TI2 external clock connection example



1. Codes ranging from 01000 to 11111 are reserved

For example, to configure the upcounter to count in response to a rising edge on the TI2 input, use the following procedure:

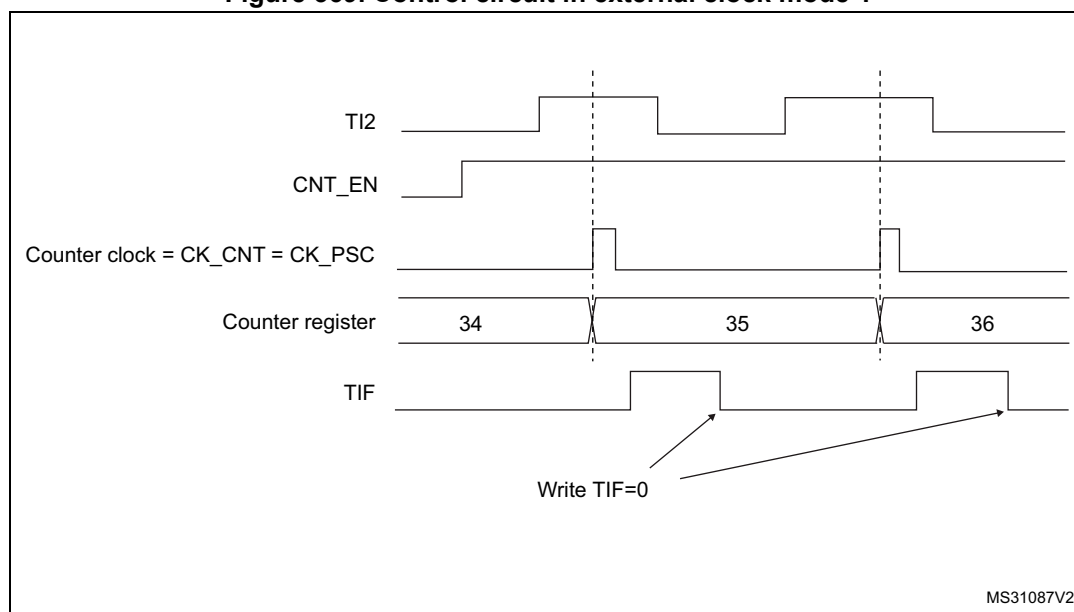
1. Select the proper TI2x source (internal or external) with the TI2SEL[3:0] bits in the TIMx_TISEL register.
2. Configure channel 2 to detect rising edges on the TI2 input by writing CC2S = '01' in the TIMx_CCMR1 register.
3. Configure the input filter duration by writing the IC2F[3:0] bits in the TIMx_CCMR1 register (if no filter is needed, keep IC2F=0000).
4. Select rising edge polarity by writing CC2P=0 and CC2NP=0 in the TIMx_CCER register.
5. Configure the timer in external clock mode 1 by writing SMS=111 in the TIMx_SMCR register.
6. Select TI2 as the trigger input source by writing TS=00110 in the TIMx_SMCR register.
7. Enable the counter by writing CEN=1 in the TIMx_CR1 register.

Note: The capture prescaler is not used for triggering, so the user does not need to configure it.

When a rising edge occurs on TI2, the counter counts once and the TIF flag is set.

The delay between the rising edge on TI2 and the actual clock of the counter is due to the resynchronization circuit on TI2 input.

Figure 369. Control circuit in external clock mode 1



MS31087V2

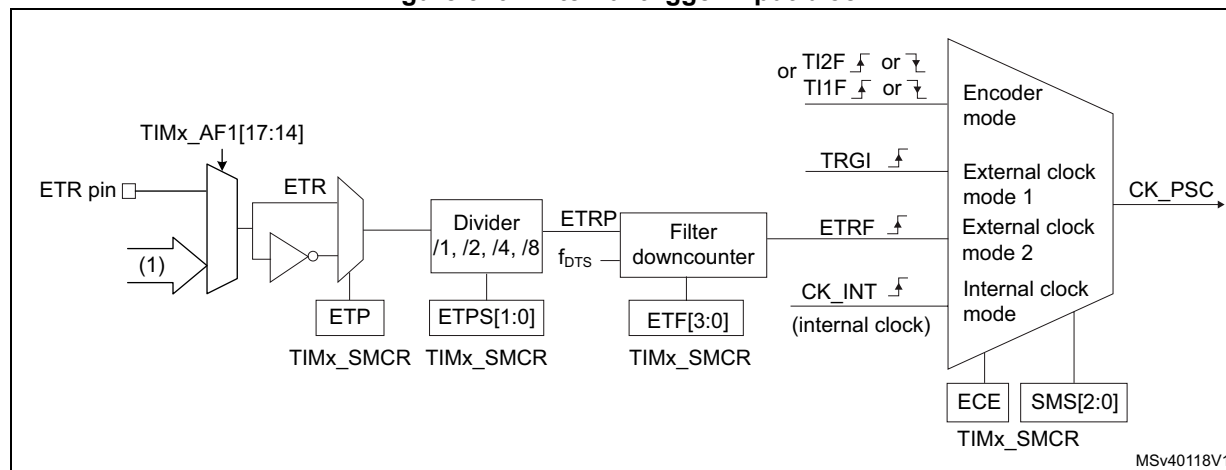
External clock source mode 2

This mode is selected by writing ECE=1 in the TIMx_SMCR register.

The counter can count at each rising or falling edge on the external trigger input ETR.

The [Figure 370](#) gives an overview of the external trigger input block.

Figure 370. External trigger input block



MSv40118V1

1. Refer to [Figure 366: TIM1/TIM8 ETR input circuitry](#).

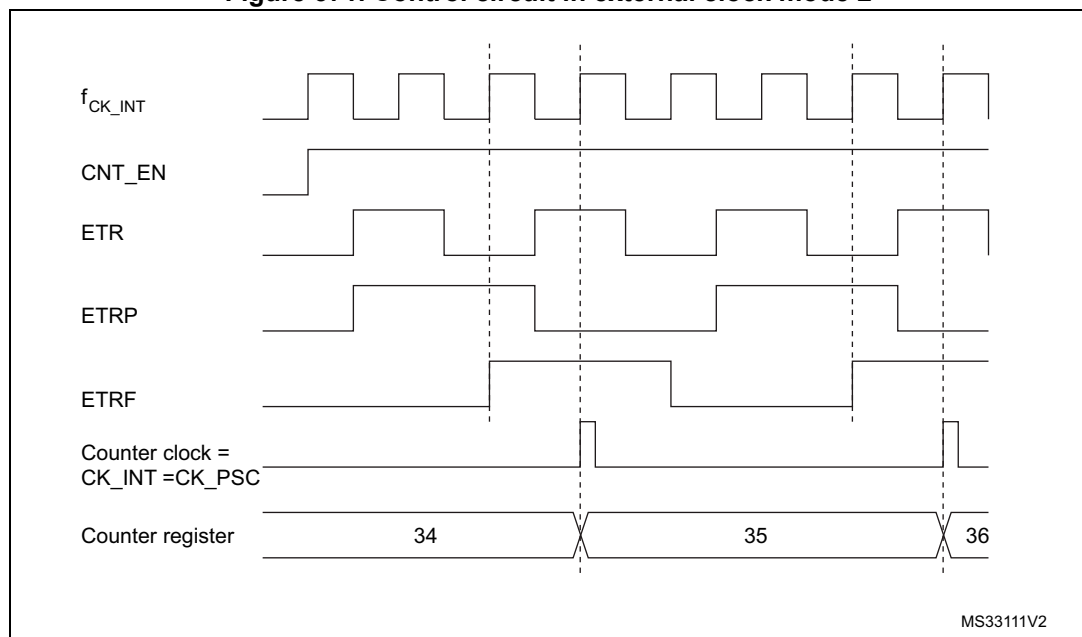
For example, to configure the upcounter to count each 2 rising edges on ETR, use the following procedure:

1. As no filter is needed in this example, write ETRF[3:0]=0000 in the TIMx_SMCR register.
2. Set the prescaler by writing ETPS[1:0]=01 in the TIMx_SMCR register
3. Select rising edge detection on the ETR pin by writing ETP=0 in the TIMx_SMCR register
4. Enable external clock mode 2 by writing ECE=1 in the TIMx_SMCR register.
5. Enable the counter by writing CEN=1 in the TIMx_CR1 register.

The counter counts once each 2 ETR rising edges.

The delay between the rising edge on ETR and the actual clock of the counter is due to the resynchronization circuit on the ETRP signal. As a consequence, the maximum frequency which can be correctly captured by the counter is at most $\frac{1}{4}$ of TIMxCLK frequency. When the ETRP signal is faster, the user should apply a division of the external signal by proper ETPS prescaler setting.

Figure 371. Control circuit in external clock mode 2



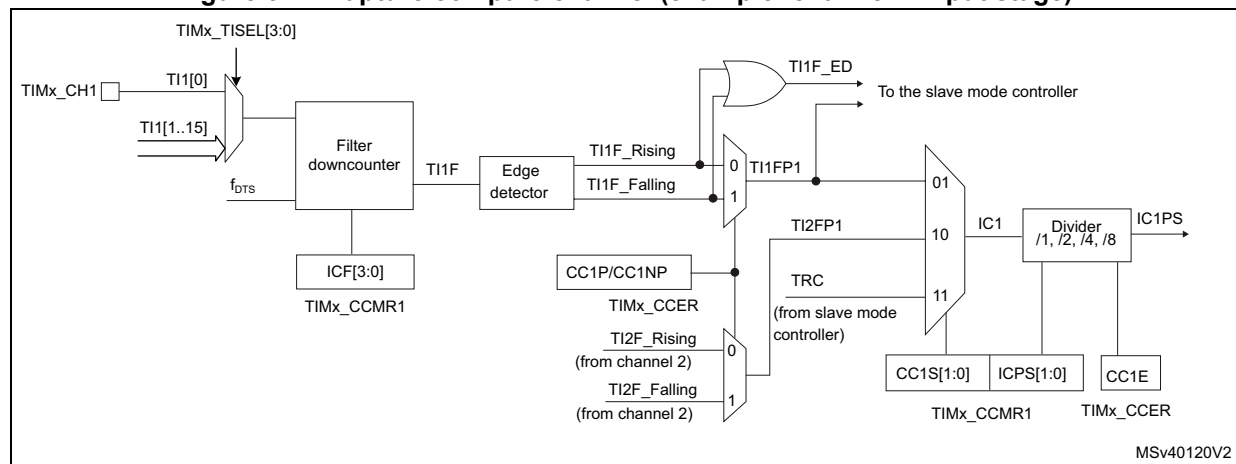
38.3.6 Capture/compare channels

Each Capture/Compare channel is built around a capture/compare register (including a shadow register), an input stage for capture (with digital filter, multiplexing, and prescaler, except for channels 5 and 6) and an output stage (with comparator and output control).

Figure 372 to *Figure 375* give an overview of one Capture/Compare channel.

The input stage samples the corresponding Tlx input to generate a filtered signal TlxF. Then, an edge detector with polarity selection generates a signal (TlxFPx) which can be used as trigger input by the slave mode controller or as the capture command. It is prescaled before the capture register (ICxPS).

Figure 372. Capture/compare channel (example: channel 1 input stage)



The output stage generates an intermediate waveform which is then used for reference: OCxRef (active high). The polarity acts at the end of the chain.

Figure 373. Capture/compare channel 1 main circuit

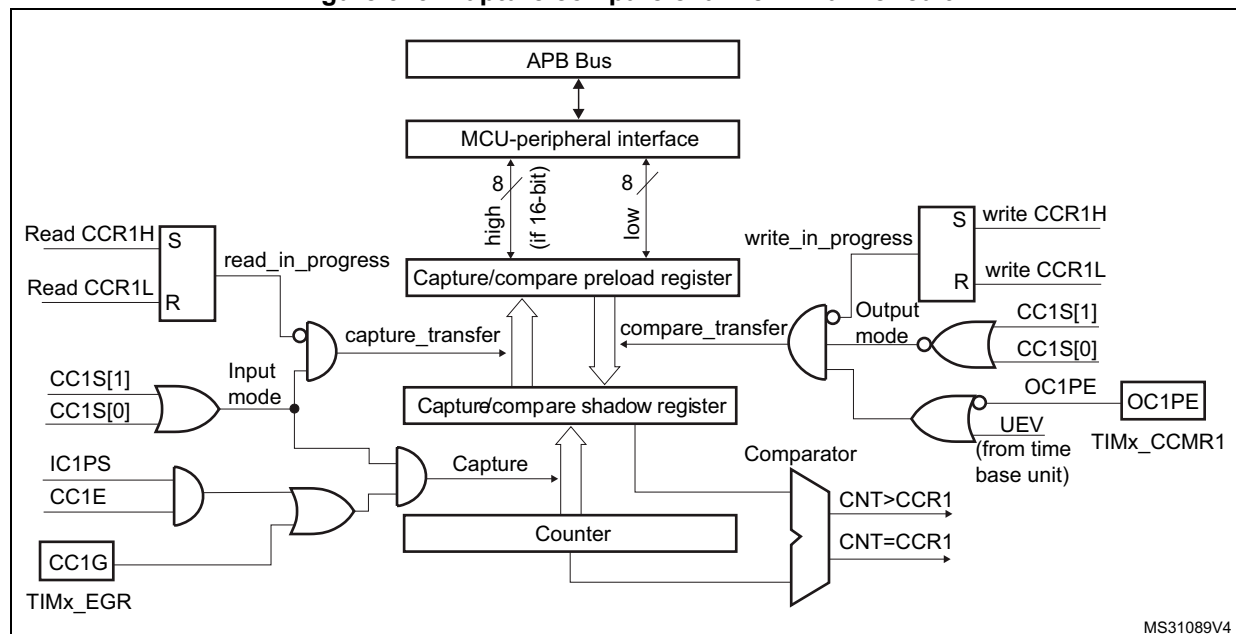
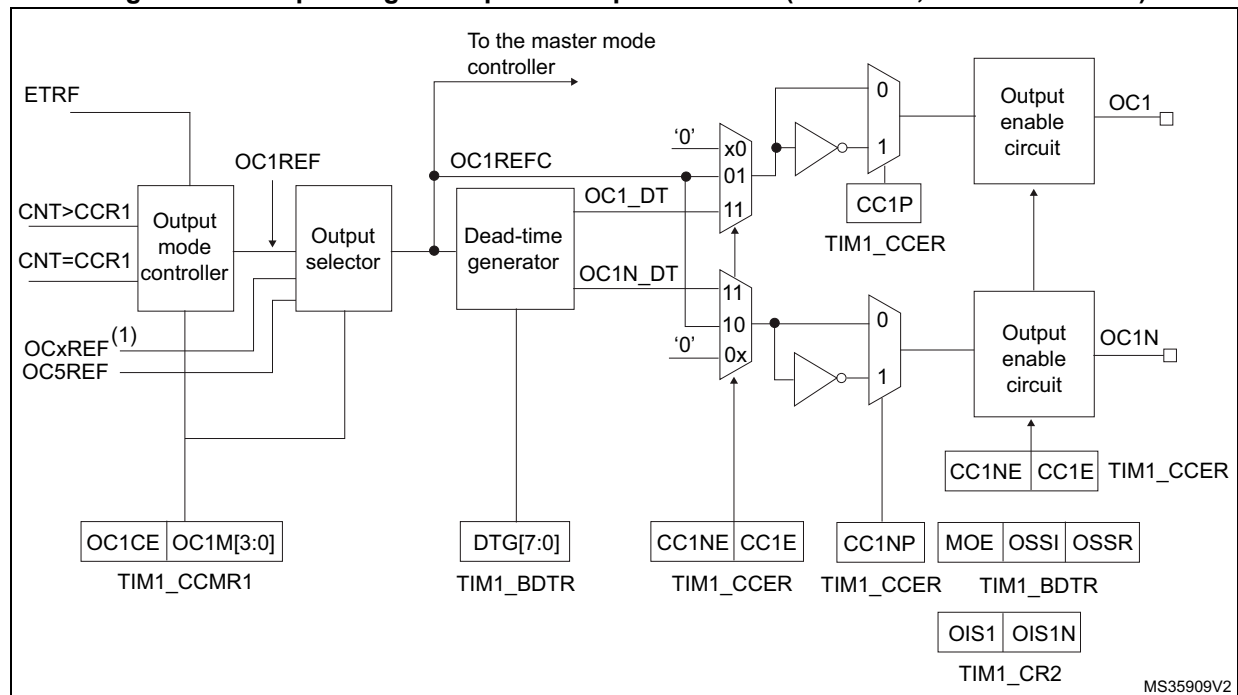


Figure 374. Output stage of capture/compare channel (channel 1, idem ch. 2 and 3)



1. OCxREF, where x is the rank of the complementary channel

Figure 375. Output stage of capture/compare channel (channel 4)

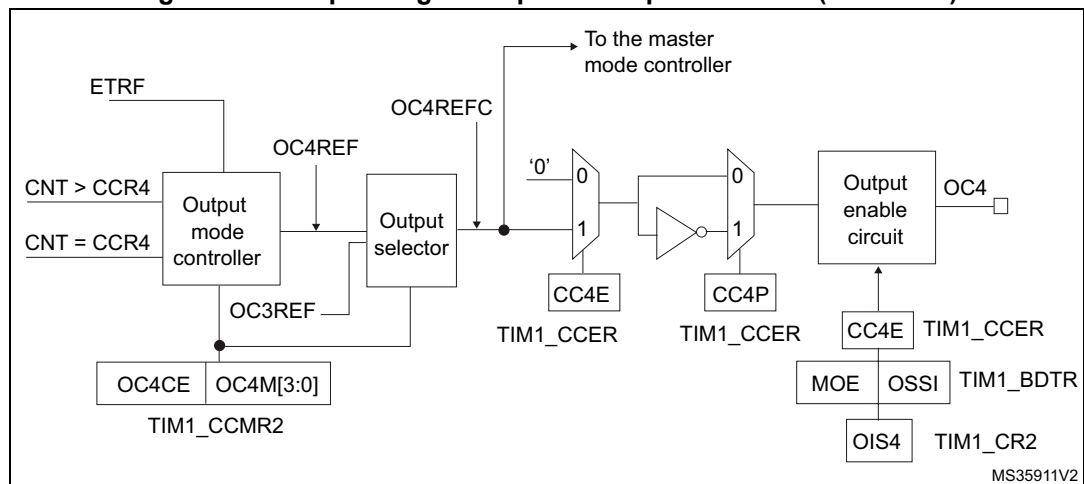
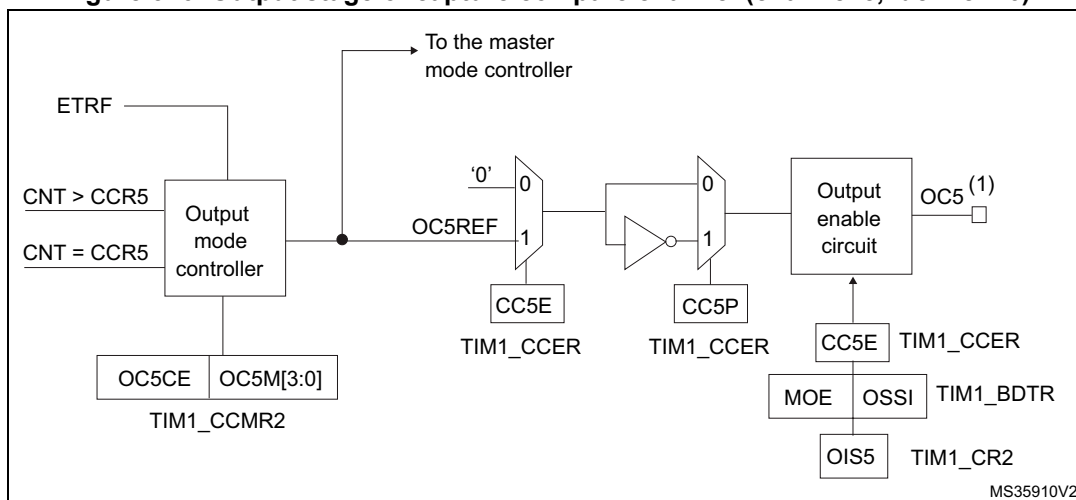


Figure 376. Output stage of capture/compare channel (channel 5, idem ch. 6)

1. Not available externally.

The capture/compare block is made of one preload register and one shadow register. Write and read always access the preload register.

In capture mode, captures are actually done in the shadow register, which is copied into the preload register.

In compare mode, the content of the preload register is copied into the shadow register which is compared to the counter.

38.3.7 Input capture mode

In Input capture mode, the Capture/Compare Registers (**TIMx_CCRx**) are used to latch the value of the counter after a transition detected by the corresponding **ICx** signal. When a capture occurs, the corresponding **CCXIF** flag (**TIMx_SR** register) is set and an interrupt or a DMA request can be sent if they are enabled. If a capture occurs while the **CCxIF** flag was already high, then the over-capture flag **CCxOF** (**TIMx_SR** register) is set. **CCxIF** can be cleared by software by writing it to '0' or by reading the captured data stored in the **TIMx_CCRx** register. **CCxOF** is cleared when written with '0'.

The following example shows how to capture the counter value in **TIMx_CCR1** when **TI1** input rises. To do this, use the following procedure:

1. Select the proper **TI1x** source (internal or external) with the **TI1SEL[3:0]** bits in the **TIMx_TISEL** register.
2. Select the active input: **TIMx_CCR1** must be linked to the **TI1** input, so write the **CC1S** bits to 01 in the **TIMx_CCMR1** register. As soon as **CC1S** becomes different from 00, the channel is configured in input and the **TIMx_CCR1** register becomes read-only.
3. Program the appropriate input filter duration in relation with the signal connected to the timer (when the input is one of the **TIx** (**ICxF** bits in the **TIMx_CCMRx** register)). Let's imagine that, when toggling, the input signal is not stable during at most 5 internal clock cycles. We must program a filter duration longer than these 5 clock cycles. We can validate a transition on **TI1** when 8 consecutive samples with the new level have been

detected (sampled at f_{DTS} frequency). Then write IC1F bits to 0011 in the TIMx_CCMR1 register.

4. Select the edge of the active transition on the TI1 channel by writing CC1P and CC1NP bits to 0 in the TIMx_CCER register (rising edge in this case).
5. Program the input prescaler. In our example, we wish the capture to be performed at each valid transition, so the prescaler is disabled (write IC1PS bits to '00' in the TIMx_CCMR1 register).
6. Enable capture from the counter into the capture register by setting the CC1E bit in the TIMx_CCER register.
7. If needed, enable the related interrupt request by setting the CC1IE bit in the TIMx_DIER register, and/or the DMA request by setting the CC1DE bit in the TIMx_DIER register.

When an input capture occurs:

- The TIMx_CCR1 register gets the value of the counter on the active transition.
- CC1IF flag is set (interrupt flag). CC1OF is also set if at least two consecutive captures occurred whereas the flag was not cleared.
- An interrupt is generated depending on the CC1IE bit.
- A DMA request is generated depending on the CC1DE bit.

In order to handle the overcapture, it is recommended to read the data before the overcapture flag. This is to avoid missing an overcapture which could happen after reading the flag and before reading the data.

Note: IC interrupt and/or DMA requests can be generated by software by setting the corresponding CCxG bit in the TIMx_EGR register.

38.3.8 PWM input mode

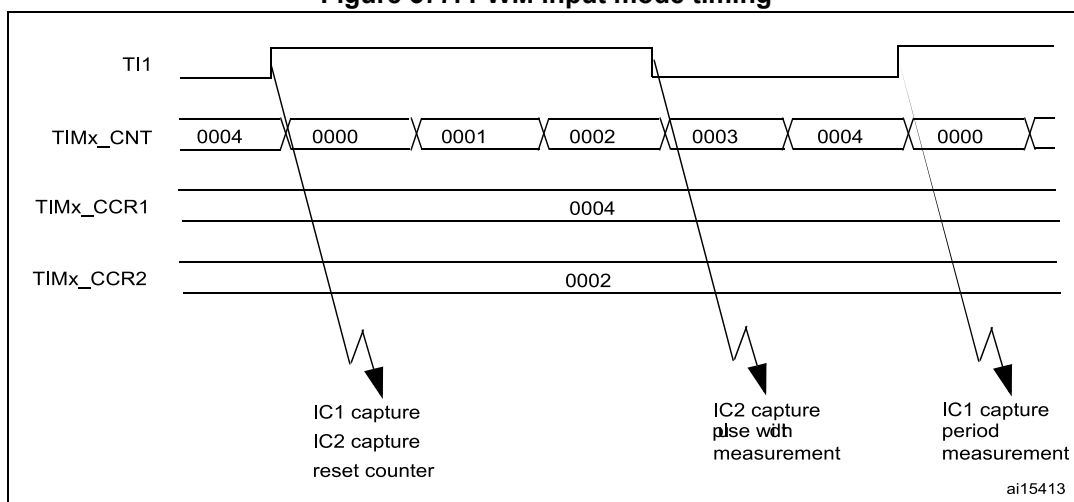
This mode is a particular case of input capture mode. The procedure is the same except:

- Two ICx signals are mapped on the same TIx input.
- These 2 ICx signals are active on edges with opposite polarity.
- One of the two TIxFP signals is selected as trigger input and the slave mode controller is configured in reset mode.

For example, the user can measure the period (in TIMx_CCR1 register) and the duty cycle (in TIMx_CCR2 register) of the PWM applied on TI1 using the following procedure (depending on CK_INT frequency and prescaler value):

1. Select the proper TI1x source (internal or external) with the TI1SEL[3:0] bits in the TIMx_TISEL register.
2. Select the active input for TIMx_CCR1: write the CC1S bits to 01 in the TIMx_CCMR1 register (TI1 selected).
3. Select the active polarity for TI1FP1 (used both for capture in TIMx_CCR1 and counter clear): write the CC1P and CC1NP bits to '0' (active on rising edge).
4. Select the active input for TIMx_CCR2: write the CC2S bits to 10 in the TIMx_CCMR1 register (TI1 selected).
5. Select the active polarity for TI1FP2 (used for capture in TIMx_CCR2): write the CC2P and CC2NP bits to CC2P/CC2NP='10' (active on falling edge).
6. Select the valid trigger input: write the TS bits to 00101 in the TIMx_SMCR register (TI1FP1 selected).
7. Configure the slave mode controller in reset mode: write the SMS bits to 0100 in the TIMx_SMCR register.
8. Enable the captures: write the CC1E and CC2E bits to '1' in the TIMx_CCER register.

Figure 377. PWM input mode timing



38.3.9 Forced output mode

In output mode (CCxS bits = 00 in the TIMx_CCMRx register), each output compare signal (OCxREF and then OCx/OCxN) can be forced to active or inactive level directly by software, independently of any comparison between the output compare register and the counter.

To force an output compare signal (OCxREF/OCx) to its active level, user just needs to write 0101 in the OCxM bits in the corresponding TIMx_CCMRx register. Thus OCxREF is forced high (OCxREF is always active high) and OCx get opposite value to CCxP polarity bit.

For example: CCxP=0 (OCx active high) => OCx is forced to high level.

The OCxREF signal can be forced low by writing the OCxM bits to 0100 in the TIMx_CCMRx register.

Anyway, the comparison between the TIMx_CCRx shadow register and the counter is still performed and allows the flag to be set. Interrupt and DMA requests can be sent accordingly. This is described in the output compare mode section below.

38.3.10 Output compare mode

This function is used to control an output waveform or indicate when a period of time has elapsed. Channels 1 to 4 can be output, while Channel 5 and 6 are only available inside the device (for instance, for compound waveform generation or for ADC triggering).

When a match is found between the capture/compare register and the counter, the output compare function:

- Assigns the corresponding output pin to a programmable value defined by the output compare mode (OCxM bits in the TIMx_CCMRx register) and the output polarity (CCxP bit in the TIMx_CCER register). The output pin can keep its level (OCxM=0000), be set active (OCxM=0001), be set inactive (OCxM=0010) or can toggle (OCxM=0011) on match.
- Sets a flag in the interrupt status register (CCxIF bit in the TIMx_SR register).
- Generates an interrupt if the corresponding interrupt mask is set (CCXIE bit in the TIMx_DIER register).
- Sends a DMA request if the corresponding enable bit is set (CCxDE bit in the TIMx_DIER register, CCDS bit in the TIMx_CR2 register for the DMA request selection).

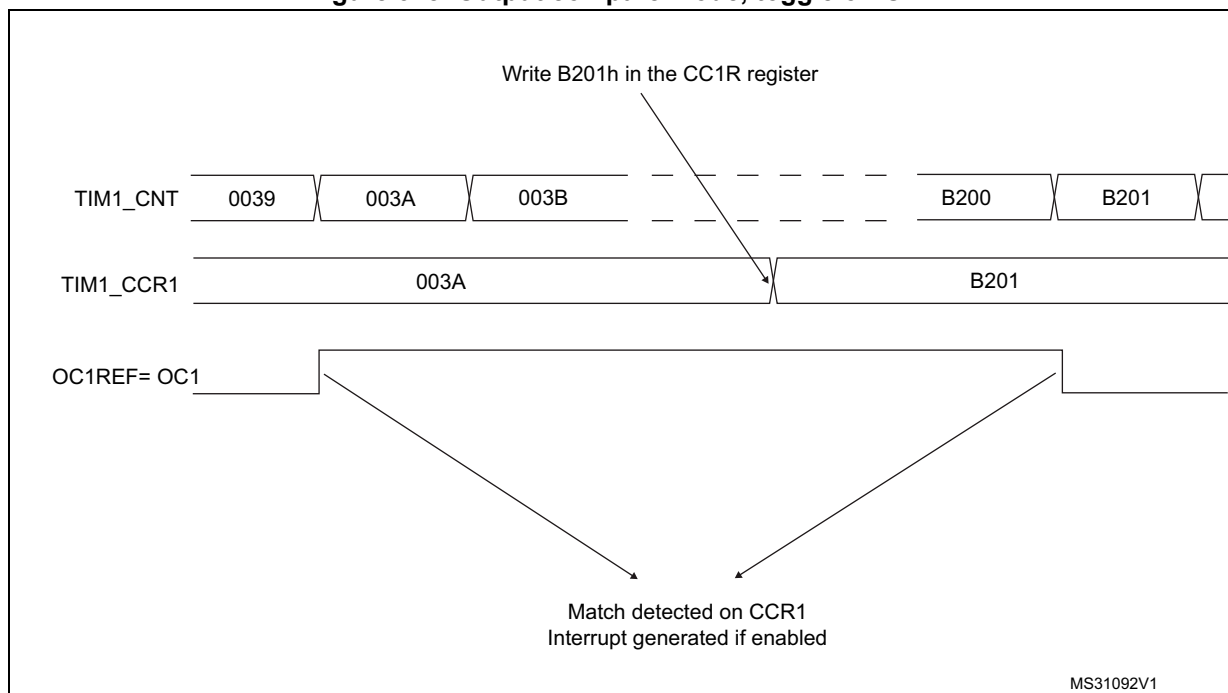
The TIMx_CCRx registers can be programmed with or without preload registers using the OCxPE bit in the TIMx_CCMRx register.

In output compare mode, the update event UEV has no effect on OCxREF and OCx output. The timing resolution is one count of the counter. Output compare mode can also be used to output a single pulse (in One Pulse mode).

Procedure

1. Select the counter clock (internal, external, prescaler).
2. Write the desired data in the TIMx_ARR and TIMx_CCRx registers.
3. Set the CCXIE bit if an interrupt request is to be generated.
4. Select the output mode. For example:
 - Write OCxM = 0011 to toggle OCx output pin when CNT matches CCRx
 - Write OCxPE = 0 to disable preload register
 - Write CCxP = 0 to select active high polarity
 - Write CCxE = 1 to enable the output
5. Enable the counter by setting the CEN bit in the TIMx_CR1 register.

The TIMx_CCRx register can be updated at any time by software to control the output waveform, provided that the preload register is not enabled (OCxPE='0', else TIMx_CCRx shadow register is updated only at the next update event UEV). An example is given in [Figure 378](#).

Figure 378. Output compare mode, toggle on OC1

38.3.11 PWM mode

Pulse Width Modulation mode allows a signal to be generated with a frequency determined by the value of the TIMx_ARR register and a duty cycle determined by the value of the TIMx_CCRx register.

The PWM mode can be selected independently on each channel (one PWM per OCx output) by writing '0110' (PWM mode 1) or '0111' (PWM mode 2) in the OCxM bits in the TIMx_CCMRx register. The corresponding preload register must be enabled by setting the OCxPE bit in the TIMx_CCMRx register, and eventually the auto-reload preload register (in upcounting or center-aligned modes) by setting the ARPE bit in the TIMx_CR1 register.

As the preload registers are transferred to the shadow registers only when an update event occurs, before starting the counter, all registers must be initialized by setting the UG bit in the TIMx_EGR register.

OCx polarity is software programmable using the CCxP bit in the TIMx_CCER register. It can be programmed as active high or active low. OCx output is enabled by a combination of the CCxE, CCxNE, MOE, OSSI and OSSR bits (TIMx_CCER and TIMx_BDTR registers). Refer to the TIMx_CCER register description for more details.

In PWM mode (1 or 2), TIMx_CNT and TIMx_CCRx are always compared to determine whether $TIMx_CCRx \leq TIMx_CNT$ or $TIMx_CNT \leq TIMx_CCRx$ (depending on the direction of the counter).

The timer is able to generate PWM in edge-aligned mode or center-aligned mode depending on the CMS bits in the TIMx_CR1 register.

PWM edge-aligned mode

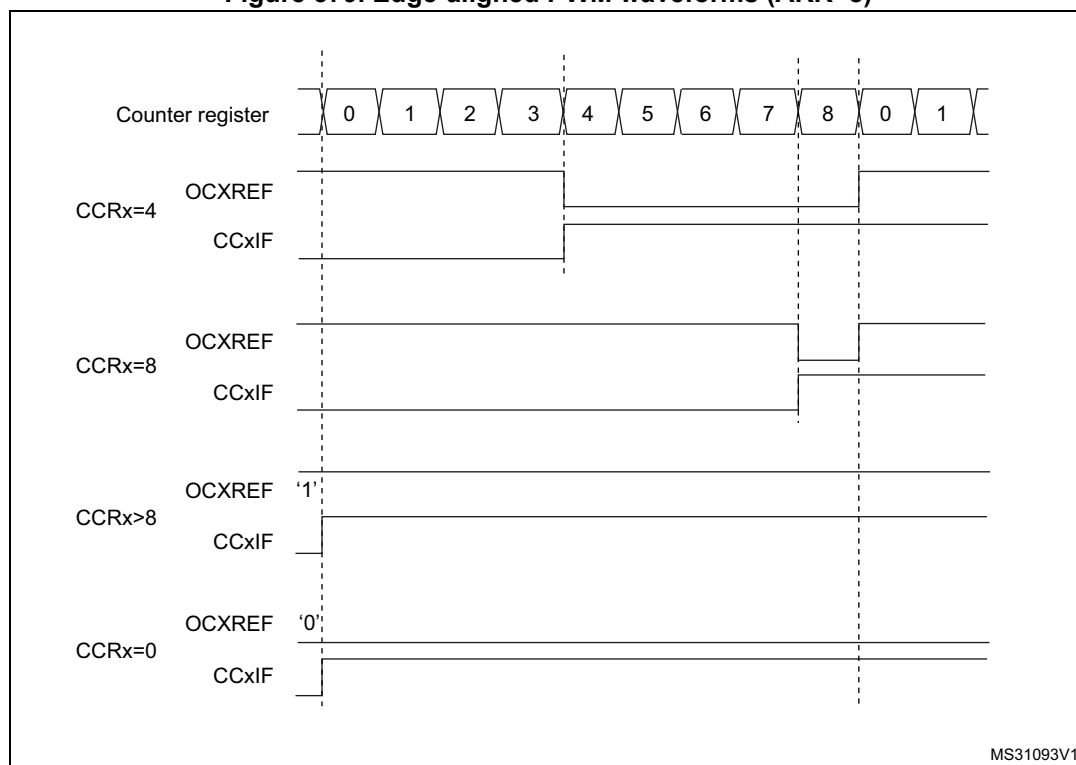
- Upcounting configuration

Upcounting is active when the DIR bit in the TIMx_CR1 register is low. Refer to the [Upcounting mode on page 1550](#).

In the following example, we consider PWM mode 1. The reference PWM signal OCxREF is high as long as TIMx_CNT < TIMx_CCRx else it becomes low. If the compare value in TIMx_CCRx is greater than the auto-reload value (in TIMx_ARR) then OCxREF is held at '1'. If the compare value is 0 then OCxRef is held at '0'.

[Figure 379](#) shows some edge-aligned PWM waveforms in an example where TIMx_ARR=8.

Figure 379. Edge-aligned PWM waveforms (ARR=8)



- Downcounting configuration

Downcounting is active when DIR bit in TIMx_CR1 register is high. Refer to the [Downcounting mode on page 1554](#)

In PWM mode 1, the reference signal OCxRef is low as long as TIMx_CNT > TIMx_CCRx else it becomes high. If the compare value in TIMx_CCRx is greater than the auto-reload value in TIMx_ARR, then OCxREF is held at '1'. 0% PWM is not possible in this mode.

PWM center-aligned mode

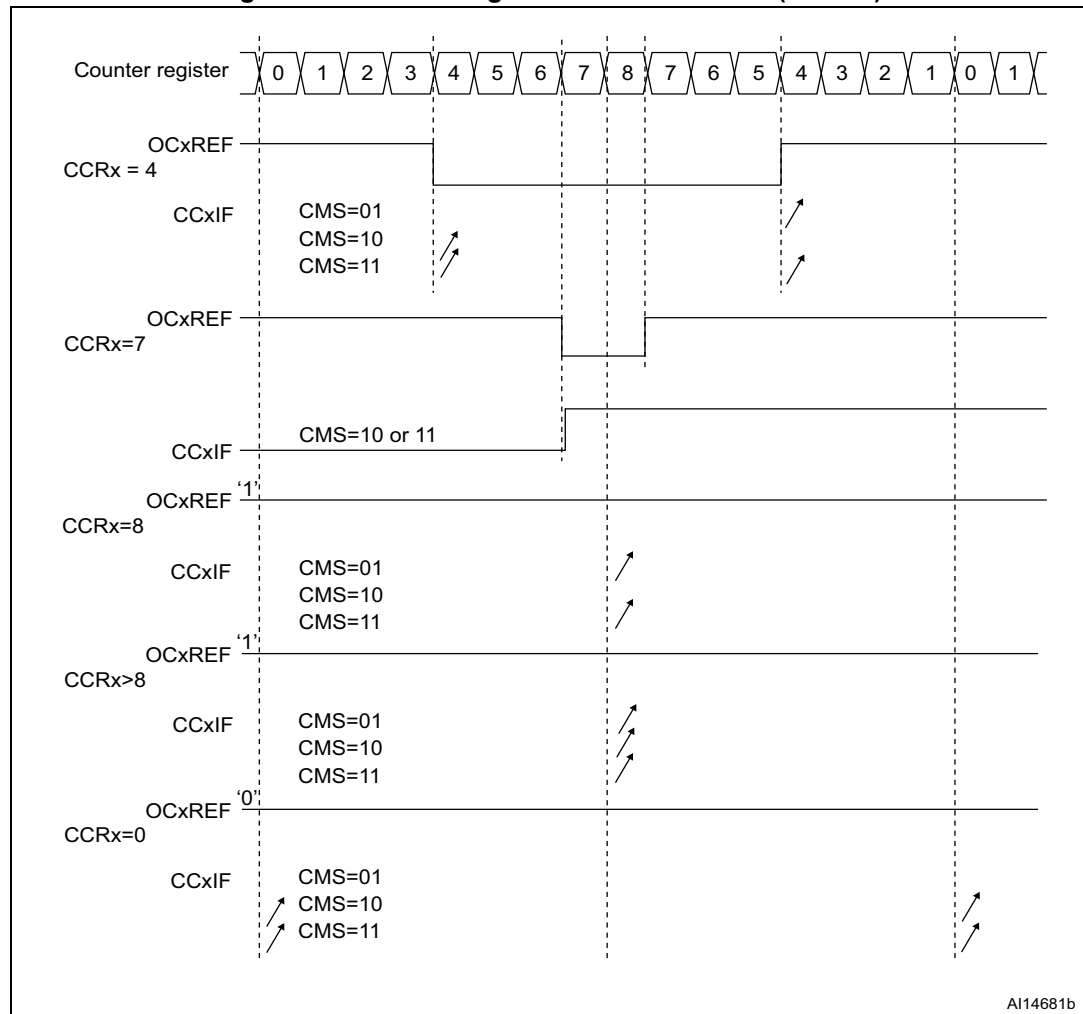
Center-aligned mode is active when the CMS bits in TIMx_CR1 register are different from '00' (all the remaining configurations having the same effect on the OCxRef/OCx signals). The compare flag is set when the counter counts up, when it counts down or both when it counts up and down depending on the CMS bits configuration. The direction bit (DIR) in the

TIMx_CR1 register is updated by hardware and must not be changed by software. Refer to the [Center-aligned mode \(up/down counting\) on page 1557](#).

Figure 380 shows some center-aligned PWM waveforms in an example where:

- TIMx_ARR=8,
- PWM mode is the PWM mode 1,
- The flag is set when the counter counts down corresponding to the center-aligned mode 1 selected for CMS=01 in TIMx_CR1 register.

Figure 380. Center-aligned PWM waveforms (ARR=8)



Hints on using center-aligned mode

- When starting in center-aligned mode, the current up-down configuration is used. It means that the counter counts up or down depending on the value written in the DIR bit

in the TIMx_CR1 register. Moreover, the DIR and CMS bits must not be changed at the same time by the software.

- Writing to the counter while running in center-aligned mode is not recommended as it can lead to unexpected results. In particular:
 - The direction is not updated if a value greater than the auto-reload value is written in the counter (TIMx_CNT > TIMx_ARR). For example, if the counter was counting up, it continues to count up.
 - The direction is updated if 0 or the TIMx_ARR value is written in the counter but no Update Event UEV is generated.
- The safest way to use center-aligned mode is to generate an update by software (setting the UG bit in the TIMx_EGR register) just before starting the counter and not to write the counter while it is running.

38.3.12 Asymmetric PWM mode

Asymmetric mode allows two center-aligned PWM signals to be generated with a programmable phase shift. While the frequency is determined by the value of the TIMx_ARR register, the duty cycle and the phase-shift are determined by a pair of TIMx_CCRx register. One register controls the PWM during up-counting, the second during down counting, so that PWM is adjusted every half PWM cycle:

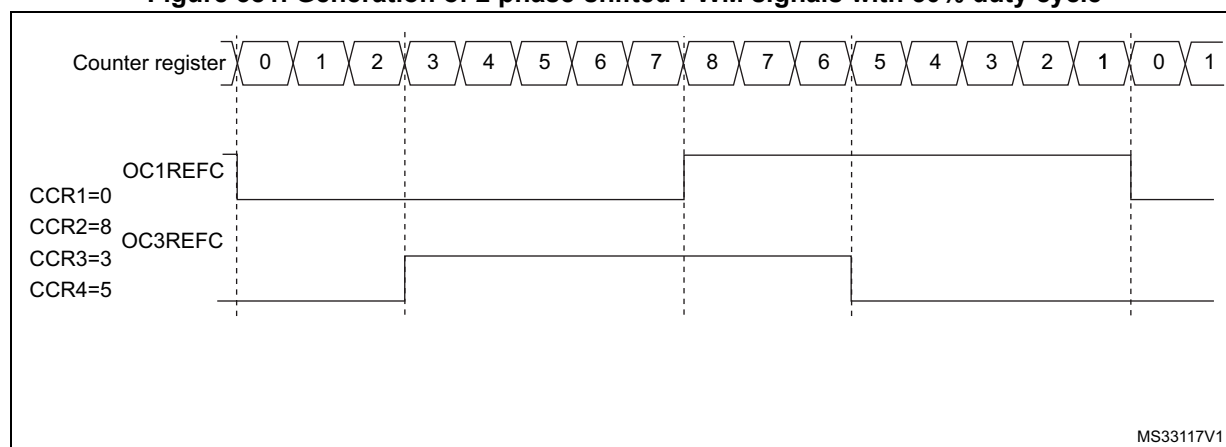
- OC1REFC (or OC2REFC) is controlled by TIMx_CCR1 and TIMx_CCR2
- OC3REFC (or OC4REFC) is controlled by TIMx_CCR3 and TIMx_CCR4

Asymmetric PWM mode can be selected independently on two channel (one OCx output per pair of CCR registers) by writing '1110' (Asymmetric PWM mode 1) or '1111' (Asymmetric PWM mode 2) in the OCxM bits in the TIMx_CCMRx register.

Note: The OCxM[3:0] bit field is split into two parts for compatibility reasons, the most significant bit is not contiguous with the 3 least significant ones.

When a given channel is used as asymmetric PWM channel, its complementary channel can also be used. For instance, if an OC1REFC signal is generated on channel 1 (Asymmetric PWM mode 1), it is possible to output either the OC2REF signal on channel 2, or an OC2REFC signal resulting from asymmetric PWM mode 1.

[Figure 381](#) represents an example of signals that can be generated using Asymmetric PWM mode (channels 1 to 4 are configured in Asymmetric PWM mode 1). Together with the deadtime generator, this allows a full-bridge phase-shifted DC to DC converter to be controlled.

Figure 381. Generation of 2 phase-shifted PWM signals with 50% duty cycle

38.3.13 Combined PWM mode

Combined PWM mode allows two edge or center-aligned PWM signals to be generated with programmable delay and phase shift between respective pulses. While the frequency is determined by the value of the TIMx_ARR register, the duty cycle and delay are determined by the two TIMx_CCRx registers. The resulting signals, OCxREFC, are made of an OR or AND logical combination of two reference PWMs:

- OC1REFC (or OC2REFC) is controlled by TIMx_CCR1 and TIMx_CCR2
- OC3REFC (or OC4REFC) is controlled by TIMx_CCR3 and TIMx_CCR4

Combined PWM mode can be selected independently on two channels (one OCx output per pair of CCR registers) by writing '1100' (Combined PWM mode 1) or '1101' (Combined PWM mode 2) in the OCxM bits in the TIMx_CCMRx register.

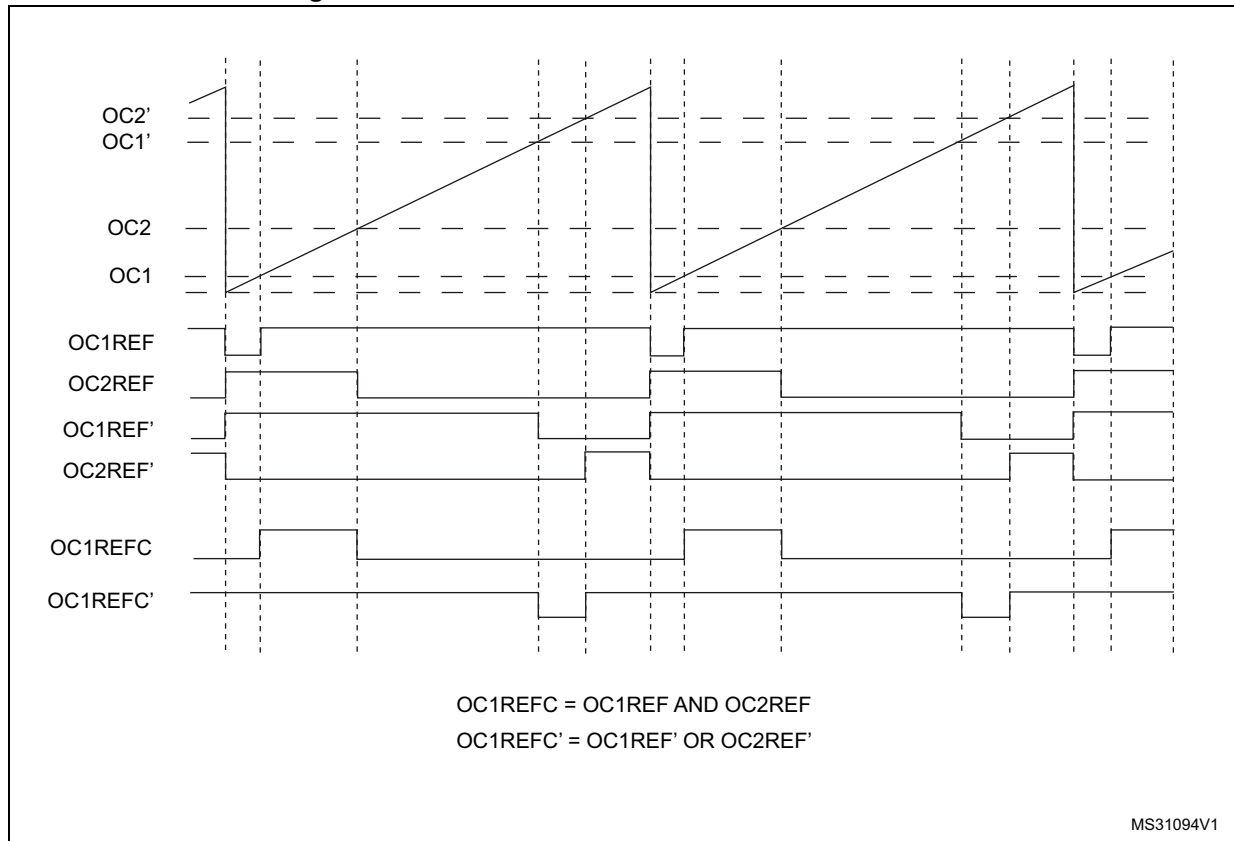
When a given channel is used as combined PWM channel, its complementary channel must be configured in the opposite PWM mode (for instance, one in Combined PWM mode 1 and the other in Combined PWM mode 2).

Note: The OCxM[3:0] bit field is split into two parts for compatibility reasons, the most significant bit is not contiguous with the 3 least significant ones.

Figure 382 represents an example of signals that can be generated using Asymmetric PWM mode, obtained with the following configuration:

- Channel 1 is configured in Combined PWM mode 2,
- Channel 2 is configured in PWM mode 1,
- Channel 3 is configured in Combined PWM mode 2,
- Channel 4 is configured in PWM mode 1.

Figure 382. Combined PWM mode on channel 1 and 3

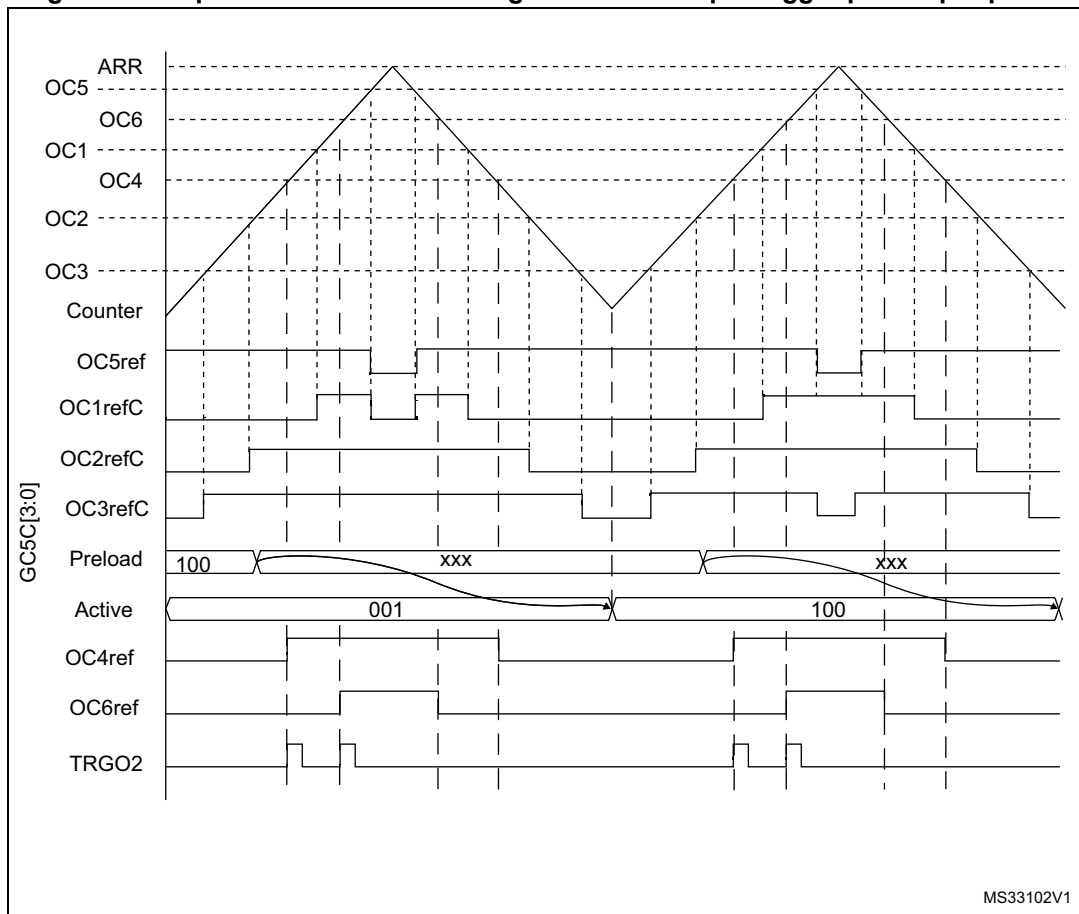


38.3.14 Combined 3-phase PWM mode

Combined 3-phase PWM mode allows one to three center-aligned PWM signals to be generated with a single programmable signal ANDed in the middle of the pulses. The OC5REF signal is used to define the resulting combined signal. The 3-bits GC5C[3:1] in the TIMx_CCR5 allow selection on which reference signal the OC5REF is combined. The resulting signals, OCxREFC, are made of an AND logical combination of two reference PWMs:

- If GC5C1 is set, OC1REFC is controlled by TIMx_CCR1 and TIMx_CCR5
- If GC5C2 is set, OC2REFC is controlled by TIMx_CCR2 and TIMx_CCR5
- If GC5C3 is set, OC3REFC is controlled by TIMx_CCR3 and TIMx_CCR5

Combined 3-phase PWM mode can be selected independently on channels 1 to 3 by setting at least one of the 3-bits GC5C[3:1].

Figure 383. 3-phase combined PWM signals with multiple trigger pulses per period

The TRGO2 waveform shows how the ADC can be synchronized on given 3-phase PWM signals. Refer to [Section 38.3.27: ADC synchronization](#) for more details.

38.3.15 Complementary outputs and dead-time insertion

The advanced-control timers (TIM1/TIM8) can output two complementary signals and manage the switching-off and the switching-on instants of the outputs.

This time is generally known as dead-time and it has to be adjusted depending on the devices that are connected to the outputs and their characteristics (intrinsic delays of level-shifters, delays due to power switches...)

The polarity of the outputs (main output OCx or complementary OCxN) can be selected independently for each output. This is done by writing to the CCxP and CCxNP bits in the TIMx_CCER register.

The complementary signals OCx and OCxN are activated by a combination of several control bits: the CCxE and CCxNE bits in the TIMx_CCER register and the MOE, OISx, OISxN, OSSI and OSSR bits in the TIMx_BDTR and TIMx_CR2 registers. Refer to [Table 330: Output control bits for complementary OCx and OCxN channels with break feature on page 1625](#) for more details. In particular, the dead-time is activated when switching to the idle state (MOE falling down to 0).

Dead-time insertion is enabled by setting both CCxE and CCxNE bits, and the MOE bit if the break circuit is present. There is one 10-bit dead-time generator for each channel. From a reference waveform OCxREF, it generates 2 outputs OCx and OCxN. If OCx and OCxN are active high:

- The OCx output signal is the same as the reference signal except for the rising edge, which is delayed relative to the reference rising edge.
- The OCxN output signal is the opposite of the reference signal except for the rising edge, which is delayed relative to the reference falling edge.

If the delay is greater than the width of the active output (OCx or OCxN) then the corresponding pulse is not generated.

The following figures show the relationships between the output signals of the dead-time generator and the reference signal OCxREF. (we suppose CCxP=0, CCxNP=0, MOE=1, CCxE=1 and CCxNE=1 in these examples)

Figure 384. Complementary output with dead-time insertion

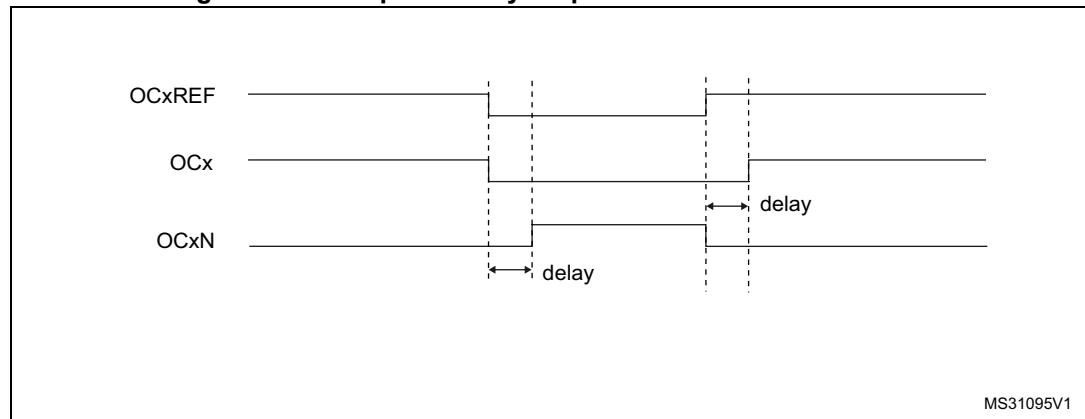


Figure 385. Dead-time waveforms with delay greater than the negative pulse

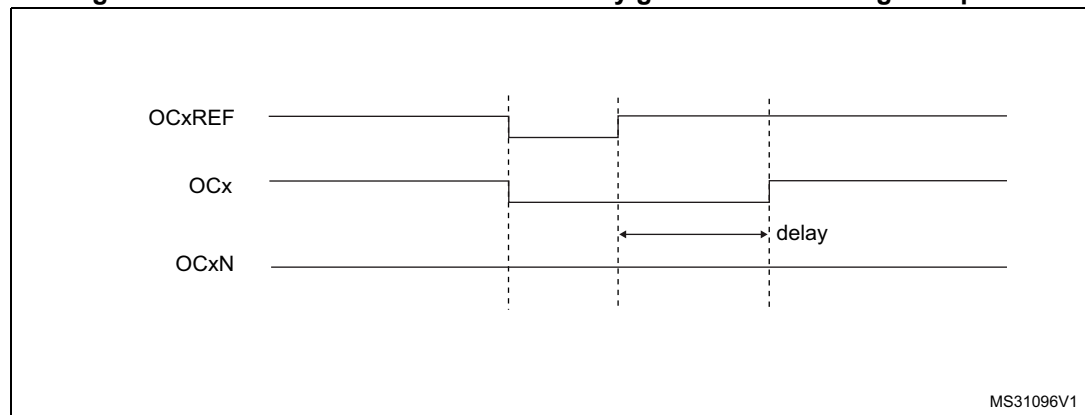
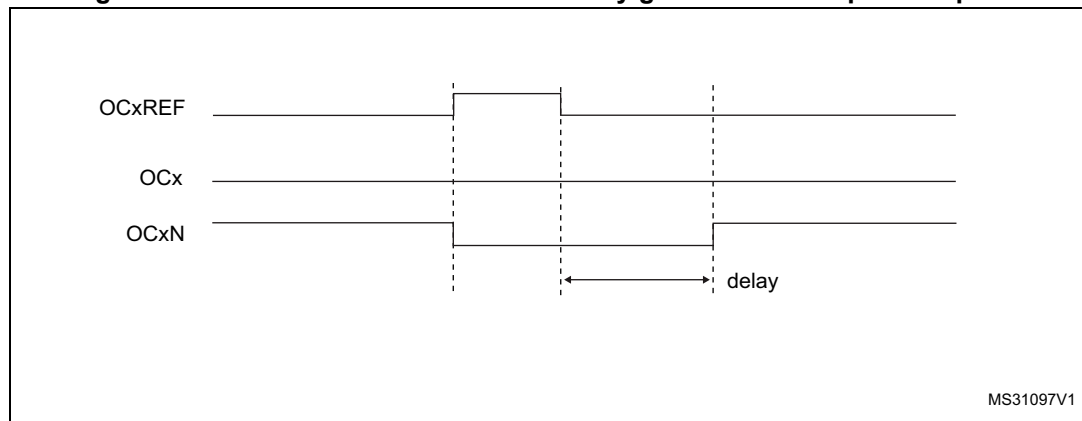


Figure 386. Dead-time waveforms with delay greater than the positive pulse

The dead-time delay is the same for each of the channels and is programmable with the DTG bits in the TIMx_BDTR register. Refer to [Section 38.4.20: TIMx break and dead-time register \(TIMx_BDTR\)\(x = 1, 8\)](#) for delay calculation.

Re-directing OCxREF to OCx or OCxN

In output mode (forced, output compare or PWM), OCxREF can be re-directed to the OCx output or to OCxN output by configuring the CCxE and CCxNE bits in the TIMx_CCER register.

This allows a specific waveform to be sent (such as PWM or static active level) on one output while the complementary remains at its inactive level. Other alternative possibilities are to have both outputs at inactive level or both outputs active and complementary with dead-time.

Note: *When only OCxN is enabled (CCxE=0, CCxNE=1), it is not complemented and becomes active as soon as OCxREF is high. For example, if CCxNP=0 then OCxN=OCxRef. On the other hand, when both OCx and OCxN are enabled (CCxE=CCxNE=1) OCx becomes active when OCxREF is high whereas OCxN is complemented and becomes active when OCxREF is low.*

38.3.16 Using the break function

The purpose of the break function is to protect power switches driven by PWM signals generated with the TIM1 and TIM8 timers. The two break inputs are usually connected to fault outputs of power stages and 3-phase inverters. When activated, the break circuitry shuts down the PWM outputs and forces them to a predefined safe state. A number of internal MCU events can also be selected to trigger an output shut-down.

The break features two channels. A break channel which gathers both system-level fault (clock failure, parity error,...) and application fault (from input pins and built-in comparator), and can force the outputs to a predefined level (either active or inactive) after a deadtime duration. A break2 channel which only includes application faults and is able to force the outputs to an inactive state.

The output enable signal and output levels during break are depending on several control bits:

- the MOE bit in TIMx_BDTR register allows the outputs to be enabled/disabled by software and is reset in case of break or break2 event.
- the OSSI bit in the TIMx_BDTR register defines whether the timer controls the output in inactive state or releases the control to the GPIO controller (typically to have it in Hi-Z mode)
- the OISx and OISxN bits in the TIMx_CR2 register which are setting the output shut-down level, either active or inactive. The OCx and OCxN outputs cannot be set both to active level at a given time, whatever the OISx and OISxN values. Refer to [Table 330: Output control bits for complementary OCx and OCxN channels with break feature on page 1625](#) for more details.

When exiting from reset, the break circuit is disabled and the MOE bit is low. The break functions can be enabled by setting the BKE and BK2E bits in the TIMx_BDTR register. The break input polarities can be selected by configuring the BKP and BK2P bits in the same register. BKE/BK2E and BKP/BK2P can be modified at the same time. When the BKE/BK2E and BKP/BK2P bits are written, a delay of 1 APB clock cycle is applied before the writing is effective. Consequently, it is necessary to wait 1 APB clock period to correctly read back the bit after the write operation.

Because MOE falling edge can be asynchronous, a resynchronization circuit has been inserted between the actual signal (acting on the outputs) and the synchronous control bit (accessed in the TIMx_BDTR register). It results in some delays between the asynchronous and the synchronous signals. In particular, if MOE is set to 1 whereas it was low, a delay must be inserted (dummy instruction) before reading it correctly. This is because the write acts on the asynchronous signal whereas the read reflects the synchronous signal.

The sources for break (BRK) channel are:

- An external source connected to one of the BKIN pin (as per selection done in the AFIO controller), with polarity selection and optional digital filtering
- An internal source:
 - the output from a comparator, with polarity selection and optional digital filtering
 - the analog watchdog output of the DFSDM1 peripheral
 - A system break:
 - the Cortex®-M7 LOCKUP output
 - the PVD output
 - all SRAM and TCM double ECC errors (AXI-SRAM, ITCM, DTCM, SRAM1, SRAM2, SRAM3, SRAM4, BKRAM, refer to SYSCFG_CFGR register for details)
 - a Flash double ECC error
 - a clock failure event generated by the CSS detector

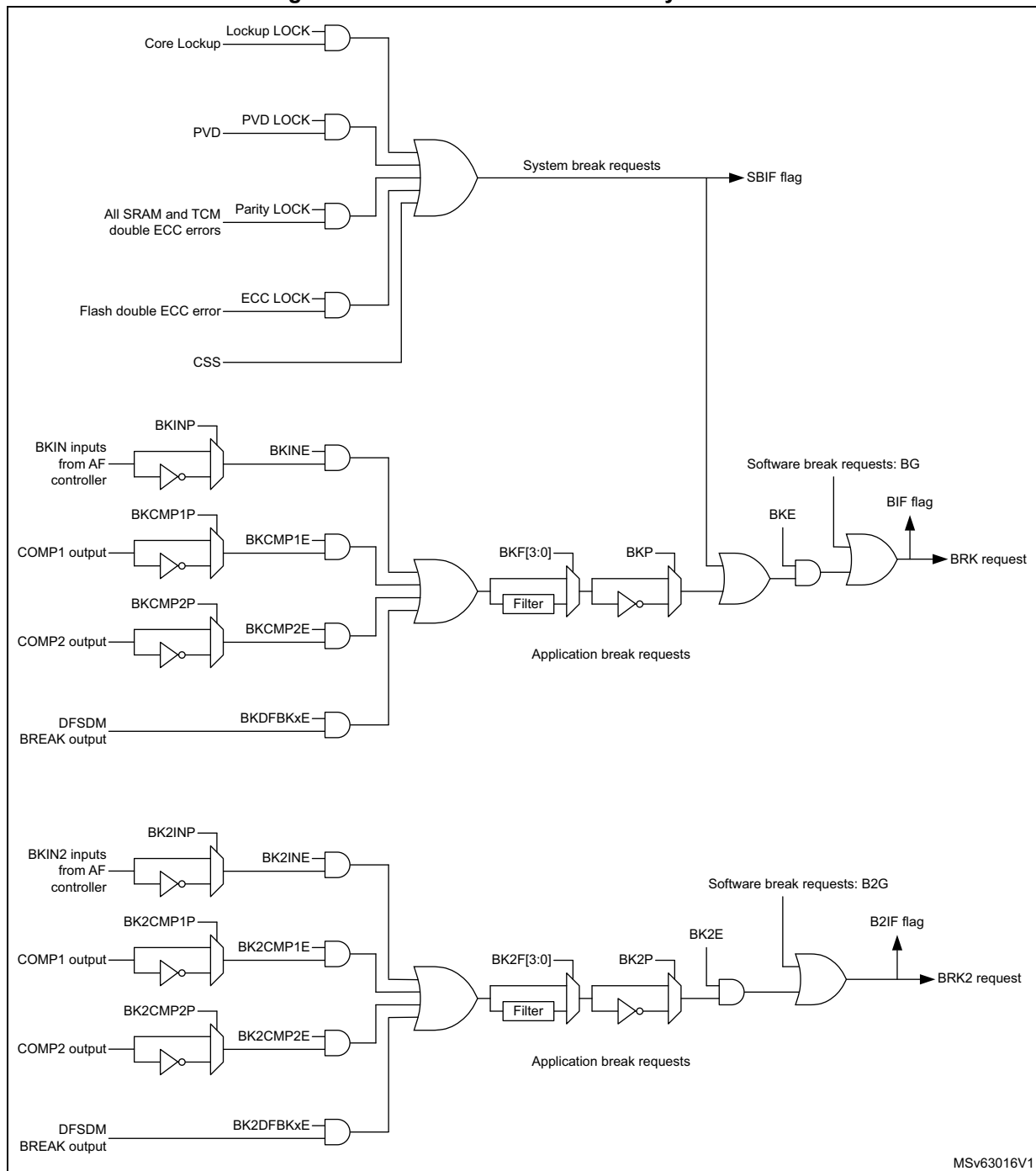
The sources for break2 (BRK2) are:

- An external source connected to one of the BKIN pin (as per selection done in the AFIO controller), with polarity selection and optional digital filtering
- An internal source coming from a comparator output.
- The analog watchdog output of the DFSDM1 peripheral

Break events can also be generated by software using BG and B2G bits in the TIMx_EGR register. The software break generation using BG and B2G is active whatever the BKE and BK2E enable bits values.

All sources are ORed before entering the timer BRK or BRK2 inputs, as per [Figure 387](#) below.

Figure 387. Break and Break2 circuitry overview



Note: An asynchronous (clockless) operation is only guaranteed when the programmable filter is disabled. If it is enabled, a fail safe clock mode (for example by using the internal PLL and/or the CSS) must be used to guarantee that break events are handled.

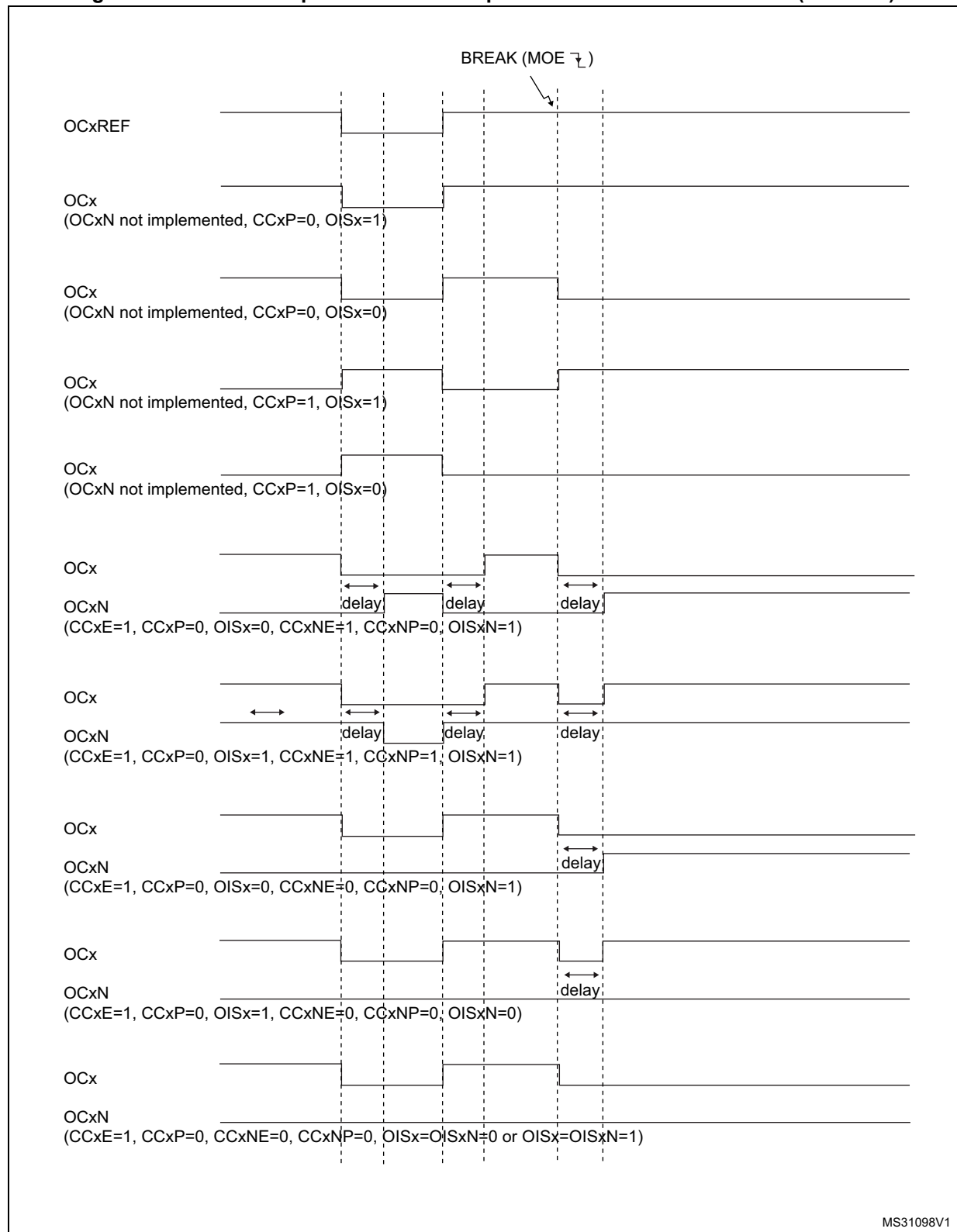
When one of the breaks occurs (selected level on one of the break inputs):

- The MOE bit is cleared asynchronously, putting the outputs in inactive state, idle state or even releasing the control to the GPIO controller (selected by the OSS1 bit). This feature is enabled even if the MCU oscillator is off.
- Each output channel is driven with the level programmed in the OISx bit in the TIMx_CR2 register as soon as MOE=0. If OSS1=0, the timer releases the output control (taken over by the GPIO controller), otherwise the enable output remains high.
- When complementary outputs are used:
 - The outputs are first put in inactive state (depending on the polarity). This is done asynchronously so that it works even if no clock is provided to the timer.
 - If the timer clock is still present, then the dead-time generator is reactivated in order to drive the outputs with the level programmed in the OISx and OISxN bits after a dead-time. Even in this case, OCx and OCxN cannot be driven to their active level together. Note that because of the resynchronization on MOE, the dead-time duration is slightly longer than usual (around 2 ck_tim clock cycles).
 - If OSS1=0, the timer releases the output control (taken over by the GPIO controller which forces a Hi-Z state), otherwise the enable outputs remain or become high as soon as one of the CCxE or CCxNE bits is high.
- The break status flag (SBIF, BIF and B2IF bits in the TIMx_SR register) is set. An interrupt is generated if the BIE bit in the TIMx_DIER register is set.
- If the AOE bit in the TIMx_BDTR register is set, the MOE bit is automatically set again at the next update event (UEV). As an example, this can be used to perform a regulation. Otherwise, MOE remains low until the application sets it to '1' again. In this case, it can be used for security and the break input can be connected to an alarm from power drivers, thermal sensors or any security components.

Note: *The break inputs are active on level. Thus, the MOE cannot be set while the break input is active (neither automatically nor by software). In the meantime, the status flag BIF and B2IF cannot be cleared.*

In addition to the break input and the output management, a write protection has been implemented inside the break circuit to safeguard the application. It allows the configuration of several parameters to be frozen (dead-time duration, OCx/OCxN polarities and state when disabled, OCxM configurations, break enable and polarity). The application can choose from 3 levels of protection selected by the LOCK bits in the TIMx_BDTR register. Refer to [Section 38.4.20: TIMx break and dead-time register \(TIMx_BDTR\)\(x = 1, 8\)](#). The LOCK bits can be written only once after an MCU reset.

[Figure 388](#) shows an example of behavior of the outputs in response to a break.

Figure 388. Various output behavior in response to a break event on BRK (OSS1 = 1)

The two break inputs have different behaviors on timer outputs:

- The BRK input can either disable (inactive state) or force the PWM outputs to a predefined safe state.
- BRK2 can only disable (inactive state) the PWM outputs.

The BRK has a higher priority than BRK2 input, as described in [Table 327](#).

Note: *BRK2 must only be used with OSSR = OSSl = 1.*

Table 327. Behavior of timer outputs versus BRK/BRK2 inputs

BRK	BRK2	Timer outputs state	Typical use case	
			OCxN output (low side switches)	OCx output (high side switches)
Active	X	– Inactive then forced output state (after a deadtime) – Outputs disabled if OSSl = 0 (control taken over by GPIO logic)	ON after deadtime insertion	OFF
Inactive	Active	Inactive	OFF	OFF

[Figure 389](#) gives an example of OCx and OCxN output behavior in case of active signals on BRK and BRK2 inputs. In this case, both outputs have active high polarities (CCxP = CCxNP = 0 in TIMx_CCER register).

Figure 389. PWM output state following BRK and BRK2 pins assertion (OSSl=1)

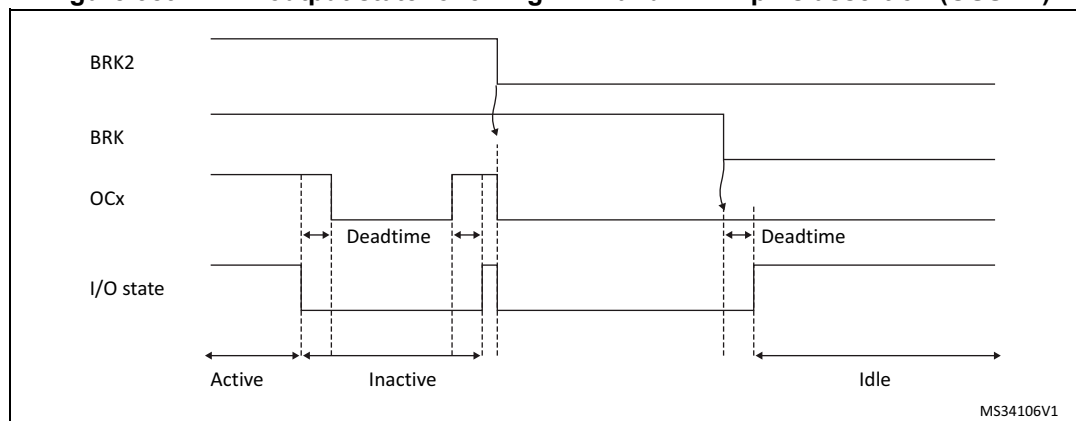
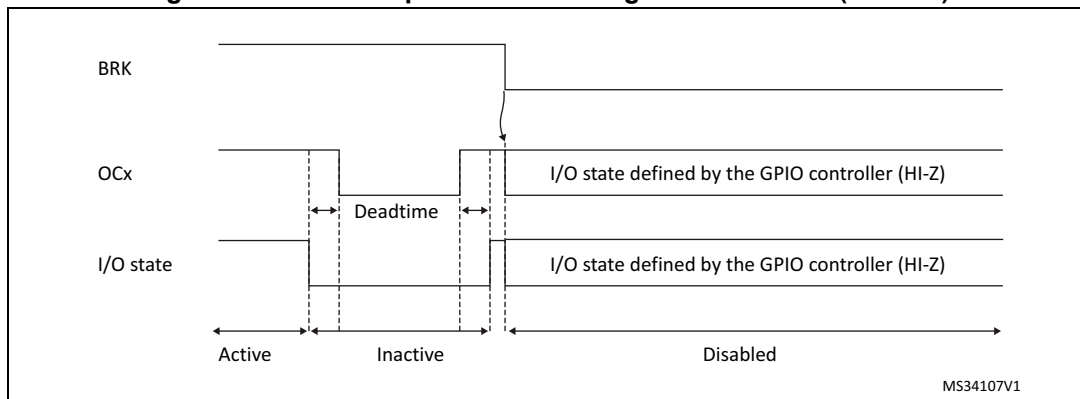


Figure 390. PWM output state following BRK assertion (OSSI=0)



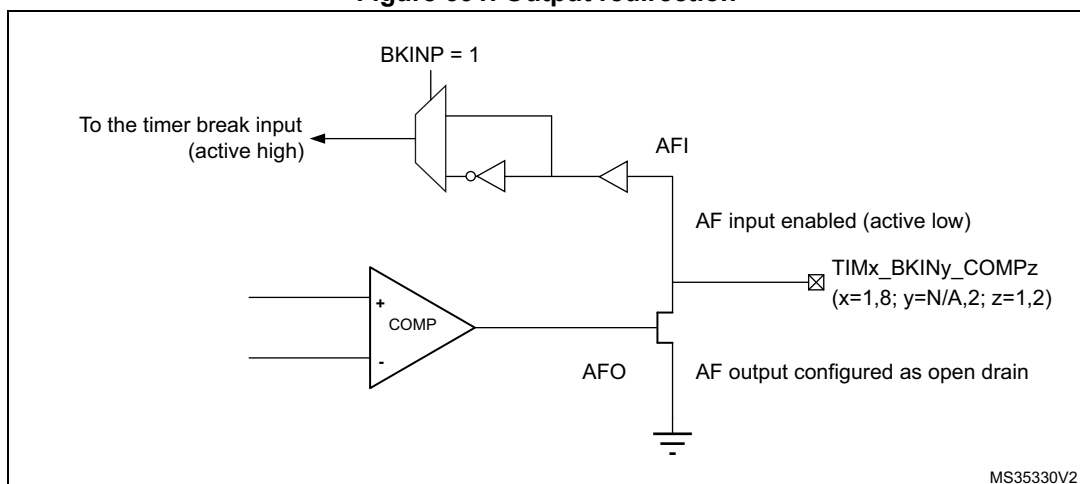
38.3.17 Bidirectional break inputs

Beside regular digital break inputs and internal break events coming from the comparators, the timer 1 and 8 are featuring bidirectional break inputs/outputs combining the two sources, as represented on [Figure 391](#).

The TIMx_BKINy_COMPz pins are combining the COMPz output (to be configured in open drain) and the Timerx's TIMx_BKINy input. They allow to have:

- A global break information available for external MCUs or gate drivers shut down inputs, with a single-pin.
- An internal comparator and multiple external open drain comparators outputs OREd together and triggering a break event, when the multiple internal and external break inputs must be merged.

Figure 391. Output redirection



38.3.18 Clearing the OCxREF signal on an external event

The OCxREF signal of a given channel can be cleared when a high level is applied on the ocref_clr_int input (OCxCE enable bit in the corresponding TIMx_CCMRx register set to 1). OCxREF remains low until the next update event (UEV) occurs. This function can only be

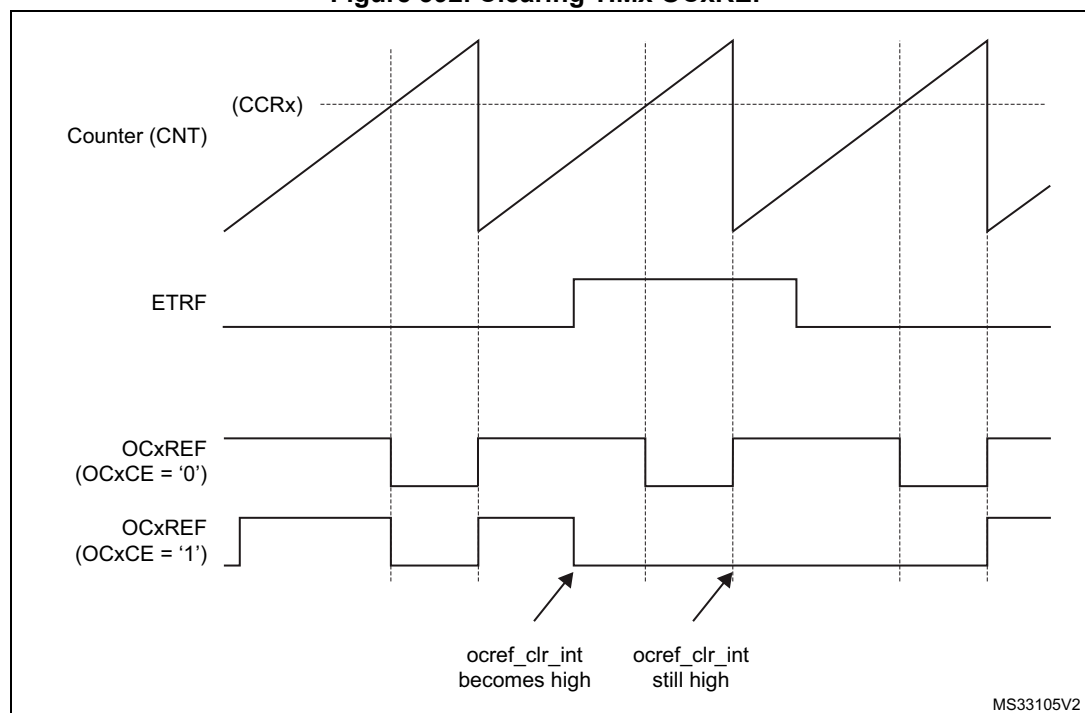
used in Output compare and PWM modes. It does not work in Forced mode. The `ocref_clr_int` is connected to the ETRF signal (ETRF after filtering).

When ETRF is chosen, ETR must be configured as follows:

1. The External Trigger Prescaler should be kept off: bits ETPS[1:0] of the TIMx_SMCR register set to '00'.
2. The external clock mode 2 must be disabled: bit ECE of the TIMx_SMCR register set to '0'.
3. The External Trigger Polarity (ETP) and the External Trigger Filter (ETF) can be configured according to the user needs.

Figure 392 shows the behavior of the OCxREF signal when the ETRF Input becomes High, for both values of the enable bit OCxCE. In this example, the timer TIMx is programmed in PWM mode.

Figure 392. Clearing TIMx OCxREF



Note: In case of a PWM with a 100% duty cycle (if $CCR_x > ARR$), then OCxREF is enabled again at the next counter overflow.

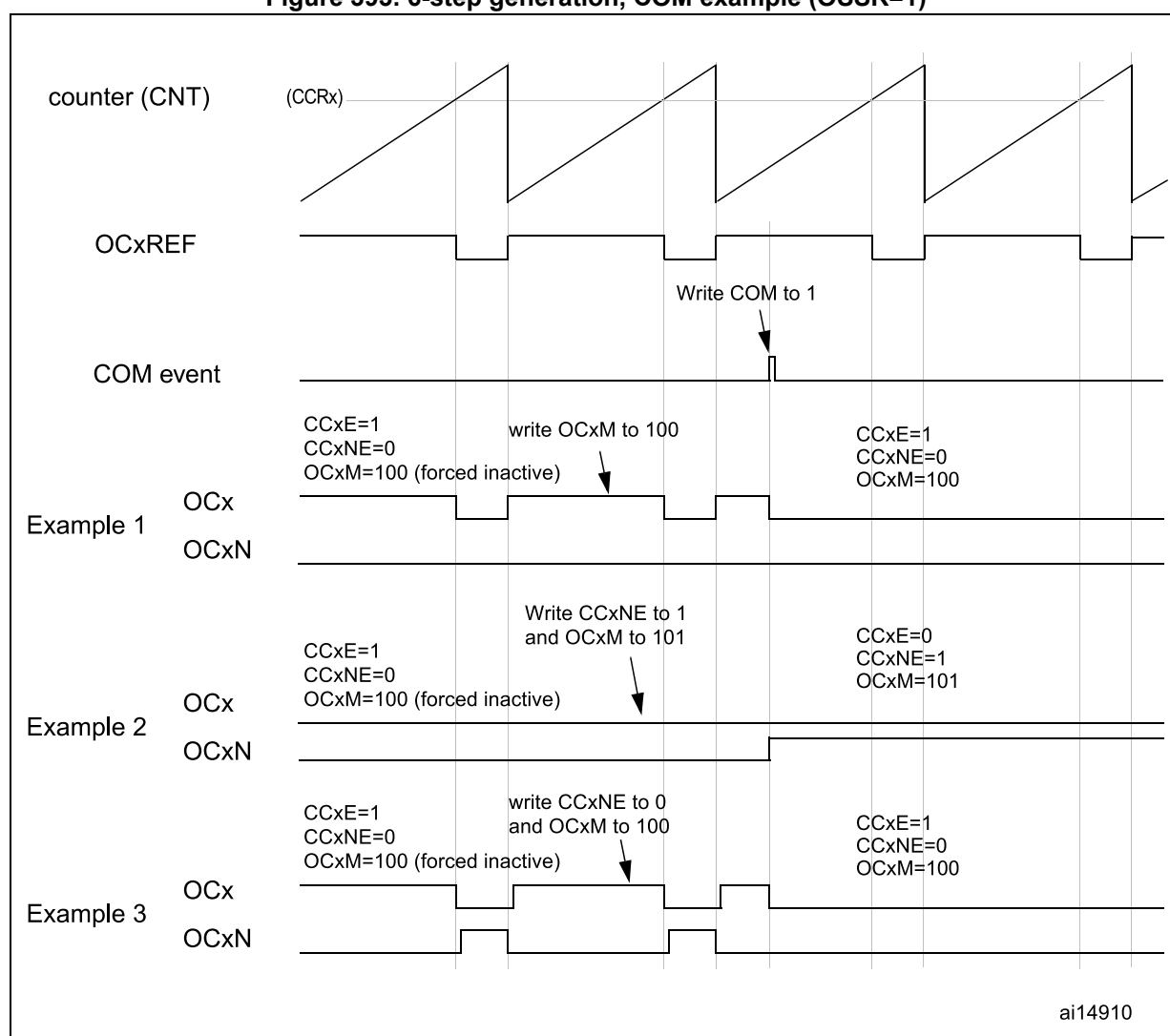
38.3.19 6-step PWM generation

When complementary outputs are used on a channel, preload bits are available on the OCxM, CCxE and CCxNE bits. The preload bits are transferred to the shadow bits at the COM commutation event. Thus one can program in advance the configuration for the next step and change the configuration of all the channels at the same time. COM can be generated by software by setting the COM bit in the TIMx_EGR register or by hardware (on TRGI rising edge).

A flag is set when the COM event occurs (COMIF bit in the TIMx_SR register), which can generate an interrupt (if the COMIE bit is set in the TIMx_DIER register) or a DMA request (if the COMDE bit is set in the TIMx_DIER register).

The [Figure 393](#) describes the behavior of the OCx and OCxN outputs when a COM event occurs, in 3 different examples of programmed configurations.

Figure 393. 6-step generation, COM example (OSSR=1)



38.3.20 One-pulse mode

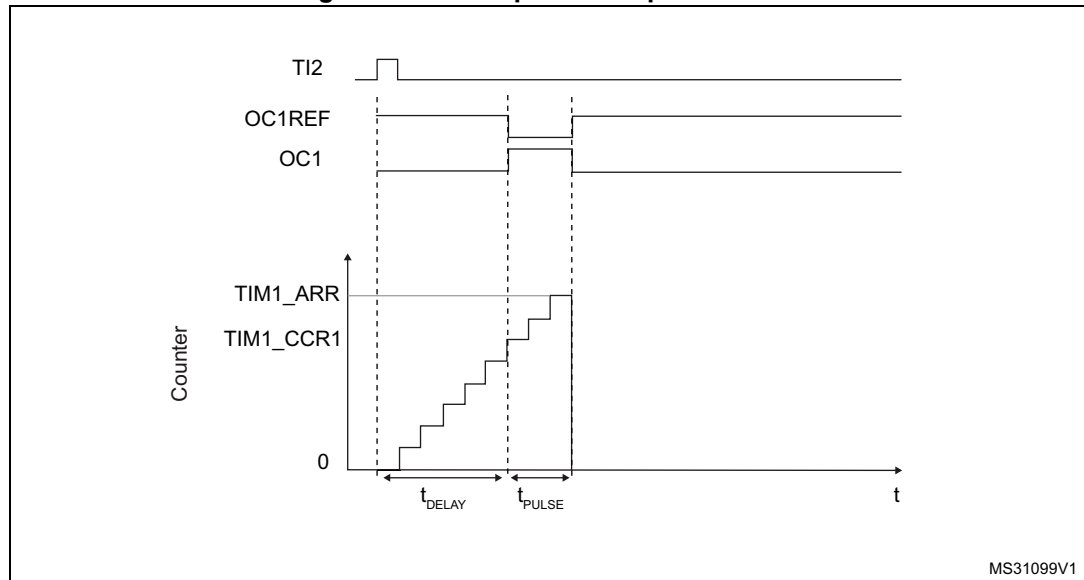
One-pulse mode (OPM) is a particular case of the previous modes. It allows the counter to be started in response to a stimulus and to generate a pulse with a programmable length after a programmable delay.

Starting the counter can be controlled through the slave mode controller. Generating the waveform can be done in output compare mode or PWM mode. One-pulse mode is selected by setting the OPM bit in the TIMx_CR1 register. This makes the counter stop automatically at the next update event UEV.

A pulse can be correctly generated only if the compare value is different from the counter initial value. Before starting (when the timer is waiting for the trigger), the configuration must be:

- In upcounting: $CNT < CCRx \leq ARR$ (in particular, $0 < CCRx$)
- In downcounting: $CNT > CCRx$

Figure 394. Example of one pulse mode.



For example one may want to generate a positive pulse on OC1 with a length of t_{PULSE} and after a delay of t_{DELAY} as soon as a positive edge is detected on the TI2 input pin.

Let's use TI2FP2 as trigger 1:

1. Select the proper TI2x source (internal or external) with the TI2SEL[3:0] bits in the TIMx_TISEL register.
2. Map TI2FP2 to TI2 by writing CC2S='01' in the TIMx_CCMR1 register.
3. TI2FP2 must detect a rising edge, write CC2P='0' and CC2NP='0' in the TIMx_CCER register.
4. Configure TI2FP2 as trigger for the slave mode controller (TRGI) by writing TS=00110 in the TIMx_SMCR register.
5. TI2FP2 is used to start the counter by writing SMS to '110' in the TIMx_SMCR register (trigger mode).

The OPM waveform is defined by writing the compare registers (taking into account the clock frequency and the counter prescaler).

- The t_{DELAY} is defined by the value written in the TIMx_CCR1 register.
- The t_{PULSE} is defined by the difference between the auto-reload value and the compare value (TIMx_ARR - TIMx_CCR1).
- Let's say one want to build a waveform with a transition from '0' to '1' when a compare match occurs and a transition from '1' to '0' when the counter reaches the auto-reload value. To do this PWM mode 2 must be enabled by writing OC1M=111 in the TIMx_CCMR1 register. Optionally the preload registers can be enabled by writing OC1PE='1' in the TIMx_CCMR1 register and ARPE in the TIMx_CR1 register. In this case one has to write the compare value in the TIMx_CCR1 register, the auto-reload value in the TIMx_ARR register, generate an update by setting the UG bit and wait for external trigger event on TI2. CC1P is written to '0' in this example.

In our example, the DIR and CMS bits in the TIMx_CR1 register should be low.

Since only 1 pulse (Single mode) is needed, a 1 must be written in the OPM bit in the TIMx_CR1 register to stop the counter at the next update event (when the counter rolls over from the auto-reload value back to 0). When OPM bit in the TIMx_CR1 register is set to '0', so the Repetitive Mode is selected.

Particular case: OCx fast enable:

In One-pulse mode, the edge detection on TIx input set the CEN bit which enables the counter. Then the comparison between the counter and the compare value makes the output toggle. But several clock cycles are needed for these operations and it limits the minimum delay $t_{\text{DELAY min}}$ we can get.

If one wants to output a waveform with the minimum delay, the OCxFE bit can be set in the TIMx_CCMRx register. Then OCxRef (and OCx) are forced in response to the stimulus, without taking in account the comparison. Its new level is the same as if a compare match had occurred. OCxFE acts only if the channel is configured in PWM1 or PWM2 mode.

38.3.21 Retriggerable one pulse mode (OPM)

This mode allows the counter to be started in response to a stimulus and to generate a pulse with a programmable length, but with the following differences with Non-retriggerable one pulse mode described in [Section 38.3.20](#):

- The pulse starts as soon as the trigger occurs (no programmable delay)
- The pulse is extended if a new trigger occurs before the previous one is completed

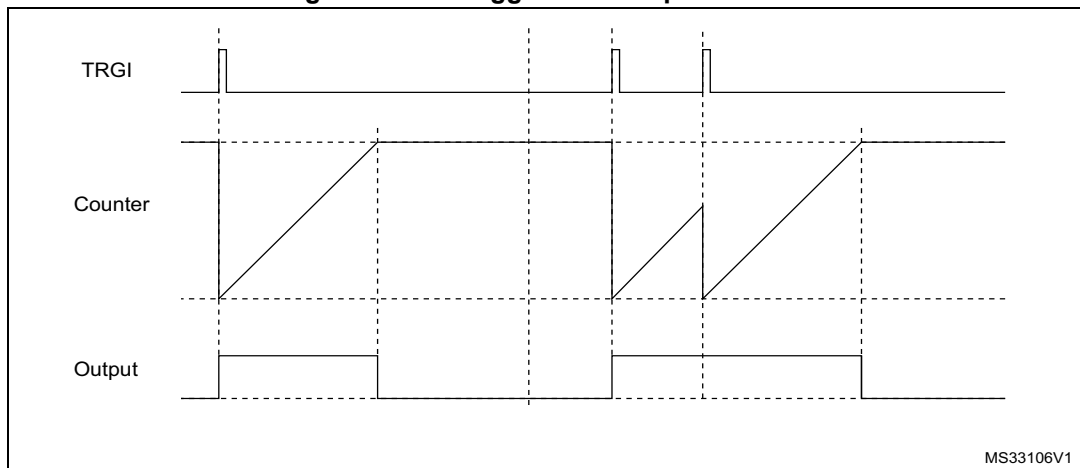
The timer must be in Slave mode, with the bits SMS[3:0] = '1000' (Combined Reset + trigger mode) in the TIMx_SMCR register, and the OCxM[3:0] bits set to '1000' or '1001' for Retriggerable OPM mode 1 or 2.

If the timer is configured in Up-counting mode, the corresponding CCRx must be set to 0 (the ARR register sets the pulse length). If the timer is configured in Down-counting mode, CCRx must be above or equal to ARR.

Note: The OCxM[3:0] and SMS[3:0] bit fields are split into two parts for compatibility reasons, the most significant bit are not contiguous with the 3 least significant ones.

This mode must not be used with center-aligned PWM modes. It is mandatory to have CMS[1:0] = 00 in TIMx_CR1.

Figure 395. Retriggerable one pulse mode



38.3.22 Encoder interface mode

To select Encoder Interface mode write SMS='001' in the TIMx_SMCR register if the counter is counting on TI2 edges only, SMS='010' if it is counting on TI1 edges only and SMS='011' if it is counting on both TI1 and TI2 edges.

Select the TI1 and TI2 polarity by programming the CC1P and CC2P bits in the TIMx_CCER register. When needed, the input filter can be programmed as well. CC1NP and CC2NP must be kept low.

The two inputs TI1 and TI2 are used to interface to a quadrature encoder. Refer to [Table 328](#). The counter is clocked by each valid transition on TI1FP1 or TI2FP2 (TI1 and TI2 after input filter and polarity selection, TI1FP1=TI1 if not filtered and not inverted, TI2FP2=TI2 if not filtered and not inverted) assuming that it is enabled (CEN bit in TIMx_CR1 register written to '1'). The sequence of transitions of the two inputs is evaluated and generates count pulses as well as the direction signal. Depending on the sequence the counter counts up or down, the DIR bit in the TIMx_CR1 register is modified by hardware accordingly. The DIR bit is calculated at each transition on any input (TI1 or TI2), whatever the counter is counting on TI1 only, TI2 only or both TI1 and TI2.

Encoder interface mode acts simply as an external clock with direction selection. This means that the counter just counts continuously between 0 and the auto-reload value in the TIMx_ARR register (0 to ARR or ARR down to 0 depending on the direction). So the TIMx_ARR must be configured before starting. In the same way, the capture, compare, repetition counter, trigger output features continue to work as normal. Encoder mode and External clock mode 2 are not compatible and must not be selected together.

Note: *The prescaler must be set to zero when encoder mode is enabled*

In this mode, the counter is modified automatically following the speed and the direction of the quadrature encoder and its content, therefore, always represents the encoder's position. The count direction correspond to the rotation direction of the connected sensor. The table summarizes the possible combinations, assuming TI1 and TI2 do not switch at the same time.

Table 328. Counting direction versus encoder signals

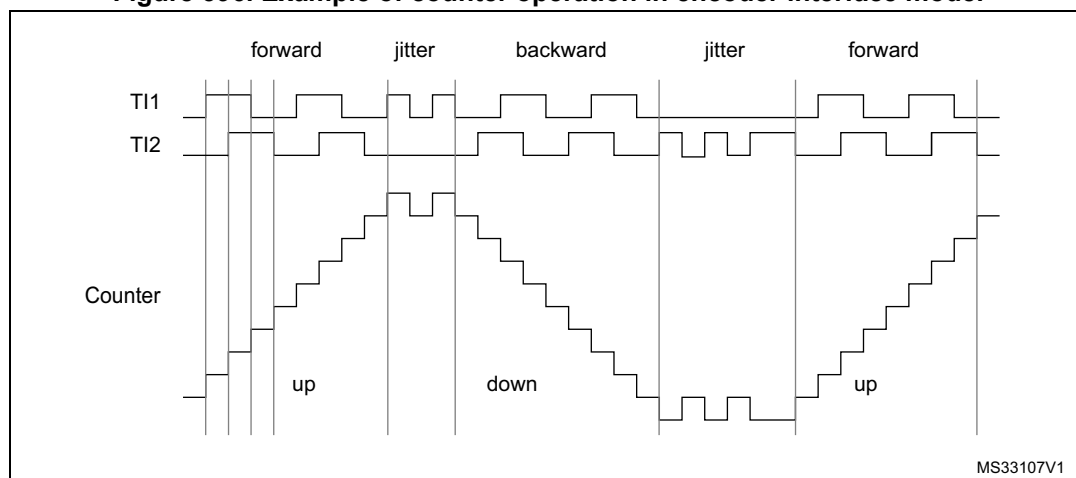
Active edge	Level on opposite signal (TI1FP1 for TI2, TI2FP2 for TI1)	TI1FP1 signal		TI2FP2 signal	
		Rising	Falling	Rising	Falling
Counting on TI1 only	High	Down	Up	No Count	No Count
	Low	Up	Down	No Count	No Count
Counting on TI2 only	High	No Count	No Count	Up	Down
	Low	No Count	No Count	Down	Up
Counting on TI1 and TI2	High	Down	Up	Up	Down
	Low	Up	Down	Down	Up

A quadrature encoder can be connected directly to the MCU without external interface logic. However, comparators are normally be used to convert the encoder's differential outputs to digital signals. This greatly increases noise immunity. The third encoder output which indicate the mechanical zero position, may be connected to an external interrupt input and trigger a counter reset.

The [Figure 396](#) gives an example of counter operation, showing count signal generation and direction control. It also shows how input jitter is compensated where both edges are selected. This might occur if the sensor is positioned near to one of the switching points. For this example we assume that the configuration is the following:

- CC1S='01' (TIMx_CCMR1 register, TI1FP1 mapped on TI1).
- CC2S='01' (TIMx_CCMR2 register, TI1FP2 mapped on TI2).
- CC1P='0' and CC1NP='0' (TIMx_CCER register, TI1FP1 non-inverted, TI1FP1=TI1).
- CC2P='0' and CC2NP='0' (TIMx_CCER register, TI1FP2 non-inverted, TI1FP2= TI2).
- SMS='011' (TIMx_SMCR register, both inputs are active on both rising and falling edges).
- CEN='1' (TIMx_CR1 register, Counter enabled).

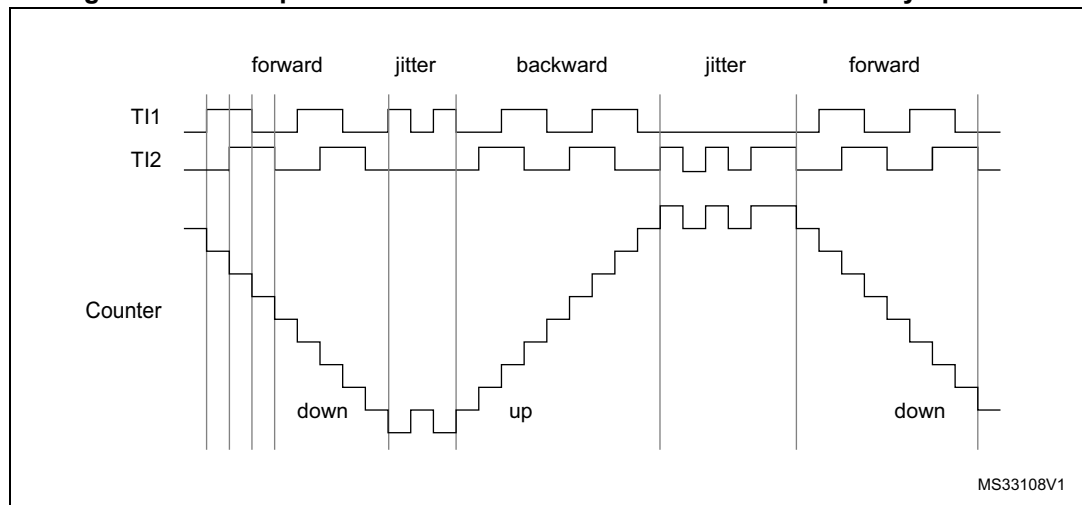
Figure 396. Example of counter operation in encoder interface mode.



MS33107V1

Figure 397 gives an example of counter behavior when TI1FP1 polarity is inverted (same configuration as above except CC1P='1').

Figure 397. Example of encoder interface mode with TI1FP1 polarity inverted.



The timer, when configured in Encoder Interface mode provides information on the sensor's current position. Dynamic information can be obtained (speed, acceleration, deceleration) by measuring the period between two encoder events using a second timer configured in capture mode. The output of the encoder which indicates the mechanical zero can be used for this purpose. Depending on the time between two events, the counter can also be read at regular times. This can be done by latching the counter value into a third input capture register if available (then the capture signal must be periodic and can be generated by another timer). when available, it is also possible to read its value through a DMA request.

The IUFREMAP bit in the TIMx_CR1 register forces a continuous copy of the update interrupt flag (UIF) into the timer counter register's bit 31 (TIMxCNT[31]). This allows both the counter value and a potential roll-over condition signaled by the UIFCPY flag to be read in an atomic way. It eases the calculation of angular speed by avoiding race conditions caused, for instance, by a processing shared between a background task (counter reading) and an interrupt (update interrupt).

There is no latency between the UIF and UIFCPY flag assertions.

In 32-bit timer implementations, when the IUFREMAP bit is set, bit 31 of the counter is overwritten by the UIFCPY flag upon read access (the counter's most significant bit is only accessible in write mode).

38.3.23 UIF bit remapping

The IUFREMAP bit in the TIMx_CR1 register forces a continuous copy of the Update Interrupt Flag UIF into the timer counter register's bit 31 (TIMxCNT[31]). This allows both the counter value and a potential roll-over condition signaled by the UIFCPY flag to be read in an atomic way. In particular cases, it can ease the calculations by avoiding race conditions, caused for instance by a processing shared between a background task (counter reading) and an interrupt (Update Interrupt).

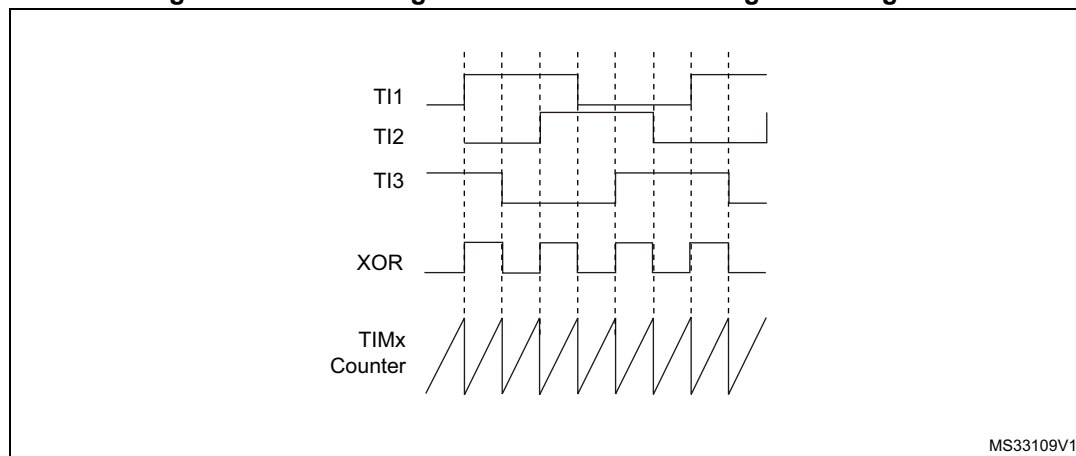
There is no latency between the UIF and UIFCPY flags assertion.

38.3.24 Timer input XOR function

The TI1S bit in the TIMx_CR2 register, allows the input filter of channel 1 to be connected to the output of an XOR gate, combining the three input pins TIMx_CH1, TIMx_CH2 and TIMx_CH3.

The XOR output can be used with all the timer input functions such as trigger or input capture. It is convenient to measure the interval between edges on two input signals, as per [Figure 398](#) below.

Figure 398. Measuring time interval between edges on 3 signals



38.3.25 Interfacing with Hall sensors

This is done using the advanced-control timers (TIM1 or TIM8) to generate PWM signals to drive the motor and another timer TIMx (TIM2, TIM3, TIM4) referred to as “interfacing timer” in [Figure 399](#). The “interfacing timer” captures the 3 timer input pins (CC1, CC2, CC3) connected through a XOR to the TI1 input channel (selected by setting the TI1S bit in the TIMx_CR2 register).

The slave mode controller is configured in reset mode; the slave input is TI1F_ED. Thus, each time one of the 3 inputs toggles, the counter restarts counting from 0. This creates a time base triggered by any change on the Hall inputs.

On the “interfacing timer”, capture/compare channel 1 is configured in capture mode, capture signal is TRC (See [Figure 372: Capture/compare channel \(example: channel 1 input stage\) on page 1568](#)). The captured value, which corresponds to the time elapsed between 2 changes on the inputs, gives information about motor speed.

The “interfacing timer” can be used in output mode to generate a pulse which changes the configuration of the channels of the advanced-control timer (TIM1 or TIM8) (by triggering a COM event). The TIM1 timer is used to generate PWM signals to drive the motor. To do this, the interfacing timer channel must be programmed so that a positive pulse is generated after a programmed delay (in output compare or PWM mode). This pulse is sent to the advanced-control timer (TIM1 or TIM8) through the TRGO output.

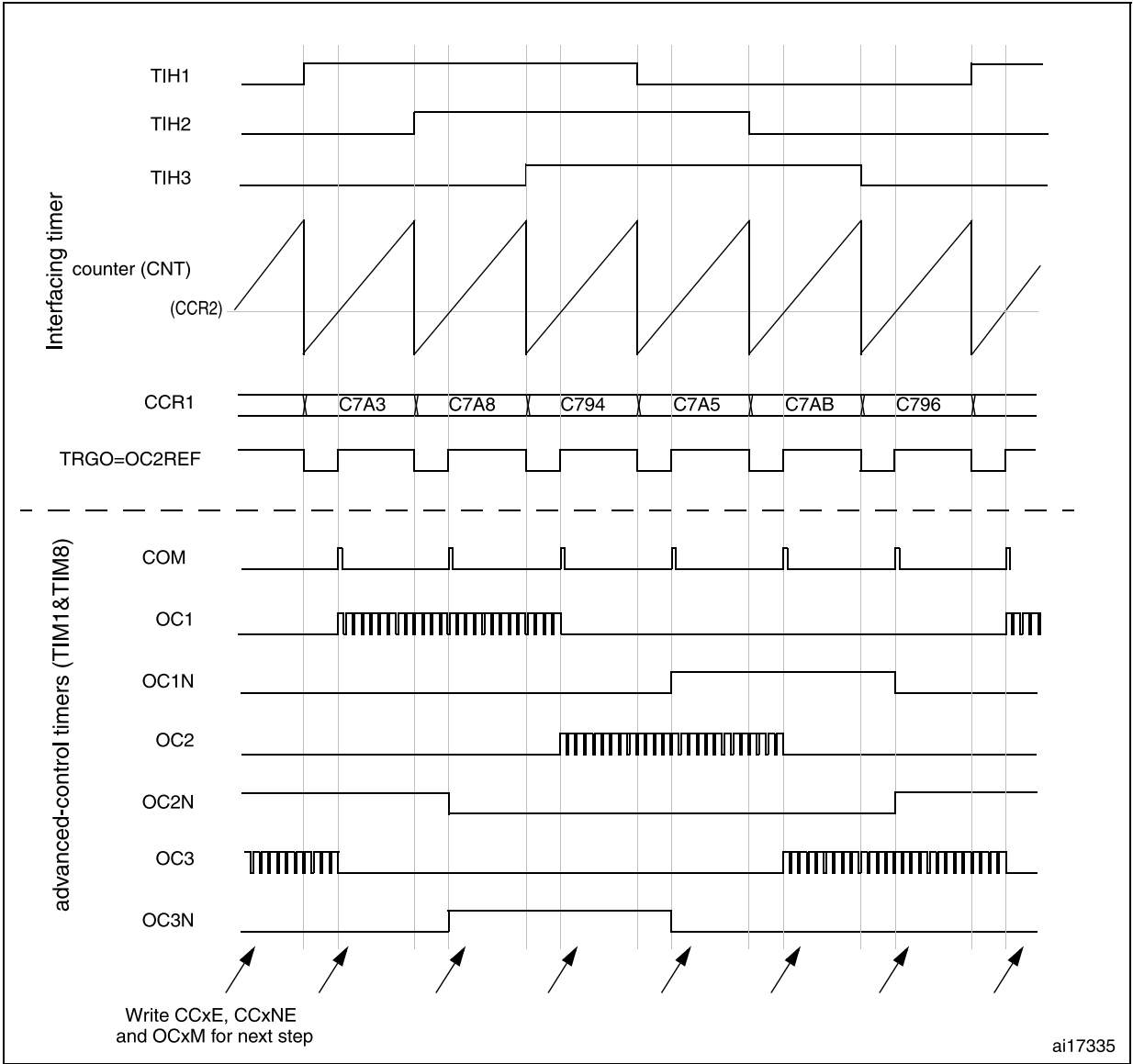
Example: one wants to change the PWM configuration of the advanced-control timer TIM1 after a programmed delay each time a change occurs on the Hall inputs connected to one of the TIMx timers.

- Configure 3 timer inputs ORed to the TI1 input channel by writing the TI1S bit in the TIMx_CR2 register to '1',
- Program the time base: write the TIMx_ARR to the max value (the counter must be cleared by the TI1 change. Set the prescaler to get a maximum counter period longer than the time between 2 changes on the sensors,
- Program the channel 1 in capture mode (TRC selected): write the CC1S bits in the TIMx_CCMR1 register to '01'. The digital filter can also be programmed if needed,
- Program the channel 2 in PWM 2 mode with the desired delay: write the OC2M bits to '111' and the CC2S bits to '00' in the TIMx_CCMR1 register,
- Select OC2REF as trigger output on TRGO: write the MMS bits in the TIMx_CR2 register to '101',

In the advanced-control timer TIM1, the right ITR input must be selected as trigger input, the timer is programmed to generate PWM signals, the capture/compare control signals are preloaded (CCPC=1 in the TIMx_CR2 register) and the COM event is controlled by the trigger input (CCUS=1 in the TIMx_CR2 register). The PWM control bits (CCxE, OCxM) are written after a COM event for the next step (this can be done in an interrupt subroutine generated by the rising edge of OC2REF).

The [Figure 399](#) describes this example.

Figure 399. Example of Hall sensor interface



38.3.26 Timer synchronization

The TIMx timers are linked together internally for timer synchronization or chaining. They can be synchronized in several modes: Reset mode, Gated mode, and Trigger mode.

Slave mode: Reset mode

The counter and its prescaler can be reinitialized in response to an event on a trigger input. Moreover, if the URS bit from the TIMx_CR1 register is low, an update event UEV is generated. Then all the preloaded registers (TIMx_ARR, TIMx_CCRx) are updated.

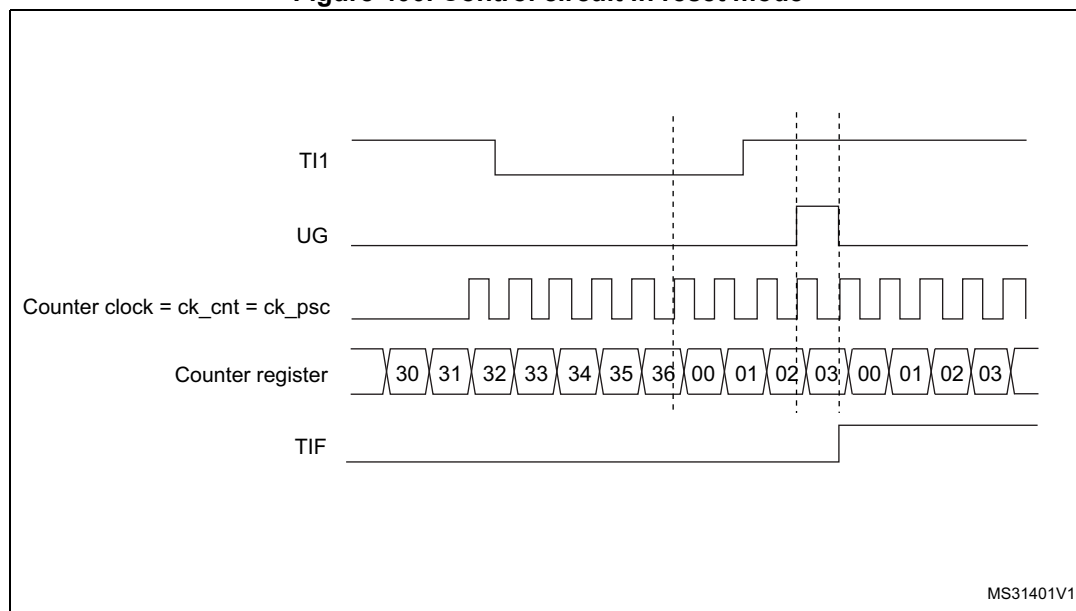
In the following example, the upcounter is cleared in response to a rising edge on TI1 input:

- Configure the channel 1 to detect rising edges on TI1. Configure the input filter duration (in this example, we do not need any filter, so we keep IC1F=0000). The capture prescaler is not used for triggering, so it does not need to be configured. The CC1S bits select the input capture source only, CC1S = 01 in the TIMx_CCMR1 register. Write CC1P=0 and CC1NP='0' in TIMx_CCER register to validate the polarity (and detect rising edges only).
- Configure the timer in reset mode by writing SMS=100 in TIMx_SMCR register. Select TI1 as the input source by writing TS=00101 in TIMx_SMCR register.
- Start the counter by writing CEN=1 in the TIMx_CR1 register.

The counter starts counting on the internal clock, then behaves normally until TI1 rising edge. When TI1 rises, the counter is cleared and restarts from 0. In the meantime, the trigger flag is set (TIF bit in the TIMx_SR register) and an interrupt request, or a DMA request can be sent if enabled (depending on the TIE and TDE bits in TIMx_DIER register).

The following figure shows this behavior when the auto-reload register TIMx_ARR=0x36. The delay between the rising edge on TI1 and the actual reset of the counter is due to the resynchronization circuit on TI1 input.

Figure 400. Control circuit in reset mode



Slave mode: Gated mode

The counter can be enabled depending on the level of a selected input.

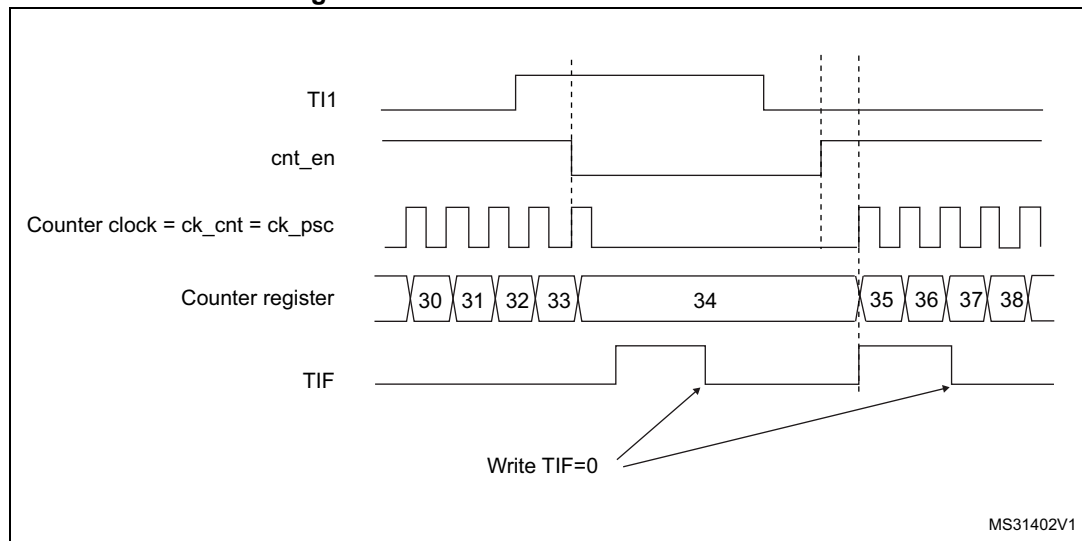
In the following example, the upcounter counts only when TI1 input is low:

- Configure the channel 1 to detect low levels on TI1. Configure the input filter duration (in this example, we do not need any filter, so we keep IC1F=0000). The capture prescaler is not used for triggering, so it does not need to be configured. The CC1S bits select the input capture source only, CC1S=01 in TIMx_CCMR1 register. Write CC1P=1 and CC1NP='0' in TIMx_CCER register to validate the polarity (and detect low level only).
- Configure the timer in gated mode by writing SMS=101 in TIMx_SMCR register. Select TI1 as the input source by writing TS=00101 in TIMx_SMCR register.
- Enable the counter by writing CEN=1 in the TIMx_CR1 register (in gated mode, the counter doesn't start if CEN=0, whatever is the trigger input level).

The counter starts counting on the internal clock as long as TI1 is low and stops as soon as TI1 becomes high. The TIF flag in the TIMx_SR register is set both when the counter starts or stops.

The delay between the rising edge on TI1 and the actual stop of the counter is due to the resynchronization circuit on TI1 input.

Figure 401. Control circuit in Gated mode



Slave mode: Trigger mode

The counter can start in response to an event on a selected input.

In the following example, the upcounter starts in response to a rising edge on TI2 input:

- Configure the channel 2 to detect rising edges on TI2. Configure the input filter duration (in this example, we do not need any filter, so we keep IC2F=0000). The capture prescaler is not used for triggering, so it does not need to be configured. The CC2S bits are configured to select the input capture source only, CC2S=01 in TIMx_CCMR1

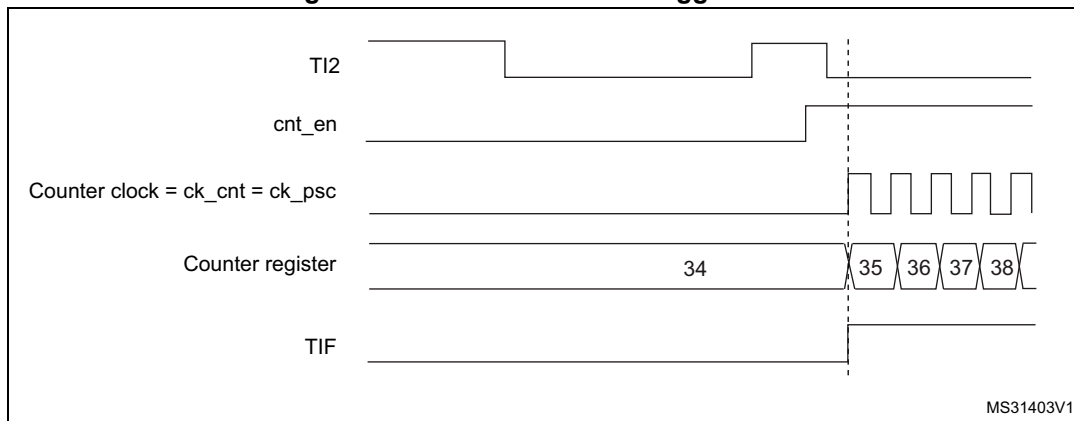
register. Write CC2P=1 and CC2NP=0 in TIMx_CCER register to validate the polarity (and detect low level only).

- Configure the timer in trigger mode by writing SMS=110 in TIMx_SMCR register. Select TI2 as the input source by writing TS=00110 in TIMx_SMCR register.

When a rising edge occurs on TI2, the counter starts counting on the internal clock and the TIF flag is set.

The delay between the rising edge on TI2 and the actual start of the counter is due to the resynchronization circuit on TI2 input.

Figure 402. Control circuit in trigger mode



Slave mode: Combined reset + trigger mode

In this case, a rising edge of the selected trigger input (TRGI) reinitializes the counter, generates an update of the registers, and starts the counter.

This mode is used for one-pulse mode.

Slave mode: external clock mode 2 + trigger mode

The external clock mode 2 can be used in addition to another slave mode (except external clock mode 1 and encoder mode). In this case, the ETR signal is used as external clock input, and another input can be selected as trigger input (in reset mode, gated mode or trigger mode). It is recommended not to select ETR as TRGI through the TS bits of TIMx_SMCR register.

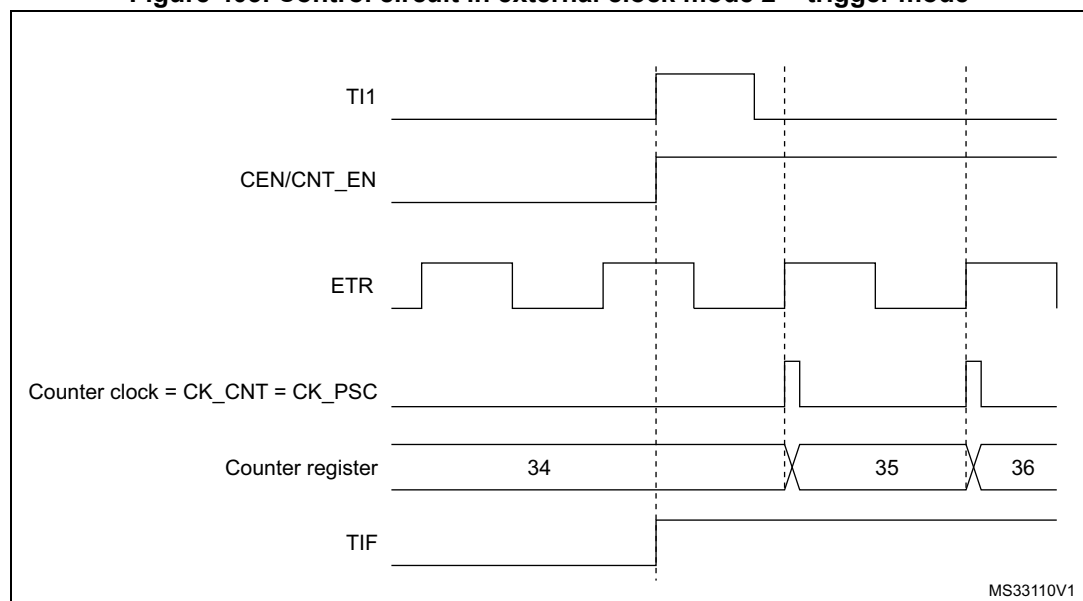
In the following example, the upcounter is incremented at each rising edge of the ETR signal as soon as a rising edge of TI1 occurs:

1. Configure the external trigger input circuit by programming the TIMx_SMCR register as follows:
 - ETF = 0000: no filter
 - ETPS=00: prescaler disabled
 - ETP=0: detection of rising edges on ETR and ECE=1 to enable the external clock mode 2.
2. Configure the channel 1 as follows, to detect rising edges on TI:
 - IC1F=0000: no filter.
 - The capture prescaler is not used for triggering and does not need to be configured.
 - CC1S=01 in TIMx_CCMR1 register to select only the input capture source
 - CC1P=0 and CC1NP='0' in TIMx_CCER register to validate the polarity (and detect rising edge only).
3. Configure the timer in trigger mode by writing SMS=110 in TIMx_SMCR register. Select TI1 as the input source by writing TS=00101 in TIMx_SMCR register.

A rising edge on TI1 enables the counter and sets the TIF flag. The counter then counts on ETR rising edges.

The delay between the rising edge of the ETR signal and the actual reset of the counter is due to the resynchronization circuit on ETRP input.

Figure 403. Control circuit in external clock mode 2 + trigger mode



Note: The clock of the slave peripherals (timer, ADC, ...) receiving the TRGO or the TRGO2 signals must be enabled prior to receive events from the master timer, and the clock frequency (prescaler) must not be changed on-the-fly while triggers are received from the master timer.

38.3.27 ADC synchronization

The timer can generate an ADC triggering event with various internal signals, such as reset, enable or compare events. It is also possible to generate a pulse issued by internal edge detectors, such as:

- Rising and falling edges of OC4ref
- Rising edge on OC5ref or falling edge on OC6ref

The triggers are issued on the TRGO2 internal line which is redirected to the ADC. There is a total of 16 possible events, which can be selected using the MMS2[3:0] bits in the TIMx_CR2 register.

An example of an application for 3-phase motor drives is given in [Figure 383 on page 1580](#).

Note: The clock of the slave peripherals (timer, ADC, ...) receiving the TRGO or the TRGO2 signals must be enabled prior to receive events from the master timer, and the clock frequency (prescaler) must not be changed on-the-fly while triggers are received from the master timer.

Note: The clock of the ADC must be enabled prior to receive events from the master timer, and must not be changed on-the-fly while triggers are received from the timer.

38.3.28 DMA burst mode

The TIMx timers have the capability to generate multiple DMA requests upon a single event. The main purpose is to be able to re-program part of the timer multiple times without software overhead, but it can also be used to read several registers in a row, at regular intervals.

The DMA controller destination is unique and must point to the virtual register TIMx_DMAR. On a given timer event, the timer launches a sequence of DMA requests (burst). Each write into the TIMx_DMAR register is actually redirected to one of the timer registers.

The DBL[4:0] bits in the TIMx_DCR register set the DMA burst length. The timer recognizes a burst transfer when a read or a write access is done to the TIMx_DMAR address, i.e. the number of transfers (either in half-words or in bytes).

The DBA[4:0] bits in the TIMx_DCR registers define the DMA base address for DMA transfers (when read/write access are done through the TIMx_DMAR address). DBA is defined as an offset starting from the address of the TIMx_CR1 register:

Example:

00000: TIMx_CR1

00001: TIMx_CR2

00010: TIMx_SMCR

As an example, the timer DMA burst feature is used to update the contents of the CCRx registers (x = 2, 3, 4) upon an update event, with the DMA transferring half words into the CCRx registers.

This is done in the following steps:

1. Configure the corresponding DMA channel as follows:
 - DMA channel peripheral address is the DMAR register address
 - DMA channel memory address is the address of the buffer in the RAM containing the data to be transferred by DMA into CCRx registers.
 - Number of data to transfer = 3 (See note below).
 - Circular mode disabled.
2. Configure the DCR register by configuring the DBA and DBL bit fields as follows:
DBL = 3 transfers, DBA = 0xE.
3. Enable the TIMx update DMA request (set the UDE bit in the DIER register).
4. Enable TIMx
5. Enable the DMA channel

This example is for the case where every CCRx register to be updated once. If every CCRx register is to be updated twice for example, the number of data to transfer should be 6. Let's take the example of a buffer in the RAM containing data1, data2, data3, data4, data5 and data6. The data is transferred to the CCRx registers as follows: on the first update DMA request, data1 is transferred to CCR2, data2 is transferred to CCR3, data3 is transferred to CCR4 and on the second update DMA request, data4 is transferred to CCR2, data5 is transferred to CCR3 and data6 is transferred to CCR4.

Note: A null value can be written to the reserved registers.

38.3.29 Debug mode

When the microcontroller enters debug mode (Cortex[®]-M7 with FPU core halted), the TIMx counter either continues to work normally or stops, depending on DBG_TIMx_STOP configuration bit in DBG module.

For safety purposes, when the counter is stopped, the outputs are disabled (as if the MOE bit was reset). The outputs can either be forced to an inactive state (OSSI bit = 1), or have their control taken over by the GPIO controller (OSSI bit = 0), typically to force a Hi-Z.

For more details, refer to section Debug support (DBG).

38.4 TIM1/TIM8 registers

Refer to for a list of abbreviations used in register descriptions.

The peripheral registers can be accessed by half-words (16-bit) or words (32-bit).

38.4.1 TIMx control register 1 (TIMx_CR1)(x = 1, 8)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	UIFRE MAP	Res.	CKD[1:0]		ARPE	CMS[1:0]		DIR	OPM	URS	UDIS	CEN
				rW		rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 15:12 Reserved, must be kept at reset value.

Bit 11 **UIFREMAP**: UIF status bit remapping

0: No remapping. UIF status bit is not copied to TIMx_CNT register bit 31.

1: Remapping enabled. UIF status bit is copied to TIMx_CNT register bit 31.

Bit 10 Reserved, must be kept at reset value.

Bits 9:8 **CKD[1:0]**: Clock division

This bit-field indicates the division ratio between the timer clock (CK_INT) frequency and the dead-time and sampling clock (t_{DTS}) used by the dead-time generators and the digital filters (ETR, TIx),

00: $t_{DTS}=t_{CK_INT}$

01: $t_{DTS}=2*t_{CK_INT}$

10: $t_{DTS}=4*t_{CK_INT}$

11: Reserved, do not program this value

Bit 7 **ARPE**: Auto-reload preload enable

0: TIMx_ARR register is not buffered

1: TIMx_ARR register is buffered

Bits 6:5 **CMS[1:0]**: Center-aligned mode selection

00: Edge-aligned mode. The counter counts up or down depending on the direction bit (DIR).

01: Center-aligned mode 1. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set only when the counter is counting down.

10: Center-aligned mode 2. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set only when the counter is counting up.

11: Center-aligned mode 3. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set both when the counter is counting up or down.

Note: Switch from edge-aligned mode to center-aligned mode as long as the counter is enabled (CEN=1) is not allowed

Bit 4 **DIR**: Direction

- 0: Counter used as upcounter
- 1: Counter used as downcounter

Note: This bit is read only when the timer is configured in Center-aligned mode or Encoder mode.

Bit 3 **OPM**: One pulse mode

- 0: Counter is not stopped at update event
- 1: Counter stops counting at the next update event (clearing the bit CEN)

Bit 2 **URS**: Update request source

This bit is set and cleared by software to select the UEV event sources.

0: Any of the following events generate an update interrupt or DMA request if enabled.
These events can be:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

1: Only counter overflow/underflow generates an update interrupt or DMA request if enabled.

Bit 1 **UDIS**: Update disable

This bit is set and cleared by software to enable/disable UEV event generation.

0: UEV enabled. The Update (UEV) event is generated by one of the following events:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

Buffered registers are then loaded with their preload values.

1: UEV disabled. The Update event is not generated, shadow registers keep their value (ARR, PSC, CCRx). However the counter and the prescaler are reinitialized if the UG bit is set or if a hardware reset is received from the slave mode controller.

Bit 0 **CEN**: Counter enable

- 0: Counter disabled
- 1: Counter enabled

Note: External clock, gated mode and encoder mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.

38.4.2 TIMx control register 2 (TIMx_CR2)(x = 1, 8)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MMS2[3:0]				Res.	OIS6	Res.	OIS5
								rw	rw	rw	rw		rw		rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	OIS4	OIS3N	OIS3	OIS2N	OIS2	OIS1N	OIS1	TI1S	MMS[2:0]			CCDS	CCUS	Res.	CCPC
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw

Bits 31:24 Reserved, must be kept at reset value.

Bits 23:20 **MMS2[3:0]**: Master mode selection 2

These bits allow the information to be sent to ADC for synchronization (TRGO2) to be selected. The combination is as follows:

0000: **Reset** - the UG bit from the TIMx_EGR register is used as trigger output (TRGO2). If the reset is generated by the trigger input (slave mode controller configured in reset mode), the signal on TRGO2 is delayed compared to the actual reset.

0001: **Enable** - the Counter Enable signal CNT_EN is used as trigger output (TRGO2). It is useful to start several timers at the same time or to control a window in which a slave timer is enabled. The Counter Enable signal is generated by a logic AND between the CEN control bit and the trigger input when configured in Gated mode. When the Counter Enable signal is controlled by the trigger input, there is a delay on TRGO2, except if the Master/Slave mode is selected (see the MSM bit description in TIMx_SMCR register).

0010: **Update** - the update event is selected as trigger output (TRGO2). For instance, a master timer can then be used as a prescaler for a slave timer.

0011: **Compare pulse** - the trigger output sends a positive pulse when the CC1IF flag is to be set (even if it was already high), as soon as a capture or compare match occurs (TRGO2).

0100: **Compare** - OC1REF signal is used as trigger output (TRGO2)

0101: **Compare** - OC2REF signal is used as trigger output (TRGO2)

0110: **Compare** - OC3REF signal is used as trigger output (TRGO2)

0111: **Compare** - OC4REF signal is used as trigger output (TRGO2)

1000: **Compare** - OC5REF signal is used as trigger output (TRGO2)

1001: **Compare** - OC6REF signal is used as trigger output (TRGO2)

1010: **Compare Pulse** - OC4REF rising or falling edges generate pulses on TRGO2

1011: **Compare Pulse** - OC6REF rising or falling edges generate pulses on TRGO2

1100: **Compare Pulse** - OC4REF or OC6REF rising edges generate pulses on TRGO2

1101: **Compare Pulse** - OC4REF rising or OC6REF falling edges generate pulses on TRGO2

1110: **Compare Pulse** - OC5REF or OC6REF rising edges generate pulses on TRGO2

1111: **Compare Pulse** - OC5REF rising or OC6REF falling edges generate pulses on TRGO2

Note: The clock of the slave timer or ADC must be enabled prior to receive events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

Bit 19 Reserved, must be kept at reset value.

Bit 18 **OIS6**: Output Idle state 6 (OC6 output)

Refer to OIS1 bit

Bit 17 Reserved, must be kept at reset value.

Bit 16 **OIS5**: Output Idle state 5 (OC5 output)

Refer to OIS1 bit

Bit 15 Reserved, must be kept at reset value.

Bit 14 **OIS4**: Output Idle state 4 (OC4 output)

Refer to OIS1 bit

Bit 13 **OIS3N**: Output Idle state 3 (OC3N output)

Refer to OIS1N bit

Bit 12 **OIS3**: Output Idle state 3 (OC3 output)

Refer to OIS1 bit

- Bit 11 **OIS2N**: Output Idle state 2 (OC2N output)
Refer to OIS1N bit
- Bit 10 **OIS2**: Output Idle state 2 (OC2 output)
Refer to OIS1 bit
- Bit 9 **OIS1N**: Output Idle state 1 (OC1N output)
0: OC1N=0 after a dead-time when MOE=0
1: OC1N=1 after a dead-time when MOE=0
Note: This bit can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).
- Bit 8 **OIS1**: Output Idle state 1 (OC1 output)
0: OC1=0 (after a dead-time if OC1N is implemented) when MOE=0
1: OC1=1 (after a dead-time if OC1N is implemented) when MOE=0
Note: This bit can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).
- Bit 7 **TI1S**: TI1 selection
0: The TIMx_CH1 pin is connected to TI1 input
1: The TIMx_CH1, CH2 and CH3 pins are connected to the TI1 input (XOR combination)
- Bits 6:4 **MMS[2:0]**: Master mode selection
These bits allow selected information to be sent in master mode to slave timers for synchronization (TRGO). The combination is as follows:
000: **Reset** - the UG bit from the TIMx_EGR register is used as trigger output (TRGO). If the reset is generated by the trigger input (slave mode controller configured in reset mode) then the signal on TRGO is delayed compared to the actual reset.
001: **Enable** - the Counter Enable signal CNT_EN is used as trigger output (TRGO). It is useful to start several timers at the same time or to control a window in which a slave timer is enable. The Counter Enable signal is generated by a logic AND between CEN control bit and the trigger input when configured in gated mode. When the Counter Enable signal is controlled by the trigger input, there is a delay on TRGO, except if the master/slave mode is selected (see the MSM bit description in TIMx_SMCR register).
010: **Update** - The update event is selected as trigger output (TRGO). For instance a master timer can then be used as a prescaler for a slave timer.
011: **Compare Pulse** - The trigger output send a positive pulse when the CC1IF flag is to be set (even if it was already high), as soon as a capture or a compare match occurred. (TRGO).
100: **Compare** - OC1REF signal is used as trigger output (TRGO)
101: **Compare** - OC2REF signal is used as trigger output (TRGO)
110: **Compare** - OC3REF signal is used as trigger output (TRGO)
111: **Compare** - OC4REF signal is used as trigger output (TRGO)
Note: The clock of the slave timer or ADC must be enabled prior to receive events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.
- Bit 3 **CCDS**: Capture/compare DMA selection
0: CCx DMA request sent when CCx event occurs
1: CCx DMA requests sent when update event occurs

Bit 2 **CCUS**: Capture/compare control update selection

0: When capture/compare control bits are preloaded (CCPC=1), they are updated by setting the COMG bit only

1: When capture/compare control bits are preloaded (CCPC=1), they are updated by setting the COMG bit or when an rising edge occurs on TRGI

Note: This bit acts only on channels that have a complementary output.

Bit 1 Reserved, must be kept at reset value.

Bit 0 **CCPC**: Capture/compare preloaded control

0: CCxE, CCxNE and OCxM bits are not preloaded

1: CCxE, CCxNE and OCxM bits are preloaded, after having been written, they are updated only when a commutation event (COM) occurs (COMG bit set or rising edge detected on TRGI, depending on the CCUS bit).

Note: This bit acts only on channels that have a complementary output.

38.4.3 TIMx slave mode control register (TIMx_SMCR)(x = 1, 8)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TS[4:3]		Res.	Res.	Res.	SMS[3]
										rw	rw				rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETP	ECE	ETPS[1:0]		ETF[3:0]				MSM	TS[2:0]			Res.	SMS[2:0]		
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw

Bits 31:22 Reserved, must be kept at reset value.

Bits 19:17 Reserved, must be kept at reset value.

Bit 15 **ETP**: External trigger polarity

This bit selects whether ETR or $\overline{\text{ETR}}$ is used for trigger operations

0: ETR is non-inverted, active at high level or rising edge.

1: ETR is inverted, active at low level or falling edge.

Bit 14 **ECE**: External clock enable

This bit enables External clock mode 2.

0: External clock mode 2 disabled

1: External clock mode 2 enabled. The counter is clocked by any active edge on the ETRF signal.

Note: Setting the ECE bit has the same effect as selecting external clock mode 1 with TRGI connected to ETRF (SMS=111 and TS=00111).

It is possible to simultaneously use external clock mode 2 with the following slave modes: reset mode, gated mode and trigger mode. Nevertheless, TRGI must not be connected to ETRF in this case (TS bits must not be 00111).

If external clock mode 1 and external clock mode 2 are enabled at the same time, the external clock input is ETRF.

Bits 13:12 **ETPS[1:0]**: External trigger prescaler

External trigger signal ETRP frequency must be at most 1/4 of TIMxCLK frequency. A prescaler can be enabled to reduce ETRP frequency. It is useful when inputting fast external clocks.

- 00: Prescaler OFF
- 01: ETRP frequency divided by 2
- 10: ETRP frequency divided by 4
- 11: ETRP frequency divided by 8

Bits 11:8 **ETF[3:0]**: External trigger filter

This bit-field then defines the frequency used to sample ETRP signal and the length of the digital filter applied to ETRP. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

- 0000: No filter, sampling is done at f_{DTS}
- 0001: $f_{SAMPLING}=f_{CK_INT}$, N=2
- 0010: $f_{SAMPLING}=f_{CK_INT}$, N=4
- 0011: $f_{SAMPLING}=f_{CK_INT}$, N=8
- 0100: $f_{SAMPLING}=f_{DTS}/2$, N=6
- 0101: $f_{SAMPLING}=f_{DTS}/2$, N=8
- 0110: $f_{SAMPLING}=f_{DTS}/4$, N=6
- 0111: $f_{SAMPLING}=f_{DTS}/4$, N=8
- 1000: $f_{SAMPLING}=f_{DTS}/8$, N=6
- 1001: $f_{SAMPLING}=f_{DTS}/8$, N=8
- 1010: $f_{SAMPLING}=f_{DTS}/16$, N=5
- 1011: $f_{SAMPLING}=f_{DTS}/16$, N=6
- 1100: $f_{SAMPLING}=f_{DTS}/16$, N=8
- 1101: $f_{SAMPLING}=f_{DTS}/32$, N=5
- 1110: $f_{SAMPLING}=f_{DTS}/32$, N=6
- 1111: $f_{SAMPLING}=f_{DTS}/32$, N=8

Bit 7 **MSM**: Master/slave mode

- 0: No action
- 1: The effect of an event on the trigger input (TRGI) is delayed to allow a perfect synchronization between the current timer and its slaves (through TRGO). It is useful if we want to synchronize several timers on a single external event.

Bits 21, 20, 6, 5, 4 **TS[4:0]**: Trigger selection

This bit-field selects the trigger input to be used to synchronize the counter.

- 00000: Internal Trigger 0 (ITR0)
- 00001: Internal Trigger 1 (ITR1)
- 00010: Internal Trigger 2 (ITR2)
- 00011: Internal Trigger 3 (ITR3)
- 00100: TI1 Edge Detector (TI1F_ED)
- 00101: Filtered Timer Input 1 (TI1FP1)
- 00110: Filtered Timer Input 2 (TI2FP2)
- 00111: External Trigger input (ETRF)
- Others: Reserved

See [Table 329: TIMx internal trigger connection on page 1611](#) for more details on ITRx meaning for each Timer.

Note: These bits must be changed only when they are not used (e.g. when SMS=000) to avoid wrong edge detections at the transition.

Bit 3 Reserved, must be kept at reset value.

Bits 16, 2, 1, 0 **SMS[3:0]**: Slave mode selection

When external signals are selected the active edge of the trigger signal (TRGI) is linked to the polarity selected on the external input (see Input Control register and Control Register description).

0000: Slave mode disabled - if CEN = '1' then the prescaler is clocked directly by the internal clock.

0001: Encoder mode 1 - Counter counts up/down on TI1FP1 edge depending on TI2FP2 level.

0010: Encoder mode 2 - Counter counts up/down on TI2FP2 edge depending on TI1FP1 level.

0011: Encoder mode 3 - Counter counts up/down on both TI1FP1 and TI2FP2 edges depending on the level of the other input.

0100: Reset Mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter and generates an update of the registers.

0101: Gated Mode - The counter clock is enabled when the trigger input (TRGI) is high. The counter stops (but is not reset) as soon as the trigger becomes low. Both start and stop of the counter are controlled.

0110: Trigger Mode - The counter starts at a rising edge of the trigger TRGI (but it is not reset). Only the start of the counter is controlled.

0111: External Clock Mode 1 - Rising edges of the selected trigger (TRGI) clock the counter.

1000: Combined reset + trigger mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter, generates an update of the registers and starts the counter.

Codes above 1000: Reserved.

Note: The gated mode must not be used if TI1F_ED is selected as the trigger input (TS=00100). Indeed, TI1F_ED outputs 1 pulse for each transition on TI1F, whereas the gated mode checks the level of the trigger signal.

Note: The clock of the slave peripherals (timer, ADC, ...) receiving the TRGO or the TRGO2 signals must be enabled prior to receive events from the master timer, and the clock frequency (prescaler) must not be changed on-the-fly while triggers are received from the master timer.

Table 329. TIMx internal trigger connection

Slave TIM	ITR0 (TS = 00000)	ITR1 (TS = 00001)	ITR2 (TS = 00010)	ITR3 (TS = 00011)
TIM1	TIM15	TIM2	TIM3	TIM4
TIM8	TIM1	TIM2	TIM4	TIM5

38.4.4 TIMx DMA/interrupt enable register (TIMx_DIER)(x = 1, 8)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	TDE	COMDE	CC4DE	CC3DE	CC2DE	CC1DE	UDE	BIE	TIE	COMIE	CC4IE	CC3IE	CC2IE	CC1IE	UIE
	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

- Bit 15 Reserved, must be kept at reset value.
- Bit 14 **TDE**: Trigger DMA request enable
0: Trigger DMA request disabled
1: Trigger DMA request enabled
- Bit 13 **COMDE**: COM DMA request enable
0: COM DMA request disabled
1: COM DMA request enabled
- Bit 12 **CC4DE**: Capture/Compare 4 DMA request enable
0: CC4 DMA request disabled
1: CC4 DMA request enabled
- Bit 11 **CC3DE**: Capture/Compare 3 DMA request enable
0: CC3 DMA request disabled
1: CC3 DMA request enabled
- Bit 10 **CC2DE**: Capture/Compare 2 DMA request enable
0: CC2 DMA request disabled
1: CC2 DMA request enabled
- Bit 9 **CC1DE**: Capture/Compare 1 DMA request enable
0: CC1 DMA request disabled
1: CC1 DMA request enabled
- Bit 8 **UDE**: Update DMA request enable
0: Update DMA request disabled
1: Update DMA request enabled
- Bit 7 **BIE**: Break interrupt enable
0: Break interrupt disabled
1: Break interrupt enabled
- Bit 6 **TIE**: Trigger interrupt enable
0: Trigger interrupt disabled
1: Trigger interrupt enabled
- Bit 5 **COMIE**: COM interrupt enable
0: COM interrupt disabled
1: COM interrupt enabled
- Bit 4 **CC4IE**: Capture/Compare 4 interrupt enable
0: CC4 interrupt disabled
1: CC4 interrupt enabled
- Bit 3 **CC3IE**: Capture/Compare 3 interrupt enable
0: CC3 interrupt disabled
1: CC3 interrupt enabled

Bit 2 **CC2IE**: Capture/Compare 2 interrupt enable

0: CC2 interrupt disabled

1: CC2 interrupt enabled

Bit 1 **CC1IE**: Capture/Compare 1 interrupt enable

0: CC1 interrupt disabled

1: CC1 interrupt enabled

Bit 0 **UIE**: Update interrupt enable

0: Update interrupt disabled

1: Update interrupt enabled

38.4.5 TIMx status register (TIMx_SR)(x = 1, 8)

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC6IF	CC5IF
														rc_w0	rc_w0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	SBIF	CC4OF	CC3OF	CC2OF	CC1OF	B2IF	B1F	TIF	COMIF	CC4IF	CC3IF	CC2IF	CC1IF	UIF
		rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0

Bits 31:18 Reserved, must be kept at reset value.

Bit 17 **CC6IF**: Compare 6 interrupt flag

Refer to CC1IF description (Note: Channel 6 can only be configured as output)

Bit 16 **CC5IF**: Compare 5 interrupt flag

Refer to CC1IF description (Note: Channel 5 can only be configured as output)

Bits 15:14 Reserved, must be kept at reset value.

Bit 13 **SBIF**: System Break interrupt flag

This flag is set by hardware as soon as the system break input goes active. It can be cleared by software if the system break input is not active.

This flag must be reset to re-start PWM operation.

0: No break event occurred.

1: An active level has been detected on the system break input. An interrupt is generated if BIE=1 in the TIMx_DIER register.

Bit 12 **CC4OF**: Capture/Compare 4 overcapture flag

Refer to CC1OF description

Bit 11 **CC3OF**: Capture/Compare 3 overcapture flag

Refer to CC1OF description

Bit 10 **CC2OF**: Capture/Compare 2 overcapture flag

Refer to CC1OF description

Bit 9 **CC1OF**: Capture/Compare 1 overcapture flag

This flag is set by hardware only when the corresponding channel is configured in input capture mode. It is cleared by software by writing it to '0'.

0: No overcapture has been detected.

1: The counter value has been captured in TIMx_CCR1 register while CC1IF flag was already set

Bit 8 **B2IF**: Break 2 interrupt flag

This flag is set by hardware as soon as the break 2 input goes active. It can be cleared by software if the break 2 input is not active.

0: No break event occurred.

1: An active level has been detected on the break 2 input. An interrupt is generated if BIE=1 in the TIMx_DIER register.

Bit 7 **BIF**: Break interrupt flag

This flag is set by hardware as soon as the break input goes active. It can be cleared by software if the break input is not active.

0: No break event occurred.

1: An active level has been detected on the break input. An interrupt is generated if BIE=1 in the TIMx_DIER register.

Bit 6 **TIF**: Trigger interrupt flag

This flag is set by hardware on the TRG trigger event (active edge detected on TRGI input when the slave mode controller is enabled in all modes but gated mode. It is set when the counter starts or stops when gated mode is selected. It is cleared by software.

0: No trigger event occurred.

1: Trigger interrupt pending.

Bit 5 **COMIF**: COM interrupt flag

This flag is set by hardware on COM event (when Capture/compare Control bits - CCxE, CCxNE, OCxM - have been updated). It is cleared by software.

0: No COM event occurred.

1: COM interrupt pending.

Bit 4 **CC4IF**: Capture/Compare 4 interrupt flag

Refer to CC1IF description

Bit 3 **CC3IF**: Capture/Compare 3 interrupt flag

Refer to CC1IF description

Bit 2 **CC2IF**: Capture/Compare 2 interrupt flag

Refer to CC1IF description

Bit 1 **CC1IF**: Capture/Compare 1 interrupt flag

This flag is set by hardware. It is cleared by software (input capture or output compare mode) or by reading the TIMx_CCR1 register (input capture mode only).

0: No compare match / No input capture occurred

1: A compare match or an input capture occurred.

If channel CC1 is configured as output: this flag is set when the content of the counter TIMx_CNT matches the content of the TIMx_CCR1 register. When the content of TIMx_CCR1 is greater than the content of TIMx_ARR, the CC1IF bit goes high on the counter overflow (in up-counting and up/down-counting modes) or underflow (in down-counting mode). There are 3 possible options for flag setting in center-aligned mode, refer to the CMS bits in the TIMx_CR1 register for the full description.

If channel CC1 is configured as input: this bit is set when counter value has been captured in TIMx_CCR1 register (an edge has been detected on IC1, as per the edge sensitivity defined with the CC1P and CC1NP bits setting, in TIMx_CCER).

Bit 0 **UIF**: Update interrupt flag

This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred.

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

- At overflow or underflow regarding the repetition counter value (update if repetition counter = 0) and if the UDIS=0 in the TIMx_CR1 register.
- When CNT is reinitialized by software using the UG bit in TIMx_EGR register, if URS=0 and UDIS=0 in the TIMx_CR1 register.
- When CNT is reinitialized by a trigger event (refer to [Section 38.4.3: TIMx slave mode control register \(TIMx_SMCR\)\(x = 1, 8\)](#)), if URS=0 and UDIS=0 in the TIMx_CR1 register.

38.4.6 TIMx event generation register (TIMx_EGR)(x = 1, 8)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	B2G	BG	TG	COMG	CC4G	CC3G	CC2G	CC1G	UG
							w	w	w	w	w	w	w	w	w

Bits 15:9 Reserved, must be kept at reset value.

Bit 8 **B2G**: Break 2 generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: A break 2 event is generated. MOE bit is cleared and B2IF flag is set. Related interrupt can occur if enabled.

Bit 7 **BG**: Break generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: A break event is generated. MOE bit is cleared and BIF flag is set. Related interrupt or DMA transfer can occur if enabled.

Bit 6 **TG**: Trigger generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: The TIF flag is set in TIMx_SR register. Related interrupt or DMA transfer can occur if enabled.

Bit 5 **COMG**: Capture/Compare control update generation

This bit can be set by software, it is automatically cleared by hardware

0: No action

1: When CCPC bit is set, it allows CCxE, CCxNE and OCxM bits to be updated.

Note: This bit acts only on channels having a complementary output.

Bit 4 **CC4G**: Capture/Compare 4 generation

Refer to CC1G description

Bit 3 **CC3G**: Capture/Compare 3 generation

Refer to CC1G description

Bit 2 **CC2G**: Capture/Compare 2 generation

Refer to CC1G description

Bit 1 **CC1G**: Capture/Compare 1 generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: A capture/compare event is generated on channel 1:

If channel CC1 is configured as output:

CC1IF flag is set, Corresponding interrupt or DMA request is sent if enabled.

If channel CC1 is configured as input:

The current value of the counter is captured in TIMx_CCR1 register. The CC1IF flag is set, the corresponding interrupt or DMA request is sent if enabled. The CC1OF flag is set if the CC1IF flag was already high.

Bit 0 **UG**: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action

1: Reinitialize the counter and generates an update of the registers. The prescaler internal counter is also cleared (the prescaler ratio is not affected). The counter is cleared if the center-aligned mode is selected or if DIR=0 (upcounting), else it takes the auto-reload value (TIMx_ARR) if DIR=1 (downcounting).

38.4.7 TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1)(x = 1, 8)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for input capture mode (this section) or for output compare mode (next section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function for input capture and for output compare modes. It is possible to combine both modes independently (e.g. channel 1 in input capture mode and channel 2 in output compare mode).

Input capture mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IC2F[3:0]				IC2PSC[1:0]		CC2S[1:0]		IC1F[3:0]				IC1PSC[1:0]		CC1S[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:12 **IC2F[3:0]**: Input capture 2 filter

Refer to IC1F[3:0] description.

Bits 11:10 **IC2PSC[1:0]**: Input capture 2 prescaler

Refer to OC1PSC[1:0] description.

Bits 9:8 **CC2S[1:0]**: Capture/Compare 2 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output

01: CC2 channel is configured as input, IC2 is mapped on TI2

10: CC2 channel is configured as input, IC2 is mapped on TI1

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC2S bits are writable only when the channel is OFF (CC2E = '0' in TIMx_CCER).

Bits 7:4 **IC1F[3:0]**: Input capture 1 filter

This bit-field defines the frequency used to sample TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, N=2

0010: $f_{SAMPLING}=f_{CK_INT}$, N=4

0011: $f_{SAMPLING}=f_{CK_INT}$, N=8

0100: $f_{SAMPLING}=f_{DTS}/2$, N=6

0101: $f_{SAMPLING}=f_{DTS}/2$, N=8

0110: $f_{SAMPLING}=f_{DTS}/4$, N=6

0111: $f_{SAMPLING}=f_{DTS}/4$, N=8

1000: $f_{SAMPLING}=f_{DTS}/8$, N=6

1001: $f_{SAMPLING}=f_{DTS}/8$, N=8

1010: $f_{SAMPLING}=f_{DTS}/16$, N=5

1011: $f_{SAMPLING}=f_{DTS}/16$, N=6

1100: $f_{SAMPLING}=f_{DTS}/16$, N=8

1101: $f_{SAMPLING}=f_{DTS}/32$, N=5

1110: $f_{SAMPLING}=f_{DTS}/32$, N=6

1111: $f_{SAMPLING}=f_{DTS}/32$, N=8

Bits 3:2 **IC1PSC[1:0]**: Input capture 1 prescaler

This bit-field defines the ratio of the prescaler acting on CC1 input (IC1). The prescaler is reset as soon as CC1E='0' (TIMx_CCER register).

00: no prescaler, capture is done each time an edge is detected on the capture input

01: capture is done once every 2 events

10: capture is done once every 4 events

11: capture is done once every 8 events

Bits 1:0 **CC1S[1:0]**: Capture/Compare 1 Selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: CC1 channel is configured as input, IC1 is mapped on TI2

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC1S bits are writable only when the channel is OFF (CC1E = '0' in TIMx_CCER).

38.4.8 TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1)(x = 1, 8)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for output compare mode (this section) or for input capture mode (previous section). The direction of a channel is defined by configuring the

corresponding CCxS bits. All the other bits of this register have a different function for input capture and for output compare modes. It is possible to combine both modes independently (e.g. channel 1 in input capture mode and channel 2 in output compare mode).

Output compare mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC2M[3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M[3]
							rw								rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2 CE	OC2M[2:0]			OC2 PE	OC2 FE	CC2S[1:0]		OC1 CE	OC1M[2:0]			OC1 PE	OC1 FE	CC1S[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bits 23:17 Reserved, must be kept at reset value.

Bit 15 **OC2CE**: Output Compare 2 clear enable
Refer to OC1CE description.

Bits 24, 14:12 **OC2M[3:0]**: Output Compare 2 mode
Refer to OC1M[3:0] description.

Bit 11 **OC2PE**: Output Compare 2 preload enable
Refer to OC1PE description.

Bit 10 **OC2FE**: Output Compare 2 fast enable
Refer to OC1FE description.

Bits 9:8 **CC2S[1:0]**: Capture/Compare 2 selection
This bit-field defines the direction of the channel (input/output) as well as the used input.
00: CC2 channel is configured as output
01: CC2 channel is configured as input, IC2 is mapped on TI2
10: CC2 channel is configured as input, IC2 is mapped on TI1
11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode is working only if an internal trigger input is selected through the TS bit (TIMx_SMCR register)
Note: CC2S bits are writable only when the channel is OFF (CC2E = '0' in TIMx_CCER).

Bit 7 **OC1CE**: Output Compare 1 clear enable
0: OC1Ref is not affected by the ETRF input
1: OC1Ref is cleared as soon as a High level is detected on ETRF input

Bits 16, 6:4 **OC1M[3:0]**: Output Compare 1 mode

These bits define the behavior of the output reference signal OC1REF from which OC1 and OC1N are derived. OC1REF is active high whereas OC1 and OC1N active level depends on CC1P and CC1NP bits.

0000: Frozen - The comparison between the output compare register TIMx_CCR1 and the counter TIMx_CNT has no effect on the outputs.(this mode is used to generate a timing base).

0001: Set channel 1 to active level on match. OC1REF signal is forced high when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

0010: Set channel 1 to inactive level on match. OC1REF signal is forced low when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

0011: Toggle - OC1REF toggles when TIMx_CNT=TIMx_CCR1.

0100: Force inactive level - OC1REF is forced low.

0101: Force active level - OC1REF is forced high.

0110: PWM mode 1 - In upcounting, channel 1 is active as long as TIMx_CNT<TIMx_CCR1 else inactive. In downcounting, channel 1 is inactive (OC1REF='0') as long as TIMx_CNT>TIMx_CCR1 else active (OC1REF='1').

0111: PWM mode 2 - In upcounting, channel 1 is inactive as long as TIMx_CNT<TIMx_CCR1 else active. In downcounting, channel 1 is active as long as TIMx_CNT>TIMx_CCR1 else inactive.

1000: Retriggerable OPM mode 1 - In up-counting mode, the channel is active until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 1 and the channels becomes active again at the next update. In down-counting mode, the channel is inactive until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 1 and the channels becomes inactive again at the next update.

1001: Retriggerable OPM mode 2 - In up-counting mode, the channel is inactive until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 2 and the channels becomes inactive again at the next update. In down-counting mode, the channel is active until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 1 and the channels becomes active again at the next update.

1010: Reserved,

1011: Reserved,

1100: Combined PWM mode 1 - OC1REF has the same behavior as in PWM mode 1. OC1REFC is the logical OR between OC1REF and OC2REF.

1101: Combined PWM mode 2 - OC1REF has the same behavior as in PWM mode 2. OC1REFC is the logical AND between OC1REF and OC2REF.

1110: Asymmetric PWM mode 1 - OC1REF has the same behavior as in PWM mode 1. OC1REFC outputs OC1REF when the counter is counting up, OC2REF when it is counting down.

1111: Asymmetric PWM mode 2 - OC1REF has the same behavior as in PWM mode 2. OC1REFC outputs OC1REF when the counter is counting up, OC2REF when it is counting down.

Note: These bits can not be modified as long as LOCK level 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S='00' (the channel is configured in output).

Note: In PWM mode, the OCREF level changes only when the result of the comparison changes or when the output compare mode switches from "frozen" mode to "PWM" mode.

Note: On channels having a complementary output, this bit field is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the OC1M active bits take the new value from the preloaded bits only when a COM event is generated.

Bit 3 **OC1PE**: Output Compare 1 preload enable

0: Preload register on TIMx_CCR1 disabled. TIMx_CCR1 can be written at anytime, the new value is taken in account immediately.

1: Preload register on TIMx_CCR1 enabled. Read/Write operations access the preload register. TIMx_CCR1 preload value is loaded in the active register at each update event.

Note: These bits can not be modified as long as LOCK level 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S='00' (the channel is configured in output).

The PWM mode can be used without validating the preload register only in one pulse mode (OPM bit set in TIMx_CR1 register). Else the behavior is not guaranteed.

Bit 2 **OC1FE**: Output Compare 1 fast enable

This bit is used to accelerate the effect of an event on the trigger in input on the CC output.

0: CC1 behaves normally depending on counter and CCR1 values even when the trigger is ON. The minimum delay to activate CC1 output when an edge occurs on the trigger input is 5 clock cycles.

1: An active edge on the trigger input acts like a compare match on CC1 output. Then, OC is set to the compare level independently from the result of the comparison. Delay to sample the trigger input and to activate CC1 output is reduced to 3 clock cycles. OCFE acts only if the channel is configured in PWM1 or PWM2 mode.

Bits 1:0 **CC1S[1:0]**: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: CC1 channel is configured as input, IC1 is mapped on TI2

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC1S bits are writable only when the channel is OFF (CC1E = '0' in TIMx_CCER).

38.4.9 TIMx capture/compare mode register 2 [alternate] (TIMx_CCMR2)(x = 1, 8)

Address offset: 0x1C

Reset value: 0x0000 0000

The same register can be used for input capture mode (this section) or for output compare mode (next section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function for input capture and for output compare modes. It is possible to combine both modes independently (e.g. channel 1 in input capture mode and channel 2 in output compare mode).

Input capture mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IC4F[3:0]				IC4PSC[1:0]		CC4S[1:0]		IC3F[3:0]				IC3PSC[1:0]		CC3S[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:12 **IC4F[3:0]**: Input capture 4 filter
Refer to IC1F[3:0] description.

Bits 11:10 **IC4PSC[1:0]**: Input capture 4 prescaler
Refer to IC1PSC[1:0] description.

Bits 9:8 **CC4S[1:0]**: Capture/Compare 4 selection
This bit-field defines the direction of the channel (input/output) as well as the used input.
00: CC4 channel is configured as output
01: CC4 channel is configured as input, IC4 is mapped on TI4
10: CC4 channel is configured as input, IC4 is mapped on TI3
11: CC4 channel is configured as input, IC4 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)
Note: CC4S bits are writable only when the channel is OFF (CC4E = '0' in TIMx_CCER).

Bits 7:4 **IC3F[3:0]**: Input capture 3 filter
Refer to IC1F[3:0] description.

Bits 3:2 **IC3PSC[1:0]**: Input capture 3 prescaler
Refer to IC1PSC[1:0] description.

Bits 1:0 **CC3S[1:0]**: Capture/compare 3 selection
This bit-field defines the direction of the channel (input/output) as well as the used input.
00: CC3 channel is configured as output
01: CC3 channel is configured as input, IC3 is mapped on TI3
10: CC3 channel is configured as input, IC3 is mapped on TI4
11: CC3 channel is configured as input, IC3 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)
Note: CC3S bits are writable only when the channel is OFF (CC3E = '0' in TIMx_CCER).

38.4.10 TIMx capture/compare mode register 2 [alternate] (TIMx_CCMR2)(x = 1, 8)

Address offset: 0x1C

Reset value: 0x0000 0000

The same register can be used for output compare mode (this section) or for input capture mode (previous section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function for input capture and for output compare modes. It is possible to combine both modes independently (e.g. channel 1 in input capture mode and channel 2 in output compare mode).

Output compare mode

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC4M[3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC3M[3]
							rw								rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC4 CE	OC4M[2:0]			OC4 PE	OC4 FE	CC4S[1:0]		OC3 CE	OC3M[2:0]			OC3 PE	OC3 FE	CC3S[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bits 23:17 Reserved, must be kept at reset value.

Bit 15 **OC4CE**: Output compare 4 clear enable
Refer to OC1CE description.

Bits 24, 14:12 **OC4M[3:0]**: Output compare 4 mode
Refer to OC3M[3:0] description.

Bit 11 **OC4PE**: Output compare 4 preload enable
Refer to OC1PE description.

Bit 10 **OC4FE**: Output compare 4 fast enable
Refer to OC1FE description.

Bits 9:8 **CC4S[1:0]**: Capture/Compare 4 selection
This bit-field defines the direction of the channel (input/output) as well as the used input.
00: CC4 channel is configured as output
01: CC4 channel is configured as input, IC4 is mapped on TI4
10: CC4 channel is configured as input, IC4 is mapped on TI3
11: CC4 channel is configured as input, IC4 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)
Note: CC4S bits are writable only when the channel is OFF (CC4E = '0' in TIMx_CCER).

Bit 7 **OC3CE**: Output compare 3 clear enable
Refer to OC1CE description.

Bits 16, 6:4 **OC3M[3:0]**: Output compare 3 mode
Refer to OC1M[3:0] description.

Bit 3 **OC3PE**: Output compare 3 preload enable
Refer to OC1PE description.

Bit 2 **OC3FE**: Output compare 3 fast enable
Refer to OC1FE description.

Bits 1:0 **CC3S[1:0]**: Capture/Compare 3 selection
This bit-field defines the direction of the channel (input/output) as well as the used input.
00: CC3 channel is configured as output
01: CC3 channel is configured as input, IC3 is mapped on TI3
10: CC3 channel is configured as input, IC3 is mapped on TI4
11: CC3 channel is configured as input, IC3 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)
Note: CC3S bits are writable only when the channel is OFF (CC3E = '0' in TIMx_CCER).

38.4.11 TIMx capture/compare enable register (TIMx_CCER)(x = 1, 8)

Address offset: 0x20

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC6P	CC6E	Res.	Res.	CC5P	CC5E
										rw	rw			rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CC4NP	Res.	CC4P	CC4E	CC3NP	CC3NE	CC3P	CC3E	CC2NP	CC2NE	CC2P	CC2E	CC1NP	CC1NE	CC1P	CC1E
rw		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:22 Reserved, must be kept at reset value.

Bit 21 **CC6P**: Capture/Compare 6 output polarity
Refer to CC1P description

Bit 20 **CC6E**: Capture/Compare 6 output enable
Refer to CC1E description

Bits 19:18 Reserved, must be kept at reset value.

Bit 17 **CC5P**: Capture/Compare 5 output polarity
Refer to CC1P description

Bit 16 **CC5E**: Capture/Compare 5 output enable
Refer to CC1E description

Bit 15 **CC4NP**: Capture/Compare 4 complementary output polarity
Refer to CC1NP description

Bit 14 Reserved, must be kept at reset value.

Bit 13 **CC4P**: Capture/Compare 4 output polarity
Refer to CC1P description

Bit 12 **CC4E**: Capture/Compare 4 output enable
Refer to CC1E description

Bit 11 **CC3NP**: Capture/Compare 3 complementary output polarity
Refer to CC1NP description

Bit 10 **CC3NE**: Capture/Compare 3 complementary output enable
Refer to CC1NE description

Bit 9 **CC3P**: Capture/Compare 3 output polarity
Refer to CC1P description

Bit 8 **CC3E**: Capture/Compare 3 output enable
Refer to CC1E description

Bit 7 **CC2NP**: Capture/Compare 2 complementary output polarity
Refer to CC1NP description

Bit 6 **CC2NE**: Capture/Compare 2 complementary output enable
Refer to CC1NE description

Bit 5 **CC2P**: Capture/Compare 2 output polarity
Refer to CC1P description

Bit 4 **CC2E**: Capture/Compare 2 output enable
Refer to CC1E description

Bit 3 **CC1NP**: Capture/Compare 1 complementary output polarity

CC1 channel configured as output:

0: OC1N active high.

1: OC1N active low.

CC1 channel configured as input:

This bit is used in conjunction with CC1P to define the polarity of TI1FP1 and TI2FP1. Refer to CC1P description.

Note: This bit is not writable as soon as LOCK level 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S="00" (channel configured as output).

On channels having a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1NP active bit takes the new value from the preloaded bit only when a Commutation event is generated.

Bit 2 **CC1NE**: Capture/Compare 1 complementary output enable

0: Off - OC1N is not active. OC1N level is then function of MOE, OSSI, OSSR, OIS1, OIS1N and CC1E bits.

1: On - OC1N signal is output on the corresponding output pin depending on MOE, OSSI, OSSR, OIS1, OIS1N and CC1E bits.

On channels having a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1NE active bit takes the new value from the preloaded bit only when a Commutation event is generated.

Bit 1 **CC1P**: Capture/Compare 1 output polarity

0: OC1 active high (output mode) / Edge sensitivity selection (input mode, see below)

1: OC1 active low (output mode) / Edge sensitivity selection (input mode, see below)

When CC1 channel is configured as input, both CC1NP/CC1P bits select the active polarity of TI1FP1 and TI2FP1 for trigger or capture operations.

CC1NP=0, CC1P=0: non-inverted/rising edge. The circuit is sensitive to TIxFP1 rising edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode or encoder mode).

CC1NP=0, CC1P=1: inverted/falling edge. The circuit is sensitive to TIxFP1 falling edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is inverted (trigger operation in gated mode or encoder mode).

CC1NP=1, CC1P=1: non-inverted/both edges/ The circuit is sensitive to both TIxFP1 rising and falling edges (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode). This configuration must not be used in encoder mode.

CC1NP=1, CC1P=0: The configuration is reserved, it must not be used.

Note: This bit is not writable as soon as LOCK level 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

On channels having a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1P active bit takes the new value from the preloaded bit only when a Commutation event is generated.

Bit 0 **CC1E**: Capture/Compare 1 output enable

0: Capture mode disabled / OC1 is not active (see below)

1: Capture mode enabled / OC1 signal is output on the corresponding output pin

When CC1 channel is configured as output, the OC1 level depends on MOE, OSSI, OSSR, OIS1, OIS1N and CC1NE bits, regardless of the CC1E bits state. Refer to [Table 330](#) for details.

Note: On channels having a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1E active bit takes the new value from the preloaded bit only when a Commutation event is generated.

Table 330. Output control bits for complementary OCx and OCxN channels with break feature

Control bits					Output states ⁽¹⁾	
MOE bit	OSSI bit	OSSR bit	CCxE bit	CCxNE bit	OCx output state	OCxN output state
1	X	X	0	0	Output disabled (not driven by the timer: Hi-Z) OCx=0, OCxN=0	
		0	0	1	Output disabled (not driven by the timer: Hi-Z) OCx=0	OCxREF + Polarity OCxN = OCxREF xor CCxNP
		0	1	0	OCxREF + Polarity OCx=OCxREF xor CCxP	Output Disabled (not driven by the timer: Hi-Z) OCxN=0
		X	1	1	OCREF + Polarity + dead-time	Complementary to OCREF (not OCREF) + Polarity + dead-time
		1	0	1	Off-State (output enabled with inactive state) OCx=CCxP	OCxREF + Polarity OCxN = OCxREF x or CCxNP
		1	1	0	OCxREF + Polarity OCx=OCxREF xor CCxP	Off-State (output enabled with inactive state) OCxN=CCxNP
0	0	X	X	X	Output disabled (not driven by the timer anymore). The output state is defined by the GPIO controller and can be High, Low or Hi-Z.	
	0		0			
	0		1	Off-State (output enabled with inactive state) Asynchronously: OCx=CCxP, OCxN=CCxNP (if BRK or BRK2 is triggered). Then (this is valid only if BRK is triggered), if the clock is present: OCx=OISx and OCxN=OISxN after a dead-time, assuming that OISx and OISxN do not correspond to OCx and OCxN both in active state (may cause a short circuit when driving switches in half-bridge configuration). Note: BRK2 can only be used if OSSI = OSSR = 1.		
	1		0			
	1		1			

1. When both outputs of a channel are not used (control taken over by GPIO), the OISx, OISxN, CCxP and CCxNP bits must be kept cleared.

Note: *The state of the external I/O pins connected to the complementary OCx and OCxN channels depends on the OCx and OCxN channel state and the GPIO registers.*

38.4.12 TIMx counter (TIMx_CNT)(x = 1, 8)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UIF CPY	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
r															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bit 31 **UIFCPY**: UIF copy

This bit is a read-only copy of the UIF bit of the TIMx_ISR register. If the UIFREMAP bit in the TIMxCR1 is reset, bit 31 is reserved and read at 0.

Bits 30:16 Reserved, must be kept at reset value.

Bits 15:0 **CNT[15:0]**: Counter value**38.4.13 TIMx prescaler (TIMx_PSC)(x = 1, 8)**

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 15:0 **PSC[15:0]**: Prescaler value

The counter clock frequency (CK_CNT) is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded in the active prescaler register at each update event (including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in “reset mode”).

38.4.14 TIMx auto-reload register (TIMx_ARR)(x = 1, 8)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded in the actual auto-reload register.

Refer to the [Section 38.3.1: Time-base unit on page 1548](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

38.4.15 TIMx repetition counter register (TIMx_RCR)(x = 1, 8)

Address offset: 0x30

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
REP[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **REP[15:0]**: Repetition counter value

These bits allow the user to set-up the update rate of the compare registers (i.e. periodic transfers from preload to active registers) when preload registers are enable, as well as the update interrupt generation rate, if this interrupt is enable.

Each time the REP_CNT related downcounter reaches zero, an update event is generated and it restarts counting from REP value. As REP_CNT is reloaded with REP value only at the repetition update event U_RC, any write to the TIMx_RCR register is not taken in account until the next repetition update event.

It means in PWM mode (REP+1) corresponds to:
the number of PWM periods in edge-aligned mode
the number of half PWM period in center-aligned mode.

38.4.16 TIMx capture/compare register 1 (TIMx_CCR1)(x = 1, 8)

Address offset: 0x34

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR1[15:0]**: Capture/Compare 1 value

If channel CC1 is configured as output: CCR1 is the value to be loaded in the actual capture/compare 1 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (bit OC1PE). Else the preload value is copied in the active capture/compare 1 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC1 output.

If channel CC1 is configured as input: CR1 is the counter value transferred by the last input capture 1 event (IC1). The TIMx_CCR1 register is read-only and cannot be programmed.

38.4.17 TIMx capture/compare register 2 (TIMx_CCR2)(x = 1, 8)

Address offset: 0x38

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR2[15:0]**: Capture/Compare 2 value

If channel CC2 is configured as output: CCR2 is the value to be loaded in the actual capture/compare 2 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (bit OC2PE). Else the preload value is copied in the active capture/compare 2 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC2 output.

If channel CC2 is configured as input: CCR2 is the counter value transferred by the last input capture 2 event (IC2). The TIMx_CCR2 register is read-only and cannot be programmed.

38.4.18 TIMx capture/compare register 3 (TIMx_CCR3)(x = 1, 8)

Address offset: 0x3C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR3[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR3[15:0]**: Capture/Compare value

If channel CC3 is configured as output: CCR3 is the value to be loaded in the actual capture/compare 3 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR2 register (bit OC3PE). Else the preload value is copied in the active capture/compare 3 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC3 output.

If channel CC3 is configured as input: CCR3 is the counter value transferred by the last input capture 3 event (IC3). The TIMx_CCR3 register is read-only and cannot be programmed.

38.4.19 TIMx capture/compare register 4 (TIMx_CCR4)(x = 1, 8)

Address offset: 0x40

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR4[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR4[15:0]**: Capture/Compare value

If channel CC4 is configured as output: CCR4 is the value to be loaded in the actual capture/compare 4 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR2 register (bit OC4PE). Else the preload value is copied in the active capture/compare 4 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC4 output.

If channel CC4 is configured as input: CCR4 is the counter value transferred by the last input capture 4 event (IC4). The TIMx_CCR4 register is read-only and cannot be programmed.

38.4.20 TIMx break and dead-time register (TIMx_BDTR)(x = 1, 8)

Address offset: 0x44

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	BK2P	BK2E	BK2F[3:0]				BKF[3:0]			
						rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MOE	AOE	BKP	BKE	OSSR	OSSI	LOCK[1:0]		DTG[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Note: As the bits BK2P, BK2E, BK2F[3:0], BKF[3:0], AOE, BKP, BKE, OSSI, OSSR and DTG[7:0] can be write-locked depending on the LOCK configuration, it can be necessary to configure all of them during the first write access to the TIMx_BDTR register.

Bits 31:26 Reserved, must be kept at reset value.

Bit 25 **BK2P**: Break 2 polarity

0: Break input BRK2 is active low

1: Break input BRK2 is active high

Note: This bit cannot be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Note: Any write operation to this bit takes a delay of 1 APB clock cycle to become effective.

Bit 24 **BK2E**: Break 2 enable

This bit enables the complete break 2 protection (including all sources connected to bk_acth and BKIN sources, as per [Figure 387: Break and Break2 circuitry overview](#)).

0: Break2 function disabled

1: Break2 function enabled

Note: The BKIN2 must only be used with OSSR = OSSI = 1.

Note: This bit cannot be modified when LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Note: Any write operation to this bit takes a delay of 1 APB clock cycle to become effective.

Bits 23:20 **BK2F[3:0]**: Break 2 filter

This bit-field defines the frequency used to sample BRK2 input and the length of the digital filter applied to BRK2. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, BRK2 acts asynchronously

0001: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=2

0010: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=4

0011: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=8

0100: $f_{\text{SAMPLING}} = f_{\text{DTS}}/2$, N=6

0101: $f_{\text{SAMPLING}} = f_{\text{DTS}}/2$, N=8

0110: $f_{\text{SAMPLING}} = f_{\text{DTS}}/4$, N=6

0111: $f_{\text{SAMPLING}} = f_{\text{DTS}}/4$, N=8

1000: $f_{\text{SAMPLING}} = f_{\text{DTS}}/8$, N=6

1001: $f_{\text{SAMPLING}} = f_{\text{DTS}}/8$, N=8

1010: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=5

1011: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=6

1100: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=8

1101: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=5

1110: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=6

1111: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=8

Note: This bit cannot be modified when LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 19:16 **BKF[3:0]**: Break filter

This bit-field defines the frequency used to sample BRK input and the length of the digital filter applied to BRK. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, BRK acts asynchronously

0001: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=2

0010: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=4

0011: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=8

0100: $f_{\text{SAMPLING}} = f_{\text{DTS}}/2$, N=6

0101: $f_{\text{SAMPLING}} = f_{\text{DTS}}/2$, N=8

0110: $f_{\text{SAMPLING}} = f_{\text{DTS}}/4$, N=6

0111: $f_{\text{SAMPLING}} = f_{\text{DTS}}/4$, N=8

1000: $f_{\text{SAMPLING}} = f_{\text{DTS}}/8$, N=6

1001: $f_{\text{SAMPLING}} = f_{\text{DTS}}/8$, N=8

1010: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=5

1011: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=6

1100: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=8

1101: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=5

1110: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=6

1111: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=8

Note: This bit cannot be modified when LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 15 **MOE**: Main output enable

This bit is cleared asynchronously by hardware as soon as one of the break inputs is active (BRK or BRK2). It is set by software or automatically depending on the AOE bit. It is acting only on the channels which are configured in output.

0: In response to a break 2 event. OC and OCN outputs are disabled

In response to a break event or if MOE is written to 0: OC and OCN outputs are disabled or forced to idle state depending on the OSSR bit.

1: OC and OCN outputs are enabled if their respective enable bits are set (CCxE, CCxNE in TIMx_CCER register).

See OC/OCN enable description for more details ([Section 38.4.11: TIMx capture/compare enable register \(TIMx_CCER\)\(x = 1, 8\)](#)).

Bit 14 **AOE**: Automatic output enable

0: MOE can be set only by software

1: MOE can be set by software or automatically at the next update event (if none of the break inputs BRK and BRK2 is active)

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 13 **BKP**: Break polarity

0: Break input BRK is active low

1: Break input BRK is active high

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Note: Any write operation to this bit takes a delay of 1 APB clock cycle to become effective.

Bit 12 **BKE**: Break enable

This bit enables the complete break protection (including all sources connected to bk_acth and BKIN sources, as per [Figure 387: Break and Break2 circuitry overview](#)).

0: Break function disabled

1: Break function enabled

Note: This bit cannot be modified when LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Note: Any write operation to this bit takes a delay of 1 APB clock cycle to become effective.

Bit 11 **OSSR**: Off-state selection for Run mode

This bit is used when MOE=1 on channels having a complementary output which are configured as outputs. OSSR is not implemented if no complementary output is implemented in the timer.

See OC/OCN enable description for more details ([Section 38.4.11: TIMx capture/compare enable register \(TIMx_CCER\)\(x = 1, 8\)](#)).

0: When inactive, OC/OCN outputs are disabled (the timer releases the output control which is taken over by the GPIO logic, which forces a Hi-Z state).

1: When inactive, OC/OCN outputs are enabled with their inactive level as soon as CCxE=1 or CCxNE=1 (the output is still controlled by the timer).

Note: This bit can not be modified as soon as the LOCK level 2 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 10 **OSSI**: Off-state selection for Idle mode

This bit is used when MOE=0 due to a break event or by a software write, on channels configured as outputs.

See OC/OCN enable description for more details ([Section 38.4.11: TIMx capture/compare enable register \(TIMx_CCER\)\(x = 1, 8\)](#)).

0: When inactive, OC/OCN outputs are disabled (the timer releases the output control which is taken over by the GPIO logic and which imposes a Hi-Z state).

1: When inactive, OC/OCN outputs are first forced with their inactive level then forced to their idle level after the deadtime. The timer maintains its control over the output.

Note: This bit can not be modified as soon as the LOCK level 2 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 9:8 **LOCK[1:0]**: Lock configuration

These bits offer a write protection against software errors.

00: LOCK OFF - No bit is write protected.

01: LOCK Level 1 = DTG bits in TIMx_BDTR register, OISx and OISxN bits in TIMx_CR2 register and BKE/BKP/AOE bits in TIMx_BDTR register can no longer be written.

10: LOCK Level 2 = LOCK Level 1 + CC Polarity bits (CCxP/CCxNP bits in TIMx_CCER register, as long as the related channel is configured in output through the CCxS bits) as well as OSSR and OSSI bits can no longer be written.

11: LOCK Level 3 = LOCK Level 2 + CC Control bits (OCxM and OCxPE bits in TIMx_CCMRx registers, as long as the related channel is configured in output through the CCxS bits) can no longer be written.

Note: The LOCK bits can be written only once after the reset. Once the TIMx_BDTR register has been written, their content is frozen until the next reset.

Bits 7:0 **DTG[7:0]**: Dead-time generator setup

This bit-field defines the duration of the dead-time inserted between the complementary outputs. DT correspond to this duration.

DTG[7:5]=0xx => DT=DTG[7:0]x t_{dtg} with $t_{dtg}=t_{DTS}$.

DTG[7:5]=10x => DT=(64+DTG[5:0])x t_{dtg} with $T_{dtg}=2xt_{DTS}$.

DTG[7:5]=110 => DT=(32+DTG[4:0])x t_{dtg} with $T_{dtg}=8xt_{DTS}$.

DTG[7:5]=111 => DT=(32+DTG[4:0])x t_{dtg} with $T_{dtg}=16xt_{DTS}$.

Example if $T_{DTS}=125ns$ (8MHz), dead-time possible values are:

0 to 15875 ns by 125 ns steps,

16 us to 31750 ns by 250 ns steps,

32 us to 63us by 1 us steps,

64 us to 126 us by 2 us steps

Note: This bit-field can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

38.4.21 TIMx DMA control register (TIMx_DCR)(x = 1, 8)

Address offset: 0x48

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	DBL[4:0]					Res.	Res.	Res.	DBA[4:0]				
			rw	rw	rw	rw	rw				rw	rw	rw	rw	rw

Bits 15:13 Reserved, must be kept at reset value.

Bits 12:8 **DBL[4:0]**: DMA burst length

This 5-bit vector defines the length of DMA transfers (the timer recognizes a burst transfer when a read or a write access is done to the TIMx_DMAR address), i.e. the number of transfers. Transfers can be in half-words or in bytes (see example below).

00000: 1 transfer

00001: 2 transfers

00010: 3 transfers

...

10001: 18 transfers

Example: Let us consider the following transfer: DBL = 7 bytes & DBA = TIMx_CR1.

- If DBL = 7 bytes and DBA = TIMx_CR1 represents the address of the byte to be transferred, the address of the transfer should be given by the following equation:

(TIMx_CR1 address) + DBA + (DMA index), where DMA index = DBL

In this example, 7 bytes are added to (TIMx_CR1 address) + DBA, which gives us the address from/to which the data is copied. In this case, the transfer is done to 7 registers starting from the following address: (TIMx_CR1 address) + DBA

According to the configuration of the DMA Data Size, several cases may occur:

- If the DMA Data Size is configured in half-words, 16-bit data is transferred to each of the 7 registers.
- If the DMA Data Size is configured in bytes, the data is also transferred to 7 registers: the first register contains the first MSB byte, the second register, the first LSB byte and so on. So with the transfer Timer, one also has to specify the size of data transferred by DMA.

Bits 7:5 Reserved, must be kept at reset value.

Bits 4:0 **DBA[4:0]**: DMA base address

This 5-bits vector defines the base-address for DMA transfers (when read/write access are done through the TIMx_DMAR address). DBA is defined as an offset starting from the address of the TIMx_CR1 register.

Example:

00000: TIMx_CR1,

00001: TIMx_CR2,

00010: TIMx_SMCR,

...

38.4.22 TIMx DMA address for full transfer (TIMx_DMAR)(x = 1, 8)

Address offset: 0x4C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMAB[31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMAB[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **DMAB[31:0]**: DMA register for burst accesses

A read or write operation to the DMAR register accesses the register located at the address (TIMx_CR1 address) + (DBA + DMA index) x 4

where TIMx_CR1 address is the address of the control register 1, DBA is the DMA base address configured in TIMx_DCR register, DMA index is automatically controlled by the DMA transfer, and ranges from 0 to DBL (DBL configured in TIMx_DCR).

38.4.23 TIMx capture/compare mode register 3 (TIMx_CCMR3)(x = 1, 8)

Address offset: 0x54

Reset value: 0x0000 0000

The channels 5 and 6 can only be configured in output.

Output compare mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC6M[3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC5M[3]
							rw								rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC6 CE	OC6M[2:0]			OC6 PE	OC6FE	Res.	Res.	OC5 CE	OC5M[2:0]			OC5PE	OC5FE	Res.	Res.
rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	rw	rw		

Bits 31:25 Reserved, must be kept at reset value.

Bits 23:17 Reserved, must be kept at reset value.

Bit 15 **OC6CE**: Output compare 6 clear enable
Refer to OC1CE description.

Bits 24, 14, 13, 12 **OC6M[3:0]**: Output compare 6 mode
Refer to OC1M description.

Bit 11 **OC6PE**: Output compare 6 preload enable
Refer to OC1PE description.

Bit 10 **OC6FE**: Output compare 6 fast enable
Refer to OC1FE description.

Bits 9:8 Reserved, must be kept at reset value.

Bit 7 **OC5CE**: Output compare 5 clear enable
Refer to OC1CE description.

Bits 16, 6, 5, 4 **OC5M[3:0]**: Output compare 5 mode
Refer to OC1M description.

Bit 3 **OC5PE**: Output compare 5 preload enable
Refer to OC1PE description.

Bit 2 **OC5FE**: Output compare 5 fast enable
Refer to OC1FE description.

Bits 1:0 Reserved, must be kept at reset value.

38.4.24 TIMx capture/compare register 5 (TIMx_CCR5)(x = 1, 8)

Address offset: 0x58

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
GC5C3	GC5C2	GC5C1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rw	rw	rw													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR5[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **GC5C3**: Group Channel 5 and Channel 3

Distortion on Channel 3 output:

0: No effect of OC5REF on OC3REFC

1: OC3REFC is the logical AND of OC3REFC and OC5REF

This bit can either have immediate effect or be preloaded and taken into account after an update event (if preload feature is selected in TIMx_CCMR2).

Note: it is also possible to apply this distortion on combined PWM signals.

Bit 30 **GC5C2**: Group Channel 5 and Channel 2

Distortion on Channel 2 output:

0: No effect of OC5REF on OC2REFC

1: OC2REFC is the logical AND of OC2REFC and OC5REF

This bit can either have immediate effect or be preloaded and taken into account after an update event (if preload feature is selected in TIMx_CCMR1).

Note: it is also possible to apply this distortion on combined PWM signals.

Bit 29 **GC5C1**: Group Channel 5 and Channel 1

Distortion on Channel 1 output:

0: No effect of OC5REF on OC1REFC5

1: OC1REFC is the logical AND of OC1REFC and OC5REF

This bit can either have immediate effect or be preloaded and taken into account after an update event (if preload feature is selected in TIMx_CCMR1).

Note: it is also possible to apply this distortion on combined PWM signals.

Bits 28:16 Reserved, must be kept at reset value.

Bits 15:0 **CCR5[15:0]**: Capture/Compare 5 value

CCR5 is the value to be loaded in the actual capture/compare 5 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR3 register (bit OC5PE). Else the preload value is copied in the active capture/compare 5 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC5 output.

38.4.25 TIMx capture/compare register 6 (TIMx_CCR6)(x = 1, 8)

Address offset: 0x5C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR6[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR6[15:0]**: Capture/Compare 6 value

CCR6 is the value to be loaded in the actual capture/compare 6 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR3 register (bit OC6PE). Else the preload value is copied in the active capture/compare 6 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC6 output.

38.4.26 TIM1 alternate function option register 1 (TIM1_AF1)

Address offset: 0x60

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETRSEL[3:2]	
														rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETRSEL[1:0]		Res.	Res.	BK CMP2P	BK CMP1P	BKINP	BKDF1 BKOE	Res.	Res.	Res.	Res.	Res.	BK CMP2E	BK CMP1E	BKINE
rw	rw			rw	rw	rw	rw						rw	rw	rw

Bits 31:18 Reserved, must be kept at reset value.

Bits 17:14 **ETRSEL[3:0]**: ETR source selection

These bits select the ETR input source.

0000: ETR input is connected to I/O

0001: COMP1 output

0010: COMP2 output

0011: ADC1 AWD1

0100: ADC1 AWD2

0101: ADC1 AWD3

0110: ADC3 AWD1

0111: ADC3 AWD2

1000: ADC3 AWD3

Others: Reserved

Note: These bits can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 13:12 Reserved, must be kept at reset value.

Bit 11 BKCMP2P: BRK COMP2 input polarity

This bit selects the COMP2 input sensitivity. It must be programmed together with the BKP polarity bit.

0: COMP2 input is active high

1: COMP2 input is active low

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 10 BKCMP1P: BRK COMP1 input polarity

This bit selects the COMP1 input sensitivity. It must be programmed together with the BKP polarity bit.

0: COMP1 input is active high

1: COMP1 input is active low

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 9 BKINP: BRK BKIN input polarity

This bit selects the BKIN alternate function input sensitivity. It must be programmed together with the BKP polarity bit.

0: BKIN input is active high

1: BKIN input is active low

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 8 BKDF1BK0E: BRK dfsdm1_break[0] enable

This bit enables the dfsdm1_break[0] for the timer's BRK input. dfsdm1_break[0] output is 'ORed' with the other BRK sources.

0: dfsdm1_break[0] input disabled

1: dfsdm1_break[0] input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 7:3 Reserved, must be kept at reset value.

Bit 2 BKCMP2E: BRK COMP2 enable

This bit enables the COMP2 for the timer's BRK input. COMP2 output is 'ORed' with the other BRK sources.

0: COMP2 input disabled

1: COMP2 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 1 BKCMP1E: BRK COMP1 enable

This bit enables the COMP1 for the timer's BRK input. COMP1 output is 'ORed' with the other BRK sources.

0: COMP1 input disabled

1: COMP1 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 0 **BKINE**: BRK BKIN input enable

This bit enables the BKIN alternate function input for the timer's BRK input. BKIN input is 'ORed' with the other BRK sources.

0: BKIN input disabled

1: BKIN input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Note: Refer to Figure 366: TIM1/TIM8 ETR input circuitry and to Figure 387: Break and Break2 circuitry overview.

38.4.27 TIM1 Alternate function register 2 (TIM1_AF2)

Address offset: 0x64

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	BK2 CMP2 P	BK2 CMP1 P	BK2 INP	BK2DF1 BK1E	Res.	Res.	Res.	Res.	Res.	BK2 CMP2E	BK2 CMP1E	BK2INE
				rw	rw	rw	rw						rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bit 11 **BK2CMP2P**: BRK2 COMP2 input polarity

This bit selects the COMP2 input sensitivity. It must be programmed together with the BK2P polarity bit.

0: COMP2 input is active low

1: COMP2 input is active high

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 10 **BK2CMP1P**: BRK2 COMP1 input polarity

This bit selects the COMP1 input sensitivity. It must be programmed together with the BK2P polarity bit.

0: COMP1 input is active low

1: COMP1 input is active high

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 9 **BK2INP**: BRK2 BKIN2 input polarity

This bit selects the BKIN2 alternate function input sensitivity. It must be programmed together with the BK2P polarity bit.

0: BKIN2 input is active low

1: BKIN2 input is active high

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 8 **BK2DF1BK1E**: BRK2 dfsdm1_break[1] enable

This bit enables the dfsdm1_break[1] for the timer's BRK2 input. dfsdm1_break[1] output is 'ORed' with the other BRK2 sources.

0: dfsdm1_break[1] input disabled

1: dfsdm1_break[1] input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 7:3 Reserved, must be kept at reset value.

Bit 2 **BK2CMP2E**: BRK2 COMP2 enable

This bit enables the COMP2 for the timer's BRK2 input. COMP2 output is 'ORed' with the other BRK2 sources.

0: COMP2 input disabled

1: COMP2 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 1 **BK2CMP1E**: BRK2 COMP1 enable

This bit enables the COMP1 for the timer's BRK2 input. COMP1 output is 'ORed' with the other BRK2 sources.

0: COMP1 input disabled

1: COMP1 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 0 **BK2INE**: BRK2 BKIN input enable

This bit enables the BKIN2 alternate function input for the timer's BRK2 input. BKIN2 input is 'ORed' with the other BRK2 sources.

0: BKIN2 input disabled

1: BKIN2 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Note: Refer to [Figure 387: Break and Break2 circuitry overview](#).

38.4.28 TIM8 Alternate function option register 1 (TIM8_AF1)

Address offset: 0x60

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETRSEL[3:2]	
														rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETRSEL[1:0]		Res.	Res.	BK CMP2 P	BK CMP1 P	BKINP	BKDF1 BK2E	Res.	Res.	Res.	Res.	Res.	BK CMP2E	BK CMP1E	BKINE
rw	rw			rw	rw	rw	rw						rw	rw	rw

Bits 31:18 Reserved, must be kept at reset value.

Bits 17:14 **ETRSEL[3:0]**: ETR source selection

These bits select the ETR input source.

0000: ETR input is connected to I/O

0001: COMP1 output

0010: COMP2 output

0011: ADC2 AWD1

0100: ADC2 AWD2

0101: ADC2 AWD3

0110: ADC3 AWD1

0111: ADC3 AWD2

1000: ADC3 AWD3

Others: Reserved

Note: These bits can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 13:12 Reserved, must be kept at reset value.

Bit 11 **BKCOMP2P**: BRK COMP2 input polarity

This bit selects the COMP2 input sensitivity. It must be programmed together with the BKP polarity bit.

0: COMP2 input is active high

1: COMP2 input is active low

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 10 **BKCOMP1P**: BRK COMP1 input polarity

This bit selects the COMP1 input sensitivity. It must be programmed together with the BKP polarity bit.

0: COMP1 input is active high

1: COMP1 input is active low

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 9 **BKINP**: BRK BKIN input polarity

This bit selects the BKIN alternate function input sensitivity. It must be programmed together with the BKP polarity bit.

0: BKIN input is active high

1: BKIN input is active low

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 8 **BKDF1BK2E**: BRK dfsdm1_break[2] enable

This bit enables the dfsdm1_break[2] for the timer's BRK input. dfsdm1_break[2] output is 'ORed' with the other BRK sources.

0: dfsdm1_break[2] input disabled

1: dfsdm1_break[2] input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 7:3 Reserved, must be kept at reset value.

Bit 2 BKCMP2E: BRK COMP2 enable

This bit enables the COMP2 for the timer's BRK input. COMP2 output is 'ORed' with the other BRK sources.

0: COMP2 input disabled

1: COMP2 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 1 BKCMP1E: BRK COMP1 enable

This bit enables the COMP1 for the timer's BRK input. COMP1 output is 'ORed' with the other BRK sources.

0: COMP1 input disabled

1: COMP1 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 0 BKINE: BRK BKIN input enable

This bit enables the BKIN alternate function input for the timer's BRK input. BKIN input is 'ORed' with the other BRK sources.

0: BKIN input disabled

1: BKIN input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Note: Refer to Figure 366: TIM1/TIM8 ETR input circuitry and to Figure 387: Break and Break2 circuitry overview.

38.4.29 TIM8 Alternate function option register 2 (TIM8_AF2)

Address offset: 0x64

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	BK2 CMP2 P	BK2 CMP1 P	BK2 INP	BK2DF1 BK3E	Res.	Res.	Res.	Res.	Res.	BK2 CMP2E	BK2 CMP1E	BK2INE
				rw	rw	rw	rw						rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bit 11 BK2CMP2P: BRK2 COMP2 input polarity

This bit selects the COMP2 input sensitivity. It must be programmed together with the BK2P polarity bit.

0: COMP2 input is active low

1: COMP2 input is active high

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 10 BK2CMP1P: BRK2 COMP1 input polarity

This bit selects the COMP1 input sensitivity. It must be programmed together with the BK2P polarity bit.

0: COMP1 input is active low

1: COMP1 input is active high

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 9 BK2INP: BRK2 BKIN2 input polarity

This bit selects the BKIN2 alternate function input sensitivity. It must be programmed together with the BK2P polarity bit.

0: BKIN2 input is active low

1: BKIN2 input is active high

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 8 BK2DF1BK3E: BRK2 dfsdm1_break[3] enable

This bit enables the dfsdm1_break[3] for the timer's BRK2 input. dfsdm1_break[3] output is 'ORed' with the other BRK2 sources.

0: dfsdm1_break[3] input disabled

1: dfsdm1_break[3] input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 7:3 Reserved, must be kept at reset value.

Bit 2 BK2CMP2E: BRK2 COMP2 enable

This bit enables the COMP2 for the timer's BRK2 input. COMP2 output is 'ORed' with the other BRK2 sources.

0: COMP2 input disabled

1: COMP2 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 1 BK2CMP1E: BRK2 COMP1 enable

This bit enables the COMP1 for the timer's BRK2 input. COMP1 output is 'ORed' with the other BRK2 sources.

0: COMP1 input disabled

1: COMP1 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 0 BK2INE: BRK2 BKIN input enable

This bit enables the BKIN2 alternate function input for the timer's BRK2 input. BKIN2 input is 'ORed' with the other BRK2 sources.

0: BKIN2 input disabled

1: BKIN2 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Note: Refer to [Figure 387: Break and Break2 circuitry overview](#).

38.4.30 TIM1 timer input selection register (TIM1_TISEL)

Address offset: 0x68

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	TI4SEL[3:0]				Res.	Res.	Res.	Res.	TI3SEL[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	TI2SEL[3:0]				Res.	Res.	Res.	Res.	TI1SEL[3:0]			
				rw	rw	rw	rw						rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:24 **TI4SEL[3:0]**: selects TI4[0] to TI4[15] input
 0000: TIM1_CH4 input
 Others: Reserved

Bits 23:20 Reserved, must be kept at reset value.

Bits 19:16 **TI3SEL[3:0]**: selects TI3[0] to TI3[15] input
 0000: TIM1_CH3 input
 Others: Reserved

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:8 **TI2SEL[3:0]**: selects TI2[0] to TI2[15] input
 0000: TIM1_CH2 input
 Others: Reserved

Bits 7:4 Reserved, must be kept at reset value.

Bits 3:0 **TI1SEL[3:0]**: selects TI1[0] to TI1[15] input
 0000: TIM1_CH1 input
 0001: COMP1 output
 Others: Reserved

38.4.31 TIM8 timer input selection register (TIM8_TISEL)

Address offset: 0x68

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	TI4SEL[3:0]				Res.	Res.	Res.	Res.	TI3SEL[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	TI2SEL[3:0]				Res.	Res.	Res.	Res.	TI1SEL[3:0]			
				rw	rw	rw	rw						rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:24 **TI4SEL[3:0]**: selects TI4[0] to TI4[15] input
 0000: TIM8_CH4 input
 Others: Reserved

Bits 23:20 Reserved, must be kept at reset value.

Bits 19:16 **TI3SEL[3:0]**: selects TI3[0] to TI3[15] input
0000: TIM8_CH3 input
Others: Reserved

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:8 **TI2SEL[3:0]**: selects TI2[0] to TI2[15] input
0000: TIM8_CH2 input
Others: Reserved

Bits 7:4 Reserved, must be kept at reset value.

Bits 3:0 **TI1SEL[3:0]**: selects TI1[0] to TI1[15] input
0000: TIM8_CH1 input
0001: COMP2 output
Others: Reserved

38.4.32 TIM1 register map

TIM1 registers are mapped as 16-bit addressable registers as described in the table below:

Table 331. TIM1 register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x00	TIM1_CR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	UJFREMAP	Res.	CKD [1:0]		ARPE	CMS [1:0]		DIR	OPM	URS	UDIS	CEN	
	Reset value																				0	0	0	0	0	0	0	0	0	0	0	0		
0x04	TIM1_CR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MMS2[3:0]				Res.	OIS6	Res.	OIS5	Res.	OIS4	OIS3N	OIS3	OIS2N	OIS2	OIS1N	OIS1	T1S	MMS [2:0]		CCDS	CCUS	Res.	CCPC		
	Reset value									0	0	0	0		0		0		0	0	0	0	0	0	0	0	0	0	0	0	0		0	
0x08	TIM1_SMCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TS [4:3]		Res.	Res.	Res.	SMS[3]	ETP	ECE	ETP S [1:0]		ETF[3:0]			MSM	TS[2:0]		Res.	SMS[2:0]					
	Reset value											0	0				0	0	0	0	0	0	0	0	0	0	0	0		0	0	0		
0x0C	TIM1_DIER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TDE	COMDE	CC4DE	CC3DE	CC2DE	CC1DE	UDE	BIE	TIE	COMIE	CC4IE	CC3IE	CC2IE	CC1IE	UIE	
	Reset value																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x10	TIM1_SR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC6IF	CC5IF	Res.	SBIF	CC4OF	CC3OF	CC2OF	CC1OF	B2IF	BIF	TIF	COMIF	CC4IF	CC3IF	CC2IF	CC1IF	UIF	
	Reset value																0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x14	TIM1_EGR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	B2G	BG	TG	COMG	CC4G	CC3G	CC2G	CC1G	UG		
	Reset value																							0	0	0	0	0	0	0	0	0	0	
0x18	TIM1_CCMR1 Output Compare mode	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC2M[3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M[3]	OC2OE	OC2M [2:0]			OC2PE	OC2FE	CC2 S [1:0]	OC1OE	OC1M [2:0]			OC1PE	OC1FE	CC1 S [1:0]			
	Reset value								0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	TIM1_CCMR1 Input Capture mode	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IC2F[3:0]			IC2 PSC [1:0]	CC2 S [1:0]	IC1F[3:0]			IC1 PSC [1:0]	CC1 S [1:0]							
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x1C	TIM1_CCMR2 Output Compare mode	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC4M[3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC3M[3]	OC4CE	OC4M [2:0]			OC4PE	OC4FE	CC4 S [1:0]	OC3CE	OC3M [2:0]			OC3PE	OC3FE	CC3 S [1:0]			
	Reset value								0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	TIM1_CCMR2 Input Capture mode	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IC4F[3:0]			IC4 PSC [1:0]	CC4 S [1:0]	IC3F[3:0]			IC3 PSC [1:0]	CC3 S [1:0]							
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x20	TIM1_CCER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC6P	CC6E	Res.	Res.	Res.	CC5P	CC5E	CC4NP	Res.	CC4P	CC4E	CC3NP	CC3NE	CC3P	CC3E	CC2NP	CC2NE	CC2P	CC2E	CC1NP	CC1NE	CC1P	CC1E
	Reset value											0	0				0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	

Table 331. TIM1 register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
0x24	TIM1_CNT	UIFCP	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CNT[15:0]																			
	Reset value	0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x28	TIM1_PSC	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PSC[15:0]																			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x2C	TIM1_ARR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	ARR[15:0]																			
	Reset value																	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1				
0x30	TIM1_RCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	REP[15:0]																			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x34	TIM1_CCR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR1[15:0]																			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x38	TIM1_CCR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR2[15:0]																			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x3C	TIM1_CCR3	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR3[15:0]																			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x40	TIM1_CCR4	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR4[15:0]																			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x44	TIM1_BDTR	Res	Res	Res	Res	Res	Res	BK2P	BK2E	BK2F[3:0]			BKF[3:0]			MOE	AOE	BKP	BKE	OSSR	OSSI	LOCK [1:0]	DT[7:0]														
	Reset value								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x48	TIM1_DCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DBL[4:0]				Res	Res	Res	DBA[4:0]									
	Reset value																				0	0	0	0	0				0	0	0	0	0	0			
0x4C	TIM1_DMAR	DMAB[31:0]																																			
	Reset value																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x54	TIM1_CCMR3 Output Compare mode	Res	Res	Res	Res	Res	Res	OC6M[3]	OC6M[2:0]	Res	Res	Res	Res	Res	Res	Res	Res	OC6M[3]	OC6OE	OC6M [2:0]		OC6PE	OC6FE	Res	Res	OC5OE	OC5M [2:0]		OC5PE	OC5FE	Res	Res					
	Reset value							0										0	0	0	0	0	0			0	0	0	0	0	0						
0x58	TIM1_CCR5	GC5C3	GC5C2	GC5C1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR5[15:0]																			
	Reset value	0	0	0														0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				

Table 331. TIM1 register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x5C	TIM1_CCR6	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CCR6[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x60	TIM1_AF1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETRSEL [3:0]				Res.	Res.	BKCOMP2P	BKCOMP1P	BKINP	BKDF1BK0E	Res.	Res.	Res.	Res.	Res.	Res.	BKCOMP2E	BKCOMP1E	
	Reset value																0	0	0	0			0	0	0	0						0	0	1
0x64	TIM1_AF2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BK2CMP2P	BK2CMP1P	BK2INP	BK2DF1BK1E	Res.	Res.	Res.	Res.	Res.	Res.	BK2CMP2E	BK2CMP1E	BK2INE
	Reset value																					0	0	0	0							0	0	1
0x68	TIM1_TISEL	Res.	Res.	Res.	Res.	TI4SEL[3:0]				Res.	Res.	Res.	Res.	TI3SEL[3:0]				Res.	Res.	Res.	Res.	TI2SEL[3:0]				Res.	Res.	Res.	Res.	TI1SEL[3:0]				
	Reset value					0	0	0	0					0	0	0	0					0	0	0	0						0	0	0	0

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.

38.4.33 TIM8 register map

TIM8 registers are mapped as 16-bit addressable registers as described in the table below:

Table 332. TIM8 register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	TIM8_CR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	UIFREMA	Res.	CKD [1:0]	Res.	ARPE	CMS [1:0]		DIR	OPM	URS	UDIS	CEN
	Reset value																					0		0	0	0	0	0	0	0	0	0	0
0x04	TIM8_CR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MMS2[3:0]				Res.	OIS6	Res.	OIS5	Res.	OIS4	OIS3N	OIS3	OIS2N	OIS2	OIS1N	OIS1	T11S	MMS [2:0]		CCDS	CCUS	Res.	CCPC	
	Reset value									0	0	0	0		0		0		0	0	0	0	0	0	0	0	0	0	0	0	0		0
0x08	TIM8_SMCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TS [4:3]		Res.	Res.	Res.	SMS[3]	ETP	ECE	ETP S [1:0]		ETF[3:0]			MSM	TS[2:0]		Res.	SMS[2:0]				
	Reset value											0	0				0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	TIM8_DIER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TDE	COMDE	CC4DE	CC3DE	CC2DE	CC1DE	UDE	BIE	TIE	COMIE	CC4IE	CC3IE	CC2IE	CC1IE	UIE
	Reset value																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x10	TIM8_SR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC6IF	CC5IF	Res.	SBIF	CC4OF	CC3OF	CC2OF	CC1OF	B2IF	B1F	TIF	COMIF	CC4IF	CC3IF	CC2IF	CC1IF	UIF	
	Reset value															0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 332. TIM8 register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x14	TIM8_EGR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	B2G	BG	TG	COMG	CC4G	CC3G	CC2G	CC1G	UG	
	Reset value																								0	0	0	0	0	0	0	0	0	
0x18	TIM8_CCMR1 Output Compare mode	Res	Res	Res	Res	Res	Res	Res	OC2M[3]	Res	Res	Res	Res	Res	Res	Res	OC1M[3]	OC2CE	OC2M[2:0]			OC2PE	OC2FE	Res	CC2S[1:0]	OC1CE	OC1M[2:0]		OC1PE	OC1FE	CC1S[1:0]			
	Reset value								0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	TIM8_CCMR1 Input Capture mode	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	IC2F[3:0]			IC2PSC[1:0]	IC2S[1:0]	IC1F[3:0]		IC1PSC[1:0]		IC1S[1:0]	CC1S[1:0]						
	Reset value																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x1C	TIM8_CCMR2 Output Compare mode	Res	Res	Res	Res	Res	Res	Res	OC4M[3]	Res	Res	Res	Res	Res	Res	Res	OC3M[3]	OC4CE	OC4M[2:0]			OC4PE	OC4FE	CC4S[1:0]	OC3CE	OC3M[2:0]		OC3PE	OC3FE	CC3S[1:0]				
	Reset value								0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	TIM8_CCMR2 Input Capture mode	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	IC4F[3:0]			IC4PSC[1:0]	IC4S[1:0]	IC3F[3:0]		IC3PSC[1:0]		IC3S[1:0]	CC3S[1:0]						
	Reset value																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x20	TIM8_CCER	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CC6P	CC6E	Res	Res	Res	CC5P	CC5E	CC4NP	Res	CC4P	CC4E	CC3NP	CC3NE	CC3P	CC3E	CC2NP	CC2NE	CC2P	CC2E	CC1NP	CC1NE	CC1P	CC1E
	Reset value											0	0	Res	Res	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x24	TIM8_CNT	UIFCPY	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CNT[15:0]																
	Reset value	0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x28	TIM8_PSC	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PSC[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x2C	TIM8_ARR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	ARR[15:0]																
	Reset value																	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
0x30	TIM8_RCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	REP[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x34	TIMx_CCR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR1[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x38	TIM8_CCR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR2[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x3C	TIM8_CCR3	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR3[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x40	TIM8_CCR4	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR4[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Table 332. TIM8 register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x44	TIM8_BDTR	Res	Res	Res	Res	Res	Res	BK2P	BK2E	BK2F[3:0]				BKF[3:0]				MOE	AOE	BKP	BKE	OSSR	OSSI	LOC K [1:0]	DT[7:0]								
	Reset value							0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x48	TIM8_DCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DBL[4:0]				Res	Res	Res	DBA[4:0]					
	Reset value																				0	0	0	0	0				0	0	0	0	0
0x4C	TIM8_DMAR	DMAB[31:0]																															
	Reset value																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x54	TIM8_CCMR3 Output Compare mode	Res	Res	Res	Res	Res	Res	Res	OC6M[3]	Res	Res	Res	Res	Res	Res	Res	OC5M[3]	OC6CE	OC6M [2:0]		OC6PE	OC6FE	Res	Res	OC5CE	OC5M [2:0]		OC5PE	OC5FE	Res	Res		
	Reset value								0								0	0	0	0	0	0			0	0	0	0	0	0			
0x58	TIM8_CCR5	GC5C3	GC5C2	GC5C1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR5[15:0]															
	Reset value	0	0	0														0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x5C	TIM8_CCR6	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR6[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x60	TIM8_AF1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	ETRSEL [3:0]				Res	Res	BKCOMP2P	BKCOMP1P	BKINP	BKDF1BK2E	Res	Res	Res	Res	Res	Res	BKCOMP2E	BKCOMP1E	BKINE	
	Reset value															0	0	0	0			0	0	0	0					0	0	1	
0x64	TIM8_AF2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	BK2CMP2P	BK2CMP1P	BK2INP	BK2DF1BK3E	Res	Res	Res	Res	Res	Res	BK2CMP2E	BK2CMP1E	BK2INE
	Reset value																				0	0	0	0							0	0	1
0x68	TIM8_TISEL	Res	Res	Res	Res	TI4SEL[3:0]				Res	Res	Res	Res	TI3SEL[3:0]				Res	Res	Res	Res	TI2SEL[3:0]				Res	Res	Res	Res	TI1SEL[3:0]			
	Reset value					0	0	0	0					0	0	0	0					0	0	0	0					0	0	0	0

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.

39 General-purpose timers (TIM2/TIM3/TIM4/TIM5)

39.1 TIM2/TIM3/TIM4/TIM5 introduction

The general-purpose timers consist of a 16-bit/32-bit auto-reload counter driven by a programmable prescaler.

They may be used for a variety of purposes, including measuring the pulse lengths of input signals (*input capture*) or generating output waveforms (*output compare and PWM*).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

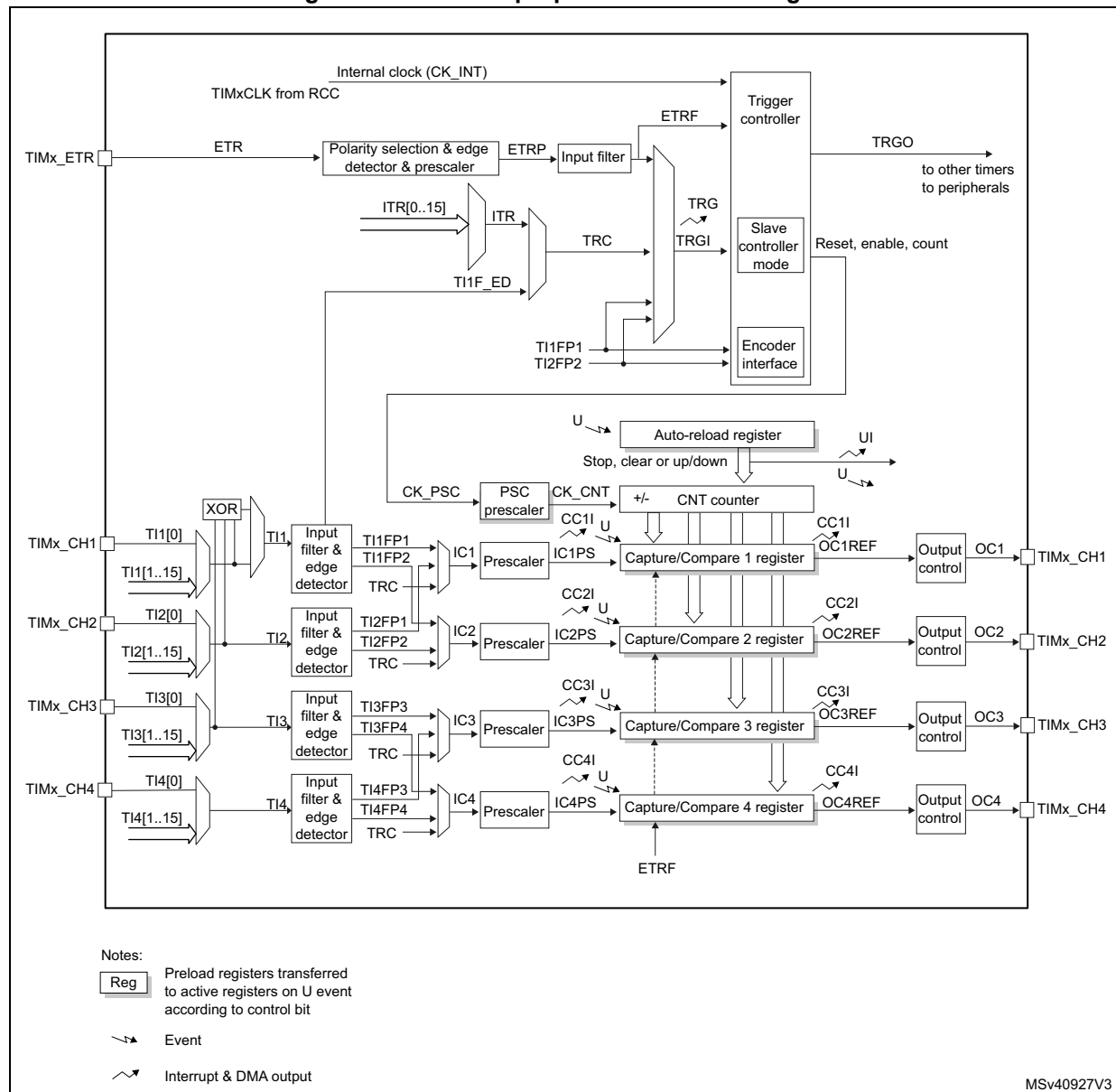
The timers are completely independent, and do not share any resources. They can be synchronized together as described in [Section 39.3.19: Timer synchronization](#).

39.2 TIM2/TIM3/TIM4/TIM5 main features

General-purpose TIMx timer features include:

- 16-bit (TIM3, TIM4) or 32-bit (TIM2 and TIM5) up, down, up/down auto-reload counter.
- 16-bit programmable prescaler used to divide (also “on the fly”) the counter clock frequency by any factor between 1 and 65535.
- Up to 4 independent channels for:
 - Input capture
 - Output compare
 - PWM generation (Edge- and Center-aligned modes)
 - One-pulse mode output
- Synchronization circuit to control the timer with external signals and to interconnect several timers.
- Interrupt/DMA generation on the following events:
 - Update: counter overflow/underflow, counter initialization (by software or internal/external trigger)
 - Trigger event (counter start, stop, initialization or count by internal/external trigger)
 - Input capture
 - Output compare
- Supports incremental (quadrature) encoder and hall-sensor circuitry for positioning purposes
- Trigger input for external clock or cycle-by-cycle current management

Figure 404. General-purpose timer block diagram



39.3 TIM2/TIM3/TIM4/TIM5 functional description

39.3.1 Time-base unit

The main block of the programmable timer is a 16-bit/32-bit counter with its related auto-reload register. The counter can count up, down or both up and down but also down or both up and down. The counter clock can be divided by a prescaler.

The counter, the auto-reload register and the prescaler register can be written or read by software. This is true even when the counter is running.

The time-base unit includes:

- Counter Register (TIMx_CNT)
- Prescaler Register (TIMx_PSC)
- Auto-Reload Register (TIMx_ARR)

The auto-reload register is preloaded. Writing to or reading from the auto-reload register accesses the preload register. The content of the preload register are transferred into the shadow register permanently or at each update event (UEV), depending on the auto-reload preload enable bit (ARPE) in TIMx_CR1 register. The update event is sent when the counter reaches the overflow (or underflow when downcounting) and if the UDIS bit equals 0 in the TIMx_CR1 register. It can also be generated by software. The generation of the update event is described in detail for each configuration.

The counter is clocked by the prescaler output CK_CNT, which is enabled only when the counter enable bit (CEN) in TIMx_CR1 register is set (refer also to the slave mode controller description to get more details on counter enabling).

Note that the actual counter enable signal CNT_EN is set 1 clock cycle after CEN.

Prescaler description

The prescaler can divide the counter clock frequency by any factor between 1 and 65536. It is based on a 16-bit counter controlled through a 16-bit/32-bit register (in the TIMx_PSC register). It can be changed on the fly as this control register is buffered. The new prescaler ratio is taken into account at the next update event.

[Figure 405](#) and [Figure 406](#) give some examples of the counter behavior when the prescaler ratio is changed on the fly:

Figure 405. Counter timing diagram with prescaler division change from 1 to 2

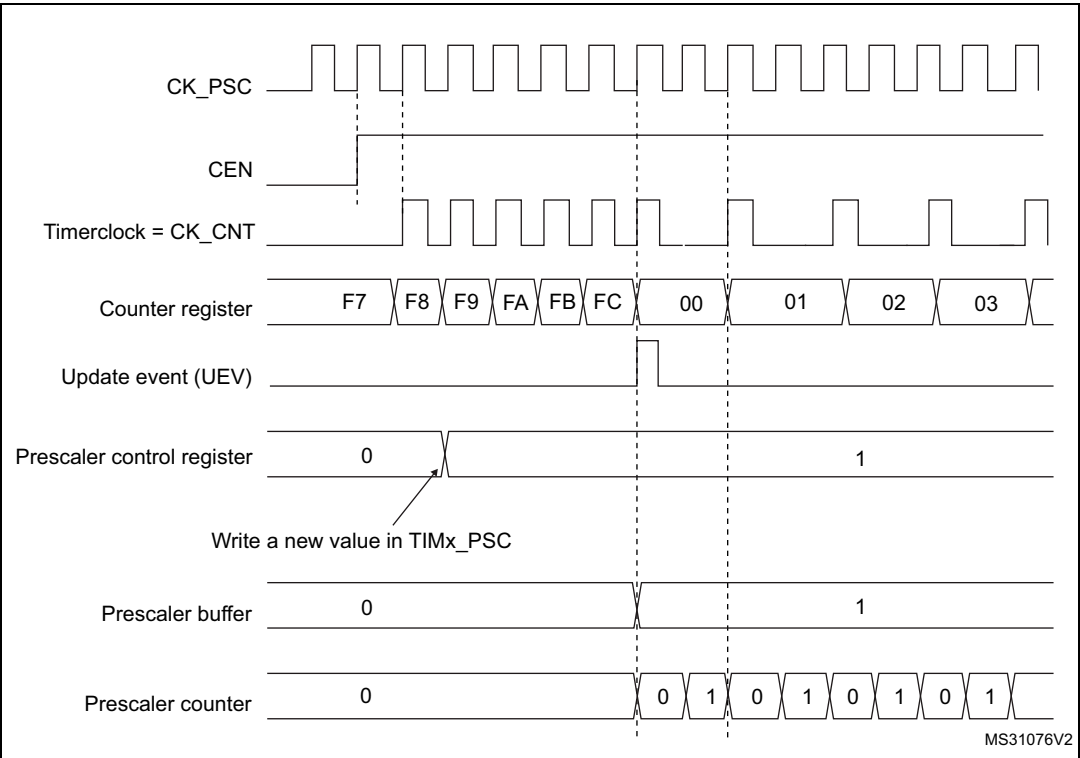
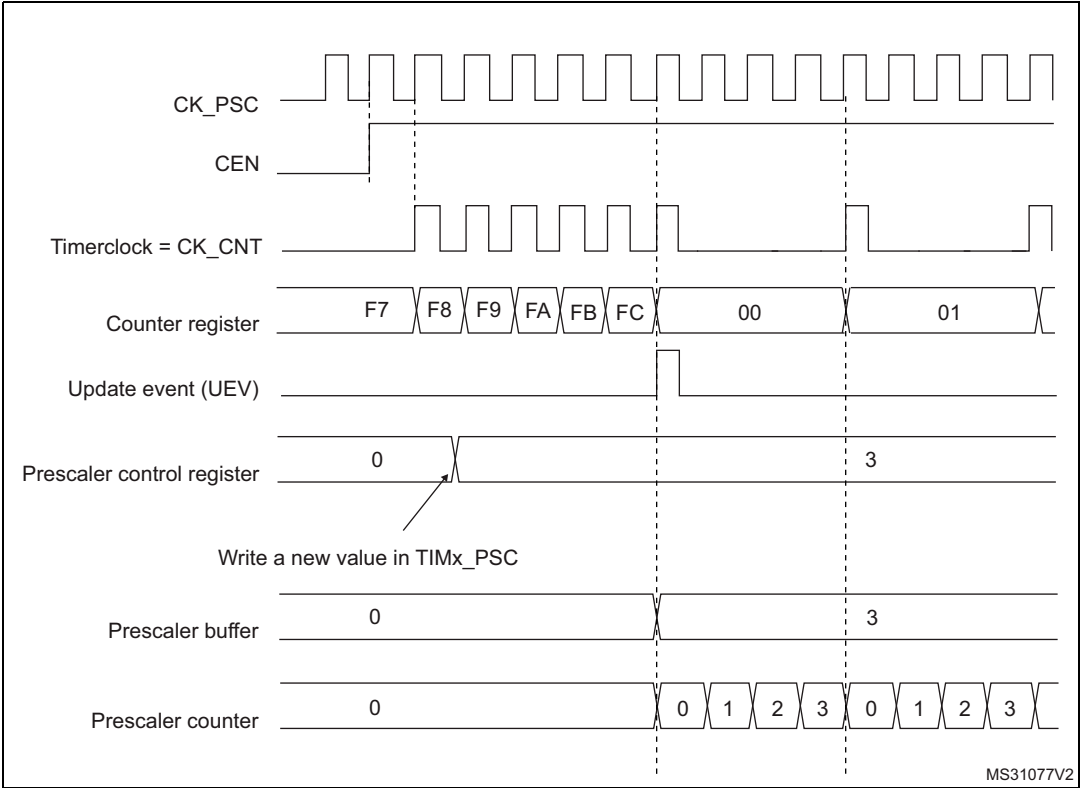


Figure 406. Counter timing diagram with prescaler division change from 1 to 4



39.3.2 Counter modes

Upcounting mode

In upcounting mode, the counter counts from 0 to the auto-reload value (content of the TIMx_ARR register), then restarts from 0 and generates a counter overflow event.

An Update event can be generated at each counter overflow or by setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller).

The UEV event can be disabled by software by setting the UDIS bit in TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until the UDIS bit has been written to 0. However, the counter restarts from 0, as well as the counter of the prescaler (but the prescale rate does not change). In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt or DMA request is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register)
- The auto-reload shadow register is updated with the preload value (TIMx_ARR)

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR=0x36.

Figure 407. Counter timing diagram, internal clock divided by 1

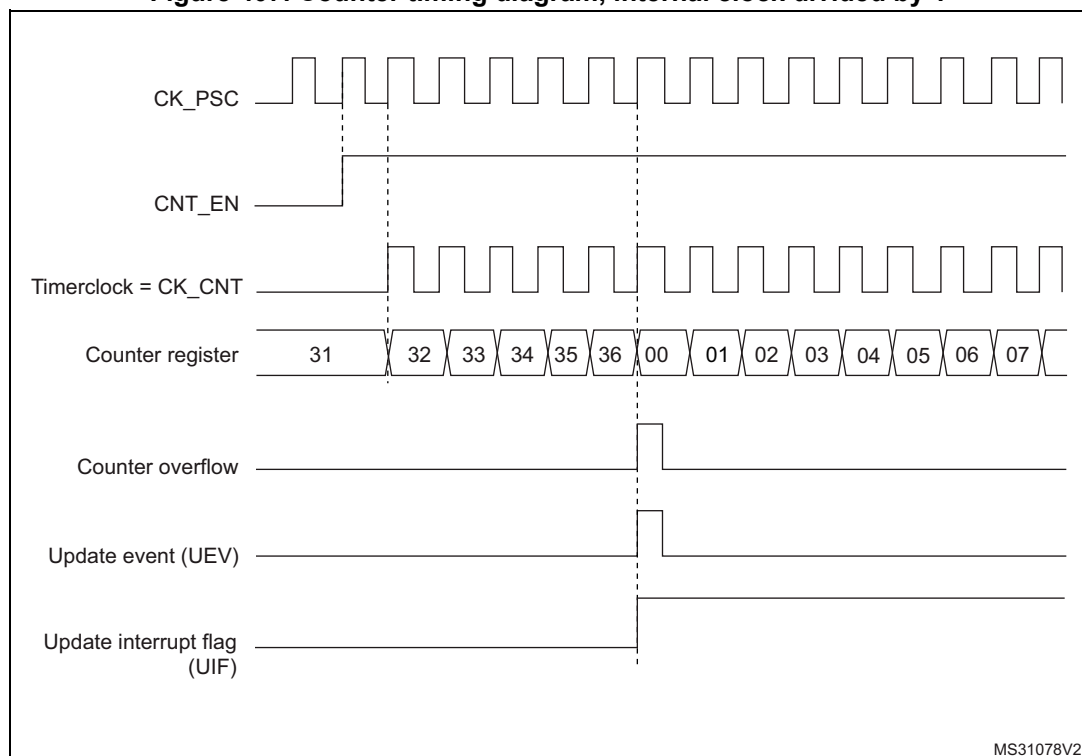


Figure 408. Counter timing diagram, internal clock divided by 2

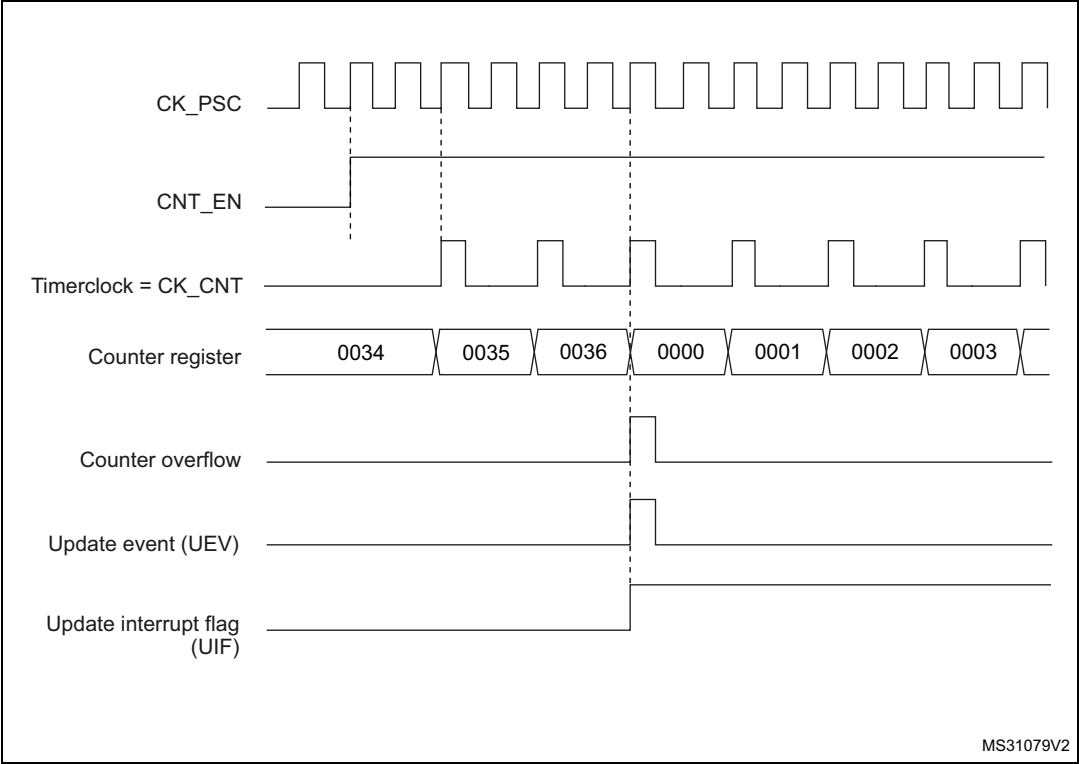


Figure 409. Counter timing diagram, internal clock divided by 4

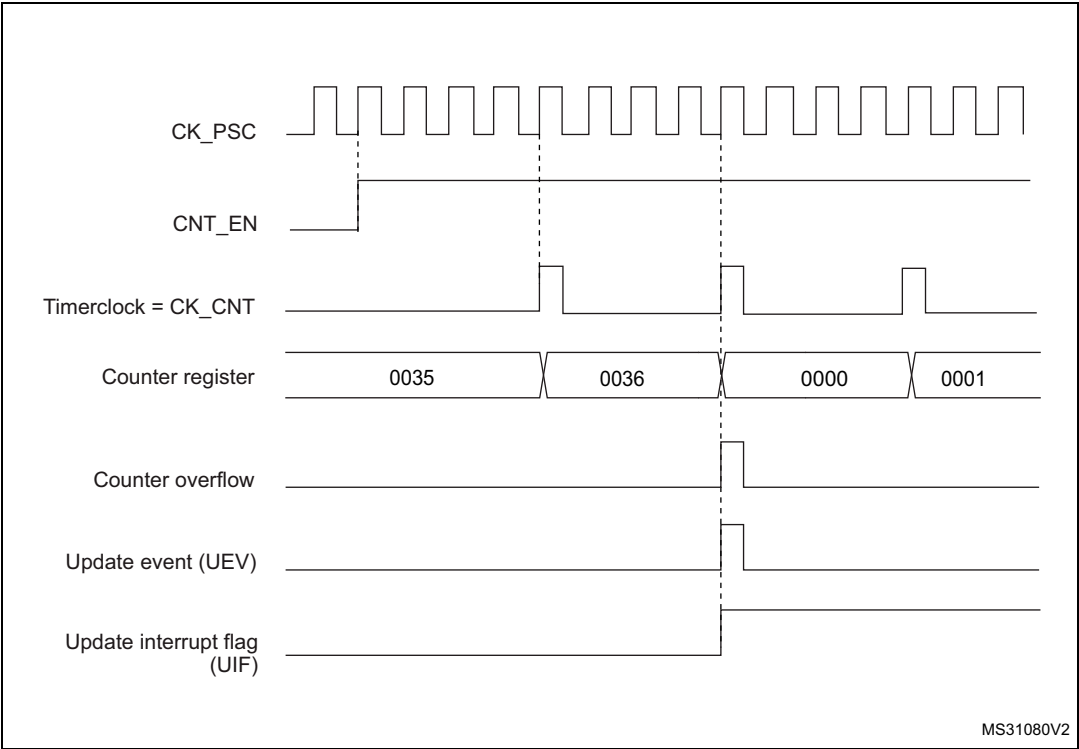


Figure 410. Counter timing diagram, internal clock divided by N

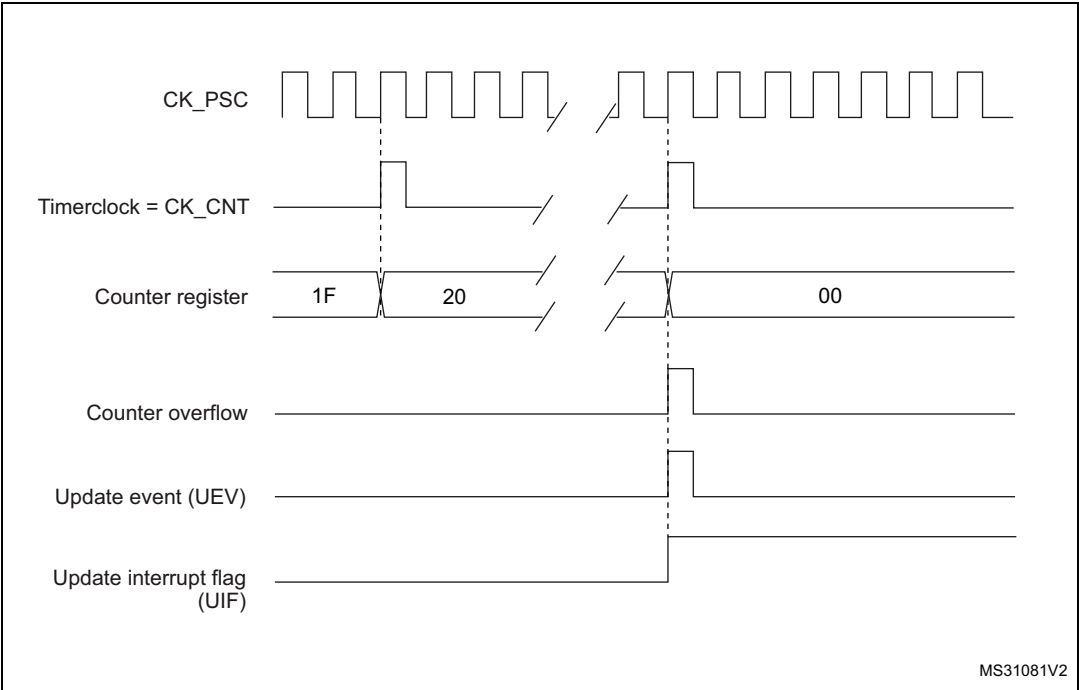


Figure 411. Counter timing diagram, Update event when ARPE=0 (TIMx_ARR not preloaded)

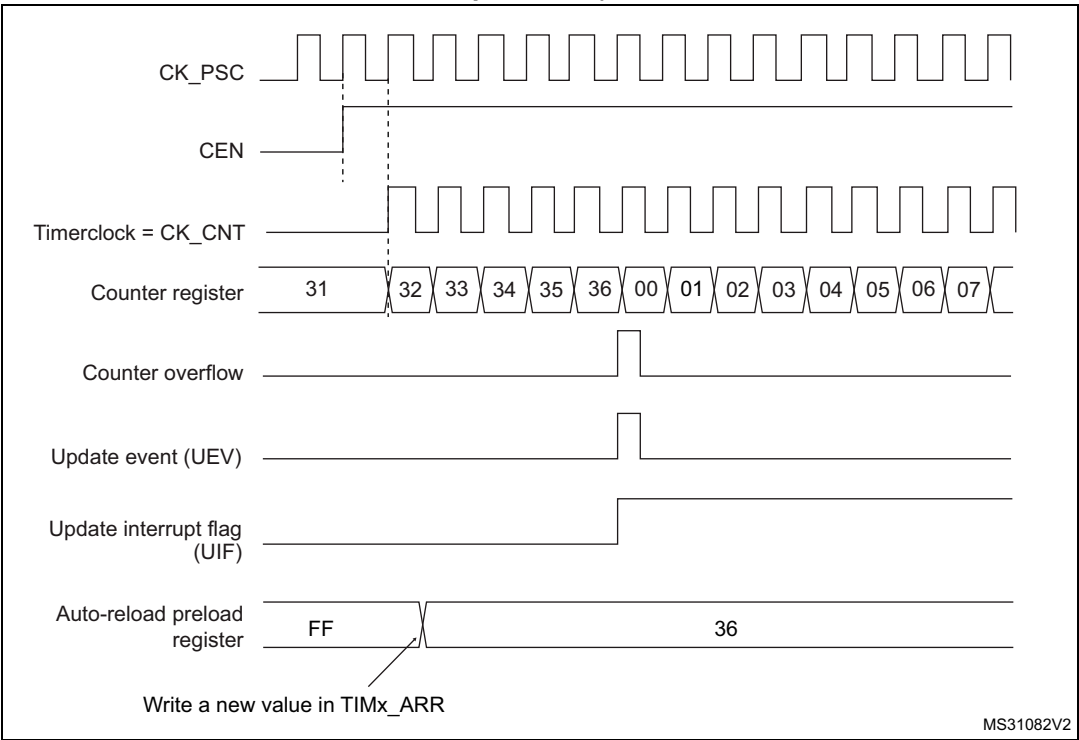
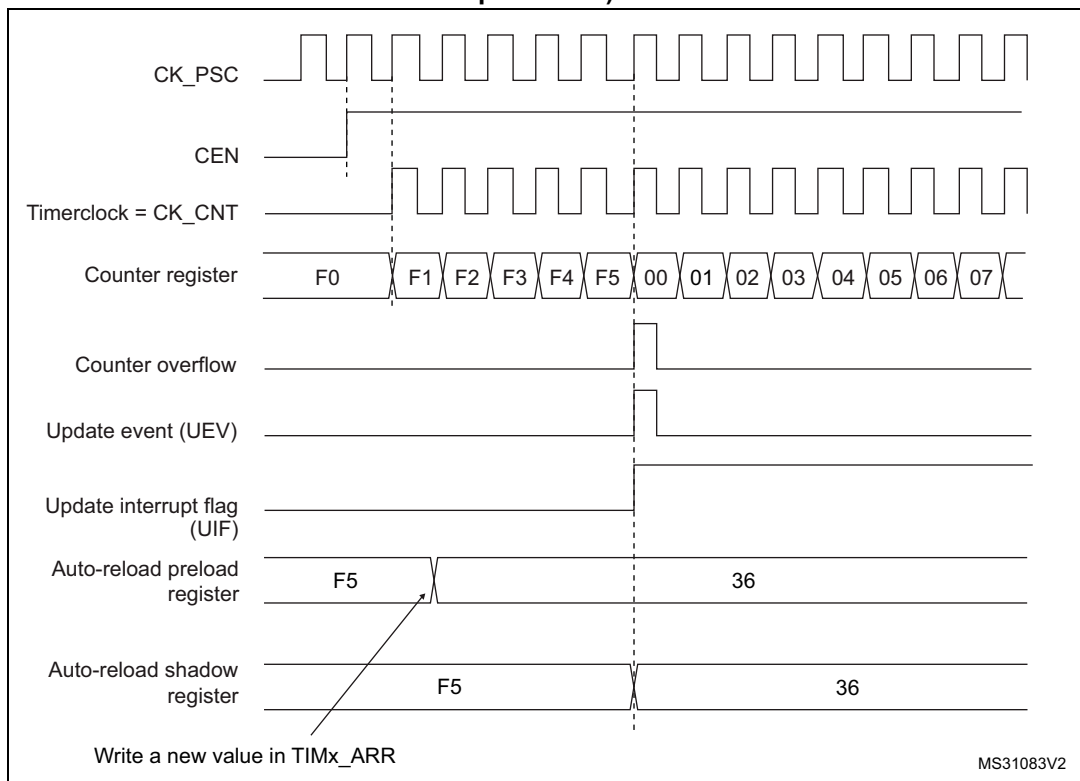


Figure 412. Counter timing diagram, Update event when ARPE=1 (TIMx_ARR preloaded)

Downcounting mode

In downcounting mode, the counter counts from the auto-reload value (content of the TIMx_ARR register) down to 0, then restarts from the auto-reload value and generates a counter underflow event.

An Update event can be generated at each counter underflow or by setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller)

The UEV update event can be disabled by software by setting the UDIS bit in TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until UDIS bit has been written to 0. However, the counter restarts from the current auto-reload value, whereas the counter of the prescaler restarts from 0 (but the prescale rate doesn't change).

In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt or DMA request is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register).
- The auto-reload active register is updated with the preload value (content of the TIMx_ARR register). Note that the auto-reload is updated before the counter is reloaded, so that the next period is the expected one.

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR=0x36.

Figure 413. Counter timing diagram, internal clock divided by 1

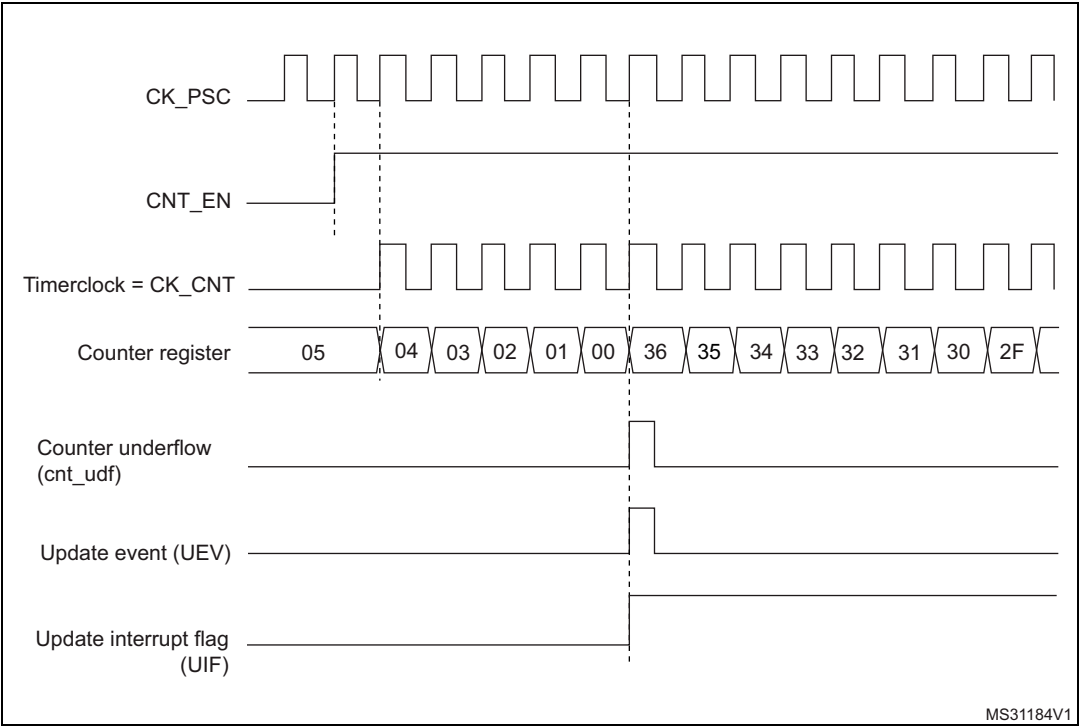


Figure 414. Counter timing diagram, internal clock divided by 2

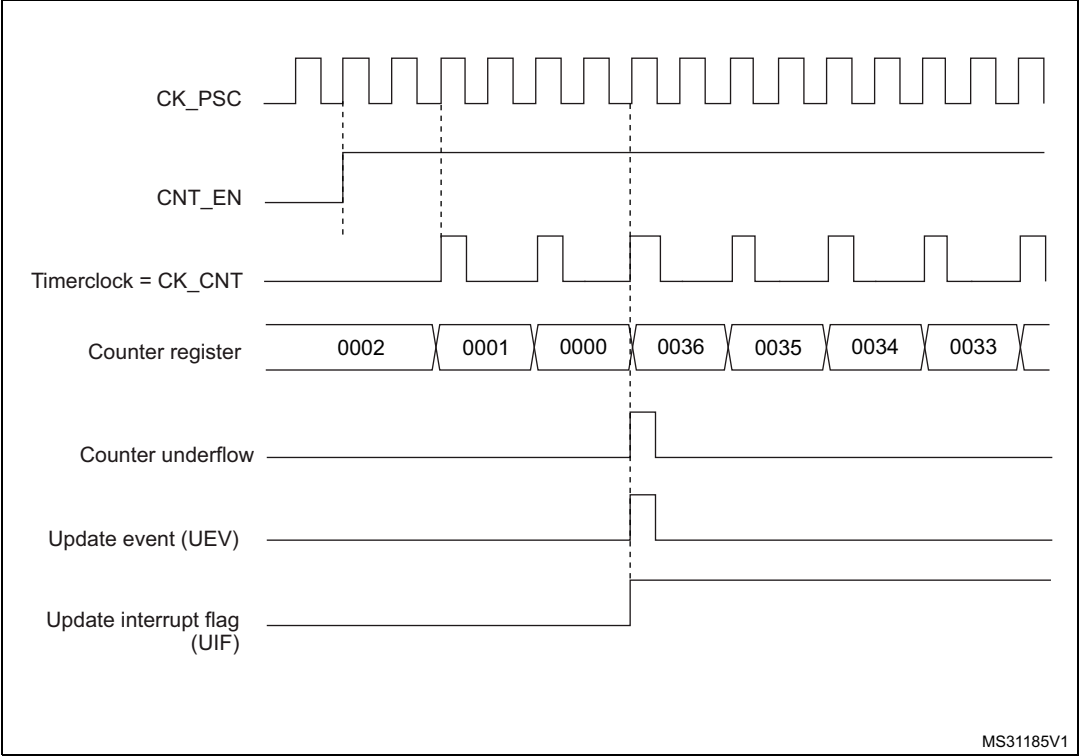


Figure 415. Counter timing diagram, internal clock divided by 4

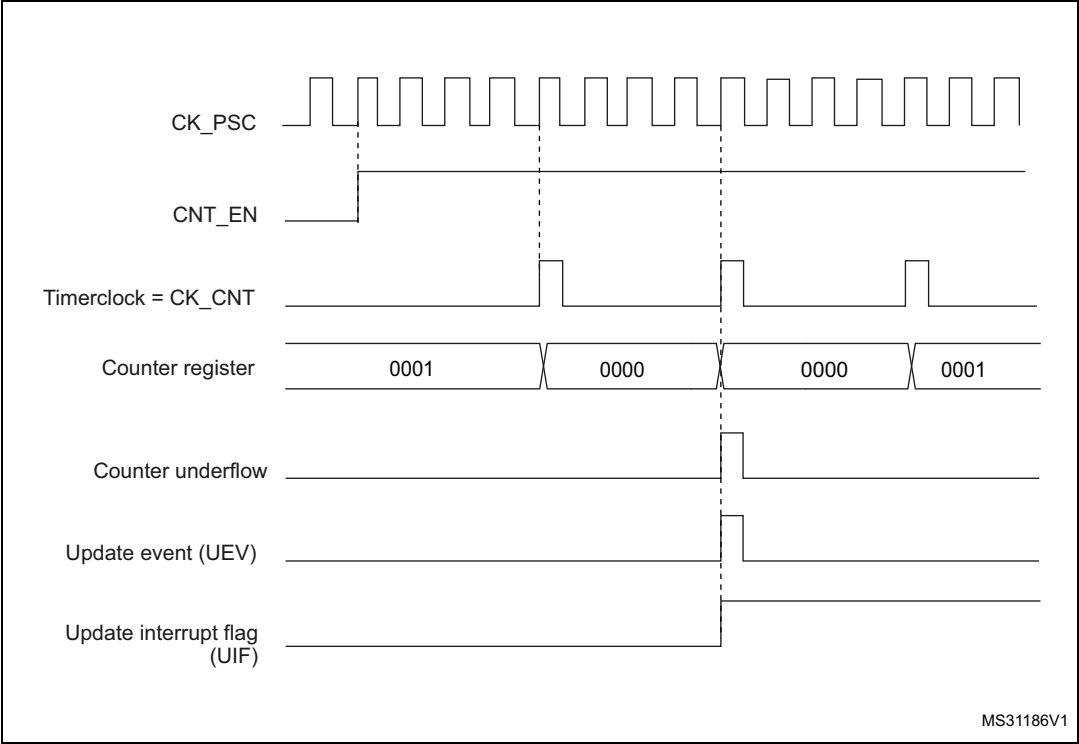


Figure 416. Counter timing diagram, internal clock divided by N

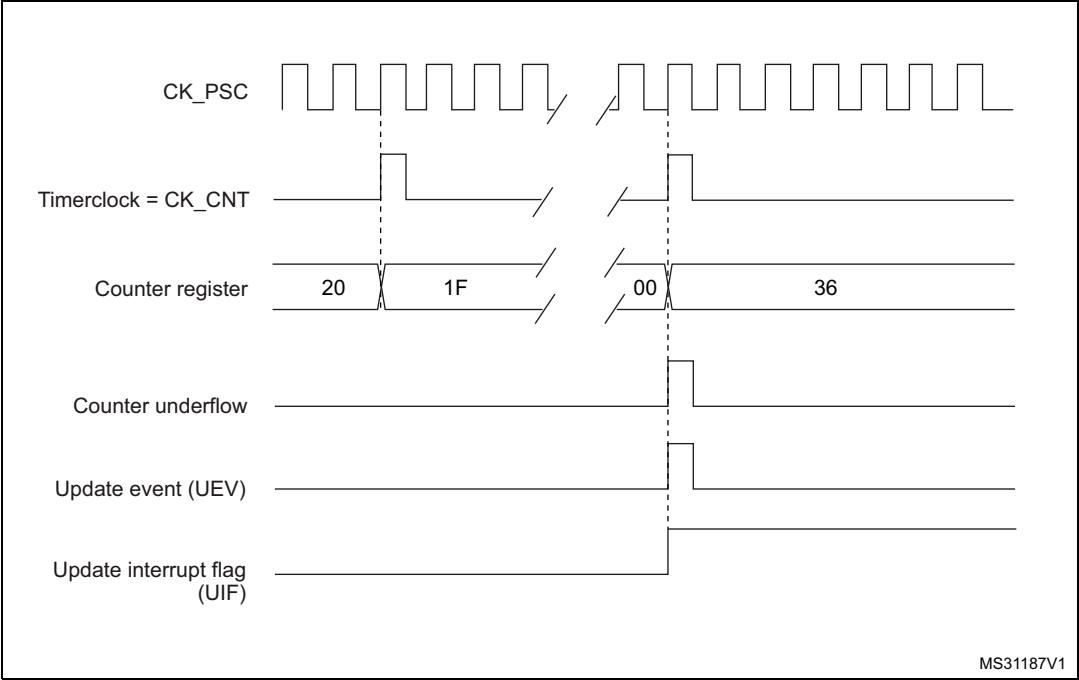
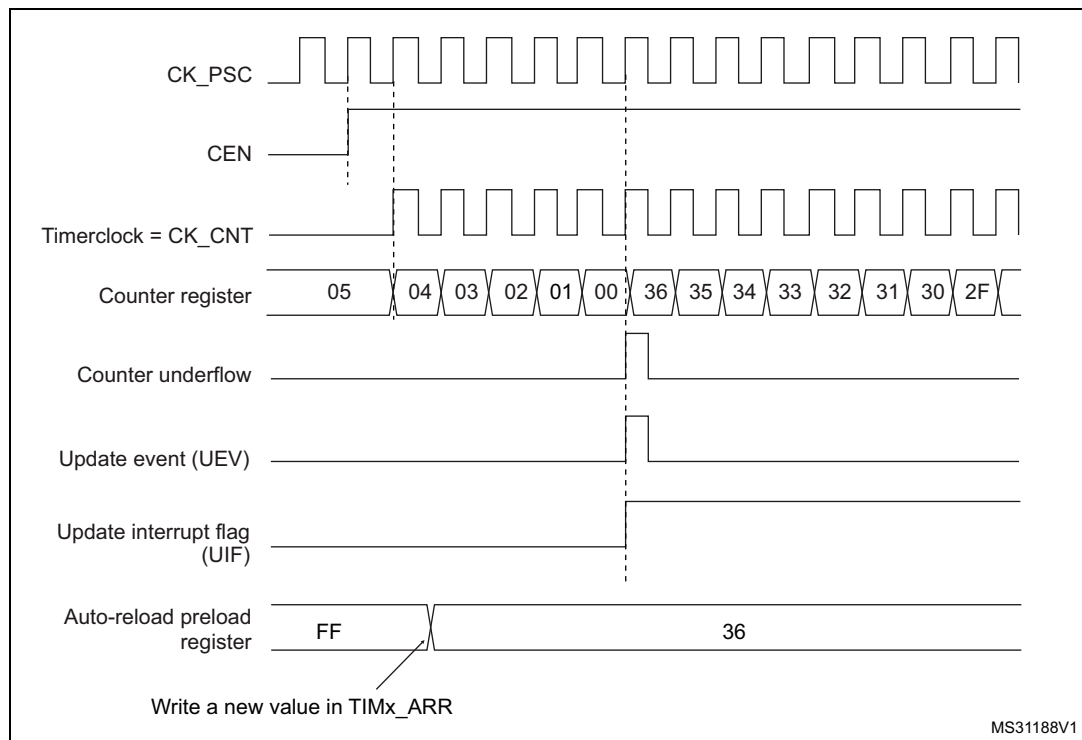


Figure 417. Counter timing diagram, Update event when repetition counter is not used**Center-aligned mode (up/down counting)**

In center-aligned mode, the counter counts from 0 to the auto-reload value (content of the TIMx_ARR register) – 1, generates a counter overflow event, then counts from the auto-reload value down to 1 and generates a counter underflow event. Then it restarts counting from 0.

Center-aligned mode is active when the CMS bits in TIMx_CR1 register are not equal to '00'. The Output compare interrupt flag of channels configured in output is set when: the counter counts down (Center aligned mode 1, CMS = "01"), the counter counts up (Center aligned mode 2, CMS = "10") the counter counts up and down (Center aligned mode 3, CMS = "11").

In this mode, the direction bit (DIR from TIMx_CR1 register) cannot be written. It is updated by hardware and gives the current direction of the counter.

The update event can be generated at each counter overflow and at each counter underflow or by setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller) also generates an update event. In this case, the counter restarts counting from 0, as well as the counter of the prescaler.

The UEV update event can be disabled by software by setting the UDIS bit in TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until the UDIS bit has been written to 0. However, the counter continues counting up and down, based on the current auto-reload value.

In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt or

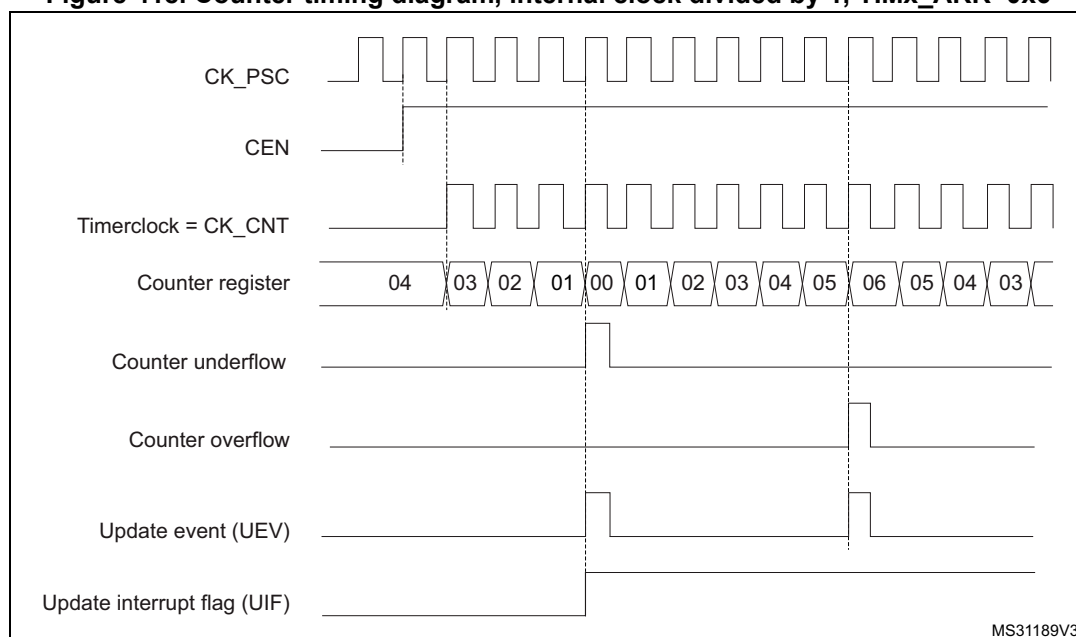
DMA request is sent). This is to avoid generating both update and capture interrupt when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register).
- The auto-reload active register is updated with the preload value (content of the TIMx_ARR register). Note that if the update source is a counter overflow, the auto-reload is updated before the counter is reloaded, so that the next period is the expected one (the counter is loaded with the new value).

The following figures show some examples of the counter behavior for different clock frequencies.

Figure 418. Counter timing diagram, internal clock divided by 1, TIMx_ARR=0x6



1. Here, center-aligned mode 1 is used (for more details refer to [Section 39.4.1: TIMx control register 1 \(TIMx_CR1\)\(x = 2 to 5\) on page 1694](#)).

Figure 419. Counter timing diagram, internal clock divided by 2

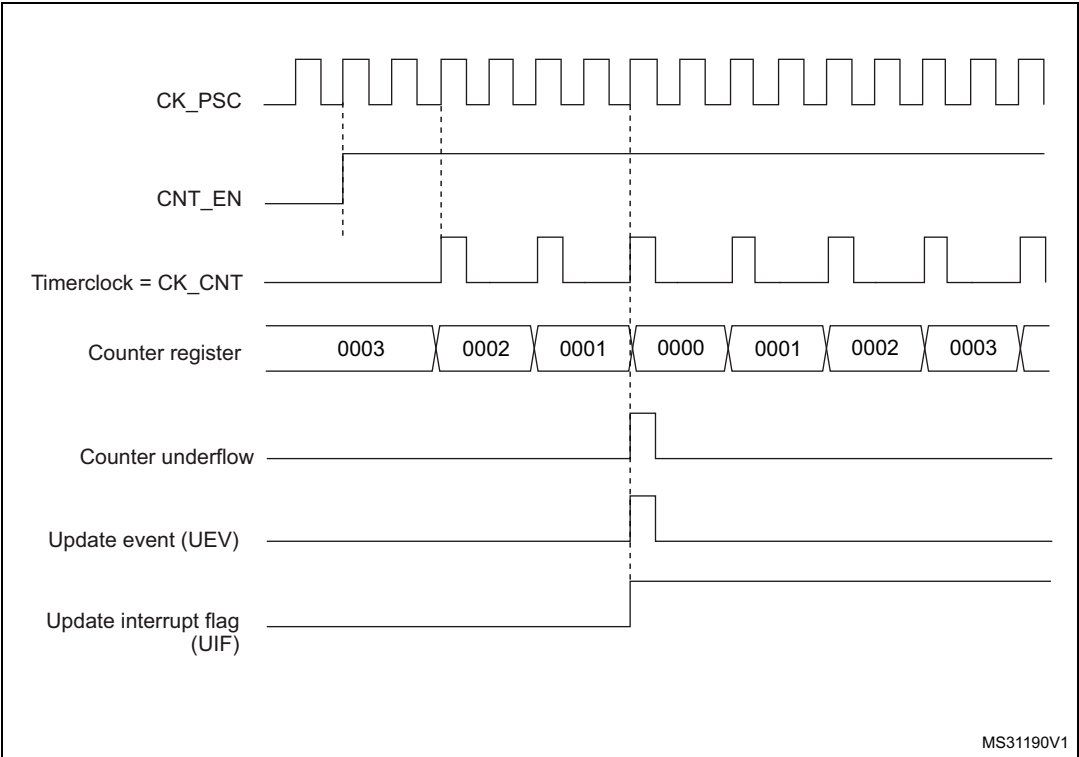
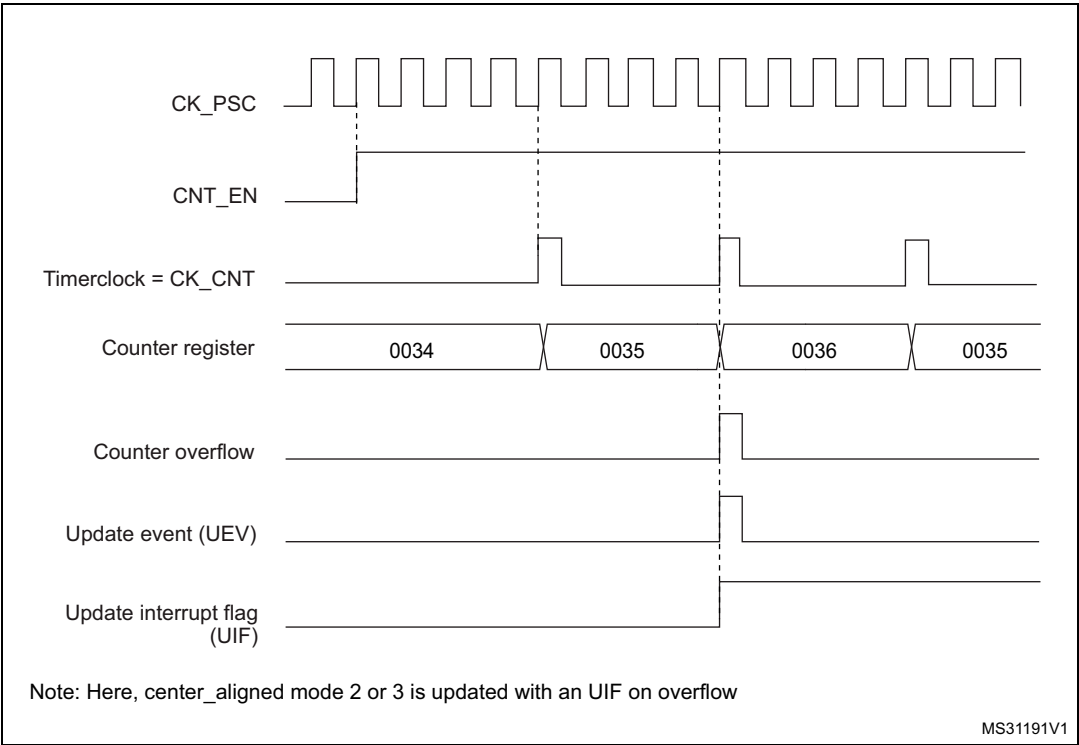


Figure 420. Counter timing diagram, internal clock divided by 4, TIMx_ARR=0x36



1. Center-aligned mode 2 or 3 is used with an UIF on overflow.

Figure 421. Counter timing diagram, internal clock divided by N

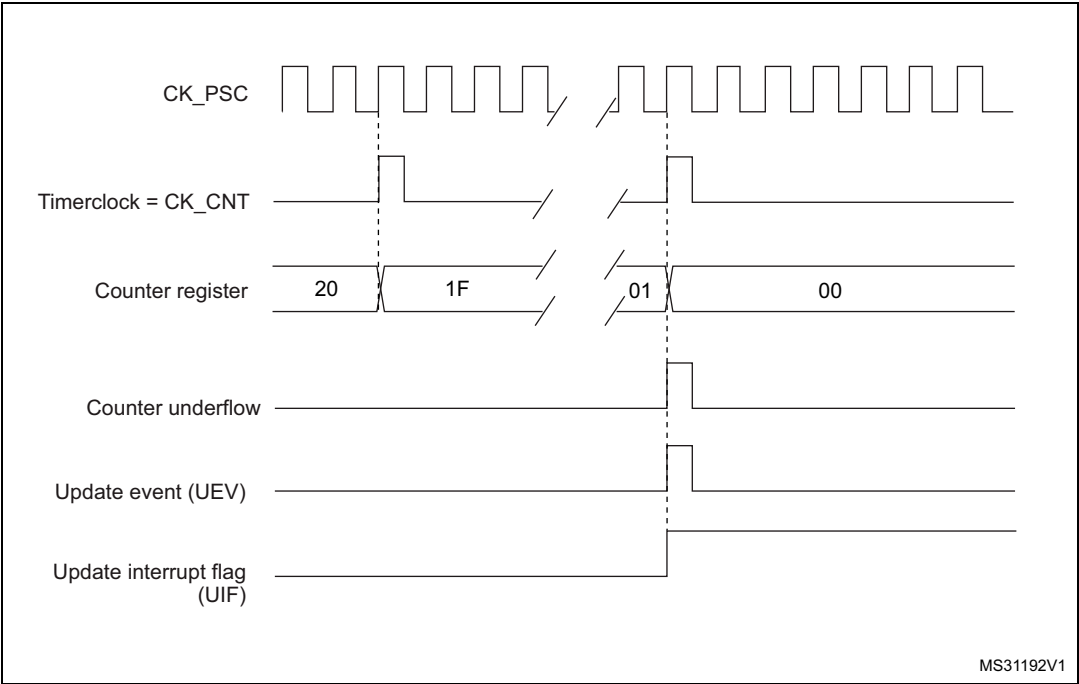


Figure 422. Counter timing diagram, Update event with ARPE=1 (counter underflow)

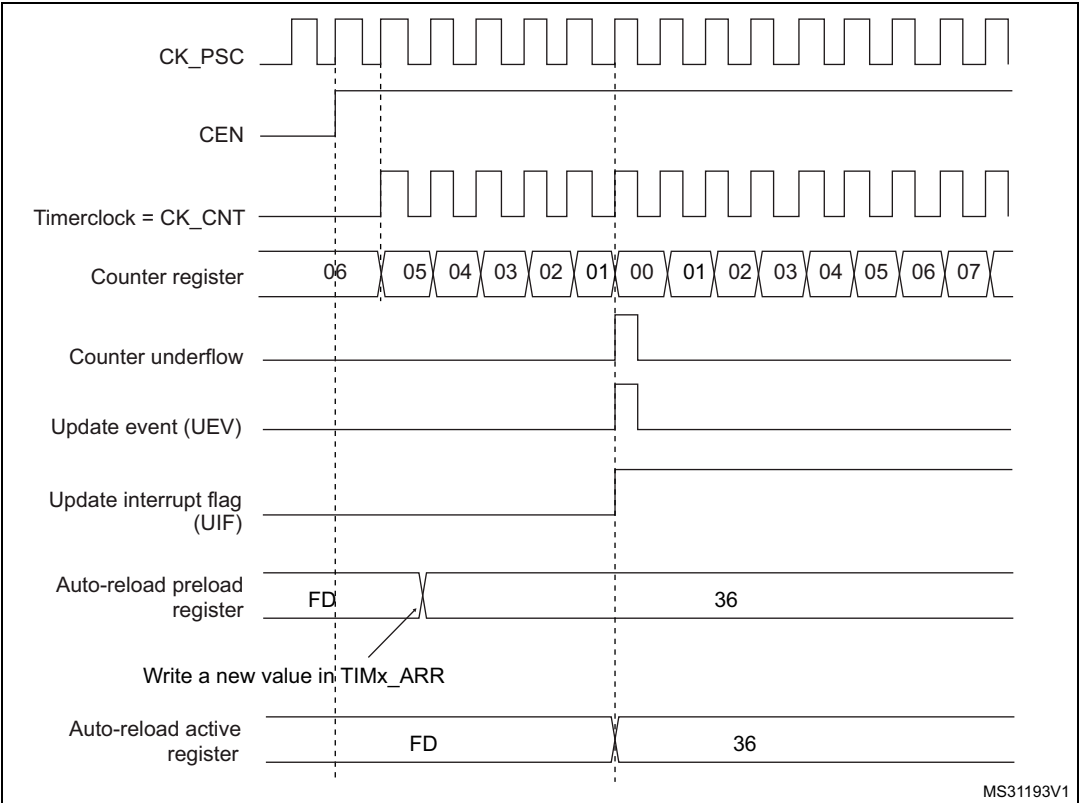
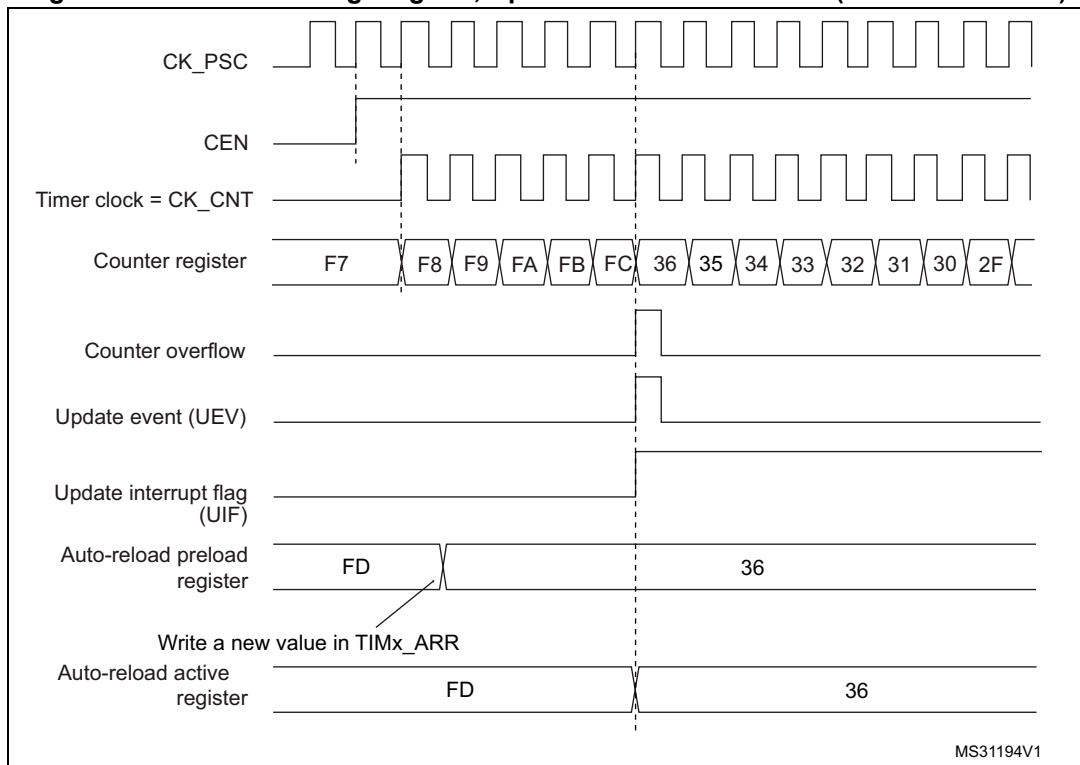


Figure 423. Counter timing diagram, Update event with ARPE=1 (counter overflow)

39.3.3 Clock selection

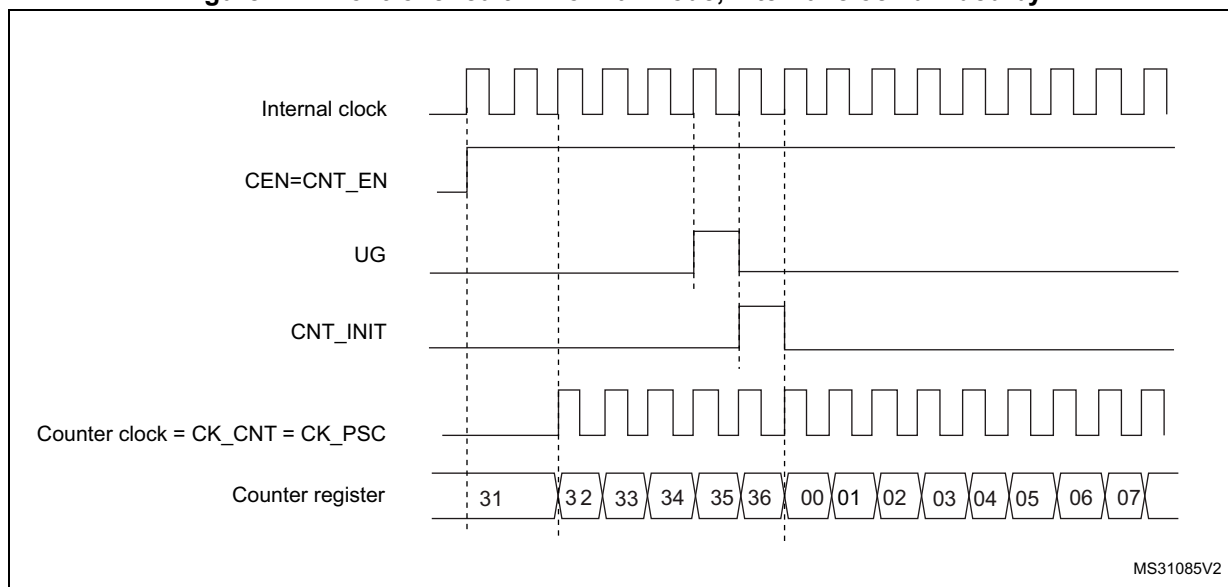
The counter clock can be provided by the following clock sources:

- Internal clock (CK_INT)
- External clock mode1: external input pin (TIx)
- External clock mode2: external trigger input (ETR)
- Internal trigger inputs (ITRx): using one timer as prescaler for another timer, for example, Timer X can be configured to act as a prescaler for Timer Y. Refer to : [Using one timer as prescaler for another timer on page 1689](#) for more details.

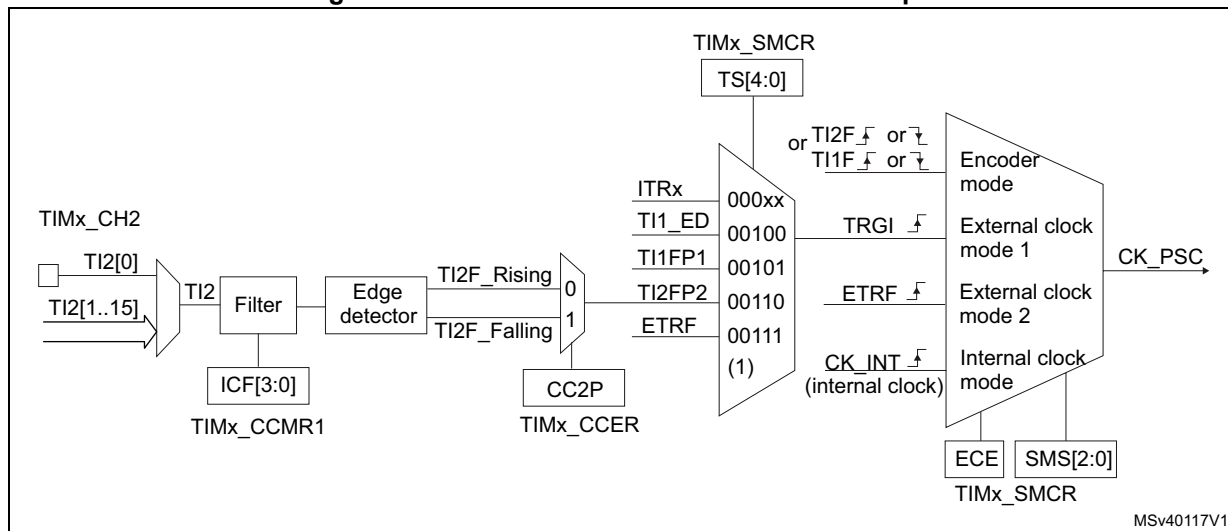
Internal clock source (CK_INT)

If the slave mode controller is disabled (SMS=000 in the TIMx_SMCR register), then the CEN, DIR (in the TIMx_CR1 register) and UG bits (in the TIMx_EGR register) are actual control bits and can be changed only by software (except UG which remains cleared automatically). As soon as the CEN bit is written to 1, the prescaler is clocked by the internal clock CK_INT.

[Figure 424](#) shows the behavior of the control circuit and the upcounter in normal mode, without prescaler.

Figure 424. Control circuit in normal mode, internal clock divided by 1**External clock source mode 1**

This mode is selected when SMS=111 in the TIMx_SMCR register. The counter can count at each rising or falling edge on a selected input.

Figure 425. TI2 external clock connection example

1. Codes ranging from 01000 to 11111: ITRy.

For example, to configure the upcounter to count in response to a rising edge on the TI2 input, use the following procedure:

1. Select the proper TI2x source (internal or external) with the TI2SEL[3:0] bits in the TIMx_TISEL register.
2. Configure channel 2 to detect rising edges on the TI2 input by writing CC2S= '01 in the TIMx_CCMR1 register.
3. Configure the input filter duration by writing the IC2F[3:0] bits in the TIMx_CCMR1 register (if no filter is needed, keep IC2F=0000).

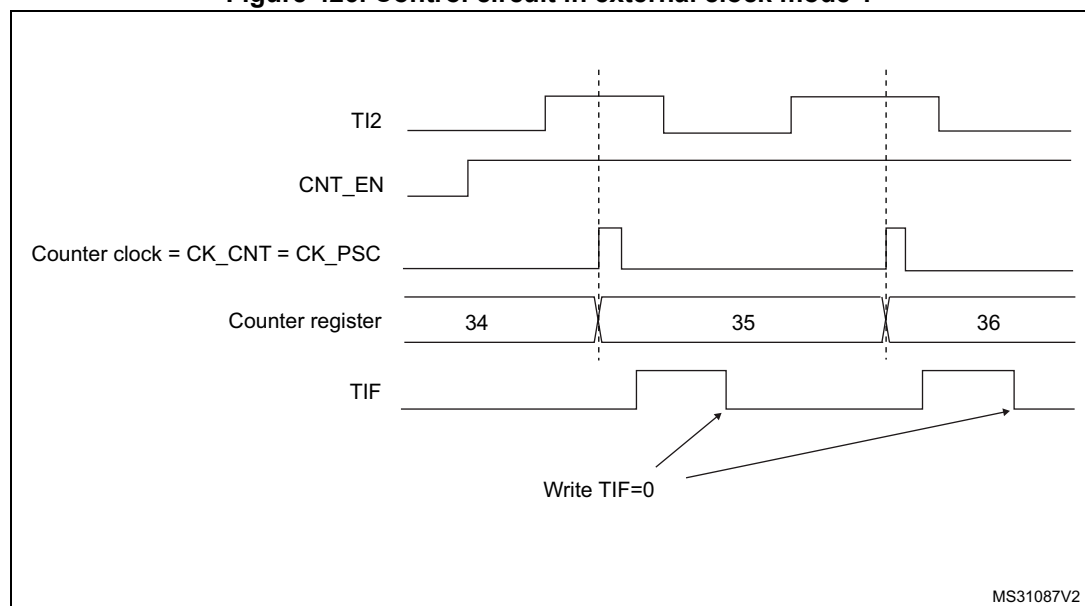
Note: The capture prescaler is not used for triggering, so it does not need to be configured.

4. Select rising edge polarity by writing CC2P=0 and CC2NP=0 and CC2NP=0 in the TIMx_CCER register.
5. Configure the timer in external clock mode 1 by writing SMS=111 in the TIMx_SMCR register.
6. Select TI2 as the input source by writing TS=00110 in the TIMx_SMCR register.
7. Enable the counter by writing CEN=1 in the TIMx_CR1 register.

When a rising edge occurs on TI2, the counter counts once and the TIF flag is set.

The delay between the rising edge on TI2 and the actual clock of the counter is due to the resynchronization circuit on TI2 input.

Figure 426. Control circuit in external clock mode 1



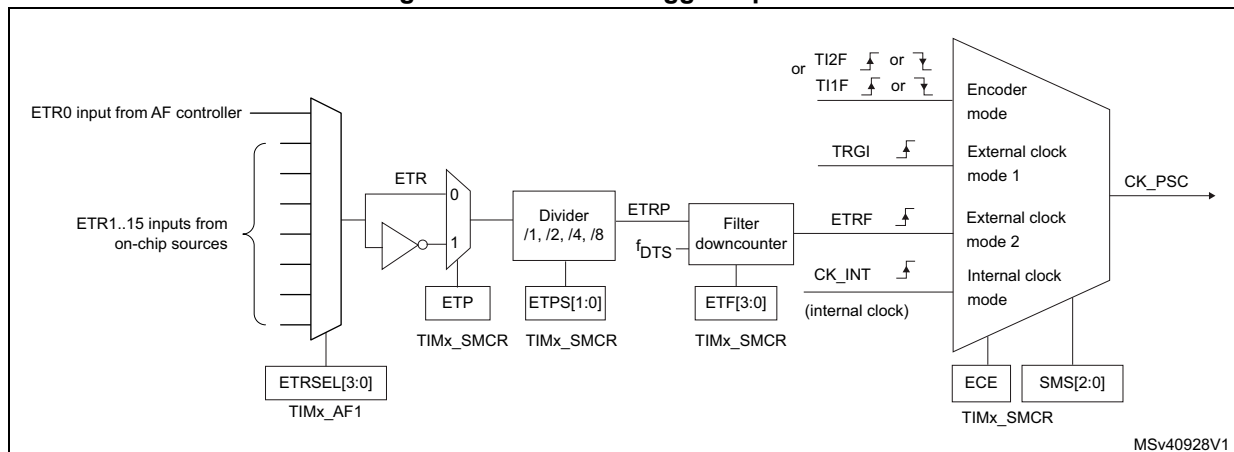
External clock source mode 2

This mode is selected by writing ECE=1 in the TIMx_SMCR register.

The counter can count at each rising or falling edge on the external trigger input ETR.

[Figure 427](#) gives an overview of the external trigger input block.

Figure 427. External trigger input block



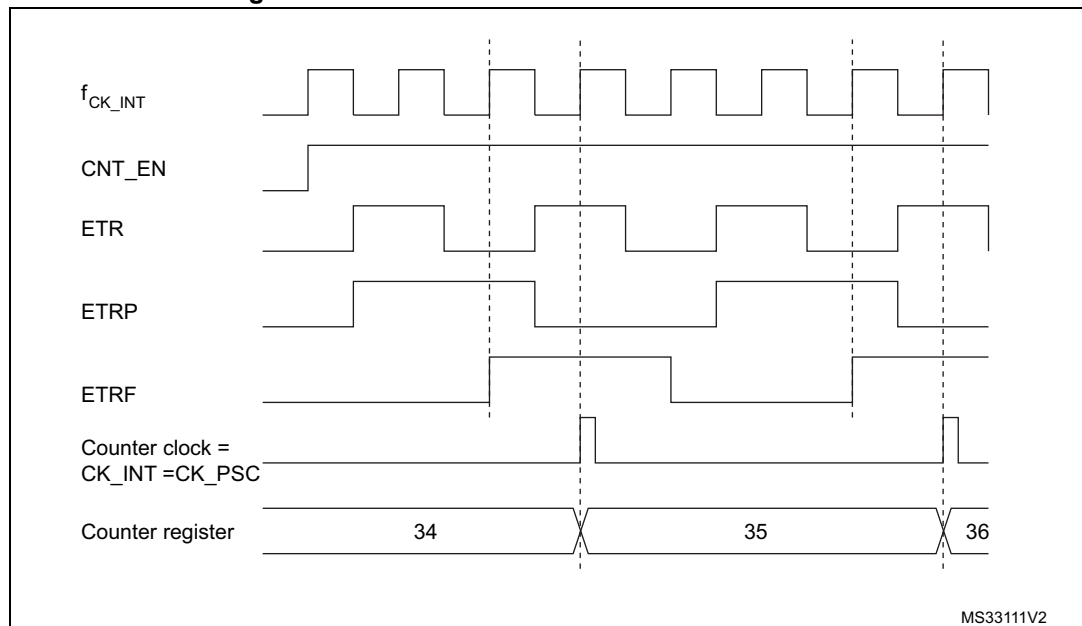
For example, to configure the upcounter to count each 2 rising edges on ETR, use the following procedure:

1. Select the proper ETR source (internal or external) with the ETRSEL[3:0] bits in the TIMx_AF1 register.
2. As no filter is needed in this example, write ETF[3:0]=0000 in the TIMx_SMCR register.
3. Set the prescaler by writing ETPS[1:0]=01 in the TIMx_SMCR register
4. Select rising edge detection on the ETR pin by writing ETP=0 in the TIMx_SMCR register
5. Enable external clock mode 2 by writing ECE=1 in the TIMx_SMCR register.
6. Enable the counter by writing CEN=1 in the TIMx_CR1 register.

The counter counts once each 2 ETR rising edges.

The delay between the rising edge on ETR and the actual clock of the counter is due to the resynchronization circuit on the ETRP signal. As a consequence, the maximum frequency which can be correctly captured by the counter is at most $\frac{1}{4}$ of TIMxCLK frequency. When the ETRP signal is faster, the user should apply a division of the external signal by a proper ETPS prescaler setting.

Figure 428. Control circuit in external clock mode 2



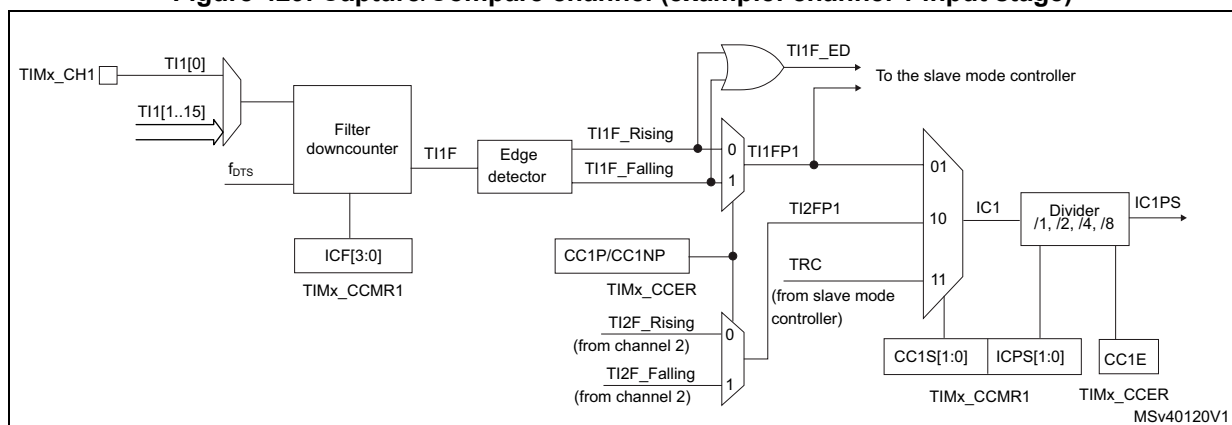
39.3.4 Capture/Compare channels

Each Capture/Compare channel is built around a capture/compare register (including a shadow register), a input stage for capture (with digital filter, multiplexing and prescaler) and an output stage (with comparator and output control).

The following figure gives an overview of one Capture/Compare channel.

The input stage samples the corresponding TIMx input to generate a filtered signal TIxF. Then, an edge detector with polarity selection generates a signal (TIxFPx) which can be used as trigger input by the slave mode controller or as the capture command. It is prescaled before the capture register (ICxPS).

Figure 429. Capture/Compare channel (example: channel 1 input stage)



The output stage generates an intermediate waveform which is then used for reference: OCxRef (active high). The polarity acts at the end of the chain.

Figure 430. Capture/Compare channel 1 main circuit

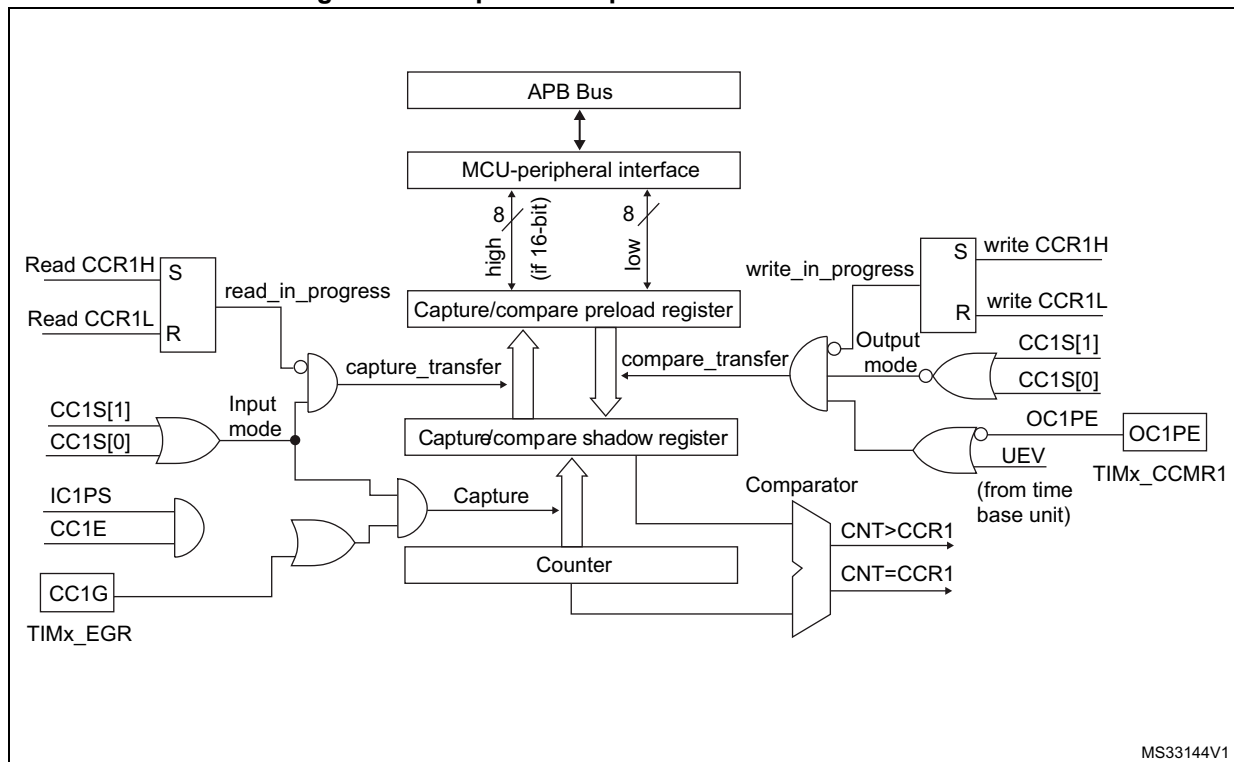
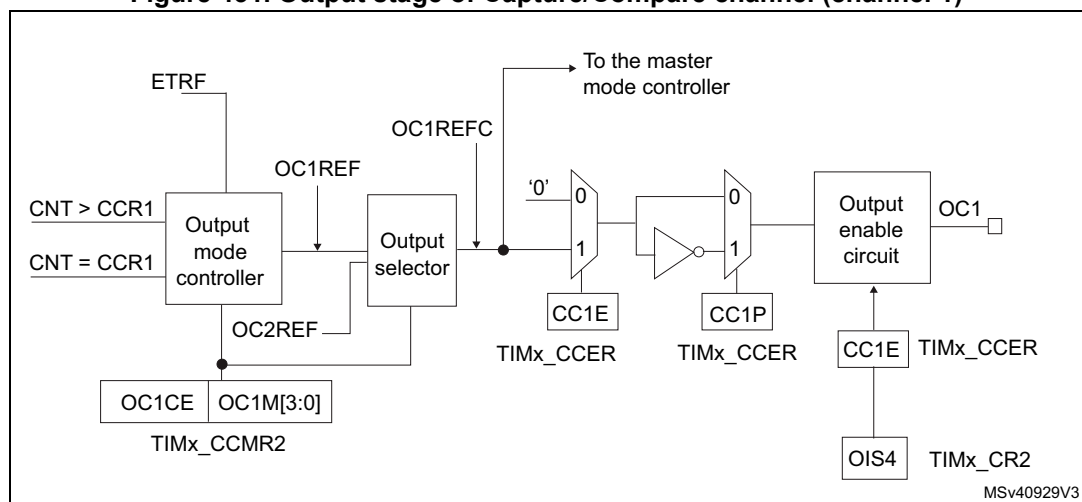


Figure 431. Output stage of Capture/Compare channel (channel 1)



The capture/compare block is made of one preload register and one shadow register. Write and read always access the preload register.

In capture mode, captures are actually done in the shadow register, which is copied into the preload register.

In compare mode, the content of the preload register is copied into the shadow register which is compared to the counter.

39.3.5 Input capture mode

In Input capture mode, the Capture/Compare Registers (TIMx_CCRx) are used to latch the value of the counter after a transition detected by the corresponding ICx signal. When a capture occurs, the corresponding CCxIF flag (TIMx_SR register) is set and an interrupt or a DMA request can be sent if they are enabled. If a capture occurs while the CCxIF flag was already high, then the over-capture flag CCxOF (TIMx_SR register) is set. CCxIF can be cleared by software by writing it to 0 or by reading the captured data stored in the TIMx_CCRx register. CCxOF is cleared when it is written with 0.

The following example shows how to capture the counter value in TIMx_CCR1 when TI1 input rises. To do this, use the following procedure:

1. Select the proper TI1x source (internal or external) with the TI1SEL[3:0] bits in the TIMx_TISEL register.
2. Select the active input: TIMx_CCR1 must be linked to the TI1 input, so write the CC1S bits to 01 in the TIMx_CCMR1 register. As soon as CC1S becomes different from 00, the channel is configured in input and the TIMx_CCR1 register becomes read-only.
3. Program the appropriate input filter duration in relation with the signal connected to the timer (when the input is one of the TIx (ICxF bits in the TIMx_CCMRx register). Let's imagine that, when toggling, the input signal is not stable during at most 5 internal clock cycles. We must program a filter duration longer than these 5 clock cycles. We can validate a transition on TI1 when 8 consecutive samples with the new level have been detected (sampled at f_{DTS} frequency). Then write IC1F bits to 0011 in the TIMx_CCMR1 register.
4. Select the edge of the active transition on the TI1 channel by writing the CC1P and CC1NP and CC1NP bits to 000 in the TIMx_CCER register (rising edge in this case).
5. Program the input prescaler. In our example, we wish the capture to be performed at each valid transition, so the prescaler is disabled (write IC1PS bits to 00 in the TIMx_CCMR1 register).
6. Enable capture from the counter into the capture register by setting the CC1E bit in the TIMx_CCER register.
7. If needed, enable the related interrupt request by setting the CC1IE bit in the TIMx_DIER register, and/or the DMA request by setting the CC1DE bit in the TIMx_DIER register.

When an input capture occurs:

- The TIMx_CCR1 register gets the value of the counter on the active transition.
- CC1IF flag is set (interrupt flag). CC1OF is also set if at least two consecutive captures occurred whereas the flag was not cleared.
- An interrupt is generated depending on the CC1IE bit.
- A DMA request is generated depending on the CC1DE bit.

In order to handle the overcapture, it is recommended to read the data before the overcapture flag. This is to avoid missing an overcapture which could happen after reading the flag and before reading the data.

Note: *IC interrupt and/or DMA requests can be generated by software by setting the corresponding CCxG bit in the TIMx_EGR register.*

39.3.6 PWM input mode

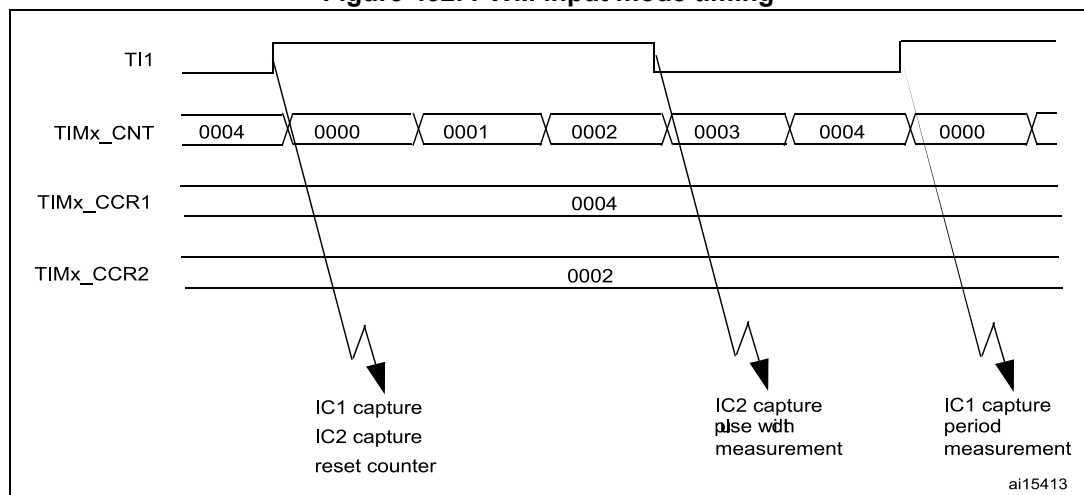
This mode is a particular case of input capture mode. The procedure is the same except:

- Two ICx signals are mapped on the same TIx input.
- These 2 ICx signals are active on edges with opposite polarity.
- One of the two TIxFP signals is selected as trigger input and the slave mode controller is configured in reset mode.

For example, one can measure the period (in TIMx_CCR1 register) and the duty cycle (in TIMx_CCR2 register) of the PWM applied on TI1 using the following procedure (depending on CK_INT frequency and prescaler value):

1. Select the proper TI1x source (internal or external) with the TI1SEL[3:0] bits in the TIMx_TISEL register.
2. Select the active input for TIMx_CCR1: write the CC1S bits to 01 in the TIMx_CCMR1 register (TI1 selected).
3. Select the active polarity for TI1FP1 (used both for capture in TIMx_CCR1 and counter clear): write the CC1P to '0' and the CC1NP bit to '0' (active on rising edge).
4. Select the active input for TIMx_CCR2: write the CC2S bits to 10 in the TIMx_CCMR1 register (TI1 selected).
5. Select the active polarity for TI1FP2 (used for capture in TIMx_CCR2): write the CC2P bit to '1' and the CC2NP bit to '0' (active on falling edge).
6. Select the valid trigger input: write the TS bits to 00101 in the TIMx_SMCR register (TI1FP1 selected).
7. Configure the slave mode controller in reset mode: write the SMS bits to 100 in the TIMx_SMCR register.
8. Enable the captures: write the CC1E and CC2E bits to '1' in the TIMx_CCER register.

Figure 432. PWM input mode timing



1. The PWM input mode can be used only with the TIMx_CH1/TIMx_CH2 signals due to the fact that only TI1FP1 and TI2FP2 are connected to the slave mode controller.

39.3.7 Forced output mode

In output mode (CCxS bits = 00 in the TIMx_CCMRx register), each output compare signal (OCxREF and then OCx) can be forced to active or inactive level directly by software, independently of any comparison between the output compare register and the counter.

To force an output compare signal (ocxref/OCx) to its active level, one just needs to write 101 in the OCxM bits in the corresponding TIMx_CCMRx register. Thus ocxref is forced high (OCxREF is always active high) and OCx get opposite value to CCxP polarity bit.

e.g.: CCxP=0 (OCx active high) => OCx is forced to high level.

ocxref signal can be forced low by writing the OCxM bits to 100 in the TIMx_CCMRx register.

Anyway, the comparison between the TIMx_CCRx shadow register and the counter is still performed and allows the flag to be set. Interrupt and DMA requests can be sent accordingly. This is described in the Output Compare Mode section.

39.3.8 Output compare mode

This function is used to control an output waveform or indicating when a period of time has elapsed.

When a match is found between the capture/compare register and the counter, the output compare function:

- Assigns the corresponding output pin to a programmable value defined by the output compare mode (OCxM bits in the TIMx_CCMRx register) and the output polarity (CCxP bit in the TIMx_CCER register). The output pin can keep its level (OCxM=000), be set active (OCxM=001), be set inactive (OCxM=010) or can toggle (OCxM=011) on match.
- Sets a flag in the interrupt status register (CCxIF bit in the TIMx_SR register).
- Generates an interrupt if the corresponding interrupt mask is set (CCxIE bit in the TIMx_DIER register).
- Sends a DMA request if the corresponding enable bit is set (CCxDE bit in the TIMx_DIER register, CCDS bit in the TIMx_CR2 register for the DMA request selection).

The TIMx_CCRx registers can be programmed with or without preload registers using the OCxPE bit in the TIMx_CCMRx register.

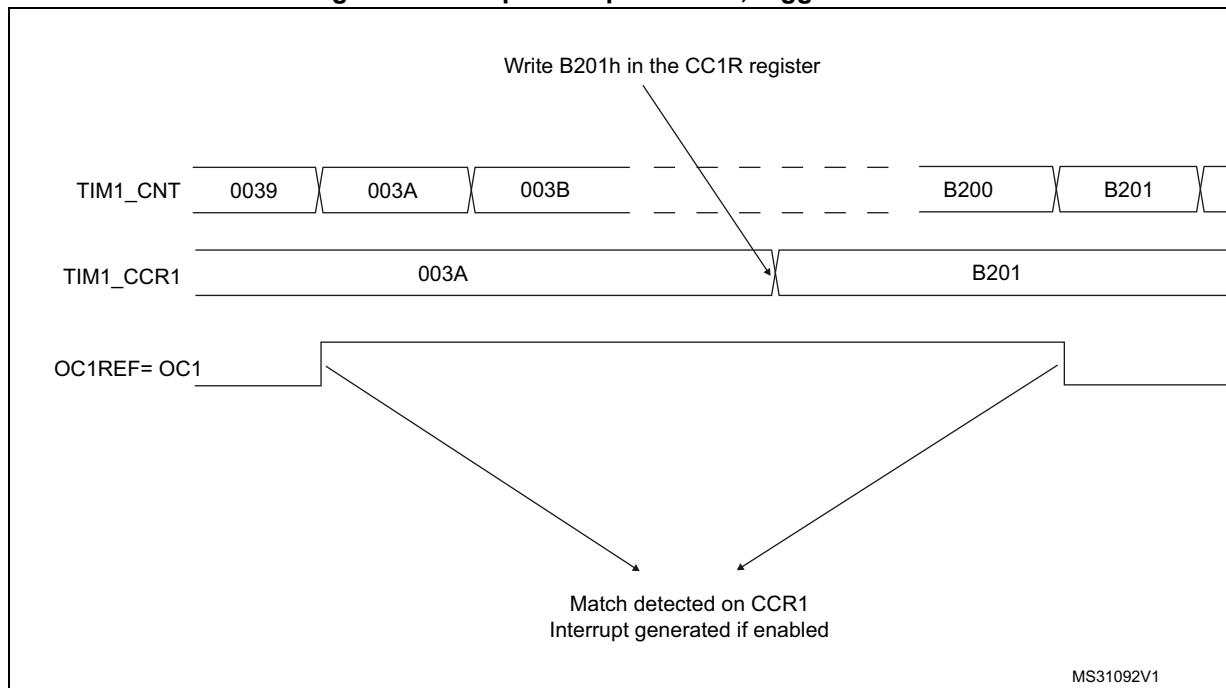
In output compare mode, the update event UEV has no effect on ocxref and OCx output. The timing resolution is one count of the counter. Output compare mode can also be used to output a single pulse (in One-pulse mode).

Procedure

1. Select the counter clock (internal, external, prescaler).
2. Write the desired data in the TIMx_ARR and TIMx_CCRx registers.
3. Set the CCxIE and/or CCxDE bits if an interrupt and/or a DMA request is to be generated.
4. Select the output mode. For example, one must write OCxM=011, OCxPE=0, CCxP=0 and CCxE=1 to toggle OCx output pin when CNT matches CCRx, CCRx preload is not used, OCx is enabled and active high.
5. Enable the counter by setting the CEN bit in the TIMx_CR1 register.

The TIMx_CCRx register can be updated at any time by software to control the output waveform, provided that the preload register is not enabled (OCxPE=0, else TIMx_CCRx shadow register is updated only at the next update event UEV). An example is given in [Figure 433](#).

Figure 433. Output compare mode, toggle on OC1



39.3.9 PWM mode

Pulse width modulation mode permits to generate a signal with a frequency determined by the value of the TIMx_ARR register and a duty cycle determined by the value of the TIMx_CCRx register.

The PWM mode can be selected independently on each channel (one PWM per OCx output) by writing 110 (PWM mode 1) or '111 (PWM mode 2) in the OCxM bits in the TIMx_CCMRx register. The corresponding preload register must be enabled by setting the OCxPE bit in the TIMx_CCMRx register, and eventually the auto-reload preload register (in upcounting or center-aligned modes) by setting the ARPE bit in the TIMx_CR1 register.

As the preload registers are transferred to the shadow registers only when an update event occurs, before starting the counter, all registers must be initialized by setting the UG bit in the TIMx_EGR register.

OCx polarity is software programmable using the CCxP bit in the TIMx_CCER register. It can be programmed as active high or active low. OCx output is enabled by the CCxE bit in the TIMx_CCER register. Refer to the TIMx_CCERx register description for more details.

In PWM mode (1 or 2), TIMx_CNT and TIMx_CCRx are always compared to determine whether $TIMx_CCRx \leq TIMx_CNT$ or $TIMx_CNT \leq TIMx_CCRx$ (depending on the direction of the counter). However, to comply with the OCREF_CLR functionality (OCREF can be

cleared by an external event through the ETR signal until the next PWM period), the OCREF signal is asserted only:

- When the result of the comparison or
- When the output compare mode (OCxM bits in TIMx_CCMRx register) switches from the “frozen” configuration (no comparison, OCxM=‘000) to one of the PWM modes (OCxM=‘110 or ‘111).

This forces the PWM by software while the timer is running.

The timer is able to generate PWM in edge-aligned mode or center-aligned mode depending on the CMS bits in the TIMx_CR1 register.

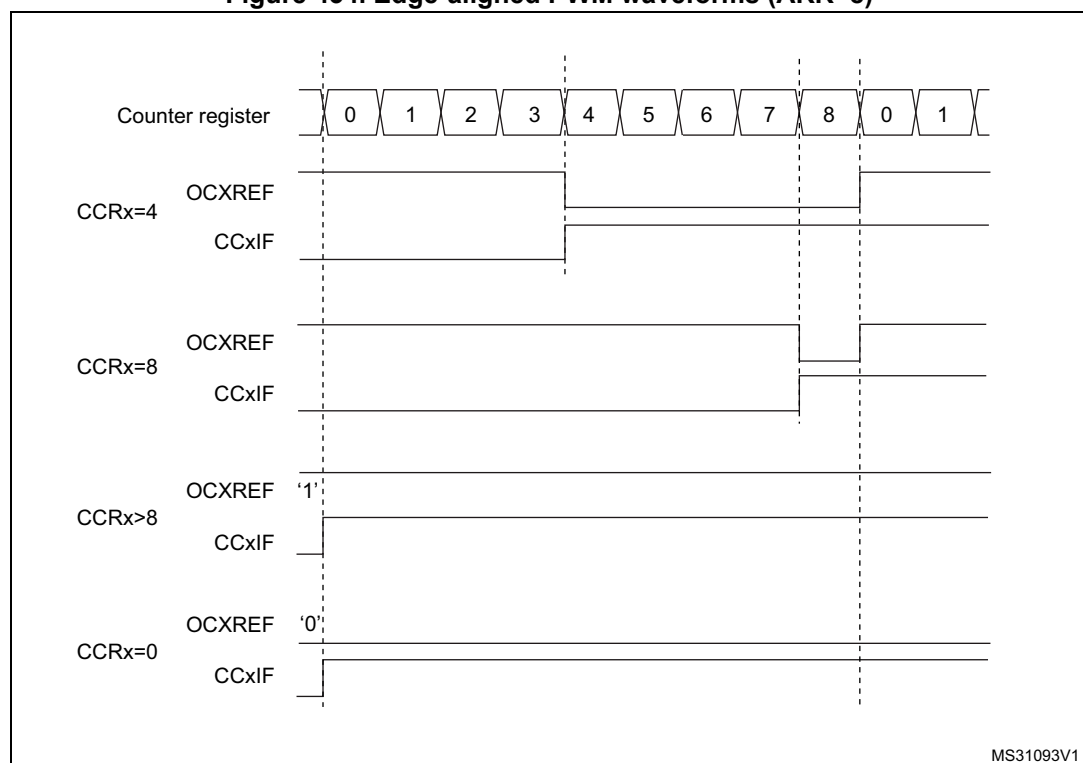
PWM edge-aligned mode

Upcounting configuration

Upcounting is active when the DIR bit in the TIMx_CR1 register is low. Refer to [Upcounting mode on page 1654](#).

In the following example, we consider PWM mode 1. The reference PWM signal OCxREF is high as long as $TIMx_CNT < TIMx_CCRx$ else it becomes low. If the compare value in TIMx_CCRx is greater than the auto-reload value (in TIMx_ARR) then OCxREF is held at ‘1’. If the compare value is 0 then OCxREF is held at ‘0’. [Figure 434](#) shows some edge-aligned PWM waveforms in an example where $TIMx_ARR=8$.

Figure 434. Edge-aligned PWM waveforms (ARR=8)



Downcounting configuration

Downcounting is active when DIR bit in TIMx_CR1 register is high. Refer to [Downcounting mode on page 1657](#).

In PWM mode 1, the reference signal ocxref is low as long as TIMx_CNT > TIMx_CCRx else it becomes high. If the compare value in TIMx_CCRx is greater than the auto-reload value in TIMx_ARR, then ocxref is held at 100%. PWM is not possible in this mode.

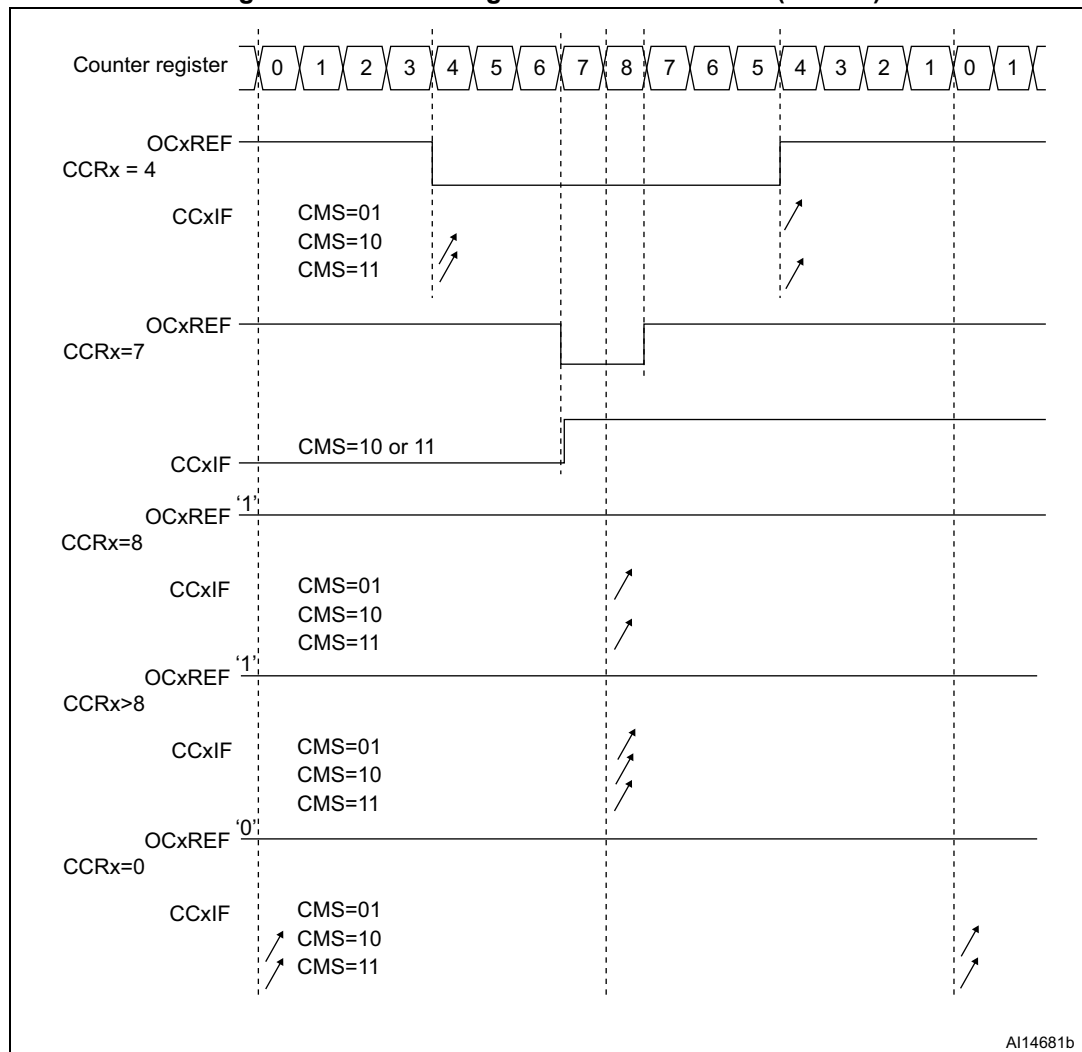
PWM center-aligned mode

Center-aligned mode is active when the CMS bits in TIMx_CR1 register are different from '00 (all the remaining configurations having the same effect on the ocxref/OCx signals). The compare flag is set when the counter counts up, when it counts down or both when it counts up and down depending on the CMS bits configuration. The direction bit (DIR) in the TIMx_CR1 register is updated by hardware and must not be changed by software. Refer to [Center-aligned mode \(up/down counting\) on page 1660](#).

[Figure 435](#) shows some center-aligned PWM waveforms in an example where:

- TIMx_ARR=8,
- PWM mode is the PWM mode 1,
- The flag is set when the counter counts down corresponding to the center-aligned mode 1 selected for CMS=01 in TIMx_CR1 register.

Figure 435. Center-aligned PWM waveforms (ARR=8)



Hints on using center-aligned mode:

- When starting in center-aligned mode, the current up-down configuration is used. It means that the counter counts up or down depending on the value written in the DIR bit in the TIMx_CR1 register. Moreover, the DIR and CMS bits must not be changed at the same time by the software.
- Writing to the counter while running in center-aligned mode is not recommended as it can lead to unexpected results. In particular:
 - The direction is not updated if a value greater than the auto-reload value is written in the counter (TIMx_CNT > TIMx_ARR). For example, if the counter was counting up, it continues to count up.
 - The direction is updated if 0 or the TIMx_ARR value is written in the counter but no Update Event UEV is generated.
- The safest way to use center-aligned mode is to generate an update by software (setting the UG bit in the TIMx_EGR register) just before starting the counter and not to write the counter while it is running.

39.3.10 Asymmetric PWM mode

Asymmetric mode allows two center-aligned PWM signals to be generated with a programmable phase shift. While the frequency is determined by the value of the TIMx_ARR register, the duty cycle and the phase-shift are determined by a pair of TIMx_CCRx registers. One register controls the PWM during up-counting, the second during down counting, so that PWM is adjusted every half PWM cycle:

- OC1REFC (or OC2REFC) is controlled by TIMx_CCR1 and TIMx_CCR2
- OC3REFC (or OC4REFC) is controlled by TIMx_CCR3 and TIMx_CCR4

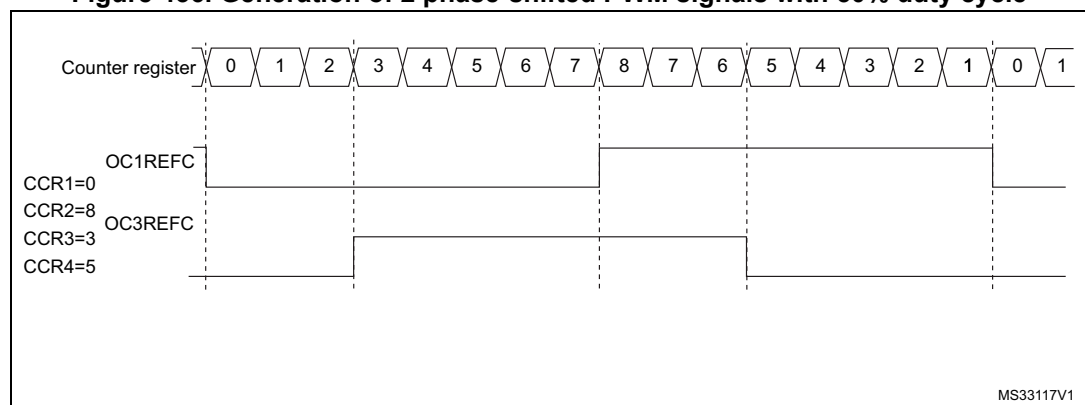
Asymmetric PWM mode can be selected independently on two channels (one OCx output per pair of CCR registers) by writing '1110' (Asymmetric PWM mode 1) or '1111' (Asymmetric PWM mode 2) in the OCxM bits in the TIMx_CCMRx register.

Note: The OCxM[3:0] bit field is split into two parts for compatibility reasons, the most significant bit is not contiguous with the 3 least significant ones.

When a given channel is used as asymmetric PWM channel, its secondary channel can also be used. For instance, if an OC1REFC signal is generated on channel 1 (Asymmetric PWM mode 1), it is possible to output either the OC2REF signal on channel 2, or an OC2REFC signal resulting from asymmetric PWM mode 2.

Figure 436 shows an example of signals that can be generated using Asymmetric PWM mode (channels 1 to 4 are configured in Asymmetric PWM mode 1).

Figure 436. Generation of 2 phase-shifted PWM signals with 50% duty cycle



39.3.11 Combined PWM mode

Combined PWM mode allows two edge or center-aligned PWM signals to be generated with programmable delay and phase shift between respective pulses. While the frequency is determined by the value of the TIMx_ARR register, the duty cycle and delay are determined by the two TIMx_CCRx registers. The resulting signals, OCxREFC, are made of an OR or AND logical combination of two reference PWMs:

- OC1REFC (or OC2REFC) is controlled by TIMx_CCR1 and TIMx_CCR2
- OC3REFC (or OC4REFC) is controlled by TIMx_CCR3 and TIMx_CCR4

Combined PWM mode can be selected independently on two channels (one OCx output per pair of CCR registers) by writing '1100' (Combined PWM mode 1) or '1101' (Combined PWM mode 2) in the OCxM bits in the TIMx_CCMRx register.

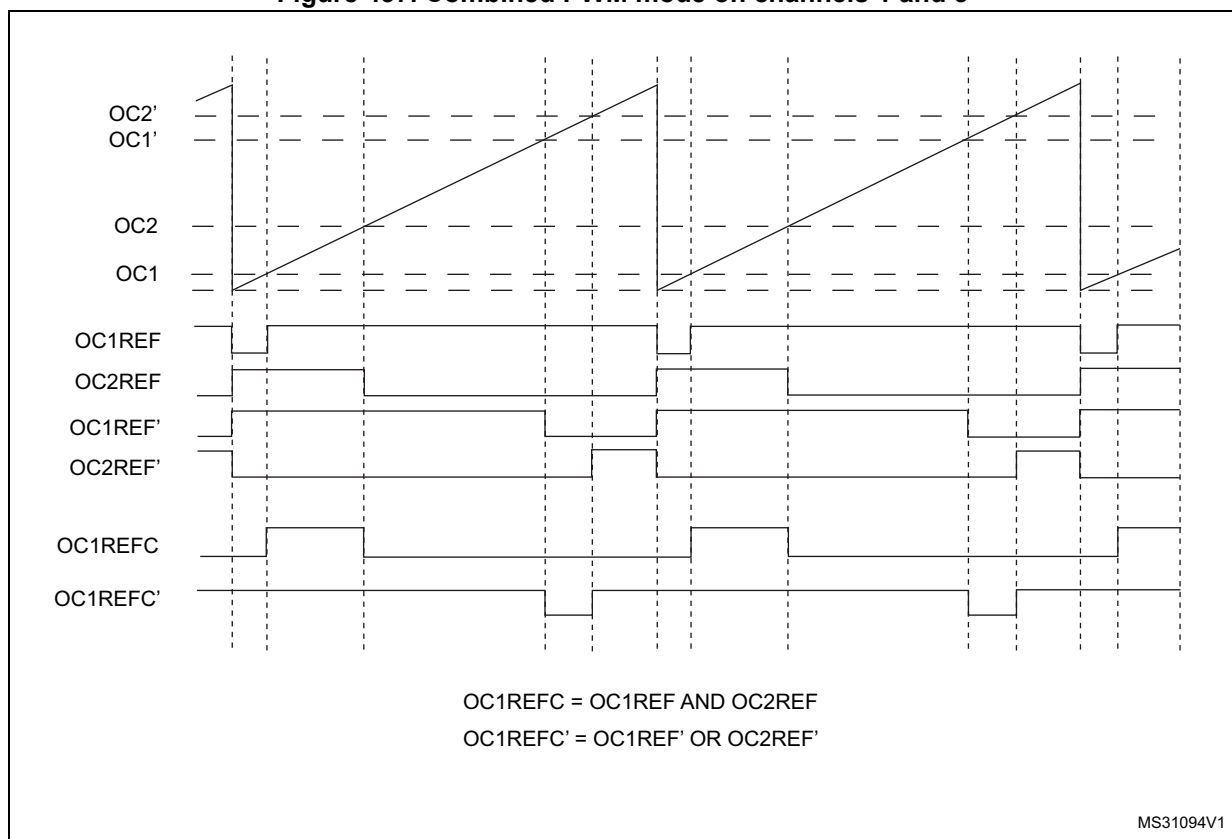
When a given channel is used as combined PWM channel, its secondary channel must be configured in the opposite PWM mode (for instance, one in Combined PWM mode 1 and the other in Combined PWM mode 2).

Note: The OCxM[3:0] bit field is split into two parts for compatibility reasons, the most significant bit is not contiguous with the 3 least significant ones.

Figure 437 shows an example of signals that can be generated using Asymmetric PWM mode, obtained with the following configuration:

- Channel 1 is configured in Combined PWM mode 2,
- Channel 2 is configured in PWM mode 1,
- Channel 3 is configured in Combined PWM mode 2,
- Channel 4 is configured in PWM mode 1

Figure 437. Combined PWM mode on channels 1 and 3



39.3.12 Clearing the OCxREF signal on an external event

The OCxREF signal of a given channel can be cleared when a high level is applied on the ocref_clr_int input (OCxCE enable bit in the corresponding TIMx_CCMRx register set to 1). OCxREF remains low until the next update event (UEV) occurs. This function can only be used in Output compare and PWM modes. It does not work in Forced mode.

The ocref_clr_int is connected to the ETRF signal (ETR after filtering).

The OCxREF signal for a given channel can be reset by applying a high level on the ETRF input (OCxCE enable bit set to 1 in the corresponding TIMx_CCMRx register). OCxREF remains low until the next update event (UEV) occurs.

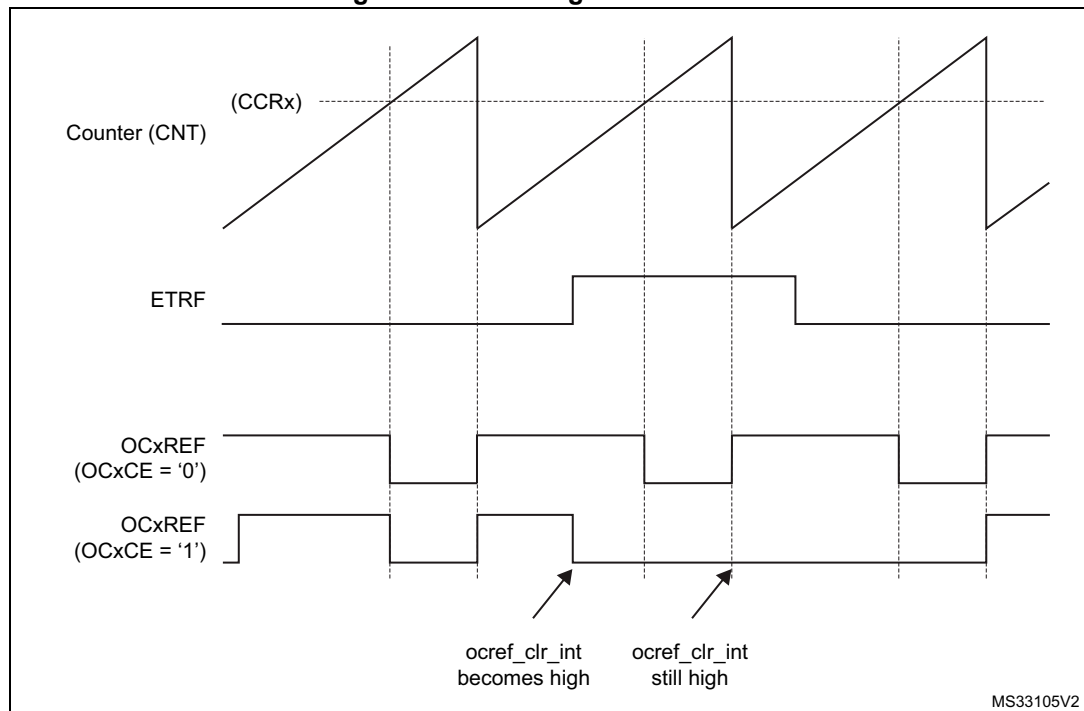
This function can be used only in the output compare and PWM modes. It does not work in forced mode.

For example, the OCxREF signal can be connected to the output of a comparator to be used for current handling. In this case, ETR must be configured as follows:

1. The external trigger prescaler should be kept off: bits ETPS[1:0] in the TIMx_SMCR register are cleared to 00.
2. The external clock mode 2 must be disabled: bit ECE in the TIM1_SMCR register is cleared to 0.
3. The external trigger polarity (ETP) and the external trigger filter (ETF) can be configured according to the application's needs.

Figure 438 shows the behavior of the OCxREF signal when the ETRF input becomes high, for both values of the OCxCE enable bit. In this example, the timer TIMx is programmed in PWM mode.

Figure 438. Clearing TIMx OCxREF



Note: In case of a PWM with a 100% duty cycle (if $CCR_x > ARR$), OCxREF is enabled again at the next counter overflow.

39.3.13 One-pulse mode

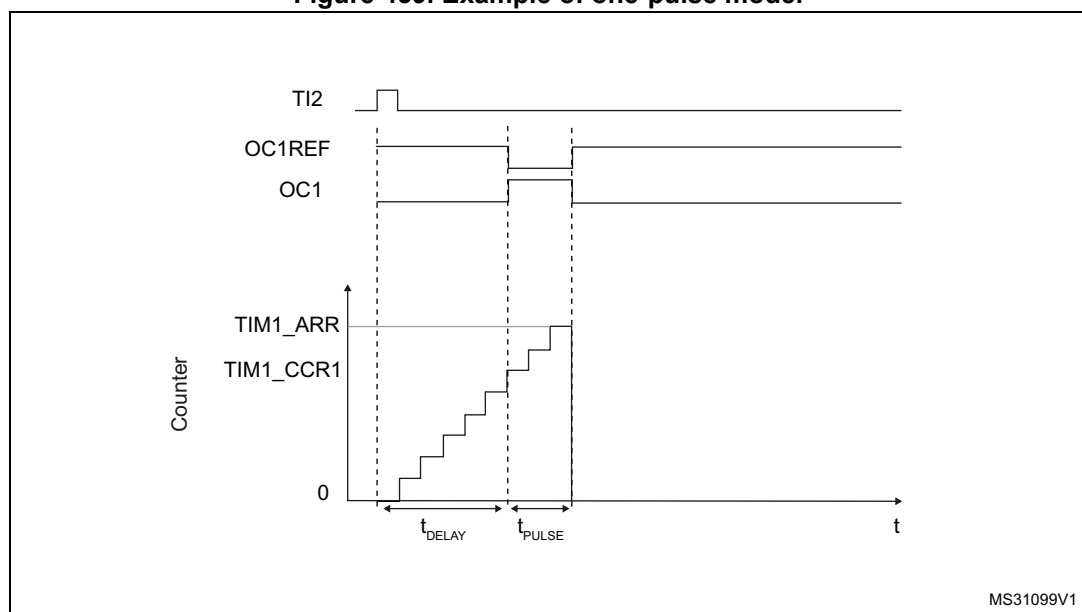
One-pulse mode (OPM) is a particular case of the previous modes. It allows the counter to be started in response to a stimulus and to generate a pulse with a programmable length after a programmable delay.

Starting the counter can be controlled through the slave mode controller. Generating the waveform can be done in output compare mode or PWM mode. One-pulse mode is selected by setting the OPM bit in the TIMx_CR1 register. This makes the counter stop automatically at the next update event UEV.

A pulse can be correctly generated only if the compare value is different from the counter initial value. Before starting (when the timer is waiting for the trigger), the configuration must be:

- $CNT < CCRx \leq ARR$ (in particular, $0 < CCRx$),

Figure 439. Example of one-pulse mode.



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For example one may want to generate a positive pulse on OC1 with a length of t_{PULSE} and after a delay of t_{DELAY} as soon as a positive edge is detected on the TI2 input pin.

Let's use TI2FP2 as trigger 1:

1. Select the proper TI2x source (internal or external) with the TI2SEL[3:0] bits in the TIMx_TISEL register.
2. Map TI2FP2 on TI2 by writing CC2S=01 in the TIMx_CCMR1 register.
3. TI2FP2 must detect a rising edge, write CC2P=0 and CC2NP='0' in the TIMx_CCER register.
4. Configure TI2FP2 as trigger for the slave mode controller (TRGI) by writing TS=00110 in the TIMx_SMCR register.
5. TI2FP2 is used to start the counter by writing SMS to '110 in the TIMx_SMCR register (trigger mode).

The OPM waveform is defined by writing the compare registers (taking into account the clock frequency and the counter prescaler).

- The t_{DELAY} is defined by the value written in the TIMx_CCR1 register.
- The t_{PULSE} is defined by the difference between the auto-reload value and the compare value (TIMx_ARR - TIMx_CCR1).
- Let's say one want to build a waveform with a transition from '0' to '1' when a compare match occurs and a transition from '1' to '0' when the counter reaches the auto-reload value. To do this PWM mode 2 must be enabled by writing OC1M=111 in the TIMx_CCMR1 register. Optionally the preload registers can be enabled by writing OC1PE=1 in the TIMx_CCMR1 register and ARPE in the TIMx_CR1 register. In this case one has to write the compare value in the TIMx_CCR1 register, the auto-reload value in the TIMx_ARR register, generate an update by setting the UG bit and wait for external trigger event on TI2. CC1P is written to '0' in this example.

In our example, the DIR and CMS bits in the TIMx_CR1 register should be low.

Since only 1 pulse (Single mode) is needed, a 1 must be written in the OPM bit in the TIMx_CR1 register to stop the counter at the next update event (when the counter rolls over from the auto-reload value back to 0). When OPM bit in the TIMx_CR1 register is set to '0', so the Repetitive Mode is selected.

Particular case: OCx fast enable:

In One-pulse mode, the edge detection on TIx input set the CEN bit which enables the counter. Then the comparison between the counter and the compare value makes the output toggle. But several clock cycles are needed for these operations and it limits the minimum delay t_{DELAY} min we can get.

If one wants to output a waveform with the minimum delay, the OCxFE bit can be set in the TIMx_CCMRx register. Then OCxRef (and OCx) is forced in response to the stimulus, without taking in account the comparison. Its new level is the same as if a compare match had occurred. OCxFE acts only if the channel is configured in PWM1 or PWM2 mode.

39.3.14 Retriggerable one pulse mode (OPM)

This mode allows the counter to be started in response to a stimulus and to generate a pulse with a programmable length, but with the following differences with Non-retriggerable one pulse mode described in [Section 39.3.13](#):

- The pulse starts as soon as the trigger occurs (no programmable delay)
- The pulse is extended if a new trigger occurs before the previous one is completed

The timer must be in Slave mode, with the bits SMS[3:0] = '1000' (Combined Reset + trigger mode) in the TIMx_SMCR register, and the OCxM[3:0] bits set to '1000' or '1001' for Retriggerable OPM mode 1 or 2.

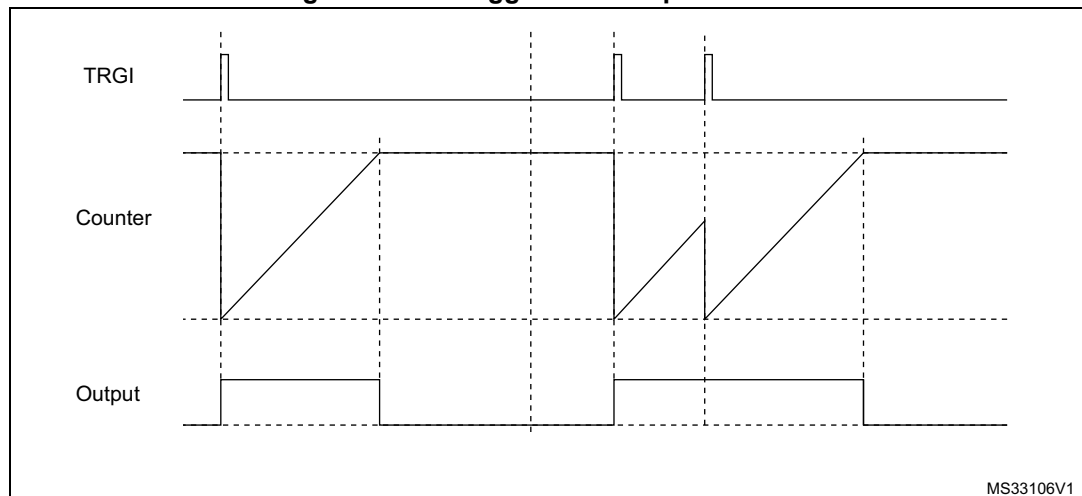
If the timer is configured in Up-counting mode, the corresponding CCRx must be set to 0 (the ARR register sets the pulse length). If the timer is configured in Down-counting mode CCRx must be above or equal to ARR.

Note: In retriggerable one pulse mode, the CCxIF flag is not significant.

The OCxM[3:0] and SMS[3:0] bit fields are split into two parts for compatibility reasons, the most significant bit is not contiguous with the 3 least significant ones.

This mode must not be used with center-aligned PWM modes. It is mandatory to have CMS[1:0] = 00 in TIMx_CR1.

Figure 440. Retriggerable one-pulse mode.



39.3.15 Encoder interface mode

To select Encoder Interface mode write SMS='001 in the TIMx_SMCR register if the counter is counting on TI2 edges only, SMS=010 if it is counting on TI1 edges only and SMS=011 if it is counting on both TI1 and TI2 edges.

Select the TI1 and TI2 polarity by programming the CC1P and CC2P bits in the TIMx_CCER register. CC1NP and CC2NP must be kept cleared. When needed, the input filter can be programmed as well. CC1NP and CC2NP must be kept low.

The two inputs TI1 and TI2 are used to interface to an incremental encoder. Refer to [Table 333](#). The counter is clocked by each valid transition on TI1FP1 or TI2FP2 (TI1 and TI2 after input filter and polarity selection, TI1FP1=TI1 if not filtered and not inverted, TI2FP2=TI2 if not filtered and not inverted) assuming that it is enabled (CEN bit in TIMx_CR1 register written to '1'). The sequence of transitions of the two inputs is evaluated and generates count pulses as well as the direction signal. Depending on the sequence the counter counts up or down, the DIR bit in the TIMx_CR1 register is modified by hardware accordingly. The DIR bit is calculated at each transition on any input (TI1 or TI2), whatever the counter is counting on TI1 only, TI2 only or both TI1 and TI2.

Encoder interface mode acts simply as an external clock with direction selection. This means that the counter just counts continuously between 0 and the auto-reload value in the TIMx_ARR register (0 to ARR or ARR down to 0 depending on the direction). So the TIMx_ARR must be configured before starting. In the same way, the capture, compare, prescaler, trigger output features continue to work as normal.

In this mode, the counter is modified automatically following the speed and the direction of the-quadrature encoder and its content, therefore, always represents the encoder's position. The count direction correspond to the rotation direction of the connected sensor. The table summarizes the possible combinations, assuming TI1 and TI2 do not switch at the same time.

Table 333. Counting direction versus encoder signals

Active edge	Level on opposite signal (TI1FP1 for TI2, TI2FP2 for TI1)	TI1FP1 signal		TI2FP2 signal	
		Rising	Falling	Rising	Falling
Counting on TI1 only	High	Down	Up	No Count	No Count
	Low	Up	Down	No Count	No Count
Counting on TI2 only	High	No Count	No Count	Up	Down
	Low	No Count	No Count	Down	Up
Counting on TI1 and TI2	High	Down	Up	Up	Down
	Low	Up	Down	Down	Up

An external incremental encoder can be connected directly to the MCU without external interface logic. However, comparators are normally be used to convert the encoder's differential outputs to digital signals. This greatly increases noise immunity. The third encoder output which indicate the mechanical zero position, may be connected to an external interrupt input and trigger a counter reset.

Figure 441 gives an example of counter operation, showing count signal generation and direction control. It also shows how input jitter is compensated where both edges are selected. This might occur if the sensor is positioned near to one of the switching points. For this example we assume that the configuration is the following:

- CC1S= 01 (TIMx_CCMR1 register, TI1FP1 mapped on TI1)
- CC2S= 01 (TIMx_CCMR2 register, TI2FP2 mapped on TI2)
- CC1P and CC1NP = '0' (TIMx_CCER register, TI1FP1 noninverted, TI1FP1=TI1)
- CC2P and CC2NP = '0' (TIMx_CCER register, TI2FP2 noninverted, TI2FP2=TI2)
- SMS= 011 (TIMx_SMCR register, both inputs are active on both rising and falling edges)
- CEN= 1 (TIMx_CR1 register, Counter is enabled)

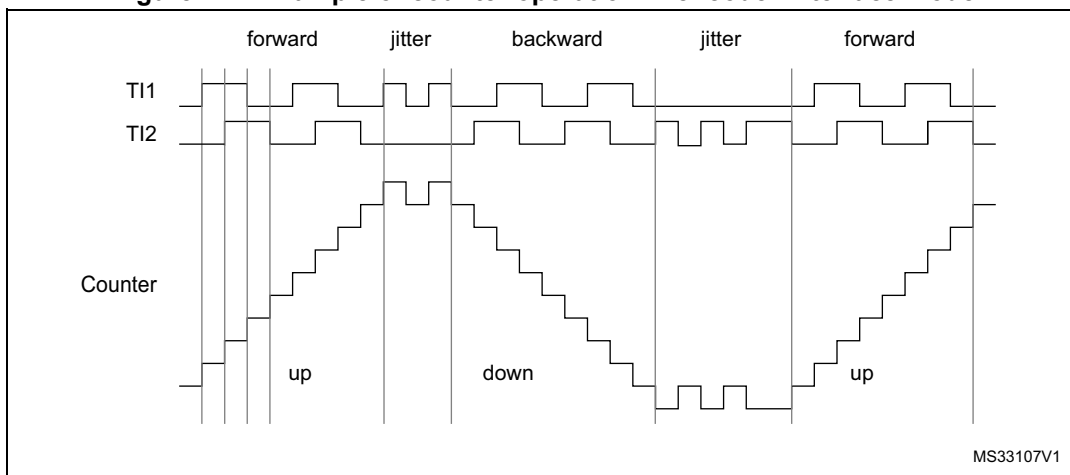
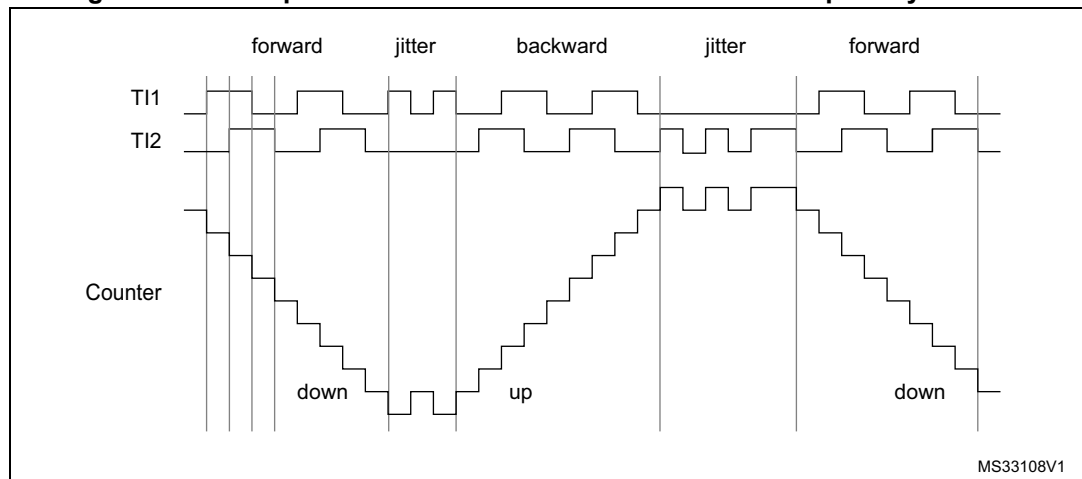
Figure 441. Example of counter operation in encoder interface mode

Figure 442 gives an example of counter behavior when TI1FP1 polarity is inverted (same configuration as above except CC1P=1).

Figure 442. Example of encoder interface mode with TI1FP1 polarity inverted

The timer, when configured in Encoder Interface mode provides information on the sensor's current position. Dynamic information can be obtained (speed, acceleration, deceleration) by measuring the period between two encoder events using a second timer configured in capture mode. The output of the encoder which indicates the mechanical zero can be used for this purpose. Depending on the time between two events, the counter can also be read at regular times. This can be done by latching the counter value into a third input capture register if available (then the capture signal must be periodic and can be generated by another timer). when available, it is also possible to read its value through a DMA request generated by a Real-Time clock.

39.3.16 UIF bit remapping

The IUFREMAP bit in the TIMx_CR1 register forces a continuous copy of the update interrupt flag (UIF) into bit 31 of the timer counter register's bit 31 (TIMxCNT[31]). This permits to atomically read both the counter value and a potential roll-over condition signaled by the UIFCPY flag. It eases the calculation of angular speed by avoiding race conditions caused, for instance, by a processing shared between a background task (counter reading) and an interrupt (update interrupt).

There is no latency between the UIF and UIFCPY flag assertions.

In 32-bit timer implementations, when the IUFREMAP bit is set, bit 31 of the counter is overwritten by the UIFCPY flag upon read access (the counter's most significant bit is only accessible in write mode).

39.3.17 Timer input XOR function

The TI1S bit in the TIM1xx_CR2 register, allows the input filter of channel 1 to be connected to the output of a XOR gate, combining the three input pins TIMx_CH1 to TIMx_CH3.

The XOR output can be used with all the timer input functions such as trigger or input capture.

An example of this feature used to interface Hall sensors is given in [Section 38.3.25: Interfacing with Hall sensors on page 1596](#).

39.3.18 Timers and external trigger synchronization

The TIMx Timers can be synchronized with an external trigger in several modes: Reset mode, Gated mode and Trigger mode.

Slave mode: Reset mode

The counter and its prescaler can be reinitialized in response to an event on a trigger input. Moreover, if the URS bit from the TIMx_CR1 register is low, an update event UEV is generated. Then all the preloaded registers (TIMx_ARR, TIMx_CCRx) are updated.

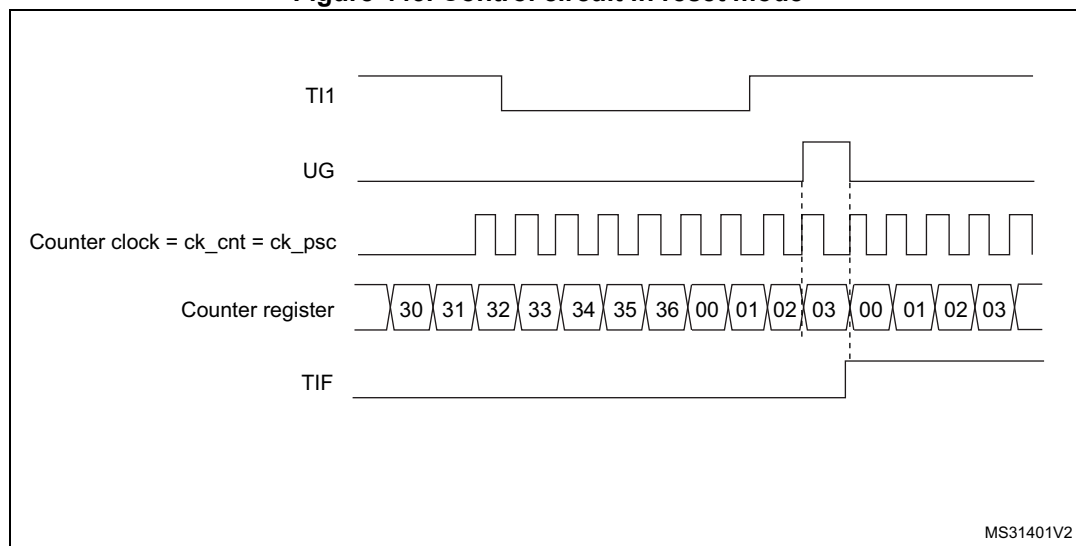
In the following example, the upcounter is cleared in response to a rising edge on TI1 input:

1. Configure the channel 1 to detect rising edges on TI1. Configure the input filter duration (in this example, we do not need any filter, so we keep IC1F=0000). The capture prescaler is not used for triggering, so it does not need to be configured. The CC1S bits select the input capture source only, CC1S = 01 in the TIMx_CCMR1 register. Write CC1P=0 and CC1NP=0 in TIMx_CCER register to validate the polarity (and detect rising edges only).
2. Configure the timer in reset mode by writing SMS=100 in TIMx_SMCR register. Select TI1 as the input source by writing TS=00101 in TIMx_SMCR register.
3. Start the counter by writing CEN=1 in the TIMx_CR1 register.

The counter starts counting on the internal clock, then behaves normally until TI1 rising edge. When TI1 rises, the counter is cleared and restarts from 0. In the meantime, the trigger flag is set (TIF bit in the TIMx_SR register) and an interrupt request, or a DMA request can be sent if enabled (depending on the TIE and TDE bits in TIMx_DIER register).

The following figure shows this behavior when the auto-reload register TIMx_ARR=0x36. The delay between the rising edge on TI1 and the actual reset of the counter is due to the resynchronization circuit on TI1 input.

Figure 443. Control circuit in reset mode



Slave mode: Gated mode

The counter can be enabled depending on the level of a selected input.

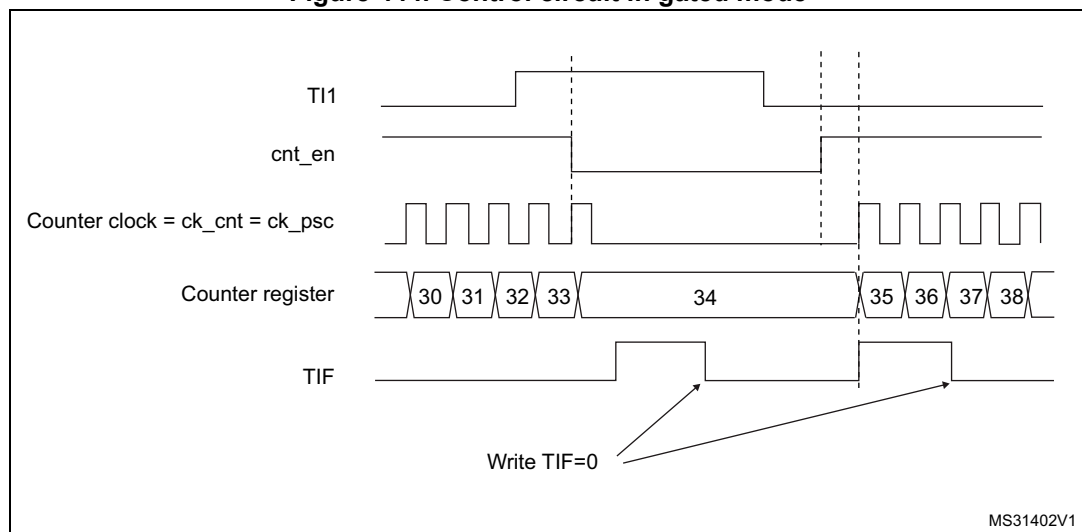
In the following example, the upcounter counts only when TI1 input is low:

1. Configure the channel 1 to detect low levels on TI1. Configure the input filter duration (in this example, we do not need any filter, so we keep IC1F=0000). The capture prescaler is not used for triggering, so it does not need to be configured. The CC1S bits select the input capture source only, CC1S=01 in TIMx_CCMR1 register. Write CC1P=1 and CC1NP=0 in TIMx_CCER register to validate the polarity (and detect low level only).
2. Configure the timer in gated mode by writing SMS=101 in TIMx_SMCR register. Select TI1 as the input source by writing TS=00101 in TIMx_SMCR register.
3. Enable the counter by writing CEN=1 in the TIMx_CR1 register (in gated mode, the counter doesn't start if CEN=0, whatever is the trigger input level).

The counter starts counting on the internal clock as long as TI1 is low and stops as soon as TI1 becomes high. The TIF flag in the TIMx_SR register is set both when the counter starts or stops.

The delay between the rising edge on TI1 and the actual stop of the counter is due to the resynchronization circuit on TI1 input.

Figure 444. Control circuit in gated mode



1. The configuration "CCxP=CCxNP=1" (detection of both rising and falling edges) does not have any effect in gated mode because gated mode acts on a level and not on an edge.

Note: *The configuration "CCxP=CCxNP=1" (detection of both rising and falling edges) does not have any effect in gated mode because gated mode acts on a level and not on an edge.*

Slave mode: Trigger mode

The counter can start in response to an event on a selected input.

In the following example, the upcounter starts in response to a rising edge on TI2 input:

1. Configure the channel 2 to detect rising edges on TI2. Configure the input filter duration (in this example, we do not need any filter, so we keep IC2F=0000). The capture prescaler is not used for triggering, so it does not need to be configured. CC2S bits are selecting the input capture source only, CC2S=01 in TIMx_CCMR1 register. Write

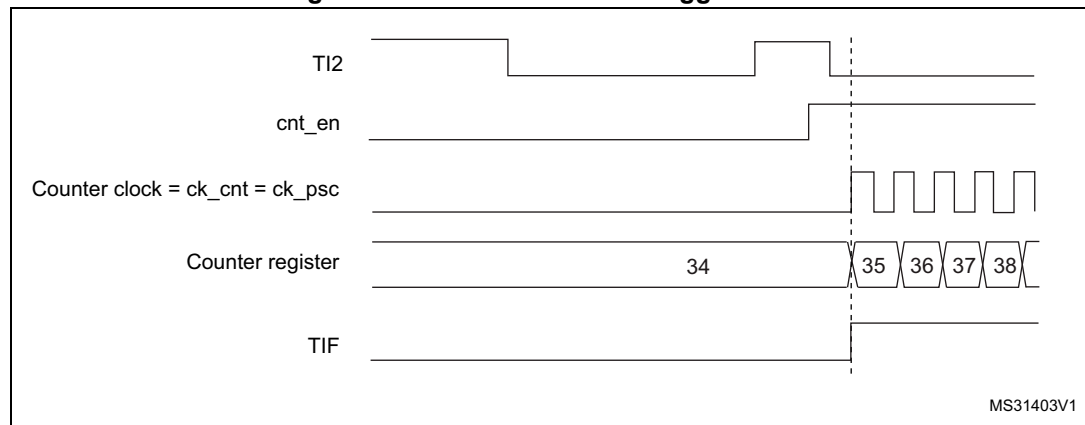
CC2P=1 and CC2NP=0 in TIMx_CCER register to validate the polarity (and detect low level only).

2. Configure the timer in trigger mode by writing SMS=110 in TIMx_SMCR register. Select TI2 as the input source by writing TS=00110 in TIMx_SMCR register.

When a rising edge occurs on TI2, the counter starts counting on the internal clock and the TIF flag is set.

The delay between the rising edge on TI2 and the actual start of the counter is due to the resynchronization circuit on TI2 input.

Figure 445. Control circuit in trigger mode



Slave mode: External Clock mode 2 + trigger mode

The external clock mode 2 can be used in addition to another slave mode (except external clock mode 1 and encoder mode). In this case, the ETR signal is used as external clock input, and another input can be selected as trigger input when operating in reset mode, gated mode or trigger mode. It is recommended not to select ETR as TRGI through the TS bits of TIMx_SMCR register.

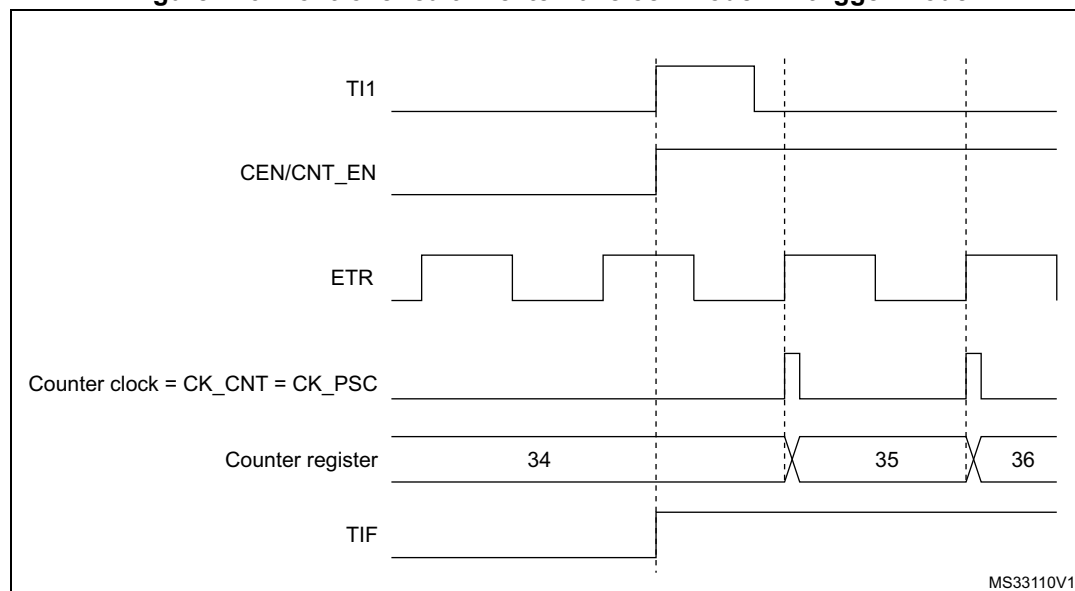
In the following example, the upcounter is incremented at each rising edge of the ETR signal as soon as a rising edge of TI1 occurs:

1. Configure the external trigger input circuit by programming the TIMx_SMCR register as follows:
 - ETF = 0000: no filter
 - ETPS=00: prescaler disabled
 - ETP=0: detection of rising edges on ETR and ECE=1 to enable the external clock mode 2.
2. Configure the channel 1 as follows, to detect rising edges on TI1:
 - IC1F=0000: no filter.
 - The capture prescaler is not used for triggering and does not need to be configured.
 - CC1S=01 in TIMx_CCMR1 register to select only the input capture source
 - CC1P=0 and CC1NP=0 in TIMx_CCER register to validate the polarity (and detect rising edge only).
3. Configure the timer in trigger mode by writing SMS=110 in TIMx_SMCR register. Select TI1 as the input source by writing TS=00101 in TIMx_SMCR register.

A rising edge on TI1 enables the counter and sets the TIF flag. The counter then counts on ETR rising edges.

The delay between the rising edge of the ETR signal and the actual reset of the counter is due to the resynchronization circuit on ETRP input.

Figure 446. Control circuit in external clock mode 2 + trigger mode

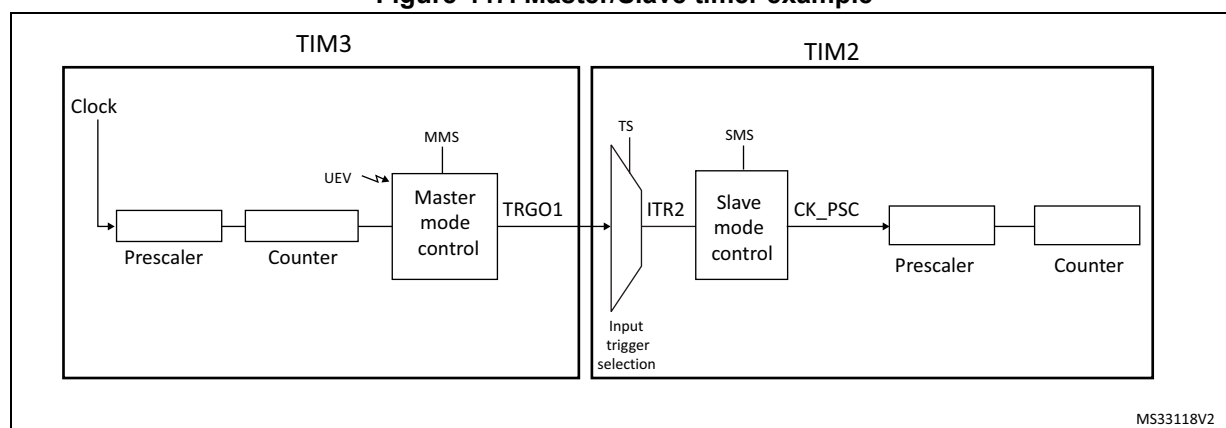


39.3.19 Timer synchronization

The TIMx timers are linked together internally for timer synchronization or chaining. When one Timer is configured in Master Mode, it can reset, start, stop or clock the counter of another Timer configured in Slave Mode.

Figure 447: Master/Slave timer example presents an overview of the trigger selection and the master mode selection blocks.

Figure 447. Master/Slave timer example



Using one timer as prescaler for another timer

For example, TIM3 can be configured to act as a prescaler for TIM2. Refer to [Figure 447](#). To do this:

1. Configure TIM3 in master mode so that it outputs a periodic trigger signal on each update event UEV. If MMS=010 is written in the TIM3_CR2 register, a rising edge is output on TRGO each time an update event is generated.
2. To connect the TRGO output of TIM3 to TIM2, TIM2 must be configured in slave mode using ITR2 as internal trigger. This is selected through the TS bits in the TIM2_SMCR register (writing TS=00010).
3. Then the slave mode controller must be put in external clock mode 1 (write SMS=111 in the TIM2_SMCR register). This causes TIM2 to be clocked by the rising edge of the periodic TIM3 trigger signal (which correspond to the TIM3 counter overflow).
4. Finally both timers must be enabled by setting their respective CEN bits (TIMx_CR1 register).

Note: If OCx is selected on TIM3 as the trigger output (MMS=1xx), its rising edge is used to clock the counter of TIM2.

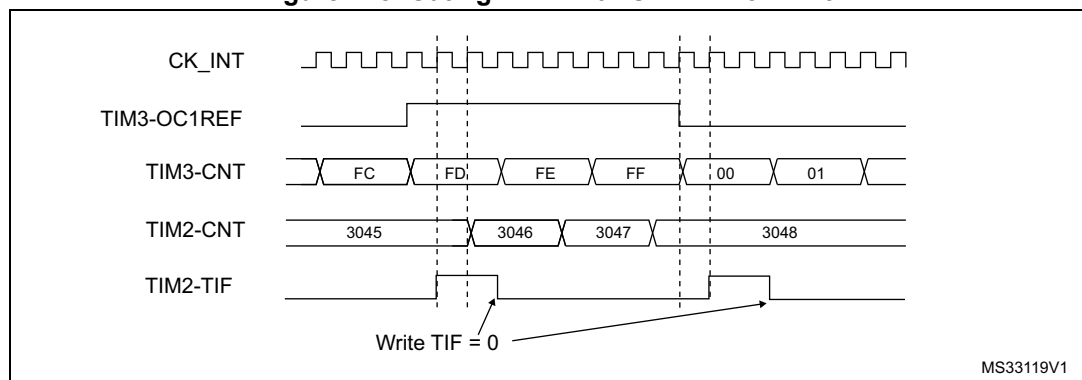
Using one timer to enable another timer

In this example, we control the enable of TIM2 with the output compare 1 of Timer 3. Refer to [Figure 447](#) for connections. TIM2 counts on the divided internal clock only when OC1REF of TIM3 is high. Both counter clock frequencies are divided by 3 by the prescaler compared to CK_INT ($f_{CK_CNT} = f_{CK_INT}/3$).

1. Configure TIM3 master mode to send its Output Compare 1 Reference (OC1REF) signal as trigger output (MMS=100 in the TIM3_CR2 register).
2. Configure the TIM3 OC1REF waveform (TIM3_CCMR1 register).
3. Configure TIM2 to get the input trigger from TIM3 (TS=00010 in the TIM2_SMCR register).
4. Configure TIM2 in gated mode (SMS=101 in TIM2_SMCR register).
5. Enable TIM2 by writing '1' in the CEN bit (TIM2_CR1 register).
6. Start TIM3 by writing '1' in the CEN bit (TIM3_CR1 register).

Note: The counter 2 clock is not synchronized with counter 1, this mode only affects the TIM2 counter enable signal.

Figure 448. Gating TIM2 with OC1REF of TIM3



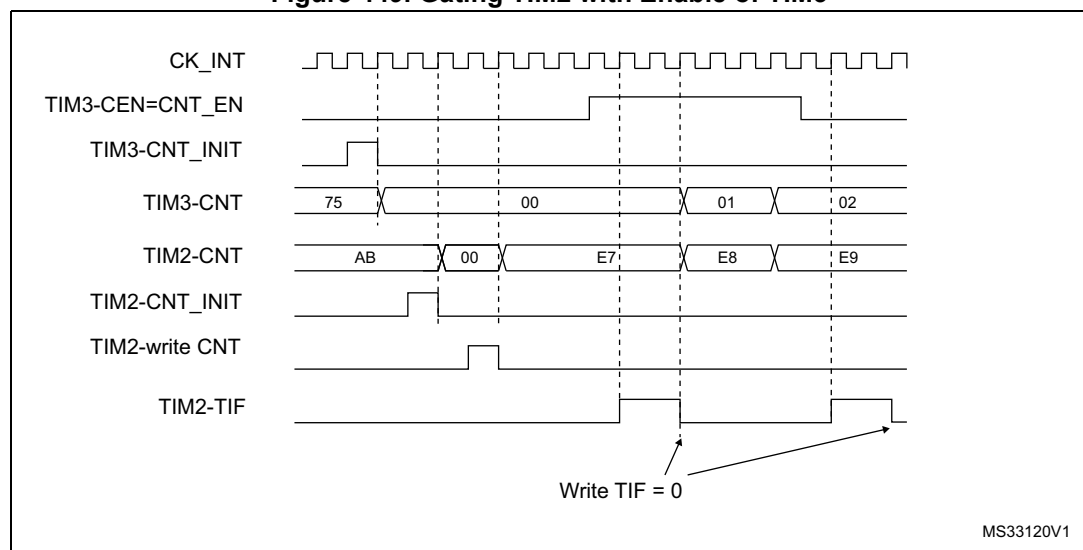
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In the example in [Figure 448](#), the TIM2 counter and prescaler are not initialized before being started. So they start counting from their current value. It is possible to start from a given value by resetting both timers before starting TIM3. Then any value can be written in the timer counters. The timers can easily be reset by software using the UG bit in the TIMx_EGR registers.

In the next example (refer to [Figure 449](#)), we synchronize TIM3 and TIM2. TIM3 is the master and starts from 0. TIM2 is the slave and starts from 0xE7. The prescaler ratio is the same for both timers. TIM2 stops when TIM3 is disabled by writing '0 to the CEN bit in the TIM3_CR1 register:

1. Configure TIM3 master mode to send its Output Compare 1 Reference (OC1REF) signal as trigger output (MMS=100 in the TIM3_CR2 register).
2. Configure the TIM3 OC1REF waveform (TIM3_CCMR1 register).
3. Configure TIM2 to get the input trigger from TIM3 (TS=00010 in the TIM2_SMCR register).
4. Configure TIM2 in gated mode (SMS=101 in TIM2_SMCR register).
5. Reset TIM3 by writing '1 in UG bit (TIM3_EGR register).
6. Reset TIM2 by writing '1 in UG bit (TIM2_EGR register).
7. Initialize TIM2 to 0xE7 by writing '0xE7' in the TIM2 counter (TIM2_CNT).
8. Enable TIM2 by writing '1 in the CEN bit (TIM2_CR1 register).
9. Start TIM3 by writing '1 in the CEN bit (TIM3_CR1 register).
10. Stop TIM3 by writing '0 in the CEN bit (TIM3_CR1 register).

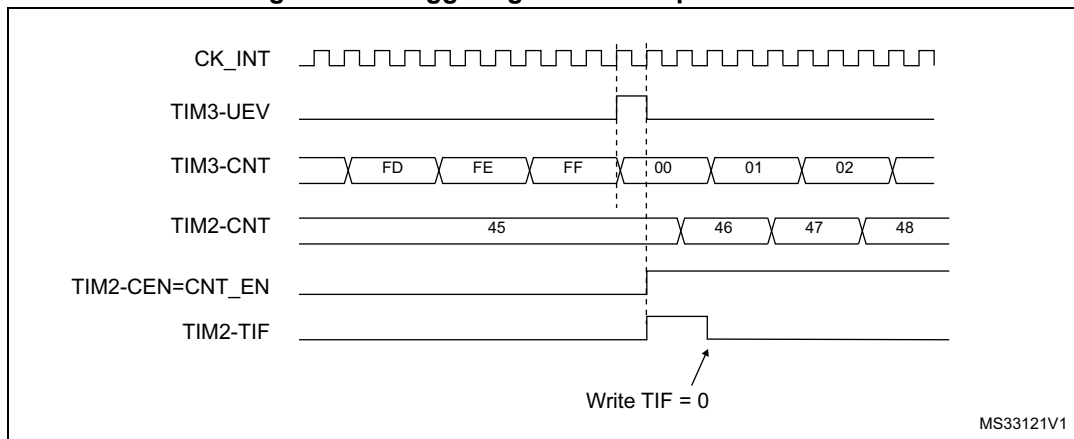
Figure 449. Gating TIM2 with Enable of TIM3



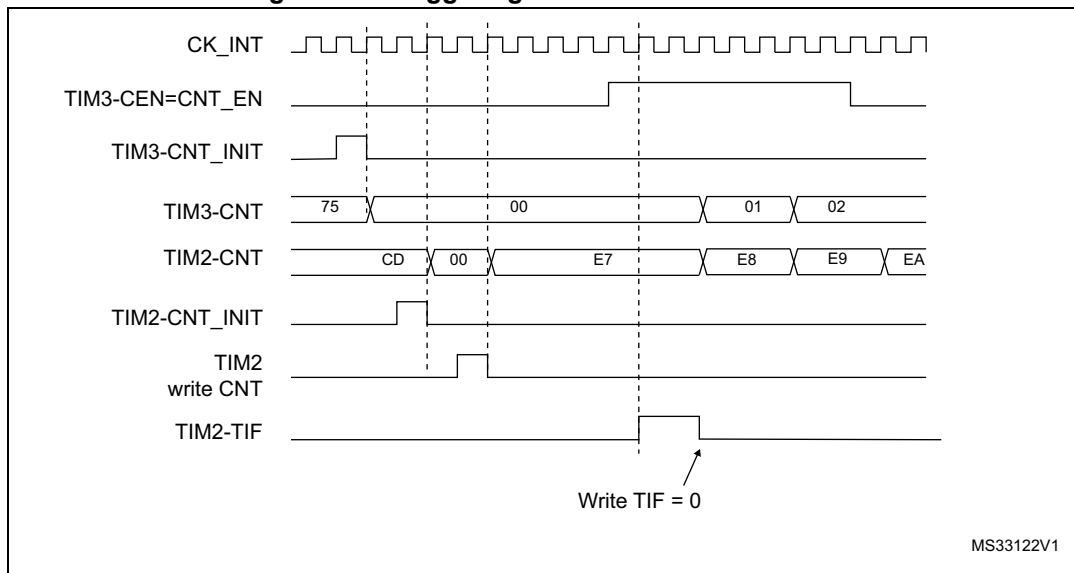
Using one timer to start another timer

In this example, we set the enable of Timer 2 with the update event of Timer 3. Refer to [Figure 447](#) for connections. Timer 2 starts counting from its current value (which can be non-zero) on the divided internal clock as soon as the update event is generated by Timer 1. When Timer 2 receives the trigger signal its CEN bit is automatically set and the counter counts until we write '0 to the CEN bit in the TIM2_CR1 register. Both counter clock frequencies are divided by 3 by the prescaler compared to CK_INT ($f_{CK_CNT} = f_{CK_INT}/3$).

1. Configure TIM3 master mode to send its Update Event (UEV) as trigger output (MMS=010 in the TIM3_CR2 register).
2. Configure the TIM3 period (TIM3_ARR registers).
3. Configure TIM2 to get the input trigger from TIM3 (TS=00010 in the TIM2_SMCR register).
4. Configure TIM2 in trigger mode (SMS=110 in TIM2_SMCR register).
5. Start TIM3 by writing '1' in the CEN bit (TIM3_CR1 register).

Figure 450. Triggering TIM2 with update of TIM3

As in the previous example, both counters can be initialized before starting counting. [Figure 451](#) shows the behavior with the same configuration as in [Figure 450](#) but in trigger mode instead of gated mode (SMS=110 in the TIM2_SMCR register).

Figure 451. Triggering TIM2 with Enable of TIM3

Starting 2 timers synchronously in response to an external trigger

In this example, we set the enable of TIM3 when its TI1 input rises, and the enable of TIM2 with the enable of TIM3. Refer to [Figure 447](#) for connections. To ensure the counters are

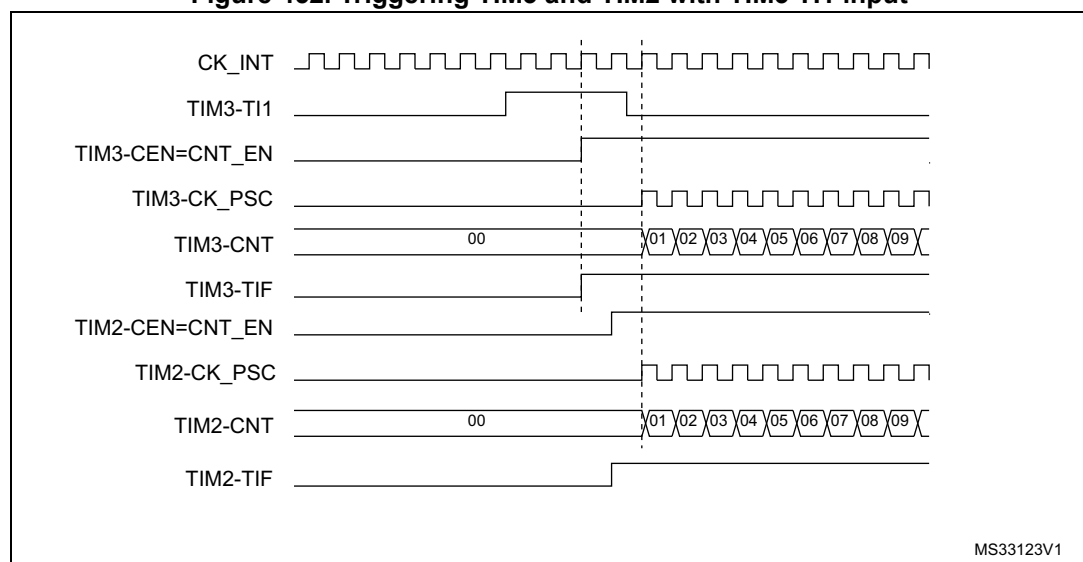
aligned, TIM3 must be configured in Master/Slave mode (slave with respect to TI1, master with respect to TIM2):

1. Configure TIM3 master mode to send its Enable as trigger output (MMS=001 in the TIM3_CR2 register).
2. Configure TIM3 slave mode to get the input trigger from TI1 (TS=00100 in the TIM3_SMCR register).
3. Configure TIM3 in trigger mode (SMS=110 in the TIM3_SMCR register).
4. Configure the TIM3 in Master/Slave mode by writing MSM=1 (TIM3_SMCR register).
5. Configure TIM2 to get the input trigger from TIM3 (TS=00000 in the TIM2_SMCR register).
6. Configure TIM2 in trigger mode (SMS=110 in the TIM2_SMCR register).

When a rising edge occurs on TI1 (TIM3), both counters starts counting synchronously on the internal clock and both TIF flags are set.

Note: *In this example both timers are initialized before starting (by setting their respective UG bits). Both counters starts from 0, but an offset can easily be inserted between them by writing any of the counter registers (TIMx_CNT). One can see that the master/slave mode insert a delay between CNT_EN and CK_PSC on TIM3.*

Figure 452. Triggering TIM3 and TIM2 with TIM3 TI1 input



Note: *The clock of the slave peripherals (timer, ADC, ...) receiving the TRGO or the TRGO2 signals must be enabled prior to receive events from the master timer, and the clock frequency (prescaler) must not be changed on-the-fly while triggers are received from the master timer.*

39.3.20 DMA burst mode

The TIMx timers have the capability to generate multiple DMA requests upon a single event. The main purpose is to be able to re-program part of the timer multiple times without software overhead, but it can also be used to read several registers in a row, at regular intervals.

The DMA controller destination is unique and must point to the virtual register TIMx_DMAR. On a given timer event, the timer launches a sequence of DMA requests (burst). Each write into the TIMx_DMAR register is actually redirected to one of the timer registers.

The DBL[4:0] bits in the TIMx_DCR register set the DMA burst length. The timer recognizes a burst transfer when a read or a write access is done to the TIMx_DMAR address, i.e. the number of transfers (either in half-words or in bytes).

The DBA[4:0] bits in the TIMx_DCR registers define the DMA base address for DMA transfers (when read/write access are done through the TIMx_DMAR address). DBA is defined as an offset starting from the address of the TIMx_CR1 register:

Example:

00000: TIMx_CR1

00001: TIMx_CR2

00010: TIMx_SMCR

As an example, the timer DMA burst feature is used to update the contents of the CCRx registers (x = 2, 3, 4) upon an update event, with the DMA transferring half words into the CCRx registers.

This is done in the following steps:

1. Configure the corresponding DMA channel as follows:
 - DMA channel peripheral address is the DMAR register address
 - DMA channel memory address is the address of the buffer in the RAM containing the data to be transferred by DMA into CCRx registers.
 - Number of data to transfer = 3 (See note below).
 - Circular mode disabled.
2. Configure the DCR register by configuring the DBA and DBL bit fields as follows:
DBL = 3 transfers, DBA = 0xE.
3. Enable the TIMx update DMA request (set the UDE bit in the DIER register).
4. Enable TIMx
5. Enable the DMA channel

This example is for the case where every CCRx register has to be updated once. If every CCRx register is to be updated twice for example, the number of data to transfer should be 6. Let's take the example of a buffer in the RAM containing data1, data2, data3, data4, data5 and data6. The data is transferred to the CCRx registers as follows: on the first update DMA request, data1 is transferred to CCR2, data2 is transferred to CCR3, data3 is transferred to CCR4 and on the second update DMA request, data4 is transferred to CCR2, data5 is transferred to CCR3 and data6 is transferred to CCR4.

Note: A null value can be written to the reserved registers.

39.3.21 Debug mode

When the microcontroller enters debug mode (Cortex[®]-M7 with FPU core halted), the TIMx counter either continues to work normally or stops, depending on TIMx configuration bit in DBGMCU module. For more details, refer to [Section 60.5.8: Microcontroller debug unit \(DBGMCU\)](#).

For safety purposes, when the counter is stopped (TIMx = 1 in DBGMCU_APB1FZ2), the outputs are disabled.

39.4 TIM2/TIM3/TIM4/TIM5 registers

Refer to [Section 1.2](#) for a list of abbreviations used in register descriptions.

The peripheral registers can be accessed by half-words (16-bit) or words (32-bit).

39.4.1 TIMx control register 1 (TIMx_CR1)(x = 2 to 5)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	UIFRE MAP	Res.	CKD[1:0]		ARPE	CMS[1:0]		DIR	OPM	URS	UDIS	CEN
				rw		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:12 Reserved, must be kept at reset value.

Bit 11 **UIFREMAP**: UIF status bit remapping

0: No remapping. UIF status bit is not copied to TIMx_CNT register bit 31.

1: Remapping enabled. UIF status bit is copied to TIMx_CNT register bit 31.

Bit 10 Reserved, must be kept at reset value.

Bits 9:8 **CKD[1:0]**: Clock division

This bit-field indicates the division ratio between the timer clock (CK_INT) frequency and sampling clock used by the digital filters (ETR, Tlx),

00: $t_{DTS} = t_{CK_INT}$

01: $t_{DTS} = 2 \times t_{CK_INT}$

10: $t_{DTS} = 4 \times t_{CK_INT}$

11: Reserved

Bit 7 **ARPE**: Auto-reload preload enable

0: TIMx_ARR register is not buffered

1: TIMx_ARR register is buffered

Bits 6:5 **CMS[1:0]**: Center-aligned mode selection

00: Edge-aligned mode. The counter counts up or down depending on the direction bit (DIR).

01: Center-aligned mode 1. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set only when the counter is counting down.

10: Center-aligned mode 2. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set only when the counter is counting up.

11: Center-aligned mode 3. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set both when the counter is counting up or down.

Note: It is not allowed to switch from edge-aligned mode to center-aligned mode as long as the counter is enabled (CEN=1)

Bit 4 **DIR**: Direction

0: Counter used as upcounter

1: Counter used as downcounter

Note: This bit is read only when the timer is configured in Center-aligned mode or Encoder mode.

Bit 3 **OPM**: One-pulse mode

0: Counter is not stopped at update event

1: Counter stops counting at the next update event (clearing the bit CEN)

Bit 2 **URS**: Update request source

This bit is set and cleared by software to select the UEV event sources.

0: Any of the following events generate an update interrupt or DMA request if enabled.

These events can be:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

1: Only counter overflow/underflow generates an update interrupt or DMA request if enabled.

Bit 1 **UDIS**: Update disable

This bit is set and cleared by software to enable/disable UEV event generation.

0: UEV enabled. The Update (UEV) event is generated by one of the following events:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

Buffered registers are then loaded with their preload values.

1: UEV disabled. The Update event is not generated, shadow registers keep their value (ARR, PSC, CCRx). However the counter and the prescaler are reinitialized if the UG bit is set or if a hardware reset is received from the slave mode controller.

Bit 0 **CEN**: Counter enable

0: Counter disabled

1: Counter enabled

Note: External clock, gated mode and encoder mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.

CEN is cleared automatically in one-pulse mode, when an update event occurs.

39.4.2 TIMx control register 2 (TIMx_CR2)(x = 2 to 5)

Address offset: 0x04

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TI1S	MMS[2:0]			CCDS	Res.	Res.	Res.
								rw	rw	rw	rw	rw			

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **TI1S**: TI1 selection

0: The TIMx_CH1 pin is connected to TI1 input

1: The TIMx_CH1, CH2 and CH3 pins are connected to the TI1 input (XOR combination)

See also [Section 38.3.25: Interfacing with Hall sensors on page 1596](#)

Bits 6:4 **MMS[2:0]**: Master mode selection

These bits permit to select the information to be sent in master mode to slave timers for synchronization (TRGO). The combination is as follows:

000: **Reset** - the UG bit from the TIMx_EGR register is used as trigger output (TRGO). If the reset is generated by the trigger input (slave mode controller configured in reset mode) then the signal on TRGO is delayed compared to the actual reset.

001: **Enable** - the Counter enable signal, CNT_EN, is used as trigger output (TRGO). It is useful to start several timers at the same time or to control a window in which a slave timer is enabled. The Counter Enable signal is generated by a logic AND between CEN control bit and the trigger input when configured in gated mode.

When the Counter Enable signal is controlled by the trigger input, there is a delay on TRGO, except if the master/slave mode is selected (see the MSM bit description in TIMx_SMCR register).

010: **Update** - The update event is selected as trigger output (TRGO). For instance a master timer can then be used as a prescaler for a slave timer.

011: **Compare Pulse** - The trigger output send a positive pulse when the CC1IF flag is to be set (even if it was already high), as soon as a capture or a compare match occurred. (TRGO)

100: **Compare** - OC1REF signal is used as trigger output (TRGO)

101: **Compare** - OC2REF signal is used as trigger output (TRGO)

110: **Compare** - OC3REF signal is used as trigger output (TRGO)

111: **Compare** - OC4REF signal is used as trigger output (TRGO)

Note: The clock of the slave timer or ADC must be enabled prior to receive events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

Bit 3 **CCDS**: Capture/compare DMA selection

0: CCx DMA request sent when CCx event occurs

1: CCx DMA requests sent when update event occurs

Bits 2:0 Reserved, must be kept at reset value.

39.4.3 TIMx slave mode control register (TIMx_SMCR)(x = 2 to 5)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TS[4:3]		Res.	Res.	Res.	SMS[3]
										rw	rw				rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETP	ECE	ETPS[1:0]		ETF[3:0]				MSM	TS[2:0]			Res.	SMS[2:0]		
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw

Bits 31:22 Reserved, must be kept at reset value.

Bits 19:17 Reserved, must be kept at reset value.

Bit 15 **ETP**: External trigger polarity

This bit selects whether ETR or $\overline{\text{ETR}}$ is used for trigger operations

0: ETR is non-inverted, active at high level or rising edge

1: ETR is inverted, active at low level or falling edge

Bit 14 **ECE**: External clock enable

This bit enables External clock mode 2.

0: External clock mode 2 disabled

1: External clock mode 2 enabled. The counter is clocked by any active edge on the ETRF signal.

Note: Setting the ECE bit has the same effect as selecting external clock mode 1 with TRGI connected to ETRF (SMS=111 and TS=00111).

It is possible to simultaneously use external clock mode 2 with the following slave modes: reset mode, gated mode and trigger mode. Nevertheless, TRGI must not be connected to ETRF in this case (TS bits must not be 00111).

If external clock mode 1 and external clock mode 2 are enabled at the same time, the external clock input is ETRF.

Bits 13:12 **ETPS[1:0]**: External trigger prescaler

External trigger signal ETRP frequency must be at most 1/4 of CK_INT frequency. A prescaler can be enabled to reduce ETRP frequency. It is useful when inputting fast external clocks.

00: Prescaler OFF

01: ETRP frequency divided by 2

10: ETRP frequency divided by 4

11: ETRP frequency divided by 8

Bits 11:8 **ETF[3:0]**: External trigger filter

This bit-field then defines the frequency used to sample ETRP signal and the length of the digital filter applied to ETRP. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, $N=2$

0010: $f_{SAMPLING}=f_{CK_INT}$, $N=4$

0011: $f_{SAMPLING}=f_{CK_INT}$, $N=8$

0100: $f_{SAMPLING}=f_{DTS}/2$, $N=6$

0101: $f_{SAMPLING}=f_{DTS}/2$, $N=8$

0110: $f_{SAMPLING}=f_{DTS}/4$, $N=6$

0111: $f_{SAMPLING}=f_{DTS}/4$, $N=8$

1000: $f_{SAMPLING}=f_{DTS}/8$, $N=6$

1001: $f_{SAMPLING}=f_{DTS}/8$, $N=8$

1010: $f_{SAMPLING}=f_{DTS}/16$, $N=5$

1011: $f_{SAMPLING}=f_{DTS}/16$, $N=6$

1100: $f_{SAMPLING}=f_{DTS}/16$, $N=8$

1101: $f_{SAMPLING}=f_{DTS}/32$, $N=5$

1110: $f_{SAMPLING}=f_{DTS}/32$, $N=6$

1111: $f_{SAMPLING}=f_{DTS}/32$, $N=8$

Bit 7 **MSM**: Master/Slave mode

0: No action

1: The effect of an event on the trigger input (TRGI) is delayed to allow a perfect synchronization between the current timer and its slaves (through TRGO). It is useful if we want to synchronize several timers on a single external event.

Bits 21, 20, 6, 5, 4 **TS[4:0]**: Trigger selection

This bit-field selects the trigger input to be used to synchronize the counter.

00000: Internal Trigger 0 (ITR0)
 00001: Internal Trigger 1 (ITR1)
 00010: Internal Trigger 2 (ITR2)
 00011: Internal Trigger 3 (ITR3)
 00100: TI1 Edge Detector (TI1F_ED)
 00101: Filtered Timer Input 1 (TI1FP1)
 00110: Filtered Timer Input 2 (TI2FP2)
 00111: External Trigger input (ETRF)
 01000: Internal Trigger 4 (ITR4)
 01001: Internal Trigger 5 (ITR5)
 01010: Internal Trigger 6 (ITR6)
 01011: Internal Trigger 7 (ITR7)
 01100: Internal Trigger 8 (ITR8)
 Others: Reserved

See [Table 334: TIMx internal trigger connection on page 1700](#) for more details on ITRx meaning for each Timer.

Note: These bits must be changed only when they are not used (e.g. when SMS=000) to avoid wrong edge detections at the transition.

Bit 3 Reserved, must be kept at reset value.

Bits 16, 2, 1, 0 **SMS[3:0]**: Slave mode selection

When external signals are selected the active edge of the trigger signal (TRGI) is linked to the polarity selected on the external input (see Input Control register and Control Register description).

0000: Slave mode disabled - if CEN = '1 then the prescaler is clocked directly by the internal clock.
 0001: Encoder mode 1 - Counter counts up/down on TI1FP1 edge depending on TI2FP2 level.
 0010: Encoder mode 2 - Counter counts up/down on TI2FP2 edge depending on TI1FP1 level.
 0011: Encoder mode 3 - Counter counts up/down on both TI1FP1 and TI2FP2 edges depending on the level of the other input.
 0100: Reset Mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter and generates an update of the registers.
 0101: Gated Mode - The counter clock is enabled when the trigger input (TRGI) is high. The counter stops (but is not reset) as soon as the trigger becomes low. Both start and stop of the counter are controlled.
 0110: Trigger Mode - The counter starts at a rising edge of the trigger TRGI (but it is not reset). Only the start of the counter is controlled.
 0111: External Clock Mode 1 - Rising edges of the selected trigger (TRGI) clock the counter.
 1000: Combined reset + trigger mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter, generates an update of the registers and starts the counter.

Note: The gated mode must not be used if TI1F_ED is selected as the trigger input (TS=00100). Indeed, TI1F_ED outputs 1 pulse for each transition on TI1F, whereas the gated mode checks the level of the trigger signal.

Note: The clock of the slave peripherals (timer, ADC, ...) receiving the TRGO or the TRGO2 signals must be enabled prior to receive events from the master timer, and the clock frequency (prescaler) must not be changed on-the-fly while triggers are received from the master timer.

Table 334. TIMx internal trigger connection

Slave TIM	ITR0	ITR1	ITR2	ITR3	ITR4	ITR5	ITR6	ITR7	ITR8
TIM2	TIM1	TIM8	TIM3	TIM4	ETH PPS	USB1 OTG_HS_SOF	USB2 OTG_FS_SOF	-	-
TIM3	TIM1	TIM2	TIM15	TIM4	ETH PPS	-	-	-	-
TIM4	TIM1	TIM2	TIM3	TIM8	-	-	-	-	-
TIM5	TIM1	TIM8	TIM3	TIM4	-	-	fdcan1_soc	USB1 OTG_HS_SOF	USB2 OTG_FS_SOF

39.4.4 TIMx DMA/Interrupt enable register (TIMx_DIER)(x = 2 to 5)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	TDE	Res.	CC4DE	CC3DE	CC2DE	CC1DE	UDE	Res.	TIE	Res.	CC4IE	CC3IE	CC2IE	CC1IE	UIE
	rw		rw	rw	rw	rw	rw		rw		rw	rw	rw	rw	rw

Bit 15 Reserved, must be kept at reset value.

Bit 14 **TDE**: Trigger DMA request enable
 0: Trigger DMA request disabled.
 1: Trigger DMA request enabled.

Bit 13 Reserved, must be kept at reset value.

Bit 12 **CC4DE**: Capture/Compare 4 DMA request enable
 0: CC4 DMA request disabled.
 1: CC4 DMA request enabled.

Bit 11 **CC3DE**: Capture/Compare 3 DMA request enable
 0: CC3 DMA request disabled.
 1: CC3 DMA request enabled.

Bit 10 **CC2DE**: Capture/Compare 2 DMA request enable
 0: CC2 DMA request disabled.
 1: CC2 DMA request enabled.

Bit 9 **CC1DE**: Capture/Compare 1 DMA request enable
 0: CC1 DMA request disabled.
 1: CC1 DMA request enabled.

Bit 8 **UDE**: Update DMA request enable
 0: Update DMA request disabled.
 1: Update DMA request enabled.

Bit 7 Reserved, must be kept at reset value.

- Bit 6 **TIE**: Trigger interrupt enable
 0: Trigger interrupt disabled.
 1: Trigger interrupt enabled.
- Bit 5 Reserved, must be kept at reset value.
- Bit 4 **CC4IE**: Capture/Compare 4 interrupt enable
 0: CC4 interrupt disabled.
 1: CC4 interrupt enabled.
- Bit 3 **CC3IE**: Capture/Compare 3 interrupt enable
 0: CC3 interrupt disabled.
 1: CC3 interrupt enabled.
- Bit 2 **CC2IE**: Capture/Compare 2 interrupt enable
 0: CC2 interrupt disabled.
 1: CC2 interrupt enabled.
- Bit 1 **CC1IE**: Capture/Compare 1 interrupt enable
 0: CC1 interrupt disabled.
 1: CC1 interrupt enabled.
- Bit 0 **UIE**: Update interrupt enable
 0: Update interrupt disabled.
 1: Update interrupt enabled.

39.4.5 TIMx status register (TIMx_SR)(x = 2 to 5)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	CC4OF	CC3OF	CC2OF	CC1OF	Res.	Res.	TIF	Res.	CC4IF	CC3IF	CC2IF	CC1IF	UIF
			rc_w0	rc_w0	rc_w0	rc_w0			rc_w0		rc_w0	rc_w0	rc_w0	rc_w0	rc_w0

Bits 15:13 Reserved, must be kept at reset value.

Bit 12 **CC4OF**: Capture/Compare 4 overcapture flag
 refer to CC1OF description

Bit 11 **CC3OF**: Capture/Compare 3 overcapture flag
 refer to CC1OF description

Bit 10 **CC2OF**: Capture/compare 2 overcapture flag
 refer to CC1OF description

Bit 9 **CC1OF**: Capture/Compare 1 overcapture flag
 This flag is set by hardware only when the corresponding channel is configured in input capture mode. It is cleared by software by writing it to '0'.
 0: No overcapture has been detected.
 1: The counter value has been captured in TIMx_CCR1 register while CC1IF flag was already set

Bits 8:7 Reserved, must be kept at reset value.

Bit 6 **TIF**: Trigger interrupt flag

This flag is set by hardware on the TRG trigger event (active edge detected on TRGI input when the slave mode controller is enabled in all modes but gated mode. It is set when the counter starts or stops when gated mode is selected. It is cleared by software.

0: No trigger event occurred.

1: Trigger interrupt pending.

Bit 5 Reserved, must be kept at reset value.

Bit 4 **CC4IF**: Capture/Compare 4 interrupt flag

Refer to CC1IF description

Bit 3 **CC3IF**: Capture/Compare 3 interrupt flag

Refer to CC1IF description

Bit 2 **CC2IF**: Capture/Compare 2 interrupt flag

Refer to CC1IF description

Bit 1 **CC1IF**: Capture/compare 1 interrupt flag

This flag is set by hardware. It is cleared by software (input capture or output compare mode) or by reading the TIMx_CCR1 register (input capture mode only).

0: No compare match / No input capture occurred

1: A compare match or an input capture occurred

If channel CC1 is configured as output: this flag is set when the content of the counter TIMx_CNT matches the content of the TIMx_CCR1 register. When the content of TIMx_CCR1 is greater than the content of TIMx_ARR, the CC1IF bit goes high on the counter overflow (in up-counting and up/down-counting modes) or underflow (in down-counting mode). There are 3 possible options for flag setting in center-aligned mode, refer to the CMS bits in the TIMx_CR1 register for the full description.

If channel CC1 is configured as input: this bit is set when counter value has been captured in TIMx_CCR1 register (an edge has been detected on IC1, as per the edge sensitivity defined with the CC1P and CC1NP bits setting, in TIMx_CCER).

Bit 0 **UIF**: Update interrupt flag

This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

At overflow or underflow (for TIM2 to TIM4) and if UDIS=0 in the TIMx_CR1 register.

When CNT is reinitialized by software using the UG bit in TIMx_EGR register, if URS=0 and UDIS=0 in the TIMx_CR1 register.

When CNT is reinitialized by a trigger event (refer to the synchro control register description), if URS=0 and UDIS=0 in the TIMx_CR1 register.

39.4.6 TIMx event generation register (TIMx_EGR)(x = 2 to 5)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TG	Res.	CC4G	CC3G	CC2G	CC1G	UG
									w		w	w	w	w	w

Bits 15:7 Reserved, must be kept at reset value.

Bit 6 **TG**: Trigger generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: The TIF flag is set in TIMx_SR register. Related interrupt or DMA transfer can occur if enabled.

Bit 5 Reserved, must be kept at reset value.

Bit 4 **CC4G**: Capture/compare 4 generation

Refer to CC1G description

Bit 3 **CC3G**: Capture/compare 3 generation

Refer to CC1G description

Bit 2 **CC2G**: Capture/compare 2 generation

Refer to CC1G description

Bit 1 **CC1G**: Capture/compare 1 generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: A capture/compare event is generated on channel 1:

If channel CC1 is configured as output:

CC1IF flag is set, Corresponding interrupt or DMA request is sent if enabled.

If channel CC1 is configured as input:

The current value of the counter is captured in TIMx_CCR1 register. The CC1IF flag is set, the corresponding interrupt or DMA request is sent if enabled. The CC1OF flag is set if the CC1IF flag was already high.

Bit 0 **UG**: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action

1: Re-initialize the counter and generates an update of the registers. Note that the prescaler counter is cleared too (anyway the prescaler ratio is not affected). The counter is cleared if the center-aligned mode is selected or if DIR=0 (upcounting), else it takes the auto-reload value (TIMx_ARR) if DIR=1 (downcounting).

39.4.7 TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1) (x = 2 to 5)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for input capture mode (this section) or for output compare mode (next section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode.

Input capture mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IC2F[3:0]				IC2PSC[1:0]		CC2S[1:0]		IC1F[3:0]				IC1PSC[1:0]		CC1S[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:12 **IC2F[3:0]**: Input capture 2 filter

Bits 11:10 **IC2PSC[1:0]**: Input capture 2 prescaler

Bits 9:8 **CC2S[1:0]**: Capture/compare 2 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output.

01: CC2 channel is configured as input, IC2 is mapped on TI2.

10: CC2 channel is configured as input, IC2 is mapped on TI1.

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC2S bits are writable only when the channel is OFF (CC2E = 0 in TIMx_CCER).

Bits 7:4 **IC1F[3:0]**: Input capture 1 filter

This bit-field defines the frequency used to sample TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, N=2

0010: $f_{SAMPLING}=f_{CK_INT}$, N=4

0011: $f_{SAMPLING}=f_{CK_INT}$, N=8

0100: $f_{SAMPLING}=f_{DTS}/2$, N=6

0101: $f_{SAMPLING}=f_{DTS}/2$, N=8

0110: $f_{SAMPLING}=f_{DTS}/4$, N=6

0111: $f_{SAMPLING}=f_{DTS}/4$, N=8

1000: $f_{SAMPLING}=f_{DTS}/8$, N=6

1001: $f_{SAMPLING}=f_{DTS}/8$, N=8

1010: $f_{SAMPLING}=f_{DTS}/16$, N=5

1011: $f_{SAMPLING}=f_{DTS}/16$, N=6

1100: $f_{SAMPLING}=f_{DTS}/16$, N=8

1101: $f_{SAMPLING}=f_{DTS}/32$, N=5

1110: $f_{SAMPLING}=f_{DTS}/32$, N=6

1111: $f_{SAMPLING}=f_{DTS}/32$, N=8

Bits 3:2 **IC1PSC[1:0]**: Input capture 1 prescaler

This bit-field defines the ratio of the prescaler acting on CC1 input (IC1). The prescaler is reset as soon as CC1E=0 (TIMx_CCER register).

00: no prescaler, capture is done each time an edge is detected on the capture input

01: capture is done once every 2 events

10: capture is done once every 4 events

11: capture is done once every 8 events

Bits 1:0 **CC1S[1:0]**: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: CC1 channel is configured as input, IC1 is mapped on TI2

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

39.4.8 TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1) (x = 2 to 5)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for output compare mode (this section) or for input capture mode (previous section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode.

Output compare mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC2M[3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M[3]
							rw								rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2CE	OC2M[2:0]			OC2PE	OC2FE	CC2S[1:0]		OC1CE	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bits 23:17 Reserved, must be kept at reset value.

Bit 15 **OC2CE**: Output compare 2 clear enable

Bits 24, 14:12 **OC2M[3:0]**: Output compare 2 mode
refer to OC1M description on bits 6:4

Bit 11 **OC2PE**: Output compare 2 preload enable

Bit 10 **OC2FE**: Output compare 2 fast enable

Bits 9:8 **CC2S[1:0]**: Capture/Compare 2 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output

01: CC2 channel is configured as input, IC2 is mapped on TI2

10: CC2 channel is configured as input, IC2 is mapped on TI1

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode is working only if an internal trigger input is selected through the TS bit (TIMx_SMCR register)

Note: CC2S bits are writable only when the channel is OFF (CC2E = 0 in TIMx_CCER).

Bit 7 **OC1CE**: Output compare 1 clear enable

0: OC1Ref is not affected by the ETRF input

1: OC1Ref is cleared as soon as a High level is detected on ETRF input

Bits 16, 6:4 **OC1M[3:0]**: Output compare 1 mode

These bits define the behavior of the output reference signal OC1REF from which OC1 and OC1N are derived. OC1REF is active high whereas OC1 and OC1N active level depends on CC1P and CC1NP bits.

0000: Frozen - The comparison between the output compare register TIMx_CCR1 and the counter TIMx_CNT has no effect on the outputs.(this mode is used to generate a timing base).

0001: Set channel 1 to active level on match. OC1REF signal is forced high when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

0010: Set channel 1 to inactive level on match. OC1REF signal is forced low when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

0011: Toggle - OC1REF toggles when TIMx_CNT=TIMx_CCR1.

0100: Force inactive level - OC1REF is forced low.

0101: Force active level - OC1REF is forced high.

0110: PWM mode 1 - In upcounting, channel 1 is active as long as TIMx_CNT<TIMx_CCR1 else inactive. In downcounting, channel 1 is inactive (OC1REF=0) as long as TIMx_CNT>TIMx_CCR1 else active (OC1REF=1).

0111: PWM mode 2 - In upcounting, channel 1 is inactive as long as TIMx_CNT<TIMx_CCR1 else active. In downcounting, channel 1 is active as long as TIMx_CNT>TIMx_CCR1 else inactive.

1000: Retriggerable OPM mode 1 - In up-counting mode, the channel is active until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 1 and the channels becomes inactive again at the next update. In down-counting mode, the channel is inactive until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 1 and the channels becomes inactive again at the next update.

1001: Retriggerable OPM mode 2 - In up-counting mode, the channel is inactive until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 2 and the channels becomes inactive again at the next update. In down-counting mode, the channel is active until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 1 and the channels becomes active again at the next update.

1010: Reserved,

1011: Reserved,

1100: Combined PWM mode 1 - OC1REF has the same behavior as in PWM mode 1. OC1REFC is the logical OR between OC1REF and OC2REF.

1101: Combined PWM mode 2 - OC1REF has the same behavior as in PWM mode 2. OC1REFC is the logical AND between OC1REF and OC2REF.

1110: Asymmetric PWM mode 1 - OC1REF has the same behavior as in PWM mode 1. OC1REFC outputs OC1REF when the counter is counting up, OC2REF when it is counting down.

1111: Asymmetric PWM mode 2 - OC1REF has the same behavior as in PWM mode 2. OC1REFC outputs OC1REF when the counter is counting up, OC2REF when it is counting down.

Note: In PWM mode, the OCREF level changes only when the result of the comparison changes or when the output compare mode switches from "frozen" mode to "PWM" mode.

Bit 3 **OC1PE**: Output compare 1 preload enable

0: Preload register on TIMx_CCR1 disabled. TIMx_CCR1 can be written at anytime, the new value is taken in account immediately.

1: Preload register on TIMx_CCR1 enabled. Read/Write operations access the preload register. TIMx_CCR1 preload value is loaded in the active register at each update event.

Note: The PWM mode can be used without validating the preload register only in one-pulse mode (OPM bit set in TIMx_CR1 register). Else the behavior is not guaranteed.

Bit 2 **OC1FE**: Output compare 1 fast enable

This bit is used to accelerate the effect of an event on the trigger in input on the CC output.
0: CC1 behaves normally depending on counter and CCR1 values even when the trigger is ON. The minimum delay to activate CC1 output when an edge occurs on the trigger input is 5 clock cycles.

1: An active edge on the trigger input acts like a compare match on CC1 output. Then, OC is set to the compare level independently from the result of the comparison. Delay to sample the trigger input and to activate CC1 output is reduced to 3 clock cycles. OCFE acts only if the channel is configured in PWM1 or PWM2 mode.

Bits 1:0 **CC1S[1:0]**: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output.

01: CC1 channel is configured as input, IC1 is mapped on TI1.

10: CC1 channel is configured as input, IC1 is mapped on TI2.

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

39.4.9 TIMx capture/compare mode register 2 [alternate] (TIMx_CCMR2) (x = 2 to 5)

Address offset: 0x1C

Reset value: 0x0000 0000

The same register can be used for input capture mode (this section) or for output compare mode (next section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode.

Input capture mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IC4F[3:0]				IC4PSC[1:0]		CC4S[1:0]		IC3F[3:0]				IC3PSC[1:0]		CC3S[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:12 **IC4F[3:0]**: Input capture 4 filter

Bits 11:10 **IC4PSC[1:0]**: Input capture 4 prescaler

Bits 9:8 **CC4S[1:0]**: Capture/Compare 4 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC4 channel is configured as output

01: CC4 channel is configured as input, IC4 is mapped on TI4

10: CC4 channel is configured as input, IC4 is mapped on TI3

11: CC4 channel is configured as input, IC4 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC4S bits are writable only when the channel is OFF (CC4E = 0 in TIMx_CCER).

Bits 7:4 **IC3F[3:0]**: Input capture 3 filter

Bits 3:2 **IC3PSC[1:0]**: Input capture 3 prescaler

Bits 1:0 **CC3S[1:0]**: Capture/Compare 3 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC3 channel is configured as output

01: CC3 channel is configured as input, IC3 is mapped on TI3

10: CC3 channel is configured as input, IC3 is mapped on TI4

11: CC3 channel is configured as input, IC3 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC3S bits are writable only when the channel is OFF (CC3E = 0 in TIMx_CCER).

39.4.10 TIMx capture/compare mode register 2 [alternate] (TIMx_CCMR2) (x = 2 to 5)

Address offset: 0x1C

Reset value: 0x0000 0000

The same register can be used for output compare mode (this section) or for input capture mode (previous section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode.

Output compare mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC4M[3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC3M[3]
							rw								rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC4CE	OC4M[2:0]			OC4PE	OC4FE	CC4S[1:0]		OC3CE	OC3M[2:0]			OC3PE	OC3FE	CC3S[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bits 23:17 Reserved, must be kept at reset value.

Bit 15 **OC4CE**: Output compare 4 clear enable

Bits 24, 14:12 **OC4M[3:0]**: Output compare 4 mode

Refer to OC1M description (bits 6:4 in TIMx_CCMR1 register)

Bit 11 **OC4PE**: Output compare 4 preload enable

Bit 10 **OC4FE**: Output compare 4 fast enable

Bits 9:8 **CC4S[1:0]**: Capture/Compare 4 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC4 channel is configured as output

01: CC4 channel is configured as input, IC4 is mapped on TI4

10: CC4 channel is configured as input, IC4 is mapped on TI3

11: CC4 channel is configured as input, IC4 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC4S bits are writable only when the channel is OFF (CC4E = 0 in TIMx_CCER).

Bit 7 **OC3CE**: Output compare 3 clear enable

Bits 16, 6:4 **OC3M[3:0]**: Output compare 3 mode

Refer to OC1M description (bits 6:4 in TIMx_CCMR1 register)

Bit 3 **OC3PE**: Output compare 3 preload enable

Bit 2 **OC3FE**: Output compare 3 fast enable

Bits 1:0 **CC3S[1:0]**: Capture/Compare 3 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC3 channel is configured as output

01: CC3 channel is configured as input, IC3 is mapped on TI3

10: CC3 channel is configured as input, IC3 is mapped on TI4

11: CC3 channel is configured as input, IC3 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC3S bits are writable only when the channel is OFF (CC3E = 0 in TIMx_CCER).

39.4.11 TIMx capture/compare enable register (TIMx_CCER)(x = 2 to 5)

Address offset: 0x20

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CC4NP	Res.	CC4P	CC4E	CC3NP	Res.	CC3P	CC3E	CC2NP	Res.	CC2P	CC2E	CC1NP	Res.	CC1P	CC1E
rw		rw	rw	rw		rw	rw	rw		rw	rw	rw		rw	rw

Bit 15 **CC4NP**: Capture/Compare 4 output Polarity.

Refer to CC1NP description

Bit 14 Reserved, must be kept at reset value.

Bit 13 **CC4P**: Capture/Compare 4 output Polarity.

Refer to CC1P description

Bit 12 **CC4E**: Capture/Compare 4 output enable.

refer to CC1E description

Bit 11 **CC3NP**: Capture/Compare 3 output Polarity.

Refer to CC1NP description

Bit 10 Reserved, must be kept at reset value.

Bit 9 **CC3P**: Capture/Compare 3 output Polarity.

Refer to CC1P description

Bit 8 **CC3E**: Capture/Compare 3 output enable.

Refer to CC1E description

Bit 7 **CC2NP**: *Capture/Compare 2 output Polarity.*

Refer to CC1NP description

Bit 6 Reserved, must be kept at reset value.

Bit 5 **CC2P**: *Capture/Compare 2 output Polarity.*

refer to CC1P description

Bit 4 **CC2E**: *Capture/Compare 2 output enable.*

Refer to CC1E description

Bit 3 **CC1NP**: *Capture/Compare 1 output Polarity.*

CC1 channel configured as output: CC1NP must be kept cleared in this case.

CC1 channel configured as input: This bit is used in conjunction with CC1P to define TI1FP1/TI2FP1 polarity. refer to CC1P description.

Bit 2 Reserved, must be kept at reset value.

Bit 1 **CC1P**: *Capture/Compare 1 output Polarity.*

0: OC1 active high (output mode) / Edge sensitivity selection (input mode, see below)

1: OC1 active low (output mode) / Edge sensitivity selection (input mode, see below)

When CC1 channel is configured as input, both CC1NP/CC1P bits select the active polarity of TI1FP1 and TI2FP1 for trigger or capture operations.

CC1NP=0, CC1P=0: non-inverted/rising edge. The circuit is sensitive to TIxFP1 rising edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode or encoder mode).

CC1NP=0, CC1P=1: inverted/falling edge. The circuit is sensitive to TIxFP1 falling edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is inverted (trigger operation in gated mode or encoder mode).

CC1NP=1, CC1P=1: non-inverted/both edges. The circuit is sensitive to both TIxFP1 rising and falling edges (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode). This configuration must not be used in encoder mode.

CC1NP=1, CC1P=0: This configuration is reserved, it must not be used.

Bit 0 **CC1E**: *Capture/Compare 1 output enable.*

0: Capture mode disabled / OC1 is not active

1: Capture mode enabled / OC1 signal is output on the corresponding output pin

Table 335. Output control bit for standard OCx channels

CCxE bit	OCx output state
0	Output Disabled (OCx=0, OCx_EN=0)
1	OCx=OCxREF + Polarity, OCx_EN=1

Note: The state of the external IO pins connected to the standard OCx channels depends on the OCx channel state and the GPIO and AFIO registers.

39.4.12 TIMx counter [alternate] (TIMx_CNT)(x = 2 to 5)

Bit 31 of this register has two possible definitions depending on the value of UIFREMAP in TIMx_CR1 register:

- This section is for UIFREMAP = 0
- Next section is for UIFREMAP = 1

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNT[31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 **CNT[31:16]**: Most significant part counter value (TIM2 and TIM5)Bits 15:0 **CNT[15:0]**: Least significant part of counter value

39.4.13 TIMx counter [alternate] (TIMx_CNT)(x = 2 to 5)

Bit 31 of this register has two possible definitions depending on the value of UIFREMAP in TIMx_CR1 register:

- Previous section is for UIFREMAP = 0
- This section is for UIFREMAP = 1

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UIFCPY	CNT[30:16]														
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **UIFCPY**: UIF Copy

This bit is a read-only copy of the UIF bit of the TIMx_ISR register

Bits 30:16 **CNT[30:16]**: Most significant part counter value (TIM2 and TIM5)Bits 15:0 **CNT[15:0]**: Least significant part of counter value

39.4.14 TIMx prescaler (TIMx_PSC)(x = 2 to 5)

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **PSC[15:0]**: Prescaler value

The counter clock frequency CK_CNT is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded in the active prescaler register at each update event (including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in “reset mode”).

39.4.15 TIMx auto-reload register (TIMx_ARR)(x = 2 to 5)

Address offset: 0x2C

Reset value: 0xFFFF FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ARR[31:16]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:16 **ARR[31:16]**: High auto-reload value (TIM2 and TIM5)

Bits 15:0 **ARR[15:0]**: Low Auto-reload value

ARR is the value to be loaded in the actual auto-reload register.

Refer to the [Section 39.3.1: Time-base unit on page 1652](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

39.4.16 TIMx capture/compare register 1 (TIMx_CCR1)(x = 2 to 5)

Address offset: 0x34

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CCR1[31:16]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:16 **CCR1[31:16]**: High Capture/Compare 1 value (TIM2 and TIM5)

Bits 15:0 **CCR1[15:0]**: Low Capture/Compare 1 value

If channel CC1 is configured as output:

CCR1 is the value to be loaded in the actual capture/compare 1 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (bit OC1PE). Else the preload value is copied in the active capture/compare 1 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC1 output.

If channel CC1 is configured as input:

CCR1 is the counter value transferred by the last input capture 1 event (IC1). The TIMx_CCR1 register is read-only and cannot be programmed.

39.4.17 TIMx capture/compare register 2 (TIMx_CCR2)(x = 2 to 5)

Address offset: 0x38

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CCR2[31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 **CCR2[31:16]**: High Capture/Compare 2 value (TIM2 and TIM5)

Bits 15:0 **CCR2[15:0]**: Low Capture/Compare 2 value

If channel CC2 is configured as output:

CCR2 is the value to be loaded in the actual capture/compare 2 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (bit OC2PE). Else the preload value is copied in the active capture/compare 2 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC2 output.

If channel CC2 is configured as input:

CCR2 is the counter value transferred by the last input capture 2 event (IC2). The TIMx_CCR2 register is read-only and cannot be programmed.

39.4.18 TIMx capture/compare register 3 (TIMx_CCR3)(x = 2 to 5)

Address offset: 0x3C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CCR3[31:16]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR3[15:0]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:16 **CCR3[31:16]**: High Capture/Compare 3 value (TIM2 and TIM5)

Bits 15:0 **CCR3[15:0]**: Low Capture/Compare value

If channel CC3 is configured as output:

CCR3 is the value to be loaded in the actual capture/compare 3 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR2 register (bit OC3PE). Else the preload value is copied in the active capture/compare 3 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC3 output.

If channel CC3 is configured as input:

CCR3 is the counter value transferred by the last input capture 3 event (IC3). The TIMx_CCR3 register is read-only and cannot be programmed.

39.4.19 TIMx capture/compare register 4 (TIMx_CCR4)(x = 2 to 5)

Address offset: 0x40

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CCR4[31:16]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR4[15:0]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:16 **CCR4[31:16]**: High Capture/Compare 4 value (TIM2 and TIM5)

Bits 15:0 **CCR4[15:0]**: Low Capture/Compare value

- if CC4 channel is configured as output (CC4S bits):
CCR4 is the value to be loaded in the actual capture/compare 4 register (preload value). It is loaded permanently if the preload feature is not selected in the TIMx_CCMR2 register (bit OC4PE). Else the preload value is copied in the active capture/compare 4 register when an update event occurs.
The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC4 output.
- if CC4 channel is configured as input (CC4S bits in TIMx_CCMR4 register):
CCR4 is the counter value transferred by the last input capture 4 event (IC4). The TIMx_CCR4 register is read-only and cannot be programmed.

39.4.20 TIMx DMA control register (TIMx_DCR)(x = 2 to 5)

Address offset: 0x48

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	DBL[4:0]					Res.	Res.	Res.	DBA[4:0]				
			rw	rw	rw	rw	rw				rw	rw	rw	rw	rw

Bits 15:13 Reserved, must be kept at reset value.

Bits 12:8 **DBL[4:0]**: DMA burst length

This 5-bit vector defines the number of DMA transfers (the timer recognizes a burst transfer when a read or a write access is done to the TIMx_DMAR address).

00000: 1 transfer,

00001: 2 transfers,

00010: 3 transfers,

...

10001: 18 transfers.

Bits 7:5 Reserved, must be kept at reset value.

Bits 4:0 **DBA[4:0]**: DMA base address

This 5-bit vector defines the base-address for DMA transfers (when read/write access are done through the TIMx_DMAR address). DBA is defined as an offset starting from the address of the TIMx_CR1 register.

Example:

00000: TIMx_CR1

00001: TIMx_CR2

00010: TIMx_SMCR

...

Example: Let us consider the following transfer: DBL = 7 transfers & DBA = TIMx_CR1. In this case the transfer is done to/from 7 registers starting from the TIMx_CR1 address.

39.4.21 TIMx DMA address for full transfer (TIMx_DMAR)(x = 2 to 5)

Address offset: 0x4C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMAB[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **DMAB[15:0]**: DMA register for burst accesses

A read or write operation to the DMAR register accesses the register located at the address
 $(\text{TIMx_CR1 address}) + (\text{DBA} + \text{DMA index}) \times 4$

where TIMx_CR1 address is the address of the control register 1, DBA is the DMA base address configured in TIMx_DCR register, DMA index is automatically controlled by the DMA transfer, and ranges from 0 to DBL (DBL configured in TIMx_DCR).

39.4.22 TIM2 alternate function option register 1 (TIM2_AF1)

Address offset: 0x60

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETRSEL[3:2]	
														rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETRSEL[1:0]		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rw	rw														

Bits 31:18 Reserved, must be kept at reset value.

Bits 17:14 **ETRSEL[3:0]**: ETR source selection

These bits select the ETR input source.

0000: ETR input is connected to I/O

0001: COMP1 output

0010: COMP2 output

0011: LSE

0100: SAI1 FS_A

0101: SAI1 FS_B

Others: Reserved

Bits 13:0 Reserved, must be kept at reset value.

39.4.23 TIM3 alternate function option register 1 (TIM3_AF1)

Address offset: 0x60

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETRSEL[3:2]	
														rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETRSEL[1:0]		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rw	rw														

Bits 31:18 Reserved, must be kept at reset value.

Bits 17:14 **ETRSEL[3:0]**: ETR source selection

These bits select the ETR input source.

0000: ETR input is connected to I/O

0001: COMP1 output

Others: Reserved

Bits 13:0 Reserved, must be kept at reset value.

39.4.24 TIM4 alternate function option register 1 (TIM4_AF1)

Address offset: 0x60

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETRSEL[3:2]	
														rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETRSEL[1:0]		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rw	rw														

Bits 31:18 Reserved, must be kept at reset value.

Bits 17:14 **ETRSEL[3:0]**: ETR source selection

These bits select the ETR input source.

0000: ETR input is connected to I/O

Others: Reserved

Bits 13:0 Reserved, must be kept at reset value.

39.4.25 TIM5 alternate function option register 1 (TIM5_AF1)

Address offset: 0x60

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ETRSEL[3:2]	
														rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETRSEL[1:0]		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rw	rw														

Bits 31:18 Reserved, must be kept at reset value.

Bits 17:14 **ETRSEL[3:0]**: ETR source selection

These bits select the ETR input source.

0000: ETR input is connected to I/O

0001: SAI2 FS_A connected to ETR input

0010: SAI2 FS_B connected to ETR input

Others: Reserved

Bits 13:0 Reserved, must be kept at reset value.

39.4.26 TIM2 timer input selection register (TIM2_TISEL)

Address offset: 0x68

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	TI4SEL[3:0]				Res.	Res.	Res.	Res.	TI3SEL[3:0]			
				rw	rw	rw	rw					rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	TI2SEL[3:0]				Res.	Res.	Res.	Res.	TI1SEL[3:0]			
				rw	rw	rw	rw					rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:24 **TI4SEL[3:0]**: TI4[0] to TI4[15] input selection

These bits select the TI4[0] to TI4[15] input source.

0000: TIM2_CH4 input

0001: COMP1 output

0010: COMP2 output

0011: COMP1 output OR COMP2 output

Others: Reserved

Bits 23:20 Reserved, must be kept at reset value.

Bits 19:16 **TI3SEL[3:0]**: TI3[0] to TI3[15] input selection

These bits select the TI3[0] to TI3[15] input source.

0000: TIM2_CH3 input

Others: Reserved

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:8 **TI2SEL[3:0]**: TI2[0] to TI2[15] input selection

These bits select the TI2[0] to TI2[15] input source.

0000: TIM2_CH2 input

Others: Reserved

Bits 7:4 Reserved, must be kept at reset value.

Bits 3:0 **TI1SEL[3:0]**: TI1[0] to TI1[15] input selection

These bits select the TI1[0] to TI1[15] input source.

0000: TIM2_CH1 input

Others: Reserved

39.4.27 TIM3 timer input selection register (TIM3_TISEL)

Address offset: 0x68

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	TI4SEL[3:0]				Res.	Res.	Res.	Res.	TI3SEL[3:0]			
				rw	rw	rw	rw					rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	TI2SEL[3:0]				Res.	Res.	Res.	Res.	TI1SEL[3:0]			
				rw	rw	rw	rw					rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:24 **TI4SEL[3:0]**: TI4[0] to TI4[15] input selection
 These bits select the TI4[0] to TI4[15] input source.
 0000: TIM3_CH4 input
 Others: Reserved

Bits 23:20 Reserved, must be kept at reset value.

Bits 19:16 **TI3SEL[3:0]**: TI3[0] to TI3[15] input selection
 These bits select the TI3[0] to TI3[15] input source.
 0000: TIM3_CH3 input
 Others: Reserved

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:8 **TI2SEL[3:0]**: TI2[0] to TI2[15] input selection
 These bits select the TI2[0] to TI2[15] input source.
 0000: TIM3_CH2 input
 Others: Reserved

Bits 7:4 Reserved, must be kept at reset value.

Bits 3:0 **TI1SEL[3:0]**: TI1[0] to TI1[15] input selection
 These bits select the TI1[0] to TI1[15] input source.
 0000: TIM3_CH1 input
 0001: COMP1 output
 0010: COMP2 output
 0011: COMP1 output OR COMP2 output
 Others: Reserved

39.4.28 TIM4 timer input selection register (TIM4_TISEL)

Address offset: 0x68

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	TI4SEL[3:0]				Res.	Res.	Res.	Res.	TI3SEL[3:0]			
				rw	rw	rw	rw					rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	TI2SEL[3:0]				Res.	Res.	Res.	Res.	TI1SEL[3:0]			
				rw	rw	rw	rw					rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:24 **TI4SEL[3:0]**: TI4[0] to TI4[15] input selection
 These bits select the TI4[0] to TI4[15] input source.
 0000: TIM4_CH4 input
 Others: Reserved

Bits 23:20 Reserved, must be kept at reset value.

Bits 19:16 **TI3SEL[3:0]**: TI3[0] to TI3[15] input selection

These bits select the TI3[0] to TI3[15] input source.

0000: TIM4_CH3 input

Others: Reserved

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:8 **TI2SEL[3:0]**: TI2[0] to TI2[15] input selection

These bits select the TI2[0] to TI2[15] input source.

0000: TIM4_CH2 input

Others: Reserved

Bits 7:4 Reserved, must be kept at reset value.

Bits 3:0 **TI1SEL[3:0]**: TI1[0] to TI1[15] input selection

These bits select the TI1[0] to TI1[15] input source.

0000: TIM4_CH1 input

Others: Reserved

39.4.29 TIM5 timer input selection register (TIM5_TISEL)

Address offset: 0x68

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	TI4SEL[3:0]				Res.	Res.	Res.	Res.	TI3SEL[3:0]			
				rw	rw	rw	rw					rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	TI2SEL[3:0]				Res.	Res.	Res.	Res.	TI1SEL[3:0]			
				rw	rw	rw	rw					rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:24 **TI4SEL[3:0]**: TI4[0] to TI4[15] input selection

These bits select the TI4[0] to TI4[15] input source.

0000: TIM5_CH4 input

Others: Reserved

Bits 23:20 Reserved, must be kept at reset value.

Bits 19:16 **TI3SEL[3:0]**: TI3[0] to TI3[15] input selection

These bits select the TI3[0] to TI3[15] input source.

0000: TIM5_CH3 input

Others: Reserved

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:8 **TI2SEL[3:0]**: TI2[0] to TI2[15] input selection

These bits select the TI2[0] to TI2[15] input source.

0000: TIM5_CH2 input

Others: Reserved

Bits 7:4 Reserved, must be kept at reset value.

Bits 3:0 **TI1SEL[3:0]**: TI1[0] to TI1[15] input selection

These bits select the TI1[0] to TI1[15] input source.

0000: TIM5_CH1 input

0001: fdcan1_tmp

0010: fdcan1_rtp

Others: Reserved

39.4.30 TIMx register map

TIMx registers are mapped as described in the table below:

Table 336. TIM2/TIM3/TIM4/TIM5 register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	TIMx_CR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	UIFREMA	Res	Res	CKD [1:0]	ARPE	CMS [1:0]	DIR	OPM	URS	UDIS	CEN	
	Reset value																					0		0	0	0	0	0	0	0	0	0	0
0x04	TIMx_CR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	T1S	MMS[2:0]		CCDS	Res	Res		
	Reset value																									0	0	0	0	0			
0x08	TIMx_SMCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	TS [4:3]					SMS[3]	ETP	EOE	ETPS [1:0]	ETF[3:0]			MSM	TS[2:0]		Res	SMS[2:0]					
	Reset value										0	0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	TIMx_DIER	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	TDE	Res	CC4DE	CC3DE	CC2DE	CC1DE	UDE	Res	TIE	Res	CC4IE	CC3IE	CC2IE	CC1IE	UIE
	Reset value																		0		0	0	0	0	0		0		0	0	0	0	0
0x10	TIMx_SR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CC4OF	CC3OF	CC2OF	CC1OF	Res	Res	TIF	Res	CC4IF	CC3IF	CC2IF	CC1IF	UIF
	Reset value																				0	0	0	0			0		0	0	0	0	0
0x14	TIMx_EGR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	TG	Res	CC4G	CC3G	CC2G	CC1G	UG	
	Reset value																									0		0	0	0	0	0	0
0x18	TIMx_CCMR1 Output Compare mode	Res	Res	Res	Res	Res	Res	Res	OC2M[3]	Res	Res	Res	Res	Res	Res	Res	OC1M[3]	OC2CE	OC2M [2:0]		OC2PE	OC2FE	CC2S [1:0]		OC1CE	OC1M [2:0]		OC1PE	OC1FE	CC1S [1:0]			
	Reset value								0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	TIMx_CCMR1 Input Capture mode	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	IC2F[3:0]			IC2 PSC [1:0]	CC2S [1:0]		IC1F[3:0]			IC1 PSC [1:0]	CC1S [1:0]					
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x1C	TIMx_CCMR2 Output Compare mode	Res	Res	Res	Res	Res	Res	Res	OC4M[3]	Res	Res	Res	Res	Res	Res	Res	OC3M[3]	O24CE	OC4M [2:0]		OC4PE	OC4FE	CC4S [1:0]		OC3CE	OC3M [2:0]		OC3PE	OC3FE	CC3S [1:0]			
	Reset value								0								0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	TIMx_CCMR2 Input Capture mode	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	IC4F[3:0]			IC4 PSC [1:0]	CC4S [1:0]		IC3F[3:0]			IC3 PSC [1:0]	CC3S [1:0]					
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x20	TIMx_CCER	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CC4NP	Res	CC4P	CC4E	CC3NP	Res	CC3P	CC3E	CC2NP	Res	CC2P	CC2E	CC1NP	Res	CC1P	CC1E
	Reset value																	0		0	0	0		0	0	0		0	0		0	0	0

Table 336. TIM2/TIM3/TIM4/TIM5 register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0					
0x24	TIMx_CNT	CNT[31] or UIFCPY	CNT[30:16] (TIM2 and TIM5 only, reserved on the other timers)															CNT[15:0]																				
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x28	TIMx_PSC	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PSC[15:0]																				
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x2C	TIMx_ARR	ARR[31:16] (TIM2 and TIM5 only, reserved on the other timers)															ARR[15:0]																					
	Reset value	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1					
0x30	Reserved																																					
0x34	TIMx_CCR1	CCR1[31:16] (TIM2 and TIM5 only, reserved on the other timers)															CCR1[15:0]																					
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x38	TIMx_CCR2	CCR2[31:16] (TIM2 and TIM5 only, reserved on the other timers)															CCR2[15:0]																					
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x3C	TIMx_CCR3	CCR3[31:16] (TIM2 and TIM5 only, reserved on the other timers)															CCR3[15:0]																					
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x40	TIMx_CCR4	CCR4[31:16] (TIM2 and TIM5 only, reserved on the other timers)															CCR4[15:0]																					
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x44	Reserved																																					
0x48	TIMx_DCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DBL[4:0]				Res	Res	Res	DBA[4:0]													
	Reset value																				0	0	0	0	0				0	0	0	0	0					
0x4C	TIMx_DMAR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DMAB[15:0]																				
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x60	TIM2_AF1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	ETRSEL [3:0]		Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res				
	Reset value																0	0	0	0																		
0x60	TIM3_AF1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	ETRSEL [3:0]		Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res				
	Reset value																0	0	0	0																		

Table 336. TIM2/TIM3/TIM4/TIM5 register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x60	TIM4_AF1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	ETRSEL[3:0]				Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res
	Reset value															0	0	0	0														
0x60	TIM5_AF1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	ETRSEL[3:0]				Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res
	Reset value															0	0	0	0														
0x68	TIM2_TISEL	Res	Res	Res	Res	TI4SEL[3:0]				Res	Res	Res	Res	TI3SEL[3:0]				Res	Res	Res	Res	TI2SEL[3:0]				Res	Res	Res	Res	TI1SEL[3:0]			
	Reset value					0	0	0	0					0	0	0	0					0	0	0	0					0	0	0	0
0x68	TIM3_TISEL	Res	Res	Res	Res	TI4SEL[3:0]				Res	Res	Res	Res	TI3SEL[3:0]				Res	Res	Res	Res	TI2SEL[3:0]				Res	Res	Res	Res	TI1SEL[3:0]			
	Reset value					0	0	0	0					0	0	0	0					0	0	0	0					0	0	0	0
0x68	TIM4_TISEL	Res	Res	Res	Res	TI4SEL[3:0]				Res	Res	Res	Res	TI3SEL[3:0]				Res	Res	Res	Res	TI2SEL[3:0]				Res	Res	Res	Res	TI1SEL[3:0]			
	Reset value					0	0	0	0					0	0	0	0					0	0	0	0					0	0	0	0
0x68	TIM5_TISEL	Res	Res	Res	Res	TI4SEL[3:0]				Res	Res	Res	Res	TI3SEL[3:0]				Res	Res	Res	Res	TI2SEL[3:0]				Res	Res	Res	Res	TI1SEL[3:0]			
	Reset value					0	0	0	0					0	0	0	0					0	0	0	0					0	0	0	0

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.

40 General-purpose timers (TIM12/TIM13/TIM14)

40.1 TIM12/TIM13/TIM14 introduction

The TIM12/TIM13/TIM14 general-purpose timers consist in a 16-bit auto-reload counter driven by a programmable prescaler.

They may be used for a variety of purposes, including measuring the pulse lengths of input signals (input capture) or generating output waveforms (output compare, PWM).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

The TIM12/TIM13/TIM14 timers are completely independent, and do not share any resources. They can be synchronized together as described in [Section 40.3.17: Timer synchronization \(TIM12\)](#).

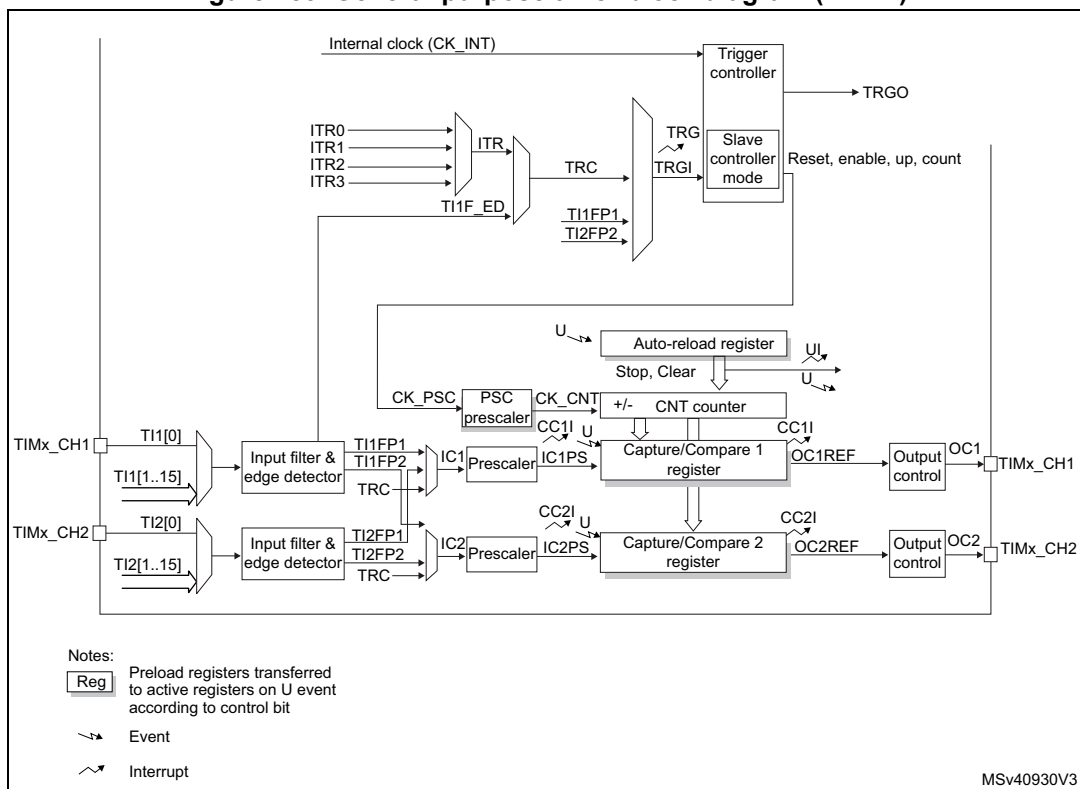
40.2 TIM12/TIM13/TIM14 main features

40.2.1 TIM12 main features

The features of the TIM12 general-purpose timer include:

- 16-bit auto-reload upcounter
- 16-bit programmable prescaler used to divide the counter clock frequency by any factor between 1 and 65536 (can be changed “on the fly”)
- Up to 2 independent channels for:
 - Input capture
 - Output compare
 - PWM generation (edge-aligned mode)
 - One-pulse mode output
- Synchronization circuit to control the timer with external signals and to interconnect several timers together
- Interrupt generation on the following events:
 - Update: counter overflow, counter initialization (by software or internal trigger)
 - Trigger event (counter start, stop, initialization or count by internal trigger)
 - Input capture
 - Output compare

Figure 453. General-purpose timer block diagram (TIM12)

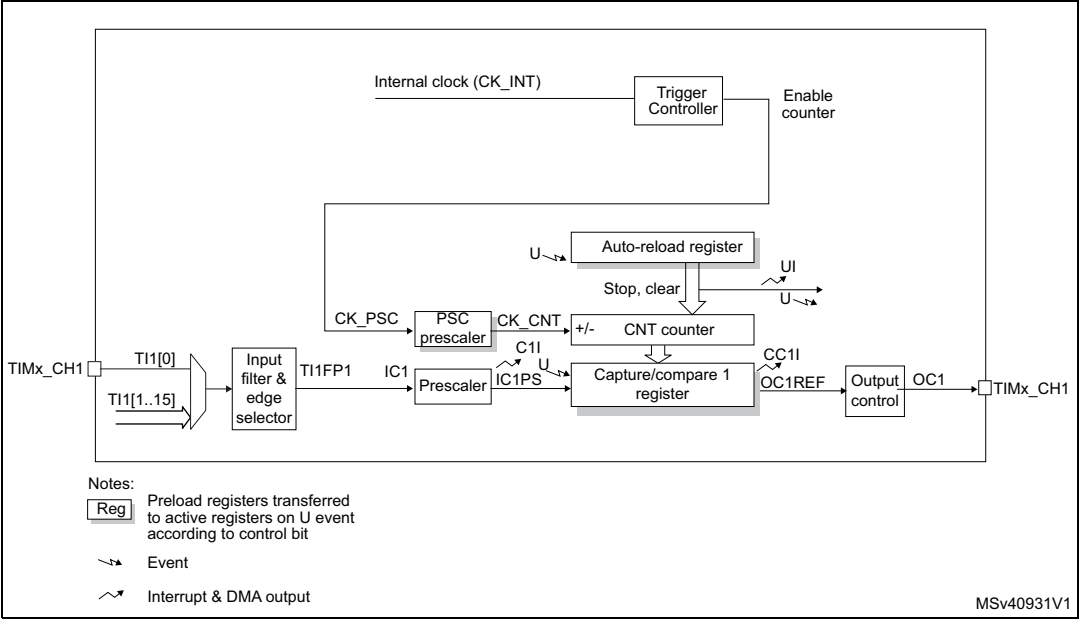


40.2.2 TIM13/TIM14 main features

The features of general-purpose timers TIM13/TIM14 include:

- 16-bit auto-reload upcounter
- 16-bit programmable prescaler used to divide the counter clock frequency by any factor between 1 and 65536 (can be changed “on the fly”)
- independent channel for:
 - Input capture
 - Output compare
 - PWM generation (edge-aligned mode)
 - One-pulse mode output
- Interrupt generation on the following events:
 - Update: counter overflow, counter initialization (by software)
 - Input capture
 - Output compare

Figure 454. General-purpose timer block diagram (TIM13/TIM14)



40.3 TIM12/TIM13/TIM14 functional description

40.3.1 Time-base unit

The main block of the timer is a 16-bit up-counter with its related auto-reload register. The counter clock can be divided by a prescaler.

The counter, the auto-reload register and the prescaler register can be written or read by software. This is true even when the counter is running.

The time-base unit includes:

- Counter register (TIMx_CNT)
- Prescaler register (TIMx_PSC)
- Auto-reload register (TIMx_ARR)

The auto-reload register is preloaded. Writing to or reading from the auto-reload register accesses the preload register. The content of the preload register are transferred into the shadow register permanently or at each update event (UEV), depending on the auto-reload preload enable bit (ARPE) in TIMx_CR1 register. The update event is sent when the counter reaches the overflow and if the UDIS bit equals 0 in the TIMx_CR1 register. It can also be generated by software. The generation of the update event is described in details for each configuration.

The counter is clocked by the prescaler output CK_CNT, which is enabled only when the counter enable bit (CEN) in TIMx_CR1 register is set (refer also to the slave mode controller description to get more details on counter enabling).

Note that the counter starts counting 1 clock cycle after setting the CEN bit in the TIMx_CR1 register.

Prescaler description

The prescaler can divide the counter clock frequency by any factor between 1 and 65536. It is based on a 16-bit counter controlled through a 16-bit register (in the TIMx_PSC register). It can be changed on the fly as this control register is buffered. The new prescaler ratio is taken into account at the next update event.

[Figure 455](#) and [Figure 456](#) give some examples of the counter behavior when the prescaler ratio is changed on the fly.

Figure 455. Counter timing diagram with prescaler division change from 1 to 2

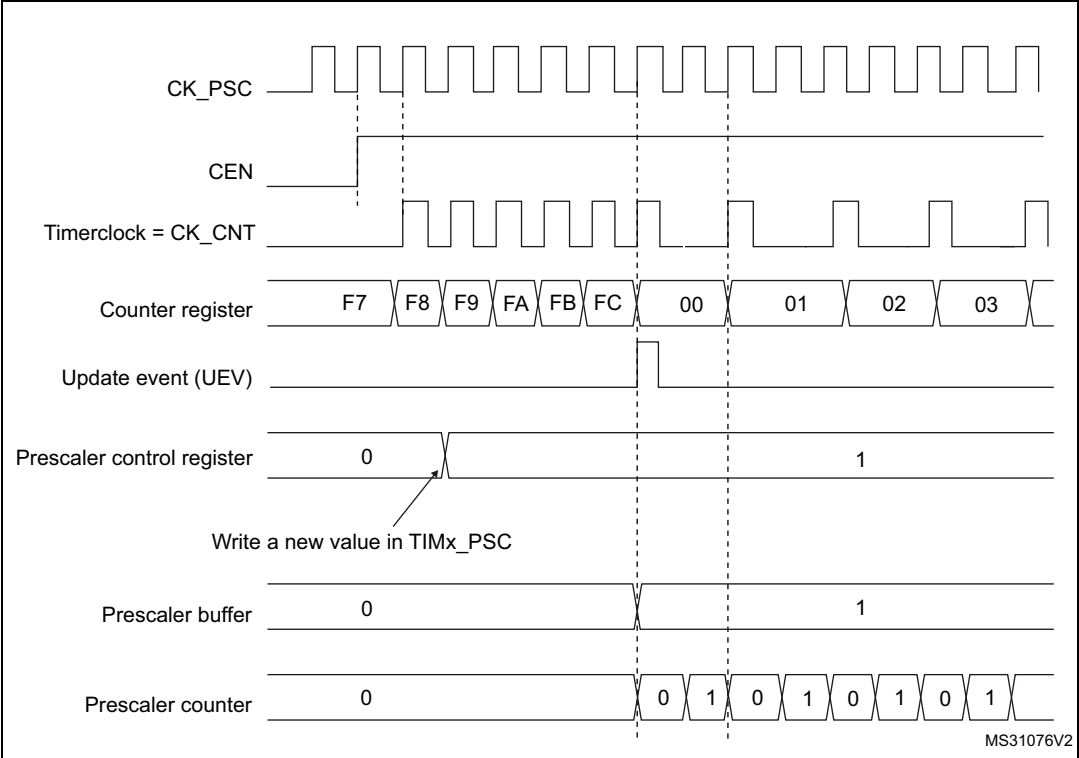
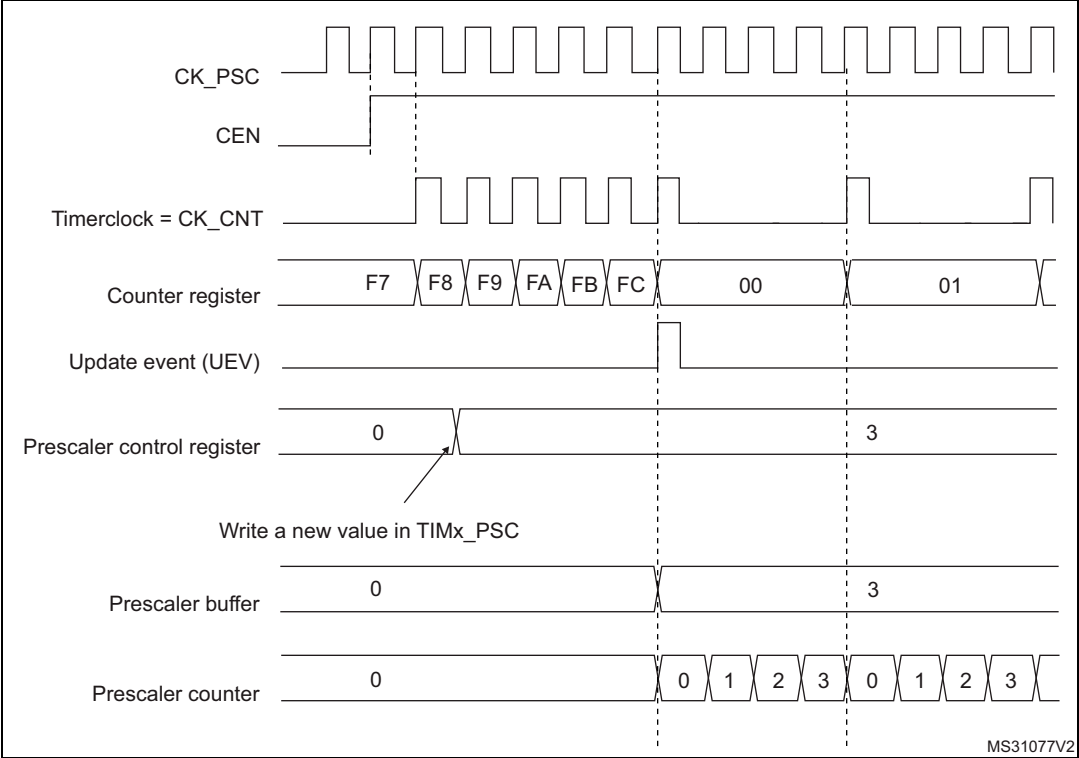


Figure 456. Counter timing diagram with prescaler division change from 1 to 4



40.3.2 Counter modes

Upcounting mode

In upcounting mode, the counter counts from 0 to the auto-reload value (content of the TIMx_ARR register), then restarts from 0 and generates a counter overflow event.

Setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller on TIM12) also generates an update event.

The UEV event can be disabled by software by setting the UDIS bit in the TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until the UDIS bit has been written to 0. However, the counter restarts from 0, as well as the counter of the prescaler (but the prescale rate does not change). In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The auto-reload shadow register is updated with the preload value (TIMx_ARR),
- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register).

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR=0x36.

Figure 457. Counter timing diagram, internal clock divided by 1

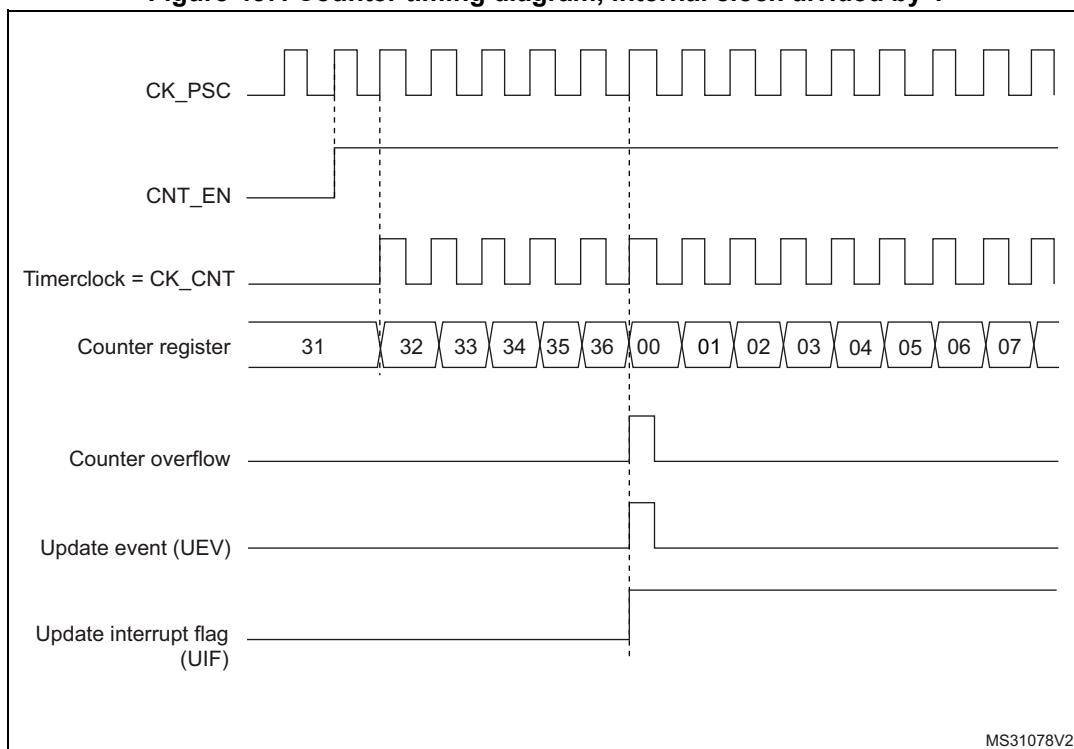


Figure 458. Counter timing diagram, internal clock divided by 2

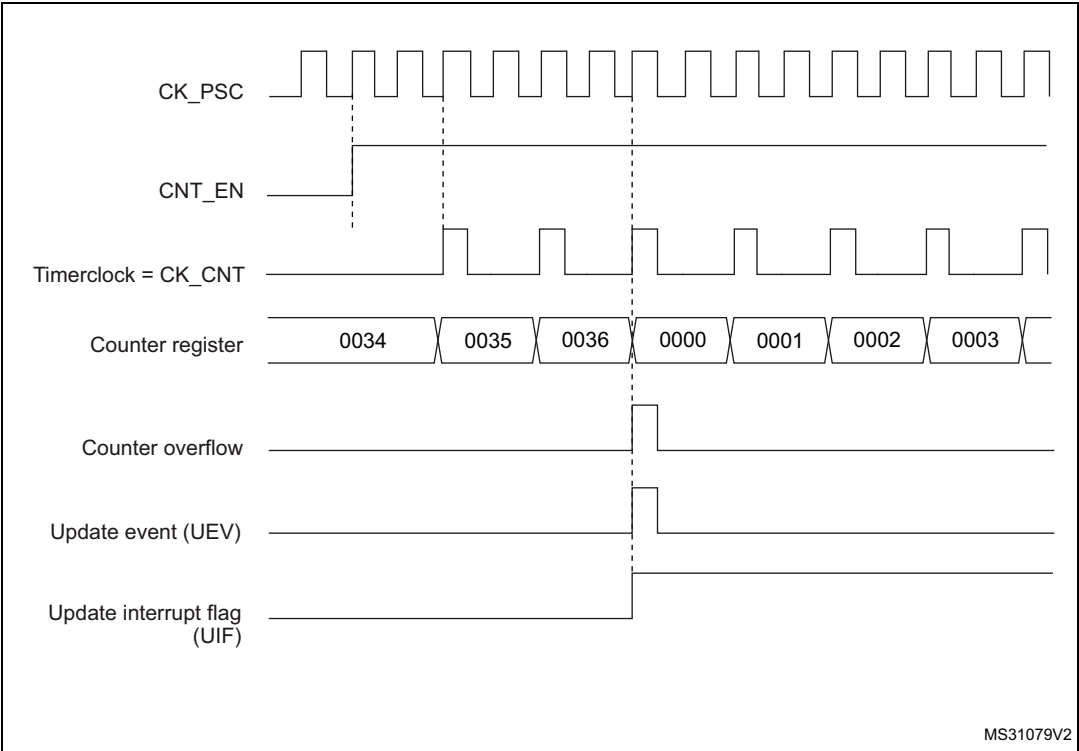


Figure 459. Counter timing diagram, internal clock divided by 4

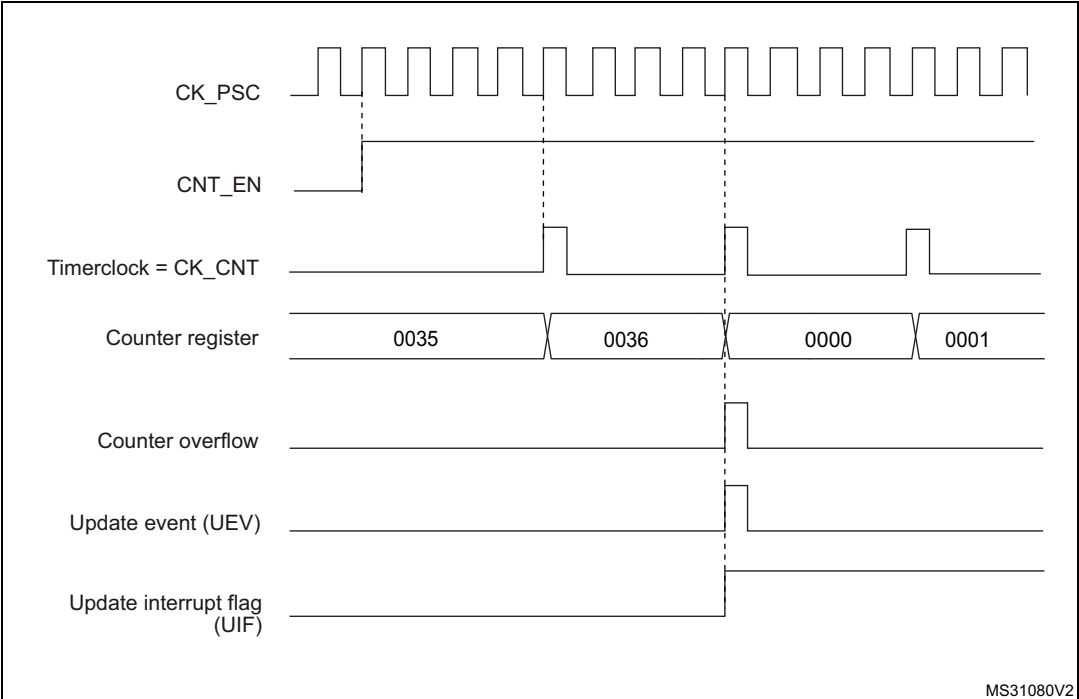


Figure 460. Counter timing diagram, internal clock divided by N

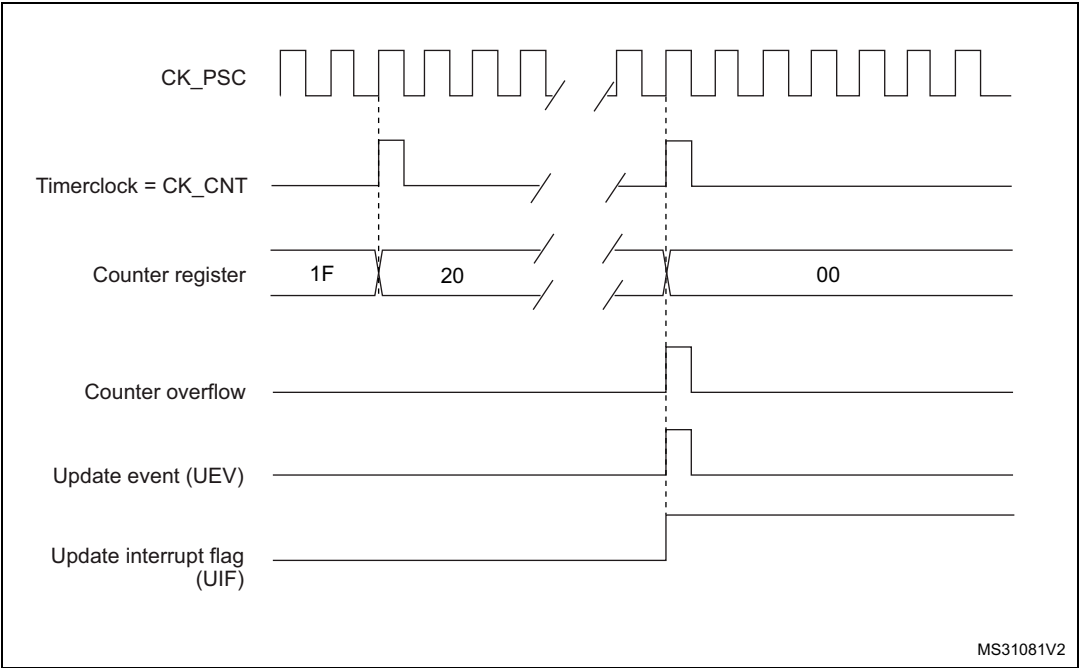


Figure 461. Counter timing diagram, update event when ARPE=0 (TIMx_ARR not preloaded)

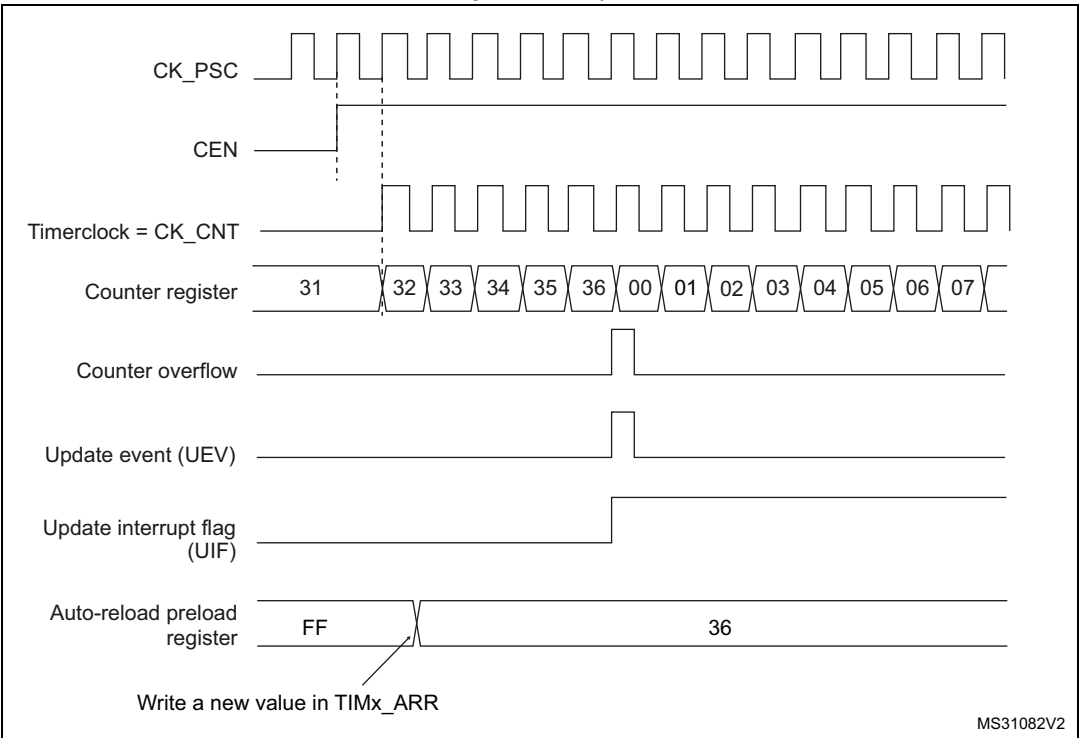
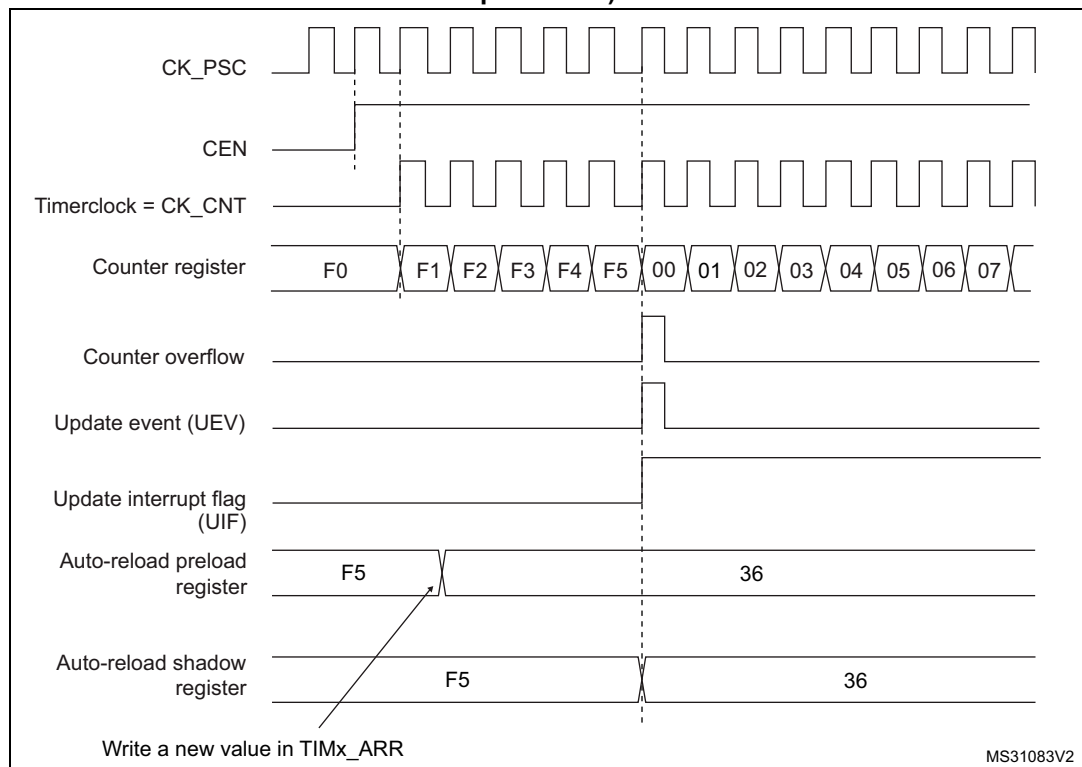


Figure 462. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded)

40.3.3 Clock selection

The counter clock can be provided by the following clock sources:

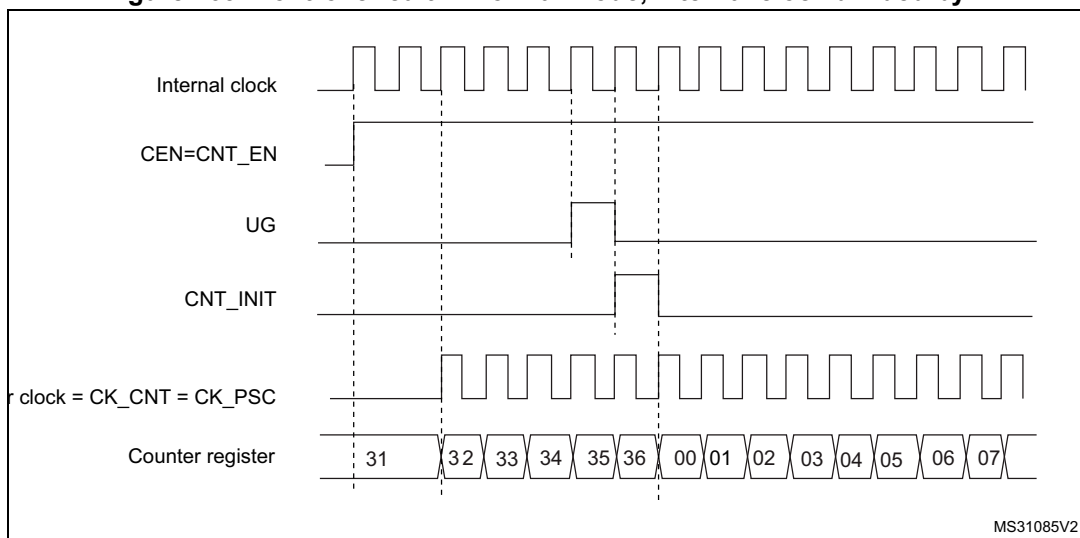
- Internal clock (CK_INT)
- External clock mode1 (for TIM12): external input pin (TIx)
- Internal trigger inputs (ITRx) (for TIM12): connecting the trigger output from another timer. For instance, another timer can be configured as a prescaler for TIM12. Refer to [Section : Using one timer as prescaler for another timer](#) for more details.

Internal clock source (CK_INT)

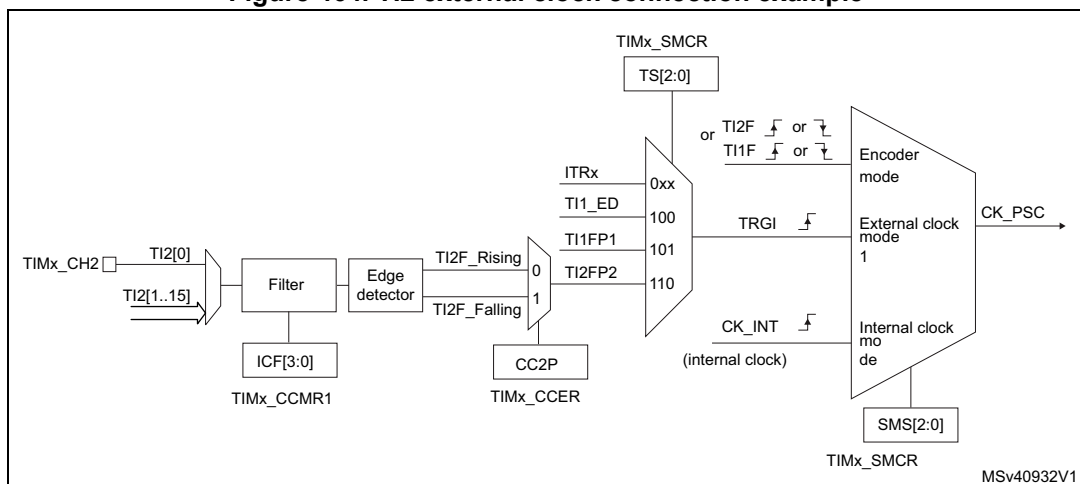
The internal clock source is the default clock source for TIM13/TIM14.

For TIM12, the internal clock source is selected when the slave mode controller is disabled (SMS='000'). The CEN bit in the TIMx_CR1 register and the UG bit in the TIMx_EGR register are then used as control bits and can be changed only by software (except for UG which remains cleared). As soon as the CEN bit is programmed to 1, the prescaler is clocked by the internal clock CK_INT.

[Figure 463](#) shows the behavior of the control circuit and the upcounter in normal mode, without prescaler.

Figure 463. Control circuit in normal mode, internal clock divided by 1**External clock source mode 1 (TIM12)**

This mode is selected when SMS='111' in the TIMx_SMCR register. The counter can count at each rising or falling edge on a selected input.

Figure 464. TI2 external clock connection example

For example, to configure the upcounter to count in response to a rising edge on the TI2 input, use the following procedure:

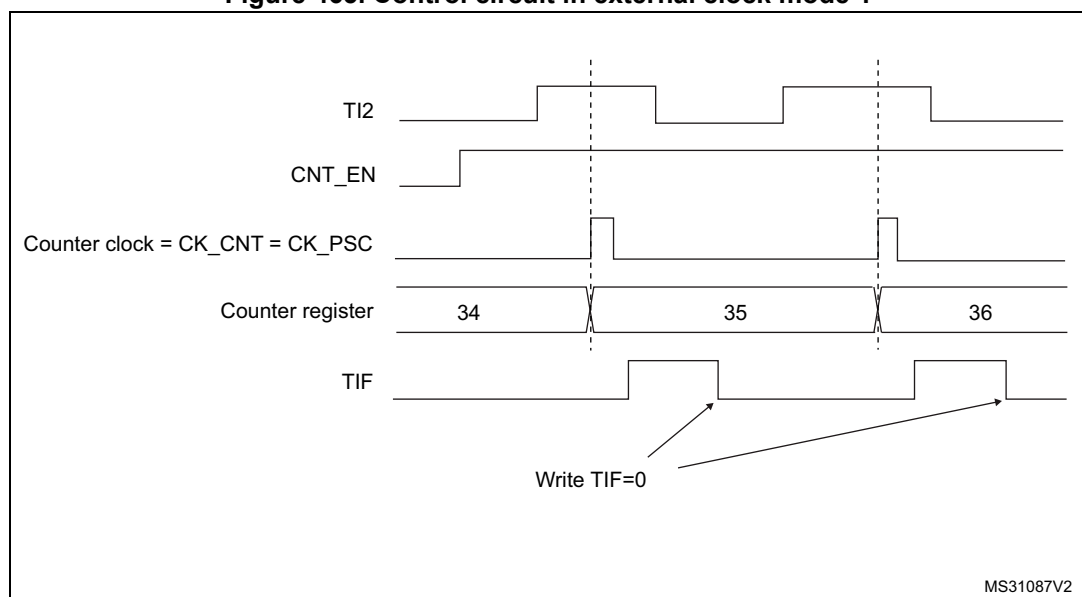
1. Select the proper TI2[x] source (internal or external) with the TI2SEL[3:0] bits in the TIMx_TISEL register.
2. Configure channel 2 to detect rising edges on the TI2 input by writing CC2S = '01' in the TIMx_CCMR1 register.
3. Configure the input filter duration by writing the IC2F[3:0] bits in the TIMx_CCMR1 register (if no filter is needed, keep IC2F='0000').
4. Select the rising edge polarity by writing CC2P='0' and CC2NP='0' in the TIMx_CCER register.
5. Configure the timer in external clock mode 1 by writing SMS='111' in the TIMx_SMCR register.
6. Select TI2 as the trigger input source by writing TS='110' in the TIMx_SMCR register.
7. Enable the counter by writing CEN='1' in the TIMx_CR1 register.

Note: *The capture prescaler is not used for triggering, so it does not need to be configured.*

When a rising edge occurs on TI2, the counter counts once and the TIF flag is set.

The delay between the rising edge on TI2 and the actual clock of the counter is due to the resynchronization circuit on TI2 input.

Figure 465. Control circuit in external clock mode 1



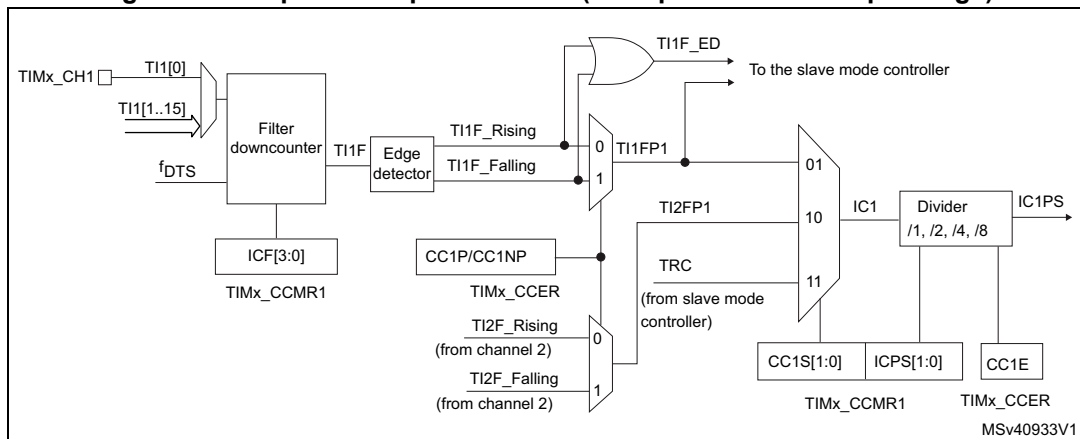
40.3.4 Capture/compare channels

Each Capture/Compare channel is built around a capture/compare register (including a shadow register), a input stage for capture (with digital filter, multiplexing and prescaler) and an output stage (with comparator and output control).

[Figure 466](#) to [Figure 468](#) give an overview of one capture/compare channel.

The input stage samples the corresponding TIx input to generate a filtered signal TIxF. Then, an edge detector with polarity selection generates a signal (TIxFPx) which can be used as trigger input by the slave mode controller or as the capture command. It is prescaled before the capture register (ICxPS).

Figure 466. Capture/compare channel (example: channel 1 input stage)



The output stage generates an intermediate waveform which is then used for reference: OCxRef (active high). The polarity acts at the end of the chain.

Figure 467. Capture/compare channel 1 main circuit

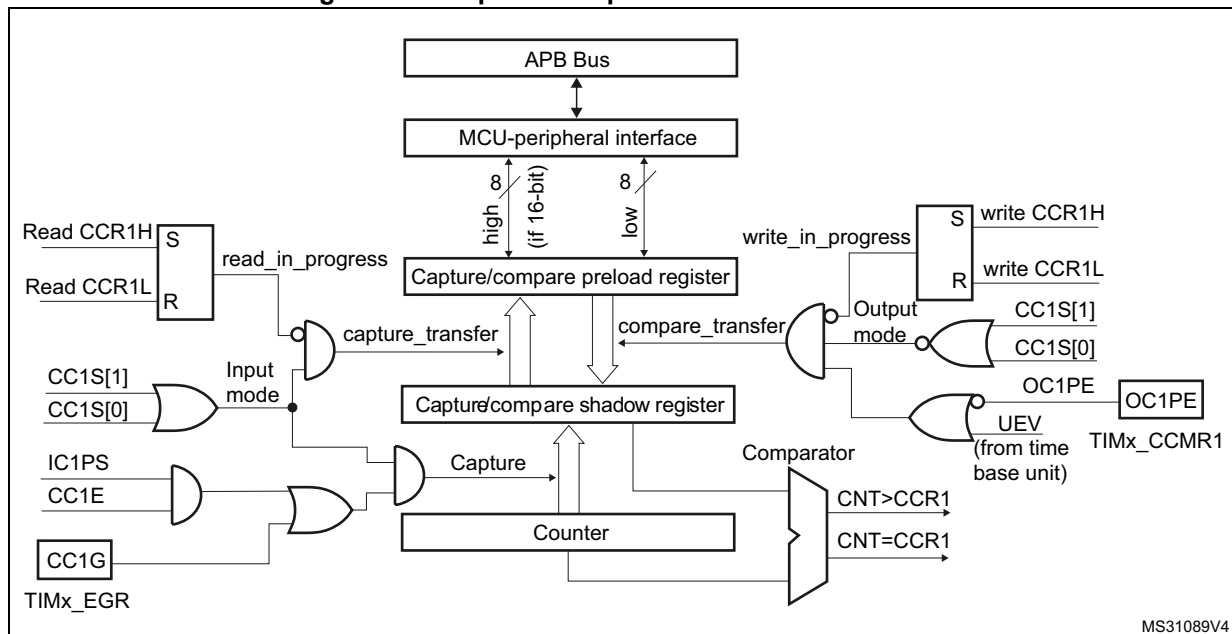
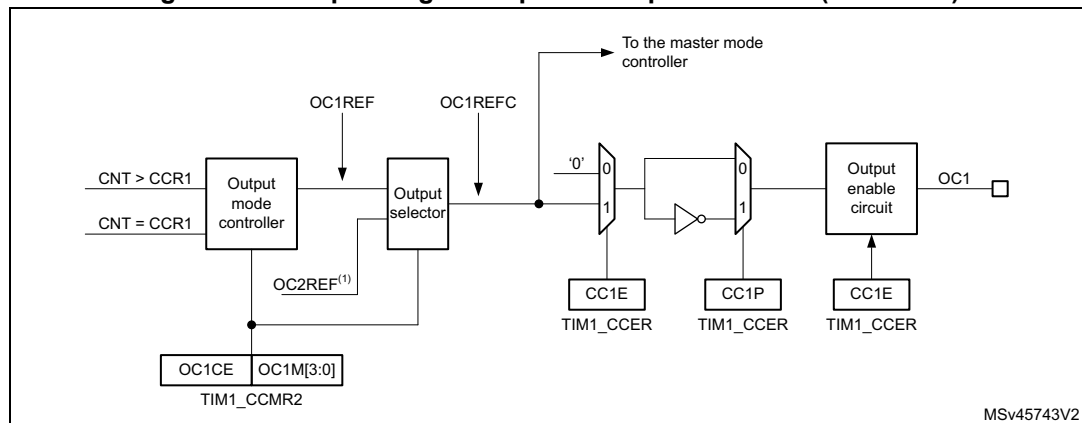


Figure 468. Output stage of capture/compare channel (channel 1)

1. Available on TIM12 only.

The capture/compare block is made of one preload register and one shadow register. Write and read always access the preload register.

In capture mode, captures are actually done in the shadow register, which is copied into the preload register.

In compare mode, the content of the preload register is copied into the shadow register which is compared to the counter.

40.3.5 Input capture mode

In Input capture mode, the Capture/Compare Registers (TIMx_CCRx) are used to latch the value of the counter after a transition detected by the corresponding ICx signal. When a capture occurs, the corresponding CCxIF flag (TIMx_SR register) is set and an interrupt or a DMA request can be sent if they are enabled. If a capture occurs while the CCxIF flag was already high, then the over-capture flag CCxOF (TIMx_SR register) is set. CCxIF can be cleared by software by writing it to '0' or by reading the captured data stored in the TIMx_CCRx register. CCxOF is cleared when it is written with 0.

The following example shows how to capture the counter value in TIMx_CCR1 when TI1 input rises. To do this, use the following procedure:

1. Select the proper TI1[x] source (internal or external) with the TI1SEL[3:0] bits in the TIMx_TISEL register.
2. Select the active input: TIMx_CCR1 must be linked to the TI1 input, so write the CC1S bits to '01' in the TIMx_CCMR1 register. As soon as CC1S becomes different from '00', the channel is configured in input mode and the TIMx_CCR1 register becomes read-only.
3. Program the appropriate input filter duration in relation with the signal connected to the timer (by programming the ICxF bits in the TIMx_CCMRx register if the input is one of the TIx inputs). Let's imagine that, when toggling, the input signal is not stable during at must 5 internal clock cycles. We must program a filter duration longer than these 5 clock cycles. We can validate a transition on TI1 when 8 consecutive samples with the

new level have been detected (sampled at f_{DTS} frequency). Then write IC1F bits to '0011' in the TIMx_CCMR1 register.

4. Select the edge of the active transition on the TI1 channel by programming CC1P and CC1NP bits to '00' in the TIMx_CCER register (rising edge in this case).
5. Program the input prescaler. In our example, we wish the capture to be performed at each valid transition, so the prescaler is disabled (write IC1PS bits to '00' in the TIMx_CCMR1 register).
6. Enable capture from the counter into the capture register by setting the CC1E bit in the TIMx_CCER register.
7. If needed, enable the related interrupt request by setting the CC1IE bit in the TIMx_DIER register.

When an input capture occurs:

- The TIMx_CCR1 register gets the value of the counter on the active transition.
- CC1IF flag is set (interrupt flag). CC1OF is also set if at least two consecutive captures occurred whereas the flag was not cleared.
- An interrupt is generated depending on the CC1IE bit.

In order to handle the overcapture, it is recommended to read the data before the overcapture flag. This is to avoid missing an overcapture which could happen after reading the flag and before reading the data.

Note: IC interrupt requests can be generated by software by setting the corresponding CCxG bit in the TIMx_EGR register.

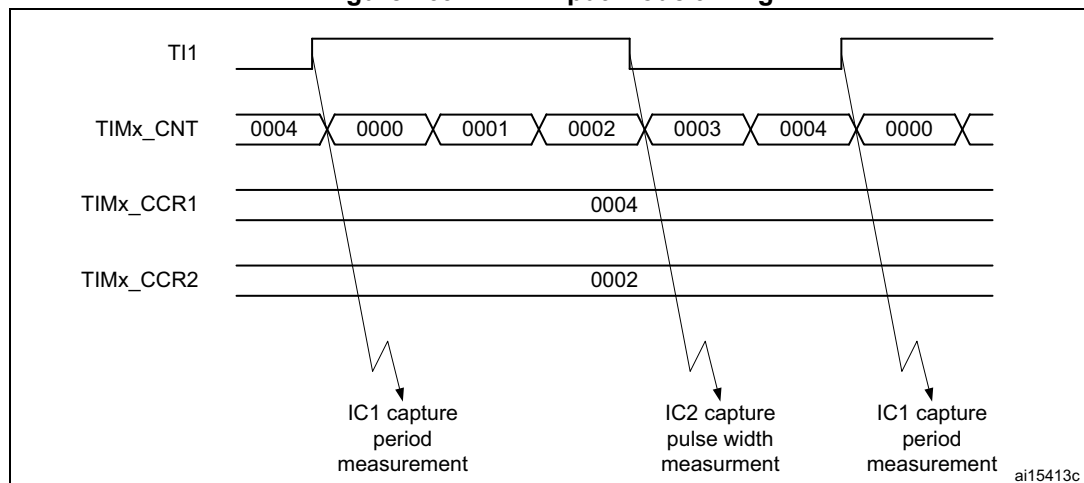
40.3.6 PWM input mode (only for TIM12)

This mode is a particular case of input capture mode. The procedure is the same except:

- Two ICx signals are mapped on the same TIx input.
- These 2 ICx signals are active on edges with opposite polarity.
- One of the two TIxFP signals is selected as trigger input and the slave mode controller is configured in reset mode.

For example, one can measure the period (in TIMx_CCR1 register) and the duty cycle (in TIMx_CCR2 register) of the PWM applied on TI1 using the following procedure (depending on CK_INT frequency and prescaler value):

1. Select the proper TI1[x] source (internal or external) with the TI1SEL[3:0] bits in the TIMx_TISEL register.
2. Select the active input for TIMx_CCR1: write the CC1S bits to '01' in the TIMx_CCMR1 register (TI1 selected).
3. Select the active polarity for TI1FP1 (used both for capture in TIMx_CCR1 and counter clear): program the CC1P and CC1NP bits to '00' (active on rising edge).
4. Select the active input for TIMx_CCR2: write the CC2S bits to '10' in the TIMx_CCMR1 register (TI1 selected).
5. Select the active polarity for TI1FP2 (used for capture in TIMx_CCR2): program the CC2P and CC2NP bits to '11' (active on falling edge).
6. Select the valid trigger input: write the TS bits to '00101' in the TIMx_SMCR register (TI1FP1 selected).
7. Configure the slave mode controller in reset mode: write the SMS bits to '100' in the TIMx_SMCR register.
8. Enable the captures: write the CC1E and CC2E bits to '1' in the TIMx_CCER register.

Figure 469. PWM input mode timing

1. The PWM input mode can be used only with the TIMx_CH1/TIMx_CH2 signals due to the fact that only TI1FP1 and TI2FP2 are connected to the slave mode controller.

40.3.7 Forced output mode

In output mode (CCxS bits = '00' in the TIMx_CCMRx register), each output compare signal (OCxREF and then OCx) can be forced to active or inactive level directly by software, independently of any comparison between the output compare register and the counter.

To force an output compare signal (OCXREF/OCx) to its active level, one just needs to write '0101' in the OCxM bits in the corresponding TIMx_CCMRx register. Thus OCXREF is forced high (OCxREF is always active high) and OCx get opposite value to CCxP polarity bit.

For example: CCxP='0' (OCx active high) => OCx is forced to high level.

The OCxREF signal can be forced low by writing the OCxM bits to '0100' in the TIMx_CCMRx register.

Anyway, the comparison between the TIMx_CCRx shadow register and the counter is still performed and allows the flag to be set. Interrupt requests can be sent accordingly. This is described in the output compare mode section below.

40.3.8 Output compare mode

This function is used to control an output waveform or indicating when a period of time has elapsed.

When a match is found between the capture/compare register and the counter, the output compare function:

1. Assigns the corresponding output pin to a programmable value defined by the output compare mode (OCxM bits in the TIMx_CCMRx register) and the output polarity (CCxP bit in the TIMx_CCER register). The output pin can keep its level (OCxM='0000'), be set active (OCxM='0001'), be set inactive (OCxM='0010') or can toggle (OCxM='0011') on match.
2. Sets a flag in the interrupt status register (CCxIF bit in the TIMx_SR register).
3. Generates an interrupt if the corresponding interrupt mask is set (CCxIE bit in the TIMx_DIER register).

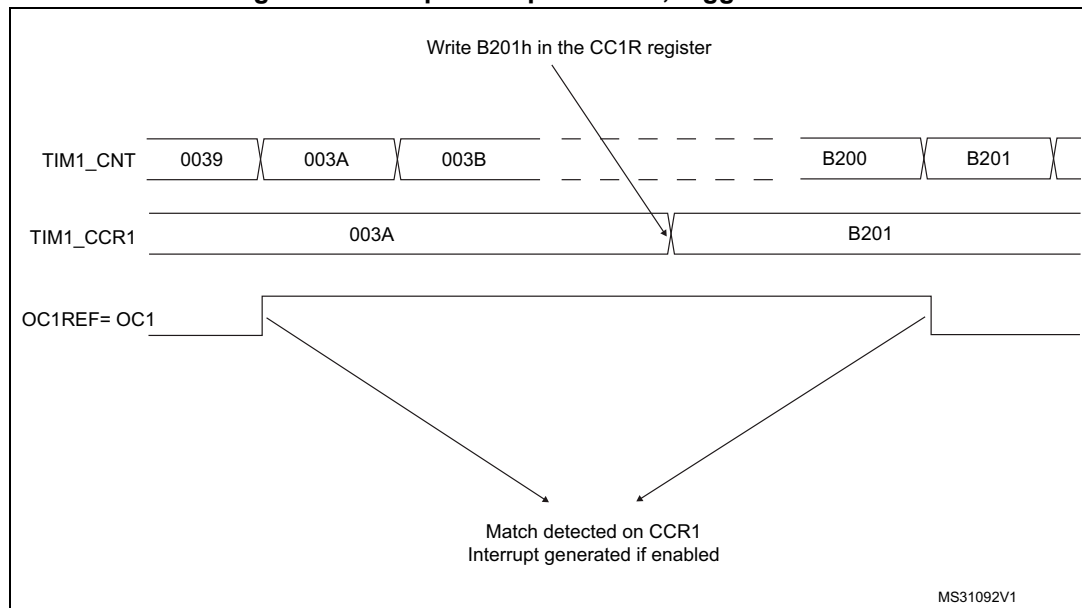
The TIMx_CCRx registers can be programmed with or without preload registers using the OCxPE bit in the TIMx_CCMRx register.

In output compare mode, the update event UEV has no effect on OCxREF and OCx output. The timing resolution is one count of the counter. Output compare mode can also be used to output a single pulse (in One-pulse mode).

Procedure:

1. Select the counter clock (internal, external, prescaler).
2. Write the desired data in the TIMx_ARR and TIMx_CCRx registers.
3. Set the CCxIE bit if an interrupt request is to be generated.
4. Select the output mode. For example:
 - Write OCxM = '0011' to toggle OCx output pin when CNT matches CCRx
 - Write OCxPE = '0' to disable preload register
 - Write CCxP = '0' to select active high polarity
 - Write CCxE = '1' to enable the output
5. Enable the counter by setting the CEN bit in the TIMx_CR1 register.

The TIMx_CCRx register can be updated at any time by software to control the output waveform, provided that the preload register is not enabled (OCxPE='0', else TIMx_CCRx shadow register is updated only at the next update event UEV). An example is given in [Figure 470](#).

Figure 470. Output compare mode, toggle on OC1.

40.3.9 PWM mode

Pulse Width Modulation mode allows to generate a signal with a frequency determined by the value of the TIMx_ARR register and a duty cycle determined by the value of the TIMx_CCRx register.

The PWM mode can be selected independently on each channel (one PWM per OCx output) by writing '0110' (PWM mode 1) or '0111' (PWM mode 2) in the OCxM bits in the TIMx_CCMRx register. The corresponding preload register must be enabled by setting the OCxPE bit in the TIMx_CCMRx register, and eventually the auto-reload preload register (in upcounting or center-aligned modes) by setting the ARPE bit in the TIMx_CR1 register.

As the preload registers are transferred to the shadow registers only when an update event occurs, before starting the counter, all registers must be initialized by setting the UG bit in the TIMx_EGR register.

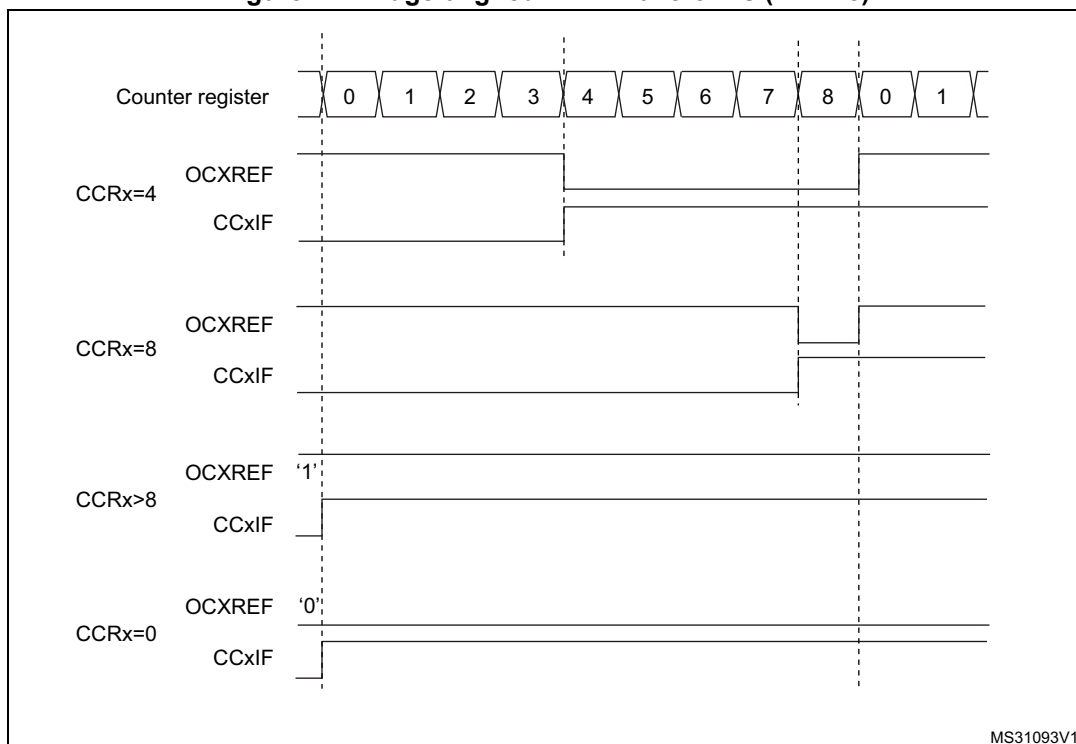
The OCx polarity is software programmable using the CCxP bit in the TIMx_CCER register. It can be programmed as active high or active low. The OCx output is enabled by the CCxE bit in the TIMx_CCER register. Refer to the TIMx_CCERx register description for more details.

In PWM mode (1 or 2), TIMx_CNT and TIMx_CCRx are always compared to determine whether $TIMx_CNT \leq TIMx_CCRx$.

The timer is able to generate PWM in edge-aligned mode only since the counter is upcounting.

In the following example, we consider PWM mode 1. The reference PWM signal OCxREF is high as long as $TIMx_CNT < TIMx_CCRx$ else it becomes low. If the compare value in TIMx_CCRx is greater than the auto-reload value (in TIMx_ARR) then OCxREF is held at '1'. If the compare value is 0 then OCxRef is held at '0'. [Figure 471](#) shows some edge-aligned PWM waveforms in an example where TIMx_ARR=8.

Figure 471. Edge-aligned PWM waveforms (ARR=8)



MS31093V1

40.3.10 Combined PWM mode (TIM12 only)

Combined PWM mode allows two edge or center-aligned PWM signals to be generated with programmable delay and phase shift between respective pulses. While the frequency is determined by the value of the TIMx_ARR register, the duty cycle and delay are determined by the two TIMx_CCRx registers. The resulting signals, OCxREFC, are made of an OR or AND logical combination of two reference PWMs:

- OC1REFC (or OC2REFC) is controlled by the TIMx_CCR1 and TIMx_CCR2 registers

Combined PWM mode can be selected independently on two channels (one OCx output per pair of CCR registers) by writing '1100' (Combined PWM mode 1) or '1101' (Combined PWM mode 2) in the OCxM bits in the TIMx_CCMRx register.

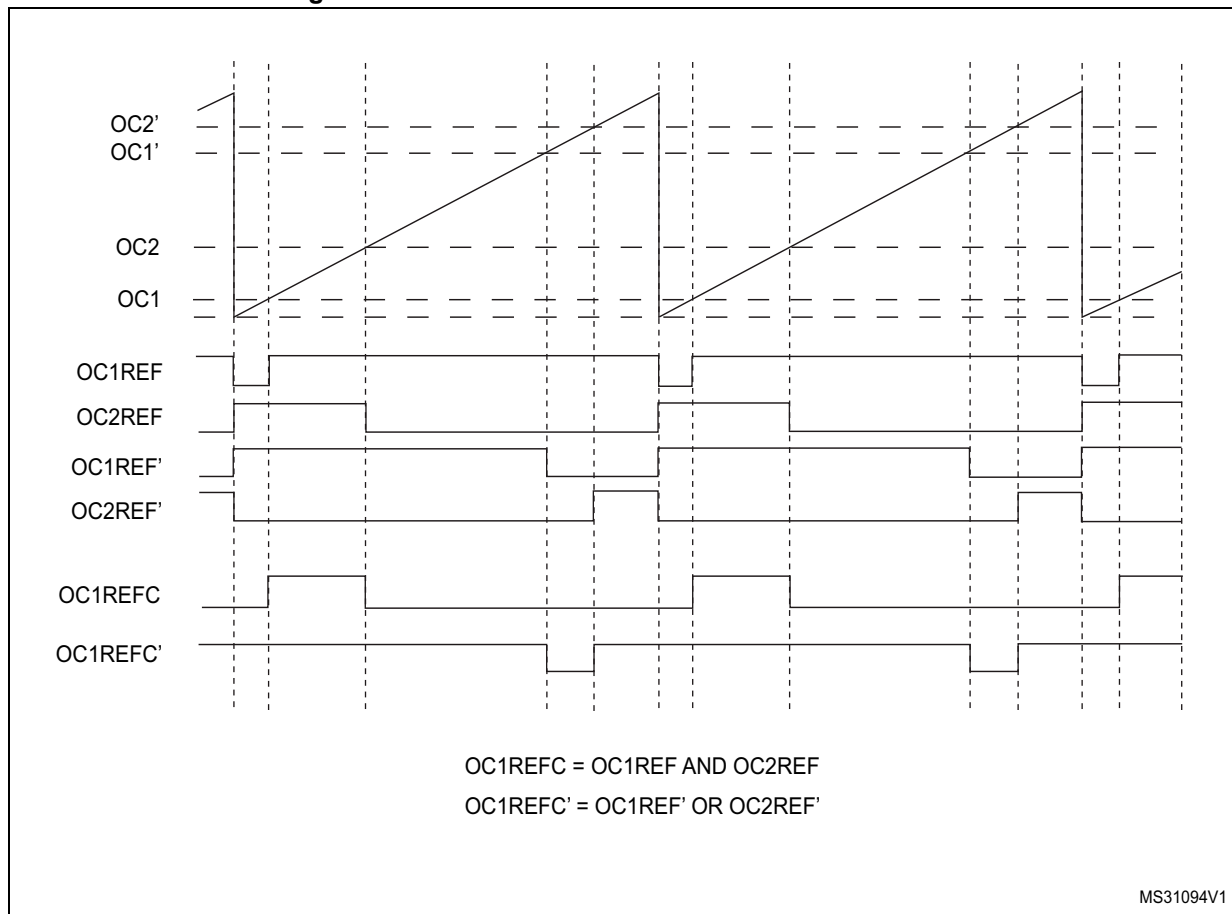
When a given channel is used as a combined PWM channel, its complementary channel must be configured in the opposite PWM mode (for instance, one in Combined PWM mode 1 and the other in Combined PWM mode 2).

Note: The OCxM[3:0] bit field is split into two parts for compatibility reasons, the most significant bit is not contiguous with the 3 least significant ones.

Figure 472 represents an example of signals that can be generated using combined PWM mode, obtained with the following configuration:

- Channel 1 is configured in Combined PWM mode 2,
- Channel 2 is configured in PWM mode 1,

Figure 472. Combined PWM mode on channel 1 and 2



40.3.11 One-pulse mode

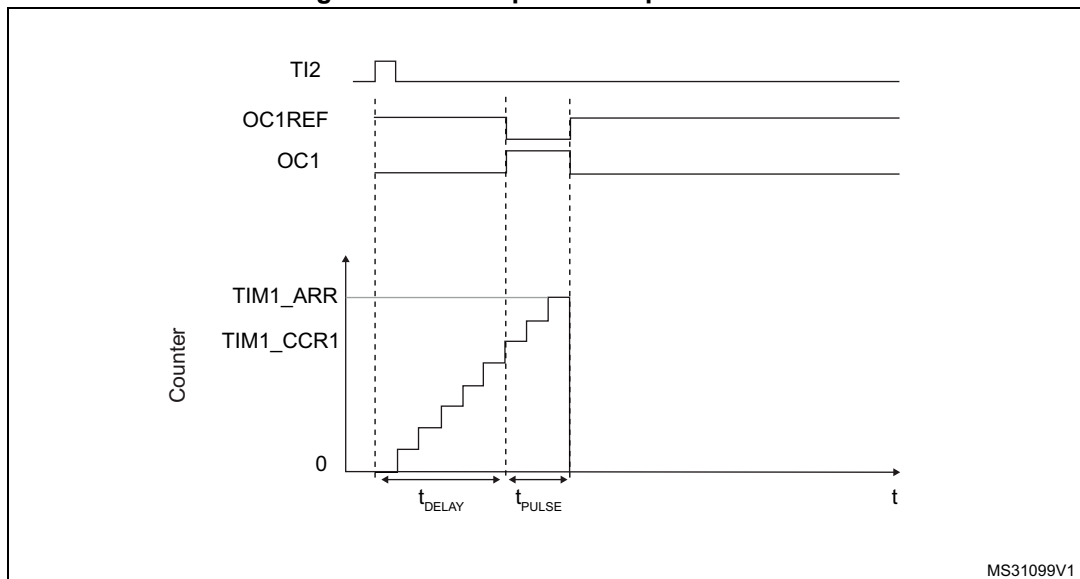
One-pulse mode (OPM) is a particular case of the previous modes. It allows the counter to be started in response to a stimulus and to generate a pulse with a programmable length after a programmable delay.

Starting the counter can be controlled through the slave mode controller. Generating the waveform can be done in output compare mode or PWM mode. One-pulse mode is selected by setting the OPM bit in the TIMx_CR1 register. This makes the counter stop automatically at the next update event UEV.

A pulse can be correctly generated only if the compare value is different from the counter initial value. Before starting (when the timer is waiting for the trigger), the configuration must be as follows:

$$CNT < CCRx \leq ARR \text{ (in particular, } 0 < CCRx)$$

Figure 473. Example of one pulse mode.



For example one may want to generate a positive pulse on OC1 with a length of t_{PULSE} and after a delay of t_{DELAY} as soon as a positive edge is detected on the TI2 input pin.

Use TI2FP2 as trigger 1:

1. Select the proper TI2[x] source (internal or external) with the TI2SEL[3:0] bits in the TIMx_TISEL register.
2. Map TI2FP2 to TI2 by writing CC2S='01' in the TIMx_CCMR1 register.
3. TI2FP2 must detect a rising edge, write CC2P='0' and CC2NP='0' in the TIMx_CCER register.
4. Configure TI2FP2 as trigger for the slave mode controller (TRGI) by writing TS='00110' in the TIMx_SMCR register.
5. TI2FP2 is used to start the counter by writing SMS to '110' in the TIMx_SMCR register (trigger mode).

The OPM waveform is defined by writing the compare registers (taking into account the clock frequency and the counter prescaler).

- The t_{DELAY} is defined by the value written in the TIMx_CCR1 register.
- The t_{PULSE} is defined by the difference between the auto-reload value and the compare value (TIMx_ARR - TIMx_CCR1).
- Let's say one want to build a waveform with a transition from '0' to '1' when a compare match occurs and a transition from '1' to '0' when the counter reaches the auto-reload value. To do this PWM mode 2 must be enabled by writing OC1M='0111' in the TIMx_CCMR1 register. Optionally the preload registers can be enabled by writing OC1PE='1' in the TIMx_CCMR1 register and ARPE in the TIMx_CR1 register. In this case one has to write the compare value in the TIMx_CCR1 register, the auto-reload value in the TIMx_ARR register, generate an update by setting the UG bit and wait for external trigger event on TI2. CC1P is written to '0' in this example.

Since only 1 pulse (Single mode) is needed, a 1 must be written in the OPM bit in the TIMx_CR1 register to stop the counter at the next update event (when the counter rolls over from the auto-reload value back to 0). When OPM bit in the TIMx_CR1 register is set to '0', so the Repetitive Mode is selected.

Particular case: OCx fast enable

In One-pulse mode, the edge detection on TIx input set the CEN bit which enables the counter. Then the comparison between the counter and the compare value makes the output toggle. But several clock cycles are needed for these operations and it limits the minimum delay $t_{\text{DELAY min}}$ we can get.

If one wants to output a waveform with the minimum delay, the OCxFE bit can be set in the TIMx_CCMRx register. Then OCxRef (and OCx) are forced in response to the stimulus, without taking in account the comparison. Its new level is the same as if a compare match had occurred. OCxFE acts only if the channel is configured in PWM1 or PWM2 mode.

40.3.12 Retriggerable one pulse mode (OPM) (TIM12 only)

This mode allows the counter to be started in response to a stimulus and to generate a pulse with a programmable length, but with the following differences with non-retriggerable one pulse mode described in [Section 40.3.11: One-pulse mode](#):

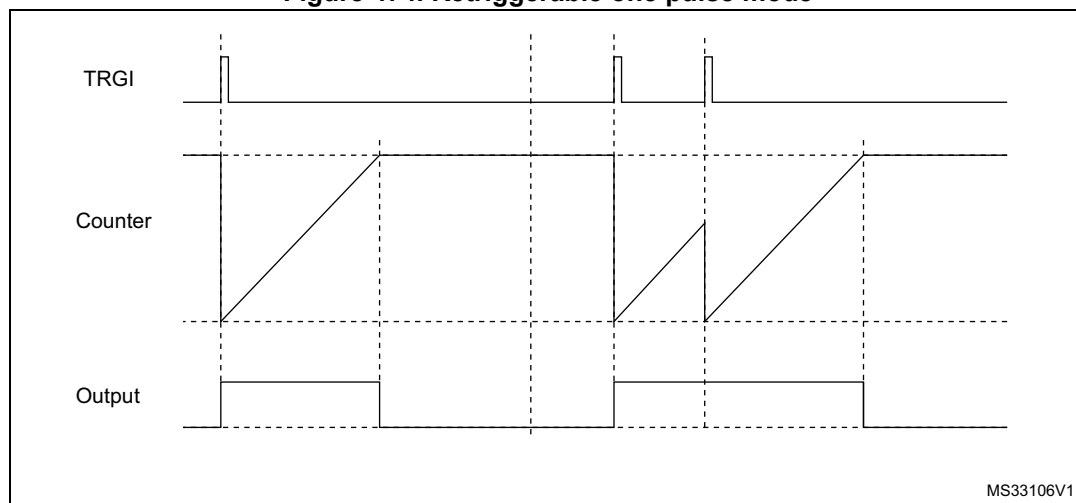
- The pulse starts as soon as the trigger occurs (no programmable delay)
- The pulse is extended if a new trigger occurs before the previous one is completed

The timer must be in Slave mode, with the bits SMS[3:0] = '1000' (Combined Reset + trigger mode) in the TIMx_SMCR register, and the OCxM[3:0] bits set to '1000' or '1001' for retriggerable OPM mode 1 or 2.

If the timer is configured in up-counting mode, the corresponding CCRx must be set to 0 (the ARR register sets the pulse length). If the timer is configured in down-counting mode, CCRx must be above or equal to ARR.

Note: The OCxM[3:0] and SMS[3:0] bit fields are split into two parts for compatibility reasons, the most significant bit are not contiguous with the 3 least significant ones.

This mode must not be used with center-aligned PWM modes. It is mandatory to have CMS[1:0] = 00 in TIMx_CR1.

Figure 474. Retriggerable one pulse mode

40.3.13 UIF bit remapping

The IUFREMAP bit in the TIMx_CR1 register forces a continuous copy of the Update Interrupt Flag UIF into bit 31 of the timer counter register (TIMxCNT[31]). This allows to atomically read both the counter value and a potential roll-over condition signaled by the UIFCPY flag. In particular cases, it can ease the calculations by avoiding race conditions caused for instance by a processing shared between a background task (counter reading) and an interrupt (Update Interrupt).

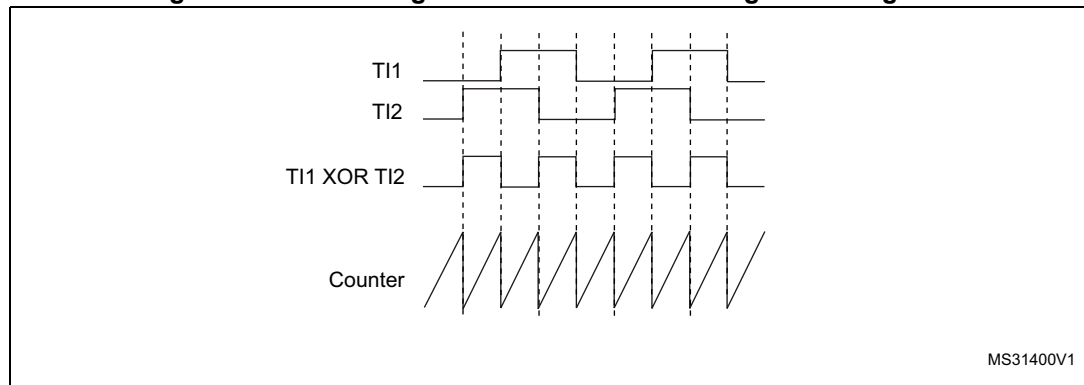
There is no latency between the assertions of the UIF and UIFCPY flags.

40.3.14 Timer input XOR function

The TI1S bit in the TIMx_CR2 register, allows the input filter of channel 1 to be connected to the output of a XOR gate, combining the two input pins TIMx_CH1 and TIMx_CH2.

The XOR output can be used with all the timer input functions such as trigger or input capture. It is useful for measuring the interval between the edges on two input signals, as shown in [Figure 475](#).

Figure 475. Measuring time interval between edges on 2 signals



40.3.15 TIM12 external trigger synchronization

The TIM12 timer can be synchronized with an external trigger in several modes: Reset mode, Gated mode and Trigger mode.

Slave mode: Reset mode

The counter and its prescaler can be reinitialized in response to an event on a trigger input. Moreover, if the URS bit from the TIMx_CR1 register is low, an update event UEV is generated. Then all the preloaded registers (TIMx_ARR, TIMx_CCRx) are updated.

In the following example, the upcounter is cleared in response to a rising edge on TI1 input:

1. Configure the channel 1 to detect rising edges on TI1. Configure the input filter duration (in this example, we do not need any filter, so we keep IC1F='0000'). The capture prescaler is not used for triggering, so it does not need to be configured. The CC1S bits select the input capture source only, CC1S = '01' in the TIMx_CCMR1 register.

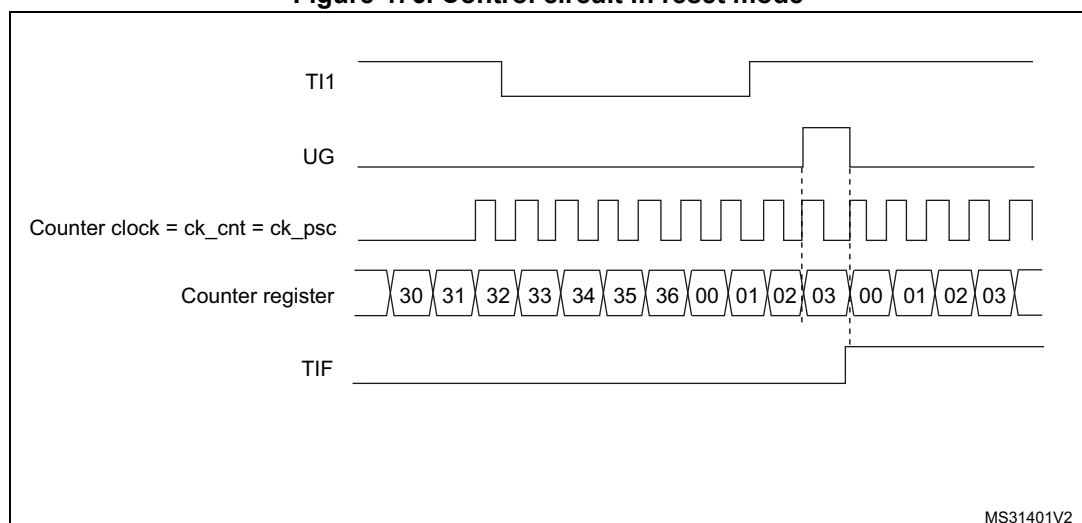
Program CC1P and CC1NP to '00' in TIMx_CCER register to validate the polarity (and detect rising edges only).

2. Configure the timer in reset mode by writing SMS='100' in TIMx_SMCR register. Select TI1 as the input source by writing TS='00101' in TIMx_SMCR register.
3. Start the counter by writing CEN='1' in the TIMx_CR1 register.

The counter starts counting on the internal clock, then behaves normally until TI1 rising edge. When TI1 rises, the counter is cleared and restarts from 0. In the meantime, the trigger flag is set (TIF bit in the TIMx_SR register) and an interrupt request can be sent if enabled (depending on the TIE bit in TIMx_DIER register).

The following figure shows this behavior when the auto-reload register TIMx_ARR=0x36. The delay between the rising edge on TI1 and the actual reset of the counter is due to the resynchronization circuit on TI1 input.

Figure 476. Control circuit in reset mode



MS31401V2

Slave mode: Gated mode

The counter can be enabled depending on the level of a selected input.

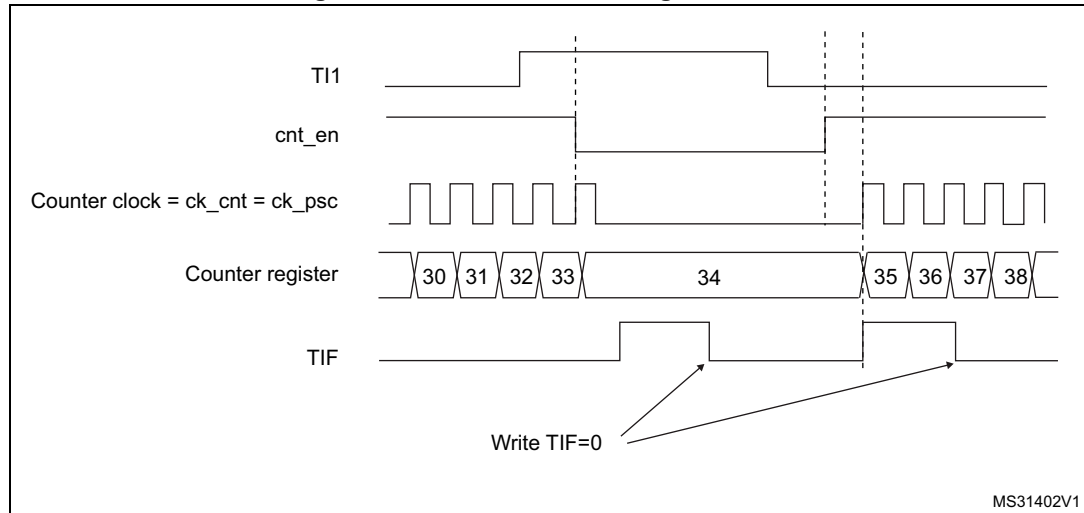
In the following example, the upcounter counts only when TI1 input is low:

1. Configure the channel 1 to detect low levels on TI1. Configure the input filter duration (in this example, we do not need any filter, so we keep IC1F='0000'). The capture prescaler is not used for triggering, so it does not need to be configured. The CC1S bits select the input capture source only, CC1S='01' in TIMx_CCMR1 register. Program CC1P='1' and CC1NP='0' in TIMx_CCER register to validate the polarity (and detect low level only).
2. Configure the timer in gated mode by writing SMS='101' in TIMx_SMCR register. Select TI1 as the input source by writing TS='00101' in TIMx_SMCR register.
3. Enable the counter by writing CEN='1' in the TIMx_CR1 register (in gated mode, the counter doesn't start if CEN='0', whatever is the trigger input level).

The counter starts counting on the internal clock as long as TI1 is low and stops as soon as TI1 becomes high. The TIF flag in the TIMx_SR register is set both when the counter starts or stops.

The delay between the rising edge on TI1 and the actual stop of the counter is due to the resynchronization circuit on TI1 input.

Figure 477. Control circuit in gated mode



Slave mode: Trigger mode

The counter can start in response to an event on a selected input.

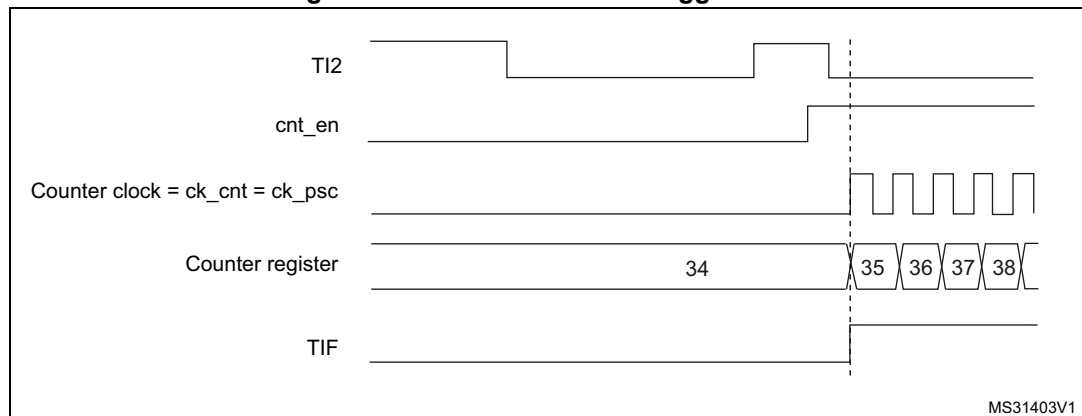
In the following example, the upcounter starts in response to a rising edge on TI2 input:

1. Configure the channel 2 to detect rising edges on TI2. Configure the input filter duration (in this example, we do not need any filter, so we keep IC2F='0000'). The capture prescaler is not used for triggering, so it does not need to be configured. The CC2S bits are configured to select the input capture source only, CC2S='01' in TIMx_CCMR1 register. Program CC2P='1' and CC2NP='0' in TIMx_CCER register to validate the polarity (and detect low level only).
2. Configure the timer in trigger mode by writing SMS='110' in TIMx_SMCR register. Select TI2 as the input source by writing TS='00110' in TIMx_SMCR register.

When a rising edge occurs on TI2, the counter starts counting on the internal clock and the TIF flag is set.

The delay between the rising edge on TI2 and the actual start of the counter is due to the resynchronization circuit on TI2 input.

Figure 478. Control circuit in trigger mode



40.3.16 Slave mode – combined reset + trigger mode

In this case, a rising edge of the selected trigger input (TRGI) reinitializes the counter, generates an update of the registers, and starts the counter.

This mode is used for one-pulse mode.

40.3.17 Timer synchronization (TIM12)

The TIM timers are linked together internally for timer synchronization or chaining. Refer to [Section 39.3.19: Timer synchronization](#) for details.

Note: *The clock of the slave timer must be enabled prior to receive events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.*

40.3.18 Debug mode

When the microcontroller enters debug mode (Cortex[®]-M7 with FPU core halted), the TIMx counter either continues to work normally or stops, depending on DBG_TIMx_STOP configuration bit in DBGMCU module. For more details, refer to [Section 60.5.8: Microcontroller debug unit \(DBGMCU\)](#).

40.4 TIM12 registers

Refer to [Section 1.2](#) for a list of abbreviations used in register descriptions.

The peripheral registers have to be written by half-words (16 bits) or words (32 bits). Read accesses can be done by bytes (8 bits), half-words (16 bits) or words (32 bits).

40.4.1 TIM12 control register 1 (TIM12_CR1)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	UIFRE MAP	Res.	CKD[1:0]		ARPE	Res.	Res.	Res.	OPM	URS	UDIS	CEN
				rw		rw	rw	rw				rw	rw	rw	rw

Bits 15:12 Reserved, must be kept at reset value.

Bit 11 **UIFREMAP**: UIF status bit remapping

0: No remapping. UIF status bit is not copied to TIMx_CNT register bit 31.

1: Remapping enabled. UIF status bit is copied to TIMx_CNT register bit 31.

Bit 10 Reserved, must be kept at reset value.

Bits 9:8 **CKD[1:0]**: Clock division

This bit-field indicates the division ratio between the timer clock (CK_INT) frequency and sampling clock used by the digital filters (Tlx),

00: $t_{DTS} = t_{CK_INT}$

01: $t_{DTS} = 2 \times t_{CK_INT}$

10: $t_{DTS} = 4 \times t_{CK_INT}$

11: Reserved

Bit 7 **ARPE**: Auto-reload preload enable

0: TIMx_ARR register is not buffered.

1: TIMx_ARR register is buffered.

Bits 6:4 Reserved, must be kept at reset value.

Bit 3 **OPM**: One-pulse mode

0: Counter is not stopped on the update event

1: Counter stops counting on the next update event (clearing the CEN bit).

Bit 2 **URS**: Update request source

This bit is set and cleared by software to select the UEV event sources.

0: Any of the following events generates an update interrupt if enabled. These events can be:

- Counter overflow
- Setting the UG bit
- Update generation through the slave mode controller

1: Only counter overflow generates an update interrupt if enabled.

Bit 1 **UDIS**: Update disable

This bit is set and cleared by software to enable/disable update event (UEV) generation.

0: UEV enabled. An UEV is generated by one of the following events:

- Counter overflow
- Setting the UG bit

Buffered registers are then loaded with their preload values.

1: UEV disabled. No UEV is generated, shadow registers keep their value (ARR, PSC, CCRx). The counter and the prescaler are reinitialized if the UG bit is set.

Bit 0 **CEN**: Counter enable

0: Counter disabled

1: Counter enabled

CEN is cleared automatically in one-pulse mode, when an update event occurs.

Note: External clock and gated mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.

40.4.2 TIM12 control register 2 (TIM12_CR2)

Address offset: 0x04

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TI1S	MMS[2:0]			Res.	Res.	Res.	Res.
								rw	rw	rw	rw				

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **TI1S**: TI1 selection

- 0: The TIM12_CH1 pin is connected to TI1 input
- 1: The TIM12_CH1, CH2 pins are connected to the TI1 input (XOR combination)

Bits 6:4 **MMS[2:0]**: Master mode selection

These bits allow to select the information to be sent in master mode to slave timers for synchronization (TRGO). The combination is as follows:

- 000: Reset - the UG bit from the TIMx_EGR register is used as trigger output (TRGO). If the reset is generated by the trigger input (slave mode controller configured in reset mode) then the signal on TRGO is delayed compared to the actual reset.
- 001: Enable - the Counter Enable signal CNT_EN is used as trigger output (TRGO). It is useful to start several timers at the same time or to control a window in which a slave timer is enable. The Counter Enable signal is generated by a logic AND between CEN control bit and the trigger input when configured in gated mode. When the Counter Enable signal is controlled by the trigger input, there is a delay on TRGO, except if the master/slave mode is selected (see the MSM bit description in TIMx_SMCR register).
- 010: Update - The update event is selected as trigger output (TRGO). For instance a master timer can then be used as a prescaler for a slave timer.
- 011: Compare Pulse - The trigger output send a positive pulse when the CC1IF flag is to be set (even if it was already high), as soon as a capture or a compare match occurred. (TRGO).
- 100: Compare - OC1REF signal is used as trigger output (TRGO).
- 101: Compare - OC2REF signal is used as trigger output (TRGO).

Bits 3:0 Reserved, must be kept at reset value.

40.4.3 TIM12 slave mode control register (TIM12_SMCR)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TS[4:3]		Res.	Res.	Res.	SMS[3]
										rw	rw				rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MSM	TS[2:0]			Res.	SMS[2:0]		
								rw	rw	rw	rw		rw	rw	rw

Bits 31:22 Reserved, must be kept at reset value.

Bits 19:17 Reserved, must be kept at reset value.

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **MSM**: Master/Slave mode

- 0: No action
- 1: The effect of an event on the trigger input (TRGI) is delayed to allow a perfect synchronization between the current timer and its slaves (through TRGO). It is useful in order to synchronize several timers on a single external event.

Bits 21, 20, 6, 5, 4 **TS[4:0]**: Trigger selection

This TS[4:0] bitfield selects the trigger input to be used to synchronize the counter.

00000: Internal Trigger 0 (ITR0)

00001: Internal Trigger 1 (ITR1)

00010: Internal Trigger 2 (ITR2)

00011: Internal Trigger 3 (ITR3)

00100: TI1 Edge Detector (TI1F_ED)

00101: Filtered Timer Input 1 (TI1FP1)

00110: Filtered Timer Input 2 (TI2FP2)

Others: Reserved

See [Table 337: TIMx internal trigger connection on page 1754](#) for more details on the meaning of ITRx for each timer.

Note: These bits must be changed only when they are not used (e.g. when SMS='000') to avoid wrong edge detections at the transition.

Bit 3 Reserved, must be kept at reset value.

Bits 16, 2, 1, 0 **SMS[3:0]**: Slave mode selection

When external signals are selected the active edge of the trigger signal (TRGI) is linked to the polarity selected on the external input (see Input Control register and Control Register description).

0000: Slave mode disabled - if CEN = '1' then the prescaler is clocked directly by the internal clock.

0001: Reserved

0010: Reserved

0011: Reserved

0100: Reset Mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter and generates an update of the registers.

0101: Gated Mode - The counter clock is enabled when the trigger input (TRGI) is high. The counter stops (but is not reset) as soon as the trigger becomes low. Both start and stop of the counter are controlled.

0110: Trigger Mode - The counter starts at a rising edge of the trigger TRGI (but it is not reset). Only the start of the counter is controlled.

0111: External Clock Mode 1 - Rising edges of the selected trigger (TRGI) clock the counter.

1000: Combined reset + trigger mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter, generates an update of the registers and starts the counter.

Other codes: reserved.

Note: The gated mode must not be used if TI1F_ED is selected as the trigger input (TS='00100'). Indeed, TI1F_ED outputs 1 pulse for each transition on TI1F, whereas the gated mode checks the level of the trigger signal.

Note: The clock of the slave timer must be enabled prior to receive events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

Table 337. TIMx internal trigger connection

Slave TIM	ITR0 (TS = '00000')	ITR1 (TS = '00001')	ITR2 (TS = '00010')	ITR3 (TS = '00011')
TIM12	TIM4	TIM5	TIM13 OC1	TIM14 OC1

40.4.4 TIM12 Interrupt enable register (TIM12_DIER)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIE	Res.	Res.	Res.	CC2IE	CC1IE	UIE
									rw				rw	rw	rw

Bits 15:7 Reserved, must be kept at reset value.

Bit 6 **TIE**: Trigger interrupt enable
 0: Trigger interrupt disabled.
 1: Trigger interrupt enabled.

Bits 5:3 Reserved, must be kept at reset value.

Bit 2 **CC2IE**: Capture/Compare 2 interrupt enable
 0: CC2 interrupt disabled.
 1: CC2 interrupt enabled.

Bit 1 **CC1IE**: Capture/Compare 1 interrupt enable
 0: CC1 interrupt disabled.
 1: CC1 interrupt enabled.

Bit 0 **UIE**: Update interrupt enable
 0: Update interrupt disabled.
 1: Update interrupt enabled.

40.4.5 TIM12 status register (TIM12_SR)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	CC2OF	CC1OF	Res.	Res.	TIF	Res.	Res.	Res.	CC2IF	CC1IF	UIF
					rc_w0	rc_w0			rc_w0				rc_w0	rc_w0	rc_w0

Bits 15:11 Reserved, must be kept at reset value.

Bit 10 **CC2OF**: Capture/compare 2 overcapture flag
 refer to CC1OF description

Bit 9 **CC1OF**: Capture/Compare 1 overcapture flag
 This flag is set by hardware only when the corresponding channel is configured in input capture mode. It is cleared by software by writing it to '0'.
 0: No overcapture has been detected.
 1: The counter value has been captured in TIMx_CCR1 register while CC1IF flag was already set

Bits 8:7 Reserved, must be kept at reset value.

Bit 6 **TIF**: Trigger interrupt flag

This flag is set by hardware on the TRG trigger event (active edge detected on TRGI input when the slave mode controller is enabled in all modes but gated mode. It is set when the counter starts or stops when gated mode is selected. It is cleared by software.

0: No trigger event occurred.

1: Trigger interrupt pending.

Bits 5:3 Reserved, must be kept at reset value.

Bit 2 **CC2IF**: Capture/Compare 2 interrupt flag

refer to CC1IF description

Bit 1 **CC1IF**: Capture/compare 1 interrupt flag

This flag is set by hardware. It is cleared by software (input capture or output compare mode) or by reading the TIMx_CCR1 register (input capture mode only).

0: No compare match / No input capture occurred

1: A compare match or an input capture occurred.

If channel CC1 is configured as output: this flag is set when the content of the counter TIMx_CNT matches the content of the TIMx_CCR1 register. When the content of TIMx_CCR1 is greater than the content of TIMx_ARR, the CC1IF bit goes high on the counter overflow (in up-counting and up/down-counting modes) or underflow (in down-counting mode). There are 3 possible options for flag setting in center-aligned mode, refer to the CMS bits in the TIMx_CR1 register for the full description.

If channel CC1 is configured as input: this bit is set when counter value has been captured in TIMx_CCR1 register (an edge has been detected on IC1, as per the edge sensitivity defined with the CC1P and CC1NP bits setting, in TIMx_CCER).

Bit 0 **UIF**: Update interrupt flag

This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred.

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

- At overflow and if UDIS='0' in the TIMx_CR1 register.
- When CNT is reinitialized by software using the UG bit in TIMx_EGR register, if URS='0' and UDIS='0' in the TIMx_CR1 register.
- When CNT is reinitialized by a trigger event (refer to the synchro control register description), if URS='0' and UDIS='0' in the TIMx_CR1 register.

40.4.6 TIM12 event generation register (TIM12_EGR)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TG	Res.	Res.	Res.	CC2G	CC1G	UG
									w				w	w	w

Bits 15:7 Reserved, must be kept at reset value.

Bit 6 **TG**: Trigger generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: The TIF flag is set in the TIMx_SR register. Related interrupt can occur if enabled

Bits 5:3 Reserved, must be kept at reset value.

Bit 2 **CC2G**: Capture/compare 2 generation
refer to CC1G description

Bit 1 **CC1G**: Capture/compare 1 generation

This bit is set by software to generate an event, it is automatically cleared by hardware.

0: No action

1: A capture/compare event is generated on channel 1:

If channel CC1 is configured as output:

the CC1IF flag is set, the corresponding interrupt is sent if enabled.

If channel CC1 is configured as input:

The current counter value is captured in the TIMx_CCR1 register. The CC1IF flag is set, the corresponding interrupt is sent if enabled. The CC1OF flag is set if the CC1IF flag was already high.

Bit 0 **UG**: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action

1: Re-initializes the counter and generates an update of the registers. The prescaler counter is also cleared and the prescaler ratio is not affected. The counter is cleared.

40.4.7 TIM12 capture/compare mode register 1 [alternate] (TIM12_CCMR1)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for input capture mode (this section) or for output compare mode (next section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits in this register have different functions in input and output modes.

Input capture mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IC2F[3:0]				IC2PSC[1:0]		CC2S[1:0]		IC1F[3:0]				IC1PSC[1:0]		CC1S[1:0]	
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:12 **IC2F[3:0]**: Input capture 2 filter

Bits 11:10 **IC2PSC[1:0]**: Input capture 2 prescaler

Bits 9:8 **CC2S[1:0]**: Capture/compare 2 selection

This bitfield defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output

01: CC2 channel is configured as input, IC2 is mapped on TI2

10: CC2 channel is configured as input, IC2 is mapped on TI1

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode works only if an internal trigger input is selected through the TS bit (TIMx_SMCR register)

Note: The CC2S bits are writable only when the channel is OFF (CC2E = 0 in TIMx_CCER).

Bits 7:4 IC1F[3:0]: Input capture 1 filter

This bitfield defines the frequency used to sample the TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, N=2

0010: $f_{SAMPLING}=f_{CK_INT}$, N=4

0011: $f_{SAMPLING}=f_{CK_INT}$, N=8

0100: $f_{SAMPLING}=f_{DTS}/2$, N=6

0101: $f_{SAMPLING}=f_{DTS}/2$, N=8

0110: $f_{SAMPLING}=f_{DTS}/4$, N=6

0111: $f_{SAMPLING}=f_{DTS}/4$, N=8

1000: $f_{SAMPLING}=f_{DTS}/8$, N=6

1001: $f_{SAMPLING}=f_{DTS}/8$, N=8

1010: $f_{SAMPLING}=f_{DTS}/16$, N=5

1011: $f_{SAMPLING}=f_{DTS}/16$, N=6

1100: $f_{SAMPLING}=f_{DTS}/16$, N=8

1101: $f_{SAMPLING}=f_{DTS}/32$, N=5

1110: $f_{SAMPLING}=f_{DTS}/32$, N=6

1111: $f_{SAMPLING}=f_{DTS}/32$, N=8

Bits 3:2 IC1PSC[1:0]: Input capture 1 prescaler

This bitfield defines the ratio of the prescaler acting on the CC1 input (IC1).

The prescaler is reset as soon as CC1E='0' (TIMx_CCER register).

00: no prescaler, capture is done each time an edge is detected on the capture input

01: capture is done once every 2 events

10: capture is done once every 4 events

11: capture is done once every 8 events

Bits 1:0 CC1S[1:0]: Capture/Compare 1 selection

This bitfield defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: CC1 channel is configured as input, IC1 is mapped on TI2

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: The CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

40.4.8 TIM12 capture/compare mode register 1 [alternate] (TIM12_CCMR1)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for output compare mode (this section) or for input capture mode (previous section). The direction of a channel is defined by configuring the

corresponding CCxS bits. All the other bits in this register have different functions in input and output modes.

Output compare mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC2M [3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M [3]
							rw								rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	OC2M[2:0]			OC2PE	OC2FE	CC2S[1:0]		Res.	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw	rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bits 23:17 Reserved, must be kept at reset value.

Bit 15 Reserved, must be kept at reset value.

Bits 24, 14:12 **OC2M[3:0]**: Output compare 2 mode

Refer to OC1M[3:0] for bit description.

Bit 11 **OC2PE**: Output compare 2 preload enable

Bit 10 **OC2FE**: Output compare 2 fast enable

Bits 9:8 **CC2S[1:0]**: Capture/Compare 2 selection

This bitfield defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output

01: CC2 channel is configured as input, IC2 is mapped on TI2

10: CC2 channel is configured as input, IC2 is mapped on TI1

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode works only if an internal trigger input is selected through the TS bit (TIMx_SMCR register)

Note: The CC2S bits are writable only when the channel is OFF (CC2E = 0 in TIMx_CCER).

Bit 7 Reserved, must be kept at reset value.

Bits 16, 6:4 **OC1M[3:0]**: Output compare 1 mode (refer to bit 16 for OC1M[3])

These bits define the behavior of the output reference signal OC1REF from which OC1 is derived. OC1REF is active high whereas the active level of OC1 depends on the CC1P.

0000: Frozen - The comparison between the output compare register TIMx_CCR1 and the counter TIMx_CNT has no effect on the outputs.(this mode is used to generate a timing base).

0001: Set channel 1 to active level on match. The OC1REF signal is forced high when the TIMx_CNT counter matches the capture/compare register 1 (TIMx_CCR1).

0010: Set channel 1 to inactive level on match. The OC1REF signal is forced low when the TIMx_CNT counter matches the capture/compare register 1 (TIMx_CCR1).

0011: Toggle - OC1REF toggles when TIMx_CNT=TIMx_CCR1

0100: Force inactive level - OC1REF is forced low

0101: Force active level - OC1REF is forced high

0110: PWM mode 1 - channel 1 is active as long as TIMx_CNT<TIMx_CCR1 else it is inactive

0111: PWM mode 2 - channel 1 is inactive as long as TIMx_CNT<TIMx_CCR1 else it is active

1000: Retriggerable OPM mode 1 - The channel is active until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 1 and the channels becomes active again at the next update.

1001: Retriggerable OPM mode 2 - The channel is inactive until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 2 and the channels becomes inactive again at the next update.

1010: Reserved,

1011: Reserved,

1100: Combined PWM mode 1 - OC1REF has the same behavior as in PWM mode 1. OC1REFC is the logical OR between OC1REF and OC2REF.

1101: Combined PWM mode 2 - OC1REF has the same behavior as in PWM mode 2. OC1REFC is the logical AND between OC1REF and OC2REF.

1110: Reserved,

1111: Reserved

Note: In PWM mode 1 or 2, the OCREF level changes only when the result of the comparison changes or when the output compare mode switches from "frozen" mode to "PWM" mode.

Bit 3 **OC1PE**: Output compare 1 preload enable

0: Preload register on TIMx_CCR1 disabled. TIMx_CCR1 can be written at anytime, the new value is taken into account immediately

1: Preload register on TIMx_CCR1 enabled. Read/Write operations access the preload register. TIMx_CCR1 preload value is loaded into the active register at each update event

Note: The PWM mode can be used without validating the preload register only in one-pulse mode (OPM bit set in the TIMx_CR1 register). Else the behavior is not guaranteed.

Bit 2 **OC1FE**: Output compare 1 fast enable

This bit decreases the latency between a trigger event and a transition on the timer output. It must be used in one-pulse mode (OPM bit set in TIMx_CR1 register), to have the output pulse starting as soon as possible after the starting trigger.

0: CC1 behaves normally depending on the counter and CCR1 values even when the trigger is ON. The minimum delay to activate the CC1 output when an edge occurs on the trigger input is 5 clock cycles

1: An active edge on the trigger input acts like a compare match on the CC1 output. Then, OC is set to the compare level independently of the result of the comparison. Delay to sample the trigger input and to activate CC1 output is reduced to 3 clock cycles. OC1FE acts only if the channel is configured in PWM1 or PWM2 mode.

Bits 1:0 **CC1S[1:0]**: Capture/Compare 1 selection

This bitfield defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: CC1 channel is configured as input, IC1 is mapped on TI2

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode works only if an internal trigger input is selected through the TS bit (TIMx_SMCR register)

Note: The CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

40.4.9 TIM12 capture/compare enable register (TIM12_CCER)

Address offset: 0x20

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC2NP	Res.	CC2P	CC2E	CC1NP	Res.	CC1P	CC1E
								rw		rw	rw	rw		rw	rw

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **CC2NP**: Capture/Compare 2 output Polarity

Refer to CC1NP description

Bit 6 Reserved, must be kept at reset value.

Bit 5 **CC2P**: Capture/Compare 2 output Polarity

Refer to CC1P description

Bit 4 **CC2E**: Capture/Compare 2 output enable

Refer to CC1E description

Bit 3 **CC1NP**: Capture/Compare 1 complementary output Polarity

CC1 channel configured as output: CC1NP must be kept cleared

CC1 channel configured as input: CC1NP is used in conjunction with CC1P to define TI1FP1/TI2FP1 polarity (refer to CC1P description).

Bit 2 Reserved, must be kept at reset value.

Bit 1 **CC1P**: Capture/Compare 1 output Polarity.

0: OC1 active high (output mode) / Edge sensitivity selection (input mode, see below)

1: OC1 active low (output mode) / Edge sensitivity selection (input mode, see below)

When CC1 channel is configured as input, both CC1NP/CC1P bits select the active polarity of TI1FP1 and TI2FP1 for trigger or capture operations.

CC1NP=0, CC1P=0: non-inverted/rising edge. The circuit is sensitive to TIxFP1 rising edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode or encoder mode).

CC1NP=0, CC1P=1: inverted/falling edge. The circuit is sensitive to TIxFP1 falling edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is inverted (trigger operation in gated mode or encoder mode).

CC1NP=1, CC1P=1: non-inverted/both edges/ The circuit is sensitive to both TIxFP1 rising and falling edges (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode). This configuration must not be used in encoder mode.

CC1NP=1, CC1P=0: This configuration is reserved, it must not be used.

Bit 0 **CC1E**: Capture/Compare 1 output enable.

0: Capture mode disabled / OC1 is not active

1: Capture mode enabled / OC1 signal is output on the corresponding output pin

Table 338. Output control bit for standard OCx channels

CCxE bit	OCx output state
0	Output disabled (OCx='0', OCx_EN='0')
1	OCx=OCxREF + Polarity, OCx_EN='1'

Note: The states of the external I/O pins connected to the standard OCx channels depend on the state of the OCx channel and on the GPIO registers.

40.4.10 TIM12 counter (TIM12_CNT)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UIF CPY	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rw															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **UIFCPY**: UIF Copy

This bit is a read-only copy of the UIF bit in the TIMx_ISR register.

Bits 30:16 Reserved, must be kept at reset value.

Bits 15:0 **CNT[15:0]**: Counter value

40.4.11 TIM12 prescaler (TIM12_PSC)

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **PSC[15:0]**: Prescaler value

The counter clock frequency CK_CNT is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded into the active prescaler register at each update event. (including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in "reset mode").

40.4.12 TIM12 auto-reload register (TIM12_ARR)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded into the actual auto-reload register.

Refer to the [Section 40.3.1: Time-base unit on page 1729](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

40.4.13 TIM12 capture/compare register 1 (TIM12_CCR1)

Address offset: 0x34

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR1[15:0]**: Capture/Compare 1 value

If channel CC1 is configured as output:

CCR1 is the value to be loaded into the actual capture/compare 1 register (preload value). It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (OC1PE bit). Else the preload value is copied into the active capture/compare 1 register when an update event occurs.

The active capture/compare register contains the value to be compared to the TIMx_CNT counter and signaled on the OC1 output.

If channel CC1 is configured as input:

CCR1 is the counter value transferred by the last input capture 1 event (IC1).

40.4.14 TIM12 capture/compare register 2 (TIM12_CCR2)

Address offset: 0x38

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR2[15:0]**: Capture/Compare 2 value**If channel CC2 is configured as output:**

CCR2 is the value to be loaded into the actual capture/compare 2 register (preload value). It is loaded permanently if the preload feature is not selected in the TIMx_CCMR2 register (OC2PE bit). Else the preload value is copied into the active capture/compare 2 register when an update event occurs.

The active capture/compare register contains the value to be compared to the TIMx_CNT counter and signalled on the OC2 output.

If channel CC2 is configured as input:

CCR2 is the counter value transferred by the last input capture 2 event (IC2).

40.4.15 TIM12 timer input selection register (TIM12_TISEL)

Address offset: 0x68

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	TI2SEL[3:0]				Res.	Res.	Res.	Res.	TI1SEL[3:0]			
				rw	rw	rw	rw					rw	rw	rw	rw

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:8 **TI2SEL[3:0]**: selects TI2[0] to TI2[15] input

0000: TIM12_CH2 input

Other: Reserved

Bits 7:4 Reserved, must be kept at reset value.

Bits 3:0 **TI1SEL[3:0]**: selects TI1[0] to TI1[15] input

0000: TIM12_CH1 input

0001: spdifrx_frame_sync

Other: Reserved

40.4.16 TIM12 register map

TIM12 registers are mapped as 16-bit addressable registers as described below:

Table 339. TIM12 register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x00	TIMx_CR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	U1FREMA	Res.	Res.	CKD [1:0]	ARPE	Res.	Res.	Res.	Res.	OPM	URS	UDIS	CEN
	Reset value																					0		0			0				0	0	0	0
0x04	TIM12_CR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	T1HS	MMS[2:0]			Res.	Res.	Res.		
	Reset value																									0	0	0	0					
0x08	TIMx_SMCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TS [4:3]	Res.	Res.	Res.	Res.	SMS[3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MSM	TS[2:0]			Res.	SMS[2:0]			
	Reset value											0	0				0									0	0	0	0		0	0	0	
0x0C	TIMx_DIER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIE	Res.	Res.	Res.	Res.	CC2IE	CC1IE	UIE	
	Reset value																									0					0	0	0	
0x10	TIMx_SR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TIF	Res.	Res.	Res.	Res.	CC2IF	CC1IF	UIF
	Reset value																									0					0	0	0	
0x14	TIMx_EGR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TG	Res.	Res.	Res.	Res.	CC2G	CC1G	UG
	Reset value																									0					0	0	0	
0x18	TIMx_CCMR1 Output Compare mode	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC2M[3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M[3]	Res.	OC2M [2:0]			OC2PE	OC2FE	CC2S [1:0]		Res.	OC1M [2:0]			OC1PE	OC1FE	CC1S [1:0]		
	Reset value								0								0		0	0	0	0	0	0		0	0	0	0	0	0	0	0	
	TIMx_CCMR1 Input Capture mode	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IC2F[3:0]			IC2 PSC [1:0]	CC2S [1:0]		IC1F[3:0]			IC1 PSC [1:0]	CC1S [1:0]						
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x1C	Reserved	Res.																																
0x20	TIMx_CCER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC2NP	Res.	CC2P	CC2E	CC1NP	Res.	CC1P	CC1E
	Reset value																									0		0	0	0		0	0	
0x24	TIMx_CNT	UIFCPY	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[15:0]																
	Reset value	0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x28	TIMx_PSC	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PSC[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Table 339. TIM12 register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0								
0x2C	TIMx_ARR	Reserved																ARR[15:0]																							
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x30	Reserved	Reserved																																							
0x34	TIMx_CCR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CCR1[15:0]																							
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x38	TIMx_CCR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CCR2[15:0]																							
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x3C to 0x64	Reserved	Res.																																							
0x68	TIM12_TISEL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TI2SEL[3:0]				Res.	Res.	Res.	Res.	TI1SEL[3:0]											
	Reset value																					0	0	0	0					0	0	0	0								

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.

40.5 TIM13/TIM14 registers

The peripheral registers have to be written by half-words (16 bits) or words (32 bits). Read accesses can be done by bytes (8 bits), half-words (16 bits) or words (32 bits).

40.5.1 TIMx control register 1 (TIMx_CR1)(x = 13 to 14)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	UIFRE MAP	Res.	CKD[1:0]		ARPE	Res.	Res.	Res.	OPM	URS	UDIS	CEN
				rw		rw	rw	rw				rw	rw	rw	rw

Bits 15:12 Reserved, must be kept at reset value.

Bit 11 **UIFREMAP**: UIF status bit remapping

0: No remapping. UIF status bit is not copied to TIMx_CNT register bit 31.

1: Remapping enabled. UIF status bit is copied to TIMx_CNT register bit 31.

Bit 10 Reserved, must be kept at reset value.

Bits 9:8 **CKD[1:0]**: Clock division

This bit-field indicates the division ratio between the timer clock (CK_INT) frequency and sampling clock used by the digital filters (Tl_x),

00: $t_{DTS} = t_{CK_INT}$

01: $t_{DTS} = 2 \times t_{CK_INT}$

10: $t_{DTS} = 4 \times t_{CK_INT}$

11: Reserved

Bit 7 **ARPE**: Auto-reload preload enable

0: TIMx_ARR register is not buffered

1: TIMx_ARR register is buffered

Bits 6:4 Reserved, must be kept at reset value.

Bit 3 **OPM**: One-pulse mode

0: Counter is not stopped on the update event

1: Counter stops counting on the next update event (clearing the CEN bit).

Bit 2 **URS**: Update request source

This bit is set and cleared by software to select the update interrupt (UEV) sources.

0: Any of the following events generate an UEV if enabled:

- Counter overflow
- Setting the UG bit

1: Only counter overflow generates an UEV if enabled.

Bit 1 **UDIS**: Update disable

This bit is set and cleared by software to enable/disable update interrupt (UEV) event generation.

0: UEV enabled. An UEV is generated by one of the following events:

- Counter overflow
- Setting the UG bit.

Buffered registers are then loaded with their preload values.

1: UEV disabled. No UEV is generated, shadow registers keep their value (ARR, PSC, CCRx). The counter and the prescaler are reinitialized if the UG bit is set.

Bit 0 **CEN**: Counter enable

0: Counter disabled

1: Counter enabled

Note: External clock and gated mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.

40.5.2 TIMx Interrupt enable register (TIMx_DIER)(x = 13 to 14)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC1IE	UIE
														rw	rw

Bits 15:2 Reserved, must be kept at reset value.

Bit 1 **CC1IE**: Capture/Compare 1 interrupt enable

0: CC1 interrupt disabled

1: CC1 interrupt enabled

Bit 0 **UIE**: Update interrupt enable

0: Update interrupt disabled

1: Update interrupt enabled

40.5.3 TIMx status register (TIMx_SR)(x = 13 to 14)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	CC1OF	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC1IF	UIF
						rc_w0								rc_w0	rc_w0

Bits 15:10 Reserved, must be kept at reset value.

Bit 9 **CC1OF**: Capture/Compare 1 overcapture flag

This flag is set by hardware only when the corresponding channel is configured in input capture mode. It is cleared by software by writing it to '0'.

0: No overcapture has been detected.

1: The counter value has been captured in TIMx_CCR1 register while CC1IF flag was already set

Bits 8:2 Reserved, must be kept at reset value.

Bit 1 **CC1IF**: Capture/compare 1 interrupt flag

This flag is set by hardware. It is cleared by software (input capture or output compare mode) or by reading the TIMx_CCR1 register (input capture mode only).

0: No compare match / No input capture occurred

1: A compare match or an input capture occurred.

If channel CC1 is configured as output: this flag is set when the content of the counter TIMx_CNT matches the content of the TIMx_CCR1 register. When the content of TIMx_CCR1 is greater than the content of TIMx_ARR, the CC1IF bit goes high on the counter overflow (in up-counting and up/down-counting modes) or underflow (in down-counting mode). There are 3 possible options for flag setting in center-aligned mode, refer to the CMS bits in the TIMx_CR1 register for the full description.

If channel CC1 is configured as input: this bit is set when counter value has been captured in TIMx_CCR1 register (an edge has been detected on IC1, as per the edge sensitivity defined with the CC1P and CC1NP bits setting, in TIMx_CCER).

Bit 0 **UIF**: Update interrupt flag

This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred.

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

- At overflow and if UDIS='0' in the TIMx_CR1 register.
- When CNT is reinitialized by software using the UG bit in TIMx_EGR register, if URS='0' and UDIS='0' in the TIMx_CR1 register.

40.5.4 TIMx event generation register (TIMx_EGR)(x = 13 to 14)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC1G	UG
														w	w

Bits 15:2 Reserved, must be kept at reset value.

Bit 1 **CC1G**: Capture/compare 1 generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: A capture/compare event is generated on channel 1:

If channel CC1 is configured as output:

CC1IF flag is set, Corresponding interrupt or is sent if enabled.

If channel CC1 is configured as input:

The current value of the counter is captured in TIMx_CCR1 register. The CC1IF flag is set, the corresponding interrupt is sent if enabled. The CC1OF flag is set if the CC1IF flag was already high.

Bit 0 **UG**: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action

1: Re-initialize the counter and generates an update of the registers. Note that the prescaler counter is cleared too (anyway the prescaler ratio is not affected). The counter is cleared.

40.5.5 TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1)(x = 13 to 14)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for input capture mode (this section) or for output compare mode (next section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode.

Input capture mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IC1F[3:0]				IC1PSC[1:0]		CC1S[1:0]	
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 IC1F[3:0]: Input capture 1 filter

This bit-field defines the frequency used to sample TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, N=2

0010: $f_{SAMPLING}=f_{CK_INT}$, N=4

0011: $f_{SAMPLING}=f_{CK_INT}$, N=8

0100: $f_{SAMPLING}=f_{DTS}/2$, N=6

0101: $f_{SAMPLING}=f_{DTS}/2$, N=8

0110: $f_{SAMPLING}=f_{DTS}/4$, N=6

0111: $f_{SAMPLING}=f_{DTS}/4$, N=8

1000: $f_{SAMPLING}=f_{DTS}/8$, N=6

1001: $f_{SAMPLING}=f_{DTS}/8$, N=8

1010: $f_{SAMPLING}=f_{DTS}/16$, N=5

1011: $f_{SAMPLING}=f_{DTS}/16$, N=6

1100: $f_{SAMPLING}=f_{DTS}/16$, N=8

1101: $f_{SAMPLING}=f_{DTS}/32$, N=5

1110: $f_{SAMPLING}=f_{DTS}/32$, N=6

1111: $f_{SAMPLING}=f_{DTS}/32$, N=8

Bits 3:2 IC1PSC[1:0]: Input capture 1 prescaler

This bit-field defines the ratio of the prescaler acting on CC1 input (IC1).

The prescaler is reset as soon as CC1E='0' (TIMx_CCER register).

00: no prescaler, capture is done each time an edge is detected on the capture input

01: capture is done once every 2 events

10: capture is done once every 4 events

11: capture is done once every 8 events

Bits 1:0 CC1S[1:0]: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: Reserved

11: Reserved

Note: CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

40.5.6 TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1)(x = 13 to 14)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for output compare mode (this section) or for input capture mode (previous section). The direction of a channel is defined by configuring the

corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode.

Output compare mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M[3]
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
									rw	rw	rw	rw	rw	rw	rw

Bits 31:17 Reserved, must be kept at reset value.

Bits 15:7 Reserved, must be kept at reset value.

Bits 16, 6:4 **OC1M[3:0]**: Output compare 1 mode (refer to bit 16 for OC1M[3])

These bits define the behavior of the output reference signal OC1REF from which OC1 is derived. OC1REF is active high whereas OC1 active level depends on CC1P bit.

0000: Frozen. The comparison between the output compare register TIMx_CCR1 and the counter TIMx_CNT has no effect on the outputs.

0001: Set channel 1 to active level on match. OC1REF signal is forced high when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

0010: Set channel 1 to inactive level on match. OC1REF signal is forced low when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

0011: Toggle - OC1REF toggles when TIMx_CNT = TIMx_CCR1.

0100: Force inactive level - OC1REF is forced low.

0101: Force active level - OC1REF is forced high.

0110: PWM mode 1 - Channel 1 is active as long as TIMx_CNT < TIMx_CCR1 else inactive.

0111: PWM mode 2 - Channel 1 is inactive as long as TIMx_CNT < TIMx_CCR1 else active

Others: Reserved

Note: In PWM mode 1 or 2, the OCREF level changes when the result of the comparison changes or when the output compare mode switches from frozen to PWM mode.

Bit 3 **OC1PE**: Output compare 1 preload enable

- 0: Preload register on TIMx_CCR1 disabled. TIMx_CCR1 can be written at anytime, the new value is taken in account immediately.
- 1: Preload register on TIMx_CCR1 enabled. Read/Write operations access the preload register. TIMx_CCR1 preload value is loaded in the active register at each update event.

Note: The PWM mode can be used without validating the preload register only in one pulse mode (OPM bit set in TIMx_CR1 register). Else the behavior is not guaranteed.

Bit 2 **OC1FE**: Output compare 1 fast enable

This bit decreases the latency between a trigger event and a transition on the timer output. It must be used in one-pulse mode (OPM bit set in TIMx_CR1 register), to have the output pulse starting as soon as possible after the starting trigger.

- 0: CC1 behaves normally depending on counter and CCR1 values even when the trigger is ON. The minimum delay to activate CC1 output when an edge occurs on the trigger input is 5 clock cycles.
- 1: An active edge on the trigger input acts like a compare match on CC1 output. OC is then set to the compare level independently of the result of the comparison. Delay to sample the trigger input and to activate CC1 output is reduced to 3 clock cycles. OC1FE acts only if the channel is configured in PWM1 or PWM2 mode.

Bits 1:0 **CC1S[1:0]**: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

- 00: CC1 channel is configured as output.
- 01: CC1 channel is configured as input, IC1 is mapped on TI1.
- 10: Reserved.
- 11: Reserved.

Note: CC1S bits are writable only when the channel is OFF (CC1E = 0 in TIMx_CCER).

40.5.7 TIMx capture/compare enable register (TIMx_CCER)(x = 13 to 14)

Address offset: 0x20

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC1NP	Res.	CC1P	CC1E
												rw		rw	rw

Bits 15:4 Reserved, must be kept at reset value.

Bit 3 **CC1NP**: Capture/Compare 1 complementary output Polarity.

CC1 channel configured as output: CC1NP must be kept cleared.

CC1 channel configured as input: CC1NP bit is used in conjunction with CC1P to define TI1FP1 polarity (refer to CC1P description).

Bit 2 Reserved, must be kept at reset value.

Bit 1 **CC1P**: Capture/Compare 1 output Polarity.

0: OC1 active high (output mode) / Edge sensitivity selection (input mode, see below)

1: OC1 active low (output mode) / Edge sensitivity selection (input mode, see below)

When CC1 channel is configured as input, both CC1NP/CC1P bits select the active polarity of TI1FP1 and TI2FP1 for trigger or capture operations.

CC1NP=0, CC1P=0: non-inverted/rising edge. The circuit is sensitive to TIxFP1 rising edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode or encoder mode).

CC1NP=0, CC1P=1: inverted/falling edge. The circuit is sensitive to TIxFP1 falling edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is inverted (trigger operation in gated mode or encoder mode).

CC1NP=1, CC1P=1: non-inverted/both edges/ The circuit is sensitive to both TIxFP1 rising and falling edges (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode). This configuration must not be used in encoder mode.

CC1NP=1, CC1P=0: This configuration is reserved, it must not be used.

Bit 0 **CC1E**: Capture/Compare 1 output enable.

0: Capture mode disabled / OC1 is not active

1: Capture mode enabled / OC1 signal is output on the corresponding output pin

Table 340. Output control bit for standard OCx channels

CCxE bit	OCx output state
0	Output Disabled (OCx='0', OCx_EN='0')
1	OCx=OCxREF + Polarity, OCx_EN='1'

Note: The state of the external I/O pins connected to the standard OCx channels depends on the OCx channel state and the GPIO registers.

40.5.8 TIMx counter (TIMx_CNT)(x = 13 to 14)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UIF CPY	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rw															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **UIFCPY**: UIF Copy

This bit is a read-only copy of the UIF bit in the TIMx_ISR register.

Bits 30:16 Reserved, must be kept at reset value.

Bits 15:0 **CNT[15:0]**: Counter value

40.5.9 TIMx prescaler (TIMx_PSC)(x = 13 to 14)

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **PSC[15:0]**: Prescaler value

The counter clock frequency CK_CNT is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded in the active prescaler register at each update event.

(including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in “reset mode”).

40.5.10 TIMx auto-reload register (TIMx_ARR)(x = 13 to 14)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded in the actual auto-reload register.

Refer to [Section 40.3.1: Time-base unit on page 1729](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

40.5.11 TIMx capture/compare register 1 (TIMx_CCR1)(x = 13 to 14)

Address offset: 0x34

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR1[15:0]**: Capture/Compare 1 value

If channel CC1 is configured as output:

CCR1 is the value to be loaded in the actual capture/compare 1 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (bit OC1PE). Else the preload value is copied in the active capture/compare 1 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC1 output.

If channel CC1 is configured as input:

CCR1 is the counter value transferred by the last input capture 1 event (IC1).

40.5.12 TIM13 timer input selection register (TIM13_TISEL)

Address offset: 0x68

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	T11SEL[3:0]			
												rw	rw	rw	rw

Bits 15:4 Reserved, must be kept at reset value.

Bits 3:0 **T11SEL[3:0]**: selects TI1[0] to TI1[15] input

0000: TIM13_CH1 input

Other: Reserved

40.5.13 TIM14 timer input selection register (TIM14_TISEL)

Address offset: 0x68

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	T11SEL[3:0]			
												rw	rw	rw	rw

Bits 15:4 Reserved, must be kept at reset value.

Bits 3:0 **T11SEL[3:0]**: selects TI1[0] to TI1[15] input

0000: TIM14_CH1 input

Other: Reserved

40.5.14 TIM13/TIM14 register map

TIMx registers are mapped as 16-bit addressable registers as described in the tables below:

Table 341. TIM13/TIM14 register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x00	TIMx_CR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	U1FREMA	Res.	Res.	CKD [1:0]	ARPE	Res.	Res.	Res.	Res.	OPM	URS	UDIS	CEN
	Reset value																					0		0			0				0	0	0	0
0x04 to 0x08	Reserved	Res.																																
0x0C	TIMx_DIER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC1IE	UIE	
	Reset value																															0	0	
0x10	TIMx_SR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC1OF	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC1IF	UIF
	Reset value																							0									0	0
0x14	TIMx_EGR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC1G	UG	
	Reset value																															0	0	
0x18	TIMx_CCMR1 Output compare mode	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M[3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M [2:0]	OC1PE	OC1FE	OC1PSC [1:0]	CC1S [1:0]				
	Reset value																0														0	0	0	0
	TIMx_CCMR1 Input capture mode	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IC1F[3:0]	IC1PSC [1:0]	CC1S [1:0]							
	Reset value																											0	0	0	0	0	0	0
0x1C	Reserved	Res.																																
0x20	TIMx_CCER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC1NP	Res.	CC1P	CC1E
	Reset value																													0		0	0	
0x24	TIMx_CNT	UIFCPY	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[15:0]																
	Reset value	0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x28	TIMx_PSC	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PSC[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x2C	TIMx_ARR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ARR[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x30	Reserved	Res.																																

Table 341. TIM13/TIM14 register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x34	TIMx_CCR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CCR1[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x38 to 0x64	Reserved	Res.																																
0x68	TIM13_TISEL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	T1SEL[3:0]			
	Reset value																														0	0	0	0
0x68	TIM14_TISEL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	T1SEL[3:0]			
	Reset value																														0	0	0	0

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.

41 General-purpose timers (TIM15/TIM16/TIM17)

41.1 TIM15/TIM16/TIM17 introduction

The TIM15/TIM16/TIM17 timers consist of a 16-bit auto-reload counter driven by a programmable prescaler.

They may be used for a variety of purposes, including measuring the pulse lengths of input signals (input capture) or generating output waveforms (output compare, PWM, complementary PWM with dead-time insertion).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

The TIM15/TIM16/TIM17 timers are completely independent, and do not share any resources. TIM15 can be synchronized as described in [Section 41.4.21: Timer synchronization \(TIM15\)](#).

41.2 TIM15 main features

TIM15 includes the following features:

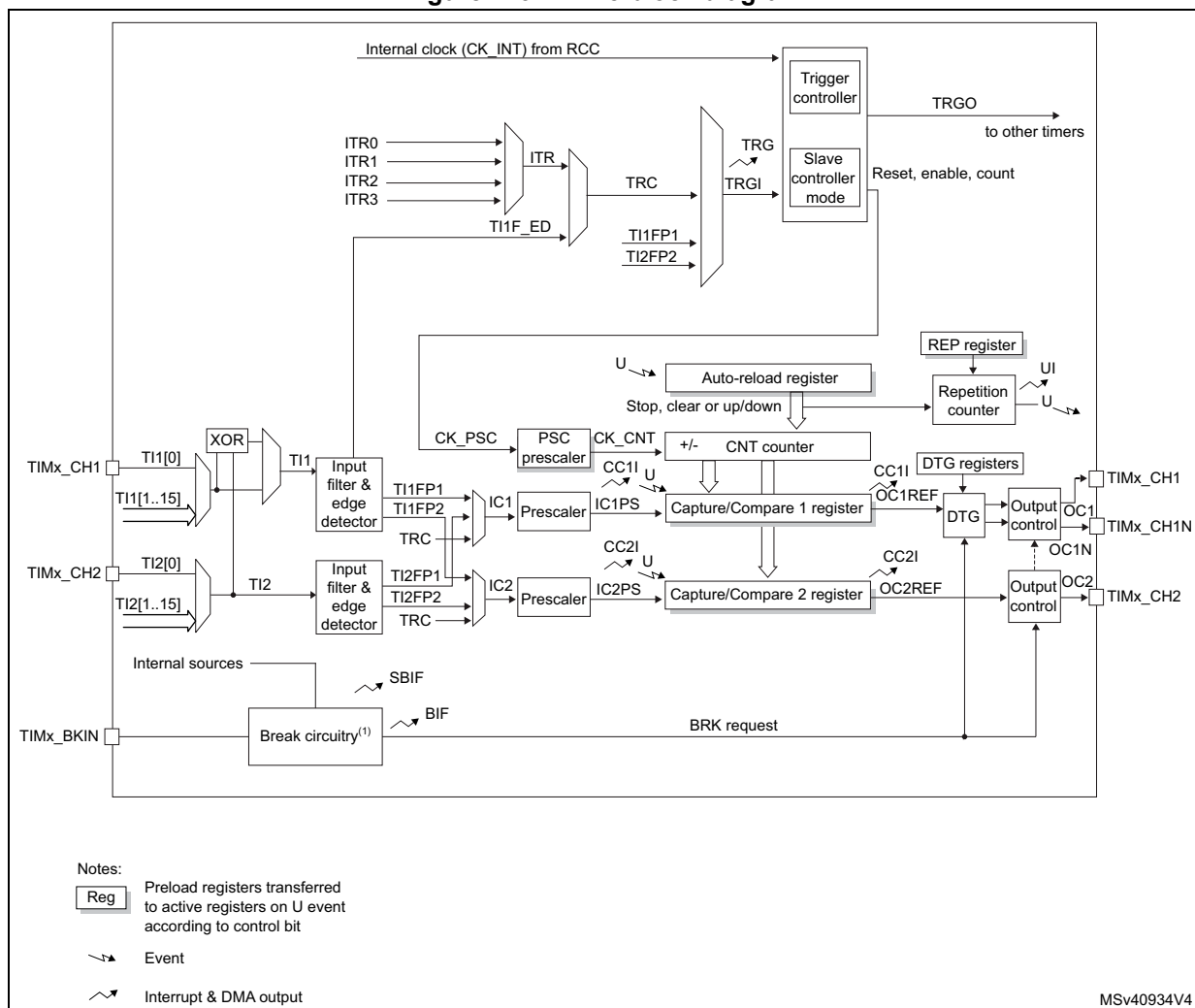
- 16-bit auto-reload upcounter
- 16-bit programmable prescaler used to divide (also “on the fly”) the counter clock frequency by any factor between 1 and 65535
- Up to 2 independent channels for:
 - Input capture
 - Output compare
 - PWM generation (edge mode)
 - One-pulse mode output
- Complementary outputs with programmable dead-time (for channel 1 only)
- Synchronization circuit to control the timer with external signals and to interconnect several timers together
- Repetition counter to update the timer registers only after a given number of cycles of the counter
- Break input to put the timer’s output signals in the reset state or a known state
- Interrupt/DMA generation on the following events:
 - Update: counter overflow, counter initialization (by software or internal/external trigger)
 - Trigger event (counter start, stop, initialization or count by internal/external trigger)
 - Input capture
 - Output compare
 - Break input (interrupt request)

41.3 TIM16/TIM17 main features

The TIM16/TIM17 timers include the following features:

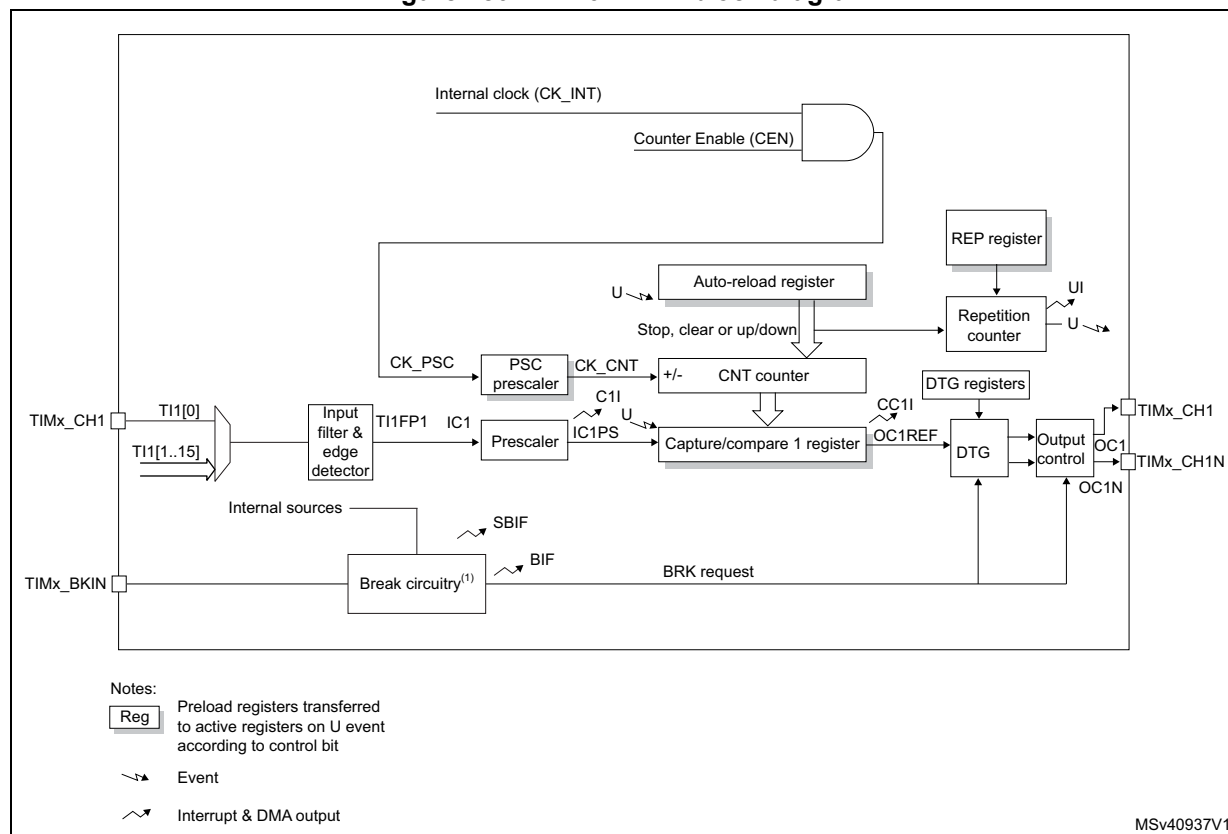
- 16-bit auto-reload upcounter
- 16-bit programmable prescaler used to divide (also “on the fly”) the counter clock frequency by any factor between 1 and 65535
- One channel for:
 - Input capture
 - Output compare
 - PWM generation (edge-aligned mode)
 - One-pulse mode output
- Complementary outputs with programmable dead-time
- Repetition counter to update the timer registers only after a given number of cycles of the counter
- Break input to put the timer’s output signals in the reset state or a known state
- Interrupt/DMA generation on the following events:
 - Update: counter overflow
 - Input capture
 - Output compare
 - Break input

Figure 479. TIM15 block diagram



1. The internal break event source can be:
 - A clock failure event generated by CSS. For further information on the CSS, refer to [Section 8.5.3: Clock Security System \(CSS\)](#)
 - A PVD output
 - all SRAM and TCM double ECC errors
 - Flash double ECC error
 - Cortex[®]-M7 with FPU LOCKUP (Hardfault) output
 - COMP output

Figure 480. TIM16/TIM17 block diagram



1. The internal break event source can be:
 - A clock failure event generated by CSS. For further information on the CSS, refer to [Section 8.5.3: Clock Security System \(CSS\)](#)
 - A PVD output
 - all SRAM and TCM double ECC errors
 - Flash double ECC error
 - Cortex[®]-M7 with FPU LOCKUP (Hardfault) output
 - COMP output

41.4 TIM15/TIM16/TIM17 functional description

41.4.1 Time-base unit

The main block of the programmable advanced-control timer is a 16-bit upcounter with its related auto-reload register. The counter clock can be divided by a prescaler.

The counter, the auto-reload register and the prescaler register can be written or read by software. This is true even when the counter is running.

The time-base unit includes:

- Counter register (TIMx_CNT)
- Prescaler register (TIMx_PSC)
- Auto-reload register (TIMx_ARR)
- Repetition counter register (TIMx_RCR)

The auto-reload register is preloaded. Writing to or reading from the auto-reload register accesses the preload register. The content of the preload register are transferred into the shadow register permanently or at each update event (UEV), depending on the auto-reload preload enable bit (ARPE) in TIMx_CR1 register. The update event is sent when the counter reaches the overflow and if the UDIS bit equals 0 in the TIMx_CR1 register. It can also be generated by software. The generation of the update event is described in detailed for each configuration.

The counter is clocked by the prescaler output CK_CNT, which is enabled only when the counter enable bit (CEN) in TIMx_CR1 register is set (refer also to the slave mode controller description to get more details on counter enabling).

Note that the counter starts counting 1 clock cycle after setting the CEN bit in the TIMx_CR1 register.

Prescaler description

The prescaler can divide the counter clock frequency by any factor between 1 and 65536. It is based on a 16-bit counter controlled through a 16-bit register (in the TIMx_PSC register). It can be changed on the fly as this control register is buffered. The new prescaler ratio is taken into account at the next update event.

Figure 481 and *Figure 482* give some examples of the counter behavior when the prescaler ratio is changed on the fly:

Figure 481. Counter timing diagram with prescaler division change from 1 to 2

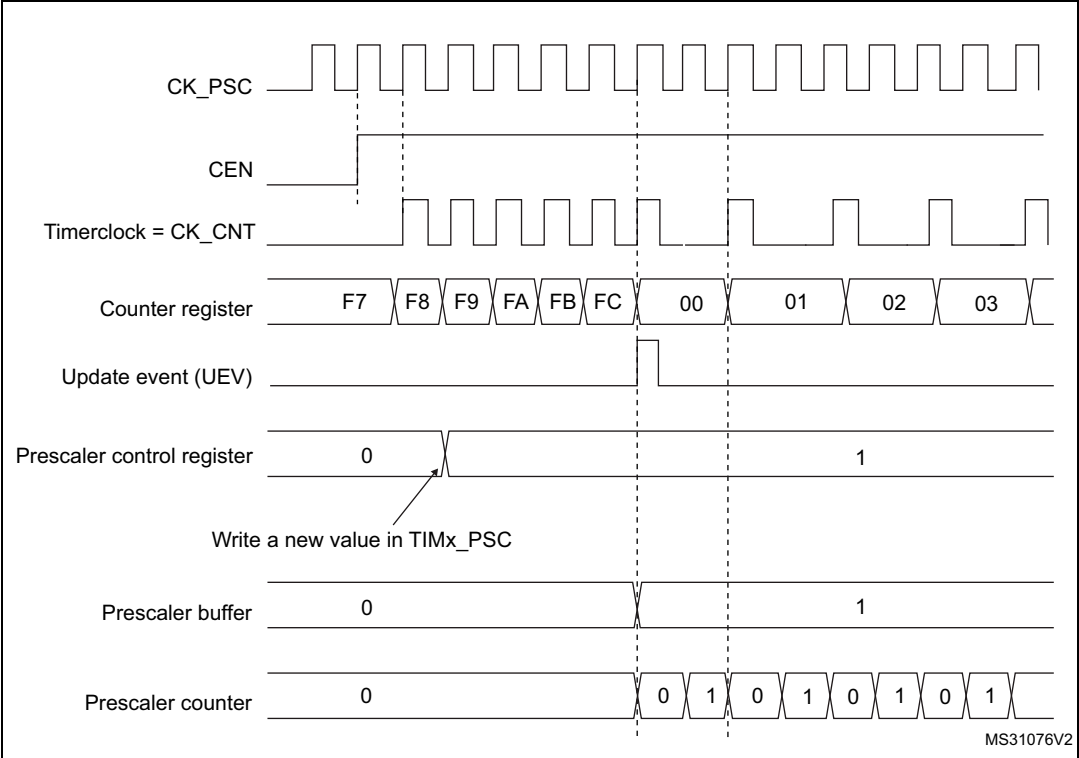
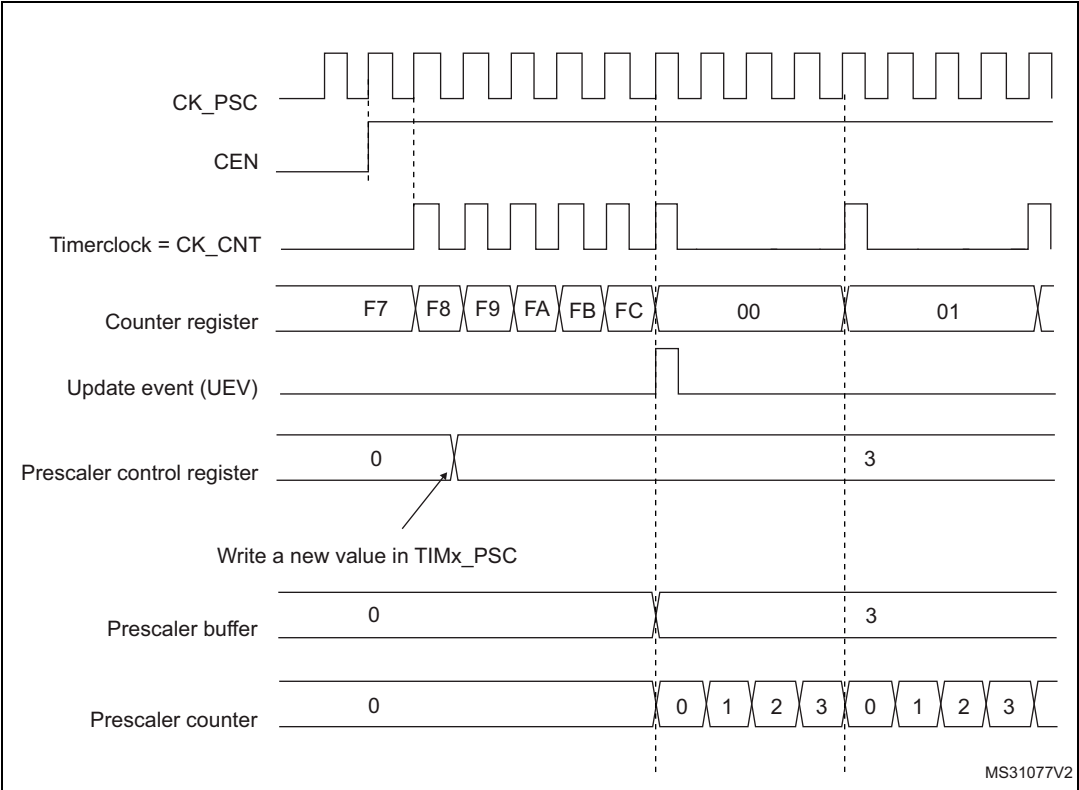


Figure 482. Counter timing diagram with prescaler division change from 1 to 4



41.4.2 Counter modes

Upcounting mode

In upcounting mode, the counter counts from 0 to the auto-reload value (content of the TIMx_ARR register), then restarts from 0 and generates a counter overflow event.

If the repetition counter is used, the update event (UEV) is generated after upcounting is repeated for the number of times programmed in the repetition counter register (TIMx_RCR). Else the update event is generated at each counter overflow.

Setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller) also generates an update event.

The UEV event can be disabled by software by setting the UDIS bit in the TIMx_CR1 register. This is to avoid updating the shadow registers while writing new values in the preload registers. Then no update event occurs until the UDIS bit has been written to 0. However, the counter restarts from 0, as well as the counter of the prescaler (but the prescale rate does not change). In addition, if the URS bit (update request selection) in TIMx_CR1 register is set, setting the UG bit generates an update event UEV but without setting the UIF flag (thus no interrupt or DMA request is sent). This is to avoid generating both update and capture interrupts when clearing the counter on the capture event.

When an update event occurs, all the registers are updated and the update flag (UIF bit in TIMx_SR register) is set (depending on the URS bit):

- The repetition counter is reloaded with the content of TIMx_RCR register,
- The auto-reload shadow register is updated with the preload value (TIMx_ARR),
- The buffer of the prescaler is reloaded with the preload value (content of the TIMx_PSC register).

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR=0x36.

Figure 483. Counter timing diagram, internal clock divided by 1

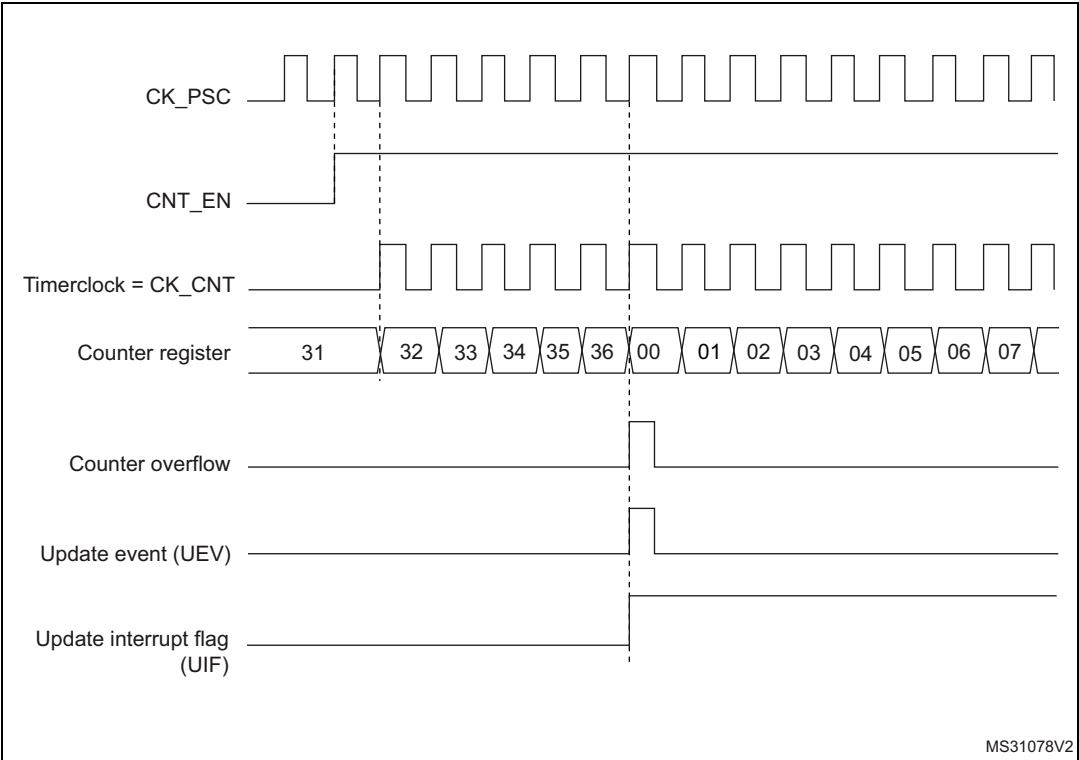


Figure 484. Counter timing diagram, internal clock divided by 2

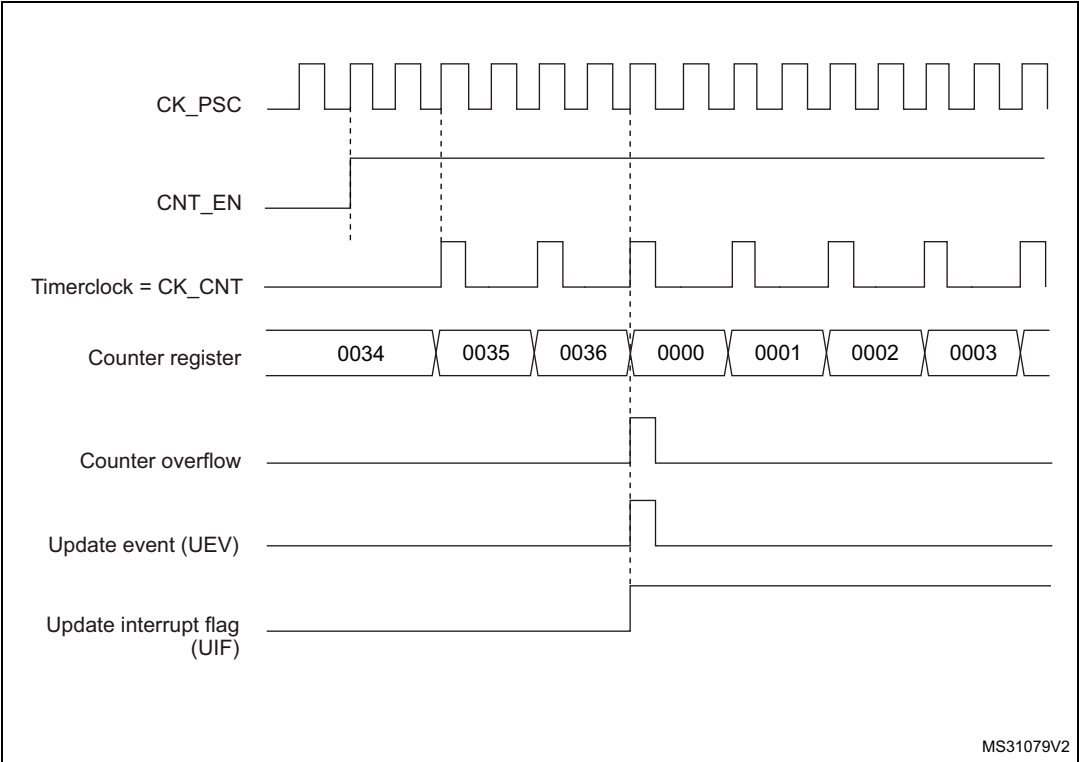


Figure 485. Counter timing diagram, internal clock divided by 4

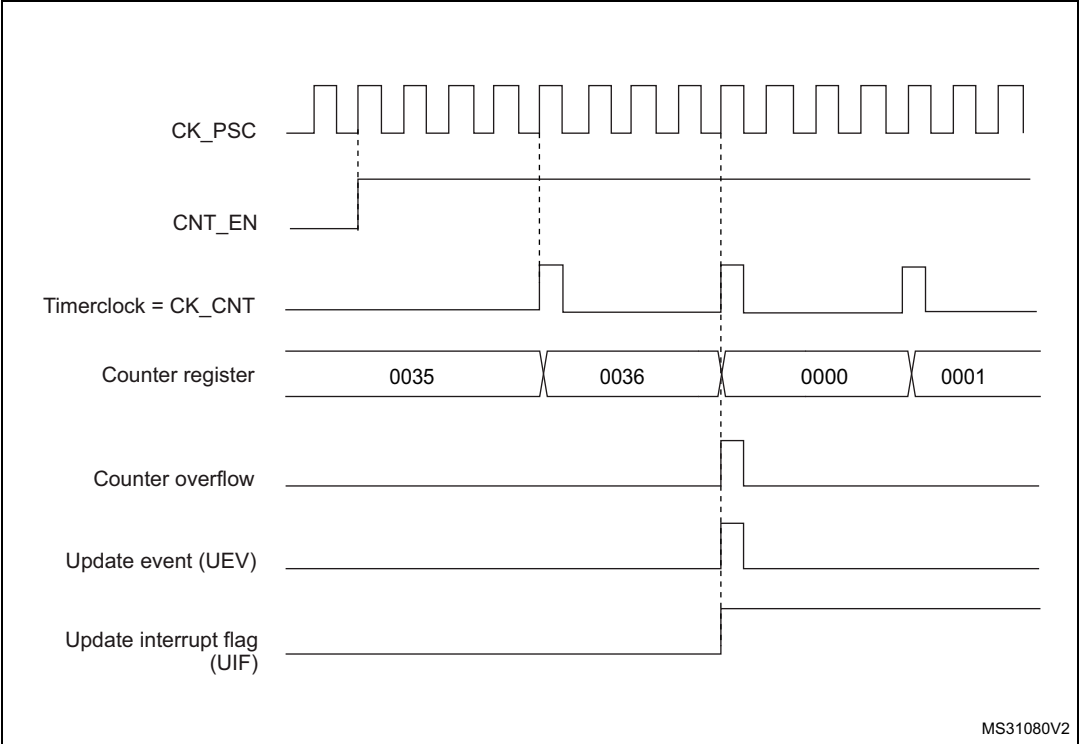


Figure 486. Counter timing diagram, internal clock divided by N

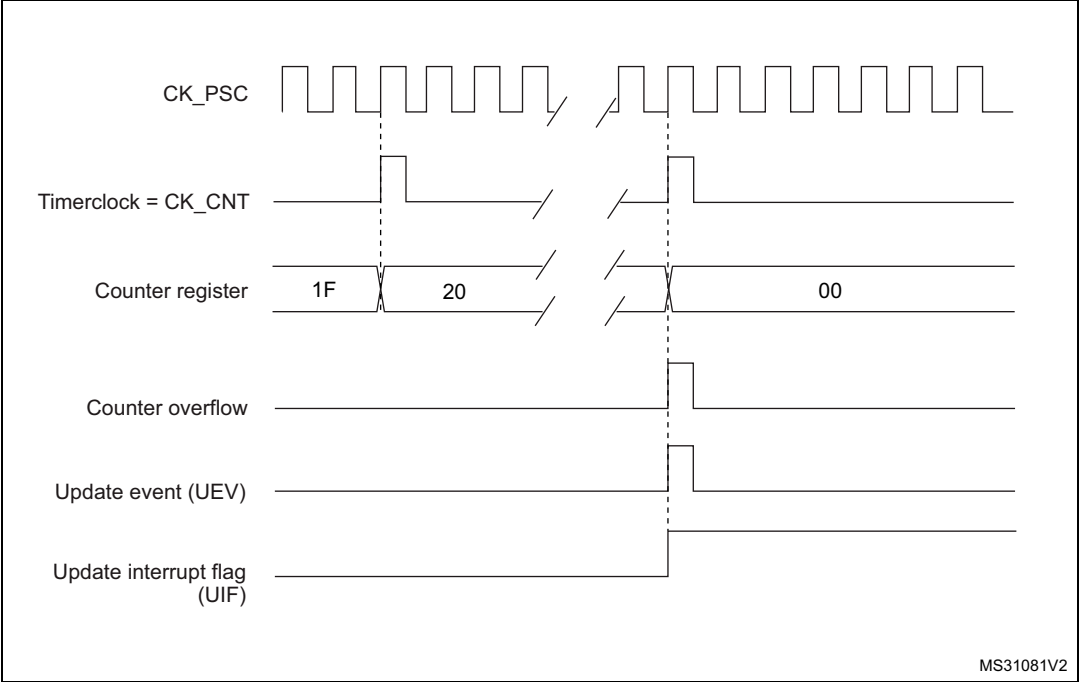


Figure 487. Counter timing diagram, update event when ARPE=0 (TIMx_ARR not preloaded)

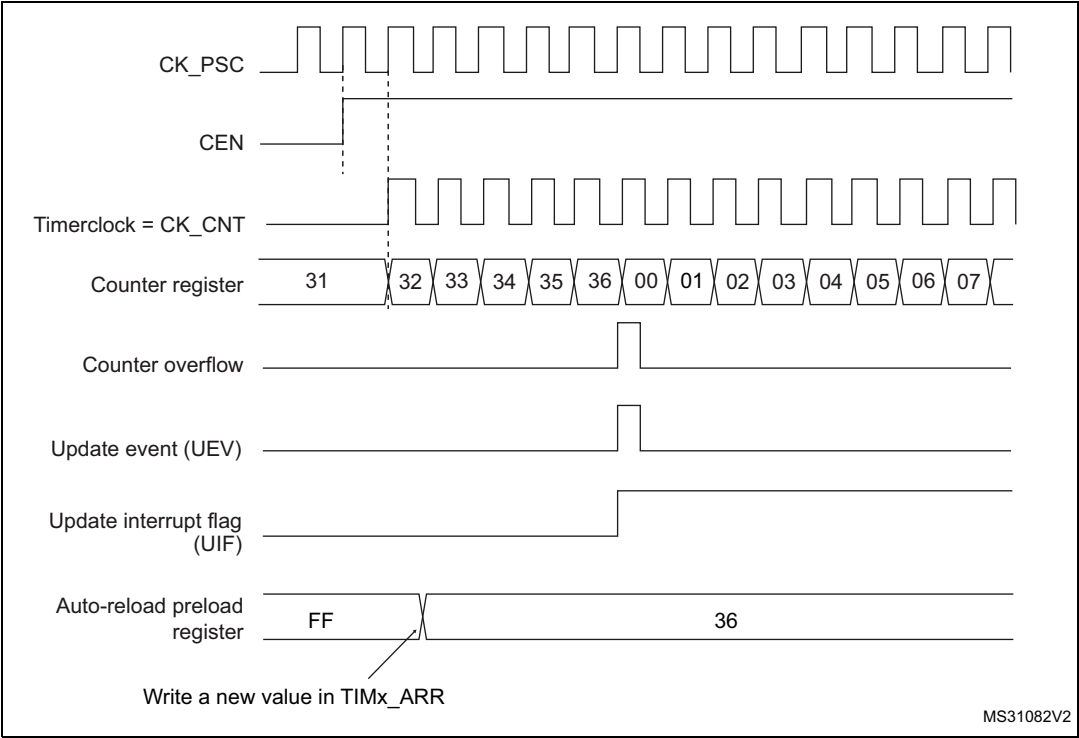
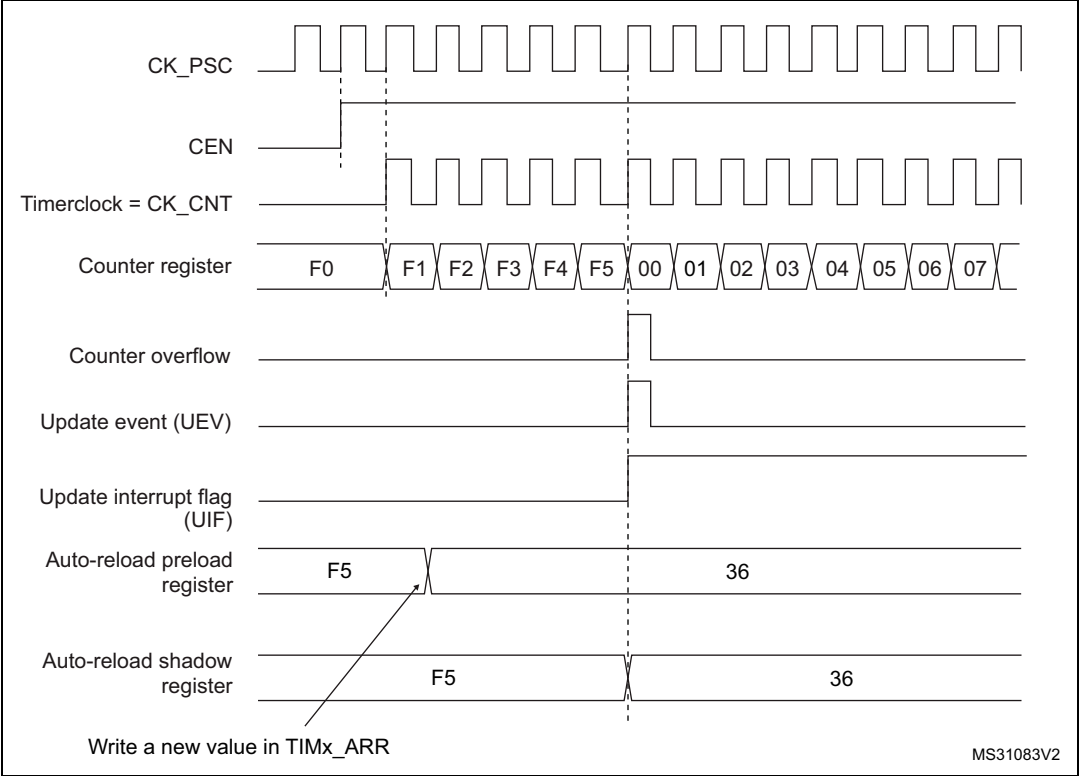


Figure 488. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded)



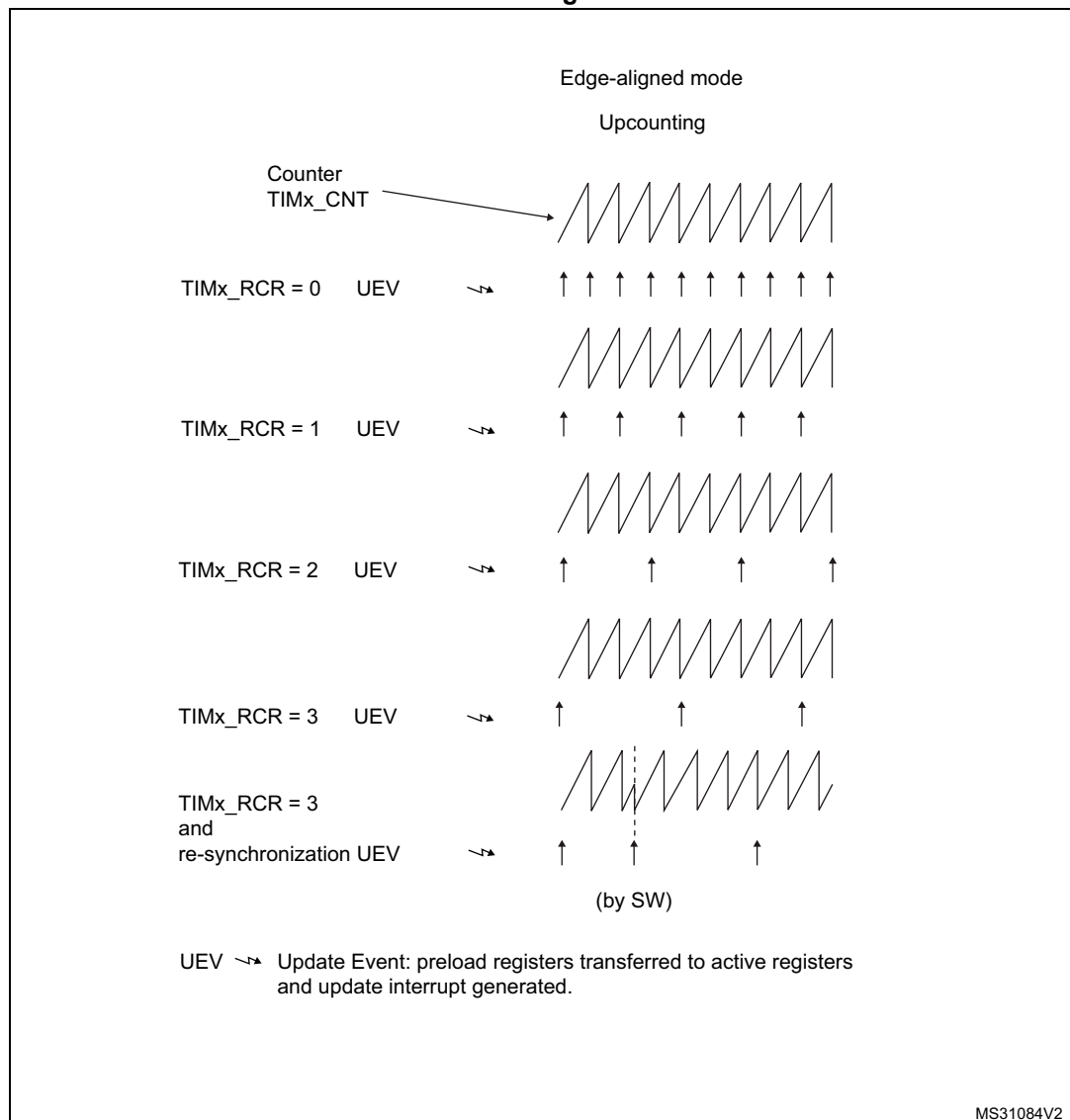
41.4.3 Repetition counter

[Section 41.4.1: Time-base unit](#) describes how the update event (UEV) is generated with respect to the counter overflows. It is actually generated only when the repetition counter has reached zero. This can be useful when generating PWM signals.

This means that data are transferred from the preload registers to the shadow registers (TIMx_ARR auto-reload register, TIMx_PSC prescaler register, but also TIMx_CCRx capture/compare registers in compare mode) every N counter overflows, where N is the value in the TIMx_RCR repetition counter register.

The repetition counter is decremented at each counter overflow.

The repetition counter is an auto-reload type; the repetition rate is maintained as defined by the TIMx_RCR register value (refer to [Figure 489](#)). When the update event is generated by software (by setting the UG bit in TIMx_EGR register) or by hardware through the slave mode controller, it occurs immediately whatever the value of the repetition counter is and the repetition counter is reloaded with the content of the TIMx_RCR register.

Figure 489. Update rate examples depending on mode and TIMx_RCR register settings

41.4.4 Clock selection

The counter clock can be provided by the following clock sources:

- Internal clock (CK_INT)
- External clock mode1: external input pin
- Internal trigger inputs (ITRx) (only for TIM15): using one timer as the prescaler for another timer, for example, TIM1 can be configured to act as a prescaler for TIM15. Refer to [Using one timer as prescaler for another timer on page 1689](#) for more details.

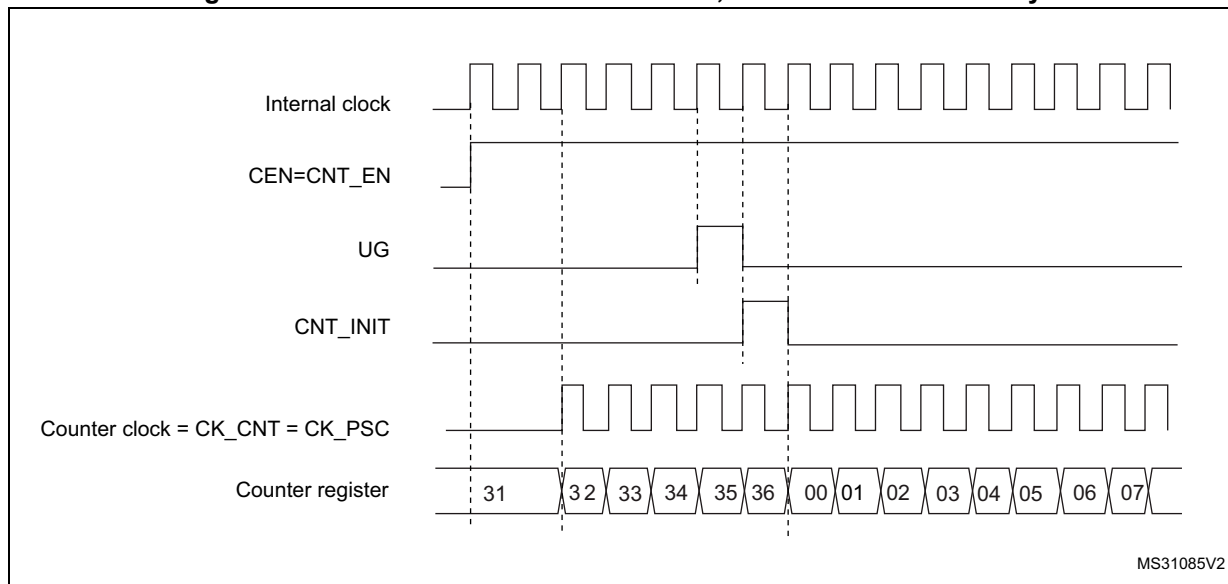
Internal clock source (CK_INT)

If the slave mode controller is disabled (SMS=000), then the CEN (in the TIMx_CR1 register) and UG bits (in the TIMx_EGR register) are actual control bits and can be changed

only by software (except UG which remains cleared automatically). As soon as the CEN bit is written to 1, the prescaler is clocked by the internal clock CK_INT.

Figure 490 shows the behavior of the control circuit and the upcounter in normal mode, without prescaler.

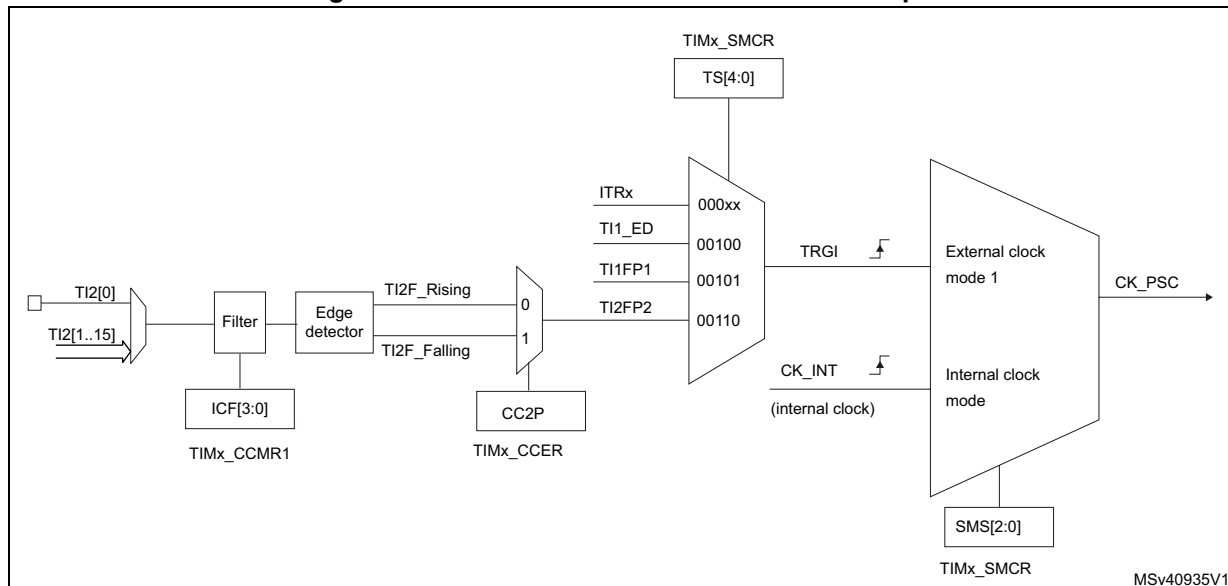
Figure 490. Control circuit in normal mode, internal clock divided by 1



External clock source mode 1

This mode is selected when SMS=111 in the TIMx_SMCR register. The counter can count at each rising or falling edge on a selected input.

Figure 491. TI2 external clock connection example



For example, to configure the upcounter to count in response to a rising edge on the TI2 input, use the following procedure:

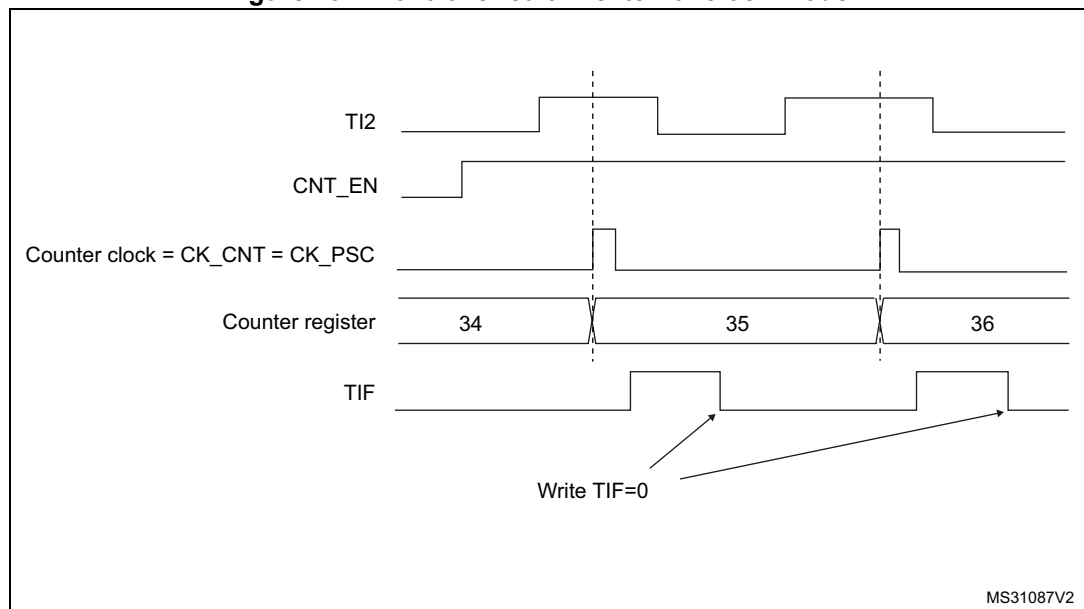
1. Select the proper TI2[x] source (internal or external) with the TI2SEL[3:0] bits in the TIMx_TISEL register.
2. Configure channel 2 to detect rising edges on the TI2 input by writing CC2S = '01' in the TIMx_CCMR1 register.
3. Configure the input filter duration by writing the IC2F[3:0] bits in the TIMx_CCMR1 register (if no filter is needed, keep IC2F=0000).
4. Select rising edge polarity by writing CC2P=0 in the TIMx_CCER register.
5. Configure the timer in external clock mode 1 by writing SMS=111 in the TIMx_SMCR register.
6. Select TI2 as the trigger input source by writing TS=00110 in the TIMx_SMCR register.
7. Enable the counter by writing CEN=1 in the TIMx_CR1 register.

Note: *The capture prescaler is not used for triggering, so it does not need to be configured.*

When a rising edge occurs on TI2, the counter counts once and the TIF flag is set.

The delay between the rising edge on TI2 and the actual clock of the counter is due to the resynchronization circuit on TI2 input.

Figure 492. Control circuit in external clock mode 1



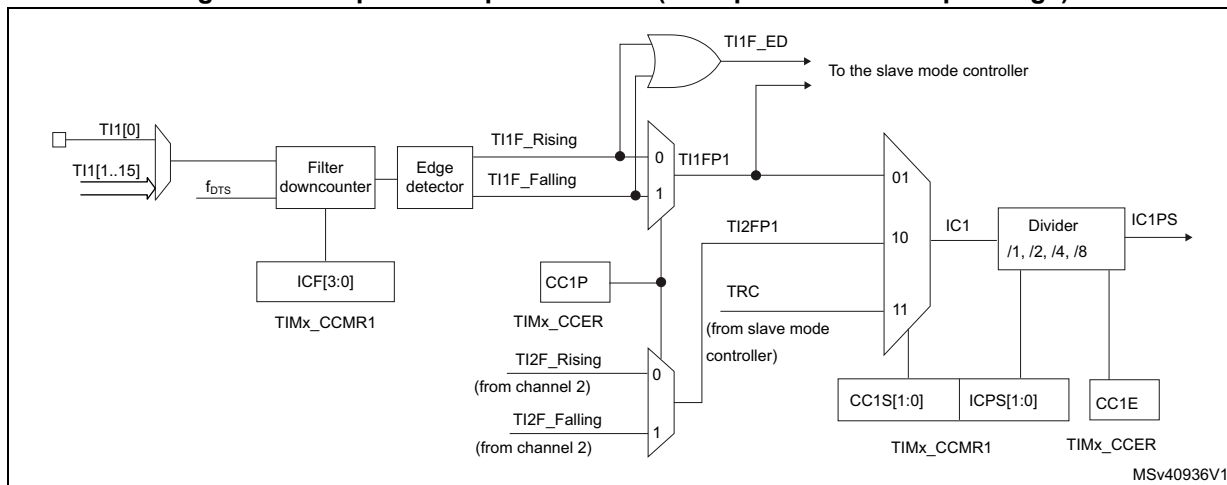
41.4.5 Capture/compare channels

Each Capture/Compare channel is built around a capture/compare register (including a shadow register), a input stage for capture (with digital filter, multiplexing and prescaler) and an output stage (with comparator and output control).

[Figure 493](#) to [Figure 496](#) give an overview of one Capture/Compare channel.

The input stage samples the corresponding TIx input to generate a filtered signal TIxF. Then, an edge detector with polarity selection generates a signal (TIxFPx) which can be used as trigger input by the slave mode controller or as the capture command. It is prescaled before the capture register (ICxPS).

Figure 493. Capture/compare channel (example: channel 1 input stage)



The output stage generates an intermediate waveform which is then used for reference: OCxRef (active high). The polarity acts at the end of the chain.

Figure 494. Capture/compare channel 1 main circuit

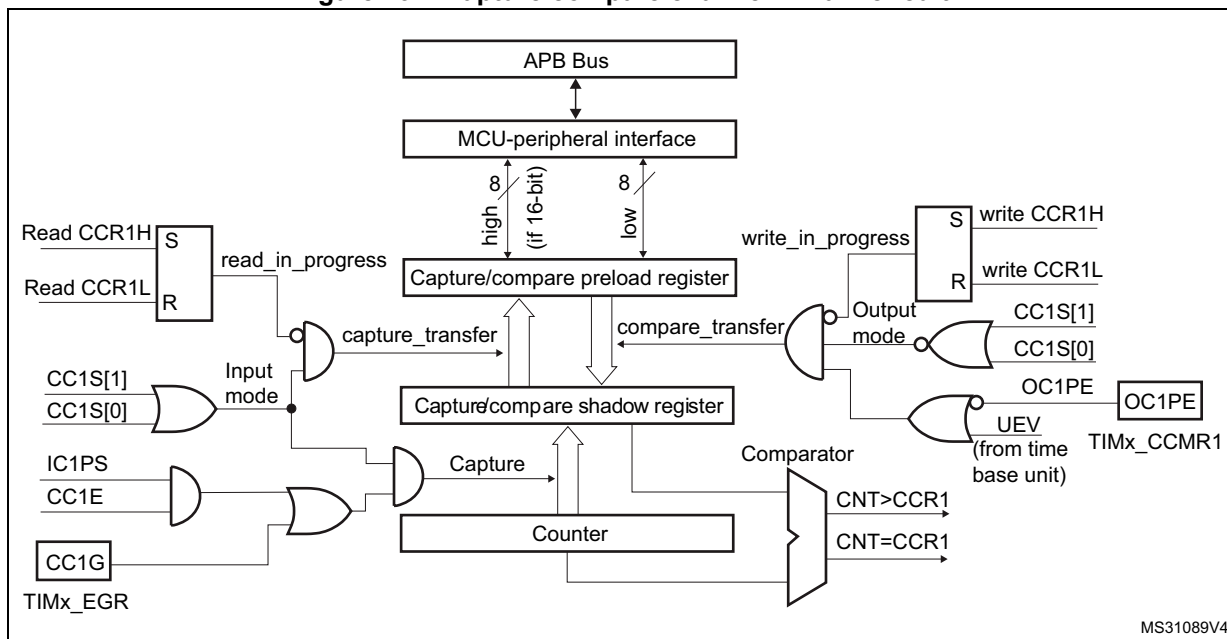


Figure 495. Output stage of capture/compare channel (channel 1)

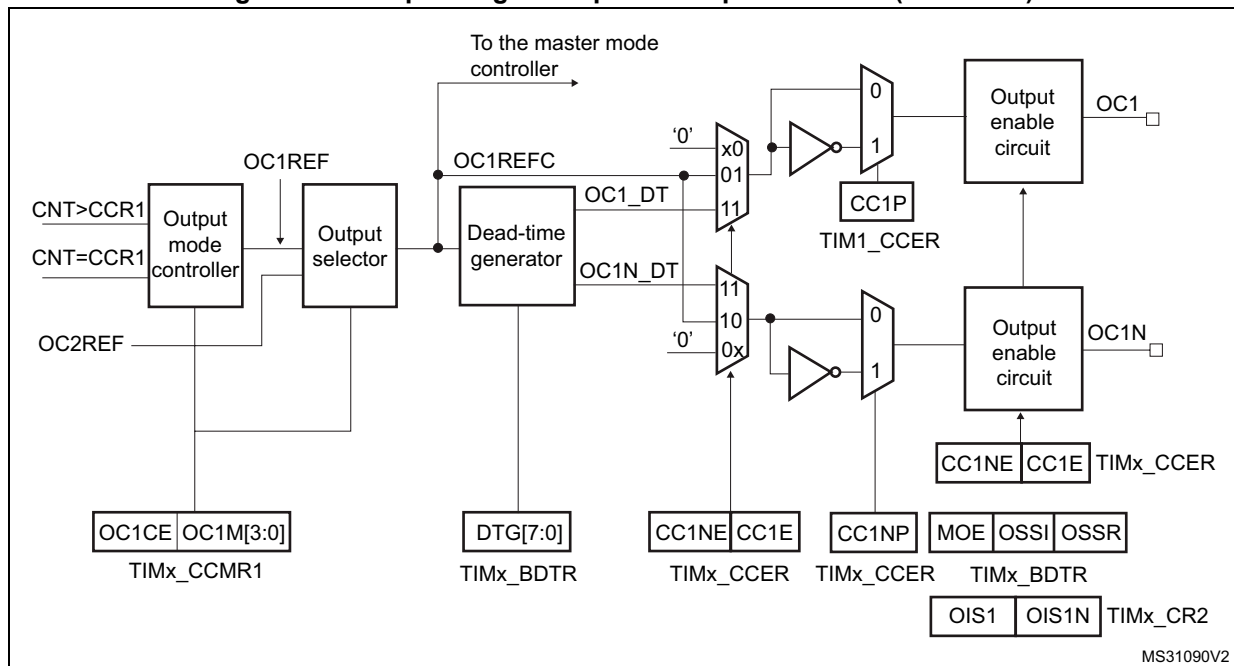
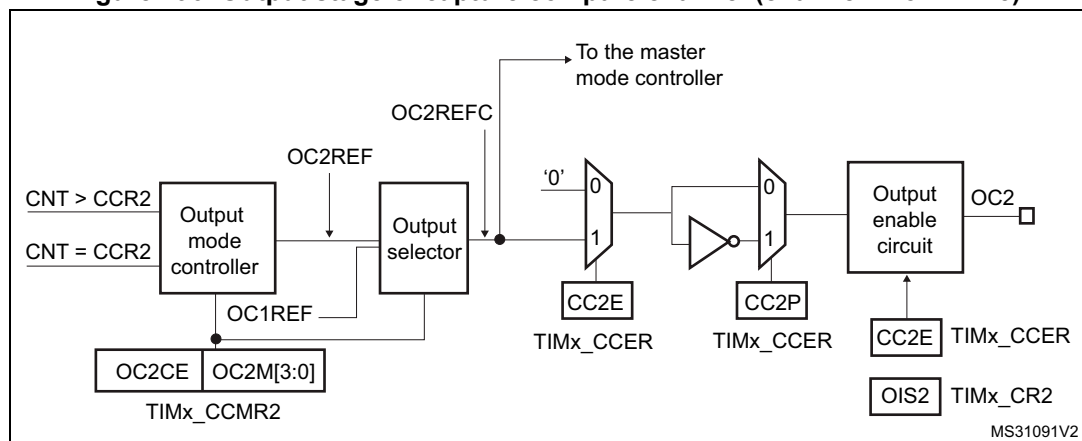


Figure 496. Output stage of capture/compare channel (channel 2 for TIM15)



The capture/compare block is made of one preload register and one shadow register. Write and read always access the preload register.

In capture mode, captures are actually done in the shadow register, which is copied into the preload register.

In compare mode, the content of the preload register is copied into the shadow register which is compared to the counter.

41.4.6 Input capture mode

In Input capture mode, the Capture/Compare Registers (TIMx_CCRx) are used to latch the value of the counter after a transition detected by the corresponding ICx signal. When a capture occurs, the corresponding CCXIF flag (TIMx_SR register) is set and an interrupt or a DMA request can be sent if they are enabled. If a capture occurs while the CCXIF flag was

already high, then the over-capture flag CCxOF (TIMx_SR register) is set. CCxIF can be cleared by software by writing it to '0' or by reading the captured data stored in the TIMx_CCRx register. CCxOF is cleared when it is written with 0.

The following example shows how to capture the counter value in TIMx_CCR1 when TI1 input rises. To do this, use the following procedure:

1. Select the proper TI1x source (internal or external) with the TI1SEL[3:0] bits in the TIMx_TISEL register.
2. Select the active input: TIMx_CCR1 must be linked to the TI1 input, so write the CC1S bits to 01 in the TIMx_CCMR1 register. As soon as CC1S becomes different from 00, the channel is configured in input and the TIMx_CCR1 register becomes read-only.
3. Program the appropriate input filter duration in relation with the signal connected to the timer (when the input is one of the TIx (ICxF bits in the TIMx_CCMRx register). Let's imagine that, when toggling, the input signal is not stable during at least 5 internal clock cycles. We must program a filter duration longer than these 5 clock cycles. We can validate a transition on TI1 when 8 consecutive samples with the new level have been detected (sampled at f_{DTS} frequency). Then write IC1F bits to 0011 in the TIMx_CCMR1 register.
4. Select the edge of the active transition on the TI1 channel by writing CC1P bit to 0 in the TIMx_CCER register (rising edge in this case).
5. Program the input prescaler. In our example, we wish the capture to be performed at each valid transition, so the prescaler is disabled (write IC1PS bits to '00' in the TIMx_CCMR1 register).
6. Enable capture from the counter into the capture register by setting the CC1E bit in the TIMx_CCER register.
7. If needed, enable the related interrupt request by setting the CC1IE bit in the TIMx_DIER register, and/or the DMA request by setting the CC1DE bit in the TIMx_DIER register.

When an input capture occurs:

- The TIMx_CCR1 register gets the value of the counter on the active transition.
- CC1IF flag is set (interrupt flag). CC1OF is also set if at least two consecutive captures occurred whereas the flag was not cleared.
- An interrupt is generated depending on the CC1IE bit.
- A DMA request is generated depending on the CC1DE bit.

In order to handle the overcapture, it is recommended to read the data before the overcapture flag. This is to avoid missing an overcapture which could happen after reading the flag and before reading the data.

Note: IC interrupt and/or DMA requests can be generated by software by setting the corresponding CCxG bit in the TIMx_EGR register.

41.4.7 PWM input mode (only for TIM15)

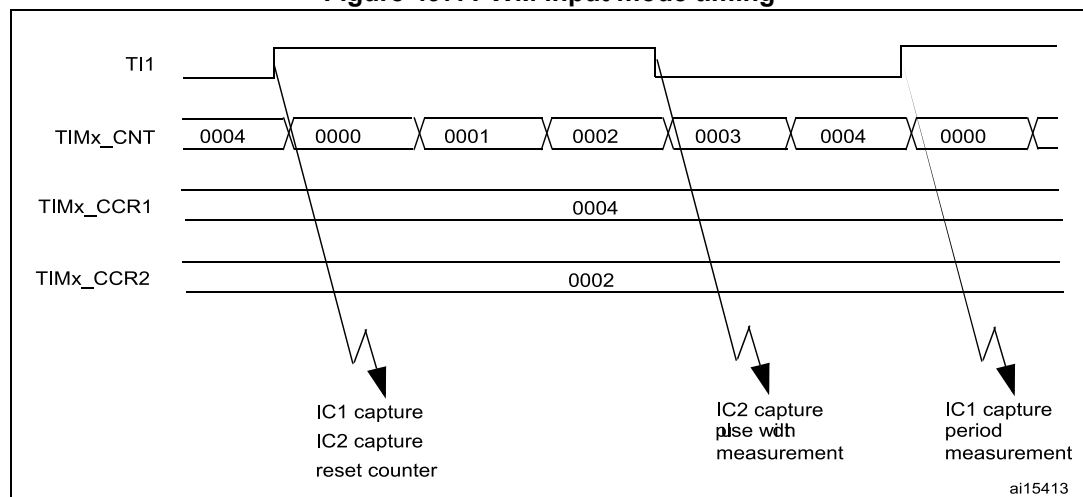
This mode is a particular case of input capture mode. The procedure is the same except:

- Two ICx signals are mapped on the same TIx input.
- These 2 ICx signals are active on edges with opposite polarity.
- One of the two TIxFP signals is selected as trigger input and the slave mode controller is configured in reset mode.

For example, one can measure the period (in TIMx_CCR1 register) and the duty cycle (in TIMx_CCR2 register) of the PWM applied on TI1 using the following procedure (depending on CK_INT frequency and prescaler value):

1. Select the proper TI1[x] source (internal or external) with the TI1SEL[3:0] bits in the TIMx_TISEL register.
2. Select the active input for TIMx_CCR1: write the CC1S bits to 01 in the TIMx_CCMR1 register (TI1 selected).
3. Select the active polarity for TI1FP1 (used both for capture in TIMx_CCR1 and counter clear): write the CC1P and CC1NP bits to '0' (active on rising edge).
4. Select the active input for TIMx_CCR2: write the CC2S bits to 10 in the TIMx_CCMR1 register (TI1 selected).
5. Select the active polarity for TI1FP2 (used for capture in TIMx_CCR2): write the CC2P and CC2NP bits to '1' (active on falling edge).
6. Select the valid trigger input: write the TS bits to 00101 in the TIMx_SMCR register (TI1FP1 selected).
7. Configure the slave mode controller in reset mode: write the SMS bits to 100 in the TIMx_SMCR register.
8. Enable the captures: write the CC1E and CC2E bits to '1' in the TIMx_CCER register.

Figure 497. PWM input mode timing



1. The PWM input mode can be used only with the TIMx_CH1/TIMx_CH2 signals due to the fact that only TI1FP1 and TI2FP2 are connected to the slave mode controller.

41.4.8 Forced output mode

In output mode (CCxS bits = 00 in the TIMx_CCMRx register), each output compare signal (OCxREF and then OCx/OCxN) can be forced to active or inactive level directly by software, independently of any comparison between the output compare register and the counter.

To force an output compare signal (OCXREF/OCx) to its active level, one just needs to write 101 in the OCxM bits in the corresponding TIMx_CCMRx register. Thus OCXREF is forced high (OCxREF is always active high) and OCx get opposite value to CCxP polarity bit.

For example: CCxP=0 (OCx active high) => OCx is forced to high level.

The OCxREF signal can be forced low by writing the OCxM bits to 100 in the TIMx_CCMRx register.

Anyway, the comparison between the TIMx_CCRx shadow register and the counter is still performed and allows the flag to be set. Interrupt and DMA requests can be sent accordingly. This is described in the output compare mode section below.

41.4.9 Output compare mode

This function is used to control an output waveform or indicating when a period of time has elapsed.

When a match is found between the capture/compare register and the counter, the output compare function:

- Assigns the corresponding output pin to a programmable value defined by the output compare mode (OCxM bits in the TIMx_CCMRx register) and the output polarity (CCxP bit in the TIMx_CCER register). The output pin can keep its level (OCxM=000), be set active (OCxM=001), be set inactive (OCxM=010) or can toggle (OCxM=011) on match.
- Sets a flag in the interrupt status register (CCxIF bit in the TIMx_SR register).
- Generates an interrupt if the corresponding interrupt mask is set (CCxIE bit in the TIMx_DIER register).
- Sends a DMA request if the corresponding enable bit is set (CCxDE bit in the TIMx_DIER register, CCDS bit in the TIMx_CR2 register for the DMA request selection).

The TIMx_CCRx registers can be programmed with or without preload registers using the OCxPE bit in the TIMx_CCMRx register.

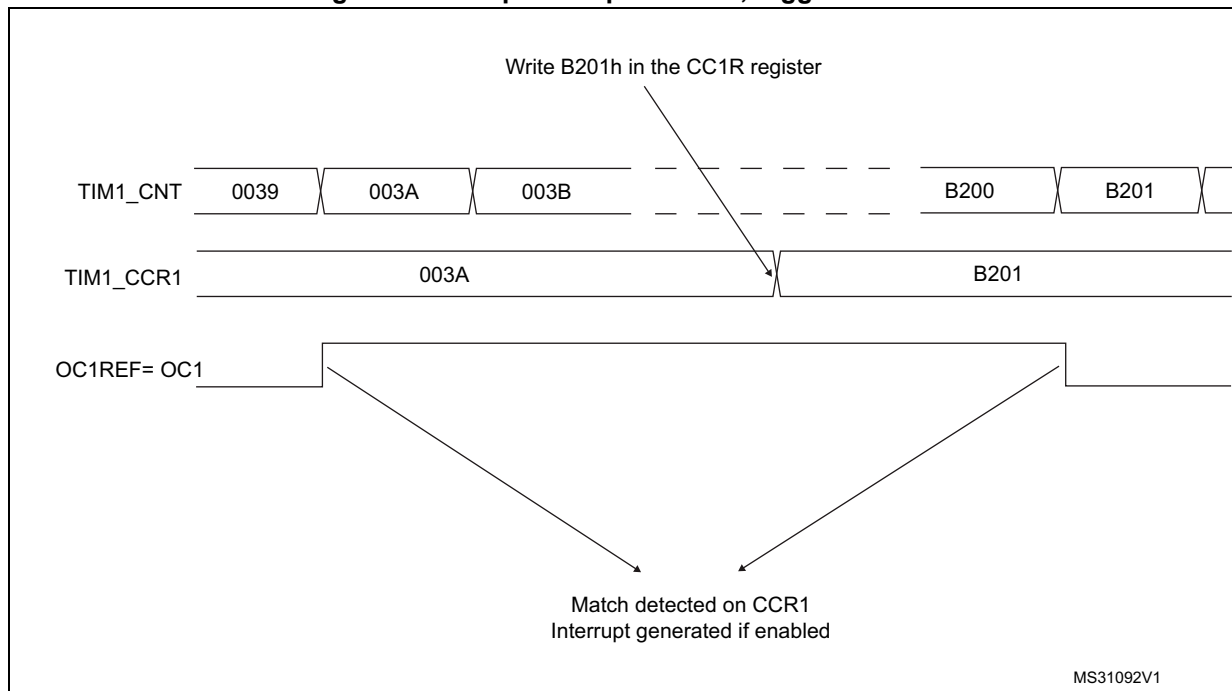
In output compare mode, the update event UEV has no effect on OCxREF and OCx output. The timing resolution is one count of the counter. Output compare mode can also be used to output a single pulse (in One-pulse mode).

Procedure

1. Select the counter clock (internal, external, prescaler).
2. Write the desired data in the TIMx_ARR and TIMx_CCRx registers.
3. Set the CCxIE bit if an interrupt request is to be generated.
4. Select the output mode. For example:
 - Write OCxM = 011 to toggle OCx output pin when CNT matches CCRx
 - Write OCxPE = 0 to disable preload register
 - Write CCxP = 0 to select active high polarity
 - Write CCxE = 1 to enable the output
5. Enable the counter by setting the CEN bit in the TIMx_CR1 register.

The TIMx_CCRx register can be updated at any time by software to control the output waveform, provided that the preload register is not enabled (OCxPE='0', else TIMx_CCRx shadow register is updated only at the next update event UEV). An example is given in [Figure 498](#).

Figure 498. Output compare mode, toggle on OC1



41.4.10 PWM mode

Pulse Width Modulation mode allows a signal to be generated with a frequency determined by the value of the TIMx_ARR register and a duty cycle determined by the value of the TIMx_CCRx register.

The PWM mode can be selected independently on each channel (one PWM per OCx output) by writing '110' (PWM mode 1) or '111' (PWM mode 2) in the OCxM bits in the TIMx_CCMRx register. The corresponding preload register must be enabled by setting the OCxPE bit in the TIMx_CCMRx register, and eventually the auto-reload preload register (in upcounting or center-aligned modes) by setting the ARPE bit in the TIMx_CR1 register.

As the preload registers are transferred to the shadow registers only when an update event occurs, before starting the counter, all registers must be initialized by setting the UG bit in the TIMx_EGR register.

OCx polarity is software programmable using the CCxP bit in the TIMx_CCER register. It can be programmed as active high or active low. OCx output is enabled by a combination of the CCxE, CCxNE, MOE, OSSI and OSSR bits (TIMx_CCER and TIMx_BDTR registers). Refer to the TIMx_CCER register description for more details.

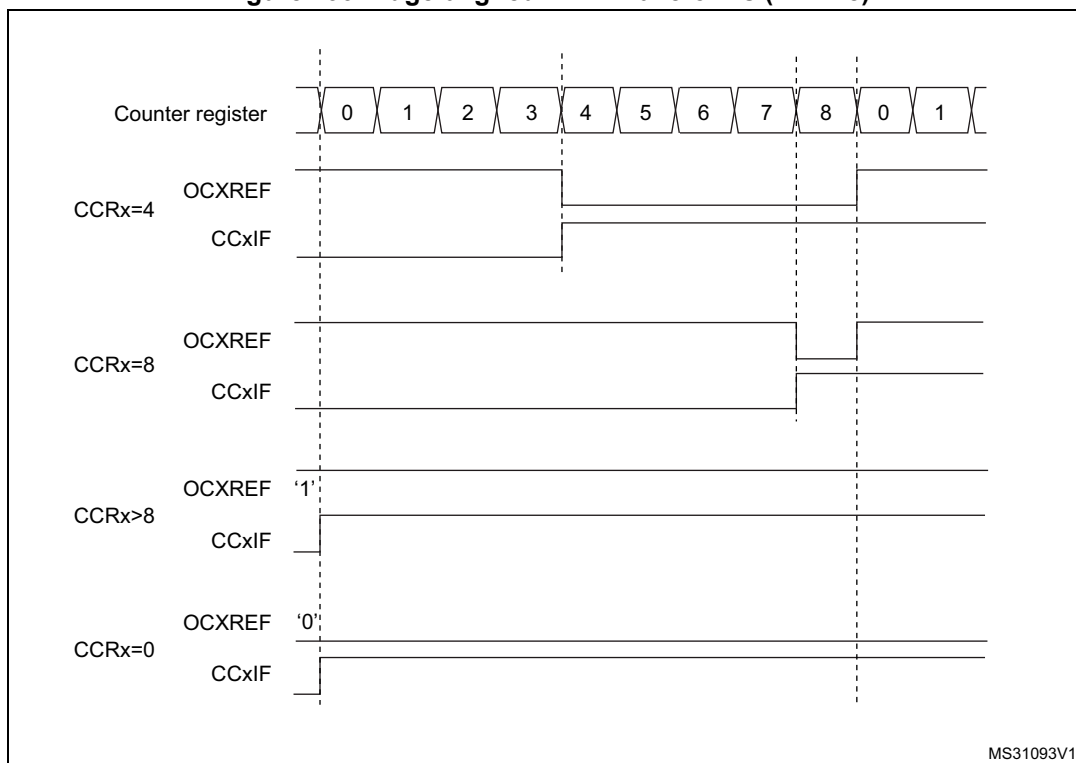
In PWM mode (1 or 2), TIMx_CNT and TIMx_CCRx are always compared to determine whether $TIMx_CCRx \leq TIMx_CNT$ or $TIMx_CNT \leq TIMx_CCRx$ (depending on the direction of the counter).

The TIM15/TIM16/TIM17 are capable of upcounting only. Refer to [Upcounting mode on page 1785](#).

In the following example, we consider PWM mode 1. The reference PWM signal OCxREF is high as long as $TIMx_CNT < TIMx_CCRx$ else it becomes low. If the compare value in TIMx_CCRx is greater than the auto-reload value (in TIMx_ARR) then OCxREF is held at

'1'. If the compare value is 0 then OCxRef is held at '0'. [Figure 499](#) shows some edge-aligned PWM waveforms in an example where TIMx_ARR=8.

Figure 499. Edge-aligned PWM waveforms (ARR=8)



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41.4.11 Combined PWM mode (TIM15 only)

Combined PWM mode allows two edge or center-aligned PWM signals to be generated with programmable delay and phase shift between respective pulses. While the frequency is determined by the value of the TIMx_ARR register, the duty cycle and delay are determined by the two TIMx_CCRx registers. The resulting signals, OCxREFC, are made of an OR or AND logical combination of two reference PWMs:

- OC1REFC (or OC2REFC) is controlled by the TIMx_CCR1 and TIMx_CCR2 registers

Combined PWM mode can be selected independently on two channels (one OCx output per pair of CCR registers) by writing '1100' (Combined PWM mode 1) or '1101' (Combined PWM mode 2) in the OCxM bits in the TIMx_CCMRx register.

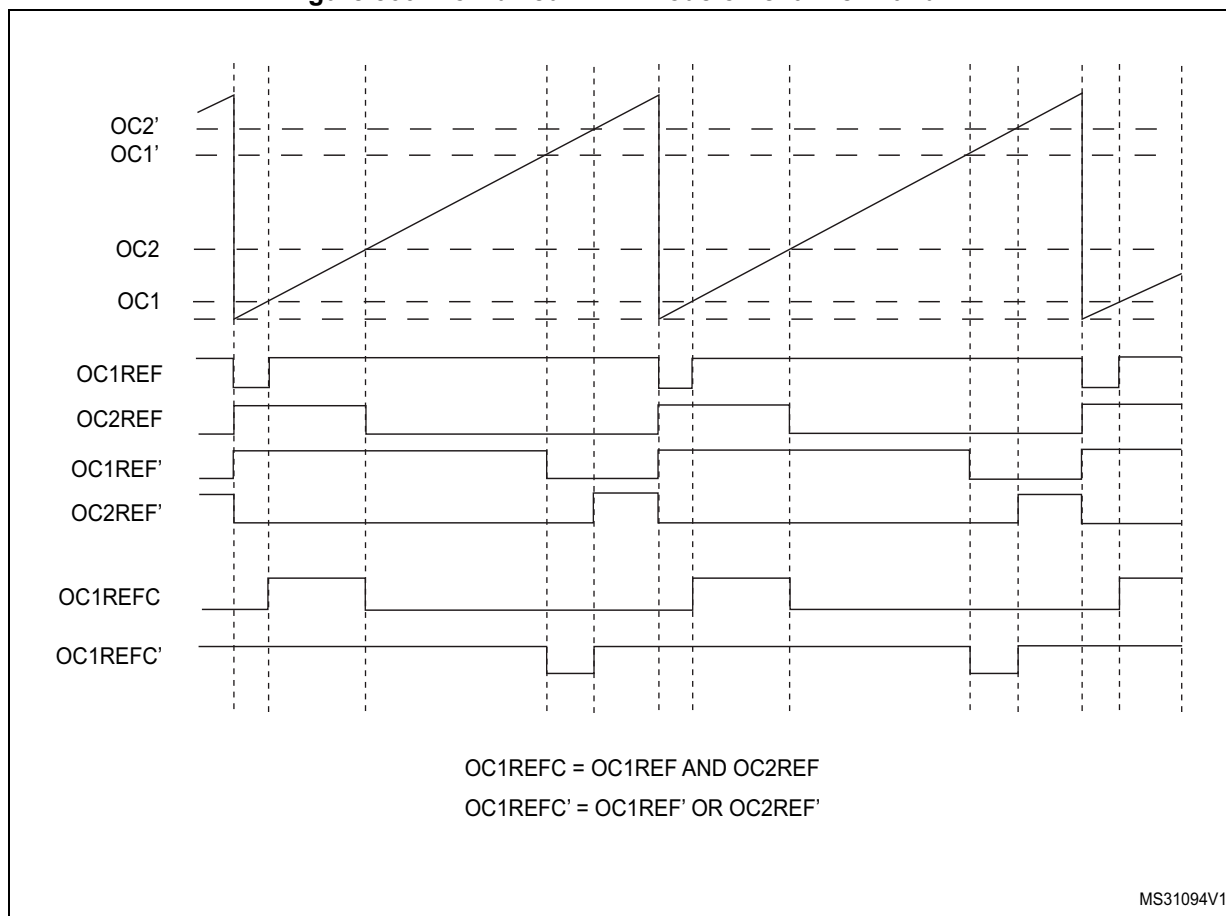
When a given channel is used as a combined PWM channel, its complementary channel must be configured in the opposite PWM mode (for instance, one in Combined PWM mode 1 and the other in Combined PWM mode 2).

Note: The OCxM[3:0] bit field is split into two parts for compatibility reasons, the most significant bit is not contiguous with the 3 least significant ones.

[Figure 500](#) represents an example of signals that can be generated using Asymmetric PWM mode, obtained with the following configuration:

- Channel 1 is configured in Combined PWM mode 2,
- Channel 2 is configured in PWM mode 1,

Figure 500. Combined PWM mode on channel 1 and 2



41.4.12 Complementary outputs and dead-time insertion

The TIM15/TIM16/TIM17 general-purpose timers can output one complementary signal and manage the switching-off and switching-on of the outputs.

This time is generally known as dead-time and it has to be adjusted depending on the devices that are connected to the outputs and their characteristics (intrinsic delays of level-shifters, delays due to power switches...)

The polarity of the outputs (main output OCx or complementary OCxN) can be selected independently for each output. This is done by writing to the CCxP and CCxNP bits in the TIMx_CCER register.

The complementary signals OCx and OCxN are activated by a combination of several control bits: the CCxE and CCxNE bits in the TIMx_CCER register and the MOE, OISx, OISxN, OSSI and OSSR bits in the TIMx_BDTR and TIMx_CR2 registers. Refer to [Table 345: Output control bits for complementary OCx and OCxN channels with break feature \(TIM16/17\) on page 1852](#) for more details. In particular, the dead-time is activated when switching to the idle state (MOE falling down to 0).

Dead-time insertion is enabled by setting both CCxE and CCxNE bits, and the MOE bit if the break circuit is present. There is one 10-bit dead-time generator for each channel. From a

reference waveform OCxREF, it generates 2 outputs OCx and OCxN. If OCx and OCxN are active high:

- The OCx output signal is the same as the reference signal except for the rising edge, which is delayed relative to the reference rising edge.
- The OCxN output signal is the opposite of the reference signal except for the rising edge, which is delayed relative to the reference falling edge.

If the delay is greater than the width of the active output (OCx or OCxN) then the corresponding pulse is not generated.

The following figures show the relationships between the output signals of the dead-time generator and the reference signal OCxREF. (we suppose CCxP=0, CCxNP=0, MOE=1, CCxE=1 and CCxNE=1 in these examples)

Figure 501. Complementary output with dead-time insertion.

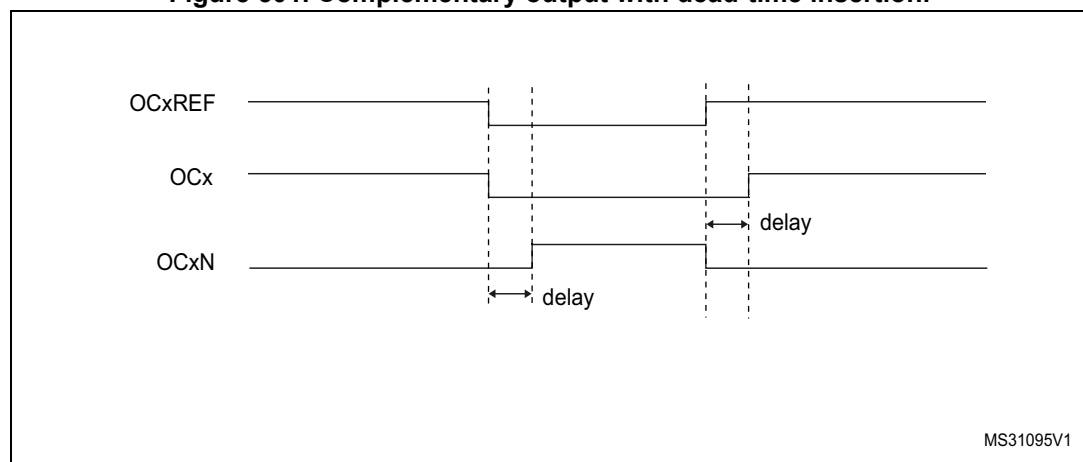


Figure 502. Dead-time waveforms with delay greater than the negative pulse.

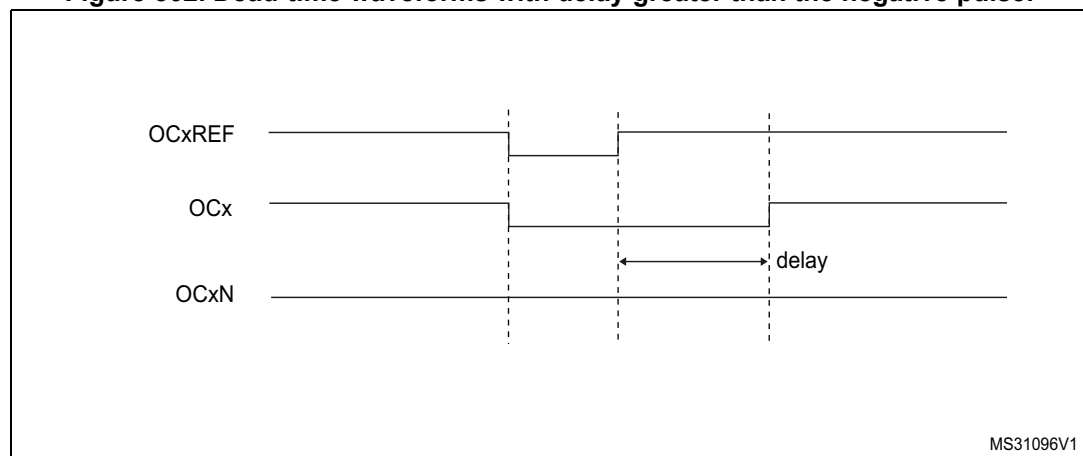
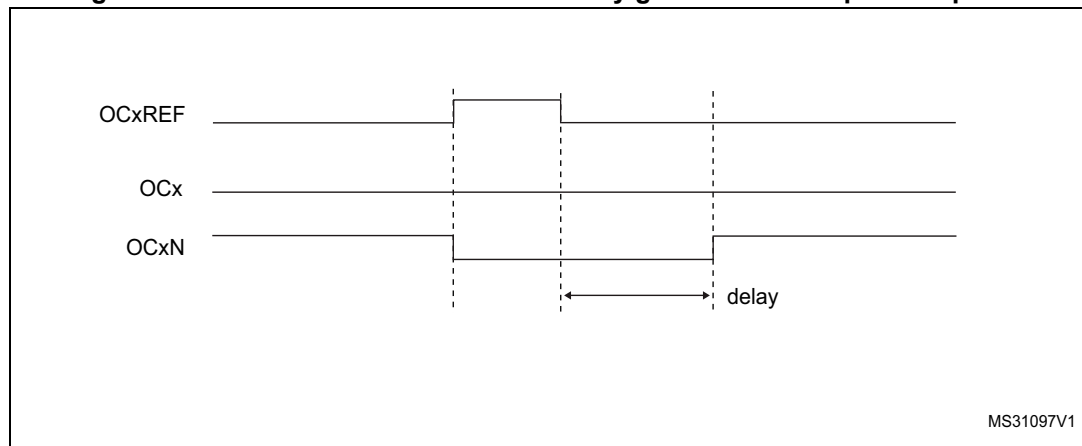


Figure 503. Dead-time waveforms with delay greater than the positive pulse.

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The dead-time delay is the same for each of the channels and is programmable with the DTG bits in the TIMx_BDTR register. Refer to [Section 41.6.14: TIMx break and dead-time register \(TIMx_BDTR\)\(x = 16 to 17\) on page 1855](#) for delay calculation.

Re-directing OCxREF to OCx or OCxN

In output mode (forced, output compare or PWM), OCxREF can be re-directed to the OCx output or to OCxN output by configuring the CCxE and CCxNE bits in the TIMx_CCER register.

This allows a specific waveform to be sent (such as PWM or static active level) on one output while the complementary remains at its inactive level. Other alternative possibilities are to have both outputs at inactive level or both outputs active and complementary with dead-time.

Note: *When only OCxN is enabled (CCxE=0, CCxNE=1), it is not complemented and becomes active as soon as OCxREF is high. For example, if CCxNP=0 then OCxN=OCxRef. On the other hand, when both OCx and OCxN are enabled (CCxE=CCxNE=1) OCx becomes active when OCxREF is high whereas OCxN is complemented and becomes active when OCxREF is low.*

41.4.13 Using the break function

The purpose of the break function is to protect power switches driven by PWM signals generated with the TIM15/TIM16/TIM17 timers. The break input is usually connected to fault outputs of power stages and 3-phase inverters. When activated, the break circuitry shuts down the PWM outputs and forces them to a predefined safe state.

The break channel gathers both system-level fault (clock failure, parity error,...) and application fault (from input pins and built-in comparator), and can force the outputs to a predefined level (either active or inactive) after a deadtime duration.

The output enable signal and output levels during break are depending on several control bits:

- the MOE bit in TIMx_BDTR register allows to enable /disable the outputs by software and is reset in case of break or break2 event.
- the OSSI bit in the TIMx_BDTR register defines whether the timer controls the output in inactive state or releases the control to the GPIO controller (typically to have it in Hi-Z mode)
- the OISx and OISxN bits in the TIMx_CR2 register which are setting the output shut-down level, either active or inactive. The OCx and OCxN outputs cannot be set both to active level at a given time, whatever the OISx and OISxN values. Refer to [Table 343: Output control bits for complementary OCx and OCxN channels with break feature \(TIM15\) on page 1831](#) for more details.

When exiting from reset, the break circuit is disabled and the MOE bit is low. The break function is enabled by setting the BKE bit in the TIMx_BDTR register. The break input polarity can be selected by configuring the BKP bit in the same register. BKE and BKP can be modified at the same time. When the BKE and BKP bits are written, a delay of 1 APB clock cycle is applied before the writing is effective. Consequently, it is necessary to wait 1 APB clock period to correctly read back the bit after the write operation.

Because MOE falling edge can be asynchronous, a resynchronization circuit has been inserted between the actual signal (acting on the outputs) and the synchronous control bit (accessed in the TIMx_BDTR register). It results in some delays between the asynchronous and the synchronous signals. In particular, if MOE is set to 1 whereas it was low, a delay must be inserted (dummy instruction) before reading it correctly. This is because the write acts on the asynchronous signal whereas the read reflects the synchronous signal.

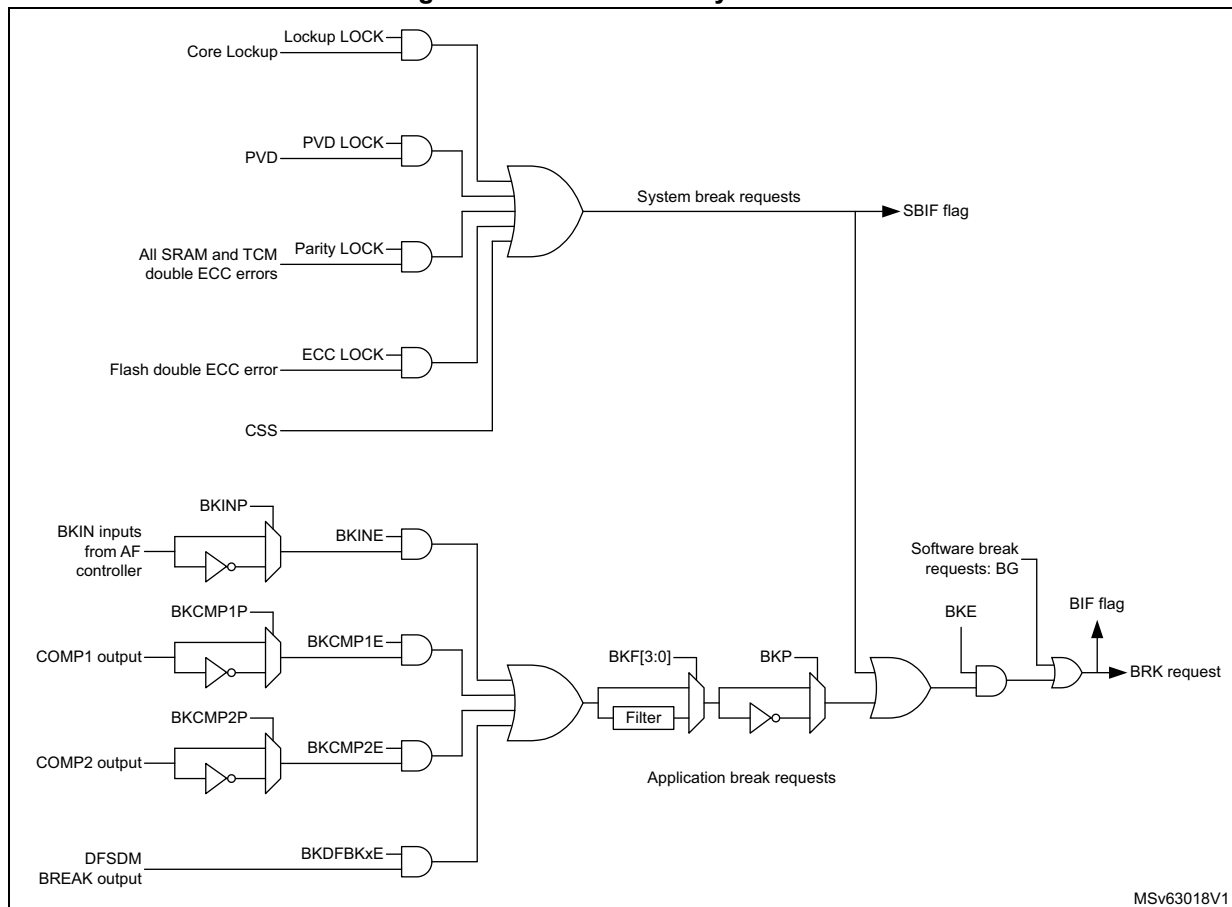
A programmable filter (BKF[3:0] bits in the TIMx_BDTR register allows to filter out spurious events.

The break can be generated from multiple sources which can be individually enabled and with programmable edge sensitivity, using the TIMx_OR2 register.

The sources for break (BRK) channel are:

- An external source connected to one of the BKIN pin (as per selection done in the AFIO controller), with polarity selection and optional digital filtering
- An internal source:
 - the output from a comparator, with polarity selection and optional digital filtering
 - the analog watchdog output of the DFSDM1 peripheral
 - A system break:
 - the Cortex®-M7 with FPU LOCKUP output
 - the PVD output
 - all SRAM and TCM double ECC errors (AXI-SRAM, ITCM, DTCM, SRAM1, SRAM2, SRAM3, SRAM4, BKRAM, refer to SYSCFG_CFGR register for details)
 - a Flash double ECC error
 - a clock failure event generated by the CSS detector

Figure 504. Break circuitry overview



Caution: An asynchronous (clockless) operation is only guaranteed when the programmable filter is disabled. If it is enabled, a fail safe clock mode (example, using the internal PLL and/or the CSS) must be used to guarantee that break events are handled.

When a break occurs (selected level on the break input):

- The MOE bit is cleared asynchronously, putting the outputs in inactive state, idle state or even releasing the control to the AFIO controller (selected by the OSS1 bit). This feature functions even if the MCU oscillator is off.
- Each output channel is driven with the level programmed in the OISx bit in the TIMx_CR2 register as soon as MOE=0. If OSS1=0, the timer releases the output control (taken over by the AFIO controller) else the enable output remains high.
- When complementary outputs are used:
 - The outputs are first put in reset state inactive state (depending on the polarity). This is done asynchronously so that it works even if no clock is provided to the timer.
 - If the timer clock is still present, then the dead-time generator is reactivated in order to drive the outputs with the level programmed in the OISx and OISxN bits after a dead-time. Even in this case, OCx and OCxN cannot be driven to their

active level together. Note that because of the resynchronization on MOE, the dead-time duration is a bit longer than usual (around 2 ck_tim clock cycles).

- If OSSR=0 then the timer releases the enable outputs (taken over by the AFIO controller which forces a Hi-Z state) else the enable outputs remain or become high as soon as one of the CCxE or CCxNE bits is high.
- The break status flag (BIF bit in the TIMx_SR register) is set. An interrupt can be generated if the BIE bit in the TIMx_DIER register is set.
- If the AOE bit in the TIMx_BDTR register is set, the MOE bit is automatically set again at the next update event UEV. This can be used to perform a regulation, for instance. Else, MOE remains low until it is written with 1 again. In this case, it can be used for security and the break input can be connected to an alarm from power drivers, thermal sensors or any security components.

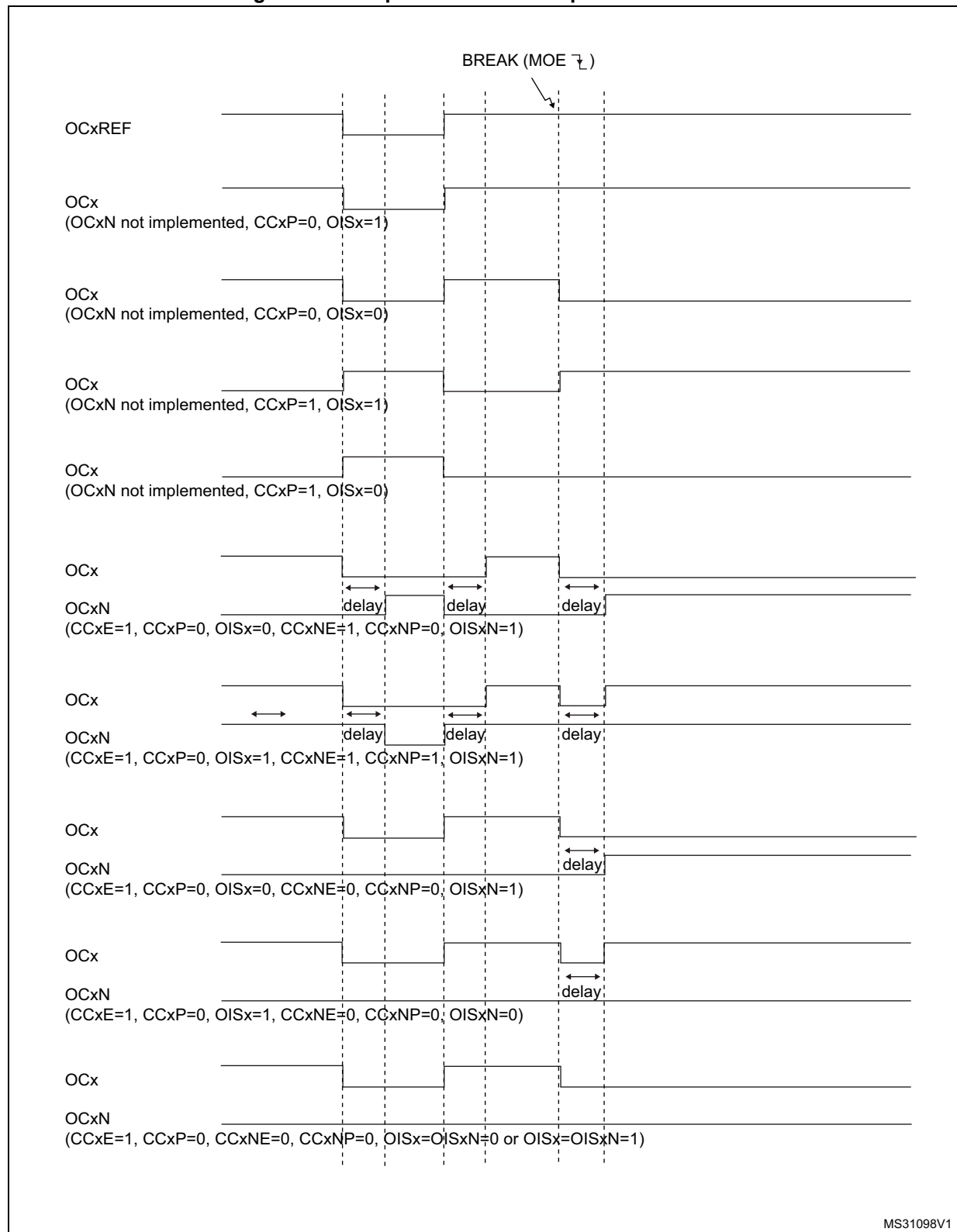
Note: *The break inputs is acting on level. Thus, the MOE cannot be set while the break input is active (neither automatically nor by software). In the meantime, the status flag BIF cannot be cleared.*

The break can be generated by the BRK input which has a programmable polarity and an enable bit BKE in the TIMx_BDTR Register.

In addition to the break input and the output management, a write protection has been implemented inside the break circuit to safeguard the application. It allows the configuration of several parameters to be frozen (dead-time duration, OCx/OCxN polarities and state when disabled, OCxM configurations, break enable and polarity). The protection can be selected among 3 levels with the LOCK bits in the TIMx_BDTR register. Refer to [Section 41.6.14: TIMx break and dead-time register \(TIMx_BDTR\)\(x = 16 to 17\) on page 1855](#). The LOCK bits can be written only once after an MCU reset.

The [Figure 505](#) shows an example of behavior of the outputs in response to a break.

Figure 505. Output behavior in response to a break



41.4.14 One-pulse mode

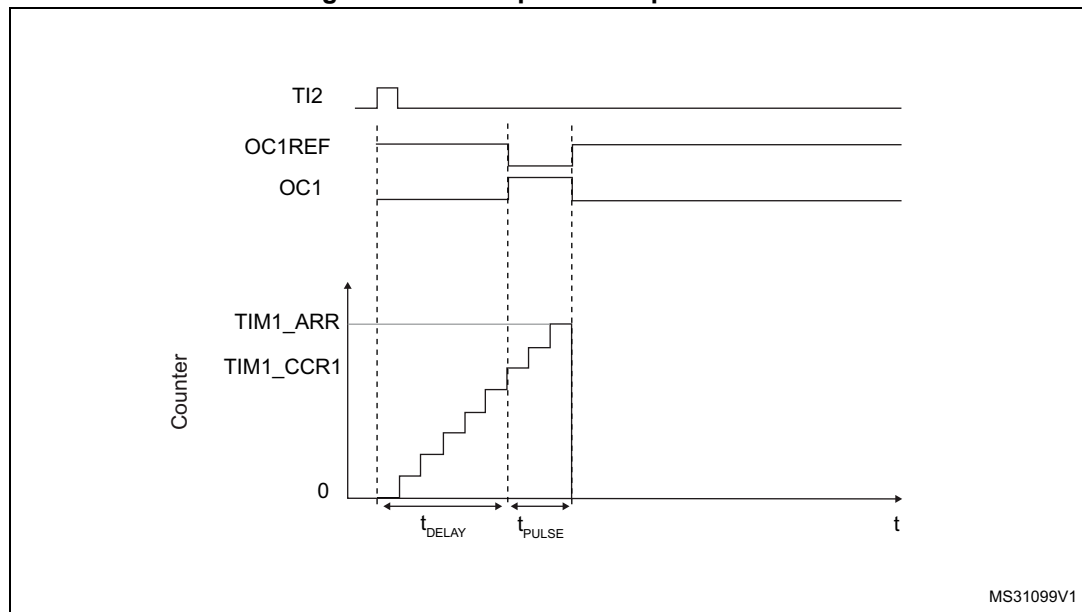
One-pulse mode (OPM) is a particular case of the previous modes. It allows the counter to be started in response to a stimulus and to generate a pulse with a programmable length after a programmable delay.

Starting the counter can be controlled through the slave mode controller. Generating the waveform can be done in output compare mode or PWM mode. One-pulse mode is selected by setting the OPM bit in the TIMx_CR1 register. This makes the counter stop automatically at the next update event UEV.

A pulse can be correctly generated only if the compare value is different from the counter initial value. Before starting (when the timer is waiting for the trigger), the configuration must be:

- $CNT < CCRx \leq ARR$ (in particular, $0 < CCRx$)

Figure 506. Example of one pulse mode



For example one may want to generate a positive pulse on OC1 with a length of t_{PULSE} and after a delay of t_{DELAY} as soon as a positive edge is detected on the TI2 input pin.

Let's use TI2FP2 as trigger 1:

1. Select the proper TI2[x] source (internal or external) with the TI2SEL[3:0] bits in the TIMx_TISEL register.
2. Map TI2FP2 to TI2 by writing CC2S='01' in the TIMx_CCMR1 register.
3. TI2FP2 must detect a rising edge, write CC2P='0' and CC2NP='0' in the TIMx_CCER register.
4. Configure TI2FP2 as trigger for the slave mode controller (TRGI) by writing TS='00110' in the TIMx_SMCR register.
5. TI2FP2 is used to start the counter by writing SMS to '110' in the TIMx_SMCR register (trigger mode).

The OPM waveform is defined by writing the compare registers (taking into account the clock frequency and the counter prescaler).

- The t_{DELAY} is defined by the value written in the TIMx_CCR1 register.
- The t_{PULSE} is defined by the difference between the auto-reload value and the compare value (TIMx_ARR - TIMx_CCR1).
- Let's say one want to build a waveform with a transition from '0' to '1' when a compare match occurs and a transition from '1' to '0' when the counter reaches the auto-reload value. To do this PWM mode 2 must be enabled by writing OC1M=111 in the TIMx_CCMR1 register. Optionally the preload registers can be enabled by writing OC1PE='1' in the TIMx_CCMR1 register and ARPE in the TIMx_CR1 register. In this case one has to write the compare value in the TIMx_CCR1 register, the auto-reload value in the TIMx_ARR register, generate an update by setting the UG bit and wait for external trigger event on TI2. CC1P is written to '0' in this example.

Since only 1 pulse is needed, a 1 must be written in the OPM bit in the TIMx_CR1 register to stop the counter at the next update event (when the counter rolls over from the auto-reload value back to 0).

Particular case: OCx fast enable

In One-pulse mode, the edge detection on TIx input set the CEN bit which enables the counter. Then the comparison between the counter and the compare value makes the output toggle. But several clock cycles are needed for these operations and it limits the minimum delay $t_{\text{DELAY min}}$ we can get.

If one wants to output a waveform with the minimum delay, the OCxFE bit can be set in the TIMx_CCMRx register. Then OCxRef (and OCx) are forced in response to the stimulus, without taking in account the comparison. Its new level is the same as if a compare match had occurred. OCxFE acts only if the channel is configured in PWM1 or PWM2 mode.

41.4.15 Retriggerable one pulse mode (OPM) (TIM15 only)

This mode allows the counter to be started in response to a stimulus and to generate a pulse with a programmable length, but with the following differences with Non-retriggerable one pulse mode described in [Section 41.4.14](#):

- The pulse starts as soon as the trigger occurs (no programmable delay)
- The pulse is extended if a new trigger occurs before the previous one is completed

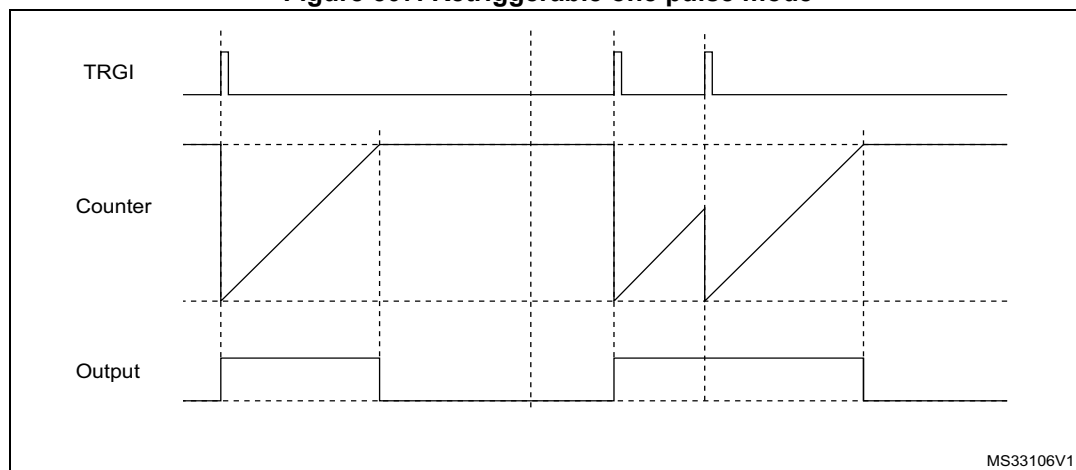
The timer must be in Slave mode, with the bits SMS[3:0] = '1000' (Combined Reset + trigger mode) in the TIMx_SMCR register, and the OCxM[3:0] bits set to '1000' or '1001' for Retriggerable OPM mode 1 or 2.

If the timer is configured in Up-counting mode, the corresponding CCRx must be set to 0 (the ARR register sets the pulse length). If the timer is configured in Down-counting mode, CCRx must be above or equal to ARR.

Note: The OCxM[3:0] and SMS[3:0] bit fields are split into two parts for compatibility reasons, the most significant bit are not contiguous with the 3 least significant ones.

This mode must not be used with center-aligned PWM modes. It is mandatory to have CMS[1:0] = 00 in TIMx_CR1.

Figure 507. Retriggerable one pulse mode



41.4.16 UIF bit remapping

The IUFREMAP bit in the TIMx_CR1 register forces a continuous copy of the Update Interrupt Flag UIF into bit 31 of the timer counter register (TIMxCNT[31]). This allows both

the counter value and a potential roll-over condition signaled by the UIFCPY flag, to be atomically read. In particular cases, it can ease the calculations by avoiding race conditions caused for instance by a processing shared between a background task (counter reading) and an interrupt (Update Interrupt).

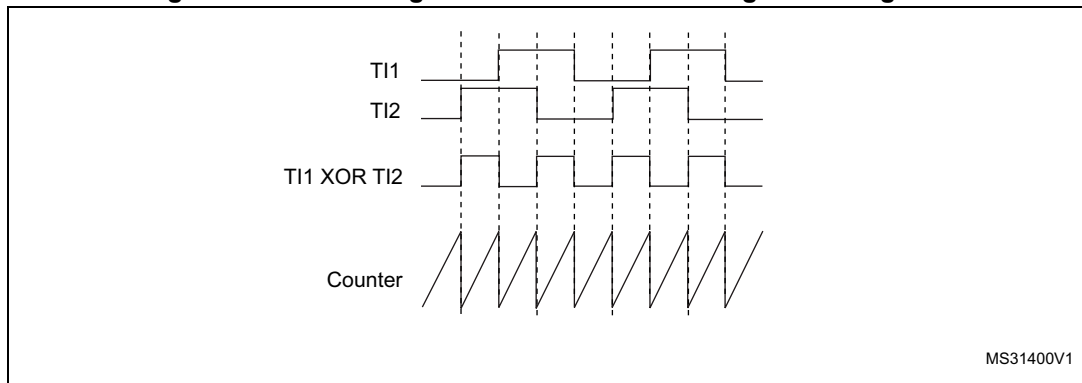
There is no latency between the assertions of the UIF and UIFCPY flags.

41.4.17 Timer input XOR function (TIM15 only)

The TI1S bit in the TIMx_CR2 register, allows the input filter of channel 1 to be connected to the output of a XOR gate, combining the two input pins TIMx_CH1 and TIMx_CH2.

The XOR output can be used with all the timer input functions such as trigger or input capture. It is useful for measuring the interval between the edges on two input signals, as shown in [Figure 508](#).

Figure 508. Measuring time interval between edges on 2 signals



41.4.18 External trigger synchronization (TIM15 only)

The TIM timers are linked together internally for timer synchronization or chaining.

The TIM15 timer can be synchronized with an external trigger in several modes: Reset mode, Gated mode and Trigger mode.

Slave mode: Reset mode

The counter and its prescaler can be reinitialized in response to an event on a trigger input. Moreover, if the URS bit from the TIMx_CR1 register is low, an update event UEV is generated. Then all the preloaded registers (TIMx_ARR, TIMx_CCRx) are updated.

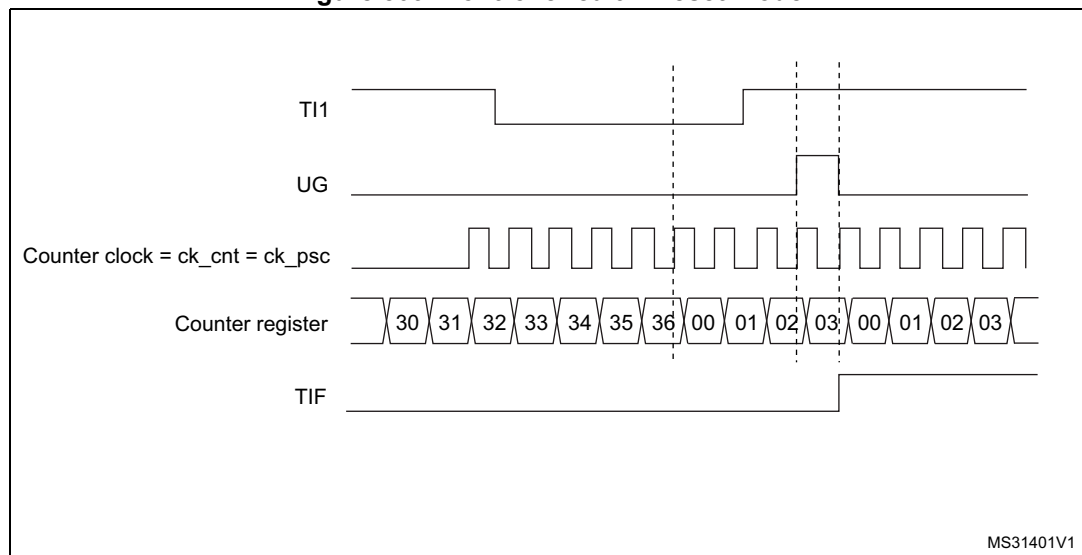
In the following example, the upcounter is cleared in response to a rising edge on TI1 input:

1. Configure the channel 1 to detect rising edges on TI1. Configure the input filter duration (in this example, we do not need any filter, so we keep IC1F=0000). The capture prescaler is not used for triggering, so it does not need to be configured. The CC1S bits select the input capture source only, CC1S = 01 in the TIMx_CCMR1 register. Write CC1P='0' and CC1NP='0' in the TIMx_CCER register to validate the polarity (and detect rising edges only).
2. Configure the timer in reset mode by writing SMS=100 in TIMx_SMCR register. Select TI1 as the input source by writing TS=00101 in TIMx_SMCR register.
3. Start the counter by writing CEN=1 in the TIMx_CR1 register.

The counter starts counting on the internal clock, then behaves normally until TI1 rising edge. When TI1 rises, the counter is cleared and restarts from 0. In the meantime, the trigger flag is set (TIF bit in the TIMx_SR register) and an interrupt request, or a DMA request can be sent if enabled (depending on the TIE and TDE bits in TIMx_DIER register).

The following figure shows this behavior when the auto-reload register TIMx_ARR=0x36. The delay between the rising edge on TI1 and the actual reset of the counter is due to the resynchronization circuit on TI1 input.

Figure 509. Control circuit in reset mode



Slave mode: Gated mode

The counter can be enabled depending on the level of a selected input.

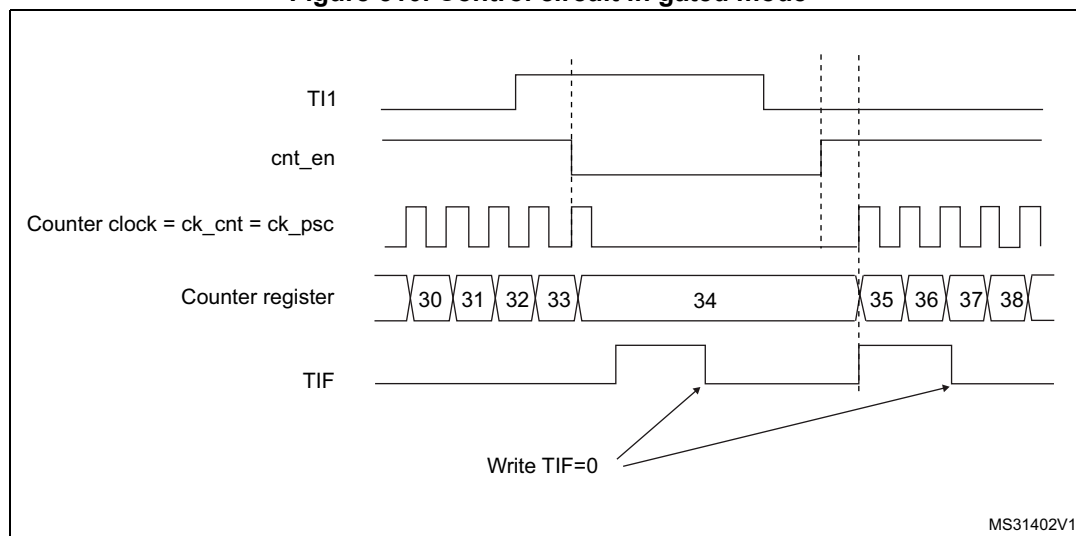
In the following example, the upcounter counts only when TI1 input is low:

1. Configure the channel 1 to detect low levels on TI1. Configure the input filter duration (in this example, we do not need any filter, so we keep IC1F=0000). The capture prescaler is not used for triggering, so it does not need to be configured. The CC1S bits select the input capture source only, CC1S=01 in TIMx_CCMR1 register. Write CC1P=1 and CC1NP = '0' in the TIMx_CCER register to validate the polarity (and detect low level only).
2. Configure the timer in gated mode by writing SMS=101 in TIMx_SMCR register. Select TI1 as the input source by writing TS=00101 in TIMx_SMCR register.
3. Enable the counter by writing CEN=1 in the TIMx_CR1 register (in gated mode, the counter doesn't start if CEN=0, whatever is the trigger input level).

The counter starts counting on the internal clock as long as TI1 is low and stops as soon as TI1 becomes high. The TIF flag in the TIMx_SR register is set both when the counter starts or stops.

The delay between the rising edge on TI1 and the actual stop of the counter is due to the resynchronization circuit on TI1 input.

Figure 510. Control circuit in gated mode



MS31402V1

Slave mode: Trigger mode

The counter can start in response to an event on a selected input.

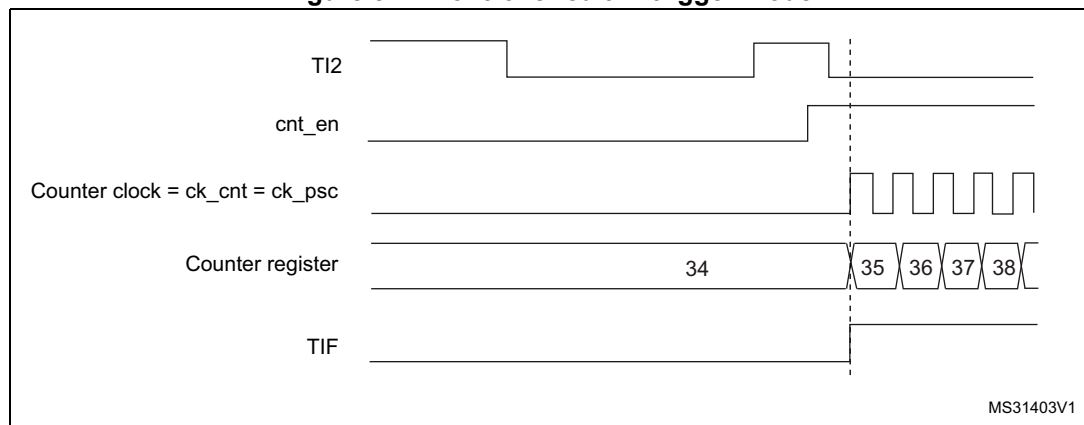
In the following example, the upcounter starts in response to a rising edge on TI2 input:

1. Configure the channel 2 to detect rising edges on TI2. Configure the input filter duration (in this example, we do not need any filter, so we keep IC2F=0000). The capture prescaler is not used for triggering, so it does not need to be configured. The CC2S bits are configured to select the input capture source only, CC2S=01 in TIMx_CCMR1 register. Write CC2P='1' and CC2NP='0' in the TIMx_CCER register to validate the polarity (and detect low level only).
2. Configure the timer in trigger mode by writing SMS=110 in the TIMx_SMCR register. Select TI2 as the input source by writing TS=00110 in the TIMx_SMCR register.

When a rising edge occurs on TI2, the counter starts counting on the internal clock and the TIF flag is set.

The delay between the rising edge on TI2 and the actual start of the counter is due to the resynchronization circuit on TI2 input.

Figure 511. Control circuit in trigger mode



41.4.19 Slave mode – combined reset + trigger mode

In this case, a rising edge of the selected trigger input (TRGI) reinitializes the counter, generates an update of the registers, and starts the counter.

This mode is used for one-pulse mode.

41.4.20 DMA burst mode

The TIMx timers have the capability to generate multiple DMA requests on a single event. The main purpose is to be able to re-program several timer registers multiple times without software overhead, but it can also be used to read several registers in a row, at regular intervals.

The DMA controller destination is unique and must point to the virtual register TIMx_DMAR. On a given timer event, the timer launches a sequence of DMA requests (burst). Each write into the TIMx_DMAR register is actually redirected to one of the timer registers.

The DBL[4:0] bits in the TIMx_DCR register set the DMA burst length. The timer recognizes a burst transfer when a read or a write access is done to the TIMx_DMAR address), i.e. the number of transfers (either in half-words or in bytes).

The DBA[4:0] bits in the TIMx_DCR registers define the DMA base address for DMA transfers (when read/write access are done through the TIMx_DMAR address). DBA is defined as an offset starting from the address of the TIMx_CR1 register.

Example:

00000: TIMx_CR1,

00001: TIMx_CR2,

00010: TIMx_SMCR,

For example, the timer DMA burst feature could be used to update the contents of the CCRx registers (x = 2, 3, 4) on an update event, with the DMA transferring half words into the CCRx registers.

This is done in the following steps:

1. Configure the corresponding DMA channel as follows:
 - DMA channel peripheral address is the DMAR register address
 - DMA channel memory address is the address of the buffer in the RAM containing the data to be transferred by DMA into the CCRx registers.
 - Number of data to transfer = 3 (See note below).
 - Circular mode disabled.
2. Configure the DCR register by configuring the DBA and DBL bit fields as follows:
DBL = 3 transfers, DBA = 0xE.
3. Enable the TIMx update DMA request (set the UDE bit in the DIER register).
4. Enable TIMx
5. Enable the DMA channel

This example is for the case where every CCRx register is to be updated once. If every CCRx register is to be updated twice for example, the number of data to transfer should be 6. Let's take the example of a buffer in the RAM containing data1, data2, data3, data4, data5 and data6. The data is transferred to the CCRx registers as follows: on the first update DMA request, data1 is transferred to CCR2, data2 is transferred to CCR3, data3 is transferred to CCR4 and on the second update DMA request, data4 is transferred to CCR2, data5 is transferred to CCR3 and data6 is transferred to CCR4.

Note: A null value can be written to the reserved registers.

41.4.21 Timer synchronization (TIM15)

The TIMx timers are linked together internally for timer synchronization or chaining. Refer to [Section 39.3.19: Timer synchronization](#) for details.

Note: The clock of the slave peripherals (timer, ADC, ...) receiving the TRGO or the TRGO2 signals must be enabled prior to receive events from the master timer, and the clock frequency (prescaler) must not be changed on-the-fly while triggers are received from the master timer.

41.4.22 Debug mode

When the microcontroller enters debug mode (Cortex[®]-M7 with FPU core halted), the TIMx counter either continues to work normally or stops, depending on TIMx bit in DBGMCU module. For more details, refer to [Section 60: Debug infrastructure](#).

For safety purposes, when the counter is stopped (TIMx = 1 in DBGMCU_APB2FZ1), the outputs are disabled (as if the MOE bit was reset). The outputs can either be forced to an inactive state (OSSI bit = 1), or have their control taken over by the GPIO controller (OSSI bit = 0) to force them to Hi-Z.

41.5 TIM15 registers

Refer to [Section 1.2](#) for a list of abbreviations used in register descriptions.

The peripheral registers can be accessed by half-words (16-bit) or words (32-bit).

41.5.1 TIM15 control register 1 (TIM15_CR1)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	UIFRE MAP	Res.	CKD[1:0]		ARPE	Res.	Res.	Res.	OPM	URS	UDIS	CEN
				rw		rw	rw	rw				rw	rw	rw	rw

Bits 15:12 Reserved, must be kept at reset value.

Bit 11 **UIFREMAP**: UIF status bit remapping

0: No remapping. UIF status bit is not copied to TIMx_CNT register bit 31.

1: Remapping enabled. UIF status bit is copied to TIMx_CNT register bit 31.

Bit 10 Reserved, must be kept at reset value.

Bits 9:8 **CKD[1:0]**: Clock division

This bitfield indicates the division ratio between the timer clock (CK_INT) frequency and the dead-time and sampling clock (t_{DTS}) used by the dead-time generators and the digital filters (TIMx)

00: $t_{DTS} = t_{CK_INT}$

01: $t_{DTS} = 2 * t_{CK_INT}$

10: $t_{DTS} = 4 * t_{CK_INT}$

11: Reserved, do not program this value

Bit 7 **ARPE**: Auto-reload preload enable

0: TIMx_ARR register is not buffered

1: TIMx_ARR register is buffered

Bits 6:4 Reserved, must be kept at reset value.

Bit 3 **OPM**: One-pulse mode

0: Counter is not stopped at update event

1: Counter stops counting at the next update event (clearing the bit CEN)

Bit 2 URS: Update request source

This bit is set and cleared by software to select the UEV event sources.

0: Any of the following events generate an update interrupt if enabled. These events can be:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

1: Only counter overflow/underflow generates an update interrupt if enabled

Bit 1 UDIS: Update disable

This bit is set and cleared by software to enable/disable UEV event generation.

0: UEV enabled. The Update (UEV) event is generated by one of the following events:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

Buffered registers are then loaded with their preload values.

1: UEV disabled. The Update event is not generated, shadow registers keep their value (ARR, PSC, CCRx). However the counter and the prescaler are reinitialized if the UG bit is set or if a hardware reset is received from the slave mode controller.

Bit 0 CEN: Counter enable

0: Counter disabled

1: Counter enabled

Note: External clock and gated mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.

41.5.2 TIM15 control register 2 (TIM15_CR2)

Address offset: 0x04

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	OIS2	OIS1N	OIS1	TI1S	MMS[2:0]			CCDS	CCUS	Res.	CCPC
					rw	rw	rw	rw	rw	rw	rw	rw	rw		rw

Bits 15:11 Reserved, must be kept at reset value.

Bit 10 OIS2: Output idle state 2 (OC2 output)

0: OC2=0 when MOE=0

1: OC2=1 when MOE=0

Note: This bit cannot be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in the TIM15_BDTR register).

Bit 9 OIS1N: Output Idle state 1 (OC1N output)

0: OC1N=0 after a dead-time when MOE=0

1: OC1N=1 after a dead-time when MOE=0

Note: This bit can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIM15_BDTR register).

Bit 8 OIS1: Output Idle state 1 (OC1 output)

0: OC1=0 (after a dead-time if OC1N is implemented) when MOE=0

1: OC1=1 (after a dead-time if OC1N is implemented) when MOE=0

Note: This bit can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIM15_BDTR register).

Bit 7 **TI1S**: TI1 selection

- 0: The TIMx_CH1 pin is connected to TI1 input
- 1: The TIMx_CH1, CH2 pins are connected to the TI1 input (XOR combination)

Bits 6:4 **MMS[2:0]**: Master mode selection

These bits allow to select the information to be sent in master mode to slave timers for synchronization (TRGO). The combination is as follows:

- 000: **Reset** - the UG bit from the TIMx_EGR register is used as trigger output (TRGO). If the reset is generated by the trigger input (slave mode controller configured in reset mode) then the signal on TRGO is delayed compared to the actual reset.
- 001: **Enable** - the Counter Enable signal CNT_EN is used as trigger output (TRGO). It is useful to start several timers at the same time or to control a window in which a slave timer is enable. The Counter Enable signal is generated by a logic AND between CEN control bit and the trigger input when configured in gated mode. When the Counter Enable signal is controlled by the trigger input, there is a delay on TRGO, except if the master/slave mode is selected (see the MSM bit description in TIMx_SMCR register).
- 010: **Update** - The update event is selected as trigger output (TRGO). For instance a master timer can then be used as a prescaler for a slave timer.
- 011: **Compare Pulse** - The trigger output send a positive pulse when the CC1IF flag is to be set (even if it was already high), as soon as a capture or a compare match occurred. (TRGO).
- 100: **Compare** - OC1REF signal is used as trigger output (TRGO).
- 101: **Compare** - OC2REF signal is used as trigger output (TRGO).

Bit 3 **CCDS**: Capture/compare DMA selection

- 0: CCx DMA request sent when CCx event occurs
- 1: CCx DMA requests sent when update event occurs

Bit 2 **CCUS**: Capture/compare control update selection

- 0: When capture/compare control bits are preloaded (CCPC=1), they are updated by setting the COMG bit only.
- 1: When capture/compare control bits are preloaded (CCPC=1), they are updated by setting the COMG bit or when an rising edge occurs on TRGI.

Note: This bit acts only on channels that have a complementary output.

Bit 1 Reserved, must be kept at reset value.

Bit 0 **CCPC**: Capture/compare preloaded control

- 0: CCxE, CCxNE and OCxM bits are not preloaded
- 1: CCxE, CCxNE and OCxM bits are preloaded, after having been written, they are updated only when a commutation event (COM) occurs (COMG bit set or rising edge detected on TRGI, depending on the CCUS bit).

Note: This bit acts only on channels that have a complementary output.

41.5.3 TIM15 slave mode control register (TIM15_SMCR)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TS[4:3]		Res.	Res.	Res.	SMS[3]
										rw	rw				rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MSM	TS[2:0]			Res.	SMS[2:0]		
								rw	rw	rw	rw		rw	rw	rw

Bits 31:22 Reserved, must be kept at reset value.

Bits 19:17 Reserved, must be kept at reset value.

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **MSM**: Master/slave mode

0: No action

1: The effect of an event on the trigger input (TRGI) is delayed to allow a perfect synchronization between the current timer and its slaves (through TRGO). It is useful if we want to synchronize several timers on a single external event.

Bits 21, 20, 6, 5, 4 **TS[4:0]**: Trigger selection

This bit field selects the trigger input to be used to synchronize the counter.

00000: Internal Trigger 0 (ITR0)

00001: Internal Trigger 1 (ITR1)

00010: Internal Trigger 2 (ITR2)

00011: Internal Trigger 3 (ITR3)

00100: TI1 Edge Detector (TI1F_ED)

00101: Filtered Timer Input 1 (TI1FP1)

00110: Filtered Timer Input 2 (TI2FP2)

Other: Reserved

See [Table 342: TIMx Internal trigger connection on page 1821](#) for more details on ITRx meaning for each Timer.

Note: These bits must be changed only when they are not used (e.g. when SMS=000) to avoid wrong edge detections at the transition.

Bit 3 Reserved, must be kept at reset value.

Bits 16, 2, 1, 0 **SMS[3:0]**: Slave mode selection

When external signals are selected the active edge of the trigger signal (TRGI) is linked to the polarity selected on the external input (see Input Control register and Control Register description).

0000: Slave mode disabled - if CEN = '1' then the prescaler is clocked directly by the internal clock.

0001: Reserved

0010: Reserved

0011: Reserved

0100: Reset Mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter and generates an update of the registers.

0101: Gated Mode - The counter clock is enabled when the trigger input (TRGI) is high. The counter stops (but is not reset) as soon as the trigger becomes low. Both start and stop of the counter are controlled.

0110: Trigger Mode - The counter starts at a rising edge of the trigger TRGI (but it is not reset). Only the start of the counter is controlled.

0111: External Clock Mode 1 - Rising edges of the selected trigger (TRGI) clock the counter.

1000: Combined reset + trigger mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter, generates an update of the registers and starts the counter.

Other codes: reserved.

Note: The gated mode must not be used if TI1F_ED is selected as the trigger input (TS='00100'). Indeed, TI1F_ED outputs 1 pulse for each transition on TI1F, whereas the gated mode checks the level of the trigger signal.

Note: The clock of the slave peripherals (timer, ADC, ...) receiving the TRGO or the TRGO2 signals must be enabled prior to receive events from the master timer, and the clock frequency (prescaler) must not be changed on-the-fly while triggers are received from the master timer.

Table 342. TIMx Internal trigger connection

Slave TIM	ITR0 (TS = 00000)	ITR1 (TS = 00001)	ITR2 (TS = 00010)	ITR3 (TS = 00011)
TIM15	TIM1	TIM3	TIM16 OC1	TIM17 OC1

41.5.4 TIM15 DMA/interrupt enable register (TIM15_DIER)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	TDE	COMDE	Res.	Res.	CC2DE	CC1DE	UDE	BIE	TIE	COMIE	Res.	Res.	CC2IE	CC1IE	UIE
	rw	rw			rw	rw	rw	rw	rw	rw			rw	rw	rw

Bit 15 Reserved, must be kept at reset value.

Bit 14 **TDE**: Trigger DMA request enable

0: Trigger DMA request disabled

1: Trigger DMA request enabled

Bit 13 **COMDE**: COM DMA request enable

0: COM DMA request disabled

1: COM DMA request enabled

Bits 12:11 Reserved, must be kept at reset value.

Bit 10 **CC2DE**: Capture/Compare 2 DMA request enable

0: CC2 DMA request disabled
1: CC2 DMA request enabled

Bit 9 **CC1DE**: Capture/Compare 1 DMA request enable

0: CC1 DMA request disabled
1: CC1 DMA request enabled

Bit 8 **UDE**: Update DMA request enable

0: Update DMA request disabled
1: Update DMA request enabled

Bit 7 **BIE**: Break interrupt enable

0: Break interrupt disabled
1: Break interrupt enabled

Bit 6 **TIE**: Trigger interrupt enable

0: Trigger interrupt disabled
1: Trigger interrupt enabled

Bit 5 **COMIE**: COM interrupt enable

0: COM interrupt disabled
1: COM interrupt enabled

Bits 4:3 Reserved, must be kept at reset value.

Bit 2 **CC2IE**: Capture/Compare 2 interrupt enable

0: CC2 interrupt disabled
1: CC2 interrupt enabled

Bit 1 **CC1IE**: Capture/Compare 1 interrupt enable

0: CC1 interrupt disabled
1: CC1 interrupt enabled

Bit 0 **UIE**: Update interrupt enable

0: Update interrupt disabled
1: Update interrupt enabled

41.5.5 TIM15 status register (TIM15_SR)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	CC2OF	CC1OF	Res.	BIF	TIF	COMIF	Res.	Res.	CC2IF	CC1IF	UIF
					rc_w0	rc_w0		rc_w0	rc_w0	rc_w0			rc_w0	rc_w0	rc_w0

Bits 15:11 Reserved, must be kept at reset value.

Bit 10 **CC2OF**: Capture/Compare 2 overcapture flag
Refer to CC1OF description

Bit 9 **CC1OF**: Capture/Compare 1 overcapture flag
This flag is set by hardware only when the corresponding channel is configured in input capture mode. It is cleared by software by writing it to '0'.
0: No overcapture has been detected
1: The counter value has been captured in TIMx_CCR1 register while CC1IF flag was already set

Bit 8 Reserved, must be kept at reset value.

Bit 7 **BIF**: Break interrupt flag
This flag is set by hardware as soon as the break input goes active. It can be cleared by software if the break input is not active.
0: No break event occurred
1: An active level has been detected on the break input

Bit 6 **TIF**: Trigger interrupt flag
This flag is set by hardware on the TRG trigger event (active edge detected on TRGI input when the slave mode controller is enabled in all modes but gated mode, both edges in case gated mode is selected). It is set when the counter starts or stops when gated mode is selected. It is cleared by software.
0: No trigger event occurred
1: Trigger interrupt pending

Bit 5 **COMIF**: COM interrupt flag
This flag is set by hardware on a COM event (once the capture/compare control bits –CCxE, CCxNE, OCxM– have been updated). It is cleared by software.
0: No COM event occurred
1: COM interrupt pending

Bits 4:3 Reserved, must be kept at reset value.

Bit 2 **CC2IF**: Capture/Compare 2 interrupt flag
refer to CC1IF description

Bit 1 **CC1IF**: Capture/Compare 1 interrupt flag

This flag is set by hardware. It is cleared by software (input capture or output compare mode) or by reading the TIMx_CCR1 register (input capture mode only).

0: No compare match / No input capture occurred

1: A compare match or an input capture occurred

If channel CC1 is configured as output: this flag is set when the content of the counter TIMx_CNT matches the content of the TIMx_CCR1 register. When the content of TIMx_CCR1 is greater than the content of TIMx_ARR, the CC1IF bit goes high on the counter overflow (in up-counting and up/down-counting modes) or underflow (in down-counting mode). There are 3 possible options for flag setting in center-aligned mode, refer to the CMS bits in the TIMx_CR1 register for the full description.

If channel CC1 is configured as input: this bit is set when counter value has been captured in TIMx_CCR1 register (an edge has been detected on IC1, as per the edge sensitivity defined with the CC1P and CC1NP bits setting, in TIMx_CCER).

Bit 0 **UIF**: Update interrupt flag

This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred.

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

- At overflow regarding the repetition counter value (update if repetition counter = 0) and if the UDIS=0 in the TIMx_CR1 register.
- When CNT is reinitialized by software using the UG bit in TIMx_EGR register, if URS=0 and UDIS=0 in the TIMx_CR1 register.
- When CNT is reinitialized by a trigger event (refer to [Section 41.5.3: TIM15 slave mode control register \(TIM15_SMCR\)](#)), if URS=0 and UDIS=0 in the TIMx_CR1 register.

41.5.6 TIM15 event generation register (TIM15_EGR)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BG	TG	COMG	Res.	Res.	CC2G	CC1G	UG
								w	w	rw			w	w	w

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **BG**: Break generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: A break event is generated. MOE bit is cleared and BIF flag is set. Related interrupt or DMA transfer can occur if enabled.

Bit 6 **TG**: Trigger generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: The TIF flag is set in TIMx_SR register. Related interrupt or DMA transfer can occur if enabled

Bit 5 **COMG**: Capture/Compare control update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action

1: When the CCPC bit is set, it is possible to update the CCxE, CCxNE and OCxM bits

Note: This bit acts only on channels that have a complementary output.

Bits 4:3 Reserved, must be kept at reset value.

Bit 2 **CC2G**: Capture/Compare 2 generation

Refer to CC1G description

Bit 1 **CC1G**: Capture/Compare 1 generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action

1: A capture/compare event is generated on channel 1:

If channel CC1 is configured as output:

CC1IF flag is set, Corresponding interrupt or DMA request is sent if enabled.

If channel CC1 is configured as input:

The current value of the counter is captured in TIMx_CCR1 register. The CC1IF flag is set, the corresponding interrupt or DMA request is sent if enabled. The CC1OF flag is set if the CC1IF flag was already high.

Bit 0 **UG**: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action

1: Reinitialize the counter and generates an update of the registers. Note that the prescaler counter is cleared too (anyway the prescaler ratio is not affected).

41.5.7 TIM15 capture/compare mode register 1 [alternate] (TIM15_CCMR1)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for input capture mode (this section) or for output compare mode (next section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode.

Input capture mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IC2F[3:0]				IC2PSC[1:0]		CC2S[1:0]		IC1F[3:0]				IC1PSC[1:0]		CC1S[1:0]	
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:12 **IC2F[3:0]**: Input capture 2 filter

Bits 11:10 **IC2PSC[1:0]**: Input capture 2 prescaler

Bits 9:8 **CC2S[1:0]**: Capture/Compare 2 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output

01: CC2 channel is configured as input, IC2 is mapped on TI2

10: CC2 channel is configured as input, IC2 is mapped on TI1

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC2S bits are writable only when the channel is OFF (CC2E = '0' in TIMx_CCER).

Bits 7:4 **IC1F[3:0]**: Input capture 1 filter

This bit-field defines the frequency used to sample TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, N=2

0010: $f_{SAMPLING}=f_{CK_INT}$, N=4

0011: $f_{SAMPLING}=f_{CK_INT}$, N=8

0100: $f_{SAMPLING}=f_{DTS}/2$, N=6

0101: $f_{SAMPLING}=f_{DTS}/2$, N=8

0110: $f_{SAMPLING}=f_{DTS}/4$, N=6

0111: $f_{SAMPLING}=f_{DTS}/4$, N=8

1000: $f_{SAMPLING}=f_{DTS}/8$, N=6

1001: $f_{SAMPLING}=f_{DTS}/8$, N=8

1010: $f_{SAMPLING}=f_{DTS}/16$, N=5

1011: $f_{SAMPLING}=f_{DTS}/16$, N=6

1100: $f_{SAMPLING}=f_{DTS}/16$, N=8

1101: $f_{SAMPLING}=f_{DTS}/32$, N=5

1110: $f_{SAMPLING}=f_{DTS}/32$, N=6

1111: $f_{SAMPLING}=f_{DTS}/32$, N=8

Bits 3:2 **IC1PSC[1:0]**: Input capture 1 prescaler

This bit-field defines the ratio of the prescaler acting on CC1 input (IC1). The prescaler is reset as soon as CC1E='0' (TIMx_CCER register).

00: no prescaler, capture is done each time an edge is detected on the capture input

01: capture is done once every 2 events

10: capture is done once every 4 events

11: capture is done once every 8 events

Bits 1:0 **CC1S[1:0]**: Capture/Compare 1 Selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

10: CC1 channel is configured as input, IC1 is mapped on TI2

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC1S bits are writable only when the channel is OFF (CC1E = '0' in TIMx_CCER).

41.5.8 TIM15 capture/compare mode register 1 [alternate] (TIM15_CCMR1)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for output compare mode (this section) or for input capture mode (previous section). The direction of a channel is defined by configuring the

corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode.

Output compare mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC2M[3]	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M[3]
							rw								rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	OC2M[2:0]			OC2PE	OC2FE	CC2S[1:0]		Res.	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw	rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bits 23:17 Reserved, must be kept at reset value.

Bit 15 Reserved, must be kept at reset value.

Bits 24, 14:12 **OC2M[3:0]**: Output Compare 2 mode

Bit 11 **OC2PE**: Output Compare 2 preload enable

Bit 10 **OC2FE**: Output Compare 2 fast enable

Bits 9:8 **CC2S[1:0]**: Capture/Compare 2 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC2 channel is configured as output.

01: CC2 channel is configured as input, IC2 is mapped on TI2.

10: CC2 channel is configured as input, IC2 is mapped on TI1.

11: CC2 channel is configured as input, IC2 is mapped on TRC. This mode is working only if an internal trigger input is selected through the TS bit (TIMx_SMCR register)

Note: CC2S bits are writable only when the channel is OFF (CC2E = '0' in TIMx_CCER).

Bit 7 Reserved, must be kept at reset value.

Bits 16, 6:4 **OC1M[3:0]**: Output Compare 1 mode

These bits define the behavior of the output reference signal OC1REF from which OC1 and OC1N are derived. OC1REF is active high whereas OC1 and OC1N active level depends on CC1P and CC1NP bits.

0000: Frozen - The comparison between the output compare register TIMx_CCR1 and the counter TIMx_CNT has no effect on the outputs.

0001: Set channel 1 to active level on match. OC1REF signal is forced high when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

0010: Set channel 1 to inactive level on match. OC1REF signal is forced low when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

0011: Toggle - OC1REF toggles when TIMx_CNT=TIMx_CCR1.

0100: Force inactive level - OC1REF is forced low.

0101: Force active level - OC1REF is forced high.

0110: PWM mode 1 - Channel 1 is active as long as TIMx_CNT<TIMx_CCR1 else inactive.

0111: PWM mode 2 - Channel 1 is inactive as long as TIMx_CNT<TIMx_CCR1 else active.

1000: Retriggerable OPM mode 1 - In up-counting mode, the channel is active until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 1 and the channels becomes active again at the next update. In down-counting mode, the channel is inactive until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 1 and the channels becomes inactive again at the next update.

1001: Retriggerable OPM mode 2 - In up-counting mode, the channel is inactive until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 2 and the channels becomes inactive again at the next update. In down-counting mode, the channel is active until a trigger event is detected (on TRGI signal). Then, a comparison is performed as in PWM mode 1 and the channels becomes active again at the next update.

1010: Reserved

1011: Reserved

1100: Combined PWM mode 1 - OC1REF has the same behavior as in PWM mode 1. OC1REFC is the logical OR between OC1REF and OC2REF.

1101: Combined PWM mode 2 - OC1REF has the same behavior as in PWM mode 2. OC1REFC is the logical AND between OC1REF and OC2REF.

1110: Reserved,

1111: Reserved,

Note: These bits can not be modified as long as LOCK level 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S='00' (the channel is configured in output).

In PWM mode, the OCREF level changes only when the result of the comparison changes or when the output compare mode switches from "frozen" mode to "PWM" mode.

On channels that have a complementary output, this bit field is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the OC1M active bits take the new value from the preloaded bits only when a COM event is generated.

Bit 3 **OC1PE**: Output Compare 1 preload enable

0: Preload register on TIMx_CCR1 disabled. TIMx_CCR1 can be written at anytime, the new value is taken in account immediately.

1: Preload register on TIMx_CCR1 enabled. Read/Write operations access the preload register. TIMx_CCR1 preload value is loaded in the active register at each update event.

Note: These bits can not be modified as long as LOCK level 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S='00' (the channel is configured in output).

The PWM mode can be used without validating the preload register only in one pulse mode (OPM bit set in TIMx_CR1 register). Else the behavior is not guaranteed.

Bit 2 **OC1FE**: Output Compare 1 fast enable

This bit decreases the latency between a trigger event and a transition on the timer output. It must be used in one-pulse mode (OPM bit set in TIMx_CR1 register), to have the output pulse starting as soon as possible after the starting trigger.

0: CC1 behaves normally depending on counter and CCR1 values even when the trigger is ON. The minimum delay to activate CC1 output when an edge occurs on the trigger input is 5 clock cycles.

1: An active edge on the trigger input acts like a compare match on CC1 output. Then, OC is set to the compare level independently of the result of the comparison. Delay to sample the trigger input and to activate CC1 output is reduced to 3 clock cycles. OC1FE acts only if the channel is configured in PWM1 or PWM2 mode.

Bits 1:0 **CC1S[1:0]**: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output.

01: CC1 channel is configured as input, IC1 is mapped on TI1.

10: CC1 channel is configured as input, IC1 is mapped on TI2.

11: CC1 channel is configured as input, IC1 is mapped on TRC. This mode is working only if an internal trigger input is selected through TS bit (TIMx_SMCR register)

Note: CC1S bits are writable only when the channel is OFF (CC1E = '0' in TIMx_CCER).

41.5.9 TIM15 capture/compare enable register (TIM15_CCER)

Address offset: 0x20

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC2NP	Res.	CC2P	CC2E	CC1NP	CC1NE	CC1P	CC1E
								rw		rw	rw	rw	rw	rw	rw

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **CC2NP**: Capture/Compare 2 complementary output polarity

Refer to CC1NP description

Bit 6 Reserved, must be kept at reset value.

Bit 5 **CC2P**: Capture/Compare 2 output polarity

Refer to CC1P description

Bit 4 **CC2E**: Capture/Compare 2 output enable

Refer to CC1E description

Bit 3 **CC1NP**: Capture/Compare 1 complementary output polarity

CC1 channel configured as output:

0: OC1N active high

1: OC1N active low

CC1 channel configured as input:

This bit is used in conjunction with CC1P to define the polarity of TI1FP1 and TI2FP1. Refer to CC1P description.

Note: This bit is not writable as soon as LOCK level 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S="00" (the channel is configured in output).

On channels that have a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1NP active bit takes the new value from the preloaded bit only when a Commutation event is generated.

Bit 2 **CC1NE**: Capture/Compare 1 complementary output enable

0: Off - OC1N is not active. OC1N level is then function of MOE, OSS1, OSSR, OIS1, OIS1N and CC1E bits.

1: On - OC1N signal is output on the corresponding output pin depending on MOE, OSS1, OSSR, OIS1, OIS1N and CC1E bits.

Bit 1 **CC1P**: Capture/Compare 1 output polarity

0: OC1 active high (output mode) / Edge sensitivity selection (input mode, see below)

1: OC1 active low (output mode) / Edge sensitivity selection (input mode, see below)

When CC1 channel is configured as input, both CC1NP/CC1P bits select the active polarity of TI1FP1 and TI2FP1 for trigger or capture operations.

CC1NP=0, CC1P=0: non-inverted/rising edge. The circuit is sensitive to TIxFP1 rising edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode or encoder mode).

CC1NP=0, CC1P=1: inverted/falling edge. The circuit is sensitive to TIxFP1 falling edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is inverted (trigger operation in gated mode or encoder mode).

CC1NP=1, CC1P=1: non-inverted/both edges/ The circuit is sensitive to both TIxFP1 rising and falling edges (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode). This configuration must not be used in encoder mode.

CC1NP=1, CC1P=0: this configuration is reserved, it must not be used.

Note: This bit is not writable as soon as LOCK level 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

On channels that have a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1P active bit takes the new value from the preloaded bit only when a Commutation event is generated.

Bit 0 **CC1E**: Capture/Compare 1 output enable

0: Capture mode disabled / OC1 is not active (see below)

1: Capture mode enabled / OC1 signal is output on the corresponding output pin

When CC1 channel is configured as output, the OC1 level depends on MOE, OSS1, OSSR, OIS1, OIS1N and CC1NE bits, regardless of the CC1E bits state. Refer to [Table 343](#) for details.

Table 343. Output control bits for complementary OCx and OCxN channels with break feature (TIM15)

Control bits					Output states ⁽¹⁾	
MOE bit	OSSI bit	OSSR bit	CCxE bit	CCxNE bit	OCx output state	OCxN output state
1	X	X	0	0	Output Disabled (not driven by the timer: Hi-Z) OCx=0 OCxN=0, OCxN_EN=0	
		0	0	1	Output Disabled (not driven by the timer: Hi-Z) OCx=0	OCxREF + Polarity OCxN=OCxREF XOR CCxNP
		0	1	0	OCxREF + Polarity OCx=OCxREF XOR CCxP	Output Disabled (not driven by the timer: Hi-Z) OCxN=0
		X	1	1	OCREF + Polarity + dead-time	Complementary to OCREF (not OCREF) + Polarity + dead-time
		1	0	1	Off-State (output enabled with inactive state) OCx=CCxP	OCxREF + Polarity OCxN=OCxREF XOR CCxNP
		1	1	0	OCxREF + Polarity OCx=OCxREF xor CCxP, OCx_EN=1	Off-State (output enabled with inactive state) OCxN=CCxNP, OCxN_EN=1
0	0	X	X	X	Output disabled (not driven by the timer anymore). The output state is defined by the GPIO controller and can be High, Low or Hi-Z.	
	1		0	0		
			0	1	Off-State (output enabled with inactive state) Asynchronously: OCx=CCxP, OCxN=CCxNP Then if the clock is present: OCx=OISx and OCxN=OISxN after a dead-time, assuming that OISx and OISxN do not correspond to OCX and OCxN both in active state	
			1	0		
			1	1		

1. When both outputs of a channel are not used (control taken over by GPIO controller), the OISx, OISxN, CCxP and CCxNP bits must be kept cleared.

Note: *The state of the external I/O pins connected to the complementary OCx and OCxN channels depends on the OCx and OCxN channel state and AFIO registers.*

41.5.10 TIM15 counter (TIM15_CNT)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UIF CPY	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
r															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bit 31 **UIFCPY**: UIF Copy

This bit is a read-only copy of the UIF bit in the TIMx_ISR register.

Bits 30:16 Reserved, must be kept at reset value.

Bits 15:0 **CNT[15:0]**: Counter value**41.5.11 TIM15 prescaler (TIM15_PSC)**

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 15:0 **PSC[15:0]**: Prescaler valueThe counter clock frequency (CK_CNT) is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded in the active prescaler register at each update event (including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in "reset mode").

41.5.12 TIM15 auto-reload register (TIM15_ARR)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded in the actual auto-reload register.

Refer to the [Section 41.4.1: Time-base unit on page 1783](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

41.5.13 TIM15 repetition counter register (TIM15_RCR)

Address offset: 0x30

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REP[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:8 Reserved, must be kept at reset value.

Bits 7:0 **REP[7:0]**: Repetition counter value

These bits allow the user to set-up the update rate of the compare registers (i.e. periodic transfers from preload to active registers) when preload registers are enable, as well as the update interrupt generation rate, if this interrupt is enable.

Each time the REP_CNT related downcounter reaches zero, an update event is generated and it restarts counting from REP value. As REP_CNT is reloaded with REP value only at the repetition update event U_RC, any write to the TIMx_RCR register is not taken in account until the next repetition update event.

It means in PWM mode (REP+1) corresponds to the number of PWM periods in edge-aligned mode.

41.5.14 TIM15 capture/compare register 1 (TIM15_CCR1)

Address offset: 0x34

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR1[15:0]**: Capture/Compare 1 value

If channel CC1 is configured as output:

CCR1 is the value to be loaded in the actual capture/compare 1 register (preload value). It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (bit OC1PE). Else the preload value is copied in the active capture/compare 1 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC1 output.

If channel CC1 is configured as input:

CCR1 is the counter value transferred by the last input capture 1 event (IC1).

41.5.15 TIM15 capture/compare register 2 (TIM15_CCR2)

Address offset: 0x38

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR2[15:0]**: Capture/Compare 2 value

If channel CC2 is configured as output:

CCR2 is the value to be loaded in the actual capture/compare 2 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR2 register (bit OC2PE). Else the preload value is copied in the active capture/compare 2 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signalled on OC2 output.

If channel CC2 is configured as input:

CCR2 is the counter value transferred by the last input capture 2 event (IC2).

41.5.16 TIM15 break and dead-time register (TIM15_BDTR)

Address offset: 0x44

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BKF[3:0]			
												rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MOE	AOE	BKP	BKE	OSSR	OSSI	LOCK[1:0]		DTG[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Note: As the BKF[3:0], AOE, BKP, BKE, OSSI, OSSR and DTG[7:0] bits may be write-locked depending on the LOCK configuration, it may be necessary to configure all of them during the first write access to the TIMx_BDTR register.

Bits 31:20 Reserved, must be kept at reset value.

Bits 19:16 **BKF[3:0]**: Break filter

This bit-field defines the frequency used to sample the BRK input signal and the length of the digital filter applied to BRK. The digital filter is made of an event counter in which N events are needed to validate a transition on the output:

0000: No filter, BRK acts asynchronously

0001: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=2

0010: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=4

0011: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=8

0100: $f_{\text{SAMPLING}} = f_{\text{DTS}}/2$, N=6

0101: $f_{\text{SAMPLING}} = f_{\text{DTS}}/2$, N=8

0110: $f_{\text{SAMPLING}} = f_{\text{DTS}}/4$, N=6

0111: $f_{\text{SAMPLING}} = f_{\text{DTS}}/4$, N=8

1000: $f_{\text{SAMPLING}} = f_{\text{DTS}}/8$, N=6

1001: $f_{\text{SAMPLING}} = f_{\text{DTS}}/8$, N=8

1010: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=5

1011: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=6

1100: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=8

1101: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=5

1110: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=6

1111: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=8

Note: This bit cannot be modified when LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 15 **MOE**: Main output enable

This bit is cleared asynchronously by hardware as soon as the break input is active. It is set by software or automatically depending on the AOE bit. It is acting only on the channels which are configured in output.

0: OC and OCN outputs are disabled or forced to idle state depending on the OSSR bit.

1: OC and OCN outputs are enabled if their respective enable bits are set (CCxE, CCxNE in TIMx_CCER register)

See OC/OCN enable description for more details ([Section 41.5.9: TIM15 capture/compare enable register \(TIM15_CCER\) on page 1829](#)).

Bit 14 **AOE**: Automatic output enable

0: MOE can be set only by software

1: MOE can be set by software or automatically at the next update event (if the break input is not be active)

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 13 **BKP**: Break polarity

0: Break input BRK is active low

1: Break input BRK is active high

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Any write operation to this bit takes a delay of 1 APB clock cycle to become effective.

Bit 12 **BKE**: Break enable

0: Break inputs (BRK and CCS clock failure event) disabled

1: Break inputs (BRK and CCS clock failure event) enabled

This bit cannot be modified when LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Note: Any write operation to this bit takes a delay of 1 APB clock cycle to become effective.

Bit 11 OSSR: Off-state selection for Run mode

This bit is used when MOE=1 on channels that have a complementary output which are configured as outputs. OSSR is not implemented if no complementary output is implemented in the timer.

See OC/OCN enable description for more details ([Section 41.5.9: TIM15 capture/compare enable register \(TIM15_CCER\) on page 1829](#)).

0: When inactive, OC/OCN outputs are disabled (the timer releases the output control which is taken over by the AFIO logic, which forces a Hi-Z state)

1: When inactive, OC/OCN outputs are enabled with their inactive level as soon as CCxE=1 or CCxNE=1 (the output is still controlled by the timer).

Note: This bit can not be modified as soon as the LOCK level 2 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 10 OSSI: Off-state selection for Idle mode

This bit is used when MOE=0 on channels configured as outputs.

See OC/OCN enable description for more details ([Section 41.5.9: TIM15 capture/compare enable register \(TIM15_CCER\) on page 1829](#)).

0: When inactive, OC/OCN outputs are disabled (OC/OCN enable output signal=0)

1: When inactive, OC/OCN outputs are forced first with their idle level as soon as CCxE=1 or CCxNE=1. OC/OCN enable output signal=1)

Note: This bit can not be modified as soon as the LOCK level 2 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 9:8 LOCK[1:0]: Lock configuration

These bits offer a write protection against software errors.

00: LOCK OFF - No bit is write protected

01: LOCK Level 1 = DTG bits in TIMx_BDTR register, OISx and OISxN bits in TIMx_CR2 register and BKE/BKP/AOE bits in TIMx_BDTR register can no longer be written

10: LOCK Level 2 = LOCK Level 1 + CC Polarity bits (CCxP/CCxNP bits in TIMx_CCER register, as long as the related channel is configured in output through the CCxS bits) as well as OSSR and OSSI bits can no longer be written.

11: LOCK Level 3 = LOCK Level 2 + CC Control bits (OCxM and OCxPE bits in TIMx_CCMRx registers, as long as the related channel is configured in output through the CCxS bits) can no longer be written.

Note: The LOCK bits can be written only once after the reset. Once the TIMx_BDTR register has been written, their content is frozen until the next reset.

Bits 7:0 DTG[7:0]: Dead-time generator setup

This bit-field defines the duration of the dead-time inserted between the complementary outputs. DT correspond to this duration.

DTG[7:5]=0xx => DT=DTG[7:0]x t_{dtg} with $t_{dtg}=t_{DTS}$

DTG[7:5]=10x => DT=(64+DTG[5:0])x t_{dtg} with $T_{dtg}=2xt_{DTS}$

DTG[7:5]=110 => DT=(32+DTG[4:0])x t_{dtg} with $T_{dtg}=8xt_{DTS}$

DTG[7:5]=111 => DT=(32+DTG[4:0])x t_{dtg} with $T_{dtg}=16xt_{DTS}$

Example if $T_{DTS}=125\text{ns}$ (8MHz), dead-time possible values are:

0 to 15875 ns by 125 ns steps,

16 μs to 31750 ns by 250 ns steps,

32 μs to 63 μs by 1 μs steps,

64 μs to 126 μs by 2 μs steps

Note: This bit-field can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

41.5.17 TIM15 DMA control register (TIM15_DCR)

Address offset: 0x48

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	DBL[4:0]					Res.	Res.	Res.	DBA[4:0]				
			rw	rw	rw	rw	rw				rw	rw	rw	rw	rw

Bits 15:13 Reserved, must be kept at reset value.

Bits 12:8 **DBL[4:0]**: DMA burst length

This 5-bit field defines the length of DMA transfers (the timer recognizes a burst transfer when a read or a write access is done to the TIMx_DMAR address).

00000: 1 transfer,

00001: 2 transfers,

00010: 3 transfers,

...

10001: 18 transfers.

Bits 7:5 Reserved, must be kept at reset value.

Bits 4:0 **DBA[4:0]**: DMA base address

This 5-bit field defines the base-address for DMA transfers (when read/write access are done through the TIMx_DMAR address). DBA is defined as an offset starting from the address of the TIMx_CR1 register.

Example:

00000: TIMx_CR1,

00001: TIMx_CR2,

00010: TIMx_SMCR,

...

41.5.18 TIM15 DMA address for full transfer (TIM15_DMAR)

Address offset: 0x4C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMAB[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **DMAB[15:0]**: DMA register for burst accesses

A read or write operation to the DMAR register accesses the register located at the address
 $(\text{TIMx_CR1 address}) + (\text{DBA} + \text{DMA index}) \times 4$

where TIMx_CR1 address is the address of the control register 1, DBA is the DMA base address configured in TIMx_DCR register, DMA index is automatically controlled by the DMA transfer, and ranges from 0 to DBL (DBL configured in TIMx_DCR).

41.5.19 TIM15 alternate register 1 (TIM15_AF1)

Address offset: 0x60

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	BKCM P2P	BKCM P1P	BKINP	BKDF1 BK0E	Res.	Res.	Res.	Res.	Res.	BKCM P2E	BKCM P1E	BKINE
				rw	rw	rw	rw						rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bit 11 BKCOMP2P: BRK COMP2 input polarity

This bit selects the COMP2 input sensitivity. It must be programmed together with the BKP polarity bit.

0: COMP2 input is active low

1: COMP2 input is active high

*Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).***Bit 10 BKCOMP1P:** BRK COMP1 input polarity

This bit selects the COMP1 input sensitivity. It must be programmed together with the BKP polarity bit.

0: COMP1 input is active low

1: COMP1 input is active high

*Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).***Bit 9 BKINP:** BRK BKIN input polarity

This bit selects the BKIN alternate function input sensitivity. It must be programmed together with the BKP polarity bit.

0: BKIN input is active low

1: BKIN input is active high

*Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).***Bit 8 BKDF1BK0E:** BRK dfsdm1_break[0] enable

This bit enables the dfsdm1_break[0] for the timer's BRK input. dfsdm1_break[0] output is 'ORed' with the other BRK sources.

0: dfsdm1_break[0]input disabled

1: dfsdm1_break[0]input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 7:3 Reserved, must be kept at reset value.

Bit 2 BKCOMP2E: BRK COMP2 enable

This bit enables the COMP2 for the timer's BRK input. COMP2 output is 'ORed' with the other BRK sources.

0: COMP2 input disabled

1: COMP2 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 1 **BKCOMP1E**: BRK COMP1 enable

This bit enables the COMP1 for the timer's BRK input. COMP1 output is 'ORed' with the other BRK sources.

0: COMP1 input disabled

1: COMP1 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 0 **BKINE**: BRK BKIN input enable

This bit enables the BKIN alternate function input for the timer's BRK input. BKIN input is 'ORed' with the other BRK sources.

0: BKIN input disabled

1: BKIN input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

41.5.20 TIM15 input selection register (TIM15_TISEL)

Address offset: 0x68

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	TI2SEL[3:0]				Res.	Res.	Res.	Res.	TI1SEL[3:0]			
				rw	rw	rw	rw					rw	rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:8 **TI2SEL[3:0]**: selects TI2[0] to TI2[15] input

0000: TIM15_CH2 input

0001: TIM2_CH2 input

0010: TIM3_CH2 input

0011: TIM4_CH2 input

Others: Reserved

Bits 7:4 Reserved, must be kept at reset value.

Bits 3:0 **TI1SEL[3:0]**: selects TI1[0] to TI1[15] input

0000: TIM15_CH1 input

0001: TIM2_CH1 input

0010: TIM3_CH1 input

0011: TIM4_CH1 input

0100: LSE

0101: CSI

0110: MCO2

Other: Reserved

41.5.21 TIM15 register map

TIM15 registers are mapped as 16-bit addressable registers as described in the table below:

Table 344. TIM15 register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x00	TIM15_CR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	UIFREMA	Res	CKD [1:0]	ARPE	Res	Res	Res	Res	OPM	URS	UDIS	CEN	
	Reset value																					0	0											0
0x04	TIM15_CR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	OIS2	OIS1N	OIS1	T1S	MMS[2:0]			CCDS	CCUS	Res	CCPC	
	Reset value																						0	0	0	0	0	0	0	0	0		0	
0x08	TIM15_SMCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	TS [4:3]	0	0	Res	Res	SMS[3]	Res	Res	Res	Res	Res	Res	Res	Res	MSM	TS[2:0]			Res	SMS[2:0]			
	Reset value																																	
0x0C	TIM15_DIER	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	TDE	COMDE	Res	Res	Res	CC2DE	CC1DE	UDE	BIE	TIE	COMIE	Res	Res	CC2IE	CC1IE	UIE
	Reset value																		0	0	Res	Res	0	0	0	0	0	0			0	0	0	
0x10	TIM15_SR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CC2OF	CC1OF	Res	BIF	TIF	COMIF	Res	Res	CC2IF	CC1IF	UIF	
	Reset value																						0	0		0	0	0			0	0	0	
0x14	TIM15_EGR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	BG	TG	COMG	Res	Res	CC2G	CC1G	UG	
	Reset value																									0	0	0			0	0	0	
0x18	TIM15_CCMR1 Output Compare mode	Res	Res	Res	Res	Res	Res	Res	OC2M[3]	Res	Res	Res	Res	Res	Res	Res	OC1M[3]	Res	OC2M [2:0]			OC2PE	OC2FE	CC2S [1:0]		Res	OC1M [2:0]			OC1PE	OC1FE	CC1S [1:0]		
	Reset value								0								0		0	0	0	0	0	0	0		0	0	0	0	0	0	0	
	TIM15_CCMR1 Input Capture mode	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	IC2F[3:0]			IC2 PSC [1:0]	CC2S [1:0]		IC1F[3:0]			IC1 PSC [1:0]	CC1S [1:0]						
	Reset value																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x20	TIM15_CCER	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CC2NP	Res	CC2P	CC2E	CC1NP	CC1NE	CC1P	CC1E
	Reset value																									0		0	0	0	0	0	0	

Table 344. TIM15 register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0			
0x24	TIM15_CNT	UIFCPY or Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[15:0]																		
	Reset value	0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x28	TIM15_PSC	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PSC[15:0]																		
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x2C	TIM15_ARR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ARR[15:0]																		
	Reset value																	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1			
0x30	TIM15_RCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REP[7:0]											
	Reset value																									0	0	0	0	0	0	0	0	0		
0x34	TIM15_CCR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CCR1[15:0]																		
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x38	TIM15_CCR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CCR2[15:0]																		
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x44	TIM15_BDTR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BKF[3:0]			MOE	AOE	BKP	BKE	OSSR	OSSI	LOCK [1:0]	DTG[7:0]												
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x48	TIM15_DCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DBL[4:0]				Res.	Res.	Res.	DBA[4:0]								
	Reset value																				0	0	0	0	0				0	0	0	0	0	0		
0x4C	TIM15_DMAR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DMAB[15:0]																		
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0x60	TIM15_AF1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BKMP2P	BKMP1P	BKINP	BKDF1BK0E	Res.	Res.	Res.	Res.	Res.	Res.	BKMP2E	BKMP1E	BKINE		
	Reset value																					0	0	0	0						0	0	0	1		
0x68	TIM15_TISEL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TI2SEL[3:0]			Res.	Res.	Res.	Res.	TI1SEL[3:0]							
	Reset value																					0	0	0	0						0	0	0	0	0	

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.

41.6 TIM16/TIM17 registers

Refer to [Section 1.2](#) for a list of abbreviations used in register descriptions.

The peripheral registers can be accessed by half-words (16-bit) or words (32-bit).

41.6.1 TIMx control register 1 (TIMx_CR1)(x = 16 to 17)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	UIFRE MAP	Res.	CKD[1:0]		ARPE	Res.	Res.	Res.	OPM	URS	UDIS	CEN
				rw		rw	rw	rw				rw	rw	rw	rw

Bits 15:12 Reserved, must be kept at reset value.

Bit 11 **UIFREMAP**: UIF status bit remapping

0: No remapping. UIF status bit is not copied to TIMx_CNT register bit 31.

1: Remapping enabled. UIF status bit is copied to TIMx_CNT register bit 31.

Bit 10 Reserved, must be kept at reset value.

Bits 9:8 **CKD[1:0]**: Clock division

This bit-field indicates the division ratio between the timer clock (CK_INT) frequency and the dead-time and sampling clock (t_{DTS}) used by the dead-time generators and the digital filters (Tlx),

00: $t_{DTS}=t_{CK_INT}$

01: $t_{DTS}=2*t_{CK_INT}$

10: $t_{DTS}=4*t_{CK_INT}$

11: Reserved, do not program this value

Bit 7 **ARPE**: Auto-reload preload enable

0: TIMx_ARR register is not buffered

1: TIMx_ARR register is buffered

Bits 6:4 Reserved, must be kept at reset value.

Bit 3 **OPM**: One pulse mode

0: Counter is not stopped at update event

1: Counter stops counting at the next update event (clearing the bit CEN)

Bit 2 **URS**: Update request source

This bit is set and cleared by software to select the UEV event sources.

0: Any of the following events generate an update interrupt or DMA request if enabled.

These events can be:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

1: Only counter overflow/underflow generates an update interrupt or DMA request if enabled.

Bit 1 **UDIS**: Update disable

This bit is set and cleared by software to enable/disable UEV event generation.

0: UEV enabled. The Update (UEV) event is generated by one of the following events:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

Buffered registers are then loaded with their preload values.

1: UEV disabled. The Update event is not generated, shadow registers keep their value (ARR, PSC, CCRx). However the counter and the prescaler are reinitialized if the UG bit is set or if a hardware reset is received from the slave mode controller.

Bit 0 **CEN**: Counter enable

0: Counter disabled

1: Counter enabled

Note: External clock and gated mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.

41.6.2 TIMx control register 2 (TIMx_CR2)(x = 16 to 17)

Address offset: 0x04

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	OIS1N	OIS1	Res.	Res.	Res.	Res.	CCDS	CCUS	Res.	CCPC
						rw	rw					rw	rw		rw

Bits 15:10 Reserved, must be kept at reset value.

Bit 9 **OIS1N**: Output Idle state 1 (OC1N output)

0: OC1N=0 after a dead-time when MOE=0

1: OC1N=1 after a dead-time when MOE=0

Note: This bit can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 8 **OIS1**: Output Idle state 1 (OC1 output)

0: OC1=0 (after a dead-time if OC1N is implemented) when MOE=0

1: OC1=1 (after a dead-time if OC1N is implemented) when MOE=0

Note: This bit can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 7:4 Reserved, must be kept at reset value.

Bit 3 **CCDS**: Capture/compare DMA selection

0: CCx DMA request sent when CCx event occurs

1: CCx DMA requests sent when update event occurs

Bit 2 **CCUS**: Capture/compare control update selection

0: When capture/compare control bits are preloaded (CCPC=1), they are updated by setting the COMG bit only.

1: When capture/compare control bits are preloaded (CCPC=1), they are updated by setting the COMG bit or when an rising edge occurs on TRGI.

Note: This bit acts only on channels that have a complementary output.

Bit 1 Reserved, must be kept at reset value.

Bit 0 **CCPC**: Capture/compare preloaded control

0: CCxE, CCxNE and OCxM bits are not preloaded

1: CCxE, CCxNE and OCxM bits are preloaded, after having been written, they are updated only when COM bit is set.

Note: This bit acts only on channels that have a complementary output.

41.6.3 TIMx DMA/interrupt enable register (TIMx_DIER)(x = 16 to 17)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	CC1DE	UDE	BIE	Res.	COMIE	Res.	Res.	Res.	CC1IE	UIE
						rw	rw	rw		rw				rw	rw

Bits 15:10 Reserved, must be kept at reset value.

Bit 9 **CC1DE**: Capture/Compare 1 DMA request enable

0: CC1 DMA request disabled

1: CC1 DMA request enabled

Bit 8 **UDE**: Update DMA request enable

0: Update DMA request disabled

1: Update DMA request enabled

Bit 7 **BIE**: Break interrupt enable

0: Break interrupt disabled

1: Break interrupt enabled

Bit 6 Reserved, must be kept at reset value.

Bit 5 **COMIE**: COM interrupt enable

0: COM interrupt disabled

1: COM interrupt enabled

Bits 4:2 Reserved, must be kept at reset value.

Bit 1 **CC1IE**: Capture/Compare 1 interrupt enable

0: CC1 interrupt disabled

1: CC1 interrupt enabled

Bit 0 **UIE**: Update interrupt enable

0: Update interrupt disabled

1: Update interrupt enabled

41.6.4 TIMx status register (TIMx_SR)(x = 16 to 17)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	CC1OF	Res.	BIF	Res.	COMIF	Res.	Res.	Res.	CC1IF	UIF
						rc_w0		rc_w0		rc_w0				rc_w0	rc_w0

Bits 15:10 Reserved, must be kept at reset value.

Bit 9 **CC1OF**: Capture/Compare 1 overcapture flag

This flag is set by hardware only when the corresponding channel is configured in input capture mode. It is cleared by software by writing it to '0'.

0: No overcapture has been detected

1: The counter value has been captured in TIMx_CCR1 register while CC1IF flag was already set

Bit 8 Reserved, must be kept at reset value.

Bit 7 **BIF**: Break interrupt flag

This flag is set by hardware as soon as the break input goes active. It can be cleared by software if the break input is not active.

0: No break event occurred

1: An active level has been detected on the break input

Bit 6 Reserved, must be kept at reset value.

Bit 5 **COMIF**: COM interrupt flag

This flag is set by hardware on a COM event (once the capture/compare control bits –CCxE, CCxNE, OCxM– have been updated). It is cleared by software.

0: No COM event occurred

1: COM interrupt pending

Bits 4:2 Reserved, must be kept at reset value.

Bit 1 **CC1IF**: Capture/Compare 1 interrupt flag

This flag is set by hardware. It is cleared by software (input capture or output compare mode) or by reading the TIMx_CCR1 register (input capture mode only).

0: No compare match / No input capture occurred

1: A compare match or an input capture occurred

If channel CC1 is configured as output: this flag is set when the content of the counter TIMx_CNT matches the content of the TIMx_CCR1 register. When the content of TIMx_CCR1 is greater than the content of TIMx_ARR, the CC1IF bit goes high on the counter overflow (in up-counting and up/down-counting modes) or underflow (in down-counting mode). There are 3 possible options for flag setting in center-aligned mode, refer to the CMS bits in the TIMx_CR1 register for the full description.

If channel CC1 is configured as input: this bit is set when counter value has been captured in TIMx_CCR1 register (an edge has been detected on IC1, as per the edge sensitivity defined with the CC1P and CC1NP bits setting, in TIMx_CCER).

Bit 0 **UIF**: Update interrupt flag

This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred.

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

- At overflow regarding the repetition counter value (update if repetition counter = 0) and if the UDIS=0 in the TIMx_CR1 register.
- When CNT is reinitialized by software using the UG bit in TIMx_EGR register, if URS=0 and UDIS=0 in the TIMx_CR1 register.

41.6.5 TIMx event generation register (TIMx_EGR)(x = 16 to 17)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BG	Res.	COMG	Res.	Res.	Res.	CC1G	UG
								w		w				w	w

Bits 15:8 Reserved, must be kept at reset value.

Bit 7 **BG**: Break generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action.

1: A break event is generated. MOE bit is cleared and BIF flag is set. Related interrupt or DMA transfer can occur if enabled.

Bit 6 Reserved, must be kept at reset value.

Bit 5 **COMG**: Capture/Compare control update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action

1: When the CCPC bit is set, it is possible to update the CCxE, CCxNE and OCxM bits

Note: This bit acts only on channels that have a complementary output.

Bits 4:2 Reserved, must be kept at reset value.

Bit 1 **CC1G**: Capture/Compare 1 generation

This bit is set by software in order to generate an event, it is automatically cleared by hardware.

0: No action.

1: A capture/compare event is generated on channel 1:

If channel CC1 is configured as output:

CC1IF flag is set, Corresponding interrupt or DMA request is sent if enabled.

If channel CC1 is configured as input:

The current value of the counter is captured in TIMx_CCR1 register. The CC1IF flag is set, the corresponding interrupt or DMA request is sent if enabled. The CC1OF flag is set if the CC1IF flag was already high.

Bit 0 **UG**: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action.

1: Reinitialize the counter and generates an update of the registers. Note that the prescaler counter is cleared too (anyway the prescaler ratio is not affected).

41.6.6 TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1) (x = 16 to 17)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for input capture mode (this section) or for output compare mode (next section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode.

Input capture mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IC1F[3:0]				IC1PSC[1:0]		CC1S[1:0]	
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **IC1F[3:0]**: Input capture 1 filter

This bit-field defines the frequency used to sample TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING}=f_{CK_INT}$, N=2

0010: $f_{SAMPLING}=f_{CK_INT}$, N=4

0011: $f_{SAMPLING}=f_{CK_INT}$, N=8

0100: $f_{SAMPLING}=f_{DTS}/2$, N=

0101: $f_{SAMPLING}=f_{DTS}/2$, N=8

0110: $f_{SAMPLING}=f_{DTS}/4$, N=6

0111: $f_{SAMPLING}=f_{DTS}/4$, N=8

1000: $f_{SAMPLING}=f_{DTS}/8$, N=6

1001: $f_{SAMPLING}=f_{DTS}/8$, N=8

1010: $f_{SAMPLING}=f_{DTS}/16$, N=5

1011: $f_{SAMPLING}=f_{DTS}/16$, N=6

1100: $f_{SAMPLING}=f_{DTS}/16$, N=8

1101: $f_{SAMPLING}=f_{DTS}/32$, N=5

1110: $f_{SAMPLING}=f_{DTS}/32$, N=6

1111: $f_{SAMPLING}=f_{DTS}/32$, N=8

Bits 3:2 **IC1PSC[1:0]**: Input capture 1 prescaler

This bit-field defines the ratio of the prescaler acting on CC1 input (IC1).

The prescaler is reset as soon as CC1E='0' (TIMx_CCER register).

00: no prescaler, capture is done each time an edge is detected on the capture input.

01: capture is done once every 2 events

10: capture is done once every 4 events

11: capture is done once every 8 events

Bits 1:0 **CC1S[1:0]**: Capture/Compare 1 Selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

Others: Reserved

Note: CC1S bits are writable only when the channel is OFF (CC1E = '0' in TIMx_CCER).

41.6.7 TIMx capture/compare mode register 1 [alternate] (TIMx_CCMR1) (x = 16 to 17)

Address offset: 0x18

Reset value: 0x0000 0000

The same register can be used for output compare mode (this section) or for input capture mode (previous section). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode.

Output compare mode:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M[3]
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
									rw	rw	rw	rw	rw	rw	rw

Bits 31:17 Reserved, must be kept at reset value.

Bits 15:7 Reserved, must be kept at reset value.

Bits 16, 6:4 **OC1M[3:0]**: Output Compare 1 mode

These bits define the behavior of the output reference signal OC1REF from which OC1 and OC1N are derived. OC1REF is active high whereas OC1 and OC1N active level depends on CC1P and CC1NP bits.

0000: Frozen - The comparison between the output compare register TIMx_CCR1 and the counter TIMx_CNT has no effect on the outputs.

0001: Set channel 1 to active level on match. OC1REF signal is forced high when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

0010: Set channel 1 to inactive level on match. OC1REF signal is forced low when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).

0011: Toggle - OC1REF toggles when TIMx_CNT=TIMx_CCR1.

0100: Force inactive level - OC1REF is forced low.

0101: Force active level - OC1REF is forced high.

0110: PWM mode 1 - Channel 1 is active as long as TIMx_CNT < TIMx_CCR1 else inactive.

0111: PWM mode 2 - Channel 1 is inactive as long as TIMx_CNT < TIMx_CCR1 else active.

All other values: Reserved

Note: These bits can not be modified as long as LOCK level 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S='00' (the channel is configured in output).

In PWM mode 1 or 2, the OCREF level changes only when the result of the comparison changes or when the output compare mode switches from "frozen" mode to "PWM" mode.

Bit 3 **OC1PE**: Output Compare 1 preload enable

0: Preload register on TIMx_CCR1 disabled. TIMx_CCR1 can be written at anytime, the new value is taken in account immediately.

1: Preload register on TIMx_CCR1 enabled. Read/Write operations access the preload register. TIMx_CCR1 preload value is loaded in the active register at each update event.

Note: These bits can not be modified as long as LOCK level 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S='00' (the channel is configured in output).

The PWM mode can be used without validating the preload register only in one pulse mode (OPM bit set in TIMx_CR1 register). Else the behavior is not guaranteed.

Bit 2 **OC1FE**: Output Compare 1 fast enable

This bit decreases the latency between a trigger event and a transition on the timer output. It must be used in one-pulse mode (OPM bit set in TIMx_CR1 register), to have the output pulse starting as soon as possible after the starting trigger.

0: CC1 behaves normally depending on counter and CCR1 values even when the trigger is ON. The minimum delay to activate CC1 output when an edge occurs on the trigger input is 5 clock cycles.

1: An active edge on the trigger input acts like a compare match on CC1 output. Then, OC is set to the compare level independently of the result of the comparison. Delay to sample the trigger input and to activate CC1 output is reduced to 3 clock cycles. OC1FE acts only if the channel is configured in PWM1 or PWM2 mode.

Bits 1:0 **CC1S[1:0]**: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) as well as the used input.

00: CC1 channel is configured as output

01: CC1 channel is configured as input, IC1 is mapped on TI1

Others: Reserved

Note: CC1S bits are writable only when the channel is OFF (CC1E = '0' in TIMx_CCER).

41.6.8 TIMx capture/compare enable register (TIMx_CCER)(x = 16 to 17)

Address offset: 0x20

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CC1NP	CC1NE	CC1P	CC1E
												rw	rw	rw	rw

Bits 15:4 Reserved, must be kept at reset value.

Bit 3 **CC1NP**: Capture/Compare 1 complementary output polarity

CC1 channel configured as output:

0: OC1N active high

1: OC1N active low

CC1 channel configured as input:

This bit is used in conjunction with CC1P to define the polarity of TI1FP1 and TI2FP1. Refer to the description of CC1P.

Note: This bit is not writable as soon as LOCK level 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register) and CC1S="00" (the channel is configured in output).

On channels that have a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1NP active bit takes the new value from the preloaded bit only when a commutation event is generated.

Bit 2 **CC1NE**: Capture/Compare 1 complementary output enable

0: Off - OC1N is not active. OC1N level is then function of MOE, OSSI, OSSR, OIS1, OIS1N and CC1E bits.

1: On - OC1N signal is output on the corresponding output pin depending on MOE, OSSI, OSSR, OIS1, OIS1N and CC1E bits.

Bit 1 **CC1P**: Capture/Compare 1 output polarity

0: OC1 active high (output mode) / Edge sensitivity selection (input mode, see below)

1: OC1 active low (output mode) / Edge sensitivity selection (input mode, see below)

When CC1 channel is configured as input, both CC1NP/CC1P bits select the active polarity of TI1FP1 and TI2FP1 for trigger or capture operations.

CC1NP=0, CC1P=0: non-inverted/rising edge. The circuit is sensitive to TIxFP1 rising edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode or encoder mode).

CC1NP=0, CC1P=1: inverted/falling edge. The circuit is sensitive to TIxFP1 falling edge (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is inverted (trigger operation in gated mode or encoder mode).

CC1NP=1, CC1P=1: non-inverted/both edges/ The circuit is sensitive to both TIxFP1 rising and falling edges (capture or trigger operations in reset, external clock or trigger mode), TIxFP1 is not inverted (trigger operation in gated mode). This configuration must not be used in encoder mode.

CC1NP=1, CC1P=0: this configuration is reserved, it must not be used.

Note: This bit is not writable as soon as LOCK level 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

On channels that have a complementary output, this bit is preloaded. If the CCPC bit is set in the TIMx_CR2 register then the CC1P active bit takes the new value from the preloaded bit only when a Commutation event is generated.

Bit 0 **CC1E**: Capture/Compare 1 output enable

0: Capture mode disabled / OC1 is not active (see below)

1: Capture mode enabled / OC1 signal is output on the corresponding output pin

When CC1 channel is configured as output, the OC1 level depends on MOE, OSSI, OSSR, OIS1, OIS1N and CC1NE bits, regardless of the CC1E bits state. Refer to [Table 345](#) for details.

Table 345. Output control bits for complementary OCx and OCxN channels with break feature (TIM16/17)

Control bits					Output states ⁽¹⁾	
MOE bit	OSSI bit	OSSR bit	CCxE bit	CCxNE bit	OCx output state	OCxN output state
1	X	X	0	0	Output Disabled (not driven by the timer: Hi-Z) OCx=0 OCxN=0, OCxN_EN=0	
		0	0	1	Output Disabled (not driven by the timer: Hi-Z) OCx=0	OCxREF + Polarity OCxN=OCxREF XOR CCxNP
		0	1	0	OCxREF + Polarity OCx=OCxREF XOR CCxP	Output Disabled (not driven by the timer: Hi-Z) OCxN=0
		X	1	1	OCREF + Polarity + dead-time	Complementary to OCREF (not OCREF) + Polarity + dead-time
		1	0	1	Off-State (output enabled with inactive state) OCx=CCxP	OCxREF + Polarity OCxN=OCxREF XOR CCxNP
		1	1	0	OCxREF + Polarity OCx=OCxREF XOR CCxP, OCx_EN=1	Off-State (output enabled with inactive state) OCxN=CCxNP, OCxN_EN=1
0	0	X	X	X	Output disabled (not driven by the timer anymore). The output state is defined by the GPIO controller and can be High, Low or Hi-Z.	
	0		0			
	0		1	Off-State (output enabled with inactive state) Asynchronously: OCx=CCxP, OCxN=CCxNP Then if the clock is present: OCx=OISx and OCxN=OISxN after a dead-time, assuming that OISx and OISxN do not correspond to OCx and OCxN both in active state		
	1		0			
	1		1			

1. When both outputs of a channel are not used (control taken over by GPIO controller), the OISx, OISxN, CCxP and CCxNP bits must be kept cleared.

Note: The state of the external I/O pins connected to the complementary OCx and OCxN channels depends on the OCx and OCxN channel state and AFIO registers.

41.6.9 TIMx counter (TIMx_CNT)(x = 16 to 17)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UIF CPY	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
r															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bit 31 **UIFCPY**: UIF Copy

This bit is a read-only copy of the UIF bit of the TIMx_ISR register. If the UIFREMAP bit in TIMx_CR1 is reset, bit 31 is reserved and read as 0.

Bits 30:16 Reserved, must be kept at reset value.

Bits 15:0 **CNT[15:0]**: Counter value

41.6.10 TIMx prescaler (TIMx_PSC)(x = 16 to 17)

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **PSC[15:0]**: Prescaler value

The counter clock frequency (CK_CNT) is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded in the active prescaler register at each update event (including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in “reset mode”).

41.6.11 TIMx auto-reload register (TIMx_ARR)(x = 16 to 17)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded in the actual auto-reload register.

Refer to the [Section 41.4.1: Time-base unit on page 1783](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

41.6.12 TIMx repetition counter register (TIMx_RCR)(x = 16 to 17)

Address offset: 0x30

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REP[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:8 Reserved, must be kept at reset value.

Bits 7:0 **REP[7:0]**: Repetition counter value

These bits allow the user to set-up the update rate of the compare registers (i.e. periodic transfers from preload to active registers) when preload registers are enable, as well as the update interrupt generation rate, if this interrupt is enable.

Each time the REP_CNT related downcounter reaches zero, an update event is generated and it restarts counting from REP value. As REP_CNT is reloaded with REP value only at the repetition update event U_RC, any write to the TIMx_RCR register is not taken in account until the next repetition update event.

It means in PWM mode (REP+1) corresponds to the number of PWM periods in edge-aligned mode.

41.6.13 TIMx capture/compare register 1 (TIMx_CCR1)(x = 16 to 17)

Address offset: 0x34

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR1[15:0]**: Capture/Compare 1 value**If channel CC1 is configured as output:**

CCR1 is the value to be loaded in the actual capture/compare 1 register (preload value).

It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (bit OC1PE). Else the preload value is copied in the active capture/compare 1 register when an update event occurs.

The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC1 output.

If channel CC1 is configured as input:

CCR1 is the counter value transferred by the last input capture 1 event (IC1).

41.6.14 TIMx break and dead-time register (TIMx_BDTR)(x = 16 to 17)

Address offset: 0x44

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BKF[3:0]			
												rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MOE	AOE	BKP	BKE	OSSR	OSSI	LOCK[1:0]		DTG[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Note: As the BKF[3:0], AOE, BKP, BKE, OSSI, OSSR and DTG[7:0] bits may be write-locked depending on the LOCK configuration, it may be necessary to configure all of them during the first write access to the TIMx_BDTR register.

Bits 31:20 Reserved, must be kept at reset value.

Bits 19:16 **BKF[3:0]**: Break filter

This bit-field defines the frequency used to sample BRK input and the length of the digital filter applied to BRK. The digital filter is made of an event counter in which N events are needed to validate a transition on the output:

0000: No filter, BRK acts asynchronously

0001: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=2

0010: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=4

0011: $f_{\text{SAMPLING}} = f_{\text{CK_INT}}$, N=8

0100: $f_{\text{SAMPLING}} = f_{\text{DTS}}/2$, N=6

0101: $f_{\text{SAMPLING}} = f_{\text{DTS}}/2$, N=8

0110: $f_{\text{SAMPLING}} = f_{\text{DTS}}/4$, N=6

0111: $f_{\text{SAMPLING}} = f_{\text{DTS}}/4$, N=8

1000: $f_{\text{SAMPLING}} = f_{\text{DTS}}/8$, N=6

1001: $f_{\text{SAMPLING}} = f_{\text{DTS}}/8$, N=8

1010: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=5

1011: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=6

1100: $f_{\text{SAMPLING}} = f_{\text{DTS}}/16$, N=8

1101: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=5

1110: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=6

1111: $f_{\text{SAMPLING}} = f_{\text{DTS}}/32$, N=8

This bit cannot be modified when LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 15 **MOE**: Main output enable

This bit is cleared asynchronously by hardware as soon as the break input is active. It is set by software or automatically depending on the AOE bit. It is acting only on the channels which are configured in output.

0: OC and OCN outputs are disabled or forced to idle state depending on the OSSI bit.

1: OC and OCN outputs are enabled if their respective enable bits are set (CCxE, CCxNE in TIMx_CCER register)

See OC/OCN enable description for more details ([Section 41.6.8: TIMx capture/compare enable register \(TIMx_CCER\)\(x = 16 to 17\) on page 1850](#)).

Bit 14 **AOE**: Automatic output enable

0: MOE can be set only by software

1: MOE can be set by software or automatically at the next update event (if the break input is not be active)

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 13 **BKP**: Break polarity

0: Break input BRK is active low

1: Break input BRK is active high

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Any write operation to this bit takes a delay of 1 APB clock cycle to become effective.

Bit 12 **BKE**: Break enable

0: Break inputs (BRK and CCS clock failure event) disabled

1: Break inputs (BRK and CCS clock failure event) enabled

Note: This bit cannot be modified when LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Any write operation to this bit takes a delay of 1 APB clock cycle to become effective.

Bit 11 **OSSR**: Off-state selection for Run mode

This bit is used when MOE=1 on channels that have a complementary output which are configured as outputs. OSSR is not implemented if no complementary output is implemented in the timer.

See OC/OCN enable description for more details ([Section 41.6.8: TIMx capture/compare enable register \(TIMx_CCER\)\(x = 16 to 17\) on page 1850](#)).

0: When inactive, OC/OCN outputs are disabled (the timer releases the output control which is taken over by the AFIO logic, which forces a Hi-Z state)

1: When inactive, OC/OCN outputs are enabled with their inactive level as soon as CCxE=1 or CCxNE=1 (the output is still controlled by the timer).

Note: This bit can not be modified as soon as the LOCK level 2 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 10 **OSSI**: Off-state selection for Idle mode

This bit is used when MOE=0 on channels configured as outputs.

See OC/OCN enable description for more details ([Section 41.6.8: TIMx capture/compare enable register \(TIMx_CCER\)\(x = 16 to 17\) on page 1850](#)).

0: When inactive, OC/OCN outputs are disabled (OC/OCN enable output signal=0)

1: When inactive, OC/OCN outputs are forced first with their idle level as soon as CCxE=1 or CCxNE=1. OC/OCN enable output signal=1)

Note: This bit can not be modified as soon as the LOCK level 2 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 9:8 **LOCK[1:0]**: Lock configuration

These bits offer a write protection against software errors.

00: LOCK OFF - No bit is write protected

01: LOCK Level 1 = DTG bits in TIMx_BDTR register, OISx and OISxN bits in TIMx_CR2 register and BKE/BKP/AOE bits in TIMx_BDTR register can no longer be written.

10: LOCK Level 2 = LOCK Level 1 + CC Polarity bits (CCxP/CCxNP bits in TIMx_CCER register, as long as the related channel is configured in output through the CCxS bits) as well as OSSR and OSSI bits can no longer be written.

11: LOCK Level 3 = LOCK Level 2 + CC Control bits (OCxM and OCxPE bits in TIMx_CCMRx registers, as long as the related channel is configured in output through the CCxS bits) can no longer be written.

Note: The LOCK bits can be written only once after the reset. Once the TIMx_BDTR register has been written, their content is frozen until the next reset.

Bits 7:0 **DTG[7:0]**: Dead-time generator setup

This bit-field defines the duration of the dead-time inserted between the complementary outputs. DT correspond to this duration.

DTG[7:5]=0xx => DT=DTG[7:0]x t_{dtg} with $t_{dtg}=t_{DTS}$

DTG[7:5]=10x => DT=(64+DTG[5:0])x t_{dtg} with $T_{dtg}=2xt_{DTS}$

DTG[7:5]=110 => DT=(32+DTG[4:0])x t_{dtg} with $T_{dtg}=8xt_{DTS}$

DTG[7:5]=111 => DT=(32+DTG[4:0])x t_{dtg} with $T_{dtg}=16xt_{DTS}$

Example if $T_{DTS}=125\text{ns}$ (8MHz), dead-time possible values are:

0 to 15875 ns by 125 ns steps,

16 μs to 31750 ns by 250 ns steps,

32 μs to 63 μs by 1 μs steps,

64 μs to 126 μs by 2 μs steps

Note: This bit-field can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

41.6.15 TIMx DMA control register (TIMx_DCR)(x = 16 to 17)

Address offset: 0x48

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	DBL[4:0]					Res.	Res.	Res.	DBA[4:0]				
			rw	rw	rw	rw	rw				rw	rw	rw	rw	rw

Bits 15:13 Reserved, must be kept at reset value.

Bits 12:8 **DBL[4:0]**: DMA burst length

This 5-bit field defines the length of DMA transfers (the timer recognizes a burst transfer when a read or a write access is done to the TIMx_DMAR address), i.e. the number of transfers. Transfers can be in half-words or in bytes (see example below).

00000: 1 transfer,

00001: 2 transfers,

00010: 3 transfers,

...

10001: 18 transfers.

Bits 7:5 Reserved, must be kept at reset value.

Bits 4:0 **DBA[4:0]**: DMA base address

This 5-bit field defines the base-address for DMA transfers (when read/write access are done through the TIMx_DMAR address). DBA is defined as an offset starting from the address of the TIMx_CR1 register.

Example:

00000: TIMx_CR1,

00001: TIMx_CR2,

00010: TIMx_SMCR,

...

Example: Let us consider the following transfer: DBL = 7 transfers and DBA = TIMx_CR1. In this case the transfer is done to/from 7 registers starting from the TIMx_CR1 address.

41.6.16 TIMx DMA address for full transfer (TIMx_DMAR)(x = 16 to 17)

Address offset: 0x4C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMAB[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **DMAB[15:0]**: DMA register for burst accesses

A read or write operation to the DMAR register accesses the register located at the address
 $(\text{TIMx_CR1 address}) + (\text{DBA} + \text{DMA index}) \times 4$

where TIMx_CR1 address is the address of the control register 1, DBA is the DMA base address configured in TIMx_DCR register, DMA index is automatically controlled by the DMA transfer, and ranges from 0 to DBL (DBL configured in TIMx_DCR).

41.6.17 TIM16 alternate function register 1 (TIM16_AF1)

Address offset: 0x60

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	BKCM P2P	BKCM P1P	BKINP	BKDF1 BK1E	Res.	Res.	Res.	Res.	Res.	BKCM P2E	BKCM P1E	BKINE
				rw	rw	rw	rw						rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bit 11 BKCM P2P: BRK COMP2 input polarity

This bit selects the COMP2 input sensitivity. It must be programmed together with the BKP polarity bit.

0: COMP2 input is active low

1: COMP2 input is active high

*Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).***Bit 10 BKCM P1P:** BRK COMP1 input polarity

This bit selects the COMP1 input sensitivity. It must be programmed together with the BKP polarity bit.

0: COMP1 input is active low

1: COMP1 input is active high

*Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).***Bit 9 BKINP:** BRK BKIN input polarity

This bit selects the BKIN alternate function input sensitivity. It must be programmed together with the BKP polarity bit.

0: BKIN input is active low

1: BKIN input is active high

*Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).***Bit 8 BKDFBK1E:** BRK dfsdm1_break[1] enable

This bit enables the dfsdm1_break[1] for the timer's BRK input. dfsdm1_break[1] output is 'ORed' with the other BRK sources.

0: dfsdm1_break[1] input disabled

1: dfsdm1_break[1] input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 7:3 Reserved, must be kept at reset value.

Bit 2 BKCOMP2E: BRK COMP2 enable

This bit enables the COMP2 for the timer's BRK input. COMP2 output is 'ORed' with the other BRK sources.

0: COMP2 input disabled

1: COMP2 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 1 BKCOMP1E: BRK COMP1 enable

This bit enables the COMP1 for the timer's BRK input. COMP1 output is 'ORed' with the other BRK sources.

0: COMP1 input disabled

1: COMP1 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 0 BKINE: BRK BKIN input enable

This bit enables the BKIN alternate function input for the timer's BRK input. BKIN input is 'ORed' with the other BRK sources.

0: BKIN input disabled

1: BKIN input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

41.6.18 TIM16 input selection register (TIM16_TISEL)

Address offset: 0x68

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	T1SEL[3:0]			
												rw	rw	rw	rw

Bits 31:4 Reserved, must be kept at reset value.

Bits 3:0 **T1SEL[3:0]**: selects T1[0] to T1[15] input

0000: TIM16_CH1 input

0001: LSI

0010: LSE

0011: WKUP_IT

Other: Reserved

41.6.19 TIM17 alternate function register 1 (TIM17_AF1)

Address offset: 0x60

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	BKCM P2P	BKCM P1P	BKINP	BKDF1 BK2E	Res.	Res.	Res.	Res.	Res.	BKCM P2E	BKCM P1E	BKINE
				rw	rw	rw	rw						rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bit 11 BKCM P2P: BRK COMP2 input polarity

This bit selects the COMP2 input sensitivity. It must be programmed together with the BKP polarity bit.

0: COMP2 input is active low

1: COMP2 input is active high

*Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).***Bit 10 BKCM P1P:** BRK COMP1 input polarity

This bit selects the COMP1 input sensitivity. It must be programmed together with the BKP polarity bit.

0: COMP1 input is active low

1: COMP1 input is active high

*Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).***Bit 9 BKINP:** BRK BKIN input polarity

This bit selects the BKIN alternate function input sensitivity. It must be programmed together with the BKP polarity bit.

0: BKIN input is active low

1: BKIN input is active high

*Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).***Bit 8 BKDF1BK2E:** BRK dfsdm1_break[2] enable

This bit enables the dfsdm1_break[2] for the timer's BRK input. dfsdm1_break[2] output is 'ORed' with the other BRK sources.

0: dfsdm1_break[2] input disabled

1: dfsdm1_break[2] input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bits 7:3 Reserved, must be kept at reset value.

Bit 2 **BKCOMP2E**: BRK COMP2 enable

This bit enables the COMP2 for the timer's BRK input. COMP2 output is 'ORed' with the other BRK sources.

0: COMP2 input disabled

1: COMP2 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 1 **BKCOMP1E**: BRK COMP1 enable

This bit enables the COMP1 for the timer's BRK input. COMP1 output is 'ORed' with the other BRK sources.

0: COMP1 input disabled

1: COMP1 input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

Bit 0 **BKINE**: BRK BKIN input enable

This bit enables the BKIN alternate function input for the timer's BRK input. BKIN input is 'ORed' with the other BRK sources.

0: BKIN input disabled

1: BKIN input enabled

Note: This bit can not be modified as long as LOCK level 1 has been programmed (LOCK bits in TIMx_BDTR register).

41.6.20 TIM17 input selection register (TIM17_TISEL)

Address offset: 0x68

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	T1SEL[3:0]			
												rw	rw	rw	rw

Bits 31:4 Reserved, must be kept at reset value.

Bits 3:0 **T1SEL[3:0]**: selects T1[0] to T1[15] input

0000: TIM17_CH1 input

0010: HSE_1MHz

0011: MCO1

Others: Reserved

41.6.21 TIM16/TIM17 register map

TIM16/TIM17 registers are mapped as 16-bit addressable registers as described in the table below:

Table 346. TIM16/TIM17 register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
0x00	TIMx_CR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	U1FREMA	Res	CKD [1:0]	OIS1	ARPE	Res	Res	Res	Res	OPM	URS	UDIS	CEN					
	Reset value																					0	0			0					0	0	0	0					
0x04	TIMx_CR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	OIS1N	OIS1	Res	Res	Res	Res	CCDS	CCUS	Res	CCPC						
	Reset value																							0	0					0	0		0						
0x0C	TIMx_DIER	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CC1DE	UDE	BIE	COMIE	Res	Res	Res	Res	CC1IE	UIE							
	Reset value																						0	0	0	0					0	0							
0x10	TIMx_SR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CC1OF	Res	BIF	COMIF	Res	Res	Res	Res	CC1IF	UIF							
	Reset value																						0		0	0					0	0							
0x14	TIMx_EGR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	BG	COMG	Res	Res	Res	Res	CC1G	UG							
	Reset value																								0	0					0	0							
0x18	TIMx_CCMR1 Output Compare mode	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	OC1M[3]	Res	Res	Res	Res	Res	Res	Res	Res	OC1M [2:0]				OC1PE	OC1FE	CC1S [1:0]							
	Reset value																0									0	0	0	0	0	0	0	0						
	TIMx_CCMR1 Input Capture mode	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	IC1F[3:0]				IC1PSC [1:0]	CC1S [1:0]									
	Reset value																									0	0	0	0	0	0	0	0						
0x20	TIMx_CCER	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CC1NP	CC1NE	CC1P	CC1E						
	Reset value																												0	0	0	0							
0x24	TIMx_CNT	UIFCPY or Res.	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CNT[15:0]															
	Reset value	0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x28	TIMx_PSC	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PSC[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x2C	TIMx_ARR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	ARR[15:0]															
	Reset value																	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1						

Table 346. TIM16/TIM17 register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
0x30	TIMx_RCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	REP[7:0]													
	Reset value																										0	0	0	0	0	0	0	0					
0x34	TIMx_CCR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CCR1[15:0]																					
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x44	TIMx_BDTR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	BKF[3:0]			MOE	AOE	BKP	BKE	OSSR	OSSI	LOCK K [1:0]	DTG[7:0]															
	Reset value													0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x48	TIMx_DCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DBL[4:0]				Res			Res	Res	DBA[4:0]									
	Reset value																				0	0	0	0	0					0	0	0	0	0					
0x4C	TIMx_DMAR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DMAB[15:0]																					
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x60	TIM16_AF1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	BKCOMP2P	BKCOMP1P	BKINP	BKDF1BK1E	Res	Res	Res	Res	Res	Res	BKCOMP2E	BKCOMP1E						
	Reset value																					0	0	0	0						0	0	1						
0x60	TIM17_AF1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	BKCOMP2P	BKCOMP1P	BKINP	BKDF1BK2E	Res	Res	Res	Res	Res	Res	BKCOMP2E	BKCOMP1E						
	Reset value																					0	0	0	0						0	0	1						
0x68	TIM16_TISEL	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	T11SEL[3:0]							
	Reset value																													0	0	0	0						
0x68	TIM17_TISEL	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	T11SEL[3:0]							
	Reset value																													0	0	0	0						

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.

42 Basic timers (TIM6/TIM7)

42.1 TIM6/TIM7 introduction

The basic timers TIM6 and TIM7 consist of a 16-bit auto-reload counter driven by a programmable prescaler.

They may be used as generic timers for time base generation but they are also specifically used to drive the digital-to-analog converter (DAC). In fact, the timers are internally connected to the DAC and are able to drive it through their trigger outputs.

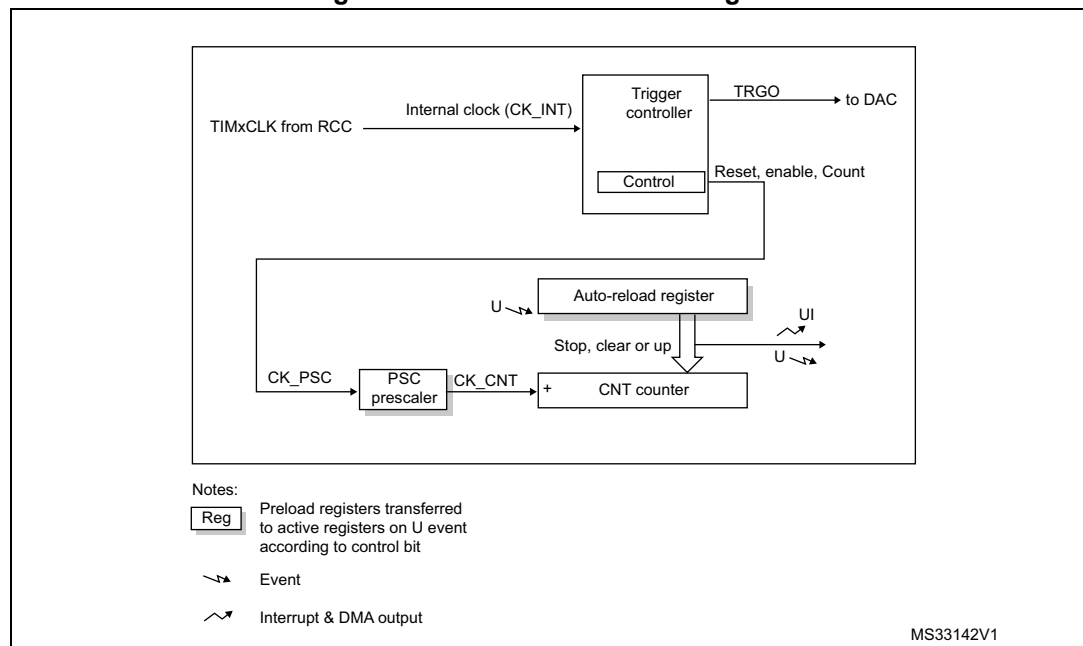
The timers are completely independent, and do not share any resources.

42.2 TIM6/TIM7 main features

Basic timer (TIM6/TIM7) features include:

- 16-bit auto-reload upcounter
- 16-bit programmable prescaler used to divide (also “on the fly”) the counter clock frequency by any factor between 1 and 65535
- Synchronization circuit to trigger the DAC
- Interrupt/DMA generation on the update event: counter overflow

Figure 512. Basic timer block diagram



42.3 TIM6/TIM7 functional description

42.3.1 Time-base unit

The main block of the programmable timer is a 16-bit upcounter with its related auto-reload register. The counter clock can be divided by a prescaler.

The counter, the auto-reload register and the prescaler register can be written or read by software. This is true even when the counter is running.

The time-base unit includes:

- Counter Register (TIMx_CNT)
- Prescaler Register (TIMx_PSC)
- Auto-Reload Register (TIMx_ARR)

The auto-reload register is preloaded. The preload register is accessed each time an attempt is made to write or read the auto-reload register. The contents of the preload register are transferred into the shadow register permanently or at each update event UEV, depending on the auto-reload preload enable bit (ARPE) in the TIMx_CR1 register. The update event is sent when the counter reaches the overflow value and if the UDIS bit equals 0 in the TIMx_CR1 register. It can also be generated by software. The generation of the update event is described in detail for each configuration.

The counter is clocked by the prescaler output CK_CNT, which is enabled only when the counter enable bit (CEN) in the TIMx_CR1 register is set.

Note that the actual counter enable signal CNT_EN is set 1 clock cycle after CEN.

Prescaler description

The prescaler can divide the counter clock frequency by any factor between 1 and 65536. It is based on a 16-bit counter controlled through a 16-bit register (in the TIMx_PSC register). It can be changed on the fly as the TIMx_PSC control register is buffered. The new prescaler ratio is taken into account at the next update event.

[Figure 513](#) and [Figure 514](#) give some examples of the counter behavior when the prescaler ratio is changed on the fly.

Figure 513. Counter timing diagram with prescaler division change from 1 to 2

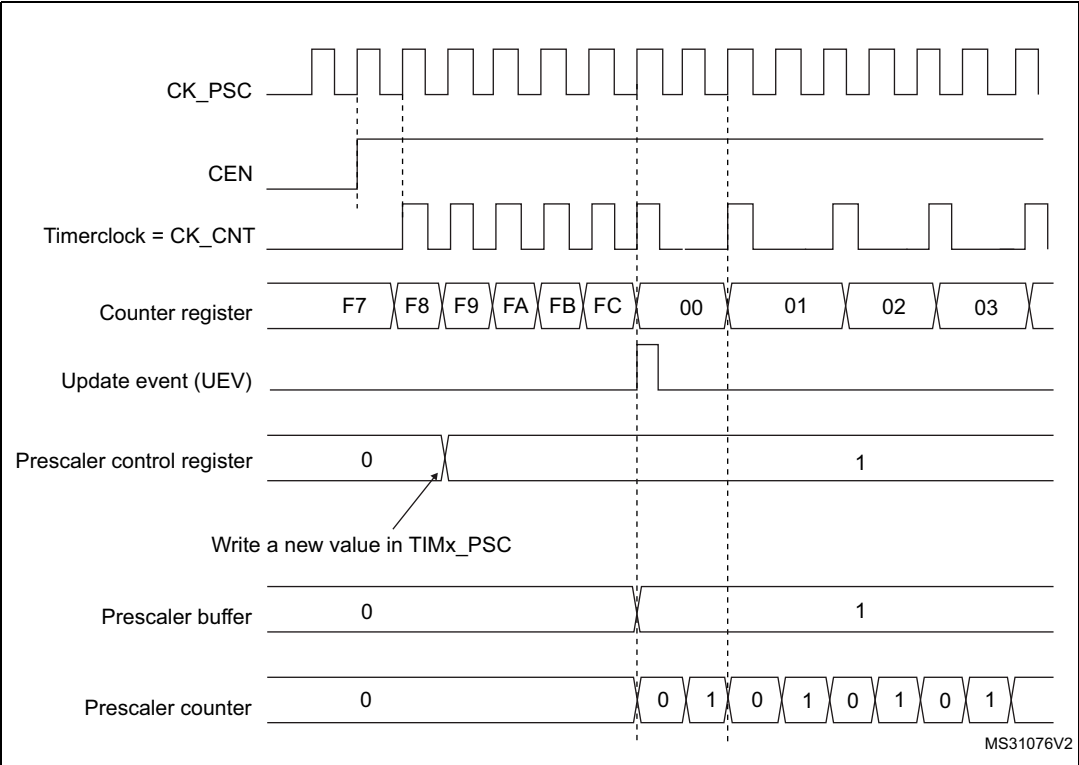
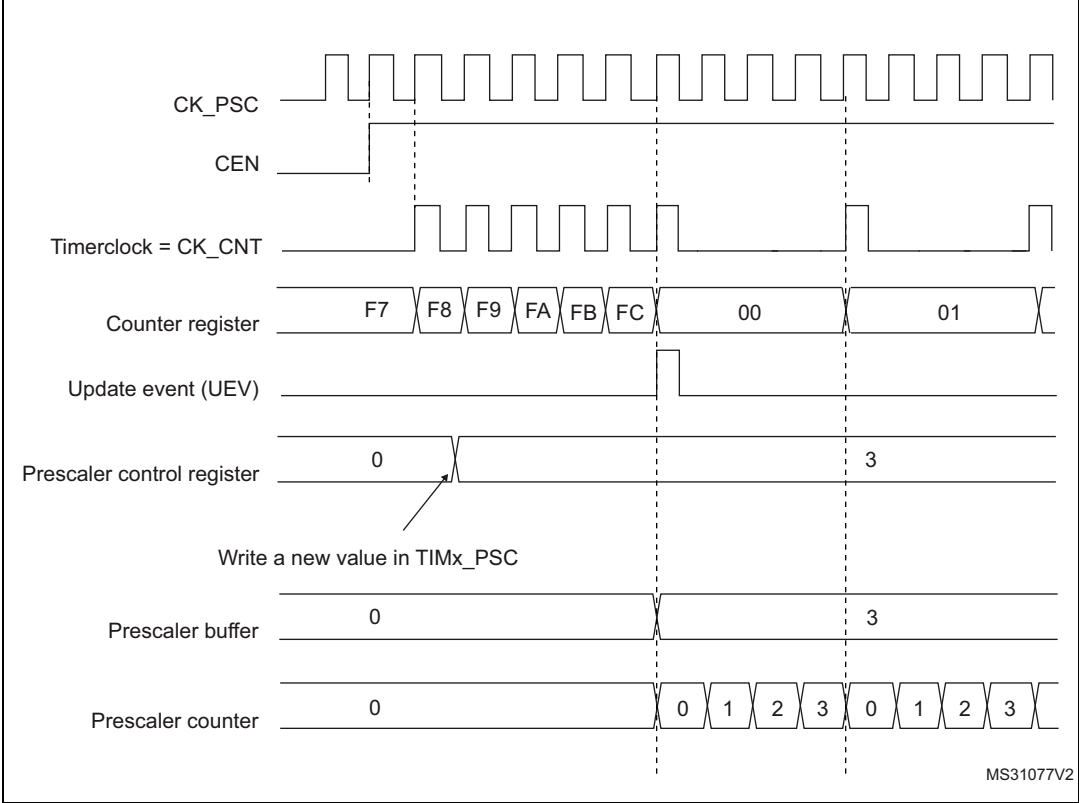


Figure 514. Counter timing diagram with prescaler division change from 1 to 4



42.3.2 Counting mode

The counter counts from 0 to the auto-reload value (contents of the TIMx_ARR register), then restarts from 0 and generates a counter overflow event.

An update event can be generated at each counter overflow or by setting the UG bit in the TIMx_EGR register (by software or by using the slave mode controller).

The UEV event can be disabled by software by setting the UDIS bit in the TIMx_CR1 register. This avoids updating the shadow registers while writing new values into the preload registers. In this way, no update event occurs until the UDIS bit has been written to 0, however, the counter and the prescaler counter both restart from 0 (but the prescale rate does not change). In addition, if the URS (update request selection) bit in the TIMx_CR1 register is set, setting the UG bit generates an update event UEV, but the UIF flag is not set (so no interrupt or DMA request is sent).

When an update event occurs, all the registers are updated and the update flag (UIF bit in the TIMx_SR register) is set (depending on the URS bit):

- The buffer of the prescaler is reloaded with the preload value (contents of the TIMx_PSC register)
- The auto-reload shadow register is updated with the preload value (TIMx_ARR)

The following figures show some examples of the counter behavior for different clock frequencies when TIMx_ARR = 0x36.

Figure 515. Counter timing diagram, internal clock divided by 1

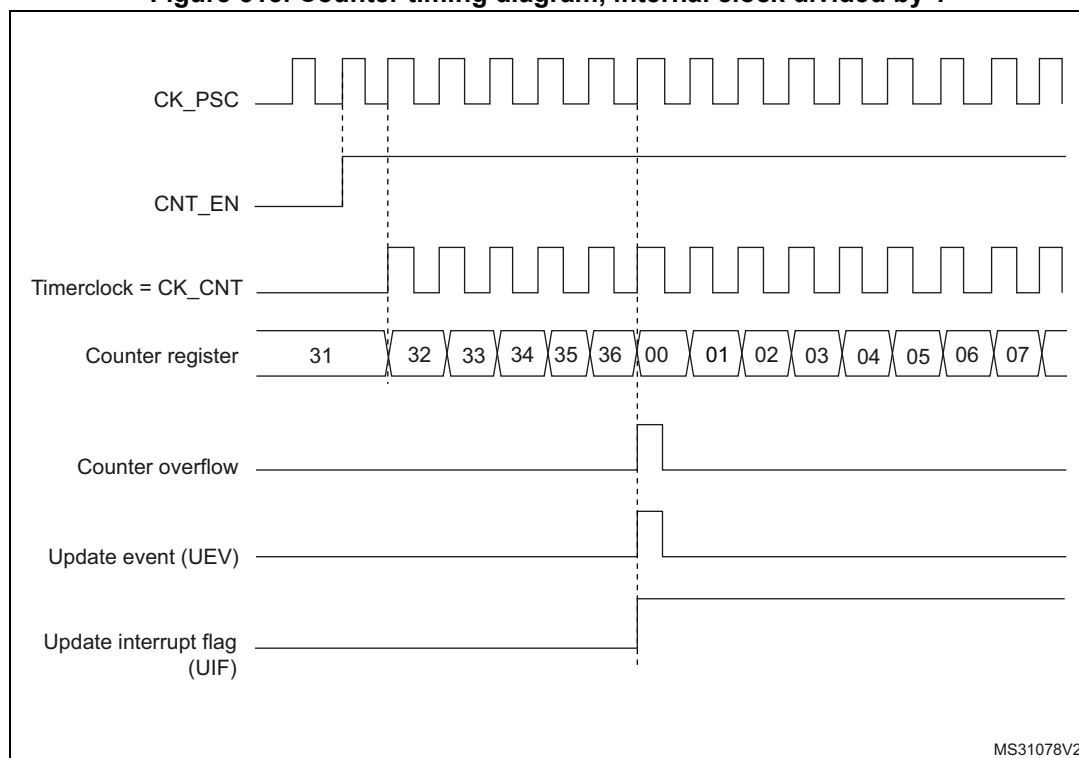


Figure 516. Counter timing diagram, internal clock divided by 2

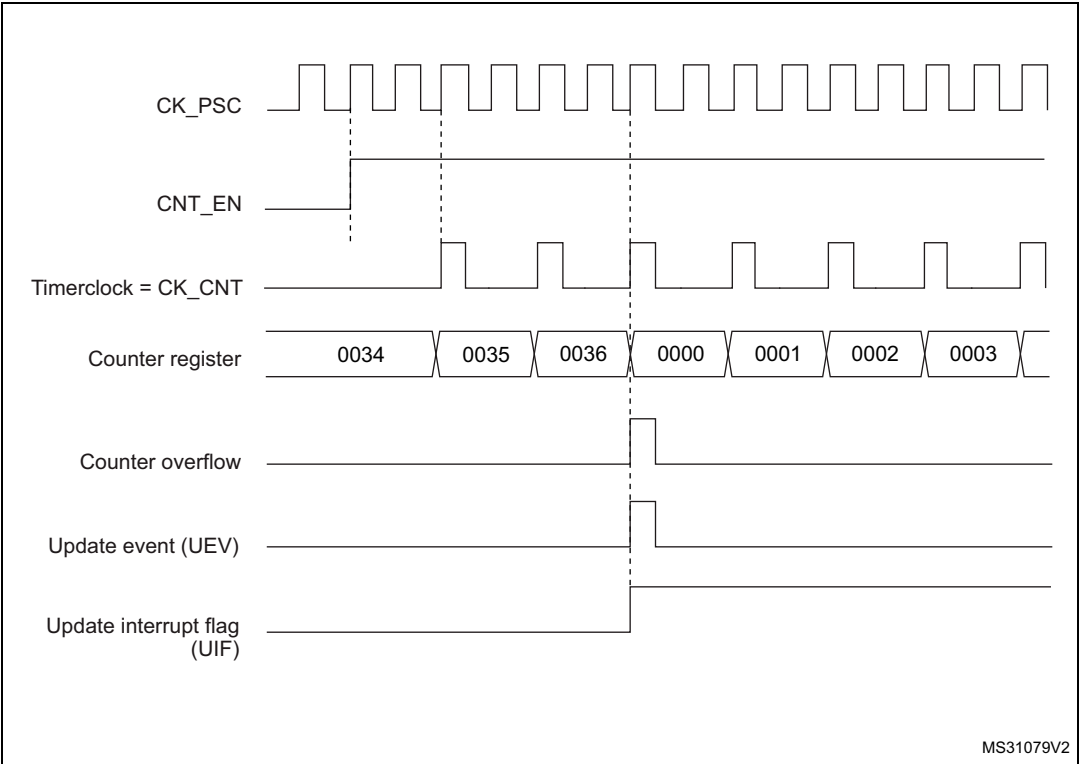


Figure 517. Counter timing diagram, internal clock divided by 4

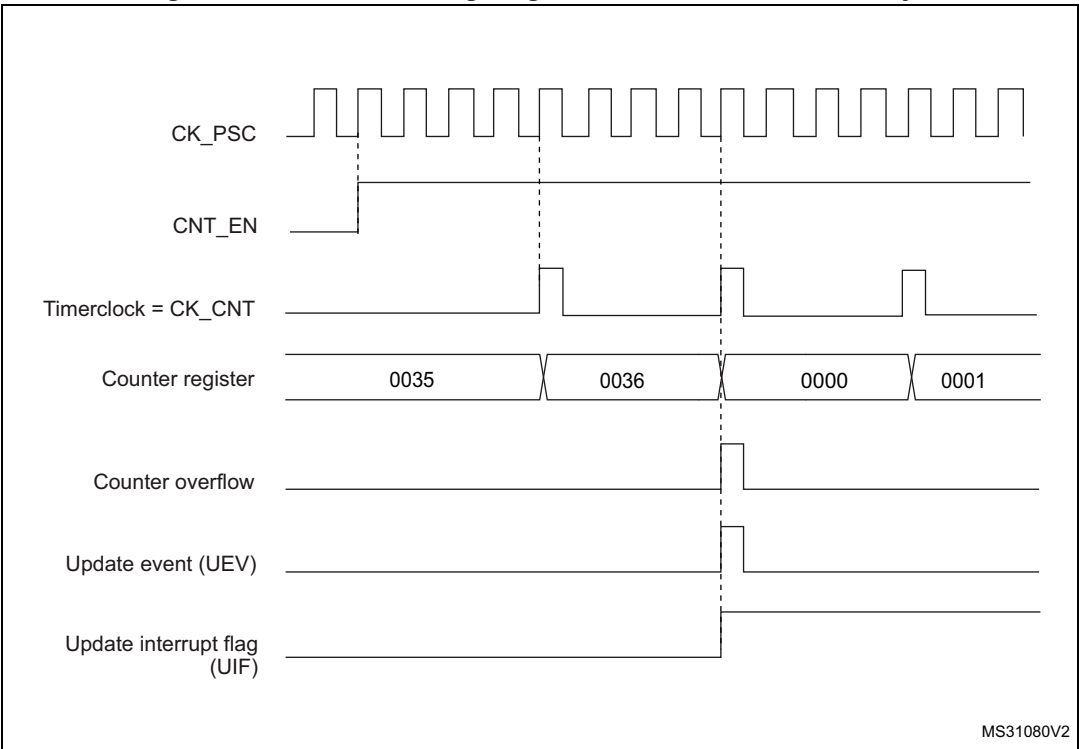


Figure 518. Counter timing diagram, internal clock divided by N

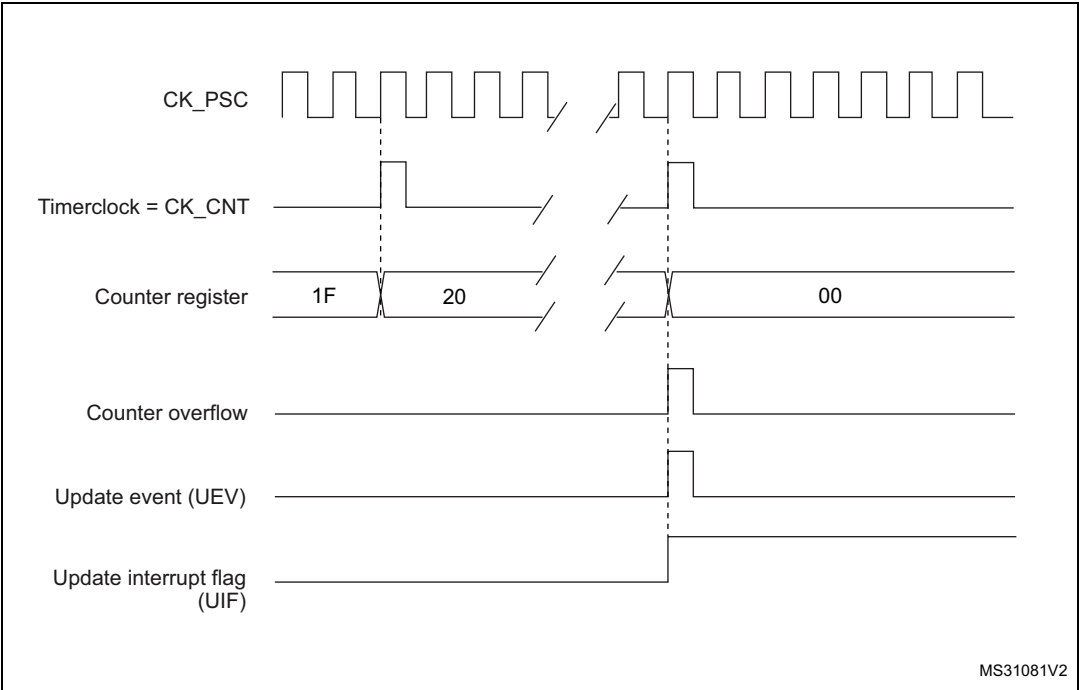


Figure 519. Counter timing diagram, update event when ARPE = 0 (TIMx_ARR not preloaded)

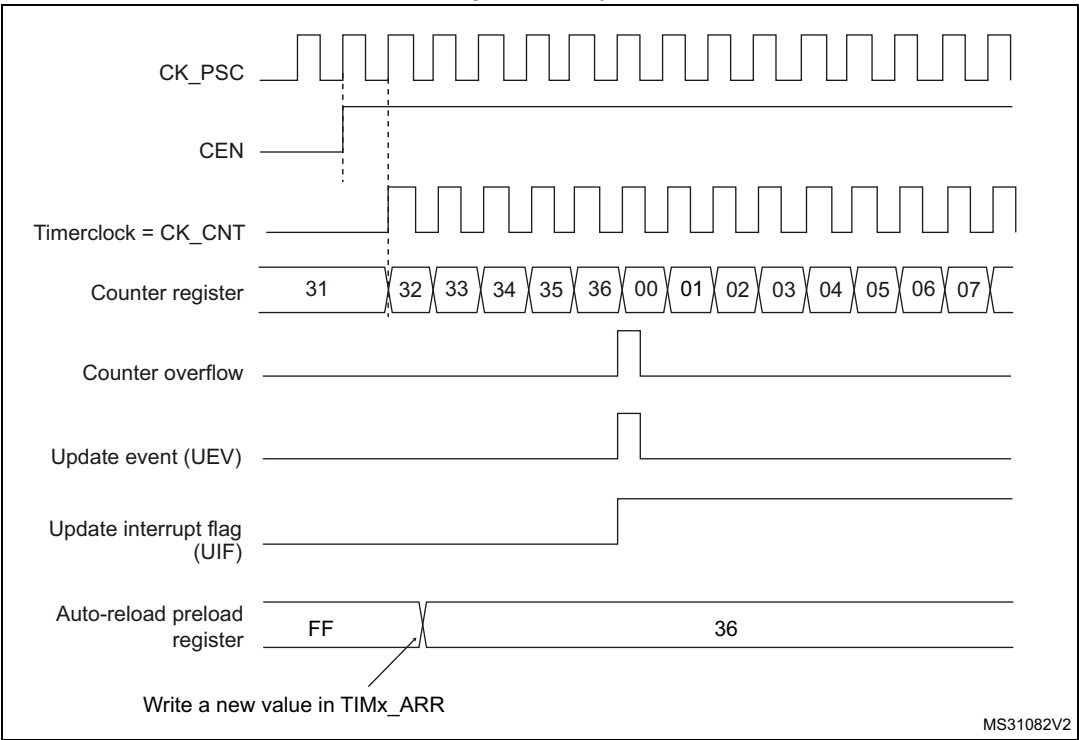
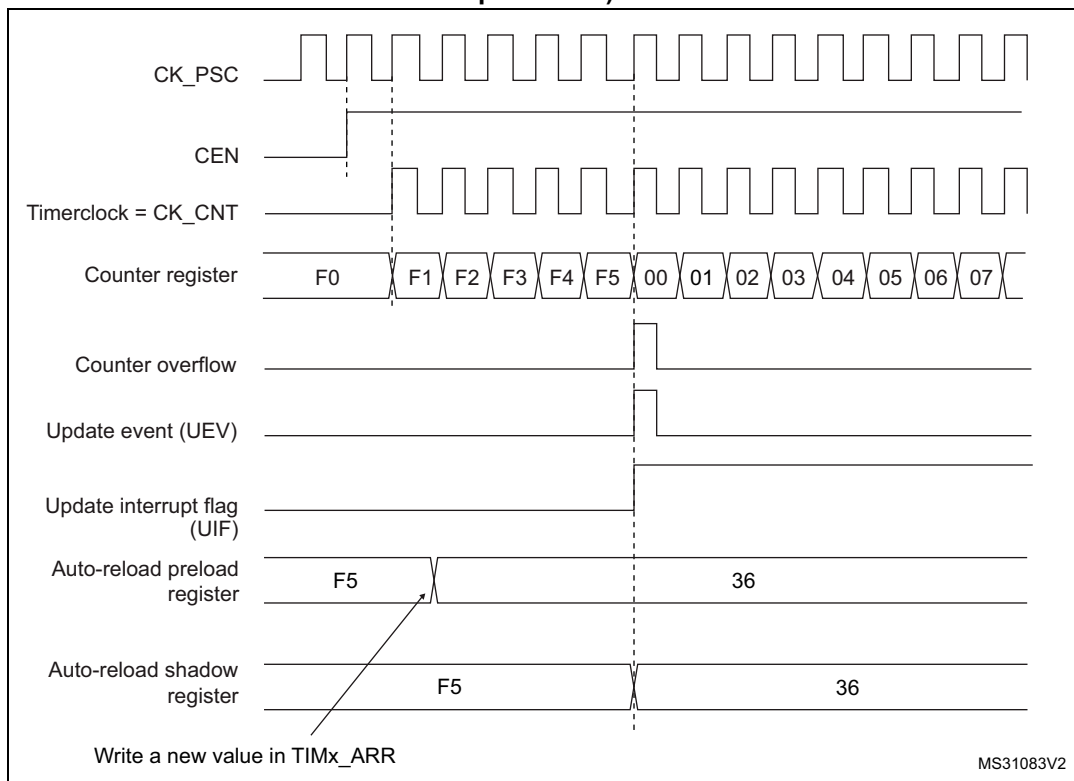


Figure 520. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded)



42.3.3 UIF bit remapping

The IUFREMAP bit in the TIMx_CR1 register forces a continuous copy of the Update Interrupt Flag UIF into the timer counter register's bit 31 (TIMxCNT[31]). This allows to atomically read both the counter value and a potential roll-over condition signaled by the UIFCPY flag. In particular cases, it can ease the calculations by avoiding race conditions caused for instance by a processing shared between a background task (counter reading) and an interrupt (Update Interrupt).

There is no latency between the assertions of the UIF and UIFCPY flags.

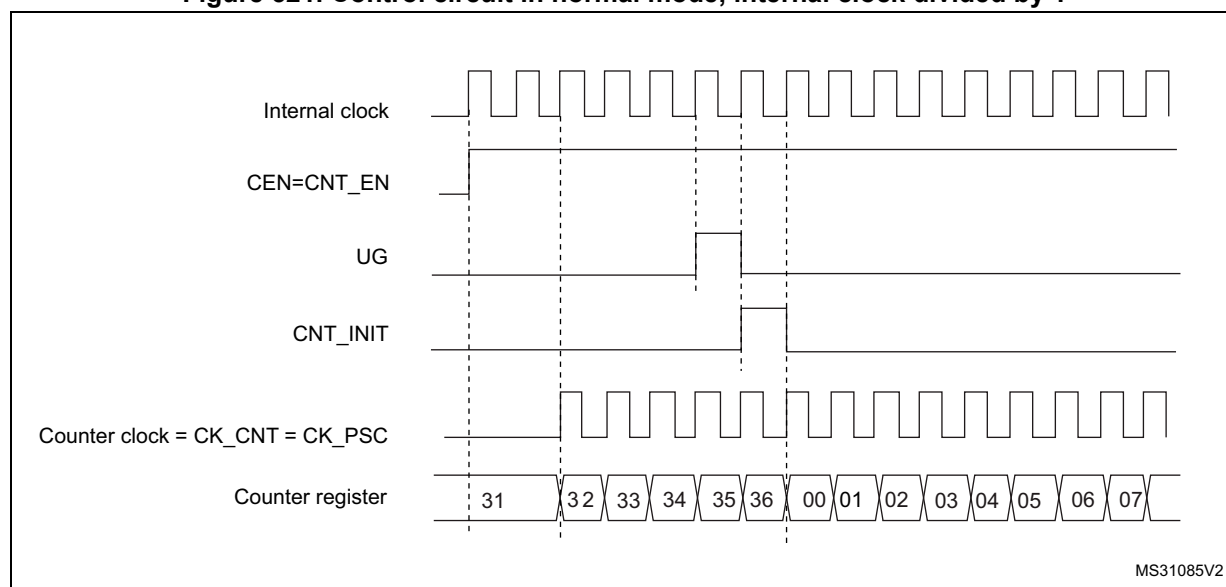
42.3.4 Clock source

The counter clock is provided by the Internal clock (CK_INT) source.

The CEN (in the TIMx_CR1 register) and UG bits (in the TIMx_EGR register) are actual control bits and can be changed only by software (except for UG that remains cleared automatically). As soon as the CEN bit is written to 1, the prescaler is clocked by the internal clock CK_INT.

[Figure 521](#) shows the behavior of the control circuit and the upcounter in normal mode, without prescaler.

Figure 521. Control circuit in normal mode, internal clock divided by 1



42.3.5 Debug mode

When the microcontroller enters the debug mode (Cortex®-M7 with FPU core - halted), the TIMx counter either continues to work normally or stops, depending on the DBG_TIMx_STOP configuration bit in the DBGMCU module. For more details, refer to [Section 60.5.8: Microcontroller debug unit \(DBGMCU\)](#).

42.4 TIM6/TIM7 registers

Refer to [Section 1.2 on page 100](#) for a list of abbreviations used in register descriptions.

The peripheral registers can be accessed by half-words (16-bit) or words (32-bit).

42.4.1 TIMx control register 1 (TIMx_CR1)(x = 6 to 7)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	UIFRE MAP	Res.	Res.	Res.	ARPE	Res.	Res.	Res.	OPM	URS	UDIS	CEN
				rw				rw				rw	rw	rw	rw

Bits 15:12 Reserved, must be kept at reset value.

Bit 11 **UIFREMAP**: UIF status bit remapping

0: No remapping. UIF status bit is not copied to TIMx_CNT register bit 31.

1: Remapping enabled. UIF status bit is copied to TIMx_CNT register bit 31.

Bits 10:8 Reserved, must be kept at reset value.

- Bit 7 **ARPE**: Auto-reload preload enable
0: TIMx_ARR register is not buffered.
1: TIMx_ARR register is buffered.

Bits 6:4 Reserved, must be kept at reset value.

- Bit 3 **OPM**: One-pulse mode
0: Counter is not stopped at update event
1: Counter stops counting at the next update event (clearing the CEN bit).

- Bit 2 **URS**: Update request source

This bit is set and cleared by software to select the UEV event sources.

0: Any of the following events generates an update interrupt or DMA request if enabled.
These events can be:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

1: Only counter overflow/underflow generates an update interrupt or DMA request if enabled.

- Bit 1 **UDIS**: Update disable

This bit is set and cleared by software to enable/disable UEV event generation.

0: UEV enabled. The Update (UEV) event is generated by one of the following events:

- Counter overflow/underflow
- Setting the UG bit
- Update generation through the slave mode controller

Buffered registers are then loaded with their preload values.

1: UEV disabled. The Update event is not generated, shadow registers keep their value (ARR, PSC). However the counter and the prescaler are reinitialized if the UG bit is set or if a hardware reset is received from the slave mode controller.

- Bit 0 **CEN**: Counter enable

0: Counter disabled
1: Counter enabled

*Note: Gated mode can work only if the CEN bit has been previously set by software.
However trigger mode can set the CEN bit automatically by hardware.*

CEN is cleared automatically in one-pulse mode, when an update event occurs.

42.4.2 TIMx control register 2 (TIMx_CR2)(x = 6 to 7)

Address offset: 0x04

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MMS[2:0]			Res.	Res.	Res.	Res.
									rw	rw	rw				

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:4 **MMS[2:0]**: Master mode selection

These bits are used to select the information to be sent in master mode to slave timers for synchronization (TRGO). The combination is as follows:

000: **Reset** - the UG bit from the TIMx_EGR register is used as a trigger output (TRGO). If reset is generated by the trigger input (slave mode controller configured in reset mode) then the signal on TRGO is delayed compared to the actual reset.

001: **Enable** - the Counter enable signal, CNT_EN, is used as a trigger output (TRGO). It is useful to start several timers at the same time or to control a window in which a slave timer is enabled. The Counter Enable signal is generated by a logic OR between CEN control bit and the trigger input when configured in gated mode.

When the Counter Enable signal is controlled by the trigger input, there is a delay on TRGO, except if the master/slave mode is selected (see the MSM bit description in the TIMx_SMCR register).

010: **Update** - The update event is selected as a trigger output (TRGO). For instance a master timer can then be used as a prescaler for a slave timer.

Note: The clock of the slave timer or ADC must be enabled prior to receive events from the master timer, and must not be changed on-the-fly while triggers are received from the master timer.

Bits 3:0 Reserved, must be kept at reset value.

42.4.3 TIMx DMA/Interrupt enable register (TIMx_DIER)(x = 6 to 7)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	UDE	Res.	Res.	Res.	Res.	Res.	Res.	Res.	UIE
							rw								rw

Bits 15:9 Reserved, must be kept at reset value.

Bit 8 **UDE**: Update DMA request enable

0: Update DMA request disabled.

1: Update DMA request enabled.

Bits 7:1 Reserved, must be kept at reset value.

Bit 0 **UIE**: Update interrupt enable

0: Update interrupt disabled.

1: Update interrupt enabled.

42.4.4 TIMx status register (TIMx_SR)(x = 6 to 7)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	UIF
															rc_w0

Bits 15:1 Reserved, must be kept at reset value.

Bit 0 **UIF**: Update interrupt flag

This bit is set by hardware on an update event. It is cleared by software.

0: No update occurred.

1: Update interrupt pending. This bit is set by hardware when the registers are updated:

- At overflow or underflow regarding the repetition counter value and if UDIS = 0 in the TIMx_CR1 register.
- When CNT is reinitialized by software using the UG bit in the TIMx_EGR register, if URS = 0 and UDIS = 0 in the TIMx_CR1 register.

42.4.5 TIMx event generation register (TIMx_EGR)(x = 6 to 7)

Address offset: 0x14

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	UG
															w

Bits 15:1 Reserved, must be kept at reset value.

Bit 0 **UG**: Update generation

This bit can be set by software, it is automatically cleared by hardware.

0: No action.

1: Re-initializes the timer counter and generates an update of the registers. Note that the prescaler counter is cleared too (but the prescaler ratio is not affected).

42.4.6 TIMx counter (TIMx_CNT)(x = 6 to 7)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UIF CPY	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
r															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bit 31 **UIFCPY**: UIF Copy

This bit is a read-only copy of the UIF bit of the TIMx_ISR register. If the UIFREMAP bit in TIMx_CR1 is reset, bit 31 is reserved and read as 0.

Bits 30:16 Reserved, must be kept at reset value.

Bits 15:0 **CNT[15:0]**: Counter value

42.4.7 TIMx prescaler (TIMx_PSC)(x = 6 to 7)

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **PSC[15:0]**: Prescaler value

The counter clock frequency CK_CNT is equal to $f_{CK_PSC} / (PSC[15:0] + 1)$.

PSC contains the value to be loaded into the active prescaler register at each update event.

(including when the counter is cleared through UG bit of TIMx_EGR register or through trigger controller when configured in “reset mode”).

42.4.8 TIMx auto-reload register (TIMx_ARR)(x = 6 to 7)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **ARR[15:0]**: Prescaler value

ARR is the value to be loaded into the actual auto-reload register.

Refer to [Section 42.3.1: Time-base unit on page 1866](#) for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

42.4.9 TIMx register map

TIMx registers are mapped as 16-bit addressable registers as described in the table below:

Table 347. TIMx register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x00	TIMx_CR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	UIFREMA	Res.	Res.	Res.	Res.	ARPE	Res.	Res.	Res.	OPM	URS	UDIS	CEN
	Reset value																					0				0				0	0	0	0	
0x04	TIMx_CR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.		MMS [2:0]			Res.	Res.	Res.	Res.	
	Reset value																									0	0	0						
0x08	Reserved																																	
0x0C	TIMx_DIER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	UDE	Res.	Res.	Res.	Res.	Res.	Res.	Res.	UIF	
	Reset value																								0								0	
0x10	TIMx_SR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	UIF	
	Reset value																																0	
0x14	TIMx_EGR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	UG	
	Reset value																																0	
0x18-0x20	Reserved																																	
0x24	TIMx_CNT	UIFCPY or Res.		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[15:0]																
	Reset value	0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x28	TIMx_PSC	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PSC[15:0]																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x2C	TIMx_ARR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ARR[15:0]																
	Reset value																	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.

43 Low-power timer (LPTIM)

43.1 Introduction

The LPTIM is a 16-bit timer that benefits from the ultimate developments in power consumption reduction. Thanks to its diversity of clock sources, the LPTIM is able to keep running in all power modes except for Standby mode. Given its capability to run even with no internal clock source, the LPTIM can be used as a “Pulse Counter” which can be useful in some applications. Also, the LPTIM capability to wake up the system from low-power modes, makes it suitable to realize “Timeout functions” with extremely low power consumption.

The LPTIM introduces a flexible clock scheme that provides the needed functionalities and performance, while minimizing the power consumption.

43.2 LPTIM main features

- 16 bit upcounter
- 3-bit prescaler with 8 possible dividing factors (1,2,4,8,16,32,64,128)
- Selectable clock
 - Internal clock sources: LSE, LSI, HSI or APB clock
 - External clock source over LPTIM input (working with no LP oscillator running, used by Pulse Counter application)
- 16 bit ARR autoreload register
- 16 bit compare register
- Continuous/One-shot mode
- Selectable software/hardware input trigger
- Programmable Digital Glitch filter
- Configurable output: Pulse, PWM
- Configurable I/O polarity
- Encoder mode

43.3 LPTIM implementation

Table 348 describes LPTIM implementation on STM32H742, STM32H743/753 and STM32H750 devices.

Table 348. STM32H742, STM32H743/753 and STM32H750 LPTIM features

LPTIM modes/features ⁽¹⁾	LPTIM1	LPTIM2	LPTIM3	LPTIM4	LPTIM5
Encoder mode	X	X	-	-	-

1. X = supported.

43.4 LPTIM functional description

43.4.1 LPTIM block diagram

Figure 522. Low-power timer block diagram (LPTIM1 and LPTIM2)

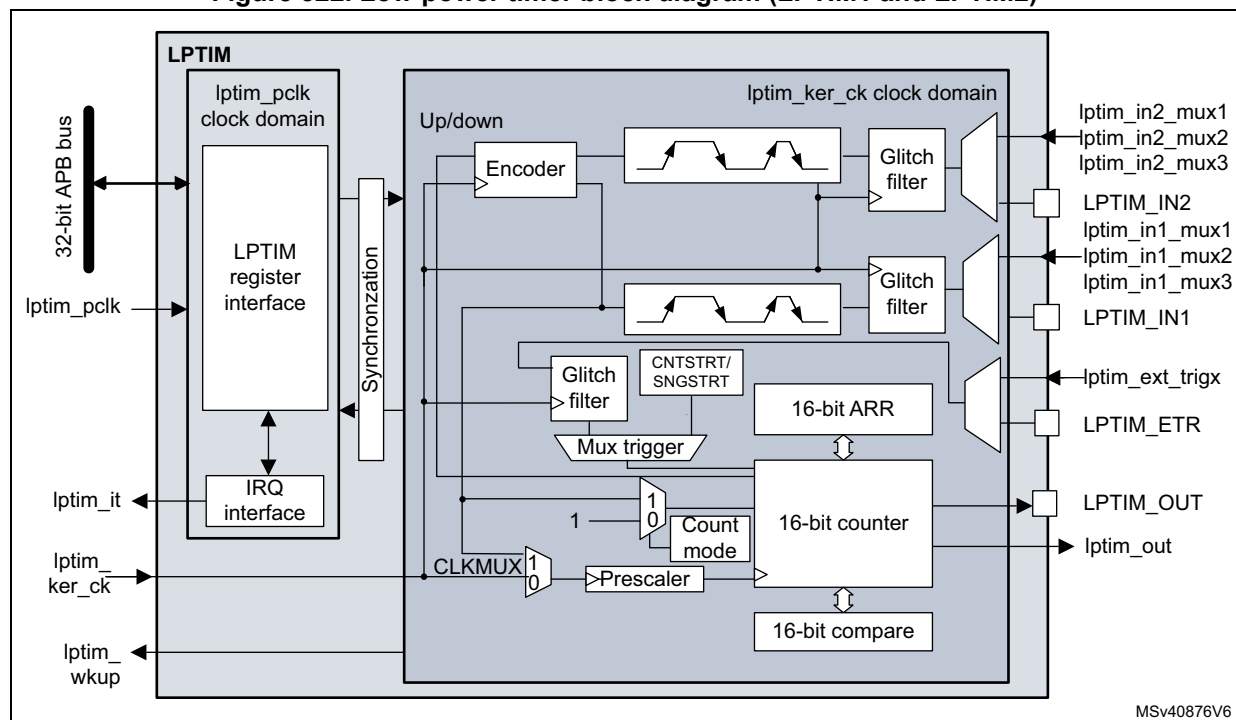


Figure 523. Low-power timer block diagram (LPTIM3)

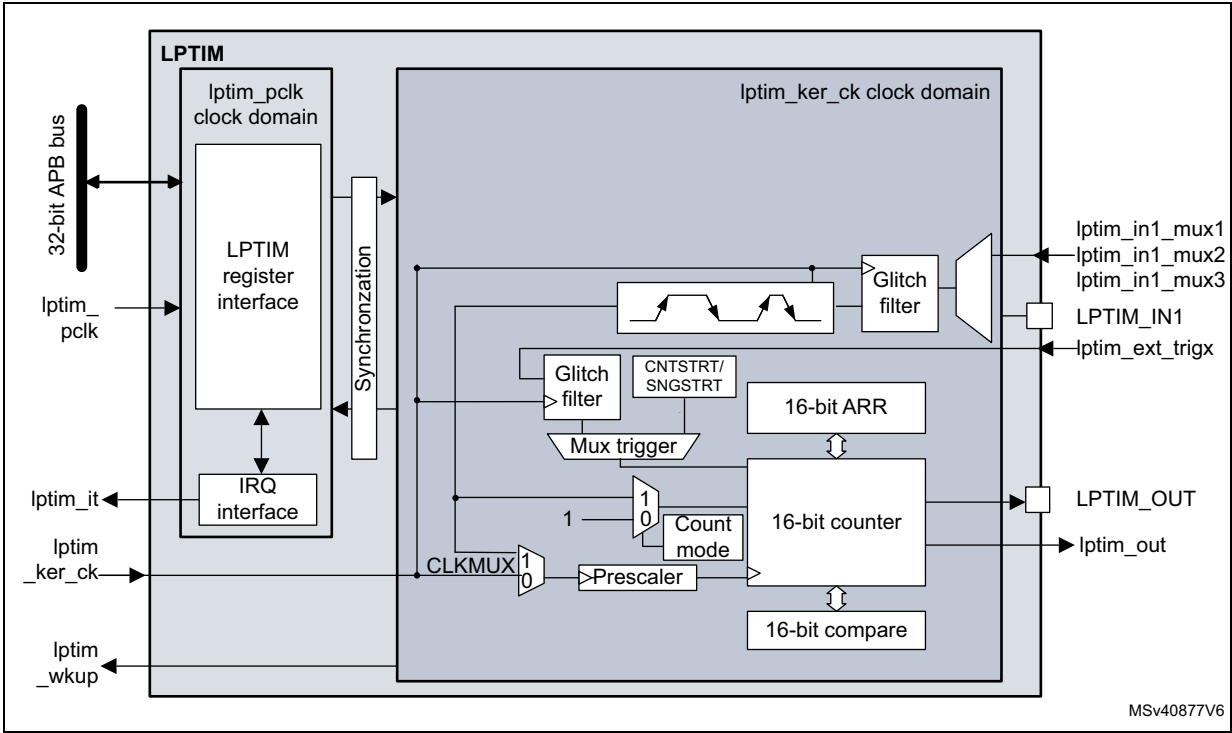
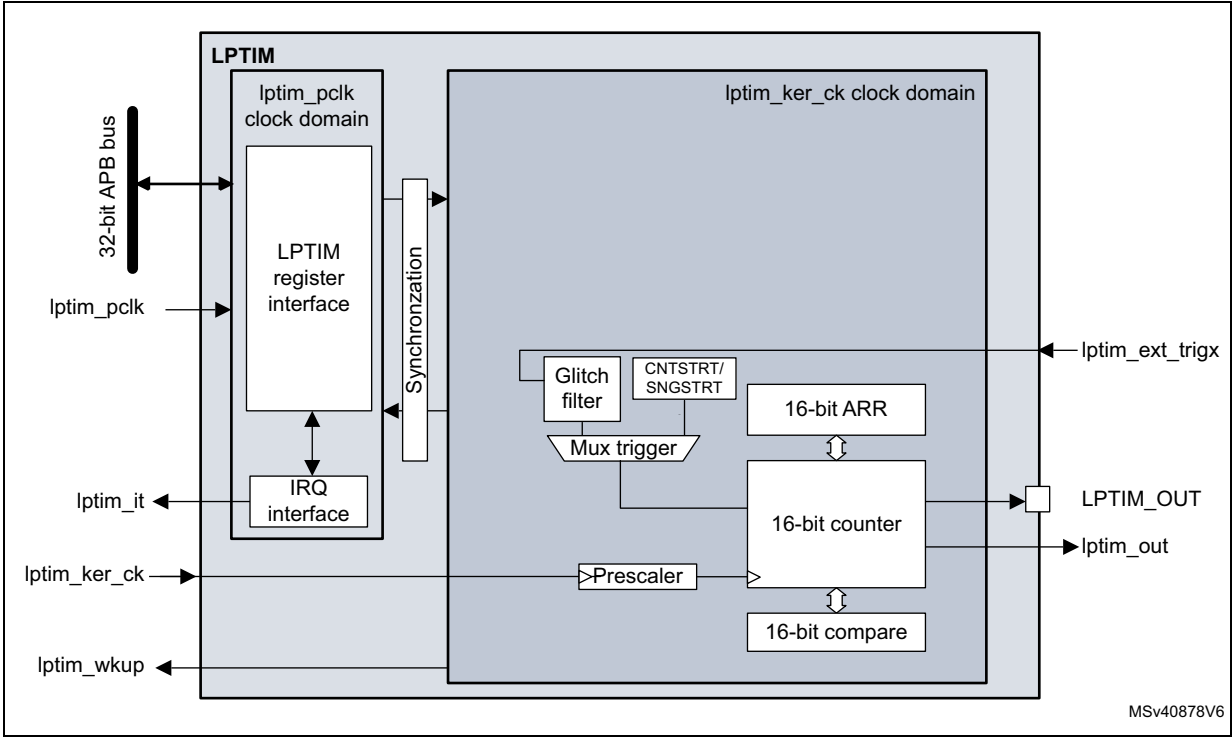


Figure 524. Low-power timer block diagram (LPTIM4 and LPTIM5)



43.4.2 LPTIM pins and internal signals

The following tables provide the list of LPTIM pins and internal signals, respectively.

Table 349. LPTIM input/output pins

Names	Signal type	Description
LPTIM_IN1	Digital input	LPTIM Input 1 from GPIO pin on mux input 0
LPTIM_IN2	Digital input	LPTIM Input 2 from GPIO pin on mux input 0
LPTIM_ETR	Digital input	LPTIM external trigger GPIO pin
LPTIM_OUT	Digital output	LPTIM Output GPIO pin

Table 350. LPTIM internal signals

Names	Signal type	Description
lptim_pclk	Digital input	LPTIM APB clock domain
lptim_ker_ck	Digital input	LPTIM kernel clock
lptim_in1_mux1	Digital input	Internal LPTIM input 1 connected to mux input 1
lptim_in1_mux2	Digital input	Internal LPTIM input 1 connected to mux input 2
lptim_in1_mux3	Digital input	Internal LPTIM input 1 connected to mux input 3
lptim_in2_mux1	Digital input	Internal LPTIM input 2 connected to mux input 1
lptim_in2_mux2	Digital input	Internal LPTIM input 2 connected to mux input 2
lptim_in2_mux3	Digital input	Internal LPTIM input 2 connected to mux input 3
lptim_ext_trigx	Digital input	LPTIM external trigger input x
lptim_out	Digital output	LPTIM counter output
lptim_it	Digital output	LPTIM global interrupt
lptim_wakeup	Digital output	LPTIM wakeup event

43.4.3 LPTIM input and trigger mapping

The LPTIM external trigger and input connections are detailed hereafter:

Table 351. LPTIM1 external trigger connection

TRIGSEL	External trigger
lptim_ext_trig0	GPIO pin as LPTIM1_ETR alternate function
lptim_ext_trig1	RTC_ALARMA
lptim_ext_trig2	RTC_ALARMB
lptim_ext_trig3	RTC_TAMP1_OUT
lptim_ext_trig4	RTC_TAMP2_OUT
lptim_ext_trig5	RTC_TAMP3_OUT

Table 351. LPTIM1 external trigger connection (continued)

TRIGSEL	External trigger
lptim_ext_trig6	COMP1_OUT
lptim_ext_trig7	COMP2_OUT

Table 352. LPTIM2 external trigger connection

TRIGSEL	External trigger
lptim_ext_trig0	GPIO pin as LPTIM2_ETR alternate function
lptim_ext_trig1	RTC_ALARM_A
lptim_ext_trig2	RTC_ALARM_B
lptim_ext_trig3	RTC_TAMP1_OUT
lptim_ext_trig4	RTC_TAMP2_OUT
lptim_ext_trig5	RTC_TAMP3_OUT
lptim_ext_trig6	COMP1_OUT
lptim_ext_trig7	COMP2_OUT

Table 353. LPTIM3 external trigger connection

TRIGSEL	External trigger
lptim_ext_trig0	LPTIM2_OUT
lptim_ext_trig1	Not connected
lptim_ext_trig2	LPTIM4_OUT
lptim_ext_trig3	LPTIM5_OUT
lptim_ext_trig4	SAI1_FS_A
lptim_ext_trig5	SAI1_FS_B
lptim_ext_trig6	Not connected
lptim_ext_trig7	Not connected

Table 354. LPTIM4 external trigger connection

TRIGSEL	External trigger
lptim_ext_trig0	LPTIM2_OUT
lptim_ext_trig1	LPTIM3_OUT
lptim_ext_trig2	Not connected
lptim_ext_trig3	LPTIM5_OUT
lptim_ext_trig4	SAI2_FS_A
lptim_ext_trig5	SAI2_FS_B

Table 354. LPTIM4 external trigger connection (continued)

TRIGSEL	External trigger
lptim_ext_trig6	Not connected
lptim_ext_trig7	Not connected

Table 355. LPTIM5 external trigger connection

TRIGSEL	External trigger
lptim_ext_trig0	LPTIM2_OUT
lptim_ext_trig1	LPTIM3_OUT
lptim_ext_trig2	LPTIM4_OUT
lptim_ext_trig3	SAI4_FS_A
lptim_ext_trig4	SAI4_FS_B
lptim_ext_trig5	Not connected
lptim_ext_trig6	Not connected
lptim_ext_trig7	Not connected

Table 356. LPTIM1 input 1 connection

lptim_in1_mux	LPTIM1 input 1 connected to
lptim_in1_mux0	GPIO pin as LPTIM1_IN1 alternate function
lptim_in1_mux1	COMP1_OUT
lptim_in1_mux2	Not connected
lptim_in1_mux3	Not connected

Table 357. LPTIM1 input 2 connection

lptim_in2_mux	LPTIM1 input 2 connected to
lptim_in2_mux0	GPIO pin as LPTIM1_IN2 alternate function
lptim_in2_mux1	COMP2_OUT
lptim_in2_mux2	Not connected
lptim_in2_mux3	Not connected

Table 358. LPTIM2 input 1 connection

lptim_in1_mux	LPTIM2 input 1 connected to
lptim_in1_mux0	GPIO pin as LPTIM2_IN1 alternate function
lptim_in1_mux1	COMP1_OUT
lptim_in1_mux2	COMP2_OUT
lptim_in1_mux3	COMP1_OUT OR COMP2_OUT

Table 359. LPTIM2 input 2 connection

lptim_in2_mux	LPTIM2 input 2 connected to
lptim_in2_mux0	GPIO pin as LPTIM2_IN2 alternate function
lptim_in2_mux1	COMP2_OUT
lptim_in2_mux2	Not connected
lptim_in2_mux3	Not connected

Table 360. LPTIM3 input 1 connection

lptim_in1_mux	LPTIM3 Input 1 connected to
lptim_in1_mux0	Not connected
lptim_in1_mux1	SAI4_FS_A
lptim_in1_mux2	SAI4_FS_B
lptim_in1_mux3	Not connected

43.4.4 LPTIM reset and clocks

The LPTIM can be clocked using several clock sources. It can be clocked using an internal clock signal which can be chosen among APB, LSI, LSE or HSI sources through the Reset and Clock controller (RCC). Also, the LPTIM can be clocked using an external clock signal injected on its external Input1. When clocked with an external clock source, the LPTIM may run in one of these two possible configurations:

- The first configuration is when the LPTIM is clocked by an external signal but in the same time an internal clock signal is provided to the LPTIM either from APB or any other embedded oscillator including LSE, LSI and HSI.
- The second configuration is when the LPTIM is solely clocked by an external clock source through its external Input1. This configuration is the one used to realize Timeout function or Pulse counter function when all the embedded oscillators are turned off after entering a low-power mode.

Programming the CKSEL and COUNTMODE bits allows controlling whether the LPTIM will use an external clock source or an internal one.

When configured to use an external clock source, the CKPOL bits are used to select the external clock signal active edge. If both edges are configured to be active ones, an internal clock signal should also be provided (first configuration). In this case, the internal clock signal frequency should be at least four times higher than the external clock signal frequency.

43.4.5 Glitch filter

The LPTIM inputs, either external (mapped to GPIOs) or internal (mapped on the chip-level to other embedded peripherals), are protected with digital filters that prevent any glitches and noise perturbations to propagate inside the LPTIM. This is in order to prevent spurious counts or triggers.

Before activating the digital filters, an internal clock source should first be provided to the LPTIM. This is necessary to guarantee the proper operation of the filters.

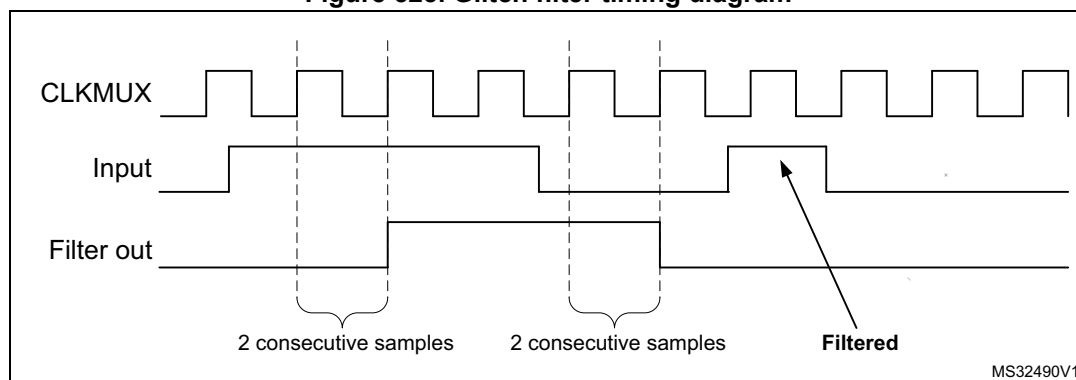
The digital filters are divided into two groups:

- The first group of digital filters protects the LPTIM external inputs. The digital filters sensitivity is controlled by the CKFLT bits
- The second group of digital filters protects the LPTIM internal trigger inputs. The digital filters sensitivity is controlled by the TRGFLT bits.

Note: *The digital filters sensitivity is controlled by groups. It is not possible to configure each digital filter sensitivity separately inside the same group.*

The filter sensitivity acts on the number of consecutive equal samples that should be detected on one of the LPTIM inputs to consider a signal level change as a valid transition. [Figure 525](#) shows an example of glitch filter behavior in case of a 2 consecutive samples programmed.

Figure 525. Glitch filter timing diagram



Note: *In case no internal clock signal is provided, the digital filter must be deactivated by setting the CKFLT and TRGFLT bits to '0'. In that case, an external analog filter may be used to protect the LPTIM external inputs against glitches.*

43.4.6 Prescaler

The LPTIM 16-bit counter is preceded by a configurable power-of-2 prescaler. The prescaler division ratio is controlled by the PRESC[2:0] 3-bit field. The table below lists all the possible division ratios:

Table 361. Prescaler division ratios

programming	dividing factor
000	/1
001	/2
010	/4
011	/8
100	/16
101	/32
110	/64
111	/128

43.4.7 Trigger multiplexer

The LPTIM counter may be started either by software or after the detection of an active edge on one of the 8 trigger inputs.

TRIGEN[1:0] is used to determine the LPTIM trigger source:

- When TRIGEN[1:0] equals '00', The LPTIM counter is started as soon as one of the CNTSTRT or the SNGSTRT bits is set by software. The three remaining possible values for the TRIGEN[1:0] are used to configure the active edge used by the trigger inputs. The LPTIM counter starts as soon as an active edge is detected.
- When TRIGEN[1:0] is different than '00', TRIGSEL[2:0] is used to select which of the 8 trigger inputs is used to start the counter.

The external triggers are considered asynchronous signals for the LPTIM. So after a trigger detection, a two-counter-clock period latency is needed before the timer starts running due to the synchronization.

If a new trigger event occurs when the timer is already started it will be ignored (unless timeout function is enabled).

Note: The timer must be enabled before setting the SNGSTRT/CNTSTRT bits. Any write on these bits when the timer is disabled will be discarded by hardware.

43.4.8 Operating mode

The LPTIM features two operating modes:

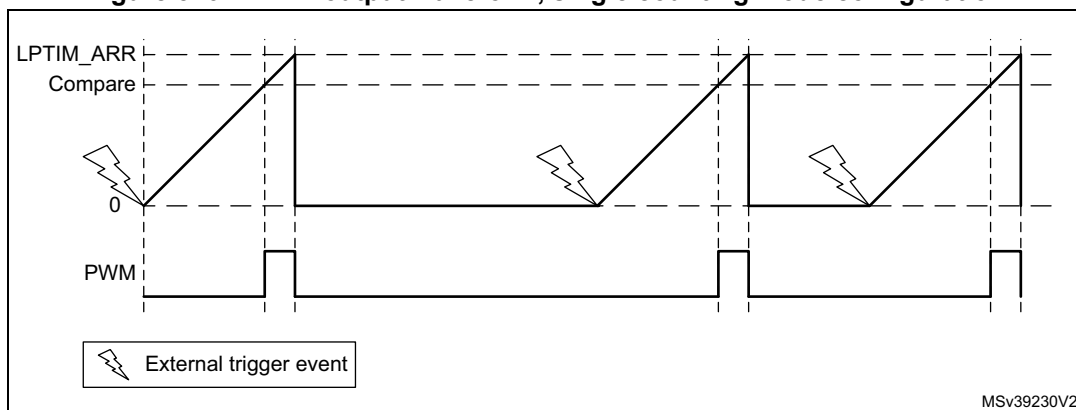
- The Continuous mode: the timer is free running, the timer is started from a trigger event and never stops until the timer is disabled
- One-shot mode: the timer is started from a trigger event and stops when reaching the ARR value.

One-shot mode

To enable the one-shot counting, the SNGSTRT bit must be set.

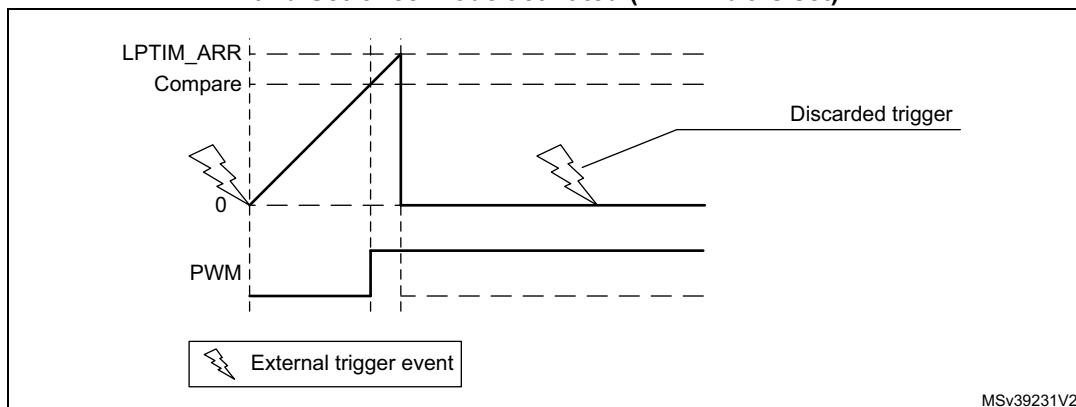
A new trigger event will re-start the timer. Any trigger event occurring after the counter starts and before the counter reaches ARR will be discarded.

In case an external trigger is selected, each external trigger event arriving after the SNGSTRT bit is set, and after the counter register has stopped (contains zero value), will start the counter for a new one-shot counting cycle as shown in [Figure 526](#).

Figure 526. LPTIM output waveform, single counting mode configuration

- Set-once mode activated:

It should be noted that when the WAVE bit-field in the LPTIM_CFGR register is set, the Set-once mode is activated. In this case, the counter is only started once following the first trigger, and any subsequent trigger event is discarded as shown in [Figure 527](#).

Figure 527. LPTIM output waveform, Single counting mode configuration and Set-once mode activated (WAVE bit is set)

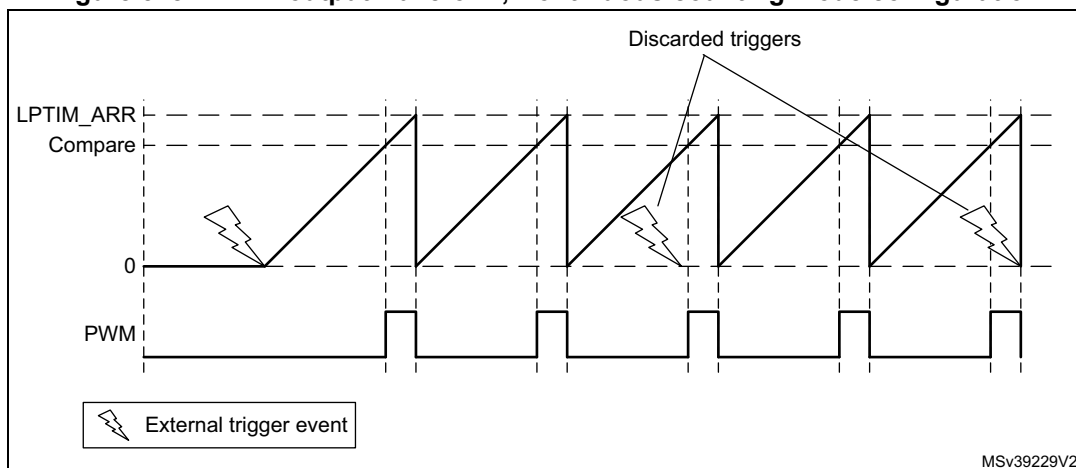
In case of software start (TRIGEN[1:0] = '00'), the SNGSTRT setting will start the counter for one-shot counting.

Continuous mode

To enable the continuous counting, the CNTSTRT bit must be set.

In case an external trigger is selected, an external trigger event arriving after CNTSTRT is set will start the counter for continuous counting. Any subsequent external trigger event will be discarded as shown in [Figure 528](#).

In case of software start (TRIGEN[1:0] = '00'), setting CNTSTRT will start the counter for continuous counting.

Figure 528. LPTIM output waveform, Continuous counting mode configuration

SNGSTRT and CNTSTRT bits can only be set when the timer is enabled (The ENABLE bit is set to '1'). It is possible to change “on the fly” from One-shot mode to Continuous mode.

If the Continuous mode was previously selected, setting SNGSTRT will switch the LPTIM to the One-shot mode. The counter (if active) will stop as soon as it reaches ARR.

If the One-shot mode was previously selected, setting CNTSTRT will switch the LPTIM to the Continuous mode. The counter (if active) will restart as soon as it reaches ARR.

43.4.9 Timeout function

The detection of an active edge on one selected trigger input can be used to reset the LPTIM counter. This feature is controlled through the TIMOUT bit.

The first trigger event will start the timer, any successive trigger event will reset the counter and the timer will restart.

A low-power timeout function can be realized. The timeout value corresponds to the compare value; if no trigger occurs within the expected time frame, the MCU is waked-up by the compare match event.

43.4.10 Waveform generation

Two 16-bit registers, the LPTIM_ARR (autoreload register) and LPTIM_CMP (compare register), are used to generate several different waveforms on LPTIM output

The timer can generate the following waveforms:

- The PWM mode: the LPTIM output is set as soon as the counter value in LPTIM_CNT exceeds the compare value in LPTIM_CMP. The LPTIM output is reset as soon as a match occurs between the LPTIM_ARR and the LPTIM_CNT registers.
- The One-pulse mode: the output waveform is similar to the one of the PWM mode for the first pulse, then the output is permanently reset
- The Set-once mode: the output waveform is similar to the One-pulse mode except that the output is kept to the last signal level (depends on the output configured polarity).

The above described modes require that the LPTIM_ARR register value be strictly greater than the LPTIM_CMP register value.

The LPTIM output waveform can be configured through the WAVE bit as follow:

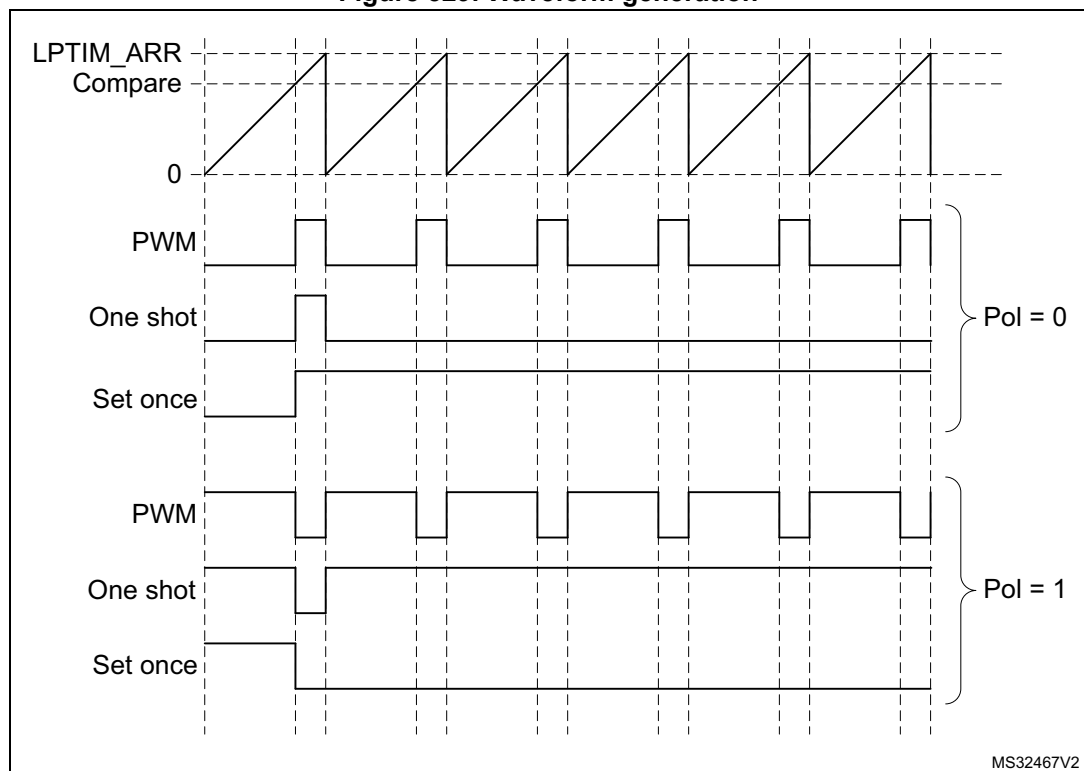
- Resetting the WAVE bit to '0' forces the LPTIM to generate either a PWM waveform or a One pulse waveform depending on which bit is set: CNTSTRT or SNGSTRT.
- Setting the WAVE bit to '1' forces the LPTIM to generate a Set-once mode waveform.

The WAVPOL bit controls the LPTIM output polarity. The change takes effect immediately, so the output default value will change immediately after the polarity is re-configured, even before the timer is enabled.

Signals with frequencies up to the LPTIM clock frequency divided by 2 can be generated.

[Figure 529](#) below shows the three possible waveforms that can be generated on the LPTIM output. Also, it shows the effect of the polarity change using the WAVPOL bit.

Figure 529. Waveform generation



43.4.11 Register update

The LPTIM_ARR register and LPTIM_CMP register are updated immediately after the APB bus write operation, or at the end of the current period if the timer is already started.

The PRELOAD bit controls how the LPTIM_ARR and the LPTIM_CMP registers are updated:

- When the PRELOAD bit is reset to '0', the LPTIM_ARR and the LPTIM_CMP registers are immediately updated after any write access.
- When the PRELOAD bit is set to '1', the LPTIM_ARR and the LPTIM_CMP registers are updated at the end of the current period, if the timer has been already started.

The LPTIM APB interface and the LPTIM kernel logic use different clocks, so there is some latency between the APB write and the moment when these values are available to the

counter comparator. Within this latency period, any additional write into these registers must be avoided.

The ARROK flag and the CMPOK flag in the LPTIM_ISR register indicate when the write operation is completed to respectively the LPTIM_ARR register and the LPTIM_CMP register.

After a write to the LPTIM_ARR register or the LPTIM_CMP register, a new write operation to the same register can only be performed when the previous write operation is completed. Any successive write before respectively the ARROK flag or the CMPOK flag be set, will lead to unpredictable results.

43.4.12 Counter mode

The LPTIM counter can be used to count external events on the LPTIM Input1 or it can be used to count internal clock cycles. The CKSEL and COUNTMODE bits control which source will be used for updating the counter.

In case the LPTIM is configured to count external events on Input1, the counter can be updated following a rising edge, falling edge or both edges depending on the value written to the CKPOL[1:0] bits.

The count modes below can be selected, depending on CKSEL and COUNTMODE values:

- CKSEL = 0: the LPTIM is clocked by an internal clock source
 - COUNTMODE = 0
The LPTIM is configured to be clocked by an internal clock source and the LPTIM counter is configured to be updated following each internal clock pulse.
 - COUNTMODE = 1
The LPTIM external Input1 is sampled with the internal clock provided to the LPTIM.
Consequently, in order not to miss any event, the frequency of the changes on the external Input1 signal should never exceed the frequency of the internal clock provided to the LPTIM. Also, the internal clock provided to the LPTIM must not be prescaled (PRESC[2:0] = 000).
- CKSEL = 1: the LPTIM is clocked by an external clock source
COUNTMODE value is don't care.
In this configuration, the LPTIM has no need for an internal clock source (except if the glitch filters are enabled). The signal injected on the LPTIM external Input1 is used as system clock for the LPTIM. This configuration is suitable for operation modes where no embedded oscillator is enabled.
For this configuration, the LPTIM counter can be updated either on rising edges or falling edges of the input1 clock signal but not on both rising and falling edges.
Since the signal injected on the LPTIM external Input1 is also used to clock the LPTIM kernel logic, there is some initial latency (after the LPTIM is enabled) before the counter is incremented. More precisely, the first five active edges on the LPTIM external Input1 (after LPTIM is enable) are lost.

43.4.13 Timer enable

The ENABLE bit located in the LPTIM_CR register is used to enable/disable the LPTIM kernel logic. After setting the ENABLE bit, a delay of two counter clock is needed before the LPTIM is actually enabled.

The LPTIM_CFGR and LPTIM_IER registers must be modified only when the LPTIM is disabled.

43.4.14 Timer counter reset

In order to reset the content of LPTIM_CNT register to zero, two reset mechanisms are implemented:

- The synchronous reset mechanism: the synchronous reset is controlled by the COUNTRST bit in the LPTIM_CR register. After setting the COUNTRST bit-field to '1', the reset signal is propagated in the LPTIM kernel clock domain. So it is important to note that a few clock pulses of the LPTIM kernel logic will elapse before the reset is taken into account. This will make the LPTIM counter count few extra pluses between the time when the reset is trigger and it become effective. Since the COUNTRST bit is located in the APB clock domain and the LPTIM counter is located in the LPTIM kernel clock domain, a delay of 3 clock cycles of the kernel clock is needed to synchronize the reset signal issued by the APB clock domain when writing '1' to the COUNTRST bit.
- The asynchronous reset mechanism: the asynchronous reset is controlled by the RSTARE bit located in the LPTIM_CR register. When this bit is set to '1', any read access to the LPTIM_CNT register will reset its content to zero. Asynchronous reset should be triggered within a timeframe in which no LPTIM core clock is provided. For example when LPTIM Input1 is used as external clock source, the asynchronous reset should be applied only when there is enough insurance that no toggle will occur on the LPTIM Input1.

It should be noted that to read reliably the content of the LPTIM_CNT register two successive read accesses must be performed and compared. A read access can be considered reliable when the value of the two read accesses is equal. Unfortunately when asynchronous reset is enabled there is no possibility to read twice the LPTIM_CNT register.

Warning: There is no mechanism inside the LPTIM that prevents the two reset mechanisms from being used simultaneously. So developer should make sure that these two mechanisms are used exclusively.

43.4.15 Encoder mode

This mode allows handling signals from quadrature encoders used to detect angular position of rotary elements. Encoder interface mode acts simply as an external clock with direction selection. This means that the counter just counts continuously between 0 and the auto-reload value programmed into the LPTIM_ARR register (0 up to ARR or ARR down to 0 depending on the direction). Therefore LPTIM_ARR must be configured before starting. From the two external input signals, Input1 and Input2, a clock signal is generated to clock the LPTIM counter. The phase between those two signals determines the counting direction.

The Encoder mode is only available when the LPTIM is clocked by an internal clock source. The signals frequency on both Input1 and Input2 inputs must not exceed the LPTIM internal clock frequency divided by 4. This is mandatory in order to guarantee a proper operation of the LPTIM.

Direction change is signaled by the two Down and Up flags in the LPTIM_ISR register. Also, an interrupt can be generated for both direction change events if enabled through the DOWNIE bit.

To activate the Encoder mode the ENC bit has to be set to '1'. The LPTIM must first be configured in Continuous mode.

When Encoder mode is active, the LPTIM counter is modified automatically following the speed and the direction of the incremental encoder. Therefore, its content always represents the encoder's position. The count direction, signaled by the Up and Down flags, correspond to the rotation direction of the encoder rotor.

According to the edge sensitivity configured using the CKPOL[1:0] bits, different counting scenarios are possible. The following table summarizes the possible combinations, assuming that Input1 and Input2 do not switch at the same time.

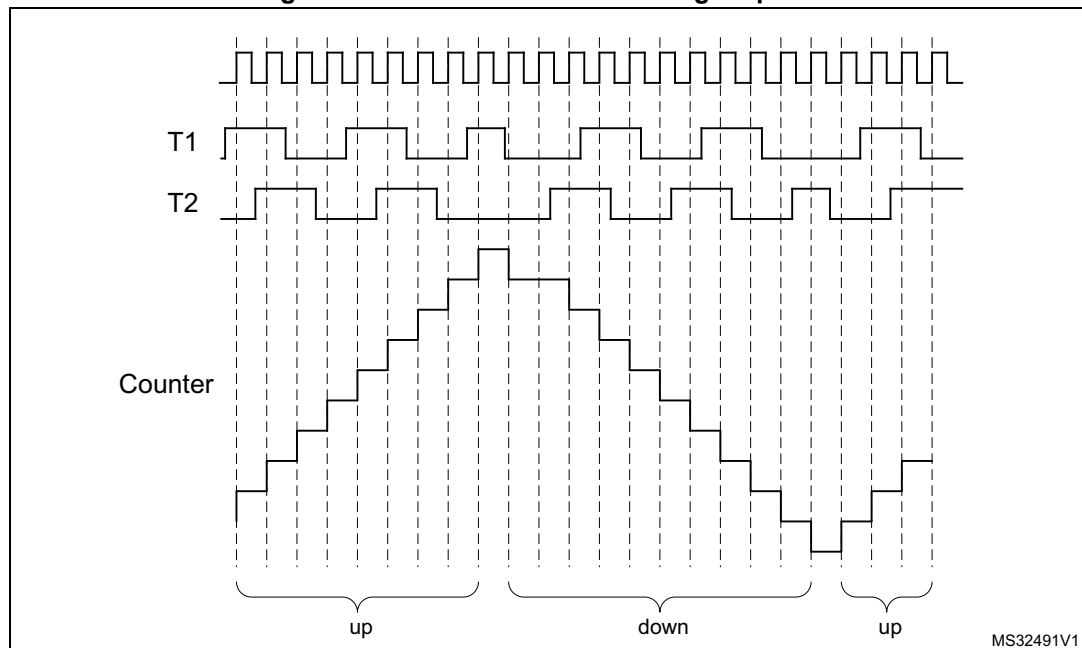
Table 362. Encoder counting scenarios

Active edge	Level on opposite signal (Input1 for Input2, Input2 for Input1)	Input1 signal		Input2 signal	
		Rising	Falling	Rising	Falling
Rising Edge	High	Down	No count	Up	No count
	Low	Up	No count	Down	No count
Falling Edge	High	No count	Up	No count	Down
	Low	No count	Down	No count	Up
Both Edges	High	Down	Up	Up	Down
	Low	Up	Down	Down	Up

The following figure shows a counting sequence for Encoder mode where both-edge sensitivity is configured.

Caution: In this mode the LPTIM must be clocked by an internal clock source, so the CKSEL bit must be maintained to its reset value which is equal to '0'. Also, the prescaler division ratio must be equal to its reset value which is 1 (PRESC[2:0] bits must be '000').

Figure 530. Encoder mode counting sequence



43.4.16 Debug mode

When the microcontroller enters debug mode (core halted), the LPTIM counter either continues to work normally or stops, depending on the DBG_LPTIM_STOP configuration bit in the DBG module.

43.5 LPTIM low-power modes

Table 363. Effect of low-power modes on the LPTIM

Mode	Description
Sleep	No effect. LPTIM interrupts cause the device to exit Sleep mode.
Stop	The LPTIM peripheral is active when it is clocked by LSE or LSI. LPTIM interrupts cause the device to exit Stop mode
Standby	The LPTIM peripheral is powered down and must be reinitialized after exiting Standby mode.

43.6 LPTIM interrupts

The following events generate an interrupt/wake-up event, if they are enabled through the LPTIM_IER register:

- Compare match
- Auto-reload match (whatever the direction if encoder mode)
- External trigger event
- Autoreload register write completed
- Compare register write completed
- Direction change (encoder mode), programmable (up / down / both).

Note: *If any bit in the LPTIM_IER register (Interrupt Enable Register) is set after that its corresponding flag in the LPTIM_ISR register (Status Register) is set, the interrupt is not asserted.*

Table 364. Interrupt events

Interrupt event	Description
Compare match	Interrupt flag is raised when the content of the Counter register (LPTIM_CNT) matches the content of the compare register (LPTIM_CMP).
Auto-reload match	Interrupt flag is raised when the content of the Counter register (LPTIM_CNT) matches the content of the Auto-reload register (LPTIM_ARR).
External trigger event	Interrupt flag is raised when an external trigger event is detected
Auto-reload register update OK	Interrupt flag is raised when the write operation to the LPTIM_ARR register is complete.
Compare register update OK	Interrupt flag is raised when the write operation to the LPTIM_CMP register is complete.
Direction change	Used in Encoder mode. Two interrupt flags are embedded to signal direction change: – UP flag signals up-counting direction change – DOWN flag signals down-counting direction change.

43.7 LPTIM registers

The peripheral registers can only be accessed by words (32-bit).

43.7.1 LPTIM interrupt and status register (LPTIM_ISR)

Address offset: 0x000

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DOWN	UP	ARR OK	CMP OK	EXT TRIG	ARRM	CMPM
									r	r	r	r	r	r	r

Bits 31:7 Reserved, must be kept at reset value.

Bit 6 **DOWN**: Counter direction change up to down

In Encoder mode, DOWN bit is set by hardware to inform application that the counter direction has changed from up to down. DOWN flag can be cleared by writing 1 to the DOWNCF bit in the LPTIM_ICR register.

Note: If the LPTIM does not support encoder mode feature, this bit is reserved. Please refer to [Section 43.3: LPTIM implementation](#).

Bit 5 **UP**: Counter direction change down to up

In Encoder mode, UP bit is set by hardware to inform application that the counter direction has changed from down to up. UP flag can be cleared by writing 1 to the UPCF bit in the LPTIM_ICR register.

Note: If the LPTIM does not support encoder mode feature, this bit is reserved. Please refer to [Section 43.3: LPTIM implementation](#).

Bit 4 **ARROK**: Autoreload register update OK

ARROK is set by hardware to inform application that the APB bus write operation to the LPTIM_ARR register has been successfully completed. ARROK flag can be cleared by writing 1 to the ARROKCF bit in the LPTIM_ICR register.

Bit 3 **CMPOK**: Compare register update OK

CMPOK is set by hardware to inform application that the APB bus write operation to the LPTIM_CMP register has been successfully completed.

Bit 2 **EXTTRIG**: External trigger edge event

EXTTRIG is set by hardware to inform application that a valid edge on the selected external trigger input has occurred. If the trigger is ignored because the timer has already started, then this flag is not set. EXTTRIG flag can be cleared by writing 1 to the EXTTRIGCF bit in the LPTIM_ICR register.

Bit 1 **ARRM**: Autoreload match

ARRM is set by hardware to inform application that LPTIM_CNT register's value reached the LPTIM_ARR register's value. ARRM flag can be cleared by writing 1 to the ARRMCF bit in the LPTIM_ICR register.

Bit 0 **CMPM**: Compare match

The CMPM bit is set by hardware to inform application that LPTIM_CNT register value reached the LPTIM_CMP register's value.

43.7.2 LPTIM interrupt clear register (LPTIM_ICR)

Address offset: 0x004

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DOWN CF	UPCF	ARRO KCF	CMPO KCF	EXTTR IGCF	ARRM CF	CMPM CF
									w	w	w	w	w	w	w

Bits 31:9 Reserved, must be kept at reset value.

Bits 8:7 Reserved, must be kept at reset value.

Bit 6 **DOWNCF**: Direction change to down clear flag

Writing 1 to this bit clear the DOWN flag in the LPTIM_ISR register.

Note: If the LPTIM does not support encoder mode feature, this bit is reserved. Please refer to [Section 43.3: LPTIM implementation](#).

Bit 5 **UPCF**: Direction change to UP clear flag

Writing 1 to this bit clear the UP flag in the LPTIM_ISR register.

Note: If the LPTIM does not support encoder mode feature, this bit is reserved. Please refer to [Section 43.3: LPTIM implementation](#).

Bit 4 **ARROKCF**: Autoreload register update OK clear flag

Writing 1 to this bit clears the ARROK flag in the LPTIM_ISR register

Bit 3 **CMPOKCF**: Compare register update OK clear flag

Writing 1 to this bit clears the CMPOK flag in the LPTIM_ISR register

Bit 2 **EXTTRIGCF**: External trigger valid edge clear flag

Writing 1 to this bit clears the EXTTRIG flag in the LPTIM_ISR register

Bit 1 **ARRMCF**: Autoreload match clear flag

Writing 1 to this bit clears the ARRM flag in the LPTIM_ISR register

Bit 0 **CMPMCF**: Compare match clear flag

Writing 1 to this bit clears the CMP flag in the LPTIM_ISR register

43.7.3 LPTIM interrupt enable register (LPTIM_IER)

Address offset: 0x008

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DOWN IE	UPIE	ARRO KIE	CMPO KIE	EXT TRIGIE	ARRM IE	CMPM IE
									r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:9 Reserved, must be kept at reset value.

Bits 8:7 Reserved, must be kept at reset value.

Bit 6 **DOWNIE**: Direction change to down Interrupt Enable

0: DOWN interrupt disabled

1: DOWN interrupt enabled

Note: If the LPTIM does not support encoder mode feature, this bit is reserved. Please refer to [Section 43.3: LPTIM implementation](#).

Bit 5 **UPIE**: Direction change to UP Interrupt Enable

0: UP interrupt disabled

1: UP interrupt enabled

Note: If the LPTIM does not support encoder mode feature, this bit is reserved. Please refer to [Section 43.3: LPTIM implementation](#).

Bit 4 **ARROKIE**: Autoreload register update OK Interrupt Enable

0: ARROK interrupt disabled

1: ARROK interrupt enabled

Bit 3 **CMPOKIE**: Compare register update OK Interrupt Enable

0: CMPOK interrupt disabled

1: CMPOK interrupt enabled

Bit 2 **EXTTRIGIE**: External trigger valid edge Interrupt Enable

0: EXTTRIG interrupt disabled

1: EXTTRIG interrupt enabled

Bit 1 **ARRMIE**: Autoreload match Interrupt Enable

0: ARRM interrupt disabled

1: ARRM interrupt enabled

Bit 0 **CMPMIE**: Compare match Interrupt Enable

0: CMPM interrupt disabled

1: CMPM interrupt enabled

Caution: The LPTIM_IER register must only be modified when the LPTIM is disabled (ENABLE bit reset to '0')

43.7.4 LPTIM configuration register (LPTIM_CFGR)

Address offset: 0x00C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	ENC	COUNT MODE	PRELOAD	WAVPOL	WAVE	TIMOUT	TRIGEN[1:0]		Res.
							rw	rw	rw	rw	rw	rw	rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRIGSEL[2:0]			Res.	PRESC[2:0]			Res.	TRGFLT[1:0]		Res.	CKFLT[1:0]		CKPOL[1:0]		CKSEL
rw	rw	rw		rw	rw	rw		rw	rw		rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 Reserved, must be kept at reset value.

Bits 28:25 Reserved, must be kept at reset value.

Bit 24 **ENC**: Encoder mode enable

The ENC bit controls the Encoder mode

- 0: Encoder mode disabled
- 1: Encoder mode enabled

Note: If the LPTIM does not support encoder mode feature, this bit is reserved. Please refer to [Section 43.3: LPTIM implementation](#).

Bit 23 **COUNTMODE**: counter mode enabled

The COUNTMODE bit selects which clock source is used by the LPTIM to clock the counter:

- 0: the counter is incremented following each internal clock pulse
- 1: the counter is incremented following each valid clock pulse on the LPTIM external Input1

Bit 22 **PRELOAD**: Registers update mode

The PRELOAD bit controls the LPTIM_ARR and the LPTIM_CMP registers update modality

- 0: Registers are updated after each APB bus write access
- 1: Registers are updated at the end of the current LPTIM period

Bit 21 **WAVPOL**: Waveform shape polarity

The WAVEPOL bit controls the output polarity

- 0: The LPTIM output reflects the compare results between LPTIM_ARR and LPTIM_CMP registers
- 1: The LPTIM output reflects the inverse of the compare results between LPTIM_ARR and LPTIM_CMP registers

Bit 20 **WAVE**: Waveform shape

The WAVE bit controls the output shape

- 0: Deactivate Set-once mode, PWM or One Pulse waveform depending on how the timer was started, CNTSTRT for PWM or SNGSTRT for One Pulse waveform.
- 1: Activate the Set-once mode

Bit 19 **TIMOUT**: Timeout enable

The TIMOUT bit controls the Timeout feature

- 0: A trigger event arriving when the timer is already started will be ignored
- 1: A trigger event arriving when the timer is already started will reset and restart the counter

Bits 18:17 **TRIGEN[1:0]**: Trigger enable and polarity

The TRIGEN bits controls whether the LPTIM counter is started by an external trigger or not. If the external trigger option is selected, three configurations are possible for the trigger active edge:

- 00: software trigger (counting start is initiated by software)
- 01: rising edge is the active edge
- 10: falling edge is the active edge
- 11: both edges are active edges

Bit 16 Reserved, must be kept at reset value.

Bits 15:13 **TRIGSEL[2:0]**: Trigger selector

The TRIGSEL bits select the trigger source that will serve as a trigger event for the LPTIM among the below 8 available sources:

000: lptim_ext_trig0

001: lptim_ext_trig1

010: lptim_ext_trig2

011: lptim_ext_trig3

100: lptim_ext_trig4

101: lptim_ext_trig5

110: lptim_ext_trig6

111: lptim_ext_trig7

See [Section 43.4.3: LPTIM input and trigger mapping](#) for details.

Bit 12 Reserved, must be kept at reset value.

Bits 11:9 **PRESC[2:0]**: Clock prescaler

The PRESC bits configure the prescaler division factor. It can be one among the following division factors:

000: /1

001: /2

010: /4

011: /8

100: /16

101: /32

110: /64

111: /128

Bit 8 Reserved, must be kept at reset value.

Bits 7:6 **TRGFLT[1:0]**: Configurable digital filter for trigger

The TRGFLT value sets the number of consecutive equal samples that should be detected when a level change occurs on an internal trigger before it is considered as a valid level transition. An internal clock source must be present to use this feature

00: any trigger active level change is considered as a valid trigger

01: trigger active level change must be stable for at least 2 clock periods before it is considered as valid trigger.

10: trigger active level change must be stable for at least 4 clock periods before it is considered as valid trigger.

11: trigger active level change must be stable for at least 8 clock periods before it is considered as valid trigger.

Bit 5 Reserved, must be kept at reset value.

Bits 4:3 **CKFLT[1:0]**: Configurable digital filter for external clock

The CKFLT value sets the number of consecutive equal samples that should be detected when a level change occurs on an external clock signal before it is considered as a valid level transition. An internal clock source must be present to use this feature

00: any external clock signal level change is considered as a valid transition

01: external clock signal level change must be stable for at least 2 clock periods before it is considered as valid transition.

10: external clock signal level change must be stable for at least 4 clock periods before it is considered as valid transition.

11: external clock signal level change must be stable for at least 8 clock periods before it is considered as valid transition.

Bits 2:1 **CKPOL[1:0]**: Clock Polarity

If LPTIM is clocked by an external clock source:

When the LPTIM is clocked by an external clock source, CKPOL bits is used to configure the active edge or edges used by the counter:

00: the rising edge is the active edge used for counting.

If the LPTIM is configured in Encoder mode (ENC bit is set), the encoder sub-mode 1 is active.

01: the falling edge is the active edge used for counting

If the LPTIM is configured in Encoder mode (ENC bit is set), the encoder sub-mode 2 is active.

10: both edges are active edges. When both external clock signal edges are considered active ones, the LPTIM must also be clocked by an internal clock source with a frequency equal to at least four times the external clock frequency.

If the LPTIM is configured in Encoder mode (ENC bit is set), the encoder sub-mode 3 is active.

11: not allowed

Refer to [Section 43.4.15: Encoder mode](#) for more details about Encoder mode sub-modes.

Bit 0 **CKSEL**: Clock selector

The CKSEL bit selects which clock source the LPTIM will use:

0: LPTIM is clocked by internal clock source (APB clock or any of the embedded oscillators)

1: LPTIM is clocked by an external clock source through the LPTIM external Input1

Caution: The LPTIM_CFGR register must only be modified when the LPTIM is disabled (ENABLE bit reset to '0').

43.7.5 LPTIM control register (LPTIM_CR)

Address offset: 0x010

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RST ARE	COUN TRST	CNT STRT	SNG STRT	ENA BLE
											w	rs	rw	rw	rw

Bits 31:5 Reserved, must be kept at reset value.

Bit 4 **RSTARE**: Reset after read enable

This bit is set and cleared by software. When RSTARE is set to '1', any read access to LPTIM_CNT register will asynchronously reset LPTIM_CNT register content.

Caution: This bitfield is write-only. This means that the bit cannot be read back to verify the value which has been written. As an example, if this bit is set to 1, attempting to read it back will return 0 even if the "Reset after read" function is enabled (due to the fact that this bitfield has previously been written to 1). To turn off the "Reset after read" or to make sure that it has already been turned off, this bit should be reset (by programming it to 0) even if it already contains 0.

Bit 3 **COUNTRST**: Counter reset

This bit is set by software and cleared by hardware. When set to '1' this bit will trigger a synchronous reset of the LPTIM_CNT counter register. Due to the synchronous nature of this reset, it only takes place after a synchronization delay of 3 LPTimer core clock cycles (LPTimer core clock may be different from APB clock).

Caution: COUNTRST must never be set to '1' by software before it is already cleared to '0' by hardware. Software should consequently check that COUNTRST bit is already cleared to '0' before attempting to set it to '1'.

Bit 2 **CNTSTRT**: Timer start in Continuous mode

This bit is set by software and cleared by hardware.

In case of software start (TRIGEN[1:0] = '00'), setting this bit starts the LPTIM in Continuous mode. If the software start is disabled (TRIGEN[1:0] different than '00'), setting this bit starts the timer in Continuous mode as soon as an external trigger is detected.

If this bit is set when a single pulse mode counting is ongoing, then the timer will not stop at the next match between the LPTIM_ARR and LPTIM_CNT registers and the LPTIM counter keeps counting in Continuous mode.

This bit can be set only when the LPTIM is enabled. It will be automatically reset by hardware.

Bit 1 **SNGSTRT**: LPTIM start in Single mode

This bit is set by software and cleared by hardware.

In case of software start (TRIGEN[1:0] = '00'), setting this bit starts the LPTIM in single pulse mode. If the software start is disabled (TRIGEN[1:0] different than '00'), setting this bit starts the LPTIM in single pulse mode as soon as an external trigger is detected.

If this bit is set when the LPTIM is in continuous counting mode, then the LPTIM will stop at the following match between LPTIM_ARR and LPTIM_CNT registers.

This bit can only be set when the LPTIM is enabled. It will be automatically reset by hardware.

Bit 0 **ENABLE**: LPTIM enable

The ENABLE bit is set and cleared by software.

0:LPTIM is disabled

1:LPTIM is enabled

43.7.6 LPTIM compare register (LPTIM_CMP)

Address offset: 0x014

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMP[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **CMP[15:0]**: Compare value

CMP is the compare value used by the LPTIM.

Caution: The LPTIM_CMP register must only be modified when the LPTIM is enabled (ENABLE bit set to '1').

43.7.7 LPTIM autoreload register (LPTIM_ARR)

Address offset: 0x018

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **ARR[15:0]**: Auto reload value

ARR is the autoreload value for the LPTIM.

This value must be strictly greater than the CMP[15:0] value.

Caution: The LPTIM_ARR register must only be modified when the LPTIM is enabled (ENABLE bit set to '1').

43.7.8 LPTIM counter register (LPTIM_CNT)

Address offset: 0x01C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **CNT[15:0]**: Counter value

When the LPTIM is running with an asynchronous clock, reading the LPTIM_CNT register may return unreliable values. So in this case it is necessary to perform two consecutive read accesses and verify that the two returned values are identical.

It should be noted that for a reliable LPTIM_CNT register read access, two consecutive read accesses must be performed and compared. A read access can be considered reliable when the values of the two consecutive read accesses are equal.

43.7.9 LPTIM configuration register 2 (LPTIM_CFGR2)

Address offset: 0x024

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IN2SEL[1:0]		Res.	Res.	IN1SEL[1:0]	
										rw	rw			rw	rw

Bits 31:6 Reserved, must be kept at reset value.

Bits 5:4 **IN2SEL[1:0]**: LPTIM input 2 selection

The IN2SEL bits control the LPTIM Input 2 multiplexer, which connect LPTIM Input 2 to one of the available inputs.

00: lptim_in2_mux0

01: lptim_in2_mux1

10: lptim_in2_mux2

11: lptim_in2_mux3

For connection details refer to [Section 43.4.3: LPTIM input and trigger mapping](#).

Note: If the LPTIM does not support encoder mode feature, these bits are reserved. Please refer to [Section 43.3: LPTIM implementation](#).

Bits 3:2 Reserved, must be kept at reset value.

Bits 1:0 **IN1SEL[1:0]**: LPTIM input 1 selection

The IN1SEL bits control the LPTIM Input 1 multiplexer, which connects LPTIM Input 1 to one of the available inputs.

00: lptim_in1_mux0

01: lptim_in1_mux1

10: lptim_in1_mux2

11: lptim_in1_mux3

For connection details refer to [Section 43.4.3: LPTIM input and trigger mapping](#).

Caution: The LPTIM_CFGR2 register must only be modified when the LPTIM is disabled (ENABLE bit reset to '0').

43.7.10 LPTIM3 configuration register 2 (LPTIM3_CFGR2)

Address offset: 0x024

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IN1SEL[1:0]	
														rw	rw

Bits 31:2 Reserved, must be kept at reset value.

Bits 1:0 **IN1SEL[1:0]**: LPTIM3 Input1 selection

The IN1SEL bits control the LPTIM3 Input 1 multiplexer, which connects LPTIM3 Input 1 to one of the available inputs.

00: lptim_in1_mux0

01: lptim_in1_mux1

10: lptim_in1_mux2

11: lptim_in1_mux3

For connection details refer to [Table 360: LPTIM3 input 1 connection](#).

Caution: The LPTIM3_CFGR2 registers must only be modified when the LPTIM is disabled (ENABLE bit reset to '0').

43.7.11 LPTIM register map

The following table summarizes the LPTIM registers.

Table 365. LPTIM register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0									
0x000	LPTIM_ISR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DOWN ⁽¹⁾	UP ⁽¹⁾	AROK	CMPOK	EXTTRIG	ARRM	CMPM									
	Reset value																										0	0	0	0	0	0	0									
0x004	LPTIM_ICR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DOWNCF ⁽¹⁾	UPCF ⁽¹⁾	ARROKCF	CMPOKCF	EXTTRIGCF	ARRMCF	CMPMCF									
	Reset value																										0	0	0	0	0	0	0									
0x008	LPTIM_IER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DOWNIE ⁽¹⁾	UPIE ⁽¹⁾	ARROKIE	CMPOKIE	EXTTRIGIE	ARRMIE	CMPMIE									
	Reset value																										0	0	0	0	0	0	0									
0x00C	LPTIM_CFGR	Res.	Res.	Res.	Res.	Res.	Res.	ENC ⁽¹⁾	COUNTMODE	PRELOAD	WAVEPOL	WAVE	TIMOUT	TRIGEN				TRIGSEL[2:0]		Res.		PRESC				TRGFLT		Res.		CKFLT		CKPOL	CKSEL									
	Reset value							0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0									
0x010	LPTIM_CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RSTARE	COUNTSTR	CNTSTRT	SNGSTRT	ENABLE									
	Reset value																											0	0	0	0	0	0									
0x014	LPTIM_CMP	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CMP[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0									
0x018	LPTIM_ARR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ARR[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1								
0x01C	LPTIM_CNT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CNT[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0									
0x024	LPTIM_CFGR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IN2SEL[1:0]				Res.	Res.	IN1SEL[1:0]								
	Reset value																										0	0				0	0	0								

Table 365. LPTIM register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x024	LPTIM3_CFGR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IN1SEL	
	Reset value																														0	0	

1. If LPTIM does not support encoder mode feature, this bit is reserved. Please refer to [Section 43.3: LPTIM implementation](#).

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.

